
MARROW ED8

Ophthalmology

Comprehensive Question Bank

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Anatomy and Development of Eye

Question 1:

The newborn eye is usually _____.

- a) Myopic
- b) Hypermetropic
- c) Emmetropic
- d) Astigmatic

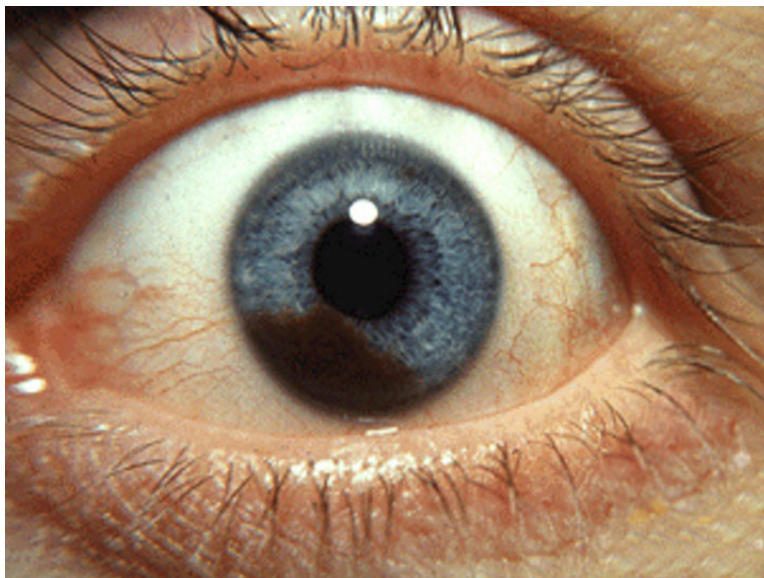
Question 2:

At what age would a child attain full visual acuity?

- a) 6 months
- b) 1 year
- c) 3 years
- d) 6 years

Question 3:

A 45-year-old lady presents with the abnormality as shown below. She is diagnosed with uveal melanoma. What is the embryological origin of the cells involved?



- a) Neural ectoderm
- b) Mesoderm
- c) Surface ectoderm
- d) Neural crest

Question 4:

Which of the following statements is false regarding the development of the eye?

- a) The macula is fully developed at birth.
- b) The lens grows throughout life.
- c) The cornea attains normal diameter by 2 years.
- d) Full visual acuity is attained by 6 years of age.

Question 5:

By how many weeks of embryonic development does the optic groove appear?

- a) 2 weeks
- b) 3 weeks
- c) 4 weeks
- d) 5 weeks

Question 6:

Embryonic development of ciliary muscles is from _____.

- a) Surface ectoderm
- b) Neural ectoderm
- c) Neural crest
- d) Mesoderm

Question 7:

After conception, the canal of Schlemm appears by _____.

- a) 3rd month
- b) 4th month
- c) 5th month
- d) 6th month

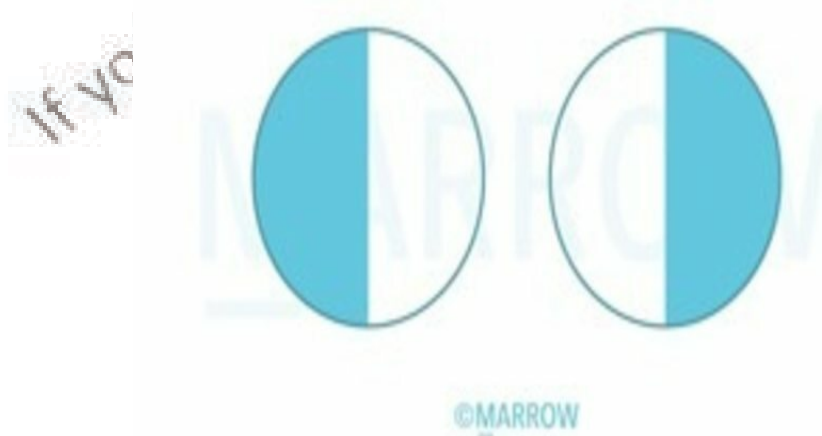
Question 8:

What is the anteroposterior diameter of the adult eyeball?

- a) 2.6 cm
- b) 2.5 cm
- c) 2.4 cm
- d) 2.1 cm

Question 9:

A 34-year-old lady comes to the clinic complaining of problems in her vision. On perimetry, a visual field defect as shown below is obtained. Which of the following vessels does not supply the anatomical location of the lesion?



- a) Anterior cerebral artery
- b) Anterior communicating artery
- c) Internal carotid artery
- d) Middle cerebral artery

Question 10:

A college student comes to the OPD for refractive error testing. Her eyes are dilated using tropicamide. Which of the following statements is false about the structure this drug acts on?

- a) It is triangular in cross section.
- b) There are around 30-40 processes in pars plana and pars plicata
- c) Its epithelium is derived from the anterior edge of the optic cup.
- d) The pars plana is approximately 4.5 mm from the limbus.

Question 11:

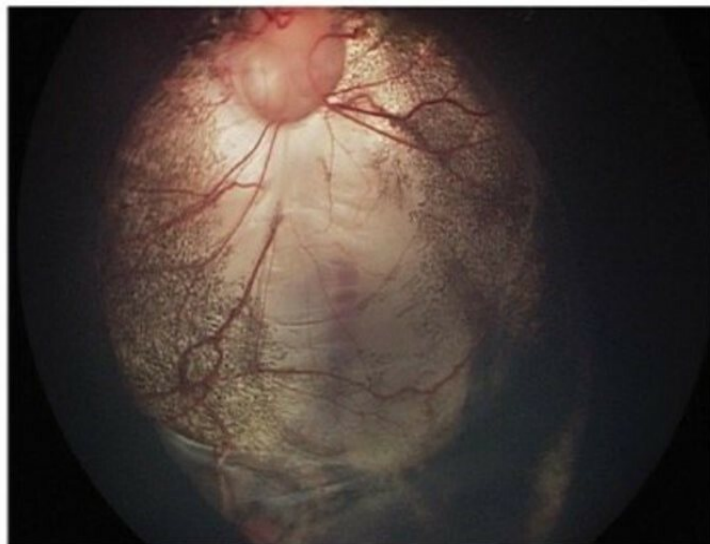
A 12-year-old girl you examined in an outreach camp is found to have this abnormality. What is the most common site where the condition is seen?



- a) Superonasal
- b) Inferonasal
- c) Superotemporal
- d) Inferotemporal

Question 12:

A child was brought with complaints of decreased vision. Fundus examination shows a developmental anomaly as shown below. What is the cause of this anomaly?



- a) Incomplete closure of the embryonic fissure
- b) Faulty migration of neural crest cells
- c) Exuberant proliferation of glial tissue
- d) Abnormal retinal differentiation

Question 13:

What is the embryological origin of the structure marked in the image?



- a) Neural ectoderm
- b) Mesoderm
- c) Surface ectoderm
- d) Neural crest

Question 14:

The optic foramen is situated between the:

- a) Lesser and greater wings of the sphenoid
- b) Lesser wing and body of the sphenoid
- c) Greater wing and body of the sphenoid
- d) Two roots of the lesser wing of the sphenoid

Question 15:

During a game, a 26-year-old baseball player is hit in the head by a baseball, which fractures the optic canal. Which of the following pairs of structures is most likely to be damaged?

- a) Optic nerve and ophthalmic vein
- b) Ophthalmic vein and ophthalmic nerve
- c) Ophthalmic artery and optic nerve
- d) Ophthalmic nerve and optic nerve

Question 16:

The anterior segment of the eye includes all except _____.

- a) Anterior chamber
- b) Posterior chamber
- c) Vitreous
- d) Lens

Question 17:

The posterior chamber in the eyeball is filled with _____.

- a) Aqueous humor
- b) Vitreous humor
- c) Lymph
- d) Blood

Question 18:

Which of the following structures is attached to Whitnall's tubercle?

- a) Check ligament of the medial rectus
- b) Superior oblique tendon
- c) Check ligament of the lateral rectus
- d) Whitnall's ligament

Question 19:

Which of the following statements is false regarding the tarsal plates?

- a) Tarsal plates are cartilaginous structures.
- b) They give structural support to the eyelid.
- c) LPS aponeurosis is attached to the superior border of the upper tarsal plate.
- d) Their medial and lateral ends form the medial and lateral palpebral ligaments, respectively.

Question 20:

A 55-year-old man is being evaluated for herpes zoster ophthalmicus. On eliciting corneal reflex, there is no response. Which of the following nerves forms the afferent limb of this reflex?

- a) Infratrochlear nerve
- b) Supratrochlear nerve
- c) Supraorbital nerve
- d) Long ciliary nerve

Question 21:

Which intrinsic ocular muscle has a parasympathetic innervation?

- a) Superior rectus
- b) Superior oblique
- c) Sphincter pupillae
- d) Dilator pupillae

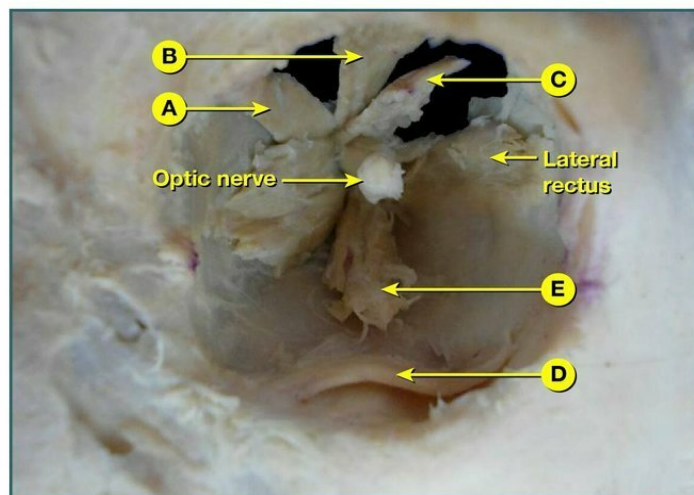
Question 22:

What is the thinnest part of the retina?

- a) Foveola
- b) Fovea centralis
- c) Optic disc
- d) Ora serrata

Question 23:

In the anatomy dissection hall, you are shown the dissected orbit with the markers as shown below. You are asked to choose the muscles causing depression of the eyeball. Which will you choose?



- a) A and E
- b) B, C, and D
- c) B, D, and E
- d) B and E

Answer Key

Question No.	Correct Option
1	b
2	d
3	d
4	a
5	b
6	c
7	b
8	c
9	d
10	b
11	b
12	a
13	c
14	d
15	c
16	c
17	a
18	c
19	a
20	d
21	c
22	d
23	a

Detailed Explanations

Solution to Question 1:

A newborn is usually hypermetropic by +2D to +3D.

The anteroposterior (AP) diameter of the eyeball at birth is about 16.5 mm, which is about 70% of the adult size. This reduced axial length causes light to converge behind the retina, leading to

hypermetropia.

Additionally, the corneal diameter at birth is 10 mm. The adult size (11.7 mm) is attained by 2 years of age. The orbit is more divergent (50°) as compared to adult (45°).

Note: The lacrimal glands are underdeveloped in a newborn and, hence, do not secrete tears.

Solution to Question 2:

Full visual acuity is attained by the age of 6 years.

Important milestones in visual development are:

- At birth, there is no central fixation, and the eyes move randomly.
- By the first month of life, the fixation reflex starts developing and becomes established by 6 months.
- By 3 months, binocular vision is established.
- By 6 months, macular stereopsis and accommodation reflex are fully developed.
- By 6 years of age, full visual acuity (6/6) is attained.

Solution to Question 3:

Uveal melanomas arise from uveal melanocytes. Uveal and conjunctival melanocytes are derived from the neural crest cells.

Solution to Question 4:

The macula is fully developed by 4–6 months of age. Apart from the macula, the rest of the retina is fully developed at birth.

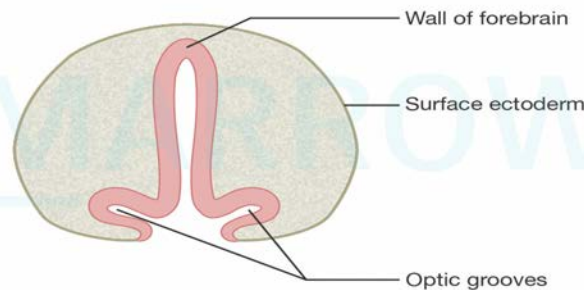
Postnatal Development of Eye:

- 3 Months - Binocular single vision develops.
- 4-6 Months - Macula fully develops. Fusional reflexes, stereopsis, and accommodation become well developed
- 6 Months - Fixation, which had started developing in the first month is now complete
- 2 Years - Cornea develops to the normal adult diameter
- 6 Years - Visual acuity fully develops and reaches 6/6
- The lens, however, continues to grow throughout the life

Solution to Question 5:

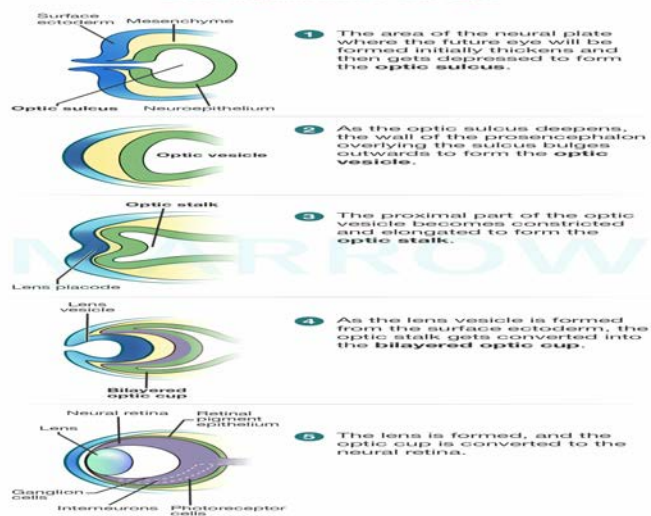
The optic groove appears by 3 weeks. By approximately day 22, 2 small grooves appear on either side of the developing forebrain in the neural folds, these are called optic grooves. As the neural tube closes, these grooves develop outpouchings called optic vesicles. These optic vesicles extend from the forebrain to the surface ectoderm.

Transverse section through the developing forebrain- Day 22



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Development of the eye



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Solution to Question 6:

Embryonic development of ciliary muscles is from the neural crest.

Solution to Question 7:

The canal of Schlemm appears by the 4th month after conception.

Solution to Question 8:

The anteroposterior (AP) diameter of the adult eye is 24 mm/2.4 cm.

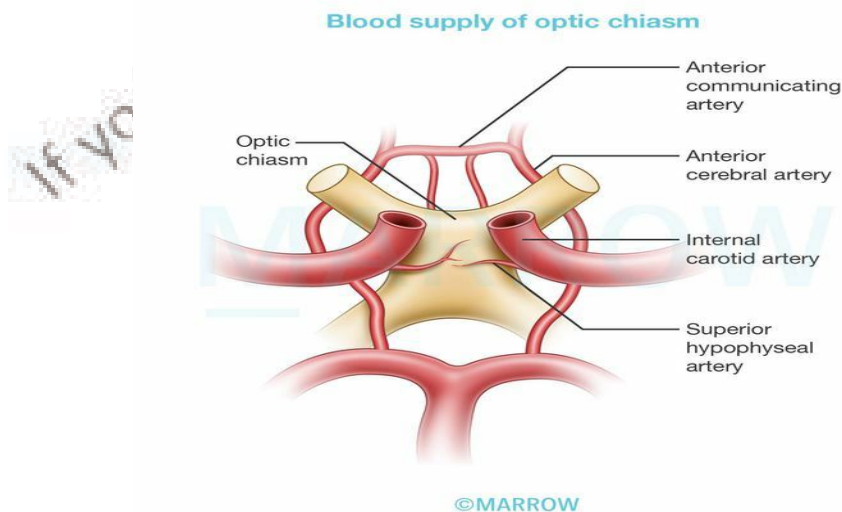
At birth, the average axial length of the eyeball is 16.5 mm. By 3 years, it grows rapidly to approximately 23 mm. From 3 to 14 years of age, the axial length gradually increases by a further 1 mm, bringing it to the adult size of 24 mm. This overall adjustment in the eye size, including the refractive power, is called emmetropization.

Solution to Question 9:

The image shows bitemporal hemianopia, a defect caused by optic chiasmal lesions. The middle cerebral artery does not supply the optic chiasma.

Blood supply of optic chiasma:

- Anterior cerebral artery - dorsal chiasma
- Anterior communicating artery - ventral chiasma
- Internal carotid artery - ventral chiasma
- Superior hypophyseal artery - ventral (branch of internal carotid artery).



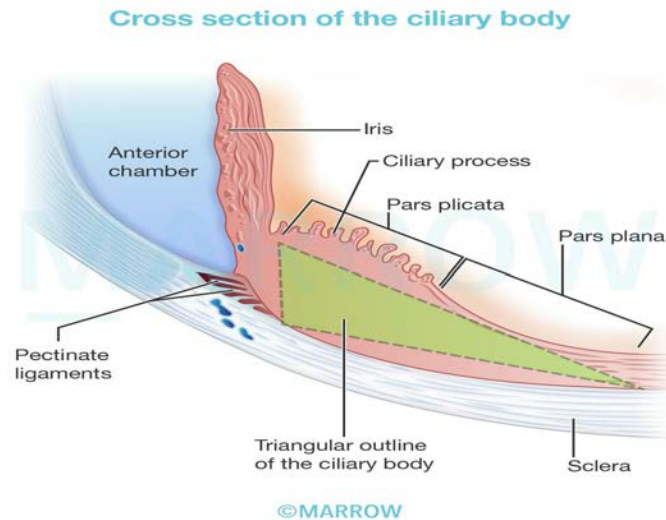
Solution to Question 10:

Tropicamide is a cycloplegic drug that acts on the ciliary body to cause mydriasis for refractory error testing. The ciliary body is a triangular structure that contains 60-70 ciliary processes in

each eye, present on the pars plicata.

The pars plana is 4.5mm from the limbus while the distance between the equator and the limbus is 14mm.

The ciliary epithelium is derived from the optic cup (neuroectoderm) by the continued growth of the outer lip's anterior edge.



Solution to Question 11:

The condition depicted in the image is a coloboma of the iris, and the most common site of coloboma is the inferonasal quadrant.

A coloboma is a congenital malformation in one of the structures in the eye, such as the iris, retina, choroid, or optic disc, due to the failure of embryonic fissure closure. Coloboma of the uveal tract may be typical (seen in the inferonasal quadrant) or atypical (occasionally found in other positions). Typical coloboma is of two types:

- Complete: Extends from pupil to optic nerve, causing a corresponding lens indentation.
- Incomplete: May involve iris alone, or iris and ciliary body (more common), or iris, ciliary body and a part of choroid.

Other embryonic remnants in the eye:

- Mittendorf's dot - Remnant of the anterior end of the hyaloid artery
- Bergmeister papilla - Remnant of the posterior end of the hyaloid artery
- Persistent hyperplastic primary vitreous (PHPV) - Failure of fetal vasculature to regress

The image below shows a funduscopy view of persistent hyperplastic primary vitreous.



Solution to Question 12:

The image shows a coloboma of the optic disc, accompanied by a large chorioretinal coloboma located below the optic disc, which is caused by incomplete closure of the embryonic fissure.

This developmental anomaly may involve the optic nerve, choroid, retina, and iris. Coloboma of the optic disc can present in two forms: a minor effect often associated with refractive errors like hypermetropia and astigmatism and a fully developed coloboma that appears as a large inferonasal whitish excavation resembling the optic disc.

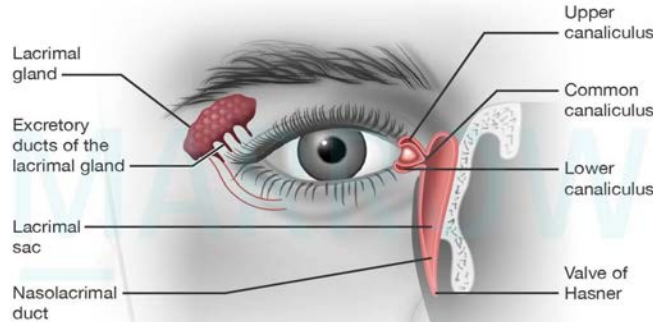
Defective vision and superior visual field defects are typically associated with fully developed coloboma. Additionally, this condition may be associated with chorioretinal coloboma, a congenital anomaly where the developing optic cup fails to fuse completely, leading to abnormal choroidal and retinal development.

Note: Faulty migration of neural crest cells results in anterior segment dysgenesis.

Solution to Question 13:

The lacrimal gland arises from the surface ectoderm.

Lacrimal Apparatus



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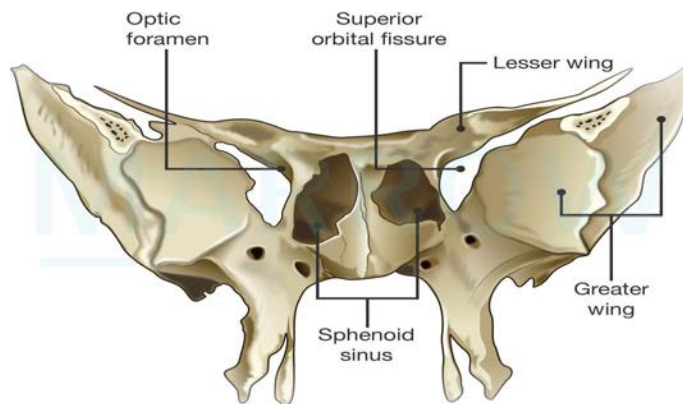
Solution to Question 14:

The optic foramen is situated between the two roots of the lesser wing of the sphenoid.

The optic foramen leads into the optic canal, which is formed medially by the body and laterally by the lesser wing of the sphenoid bone, connecting the orbit's apex to the middle cranial fossa.

The optic canal transmits the optic nerve with its meningeal coverings; the ophthalmic artery, which is below the nerve, is embedded in its dural sheath, and a few twigs from the sympathetic plexus accompanying the artery.

Sphenoid bone



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Solution to Question 15:

In the given clinical scenario, a fracture of the optic canal is most likely to damage the ophthalmic artery and optic nerve, as these are the primary structures passing through it.

The optic foramen leads into the optic canal, which is a short cylindrical structure bounded medially by the body and laterally by the lesser wing of the sphenoid bone, connecting the orbit's apex to the middle cranial fossa, serving as a pathway for the aforementioned structures along with its sympathetic plexus.

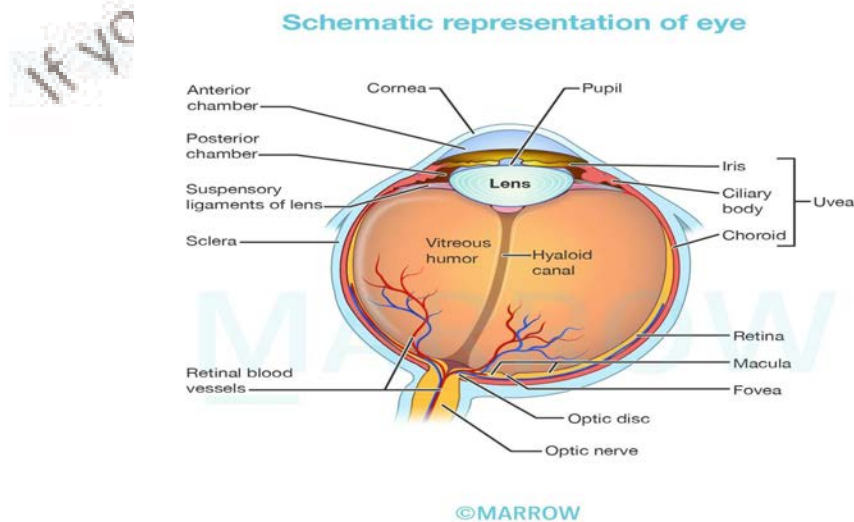
Optic canal fractures, often associated with basilar skull fractures, can lead to optic neuropathy. The intracanalicular and intracranial portion of the optic nerve is most commonly affected in head trauma.

Note: The greater and lesser wings of the sphenoid bound the superior orbital fissure.

Solution to Question 16:

The anterior segment of the eye does not include vitreous humor as it is a constituent of the posterior segment of the eye.

Anterior Segment	Posterior Segment
Cornea	Vitreous humor
Iris	Retina
Lens	Choroid
Anterior and Posterior chambers with aqueous humor	Optic Disc



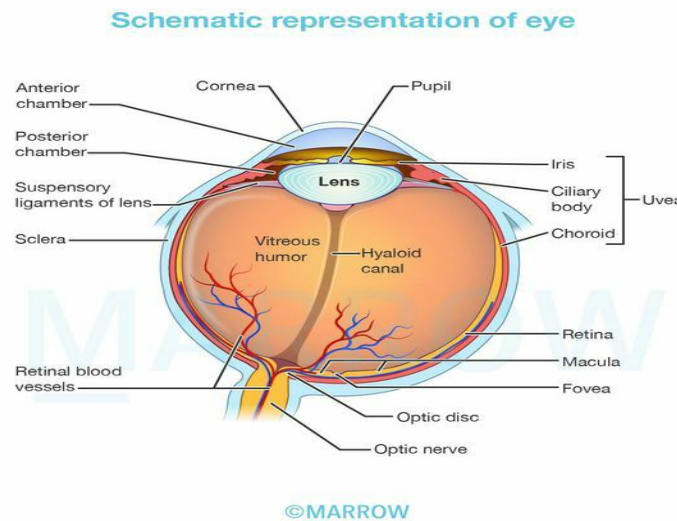
Solution to Question 17:

The posterior chamber of the eyeball is filled with aqueous humor.

The eyeball is divided into anterior and posterior segments. The anterior segment includes the lens and the structures anterior to it, such as the cornea, iris and two aqueous humor-filled spaces:

- Anterior chamber: Located between cornea and iris, contains aqueous humor and communicates with the posterior chamber through the pupil.
- Posterior chamber: Triangular-shaped space, also contains aqueous humor, bounded anteriorly by the posterior surface of the iris, laterally by the ciliary body, and posteriorly by the lens.

The posterior segment includes all structures located posterior to the lens, comprising of the vitreous humor, retina, choroid and optic disc.



Solution to Question 18:

Check ligament of the lateral rectus is attached to Whitnall's tubercle.

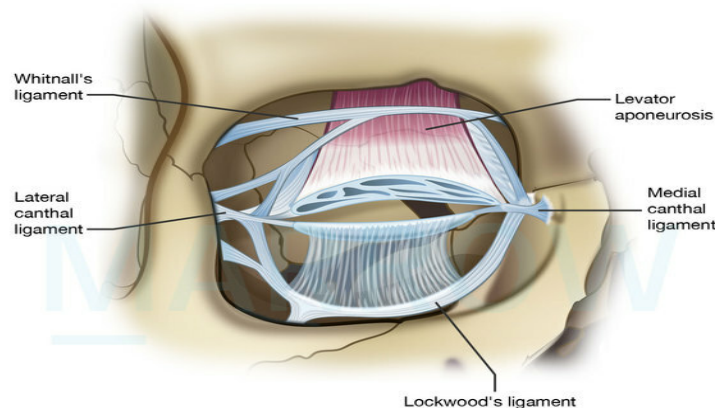
Lateral orbital tubercle or Whitnall's tubercle is a small elevation on the orbital surface of the zygomatic bone, just within the lateral orbital margin. It is about 11 mm below the fronto-zygomatic suture.

Four structures are attached to it:

- Check ligament of the lateral rectus
- Lockwood's ligament (Suspensory ligament of the eyeball)
- Lateral palpebral ligament
- Lateral horn of the levator aponeurosis

The Whitnall's ligament or the superior transverse ligament of the eye, is attached to the superior tarsal plate and the orbital bone.

Fascial expansions of the eyeball



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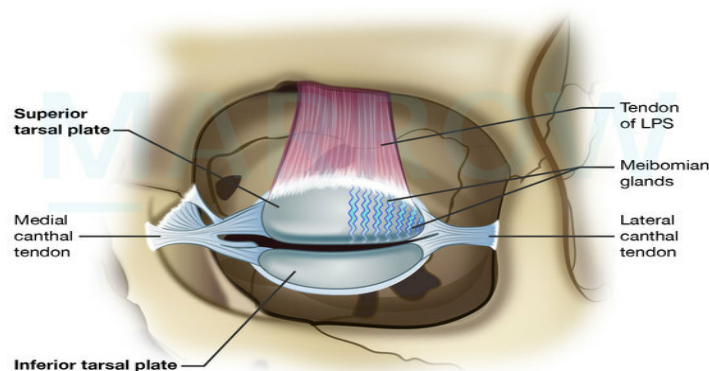
Solution to Question 19:

The tarsal plates are not cartilaginous structures but firm, dense connective tissue (fibrous tissue) plates.

The tarsal plates serve as the structural support of the eyelids. The height of the upper tarsal plate is about 10 mm, and that of the lower tarsal plate is 4 mm. Both plates have a thickness of about 1 mm. The ends form the medial and lateral palpebral ligaments (medial and lateral canthal tendons). The levator palpebrae superioris (LPS) aponeurosis is attached to the superior border of the upper tarsal plate.

The meibomian glands are located within the tarsus. These are holocrine-secreting sebaceous glands that are not associated with lash follicles.

Tarsal plates



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Solution to Question 20:

The sensory supply of the cornea comes from the long ciliary nerve (branch of the nasociliary nerve). It forms the afferent limb of the corneal reflex.

The sensory nerve supply of the eye is by the ophthalmic nerve.

- Lacrimal nerve- lacrimal gland, conjunctiva, lateral eyelid
- Frontal nerve- conjunctiva, eyelid, part of the forehead
- Nasociliary nerve
- Long ciliary nerve- cornea, ciliary body, iris
- Infra-trochlear nerve- conjunctiva, lacrimal sac, caruncle, medial part of the eyelids

Corneal sensation is reduced in leprosy and diabetes mellitus (neuropathy).

Solution to Question 21:

The intrinsic ocular muscle supplied by parasympathetic innervation is the sphincter pupillae.

1. Parasympathetic innervation:

- Sphincter pupillae: Short ciliary nerve–Ciliary ganglion
- Ciliary muscle–Accessory ganglion

2. Sympathetic innervation. Dilator pupillae and Muller's muscle

3. Motor Supply:

- 3rd nerve: All extraocular muscles (except the 2 given below)
- 6th nerve: Lateral rectus (LR6)
- 4th nerve: Superior oblique (SO4)
- 7th nerve: Orbicularis oculi

Solution to Question 22:

The retina is the thinnest (0.080 mm) at the ora serrata, and thickest, at the foveal rim (0.23, 0.320 mm).

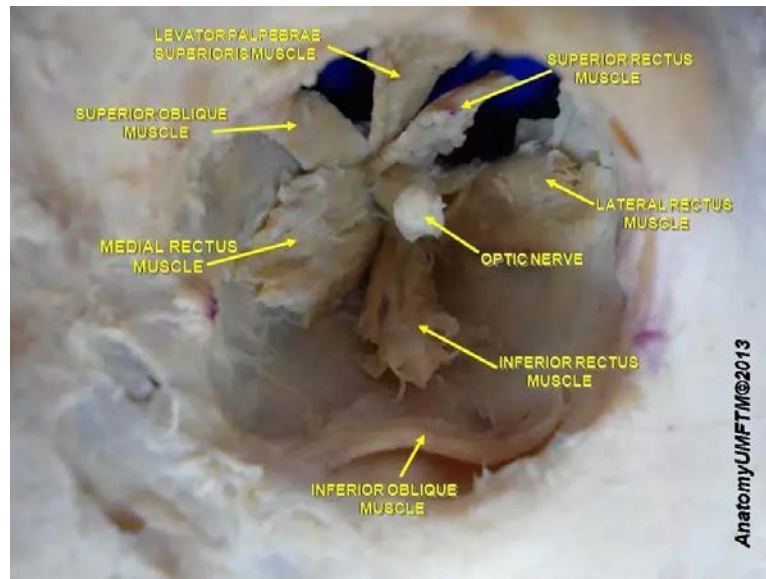
The ora serrata is the serrated junction between the retina and ciliary body.

Solution to Question 23:

The muscles responsible for the depression of the eyeball are the inferior rectus (E) and superior oblique (A).

Image markings:

- A- Superior oblique
- B- Levator palpebrae superioris
- C- Superior rectus
- D- Inferior oblique
- E- Inferior rectus



Nerve supply of extra-ocular muscles:

- Oculomotor nerve (III)- Superior rectus, inferior rectus, medial rectus, inferior oblique, and levator palpebrae superioris
- Abducens nerve (VI)- Lateral rectus
- Trochlear nerve (IV)- Superior oblique

Elementary Optics and Physiology of Vision

Question 1:

What is the most important factor influencing the focus of light rays on the retina?

- a) Anterior surface of cornea
- b) Posterior surface of cornea
- c) Anterior surface of lens
- d) Posterior surface of lens

Question 2:

The refractive index is the maximum at the _____.

- a) Anterior surface of the lens
- b) Posterior surface of the lens
- c) Cornea
- d) Centre of the lens

Question 3:

How many Purkinje images are formed in a normal eye?

- a) 3
- b) 4
- c) 5
- d) 2

Question 4:

Among the Purkinje images in the eye, which image is inverted?

- a) Image 1
- b) Image 2
- c) Image 3

d) Image 4

Question 5:

The visual axis is the line joining the _____.

- a) Centre of the cornea to the retina
- b) Object to fovea
- c) Centre of the lens to the cornea
- d) Object to center of rotation

Question 6:

What is the angle formed between the optic axis and visual axis at the nodal point?

- a) α
- b) $^\circ$
- c) κ
- d) γ

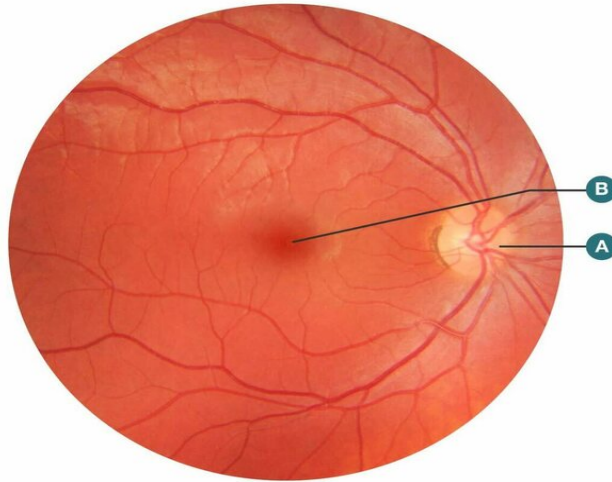
Question 7:

A 28-year-old man comes to get his power checked. The optometrist finds the focal length of his lens to be 0.75m; what will be its refractive power?

- a) 0.75 D
- b) 1.5 D
- c) 1.25 D
- d) 1.33 D

Question 8:

What is the distance between point A and B shown below?



- a) 2 DD
- b) 3 DD
- c) 1.5 DD
- d) 1 DD

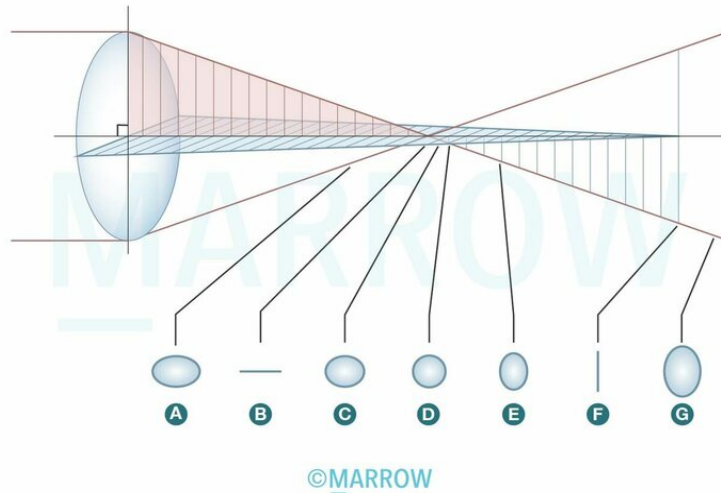
Question 9:

A 6-year-old child complains that she cannot see anything on the whiteboard clearly. On ophthalmological examination, she is found to have a 2 mm shortening of the axial length of the eyeball. How much eye power will this lead to?

- a) 3D myopia
- b) 6D myopia
- c) 3D hypermetropia
- d) 6D hypermetropia

Question 10:

You are studying the optics of a patient with regular astigmatism. Identify the point at which the clearest image is formed?



- a) C
- b) D
- c) E
- d) G

Question 11:

What is the Stiles-Crawford effect?

- a) Cutting off of the peripheral rays by the iris
- b) High refractive index of the core of nucleus of the lens than that of the peripheral cortex
- c) Low sensitivity of the peripheral retina
- d) More sensitivity of the retina to perpendicular rays than the oblique rays

Question 12:

What are the first-order neurons of the optic pathway?

- a) Photoreceptors
- b) Ganglionic cells
- c) Bipolar cells
- d) Optic radiation

Question 13:

You are tasked with studying and mapping the photoreceptor cells in a cohort of patients. Consider the following statements and pick out the false one.

- a) There are 120 million rods in each human eye.
- b) Rods produce a hyperpolarizing potential on activation.
- c) Cones predominate in the extrafoveal region.
- d) Cones produce a hyperpolarizing potential on activation.

Question 14:

Which of the following cells are retinal glial cells?

- a) Meissner's corpuscles
- b) Muller cells
- c) Amacrine cells
- d) Basket cells

Question 15:

Which of the following statements is false?

- a) Rods and cones release Ach at their synapses.
- b) Phototransduction is associated with hyperpolarization.
- c) Amacrine cells release various neurotransmitters.
- d) Bipolar cells show either depolarization or hyperpolarization.

Question 16:

Which of the following is responsible for inducing phototransduction?

- a) Metarhodopsin I
- b) Metarhodopsin II
- c) Bathorhodopsin
- d) Lumirhodopsin

Question 17:

During the dark phase of the visual cycle, which form of vitamin A combines with opsin to form rhodopsin?

- a) All trans-retinal
- b) All trans-retinol
- c) 11- cis-retinal
- d) 11- cis-retinol

Question 18:

Which of the following statements is false regarding the physiology of photoreceptors?

- a) Phototransduction results in closure of c-GMP-gated cation channels.
- b) Chromophore regeneration occurs in the outer segment of photoreceptors.
- c) Photon absorption induces photo-isomerisation of 11-cis-retinal to all-trans-retinal.
- d) 11-cis-Retinal is bound to opsins by covalent bonds.

Question 19:

What signals are carried by the parvocellular pathway from the lateral geniculate body to the visual cortex?

- a) Movement, depth, and flicker
- b) Colour vision, shape, and fine details
- c) Temporal frequency
- d) Luminance contrast

Question 20:

Which of the following statements is false regarding "blobs" in the primary visual cortex?

- a) Process colour vision
- b) Are rich in the enzyme citrate synthase
- c) Are present in layers 2 and 3 of the visual cortex
- d) Receive input from the interlaminar pathway

Question 21:

What are Konio cells involved in transmitting?

- a) Motion
- b) Stereo
- c) Form
- d) Color

Question 22:

What is the visible range of the electromagnetic spectrum of the human eye?

- a) 200-370 nm
- b) 370-740 nm
- c) 740-1140 nm
- d) 800-900 nm

Question 23:

What is the characteristic visual defect in deuteranomaly?

- a) Green color is not perceived.
- b) Blue color is not perceived.
- c) Green color perception is defective.
- d) Blue color perception is defective.

Question 24:

Majority of the energy requirement in the lens is met by the _____.

- a) Glycolysis
- b) Pentose phosphate pathway
- c) TCA cycle
- d) Sorbitol pathway

Question 25:

What is the angle subtended by the optotype on the second line of Snellen's chart when viewed from a distance of 6 meters?

- a) 5 minutes of arc
- b) 30 minutes of arc
- c) 25 minutes of arc
- d) 6 minutes of arc

Question 26:

A 42-year-old man with monocular aphakia wears contact lenses. What is the approximate percentage of aniseikonia in his eyes?

- a) 3%
- b) 8%
- c) 30%
- d) 1%

Question 27:

Which cerebellar nucleus controls the ocular saccade?

- a) Dentate
- b) Emboliform
- c) Flocculus
- d) Globose

Question 28:

Purely depolarizing cells in the optic pathway are _____.

- a) Amacrine cells
- b) Rods and cones
- c) Bipolar cells
- d) Horizontal cells

Question 29:

Most afferent fibers from the lateral geniculate body terminate in the primary visual cortex in _____.

- a) Layer 1
- b) Layers 2 and 3
- c) Layer 4
- d) Layers 5 and 6

Question 30:

What is the mean value for outflow facility of the aqueous in the normal eye?

- a) 2.3 $\mu\text{L}/\text{min}/\text{mm Hg}$
- b) 0.06 $\mu\text{L}/\text{min}/\text{mm Hg}$
- c) 2.8 $\mu\text{L}/\text{min}/\text{mm Hg}$
- d) 0.28 $\mu\text{L}/\text{min}/\text{mm Hg}$

Answer Key

Question No.	Correct Option
1	a
2	d
3	b
4	d
5	b
6	a
7	d
8	a
9	d
10	b
11	d
12	a
13	c
14	b
15	a

16	b
17	c
18	b
19	b
20	b
21	d
22	b
23	c
24	a
25	b
26	b
27	c
28	a
29	c
30	d

Detailed Explanations

Solution to Question 1:

The anterior surface of the cornea is the most important factor influencing the focus of light rays on the retina.

The refraction of light is the maximum at the cornea, due to the curvature of the cornea and the difference between the refractive index of air and tear film.

Refractive indices of various surfaces of the eye:

Surface	Refractive Index
Cornea	1.376
Lens- Total	1.40
Lens- Cortex (periphery)	1.386
Lens- Nucleus	1.406 (maximum)
Aqueous Humor	1.336
Vitreous Humor	1.336

Solution to Question 2:

The refractive index is the maximum at the center of the lens.

Refractive indices of the various surfaces of the eye:

The total dioptric power of the lens is about 18 D (16 to 20 D). Lens is composed of 64% water, 35% protein, and 1% lipid/carbohydrates/trace elements, and these components contribute to the refractive index of the lens.

Surface	Refractive Index
Cornea	1.376
Lens- Total	1.40
Lens- Cortex (periphery)	1.386
Lens- Nucleus	1.406 (maximum)
Aqueous Humor	1.336
Vitreous Humor	1.336

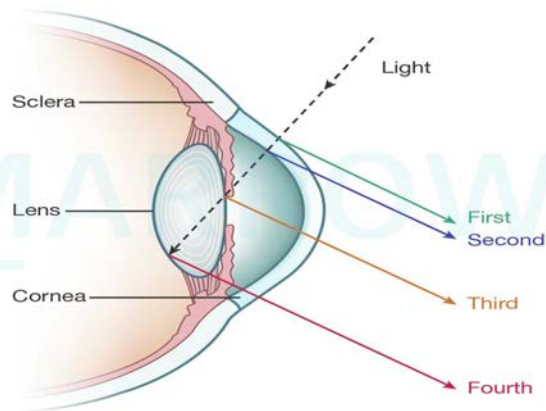
Solution to Question 3:

A total of 4 Purkinje images are formed in normal eyes.

Purkinje images are reflections of objects from the structure of the eye. They are also known as Purkinje reflexes or Purkinje–Sanson images.

Image	Location	Direction
Image 1	Anterior surface of the cornea	Erect
Image 2	Posterior surface of the cornea	Erect
Image 3	Anterior surface of the lens	Erect
Image 4	Posterior surface of the lens	Inverted

Purkinje-Sanson Images



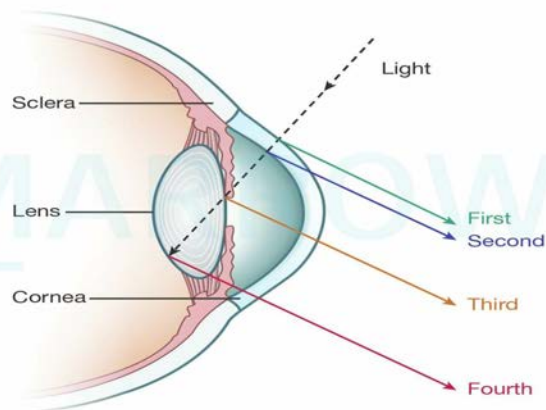
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In aphakia, images 3 and 4 are absent, while in a mature cataract, image 4 is absent.

Solution to Question 4:

Image 4, which is formed on the posterior surface of the lens, is inverted.

Purkinje-Sanson Images



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Solution to Question 5:

The visual axis is the line joining the fixation point (object), nodal point (centre of the lens), and the fovea.

Solution to Question 6:

Angle alpha (α) is the angle between the optical axis and visual axis at the nodal point.

Angle lambda ($^\circ$) is the angle between the pupillary axis and the line of sight. It is the global tilt offset of the eye when it fixates on a specific target and is a response to the eccentric displaced position of the fovea from the geometrical center of the retina. (Option B)

Angle kappa (κ) is the angle between the visual axis and the pupillary line at the cornea. It is the only angle that can be measured and is of clinical significance. (Option C)

- Positive κ is seen in pseudoexotropia
- Negative κ is seen in pseudoesotropia

Angle gamma (γ) is the angle between the optical axis and the fixation axis at the center of eyeball rotation. (Option D)

	Visual axis	Line of sight	Fixation axis
Pupillary axis	Kappa (κ)	Lambda ($^\circ$)	
Optic Axis	Alpha (α)		Gamma (γ)

Solution to Question 7:

Refractive power (dioptre) = $1/\text{Focal length in metres}$

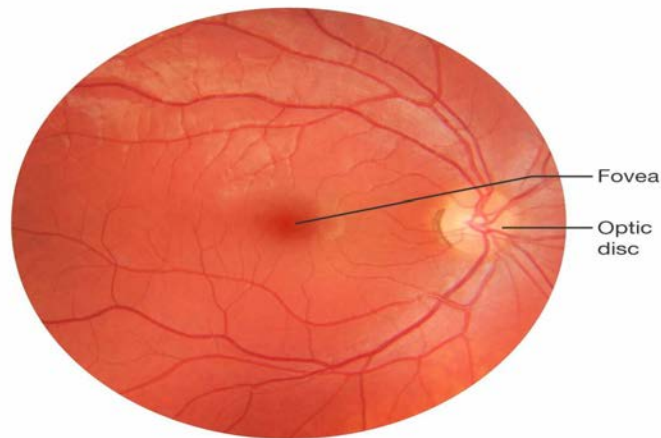
So, for the given question, refractive power (D) = $1/0.75 = 1.33$ D

Solution to Question 8:

In the image, point A is the optic disc and point B is the fovea. The fovea is situated about 2 disc diameters (3–4 mm) away from the temporal margin of the optic disc.

The optic disc is an area of about 1.5 mm. There is a blind spot / physiological scotoma, corresponding to the optic disc.

Normal fundus of the right eye



Solution to Question 9:

Shortening of 2mm of the axial length of the eyeball will lead to 6D of hypermetropia.

In axial hypermetropia, 1mm shortening of the anteroposterior diameter of the eye results in 3 dioptres of hypermetropia. Therefore, 2-mm shortening will lead to 6D of hypermetropia.

In curvature hypermetropia, a 1mm increase in radius of curvature results in 6 dioptres of hypermetropia.

Solution to Question 10:

The given figure is a Sturm's conoid. It represents a bundle of light rays formed by a regularly astigmatic surface. The image is clearest or least confusing as point D, the circle of least confusion.

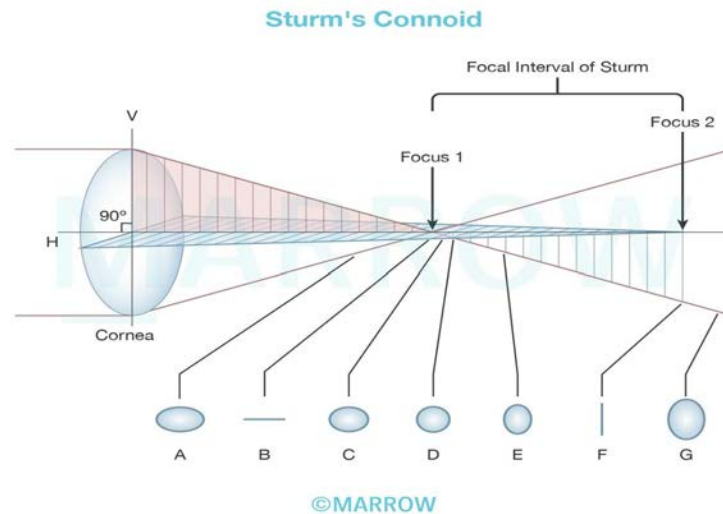
When light passes through a regularly astigmatic surface (surface with toric curvature), vertical and horizontal rays converge at different points.

The distance between the points of convergence (focus) of the vertical and horizontal rays is called the focal interval of Sturm. (distance between points B and F). The circle of least confusion/ circle of diffusion lies at the mid-point of this interval.

For the given Sturm's conoid,

- At point A, a horizontal oval is formed because vertical rays converge more rapidly than horizontal rays.
- At point B, vertical rays come to a focus, while horizontal rays are still converging; so, it will be a horizontal straight line.
- At points C, D, and E, the vertical rays are diverging and horizontal rays are still converging.

- At point D, the vertical rays have diverged as much as the horizontal rays have converged; so, we get a perfect circle called the circle of least confusion.
- At point E, a vertical oval is formed.
- At point F, a vertical straight line is formed
- At point G and beyond, both rays are diverging, and we always get a vertical oval/prolate ellipse.



Solution to Question 11:

The Stiles–Crawford effect refers to the greater sensitivity of the retina to perpendicular rays than the oblique rays. It is one of the natural mechanisms that decrease aberration in the eye. The rays of light that enter through the center of the pupil have higher sensitivity as compared to those entering from the periphery of the pupil.

Natural mechanisms that decrease aberrations in the human eye:

- Cutting-off of the peripheral rays by the iris
- The high refractive index of the core of the nucleus of the lens than that of the peripheral cortex
- Low sensitivity of the peripheral retina
- Stiles–Crawford effect

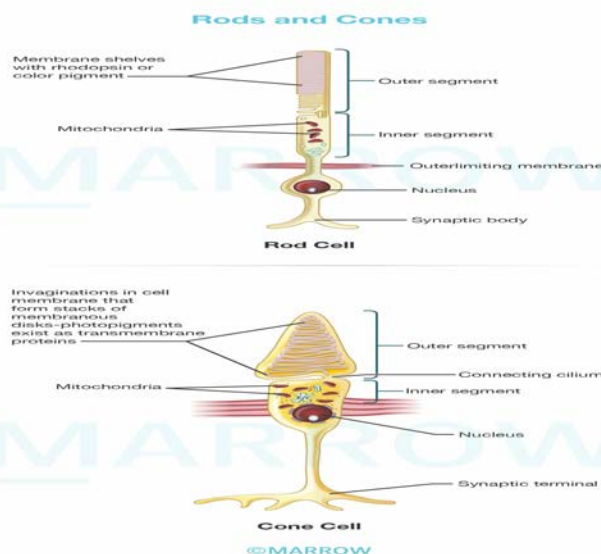
Solution to Question 12:

Photoreceptors are the first-order neurons in the visual pathway.

Solution to Question 13:

Cones predominate in the foveal region, not the extrafoveal region.

Rods	Cones
120 million in number in each human eye	6 million in number in each human eye
Receptors for dim/night vision	Receptors for day vision
Predominate in the extra-foveal region	Predominate in the foveal region
More sensitive to light	Less sensitive to light
Contain pigment rhodopsin	Contain pigment iodopsin
Produce hyperpolarizing potential changes on activation	Produce hyperpolarizing potential changes on activation



Solution to Question 14:

Müller cells are a type of glial cells found in the retina that help in the binding of the neuronal elements together. They do not take part in the neurosensory pathway of light.

Other cells in the retina:

Meissner's corpuscles: Tactile receptors localizing in the dermis between epidermal ridges, perceiving touch

Basket cells: GABAergic interneurons found in the cerebellum

Cells	Use	Potential	Comments
Rods and cones	Photoreceptor cells	Hyperpolarizing	1st order neurons

Cells	Use	Potential	Comments
Bipolar cells	Connect photoreceptors to ganglion cells	Depolarizing or hyperpolarizing	2nd order neurons
Amacrine cells	Adjust the relative color and luminosity inputs under changing light conditions.	Depolarizing	Interneurons
Ganglion cells	Form the visual pathway	Propagated action potential	3rd order neurons
Horizontal cells	Increase visual acuity by lateral inhibition	Hyperpolarizing	Interneurons

Solution to Question 15:

Rods and cones release glutamate (not acetylcholine) at their synapses.

Photoreceptors get hyperpolarized in response to stimulus, and hence, phototransduction is associated with hyperpolarization.

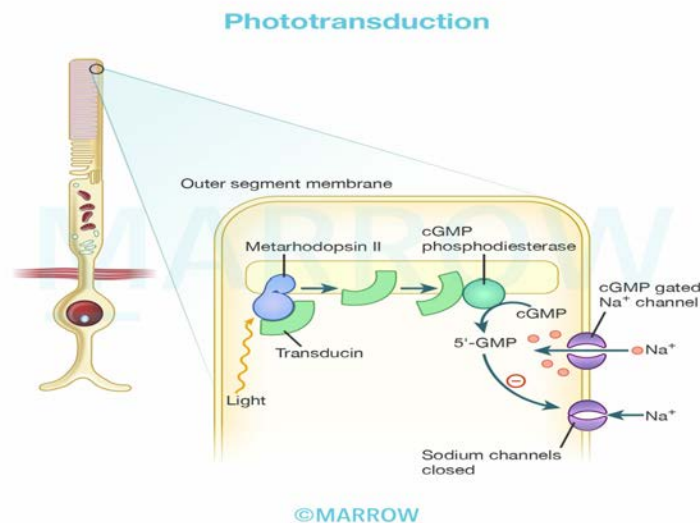
Amacrine cells secrete at least eight types of neurotransmitters, including GABA, glycine, dopamine, acetylcholine, and indolamine, all of which function as inhibitory neurotransmitters.

Bipolar cells show both depolarization/hyperpolarization.

Solution to Question 16:

Metarhodopsin II is responsible for phototransduction.

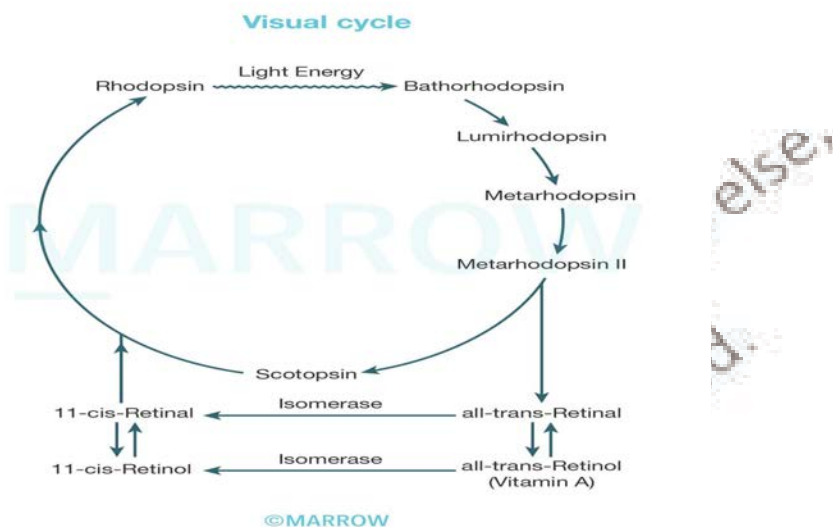
It activates transducin member of G proteins. This leads to the closure of Na⁺ channels and hyperpolarization, thereby inducing phototransduction (signal transduction in the retina).



Solution to Question 17:

Opsin combines with 11-cis-retinal to produce rhodopsin.

Only the retinal form of vitamin A is involved in the formation of the photopigment.



Solution to Question 18:

Chromophore regeneration occurs in the retinal pigment epithelial cells.

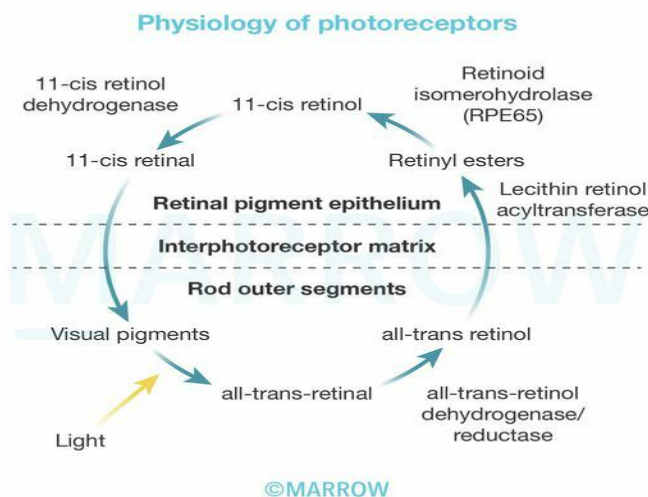
Phototransduction:

- Rhodopsin is a G-protein-coupled receptor found in the outer segments of rod photoreceptors.
- 11-cis-retinal (a chromophore) is covalently bound to opsins via a lysine residue on one of the transmembrane domains.
- The receptor is activated by photo-isomerization of prebound 11-cis-retinal to ALL-TRANS-retinal.
- Activation of the receptor causes binding and activation of the G protein, transducin, with signal transduction via cGMP phosphodiesterase, resulting in a decrease in the intracellular cGMP levels and closure of the cGMP-gated Na channels.
- This causes hyperpolarization of the photoreceptor cell, which modulates neurotransmitter release at the synapse with bipolar cells.
- This entire process triggered by the activation of rhodopsin is phototransduction, which occurs in the outer segment of photoreceptors and converts light energy into an electrical signal.

Chromophore regeneration:

- All-trans retinal is converted to all-trans-retinol.

- All-trans-retinol gets transported by interphotoreceptor retinoid-binding protein (IRBP) to the retinal pigment epithelial cells, where it is reconstituted into the 11-cis retinal isoform.



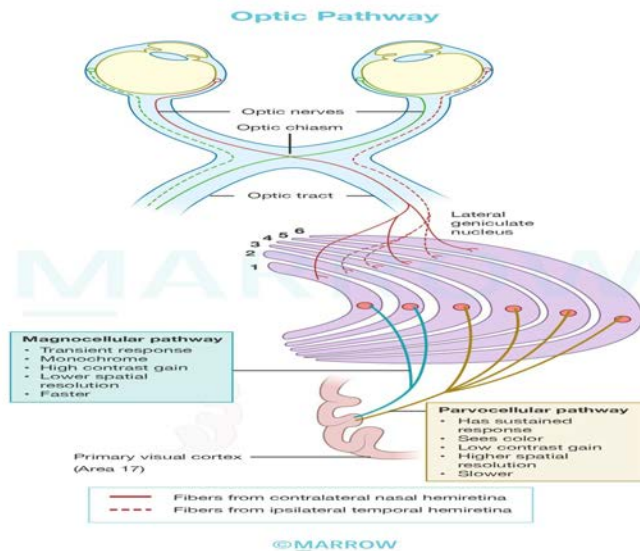
Solution to Question 19:

The parvocellular pathway gives signals for color vision, texture, shape, and fine details.

LGB: lateral geniculate body

The interlaminar pathway gives signals for color vision to layers 2 and 3 of the visual cortex having clusters of cells called blobs.

Feature	Parvocellular Pathway	Magnocellular Pathway
Arises	layers 3, 4, 5, and 6 of LGB	Layers 1 and 2 of LGB
Terminates	Layer 4 of visual cortex	Layer 4 of visual cortex
Use	Color vision, texture, shape, and fine details	Movement, depth, flicker, and spatial organization



Solution to Question 20:

Blobs in the primary visual cortex are rich in the enzyme cytochrome oxidase.

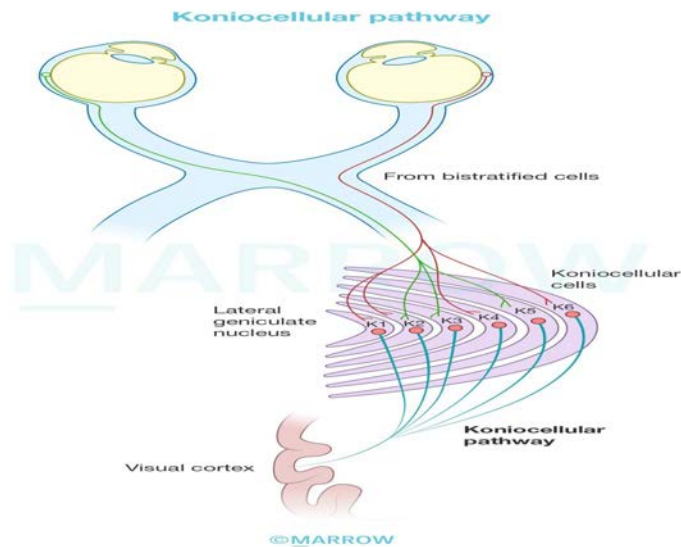
Blobs are sections of the visual cortex where groups of neurons that are sensitive to color assemble in cylindrical shapes. They are present in layers 2 and 3 of the visual cortex.

Solution to Question 21:

Konio cells are involved in the transmission of color.

The koniocellular pathway is ventral to each parvocellular and magnocellular layer of the lateral geniculate nucleus (LGN). K cells are neurochemically and anatomically distinct from M and P cells. They contain calbindin protein, whereas M cells and P cells contain parvalbumin. K1, K4, and K6 receive ipsilateral retinal inputs, and K3 and K5 receive contralateral retinal inputs. K2 receives both.

Lateral geniculate	Striate cortex	Extra-striate cortex	Higher centers for body
Magnocellulars	→	→	Motion and stereo
Parvocellulars	→	→	Form
Konio cells	→	→	Color

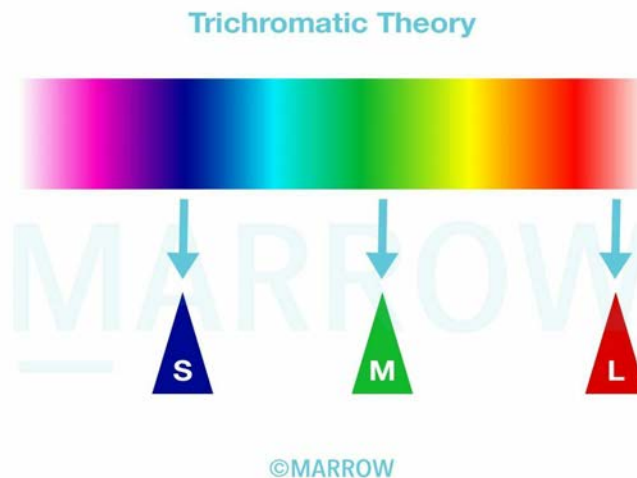


Solution to Question 22:

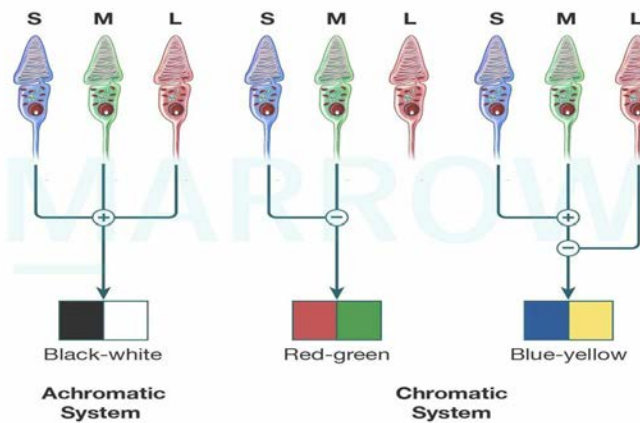
The visible range of the electromagnetic spectrum for the human eye is 400–700 nm.

Theories of color vision:

- Trichromatic theory (Young–Helmholtz theory) assumes that there are three types of color receptors. Each receptor is assumed to respond to all wavelengths, however, each has different spectral sensitivities.
- Opponent process theory, also known as the Hering theory, assumes three sets of receptor systems—red-green, blue-yellow, and black-white. Each one of these functions as an antagonistic pair (e.g., red light stimulates the red receptors and, at the same time, inhibits the green).
- Edwin Land's theory has three separate color systems (retinexes). Each is an analog to a black-and-white picture.



Opponent-Process Theory



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Solution to Question 23:

In deuteranomaly, the perception of green color is defective. It is the most common disorder of color vision.

Monochromacy: Only one of the three primary colors (red, blue, or green) can be appreciated.

Dichromacy: Two primary colors are appreciated. (anopia = absent).

- Protanopia: Red color is not perceived.
- Deuteranopia: Green color is not perceived.
- Tritanopia: Blue color is not perceived.

Trichromacy: All the three primary colors are appreciated, but one of them can be defective (anomaly = defective).

- Protanomaly: Red color perception is defective.
- Deuteranomaly: Green color perception is defective.
- Tritanomaly: Blue color perception is defective.

Achromatopsia: No color perception due to the absence of cones. It may also occur in the lesion of area V4 or V8 of the visual cortex.

92%	Normal Vision	
2.7%	Deuteranomaly	
0.66%	Protanomaly	
0.59%	Protanopia	
0.56%	Deuteranopia	
0.016%	Tritanopia	
0.01%	Tritanomaly	
<0.0001%	Achromatopsia	

Solution to Question 24:

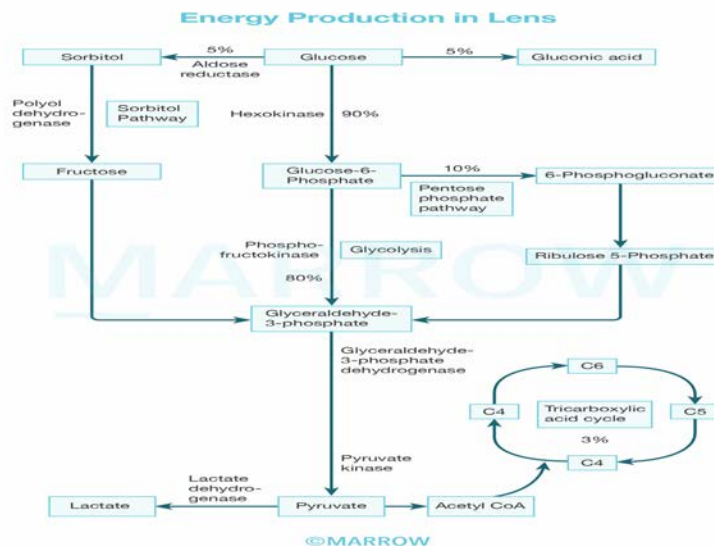
Majority of the energy requirement of the lens is met by glycolysis.

90–95 % of the glucose that enters the lens is converted to glucose 6 phosphate by hexokinase. The remaining 5% is converted to sorbitol and gluconic acid.

The glucose-6-phosphate formed is metabolized in 3 ways:

- Glycolysis: 80 %
- Pentose phosphate pathway: 10%
- TCA cycle: 3%

Energy in the lens is stored in the form of ATP.



Solution to Question 25:

The angle subtended by the optotype (individual letters on the chart) on the second line of Snellen's chart in the above case would be 30 minutes of arc.

The second line on Snellen's chart corresponds to a distance of 36 meters. When viewed from a 6m distance, it is denoted as visual acuity 6/36. Where,

- The numerator (6) denotes the distance from which the patient is actually viewing it
- The denominator (36) denotes the distance at which it should be ideally viewed. It denotes the distance at which each stroke of the letter would subtend the minimum angle of resolution (MAR) of one minute of arc.

MAR = 1/Visual acuity

Each letter is shaped such that it can be placed in a square, the sides of which are five times the breadth of individual limbs of the letters. Hence, the angle subtended by the optotype = MAR $\times$ 5

For eg: At exactly 6 meters distance from the patient, the letters on the 6/6 line shall subtend 5 minutes of arc (such that the individual limbs of the letters subtend 1 minute of arc).

The equation to calculate the minimum angle of resolution (MAR) is given as follows:

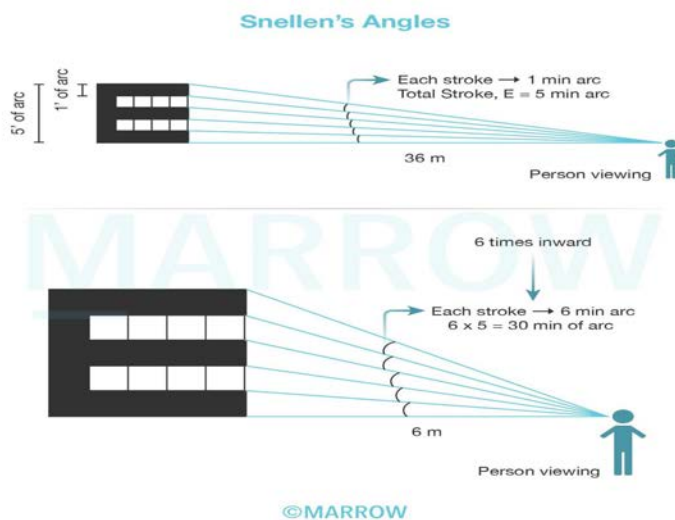
For the given question,

$$\text{MAR} = 1/6/36$$

$$\text{MAR} = 36/6$$

$$\text{MAR} = 6 \text{ minutes of arc}$$

Hence, the angle subtended by the entire letter is $6 \times 5 = 30$ minutes of arc.



Solution to Question 26:

The approximate percentage of aniseikonia in a person with monocular aphakia wearing a contact lens would be 8%.

Aniseikonia is defined as the difference in image size between the two eyes and can lead to disturbances in stereopsis. The tolerable limit of aniseikonia is 5–8%. It is reduced with contact lens (8%) compared to glasses (25–30%) in aphakia.

However, the percentage of aniseikonia is least with an intraocular lens (IOL) (1–3%), thus secondary IOL implantation is the most preferred management of unilateral aphakia.

Solution to Question 27:

The Floccular nucleus of the cerebellum controls ocular saccades.

A saccade is a rapid ballistic movement to change the point of fixation. It is the fastest of all eye movements and is initiated by the contralateral frontal lobe. Smooth pursuit is a slow tracking movement to keep the target on the fovea. It is controlled by the ipsilateral parieto-occipital lobe.

Following an object from the right to left is left pursuit and is controlled by ipsilateral (left) parieto-occipital lobe, and the corrective movement is a saccade (in this case, left to right) termed right saccade and is controlled by the contralateral (left) frontal lobe.

The floccular nucleus is found in the lobule HX of the cerebellar vermis. Its functions include matching pulse-step, gaze holding, and sustained smooth pursuit. Bilateral lesions of the flocculus result in the following:

- Post-saccadic drift- saccadic movements do not stop, but there is a brief, several hundred milliseconds drift of the eyes after each saccade.
- Saccadic pursuit- smooth tracking is replaced by simultaneous rapid jerking motions of the eye to follow an object toward the ipsilateral visual field.
- Gaze-evoked nystagmus- impaired ability to hold the eyes in the eccentric position, resulting in gaze-evoked nystagmus towards the affected side of the cerebellum.

Emboliform and globose nuclei: Generation of vertical saccades

Fastigial nuclei: Pursuit initiation

Solution to Question 28:

Amacrine cells are the cells that produce purely depolarizing potential and spikes.

Solution to Question 29:

Most afferent fibers from the lateral geniculate body (LGB) terminate in the primary visual cortex in layer 4.

Both the magnocellular and parvocellular pathways from the LGB terminate in layer 4 of the visual cortex. Only the interlaminar pathway terminates in layers 2 and 3.

Solution to Question 30:

The aqueous outflow facility is about $0.28 \mu\text{L}/\text{min}/\text{mmHg}$. Aqueous humor is produced at an average rate of $2.0\text{--}2.5 \mu\text{L}/\text{min}$.

The ease with which the aqueous humor leaves the eye is called an aqueous outflow facility.

- The outflow facility is measured using tonography.
- The normal range is between $0.22\text{--}0.28 \mu\text{L}/\text{min}/\text{mmHg}$.
- A juxtacanalicular portion of the trabecular meshwork is the primary limiting factor for the outflow facility.
- Uveoscleral outflow accounts for 10% of the total aqueous outflow facility.
- Aqueous outflow facility decreases with aging.
- Trabecular outflow facility is reduced in glaucoma.
- Laser trabeculoplasty reduces intraocular pressure (IOP) in open-angle glaucoma by increasing the trabecular outflow facility.

Myopia and Hypermetropia

Question 1:

The term ametropia refers to all except_____.

- a) Hypermetropia
- b) Emmetropia
- c) Astigmatism
- d) Myopia

Question 2:

A small kid is brought for examination due to difficulty reading from the board at school. Visual acuity was 20/60 in both eyes. His vision got corrected with concave lens spectacles. Which of the following is the most common cause of this refractive error?

- a) Increased anteroposterior diameter of the eye ball
- b) Increased curvature of the cornea
- c) Anterior placement of the lens
- d) Increase in the refractive index of lens

Question 3:

Which of the following range defines moderate myopia?

- a) -3D and -5D
- b) -3D and -6D
- c) -2D and -5D
- d) -2D and -4D

Question 4:

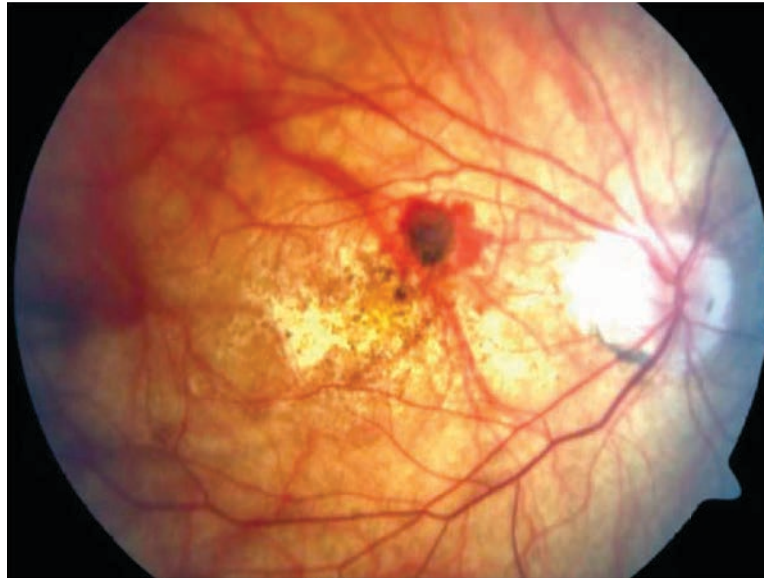
Which of the following myopias is not associated with high degree of refractive error?

- a) Degenerative myopia
- b) Congenital myopia

- c) School myopia
- d) Progressive myopia

Question 5:

A woman comes to the ophthalmology OPD because of having vision difficulties for the last 3 months. On fundoscopic examination, the following findings are seen. Which of the following is the most likely diagnosis?



- a) Hypertensive retinopathy
- b) Diabetic retinopathy
- c) Pathological myopia
- d) Sarcoidosis

Question 6:

A woman with pathological myopia visits the ophthalmology OPD for an annual eye check-up. What is the least likely finding to be seen on her ophthalmologic examination?

- a) Temporal myopic crescent
- b) Posterior staphyloma
- c) Foster Fuchs spots
- d) Shallow anterior chamber

Question 7:

Which of the following is true about degenerative myopia?

- a) It is seen in less than 6 dioptres of myopia.
- b) It is more common in males than in females.
- c) Myopic degeneration can lead to retinal detachment.
- d) The condition has no racial predilection.

Question 8:

Which of the following is the most common complication of pathological myopia?

- a) Complicated cataract
- b) Vitreous hemorrhage
- c) Macular degeneration
- d) Retinal detachment

Question 9:

While performing the following procedure on a 35-year-old male at a distance of 1 meter, you found out that the red reflex was not moving with the movement of the instrument. How will you interpret your findings?



- a) Emmetropia

- b) Hypermetropia less than 1D
- c) Myopia of 1D
- d) Myopia less than 1D

Question 10:

A worried mother brought her son for vision examination when she noticed him reading a book too closely. She says he also narrows his eyes while watching television. Examination revealed myopia. Which of the following lens should be prescribed to correct this child's vision?

- a) Convex
- b) Concave
- c) Cylindrical
- d) Concave + convex

Question 11:

Which of the following statement about the management of myopia is false?

- a) In high myopia, it is better to slightly under-correct for distance.
- b) Concave lens is used to correct myopia.
- c) Contact lens provides a wider field and larger image size as compared to glasses.
- d) Myopia must never be under-corrected with spectacles.

Question 12:

A college student is fed up with wearing spectacles for her myopia. During her annual eye checkup, she asks the ophthalmologist for eye surgery to get rid of the glasses. Which of the following is not a surgical option to correct her refractive error?

- a) SMILE
- b) Intracorneal ring implantation
- c) Radial keratotomy
- d) Conductive keratoplasty

Question 13:

Which of the following laser is used in the treatment of myopia?

- a) Argon
- b) Nd:YAG
- c) Diode
- d) Excimer

Question 14:

A mother asks for LASIK surgery for the correction of her 15-year-old daughter's myopia. Which of the following is not an eligibility criterion for this procedure?

- a) Stable refraction for the past 1 year
- b) Corneal thickness of 600 microns
- c) Age of 15 years
- d) Myopia of 4D

Question 15:

Which of the following is false about the refractive error in which the image is formed behind the retina?

- a) Long-sightedness
- b) Accommodative squint
- c) Near point comes closer
- d) Pseudopapillitis

Question 16:

You have been asked to perform fundoscopy by your senior resident in one of his patients diagnosed with hypermetropia. Which of the following options represent the findings that could be seen in this patient?

- a) Pale tessellated appearance with pseudopapillitis
- b) Shot silk appearance with pseudopapillitis.
- c) Pale tessellated appearance with papillitis
- d) Shot silk appearance with papillitis.

Question 17:

Which of the following grades of error defines high hypermetropia?

- a) $> 3D$
- b) $> 4D$
- c) $> 5D$
- d) $> 6D$

Question 18:

A 50-year-old man wants to undergo surgery for hypermetropia. All of the following are options to treat his refractive error except?

- a) Hyperopic photorefractive keratotomy
- b) Photorefractive keratectomy
- c) Conductive keratoplasty
- d) Thermal laser keratoplasty

Question 19:

A physician observed the following finding in a small child who came for an eye exam. This finding is associated with which of the following conditions?



- a) Myopia

- b) Astigmatism
- c) Hyperopia
- d) Emmetropia

Question 20:

A boy was brought by his parents for an eye exam. He has been having headaches, nausea, and visual discomfort since he started school. His ophthalmological examination revealed a refractive error of + 2D in the right eye and -1 D in the left eye. His fundus appears to be normal. What is the most likely diagnosis?

- a) Anisometropia
- b) Aniseikonia
- c) Pseudophakia
- d) Amblyopia

Question 21:

What is aniseikonia?

- a) Difference in the axial length of the two eyes
- b) Difference in the size of the image formed by the two eyes
- c) Difference in the size of the pupil in the two eyes
- d) Difference in the curvature of the cornea in the two eyes

Answer Key

Question No.	Correct Option
1	b
2	a
3	b
4	c
5	c
6	d
7	c

8	c
9	c
10	b
11	d
12	d
13	d
14	c
15	c
16	b
17	c
18	b
19	c
20	a
21	b

Detailed Explanations

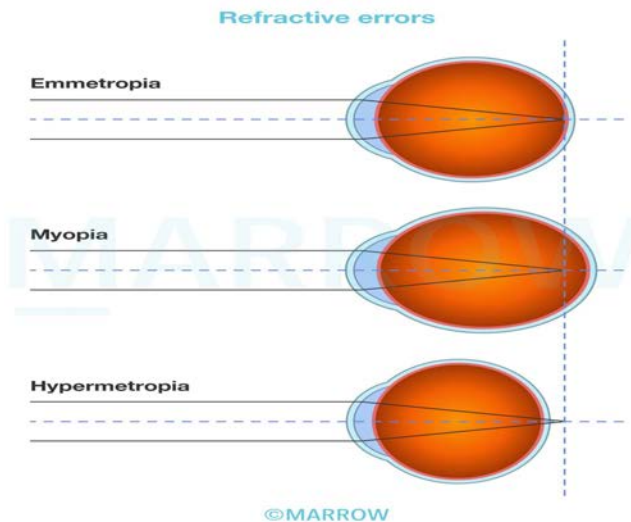
Solution to Question 1:

The term ametropia refers to conditions of refractive error.

Emmetropia refers to an optically normal eye.

The term ametropia includes conditions such as

- Myopia (shortsightedness)
- Hypermetropia (longsightedness)
- Astigmatism



Solution to Question 2:

Poor vision for distance (difficulty reading from board) that is corrected with concave lens spectacles suggest that the child has myopia. The most common cause of myopia (shortsightedness) is increased anteroposterior length of the eyeball.

Myopia is a type of refractive error seen when the parallel rays of light from infinity are focused in front of the retina when accommodation is at rest.

Solution to Question 3:

Moderate myopia is defined when the refractive error is between -3D and -6D.

Note: Myopia according to age of onset:

- Congenital: Present at birth
- Youth onset: 2–20 years of age
- Early adult onset: 20–40 years of age
- Late adult onset: ≥ 40 years of age

Solution to Question 4:

School myopia or simple myopia is associated with a low to moderate degree of refractive error.

Simple myopia is the most common type of myopia. It results from a normal biological variation in the development of the eye i.e. increased anteroposterior diameter of eyeball or increased curvature of the cornea.

Other options:

Options A and D: Pathological/Degenerative/Progressive myopia occurs due to degenerative changes in the retina, leading to a high degree of refractive error.

Option B: Congenital myopia is associated with congenital anomalies of the eye, and a high degree of refractive error.

Note: Other types of myopia include:

1. Nocturnal myopia: Accommodation is induced by low contrast in dim illumination
2. Induced myopia: Temporary and reversible myopia due to external agents (e.g., pilocarpine-induced accommodation spasm)/variations in blood sugar levels
3. Pseudomyopia: Inappropriately excessive accommodation due to overstimulation/ciliary spasm.

Solution to Question 5:

The image shows a fundus typical of pathological myopia due to the stretching of the retina.

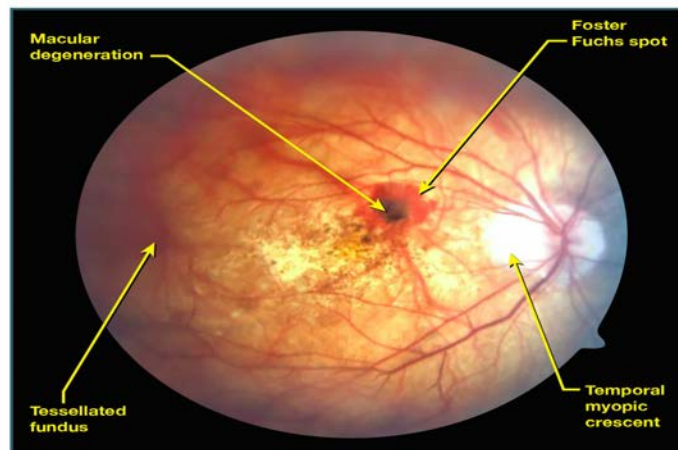
The tigroid appearance or tessellated appearance of the fundus can be appreciated. As the eye enlarges, the retinal pigment epithelium becomes thinner, resulting in:

- A tessellated (checkered) appearance of the fundus
- Increased visibility of the choroidal vasculature

Other features of the myopic fundus are as follows:

- Lacquer cracks are spontaneous ruptures of the elastic lamina of Bruch's membrane. It appears yellowish-white. it is usually located in the posterior pole.
- Annular gray shadow around the optic disc
- Lattice degeneration
- Foster Fuchs spots
- Cystoid degeneration
- Total retinal atrophy

Fundus changes in pathological myopia



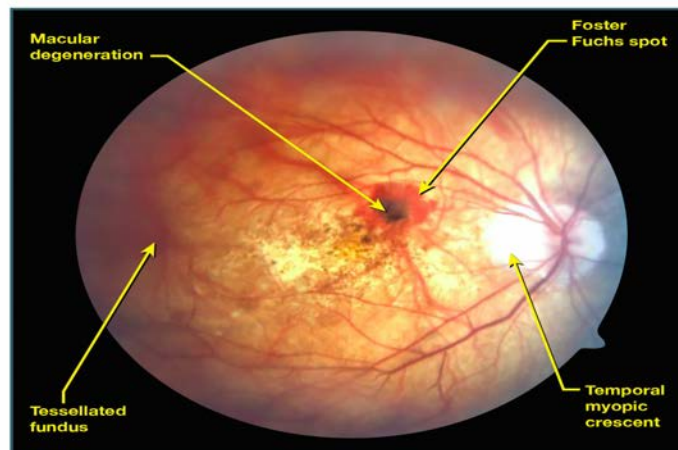
Solution to Question 6:

The shallow anterior chamber is not a feature of pathological myopia.

Signs of pathological myopia include the following:

- Prominent eyeballs
- Large cornea
- Deep anterior chamber (AC)
- Fundus picture
- Large, pale optic disc with the temporal myopic crescent
- Degenerative changes in retina and choroid
- Foster Fuchs spots (blackish spots due to retinal pigment epithelial hyperplasia)
- Lattice degeneration
- Cystoid degeneration
- Total retinal atrophy
- Posterior staphyloma
- Degenerative changes in the vitreous—posterior vitreous detachment

Fundus changes in pathological myopia



Solution to Question 7:

The statement that is true for degenerative myopia is that myopic degeneration can lead to retinal detachment.

Degenerative myopia is seen in more than 6 dioptres of myopia. It is more common in females as compared to males. The condition has a racial predilection; it is more common in Jews and Japanese people, and most cases are of genetic origin.

Changes seen in degenerative myopia are myopic crescent, rhegmatogenous retinal detachment (due to degenerative changes in the periphery of the retina with subsequent tear) and floaters (due to degenerative changes in the vitreous).

Solution to Question 8:

Macular degeneration is the most common complication of myopia, followed by rhegmatogenous retinal detachment.

Complications of pathological myopia include the following:

- Macular degeneration
- Retinal detachment
- Complicated cataract
- Vitreous hemorrhage
- Choroidal hemorrhage
- Strabismus fixus convergence

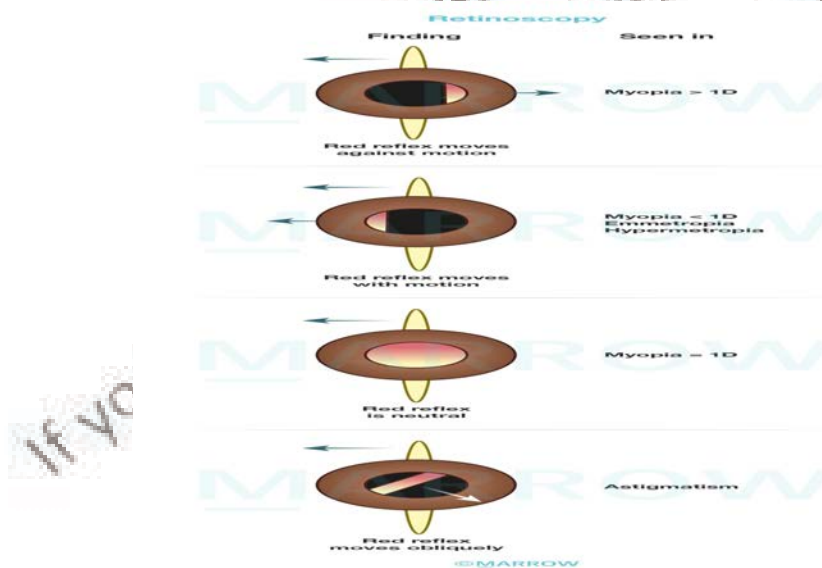
- Primary open-angle glaucoma, which is not a complication, but reported to occur in association with pathological myopia.

Solution to Question 9:

The above image shows the procedure of retinoscopy (skiascopy) being performed. As the red reflex is not moving with the movement of the retinoscope, the patient is interpreted as having myopia of 1D.

Retinoscopy is an objective method of determining the error of refraction. It is based on the principle that when light is reflected from a mirror into the eye, the direction in which the light will travel across the pupil depends upon the refractive state of the eye. Interpretation of retinoscopy done at 1m is:

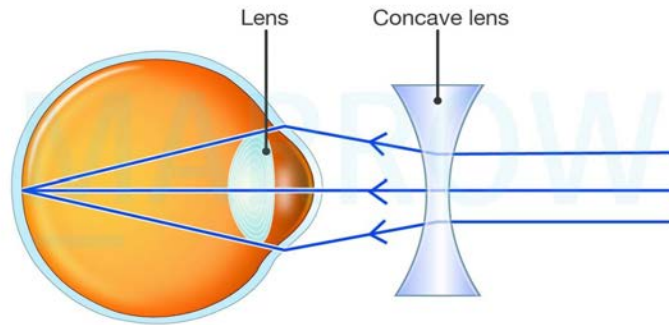
- No movement of red reflex: Myopia of 1D
- Movement of red reflex with retinoscope: Emmetropia, hypermetropia, myopia less than 1D
- Movement of red reflex against the retinoscope: Myopia more than 1D



Solution to Question 10:

Concave lenses are used for correcting myopic errors.

Myopic eye corrected with concave lens



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Other options:

Option A: Convex lenses are used in the correction of hypermetropic eye and presbyopic eye.

Option C: Cylindrical lenses are used in the correction of astigmatism.

Solution to Question 11:

The false statement is that myopia must never be under-corrected with spectacles. In fact, it is better to slightly under-correct cases of high myopia.

Concave lens are used to correct myopia. In high myopia, it is better to slightly under-correct to avoid the problem of near vision and that of minification of images. Contact lens are preferred in high myopia as they provide a wider field and larger image size as compared to glasses.

Solution to Question 12:

Conductive keratoplasty is a cornea-based surgical procedure used in the correction of hypermetropia.

Surgical procedures are undertaken after the stabilization of myopia which usually occurs after 18 years of age.

Solution to Question 13:

Excimer laser is used in the treatment of myopia.

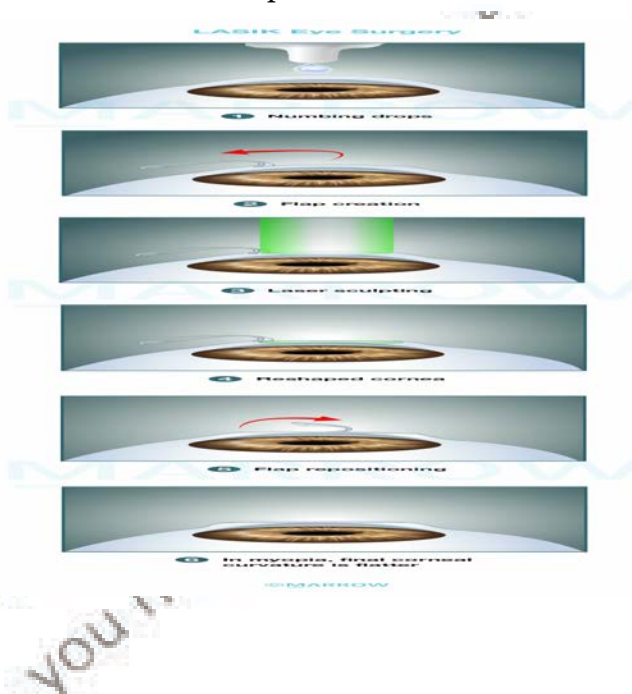
Laser-assisted in situ keratomileusis (LASIK) is the most popular treatment modality for myopia. It is performed using an excimer laser. Surgical correction is only done in individuals who have a

stable refractive error for at least 1 year. Corneal thinning disorders must be ruled out prior to this procedure.

Solution to Question 14:

Age of 15 years is not a criterion for LASIK surgery. LASIK surgery is performed after 18 years of age.

Laser-assisted in situ keratomileusis (LASIK) is performed using excimer laser in individuals aged > 18 years who have a stable refractive error for 1 year. It is done in individuals who have had unsatisfactory results with non-surgical treatment. Corneal-thinning disorders such as keratoconus must be ruled out before this procedure.



Solution to Question 15:

Hypermetropia is the refractive error where the image is formed behind the retina. In hypermetropia, near point becomes distant.

The near point or punctum proximum is the nearest point at which small objects can be seen clearly. It recedes with age.

Hypermetropia is also called long-sightedness since patients are able to focus distant objects more easily than near objects. Accommodative convergent squint in children is a complication of uncorrected long-standing hypermetropia. The fundus may show pseudopapillitis.

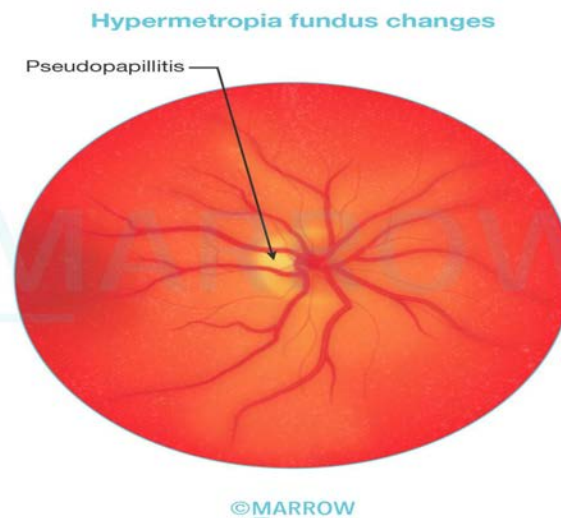
Hypermetropia is classified into the following types:

- Latent hypermetropia: Masked by the tone of the ciliary body
- Facultative hypermetropia: Masked by the voluntary accommodative effort of the person
- Absolute hypermetropia: Apparent/manifest

Solution to Question 16:

Shot silk appearance with pseudopapillitis can be seen on fundal examination of the hypermetropic eye.

Hypermetropia is a refractive error of the eye in which parallel rays of light from infinity focus behind the retina when accommodation is at rest. Fundoscopy findings in hypermetropia include shot silk appearance (retina may shine due to the greater brilliance of light reflections) and pseudopapillitis (optic disc is small, hypervascular, with ill-defined margins without disc edema).



Other signs of hypermetropia include small eyeball, small cornea and short anteroposterior length of the eyeball.

Convex lens (plus lens) is used for the correction of hypermetropia.

Solution to Question 17:

High hypermetropia is defined as an error of $>5D$.

Solution to Question 18:

Photorefractive keratectomy is not the surgical modality of the treatment of hypermetropia.

Photorefractive keratectomy is a laser ablation technique for the correction of myopia. An excimer laser is used to cause flattening of the central cornea.

Solution to Question 19:

The given image shows convergent squint or esotropia, which is commonly seen in children with hyperopia, also known as hypermetropia.

Esotropia results due to increased accommodative effort in cases where there is under correction of hypermetropia. The condition may be prevented by full hypermetropic correction. If the child is unable to wear the glasses due to flat nasal bridge or if the child is less than 2 years old, the accommodation may be treated with drugs. Miotic drugs with anticholinesterase activity such as di-isopropyl fluorophosphate (DFP) or echothiophate can be used in these children. Further treatment may include orthotropic training or surgical correction of the squint.

Solution to Question 20:

A child with refractive error of + 2D in the right eye and -1 D in the left eye has anisometropia.

Anisometropia is the condition wherein the total refractive power of the two eyes is unequal. A difference of up to 2.5 D between two eyes is well tolerated. If the anisometropia exceeds 2.5D, amblyopia may develop. Contact lenses are preferred for treatment due to less optical defects as compared to glasses.

Solution to Question 21:

Aniseikonia is a condition in which there is a difference in the size of the image formed by the two eyes.

Up to 5–8% aniseikonia is well tolerated. Treatment includes contact lens in patients with phakia and intraocular lens implantation in patients with aphakia.

Astigmatism and Errors of Accommodation

Question 1:

A 20-year-old girl started having frequent headaches and blurring of vision while studying for college exams. She says she feels better when she closes her eyes. On examination, she is found to have astigmatism. Which of the following is the most common cause responsible for this condition?

- a) Irregularity of the curvature of the cornea
- b) Irregularity of the curvature of the lens
- c) Oblique placement of the lens
- d) Oblique placement of the macula

Question 2:

Which among the following is normal astigmatism seen in eyes due to the pressure of the eyelids?

- a) With the rule
- b) Against the rule
- c) Oblique
- d) Bioblique

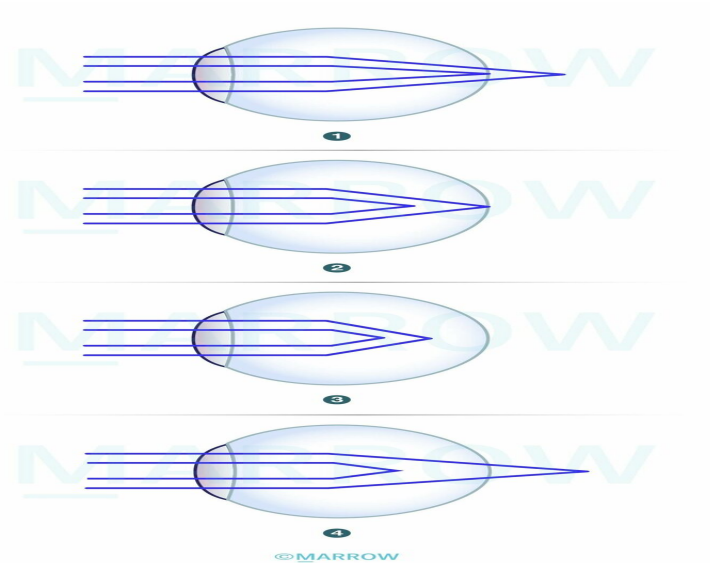
Question 3:

Which of the following statements is correct regarding with-the-rule astigmatism?

- a) The horizontal meridian is more myopic.
- b) The vertical meridian is more myopic.
- c) The vertical meridian is more hyperopic.
- d) The horizontal meridian is more hyperopic.

Question 4:

Match the images shown with the refractive errors they represent.



- a) 1 - e, 2 - c, 3 - b, 4 - a
- b) 1 - d, 2 - b, 3 - c, 4 - a
- c) 1 - d, 2 - b, 3 - a, 4 - e
- d) 1 - b, 2 - c, 3 - e, 4 - d

Question 5:

In compound myopic astigmatism, with parallel rays of light entering the eye in relaxed accommodation, the image(s) formed is/are _____.

- a) One focal point in front of the retina
- b) Two focal lines in front of the retina
- c) Two focal lines behind the retina
- d) Two focal lines, one behind and the other in front of retina

Question 6:

Which of the following would be prescribed for simple myopic astigmatism?

- a) +1.00 DS
- b) -1.00 DC X 180 degree
- c) -1.00 DS
- d) -1.00 DS -1.00 DC X 180 degree

Question 7:

The PARK procedure is used in the correction of which of the following?

- a) Myopia
- b) Hypermetropia
- c) Astigmatism
- d) Presbyopia

Question 8:

An 80-year-old man is reading a newspaper in his backyard. His accommodation while reading a newspaper is due to_____.

- a) Relaxation of the ciliary muscle
- b) Contraction of the ciliary muscle
- c) Contraction of the dilator pupillae
- d) Contraction of constrictor pupillae

Question 9:

During a routine school eye examination camp, a 5-year-old boy is able to read everything written on the board. Which of the following mechanisms aids in establishing normal distant vision in this child?

- a) Relaxation of the ciliary muscles
- b) Constriction of the pupils
- c) Shortening of the focal length of the lens
- d) Rounding of the lens

Question 10:

The condition of the accommodative power being significantly less than the normal physiological limits for the patient's age is called _____.

- a) Paralysis of accommodation
- b) Presbyopia
- c) Insufficiency of accommodation

d) Spasm of accommodation

Question 11:

What is the accommodative amplitude of a patient with a near point at 20 cm and a far point at infinity?

- a) 5 D
- b) 1 D
- c) 4 D
- d) 2.5 D

Question 12:

A 45 year old woman complains of an inability to read books and stitch garments for the past few months. Which of the following is false regarding her condition?

- a) Loss of accommodation with age
- b) Decrease in elasticity and plasticity of the crystalline lens
- c) Age-related decrease in the power of ciliary muscles
- d) It is corrected with a concave lens

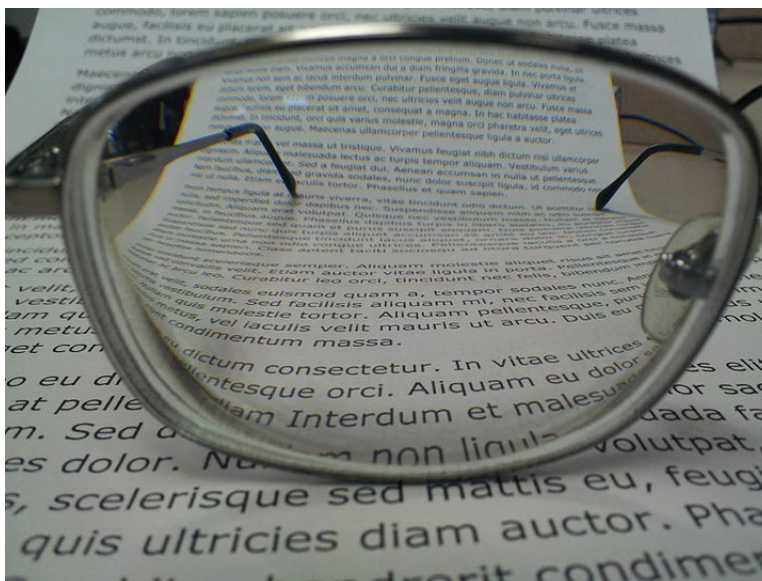
Question 13:

A 51-year-old woman comes for an ophthalmological examination. She says she is unable to thread a needle and is facing difficulty making a sweater for her grandson. She never had vision problems in her youth. What power of glasses would roughly correct her vision difficulties?

- a) +1.75D
- b) +1.5D
- c) +2D
- d) +2.25D

Question 14:

During your morning jogging routine, you found a pair of spectacles lying on the sidewalk. The spectacles are shown below. What type of presbyopic lens is present in these spectacles?



- a) Executive bifocal
- b) Trifocal
- c) Kryptok
- d) Progressive

Question 15:

A 55-year-old male with limbal scar presents to the ophthalmology clinic with marked defects in both near and far vision. Clinical examination reveals a wide and deep anterior chamber, iridodonesis, and a dark pupillary reflex. A vision of 6/6 is achieved with correcting lens of +11D. Which of the following is the most likely diagnosis?

- a) Aphakia
- b) Pseudophakia
- c) Posterior dislocation of lens
- d) Myopia

Question 16:

Which of the following is used to differentiate pseudophakia from aphakia?

- a) Deep AC chamber

- b) Purkinje image test
- c) Iridodonesis
- d) Limbal scar

Question 17:

Roving–ring scotoma is seen after the correction of aphakia by which of the following?

- a) IOL
- b) Spectacles
- c) Soft contact lens
- d) Hard contact lens

Question 18:

Soft contact lenses are made of _____.

- a) Polymethylmethacrylate
- b) Hydroxyethyl methyl acrylate
- c) Cellulose acetate butyrate
- d) Silicone

Question 19:

Which among the following is not an advantage of contact lens?

- a) It helps in eliminating peripheral aberrations.
- b) It can correct irregular astigmatism.
- c) It is helpful in dry eye.
- d) It improves corneal oxygen supply.

Question 20:

The spectacles shown in the image are not used for the management of which of the following?

15	a
16	b
17	b
18	b
19	d
20	d

Detailed Explanations

Solution to Question 1:

Abnormalities of the curvature of the cornea is the most common cause of astigmatism.

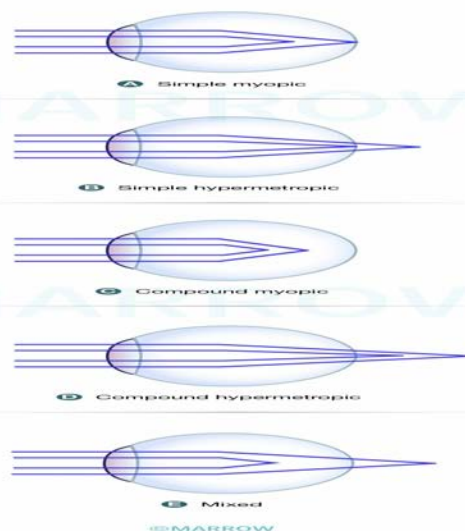
Astigmatism is a refractive error in which the rays of light entering the eye are refracted differently in different meridians of the eye. Meridia of the eye correspond to the lines of longitude on a globe.

There are two types of astigmatism. In regular astigmatism, refractive power changes uniformly from one meridian to another. In irregular astigmatism, there is irregular change of the refractive power in different meridians. It can be caused by corneal scars.

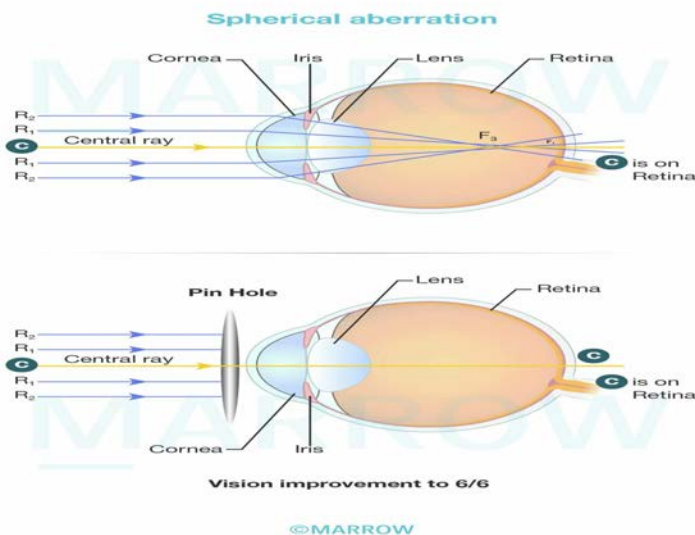
Etiology of regular astigmatism:

- Corneal: Most common cause. Occurs due to abnormalities of curvature of the cornea
- Lenticular: Occurs due to abnormalities of lens curvature and position
- Curvatural: Seen in lenticonus
- Positional: Due to tilting or oblique placement of the lens or subluxation of lens
- Retinal: Due to oblique placement of the macula

Refractive types of regular Astigmatism:



Note: Spherical aberration is a high-order refractive error of the eye. It is seen when light rays passing through the periphery of the cornea and lens focus at a slightly different location than light rays passing through the center of the cornea and lens. It improves on using a pinhole.



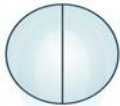
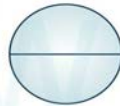
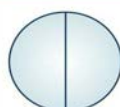
Solution to Question 2:

With-the-rule astigmatism is normally seen in the eyes due to the pressure of the eyelids.

In with-the-rule astigmatism, the vertical meridian is more curved. This is more common and often occurs due to the pressure of eyelids on the eye. In against-the-rule astigmatism (ATR) (option B), the horizontal meridian is more curved than the vertical meridian. In both these types, the principal meridians are vertical and horizontal (90 and 180 degrees) and are at right angles to each other.

The other types of astigmatism include oblique and bi-oblique astigmatism. In oblique astigmatism (option C), the two principal meridians are not the horizontal and vertical meridians, although they are at right angles to each other. In bi-oblique astigmatism (option D), the two principal meridians are not at right angles to each other; for example, one may be at 40 degrees and the other, at 100 degrees.

Clinical types of astigmatism

	Cornea	Typical age	Axis of correcting plus cylinder ($\pm 20^\circ$)	Axis of correcting minus cylinder ($\pm 20^\circ$)
With the rule astigmatism		Young		
Against the rule astigmatism		Old		

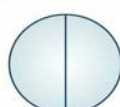
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Solution to Question 3:

In with-the-rule astigmatism, the curvature of the vertical meridian is more than the horizontal meridian. Greater the curvature of the meridian, more the myopia of that meridian. Hence, the vertical meridian is more myopic in with-the-rule astigmatism.

With-the-rule astigmatism is the type of astigmatism wherein the vertical curvature is always more than the horizontal curvature. Thus, irrespective of the type or cause of with-the-rule astigmatism, the vertical meridian is always more myopic (curved) than the horizontal meridian.

Clinical types of astigmatism

	Cornea	Typical age	Axis of correcting plus cylinder ($\pm 20^\circ$)	Axis of correcting minus cylinder ($\pm 20^\circ$)
With the rule astigmatism		Young		
Against the rule astigmatism		Old		

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Solution to Question 4:

The correct answer is 1 - d, 2 - b, 3 - c, 4 - a.

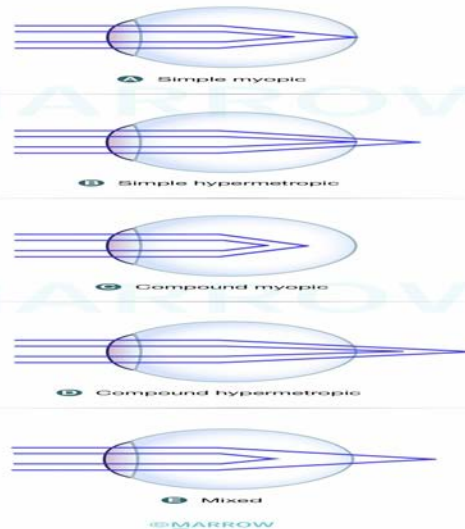
Astigmatism is a refractive error in which the rays of light entering the eye are refracted differently in different meridians of the eye. Meridia of the eye correspond to the lines of longitude on a globe. There are two types of astigmatism - regular and irregular.

In regular astigmatism, refractive power changes uniformly from one meridian to another. Thus, two principal meridians are present. The rays of light from infinity form two focal points due to the difference in curvature of the two principal meridians. Based on the relation of the two focal points to the retina, regular astigmatism has the following types:

- Simple astigmatism: One focal point lies on the retina. In simple myopic type, the second focal point lies in front of the retina while in simple hypermetropic type, the second focal point lies behind the retina.
- Compound astigmatism: In compound hypermetropic type, both focal points lie behind the retina. In compound myopic type, both focal points lie in front of the retina.
- Mixed astigmatism: One focal point lies in front of while the second focal point lies behind the retina.

Image	Refractive error
1	d. Simple hypermetropic astigmatism
2	b. Simple myopic astigmatism
3	c. Compound myopic astigmatism
4	a. Mixed astigmatism

Refractive types of regular Astigmatism:



Solution to Question 5:

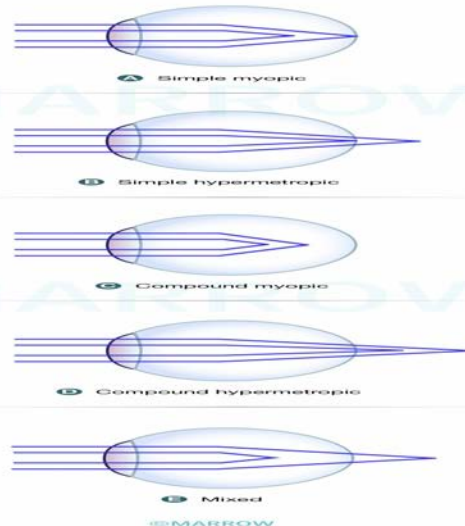
In compound myopic astigmatism, parallel rays of light entering the eye in relaxed accommodation form two focal lines in front of the retina.

Option A: One focal point in front of retina describes myopia.

Option C: Two focal lines behind the retina describes compound hypermetropic astigmatism.

Option D: Two focal lines, one behind and one in front of the retina describes mixed astigmatism.

Refractive types of regular Astigmatism:

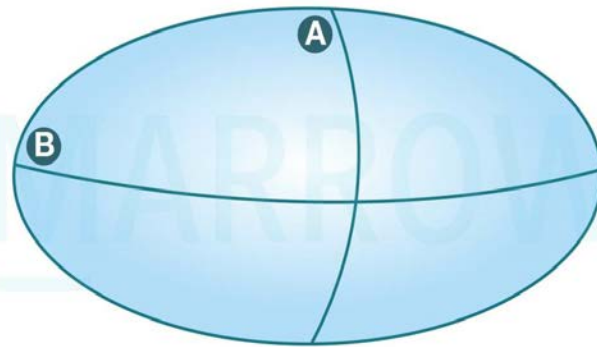


Solution to Question 6:

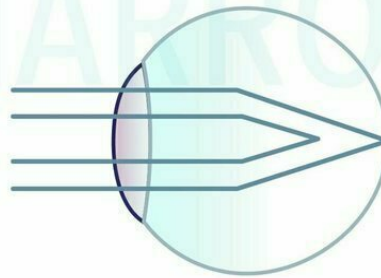
Among the above options, $-1.00 \text{ DC} \times 180$ degrees is the correct prescription for simple myopic astigmatism (SMA).

There are 3 components for prescription of glasses.

1. The first number is the refractive power of the spherical component. It is represented as convex (+) or concave (-) dioptre sphere (DS).
2. The second number denotes the power of the cylinder, represented as convex (+) or concave (-) dioptre cylinder (DC).
3. The above two components are followed by an x representing the axis (orientation) of the cylinder.



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In the most common type of SMA - with-the-rule astigmatism - meridian A (represented by 90 degrees) is more curved than meridian B (represented by 180 degrees). The light rays refracting from B focus on the retina, whereas those refracting from A focus in front of the retina.

To correct this error, a negative cylindrical lens (DC) is used to focus the myopic rays of meridian A on the retina. This correction needs to occur at 90 degrees (meridian A). The axis of a cylindrical lens is perpendicular to the required correction and thus, the spectacles will have an axis of 180 degrees.

There is no spherical refractive error since the light rays from meridian B lie on the retina. Thus DS is 0.

Solution to Question 7:

PARK is a technique used for correcting astigmatism.

Photo-astigmatic refractive keratotomy (PARK) is performed using excimer laser for the correction of astigmatism. The other surgical procedures used for correction of astigmatism are astigmatic keratotomy, LASIK and SMILE.

Solution to Question 8:

Near vision accommodation is due to the contraction of the ciliary muscle.

Accommodation is the phenomenon of increasing the convergence power. This occurs by increasing the refractive power of the lens and increasing its curvature. The capsule of the lens is more elastic than the lens itself and plays a role in increasing the curvature of the lens. When the circular fibres of the ciliary muscles contract, the ciliary body approaches the lens. This slackens the zonules so that the capsule, relieved of its tension, is able to mould the lens to its accommodative form.

Solution to Question 9:

Relaxation of the ciliary muscles would aid in establishing a normal distant vision.

Relaxation of the ciliary muscles keeps the zonules in a taut state and maintains the radius of curvature of the anterior lens surface at 10 mm. Thus, the focal length is maximum and distant objects are most clearly visualized.

Option B: The pupils constrict during accommodation to increase the depth of focus.

Option C: During accommodation, ciliary muscles contract causing relaxation of zonular fibers, rounding of the lens, and shortening of the focal length.

Option D: The radius of curvature of the anterior lens surface becomes 6 mm i.e rounder during accommodation.

Solution to Question 10:

The condition in which accommodative power is significantly less than normal physiological limits for the patient's age is called insufficiency of accommodation

Anomalies of accommodation:

Presbyopia	Insufficiency of accommodation	Paralysis of accommodation (cycloplegia)	Spasm of accommodation
Failing near vision due to age-related decrease in the amplitude of accommodation or increase in near point	Inadequacy of the accommodative power. It is significantly less than normal physiological limits for the patient's age	Complete absence of accommodation	Exertion or abnormally excessive accommodation

Presbyopia	Insufficiency of accommodation	Paralysis of accommodation (cycloplegia)	Spasm of accommodation
Causes: Age-related change in the lens- decreased elasticity and sclerosis. Age-related decline in ciliary muscle power	Causes: Premature sclerosis of the lens Weakness of the ciliary muscles due to systemic causes like anemia and malnutrition. Weakness of the ciliary muscles associated with primary open-angle glaucoma	Causes: Drug-induced Internal ophthalmoplegia due to diphtheria, diabetes mellitus, etc. As a component of 3rd nerve palsy	Causes: Drug-induced spasm, e.g., atropine, pilocarpine Spontaneous spasm: Found in children who use eyes for excessive near work in conditions of bad illumination.

Solution to Question 11:

The amplitude of accommodation for the above patient would be 5D.

Near point of accommodation (NPA) is the closest distance at which the near object is clearly focussed without blur. Amplitude of accommodation is the maximum power of the lens that permits clear near vision without blur. Its unit is dioptre (D). The amplitude of accommodation is the reciprocal of NPA in meters. 20 cm is 1/5th of a meter. Therefore, accommodative amplitude is its inverse, that is, 5D.

The NPA is measured clinically using a scale device such as the Prince, Berens, or Royal Air Force (RAF) ruler.



Solution to Question 12:

A 45 year old woman presenting with an inability to do near vision work (reading, stitching) likely has presbyopia. Presbyopia is corrected by the use of a convex lens (weakest convex lens).

Presbyopia can be caused by an age-related decline in the ciliary muscle power or age-related change in the lens including:

-
- Hardness (sclerosis) of the lens substance, which is less easily molded
- Decrease in the elasticity of lens capsule

Solution to Question 13:

The power of prescription glasses for a 51-year-old woman should be +1.5D.

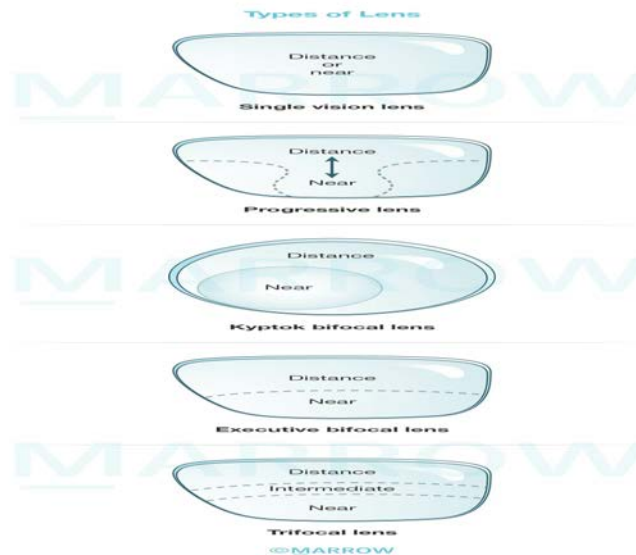
Presbyopia is the condition of decreased near vision. It is due to an age-related decrease in the amplitude of accommodation or an increase in punctum proximum (near point). For correction of presbyopia, convex lenses are used.

Roughly, the power of prescription glasses age-wise is as follows:

Age	Power
45 ye ars	+1 to +1.25 D
50 ye ars	+1.5 to 1.75 D
55 ye ars	+2 to +2.25 D
60 ye ars	+2.5 to +3D

Solution to Question 14:

The given lens in the image is a typical example of progressive lenses. Progressive lenses are identified as such due to the absence of the line of demarcation on the lens. They are gradually optimized for reading vision, distant vision, and for intermediate vision, such as computer work.



Solution to Question 15:

A limbal scar with defective near and distant vision, a wide and deep anterior chamber, iridodonesis, and dark pupillary reflex suggests aphakia.

Aphakia is the condition in which the lens is absent from the pupillary area. Signs of surgical aphakia include limbal scar, deeper anterior chamber than normal, iridodonesis (tremulousness of the iris), jet black color of the pupil. On Purkinje's test, only the first and second images are seen. A small disc is seen on fundus examination and high hypermetropia is seen on retinoscopy.

The best treatment option is secondary intraocular lens (IOL) implantation.

Solution to Question 16:

Pseudophakia shows 4 images in the Purkinje image test, whereas aphakia shows only 2 images. Other features can be seen in both.

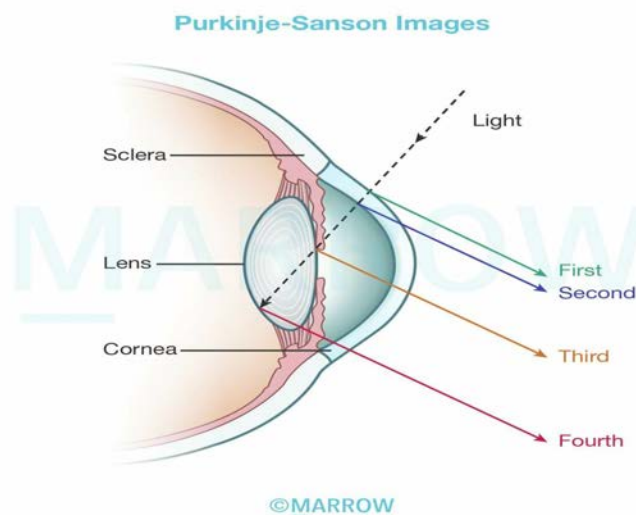
When aphakia is corrected with an intraocular lens implant (IOL), the condition is referred to as pseudophakia. Signs of pseudophakia with posterior chamber IOL include:

- Surgical scar
- Anterior chamber slightly deeper than normal
- Mild iridodonesis (tremulousness) of the iris
- Purkinje image test shows 4 images
- Pupil blackish in color but when the light is thrown in the pupillary area, shining reflex is observed

Purkinje images are reflections of objects from the structure of the eye. They are also known as Purkinje reflexes or Purkinje–Sanson images. Normally, 4 images are formed from four different

reflecting surfaces i.e.

Image	Location	Direction
Image 1	Anterior surface of the cornea	Erect
Image 2	Posterior surface of the cornea	Erect
Image 3	Anterior surface of the lens	Erect
Image 4	Posterior surface of the lens	Inverted



In aphakia, as the lens is absent, image 3 and image 4 are absent.

Solution to Question 17:

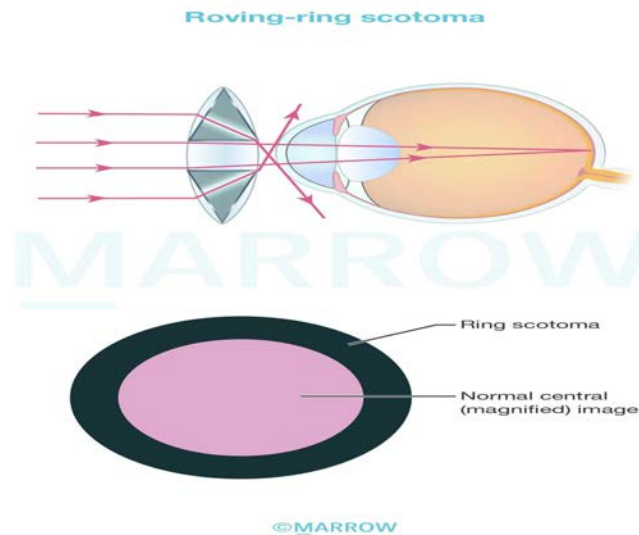
Roving-ring scotoma is seen in the correction of aphakia with spectacles due to prismatic effects at the edge of the lens.

The difficulties in the correction of aphakia by spectacles include the following:

- Image magnification by 25–30%
- Spherical aberrations producing a “pin cushion effect”
- Jack-in-the-box phenomenon from the prismatic effects at the edge of the lens
- Reduced visual fields
- Physical inconvenience

For aphakia, the best correction is obtained by an intraocular lens (IOL).

Roving-ring scotoma:



The edge of a convex lens acts as a prism. The greater the angle of the prism, the more will be the bending. In aphakic spectacles, the angle of the lens is large. Thus, the light falling at the edge of the lens bends towards the center of the lens and does not reach the pupil. This results in a ring-shaped scotoma as this prismatic effect is present all around the edges of the lens. The position of this scotoma is not fixed in the visual field because the eye keeps moving (or roving) in relation to the aphakic spectacle. Hence, the result is a roving ring scotoma.

Solution to Question 18:

Soft lenses are made up of hydroxyethyl methyl acrylate (HEMA).

Solution to Question 19:

Contact lenses decrease corneal oxygen supply.

The cornea obtains its O₂ supply from the atmosphere. The use of contact lenses decreases the O₂ supply from the atmosphere. Contact lens eliminate peripheral aberrations and can correct irregular astigmatism. Although dry eye is caused by wearing contact lenses, lenses made of hydrogel which are high in water content are beneficial in dry eye.

Rigid gas-permeable/semi-soft contact lens are used for keratoconus. Toric contact lens are used for astigmatism.

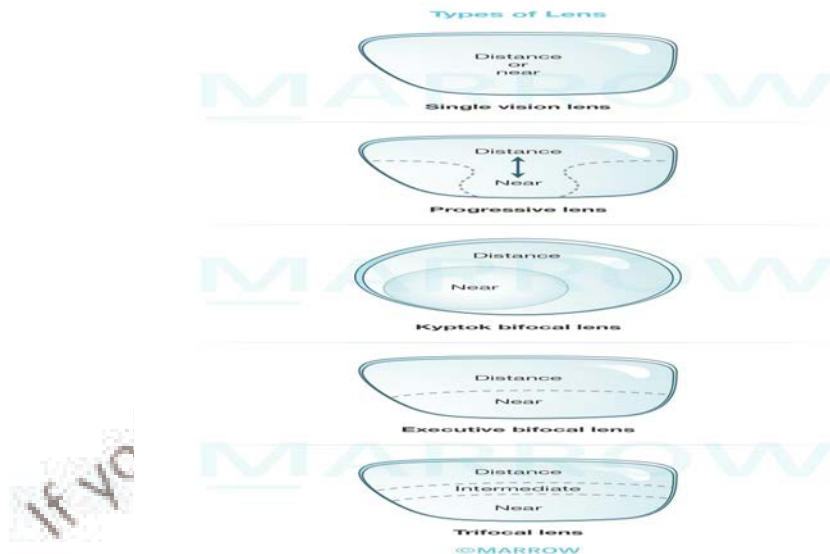
Solution to Question 20:

The given image shows spectacles made of lenses with two different optical power. Thus, it is a pair of bifocal spectacles which is not used in hypermetropia or myopia.

Bifocal glasses have different corrective powers in the upper segment for distant vision and lower segment for near vision. Thus, they are useful in conditions where both distance and near vision are affected, like the following:

1. Presbyopia (option A): It is a definitive indication. It is used to correct age-related near and far vision defects.
2. Aphakia (option B): It provides both near and far vision correction when intra-ocular lens insertion is contraindicated.
3. Paediatric pseudophakia (option C): It is used to correct refractive errors or accommodative fluctuations post intra-ocular lens (IOL) insertion.
4. Non refractive accommodative esotropia

In hypermetropia there is defect in near vision only and hence a single vision glasses having uniform corrective power through the entire surface is used instead of bifocal glasses.



Conjunctiva

Question 1:

Which of the following characteristic features would you expect to see in a histopathology slide of conjunctiva?

- a) Amacrine cells within non-keratinized stratified squamous epithelium
- b) Goblet cells within non-keratinized stratified squamous epithelium
- c) Goblet cells within keratinized stratified squamous epithelium
- d) Amacrine cells within keratinized stratified squamous epithelium

Question 2:

What is the causative organism of angular conjunctivitis?

- a) Moraxella lacunata
- b) Haemophilus influenzae
- c) Adeno virus type 32
- d) Coxsackie B virus

Question 3:

A child presents with irritation and redness of the left eye at a primary health center. There are no previous immunization records. Examination showed a greyish-white membrane over the bulbar and palpebral conjunctiva which bled on peeling. What is the likely etiology?

- a) Corynebacterium diphtheriae
- b) Moraxella lacunata
- c) Moraxella axenfeld
- d) Chlamydia trachomatis

Question 4:

A child with pharyngitis, and fever subsequently develops conjunctivitis. Initially, the left eye was affected, followed by the right eye's involvement in 2 days. Epiphora, conjunctival

hyperemia, and chemosis are noted. What is its most likely etiological agent?

- a) Picorna virus
- b) Coxsackie virus
- c) Adenovirus
- d) Enterovirus 70

Question 5:

Which of the following statements is false regarding this condition?



- a) The condition is commonly associated with a benign course.
- b) It can occur spontaneously in elders with systolic hypertension.
- c) The blood gets absorbed gradually in 6-8 weeks time without treatment.
- d) Use of NSAIDs and aspirin to treat the condition should be avoided.

Question 6:

A UNICEF volunteer presented with a bilateral red eye of 2 days duration beginning on a return flight from Ghana. The infection spread among all her family members within a day of her return. Initial examination revealed lid edema, serous discharge, and severe sub-conjunctival hemorrhage. What is the likely diagnosis?

- a) Hemorrhagic conjunctivitis
- b) Giant papillary conjunctivitis

- c) Allergic conjunctivitis
- d) Inclusion conjunctivitis

Question 7:

What is the cause of acute hemorrhagic conjunctivitis?

- a) Enterovirus 68
- b) Enterovirus 69
- c) Enterovirus 70
- d) Enterovirus 71

Question 8:

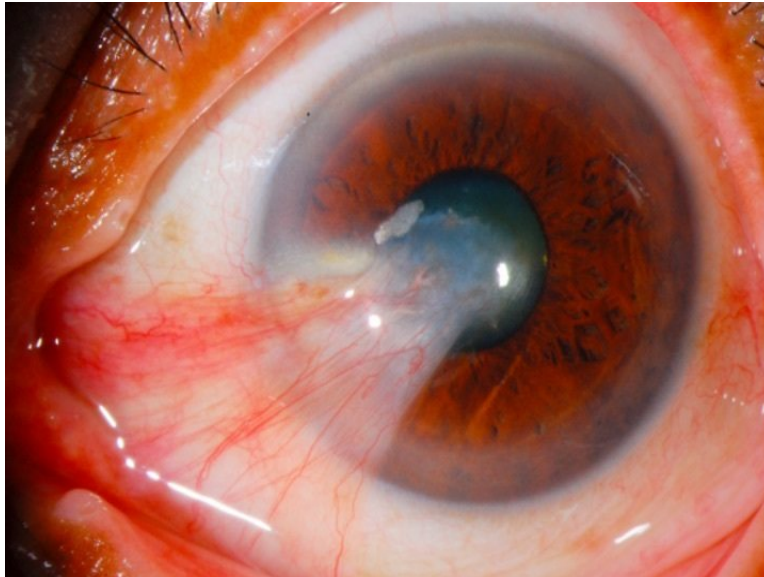
A tribal farmer, presented with a three-day history of right eye redness with pain. He had spent the prior week trekking up a mountain in order to travel to the nearby town and takes such trips frequently. The following lesion was observed on examination. What is the likely diagnosis?



- a) Retention cyst
- b) Pinguecula
- c) Pterygium
- d) Spring catarrh

Question 9:

What is the likely diagnosis from the image given below?



- a) Pinguecula
- b) Concretions
- c) Pterygium
- d) Bitots spots

Question 10:

The excision of a pinkish, triangular tissue growth at the 1 o'clock position of the eye is being performed. The patient had a history of blurry vision and persistent redness in both eyes for a month. In order to secure the graft in the scleral bed, which of the following will not be used?

- a) Vicryl suture
- b) Autologous serum
- c) Cyanoacrylate
- d) Fibrin glue

Question 11:

A man presented with red eyes associated with a sandy, gritty sensation. He was diagnosed with conjunctivitis last week, however, his symptoms persisted. He has had multiple sexual partners in the past. Examination revealed palpable preauricular nodes and a follicular

reaction involving the bulbar conjunctiva and semilunar folds. Conjunctival scrapings were sent for culture. What is the likely organism involved?

- a) Chlamydia trachomatis
- b) Chlamydia psittaci
- c) Herpes simplex
- d) Gonococcus

Question 12:

Which of the following conditions is closely related to mucus fishing syndrome?

- a) Spring catarrh
- b) Atopic keratoconjunctivitis
- c) Phlyctenular keratoconjunctivitis
- d) Giant papillary conjunctivitis

Question 13:

A young man presented with watering, redness and photophobia. He had similar complaints in the past which improved with self-medicated topical drops. His best-corrected visual acuity is 6/36 improving to 6/24. Slit-lamp examination revealed a 3 mm × 2 mm epithelial defect in the paracentral part of the cornea that stained positive with fluorescein dyes. What is the likely diagnosis?

- a) Spring catarrh
- b) Exposure keratopathy
- c) Neuroparalytic keratopathy
- d) Trachoma

Question 14:

While browsing through the news channels, Geeta came across a panel discussion on public health diseases celebrating the elimination of a peculiar condition of the eye. It was known as the most important cause of corneal blindness in India and shows the following finding. What is the likely diagnosis?



- a) Trachoma
- b) Vernal keratoconjunctivitis
- c) Phlyctenular conjunctivitis
- d) Xerophthalmia

Question 15:

Which of the following will not be seen as a corneal sign in a 6-year-old patient with trachoma?

- a) Herbert pits
- b) Pannus
- c) Opacity
- d) Arlt's line

Question 16:

A young girl has been brought in by her grandmother because she has constant pain, photosensitivity and discharge from her eyes. From the notes, you see that the child has had multiple presentations for prior eye infections. You observe severe scarring on the palpebral conjunctiva, as well as trachomatous trichiasis. Which McCallan stage will this be classified under?

- a) Stage 1
- b) Stage 2

- c) Stage 3
- d) Stage 5

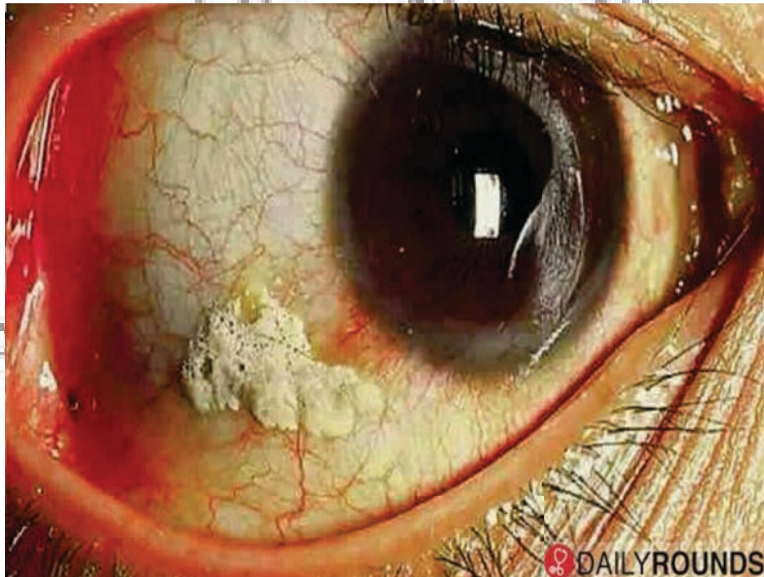
Question 17:

What is the drug of choice under the SAFE strategy for trachoma management?

- a) Azithromycin
- b) Tetracycline
- c) Doxycycline
- d) Penicillin

Question 18:

A 21-year-old man presented with the following lesion in his eye. What is the likely diagnosis?



- a) Xerophthalmia
- b) Pterygium
- c) Phlyctenular conjunctivitis
- d) Membranous conjunctivitis

Question 19:

A 10 year old malnourished boy is brought with a 2-month history of diminished night vision. His dietary history revealed a restricted diet, consisting of mainly chips and nuggets, and he would refuse to eat green leafy vegetables. What is the conjunctival change seen in this deficiency?

- a) Actinic degeneration
- b) Stromal infiltration
- c) Hyperplasia of goblet cells
- d) Hyperkeratosis of the squamous epithelium

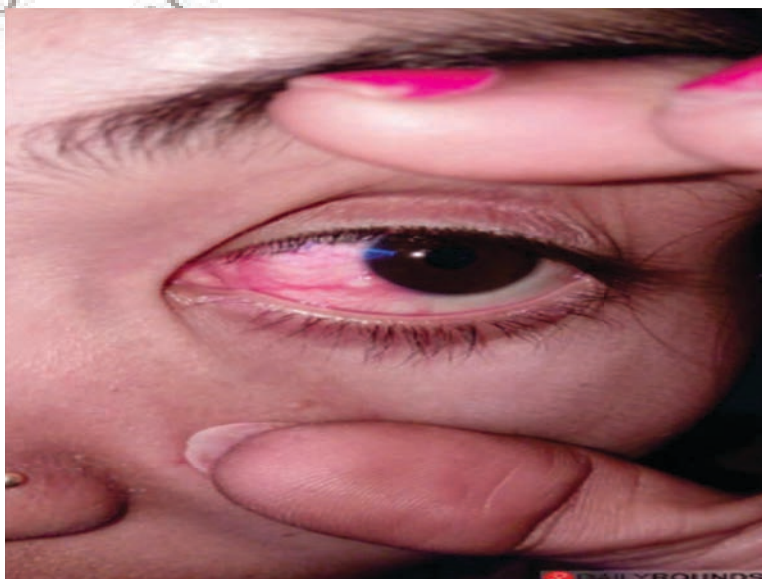
Question 20:

Ophthalmia nodosa occurs due to which of the following?

- a) Cat scratch disease
- b) Tularemia
- c) Leprosy
- d) Irritation due to caterpillar hair

Question 21:

A 28-year-old woman with redness and photophobia of the left eye was seen in the OPD. An image of a characteristic lesion of her eye is as shown below. Identify the lesion.

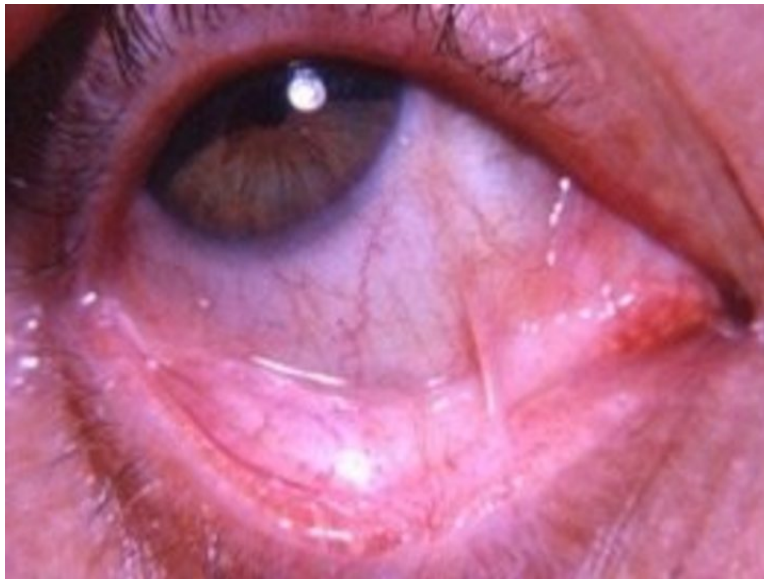


- a) Pterygium

- b) Pinguecula
- c) Phlycten
- d) Hebert's follicle

Question 22:

Which of the following should not be considered in the differential of a patient who presents with the following finding?



- a) Erythema multiforme
- b) Steven-Johnson syndrome
- c) Phlyctenular conjunctivitis
- d) Alkali injury

Question 23:

What is the most common cause of ophthalmia neonatorum in developed countries?

- a) Chlamydia
- b) Gonorrhoea
- c) Chemical conjunctivitis
- d) Staphylococcus aureus

Question 24:

An 8-year-old boy had to miss school because the matting of his eyelids made it difficult to open them in the morning and caused severe pain. His agonized mother rushed him to the hospital and reported that he had a fever and red eye for 4 days. Which group of lymph nodes is likely to be enlarged?

- a) Sublingual and submandibular nodes
- b) Parotid and sublingual nodes
- c) Submandibular and preauricular nodes
- d) Jugulo-digastric nodes

Question 25:

The image of the eye of a 32-year-old woman having superior limbic keratoconjunctivitis is given below. Which of the following is commonly associated with it?



- a) Valvular heart disease
- b) Thyroid disease
- c) Inflammatory bowel disease
- d) Systemic lupus erythematosus

Question 26:

Which of the following statements is not characteristic of pterygium?

- a) Abnormal activity of p53 is seen in the pterygium tissue.
- b) Decreased activity of tissue inhibitors of metalloproteinases is seen.
- c) Destruction of Bowman's layer occurs when the pterygium invades the cornea.
- d) Elastotic degeneration and hyalinization of the connective tissue of the conjunctiva occur.

Question 27:

A 7-year-old boy complains of intense lacrimation, associated with photophobia. On examination, the following finding is seen in his upper tarsal conjunctiva. What is the likely diagnosis?



- a) Trachoma
- b) Vernal keratoconjunctivitis
- c) Phlyctenular conjunctivitis
- d) Xerophthalmia

Question 28:

Your professor has assigned you to proofread a seminar on the anatomy of the conjunctiva made by a junior. In the explanation of conjunctival follicles, you spot a wrong statement that needs revision. What is that statement?

- a) Conjunctiva of new born does not produce follicles before 2-3 months of age.
- b) Slit lamp may be necessary to differentiate between follicles and papillae.
- c) Follicles are a hyperplasia of the normal vascular system.

d) Follicles are found in the sub-epithelial adenoid layer of the conjunctiva.

Question 29:

In the grading of trachoma, the follicular stage is defined by the presence of which of the following?

- a) 5 or more follicles in the lower tarsal conjunctiva
- b) 3 or more follicles in the upper tarsal conjunctiva
- c) 5 or more follicles in the upper tarsal conjunctiva
- d) 3 or more follicles in the lower tarsal conjunctiva

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	c
5	c
6	a
7	c
8	b
9	c
10	c
11	a
12	d
13	a
14	a
15	d
16	c
17	a
18	a
19	d
20	d

21	c
22	c
23	a
24	c
25	b
26	d
27	b
28	c
29	c

Detailed Explanations

Solution to Question 1:

The presence of goblet cells within the non-keratinized stratified squamous epithelium would help identify the conjunctiva. Goblet cells are involved in the secretion of mucin.

Amacrine cells are a type of retinal interneuron, and hence would be found in the retina, not the conjunctiva.

Solution to Question 2:

Moraxella species, a type of gram-negative diplobacilli, are involved in causing angular conjunctivitis which is a type of chronic conjunctivitis.

Angular conjunctivitis is a chronic type of conjunctivitis where there is inflammation of the conjunctiva at the lid margins and the inner and outer canthi of the eyes with red, scaly, excoriated, macerated and fissured skin at the canthi. This is because of the proteolytic enzymes secreted by the Moraxella species.

Patients with angular conjunctivitis usually experience eye irritation, a burning sensation and collection of dirty-white, foamy discharge at the angles. These patients can also have associated blepharitis and papillary and follicular conjunctivitis.

Causative organisms are Moraxella lacunata, Moraxella axenfeld and Staphylococcus aureus (Rare cause). The source of infection is typically the nasal tract or the nasal discharge of patients with angular conjunctivitis. The infection spreads from the nasal cavity to the eyes through contaminated hands or handkerchiefs. Treatment includes oxytetracycline 1% ointment for 10 to 14 days, and zinc oxide as it inhibits proteolytic enzymes.

The image given below shows the eye of a patient infected with angular conjunctivitis.



Other options:

Option B: Haemophilus influenzae causes acute conjunctivitis.

Option C: Adenovirus is the most common cause of viral conjunctivitis and it most commonly causes follicular conjunctivitis, pharyngoconjunctival fever, and epidemic keratoconjunctivitis.

Option D: Coxsackie B virus is known to cause aseptic meningitis, encephalitis, pericarditis, and myocarditis. There are no known mentions of ocular disease caused by the virus.

Note: Pathogens that cause bacterial conjunctivitis

The below-given table elaborates on the type of conjunctivitis caused by different serotypes of adenovirus.

Hyperacute(symptoms develop within 24 hours; aggressive course)	Acute(resolving in less than 4 weeks)	Chronic(of greater than 4 weeks duration)
Neisseria gonorrhoeae	Staphylococcus aureus	Staphylococcus aureus
Neisseria meningitidis	Streptococcus pneumoniae	Moraxella lacunata
	Haemophilus influenzae	Enteric bacteria

Conjunctivitis type	Adenovirus serotype
Follicular Conjunctivitis	Adenovirus serotypes 1-11 and 19
Pharyngoconjunctival Fever	Adenovirus serotypes 3, 4, and 7
Epidemic Keratoconjunctivitis	Adenovirus serotypes 8, 19, and 37

Solution to Question 3:

The of a child with irritation and redness of the eye with no previous vaccination records is suggestive of a case of membranous conjunctivitis and among the given options, *Corynebacterium diphtheriae* is the most likely etiological agent.

Membranous conjunctivitis presents with irritation and redness of the eye and is characterized by the formation of a greyish-white membrane that bleeds on peeling. Vaccines are routinely used to prevent diphtheria infection and hence, unvaccinated status increases his susceptibility to the bacteria. Moreover, the bacteria thrive in crowded populations and poor hygienic surroundings, hence a background of low socioeconomic context, such as an orphanage, acts as a risk factor in this case.

This condition is associated with the formation of fibrinous exudates on the surface and into the substance of the conjunctival epithelium. The causes include beta-hemolytic streptococci, *Streptococcus pneumoniae*, *Haemophilus aegyptius*, *Corynebacterium diphtheriae*, and *Neisseria gonorrhea*.

The image given below shows the eye of a patient with pseudomembranous conjunctivitis. It differs from membranous conjunctivitis in that the membrane can be easily peeled off in this condition leaving an intact underlying epithelium.



Solution to Question 4:

This child with pharyngitis and subsequent involvement of the eyes most likely has pharyngoconjunctival fever. This condition is commonly caused by adenovirus serotypes 3, 4, and 7.

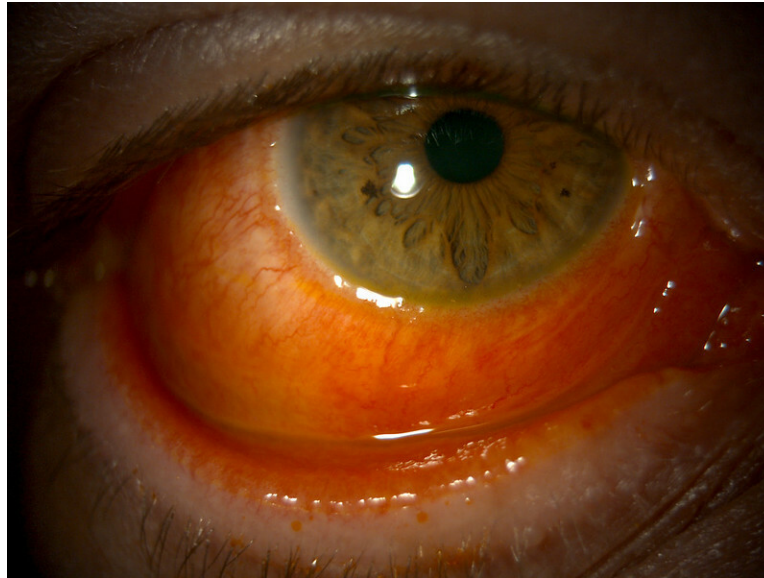
Pharyngoconjunctival fever is an acute and highly infectious illness characterized by fever, pharyngitis, acute follicular conjunctivitis, and regional lymphoid hyperplasia with tender, enlarged preauricular adenopathy. Within the classical triad, symptoms may vary in severity. It is usually bilateral, with one eye typically having onset 1-3 days prior to the second eye. It is a

self-limiting disease and treatment is mainly symptomatic, as follows:

- Cold compresses several times per day for 1-2 weeks.
- Artificial tears 4-8 times per day for 1-3 weeks.

Newcastle conjunctivitis is usually clinically indistinguishable from this condition and is derived from contact with diseased fowls. Another conjunctivitis caused by adenovirus serotypes 8, 19 and 37 is known as epidemic keratoconjunctivitis.

The image given below shows the eye of a patient with pharyngoconjunctival fever. Significant conjunctival hyperemia, petechial subconjunctival hemorrhages, and conjunctival follicles are noted.



Solution to Question 5:

The image given above shows a subconjunctival haemorrhage in the eye. In this condition, the blood gets absorbed gradually in 1-3 weeks time without treatment.

A subconjunctival hemorrhage is also called subconjunctival ecchymosis. It is due to the rupture of small blood vessels and can occur spontaneously in elders with systolic hypertension or after local trauma. The condition is commonly associated with a benign course but, if the posterior limit of the hemorrhage is not seen, one should check for the base of the skull fracture.

Recurrent episodes warrant investigations for a bleeding diathesis or leukaemia. The use of NSAIDs and aspirin to treat the condition should be avoided.

Solution to Question 6:

The given clinical scenario of a patient with bilateral red-eye and a rapid spread of infection to family members points towards a likely diagnosis of hemorrhagic conjunctivitis. Examination findings of lid edema, serous discharge, and subconjunctival hemorrhage further consolidate this

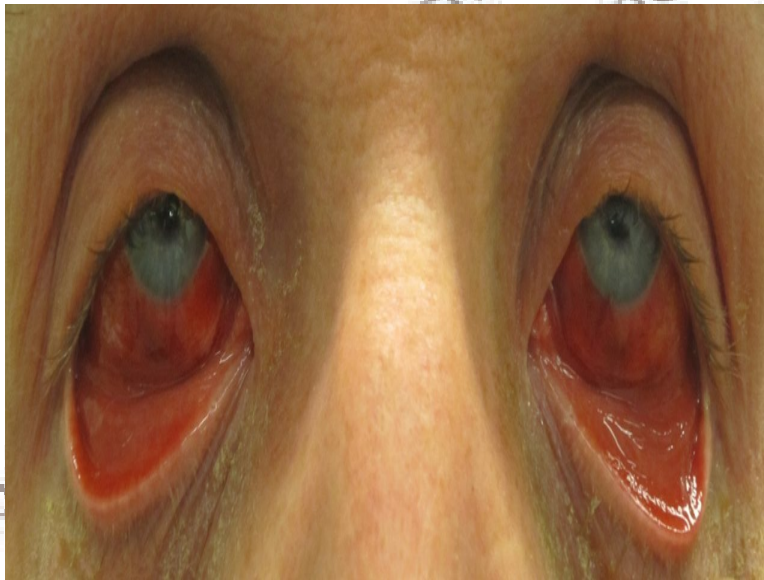
diagnosis.

Causes:

- Enterovirus 70 (most important)
- Coxsackievirus 24
- Echovirus 34
- Adenovirus
- Pneumococcus
- Haemophilus

There is no treatment currently available. The infection generally resolves itself within a five to seven-day period. The use of steroids can actually cause a corneal microbial superinfection which then requires antimicrobial therapy to eliminate.

The image given below shows the eyes of a patient with acute hemorrhagic conjunctivitis.



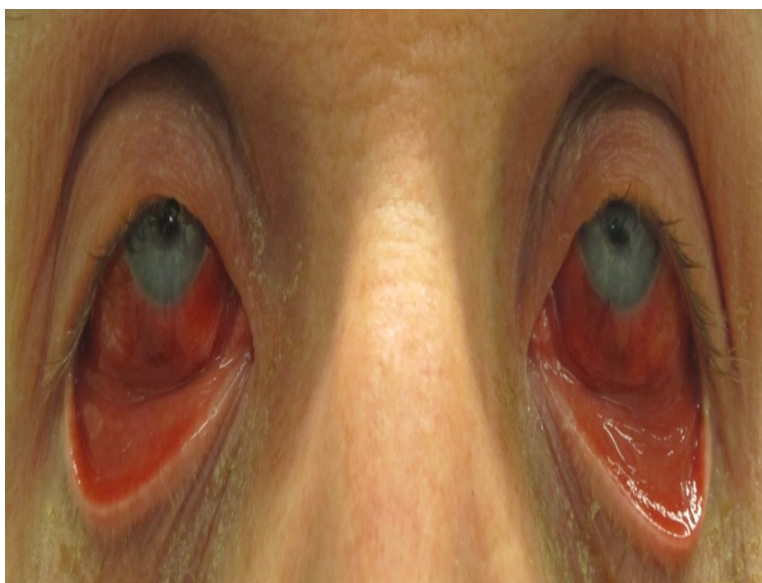
Solution to Question 7:

Acute hemorrhagic conjunctivitis is caused by picornoviruses namely enterovirus 70.

Acute hemorrhagic conjunctivitis also known as "Apollo conjunctivitis" is caused by picornoviruses such as enterovirus 70 and coxsackievirus A24. Rarely it is also caused by adenovirus type 11.

It manifests with severe conjunctival congestion, chemosis, subconjunctival hemorrhages, painful follicular conjunctivitis, tearing, lid edema and pre-auricular lymphadenopathy. The cornea may demonstrate a fine punctate keratopathy. The disease is usually self-limiting, running a course of 3-7 days.

The image given below shows the eyes of a patient with acute hemorrhagic conjunctivitis.



Enterovirus 71 (Option D) is associated with fatal meningitis, encephalitis, and paralysis. It is also implicated in Hand-Foot-Mouth disease (HFMD) and Herpangia which is papulo-vesiculo-ulcerative rash in oral cavity.

Solution to Question 8:

The given clinical scenario of right eye redness with pain, and findings of a raised whitish-yellowish lesion surrounded by mild erythema, points toward a likely diagnosis of pinguecula. A history of trekking, along with chronic sunlight exposure as a farmer, acts as a risk factor because it increases the exposure to the stronger UV radiation in the mountains.

Pinguecula has a fatty appearance and is usually bilateral. They are more often located nasally than temporally. It has horizontally oriented areas of bulbar conjunctival thickening that adjoin the limbus in the palpebral fissure area. They are associated with a 2- to 3-fold increased incidence of age-related macular degeneration. It is seen in most eyes by 70 years of age and in almost all by 80 years of age.

Pingueculitis responds to a brief course of NSAIDs or topical corticosteroids. Simple excision may be warranted in rare cases, wherein pingueculae may be chronically inflamed or cosmetically unsatisfactory.

Solution to Question 9:

The likely diagnosis, based on the above image, is pterygium.

Pterygium is a subconjunctival fibrovascular tissue encroaching the cornea. It is a degenerative condition of the sub-conjunctival tissue and is never inflammatory. It occurs due to exposure to UV-B radiation and the most common site is the nasal.

It is made up of three parts, "cap" (an avascular halo-like zone at the advancing edge), a head, and a body.

Clinical features:

- Astigmatism with the rule - most common finding
- Stocker's line - deposition of iron on the advancing edge
- Fuchs islets - small, discrete, whitish flecks consisting of clusters of pterygial epithelial cells, often present at the advancing edge.

The differential diagnosis is pseudo pterygium.

The treatment for pterygium is usually excision. To prevent a recurrence, mitomycin C or 5 fluorouracil is used. Other methods:

- Auto-grafting- Conjunctival limbal autograft (CLAG) from the superior limbal conjunctiva is the best method.
- Bare sclera technique- high recurrence rate of 30 to 80%.
- PERFECT surgery- zero rates of recurrence. (PERFECT stands for Pterygium Extended Removal Followed by Extended Conjunctival Transplant)

Pterygium	Pseudo-ptyerygium
Degenerative process	Inflammatory process
Occurs in elderly	Can occur at any age
Situated at the palpebral aperture	Can occur at any site
Either progressive, regressive, or stationary	Always stationary
A probe cannot be passed underneath	A probe can be passed under the neck

Solution to Question 10:

The given clinical vignette, with symptoms of blurry vision and persistent redness, along with findings of an abnormal triangular mass of thickened conjunctiva, is suggestive of pterygium. Cyanoacrylate is not used for pterygium surgery.

The various glues used in ophthalmology are as follows:

- Cyanoacrylate glue- Mainly used as a corneal patch to seal acute corneal perforations and improve visual outcomes, with reduced enucleation rates and need for tectonic keratoplasty
- Fibrin glue- It is a biological adhesive, which reduces the total surgical time. It is used in corneal and conjunctival surgeries, squint surgeries, cataract (to fix the haptics of intraocular lens [IOL] to the tissue in place of sutures in glued IOL), refractive surgeries, trabeculectomy, and even retinal surgeries.
- Newer tissue adhesives like bio-dendrimers and photo-cross-linkable adhesives-

- Bovine serum albumin-based bioadhesives: used in squint surgery
- Riboflavin-based bioadhesive activated on exposure to ultraviolet A: used in post-LASIK ectasia, infectious keratopathy, pseudophakic bullous keratopathy
- Hyaluronic acid-based photocatalytic glue activated with argon LASER: used in the repair of corneal lacerations

Solution to Question 11:

The given clinical scenario of red eyes, along with the examination findings of palpable preauricular nodes and follicular reaction involving the bulbar conjunctiva and semilunar folds, is suggestive of inclusion conjunctivitis. It is also known as chlamydial conjunctivitis, caused by *Chlamydia trachomatis* serotypes D–K. It is a sexually transmitted disease occurring commonly in young adults.

The modes of transmission include orogenital activities and hand-to-eye spread of infective genital secretions. Hence, it is important to treat all sexual partners simultaneously to prevent re-infection and to rule out other venereal diseases, such as gonorrhoea and syphilis. Treatment includes a single 1g dose of azithromycin or doxycycline 100 mg twice a day for 7 days.

The image given below shows the eye of a patient with chlamydial conjunctivitis.



Solution to Question 12:

Mucus fishing syndrome is closely related to giant papillary conjunctivitis.

In Mucus fishing syndrome, patients develop a chronic papillary reaction due to repetitive manual removal of mucus. The severe form of mechanically induced papillary conjunctivitis is known as giant papillary conjunctivitis (GPC). It is most frequently encountered with contact lens (CL) wear; in such cases, it is termed contact lens-associated papillary conjunctivitis (CLPC).

The increased risk in contact lens users is because of the build-up of proteinaceous deposits and cellular debris on the contact lens surface. It may also be caused by ocular prostheses, exposed sutures, and scleral buckles, corneal surface irregularity, or filtering blebs.

Solution to Question 13:

The given clinical scenario is suggestive of spring catarrh. Slit-lamp examination finding of shield ulcer is typical of vernal keratoconjunctivitis (VKC) or vernal conjunctivitis (another name of VKC).

Shield ulcers are transverse ulcers seen in the upper cornea due to epithelial macroerosions in VKC. It is a rare complication of VKC and should always be kept in mind for patients suffering from vernal conjunctivitis.

Corneal signs in VKC:

- Shield ulcers
- Vernal plaques
- Subepithelial scarring
- Pseudo-gerontoxon- resembles arcus senilis; has a cupid's bow outline

Treatment of VKC is mainly with allergy avoidance and lid hygiene. Cool compresses, topical antihistamines, topical NSAIDs and topical steroids can be used to reduce the intensity of symptoms.

The image given below shows a shield ulcer in a patient with VKC.



Solution to Question 14:

The given sago-grain appearance of the conjunctiva is seen in trachoma. It was the most important cause of corneal blindness in India affecting young children.

Conjunctival appearance:

1. In trachoma
 - Sago-grain appearance
 - A follicular reaction
2. In vernal keratoconjunctivitis
 - Cobblestone appearance
 - A papillary reaction seen in palpebral conjunctiva
3. Phlyctenular conjunctivitis
 - White-pinkish nodule in the bulbar conjunctiva
4. Xerophthalmia
 - Epithelial xerosis
 - Bitot spot in bulbar conjunctiva

Solution to Question 15:

Arlt's line is a conjunctival sign of trachoma.

Cornealsigns	Conjunctivalsigns
Superficial keratitisFollicles i n the limbal areaPannus-sup erior limbusCorneal ulcerHer berts pits-near Limbal areaC orneal opacity	Congestion- On upper tarsal, fornixFollicles (containingLe ber cellson histopathology)- On upper tarsal, fornixPapill ary hyperplasiaArlt's line-Lin ear scarring in sulcus subtars alisConcretions-Inspissated mucus, dead epithelial cells i n the glands of Henle

Solution to Question 16:

The clinical vignette of a young child with pain, photosensitivity and discharge, having repeated infections, along with findings of scarring and trichiasis is suggestive of trachoma. Cicatrising trachoma is stage 3 of trachoma and is characterized by scarring of the palpebral conjunctiva.

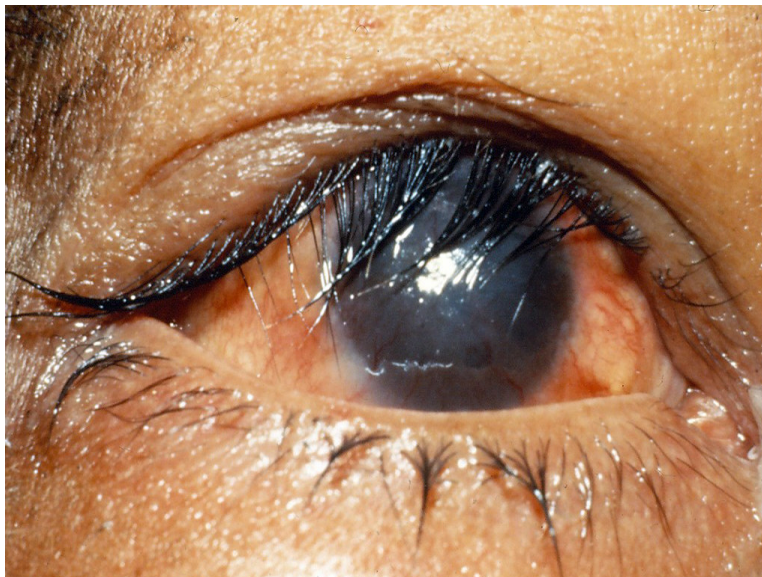
Cicatricial trachoma is the end result of scarring that occurs in individuals who suffer from severe, persistent infections in childhood. Trachomatous scarring is recognized by the presence of easily visible white bands on the underside of the upper eyelid. Severe scarring can result in contraction of the tarsal conjunctiva which causes the lid margin to roll inwards. This results in buckling of

the lid so the lashes rub on the eye known as trachomatous trichiasis. These lashes can lead to corneal opacities and scarring and eventually blindness.

The four stages of trachoma according to McCallan are as follows:

- Stage 1: Incipient - Immature follicles are present, but clinical symptoms are not obvious.
- Stage 2: Established - Typical trachomatous lesions without scarring.
- Stage 3: Cicatrising - Scarring along with active infection.
- Stage 4: Healed - Only scarring, the disease is either cured or quiet.

The image below shows trachomatous trichiasis with severe scarring-



Solution to Question 17:

Oral azithromycin is the drug of choice in the SAFE strategy of trachoma management. The adult dose is 1 g while 20 mg/kg body weight is given for children.

Oral azithromycin is slowly replacing tetracycline eye ointment as the first line drug used in trachoma treatment. The experience of Morocco, the first country that used azithromycin in a large-scale, nationwide elimination programme, has paved the way for the worldwide application of azithromycin as part of the SAFE strategy.

Solution to Question 18:

The image depicts Bitot's spot, a pathognomonic feature of xerophthalmia. Hence, the likely diagnosis in this 21-year-old man presenting with the above lesion is xerophthalmia.

Bitot's spot is a triangular, silvery-white, foamy patch on the bulbar conjunctiva which is usually bilateral. The most common site is temporal.

Solution to Question 19:

The given clinical scenario indicates that this patient has nyctalopia (diminished night vision) most likely due to vitamin A deficiency, suggested by his poor consumption of vegetables. Changes seen in the conjunctiva in vitamin A deficiency is hyperkeratosis of the squamous epithelium.

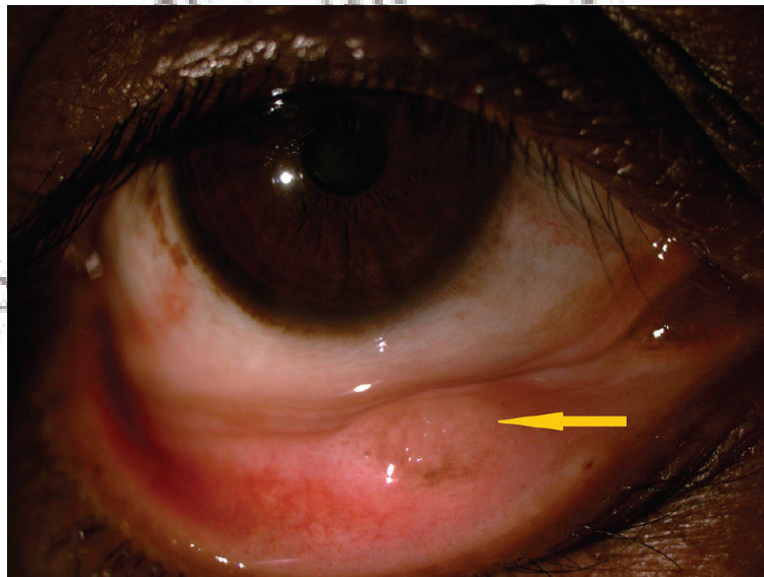
This occurs due to the metaplasia of the non-keratinized epithelium to keratinized epithelium.

Solution to Question 20:

Ophthalmia nodosa occurs due to irritation due to caterpillar hair.

It is often mistaken for tuberculosis. On examination, small, semitranslucent, reddish and yellowish-grey nodules are formed in the conjunctiva, cornea, and iris. Microscopy contains caterpillar hair surrounded by giant cells and lymphocytes. Treatment includes excision of the nodules in the conjunctiva.

The image below shows a patient's eye infected with ophthalmia nodosa.



Solution to Question 21:

The given image shows a white-pinkish nodule surrounded by hyperemia on the bulbar conjunctiva near the limbus with associated symptoms of redness and photophobia most likely has phlyctenular conjunctivitis. The given finding is referred to as a phlycten.

Phlyctenular conjunctivitis is a type 4 hypersensitivity reaction due to endogenous allergens like *Staphylococcus aureus* (most common) and *Mycobacterium tuberculosis*.

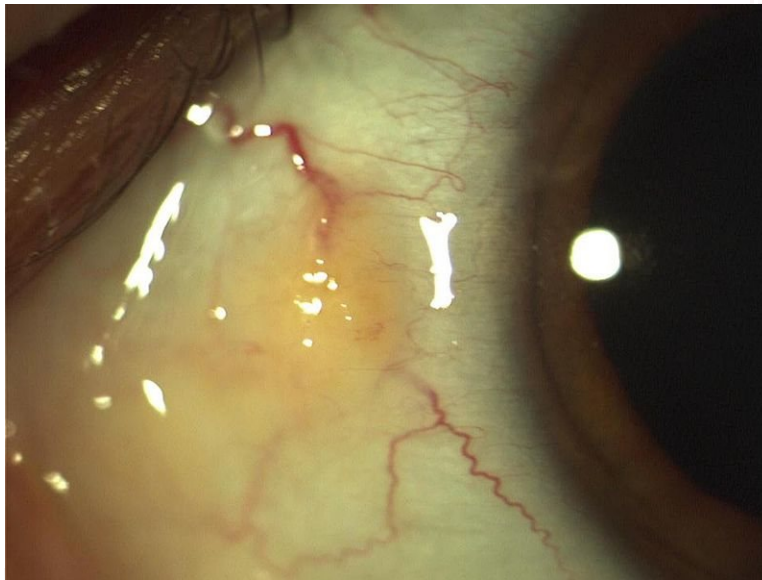
It is characterized by the following:

- Phlycten that may eventually form ring ulcers
- Shallow marginal ulcer due to the breakdown of limbal phlycten known as scrofulous ulcer
- Fascicular ulcer with a prominent parallel leash of blood vessels that heals leaving a band-shaped opacity

Treatment: Topical steroids; atropine if the cornea is involved.

Note: This can be differentiated from pinguecula by the following points:

Pinguecula is a yellow triangular patch on the conjunctiva near the limbus. It usually remains free from conjunctival congestion. The following image shows a yellow patch with tortuous blood vessels. There is no congestion as seen with a phlycten.



Solution to Question 22:

The given image shows symblepharon, which is not seen in phlyctenular conjunctivitis. It is a partial or complete adhesion of the palpebral conjunctiva of the eyelid to the bulbar conjunctiva of the eyeball.

Causes of symblepharon:

- Chemical injuries (especially alkalis)
- Burns
- Ocular cicatrical pemphigoid (OCP)
- Steven–Johnson syndrome
- Erythema multiforme
- Pemphigus bullosa
- Membranous conjunctivitis

Management:

- Lubrication
- Sweeping of fornices
- Symblepharon ring
- Symblepharon release and amniotic membrane graft

Solution to Question 23:

Chlamydia is at present the most common cause of ophthalmia neonatarum in developed countries.

Ophthalmia neonatarum also called neonatal conjunctivitis is inflammation of the conjunctiva occurring in an infant <30 days after birth. It should be ruled out in any infant having discharge or watering within the first week of birth.

Causes include:

- Chlamydia (Option A): neonatal inclusion conjunctivitis caused by serotypes D-K is at present the most common cause in developed countries.
- Gonorrhoea (Option B): effective prophylaxis against gonococcal infection and its subsequent decline has almost eliminated its incidence in developed countries.
- Staphylococcus aureus (Option D), Haemophilus species, Streptococcus haemolyticus, and Streptococcus pneumoniae are other less commonly seen causes of ophthalmia neonatarum
- Herpes simplex ophthalmia neonatrum rarely occurs and is caused by Herpes simplex type 2 virus from the mother.
- Chemical conjunctivitis (Option C) is caused by the application of prophylactic antibiotics or silver nitrate at the time of birth.

Time of presentation	Causative agent
Within 48 h	Chemical, Gonorrhoea
48–72 h	Staphylococcus aureus, Streptococcus pneumoniae
5–7 days	Herpes simplex II
>1 week	Chlamydia

Prophylactic treatment: 1% tetracycline or erythromycin eye ointment is applied to the eyes, if the mother is suspected to have gonococcal or chlamydia infection.

Curative treatment:

- Chlamydia trachomatis (D–K)- Erythromycin eye ointment
- Neisseria gonorrhoeae- Ceftriaxone injection, gentamicin drops, bacitracin eye ointment

The below image shows an infant presenting with ophthalmia neonatarum.



Solution to Question 24:

The clinical scenario of fever and red eyes, associated with matting of eyelids and pain, is suggestive of conjunctivitis. Lymph from the conjunctiva drains directly into the submandibular and preauricular nodes and hence, these group of lymph nodes are likely to be enlarged.

The lymphatic drainage of the conjunctiva is as follows-

- From the medial side, one-third of the superior and medial two-thirds of the inferior conjunctiva → Submandibular nodes
- From the lateral side, two-thirds of the superior conjunctiva and lateral one-third of the inferior conjunctiva → Preauricular nodes

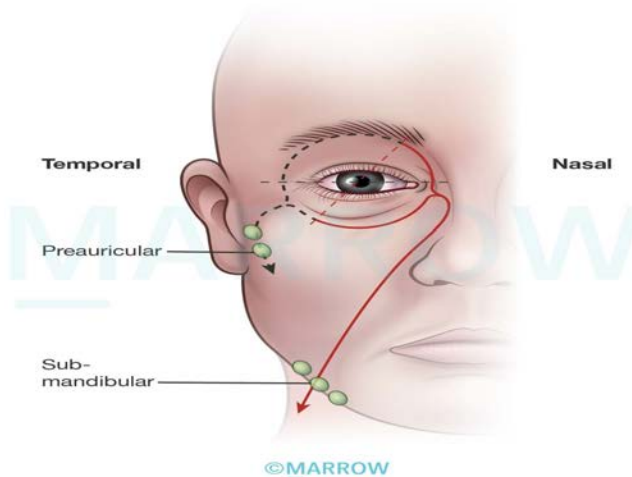
Specialized collections of T and B lymphocytes underlying a modified epithelium is known as conjunctiva-associated lymphoid tissue (CALT) present in the stroma (substantia propria). It is critical in the initiation and regulation of ocular-surface immune responses.

Note:

The stroma also consists of the accessory lacrimal glands of Krause and Wolfring which are exocrine glands responsible for the basal secretion of the middle aqueous layer of the tear film.

The image given below shows the lymphatic drainage of the conjunctiva.

Lymphatic drainage of conjunctiva



Solution to Question 25:

Superior limbic keratoconjunctivitis has a definite association with thyroid dysfunction. It affects one or both eyes of middle-aged women and about 65% of patients have abnormal thyroid function (usually hyperthyroidism).

It is a chronic disease of the superior limbus and the superior bulbar and the tarsal conjunctiva.

Clinical features:

- Papillary hypertrophy of the superior tarsal plate, often having a diffuse velvety appearance.
- Hyperemia of a radial band of the superior bulbar conjunctiva that stains with rose bengal.
-
- Superior filamentary keratitis that develops in about one-third of cases.

Management:

- Preservative-free lubricants
- Topical acetylcysteine 10% to 20%, four times daily to break down filaments
- Topical rebamipide

Solution to Question 26:

Elastotic degeneration and hyalinization of the conjunctival connective tissue occurs in pinguecula (and not pterygium).

In pterygium, there is elastotic degeneration of collagen with fibrovascular proliferation.

It is important to distinguish a pterygium from a pinguecula. A pterygium, while histologically similar to a pinguecula, is distinguished by fibrovascular tissue that grows onto the cornea.

Pingueculae do not have vascularized tissue and do not cross the limbus.

Solution to Question 27:

The above image depicts cobblestone appearance, which is a papillary reaction in the palpebral conjunctiva seen in vernal keratoconjunctivitis (VKC). No follicular reaction is seen.

Vernal conjunctivitis is also called spring catarrh or warm weather conjunctivitis. It primarily affects boys, and the onset is generally from the age of 5 years. It is primarily a type 1 hypersensitivity reaction that is IgE mediated. It also involves a type 4 hypersensitivity reaction.

Classical features of the conjunctiva in VKC:

Palpebral VKC - Cobblestone papillae are present in the upper tarsal conjunctiva associated with ropy discharge. Maxwell–Lyon sign is present which is a pseudomembrane formation of the upper lid when everted and exposed to heat

Limbal VKC - Horner – Trantas spots are white dots consisting of eosinophils and epithelial debris in the limbal area.

The image given below shows Horner-Trantas spots



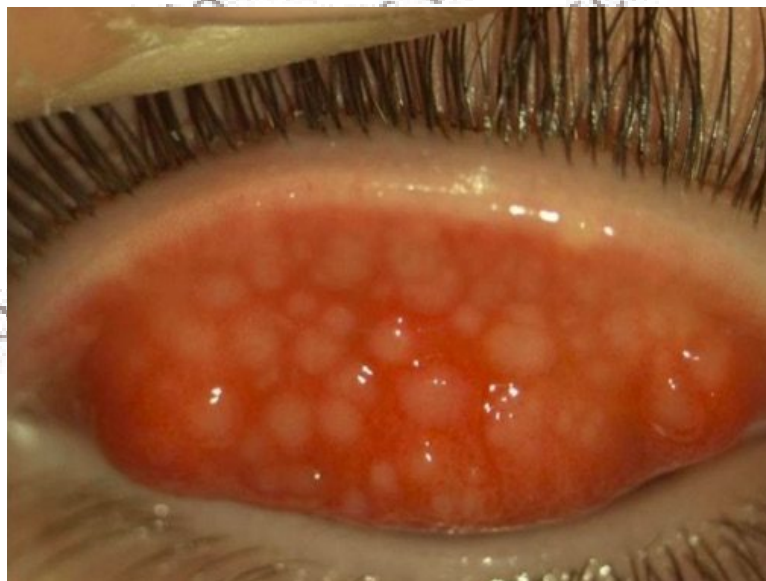
Treatment of this condition is with topical steroids and mast cell stabilizers such as cromolyn sodium. Topical cyclosporine can be used in patients who do not respond to first line treatment modalities.

Solution to Question 28:

Papillae (not follicles) are hyperplasia of the normal vascular system, with glomerulus-like branches of capillaries growing into the epithelium in inflammatory conditions. On the other hand, follicles are yellowish-white, round elevations of 1-2 mm diameter. Follicles are a localized

aggregation of lymphocytes present in the sub-epithelial adenoid layer of the conjunctiva.

Image: Conjunctival papillae (hyperplastic epithelium and capillaries) shown in the first image followed by conjunctival giant follicles (round yellowish-white elevations) shown in the second image.



Solution to Question 29:

In the grading of trachoma, follicular stage is defined by the presence of 5 or more follicles in the upper tarsal conjunctiva.

Trachoma is the leading infectious cause of blindness worldwide. It is caused by an obligate intracellular bacterium called *Chlamydia trachomatis* Serotypes A–C. The infection is transmitted by direct or indirect transfer of eye and nose discharges of infected people, particularly young children who harbour the principal reservoir of infection.

The image below shows the follicular stage of trachoma -



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If you purchased this from someone else,
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Sclera

Question 1:

What is the innermost layer of the sclera?

- a) Episclera
- b) Lamina fusca
- c) Bruch's membrane
- d) Posterior pigmented epithelium

Question 2:

The sclera is the thinnest at which of the following locations?

- a) At the limbus
- b) At the equator
- c) Anterior to attachment of the superior rectus
- d) Posterior to attachment of the superior rectus

Question 3:

A patient sustained a severe blow to the right eye from the lateral side during a fistfight. The eyeball is soft to touch and a rupture of the sclera is suspected. What is the least likely site of rupture?

- a) Insertion of recti
- b) Equator
- c) Medial limbus
- d) Lateral limbus

Question 4:

What is the predominant type of collagen in the sclera?

- a) Collagen type I
- b) Collagen type II

- c) Collagen type III
- d) Collagen type IV

Question 5:

A patient with glaucoma presents with a bulging cornea. What is the most likely diagnosis?

- a) Keratoconus
- b) Keratomalacia
- c) Staphyloma
- d) Granular dystrophy

Question 6:

Staphyloma involves _____.

- a) Uveal tissue with conjunctiva
- b) Conjunctiva with cornea
- c) Choroid with retina
- d) Uveal tissue with cornea

Question 7:

The ciliary staphyloma is due to which of the following?

- a) Myopia
- b) Choroiditis
- c) Iridocyclitis
- d) Scleritis

Question 8:

A 50-year-old female complaining of irritation in her left eye has the following finding. Which of the following condition commonly causes this?



- a) Polyarteritis nodosa
- b) Wegener's granulomatosis
- c) Rheumatoid arthritis
- d) Giant cell arteritis

Question 9:

The most common type of scleritis is _____.

- a) Non-necrotizing anterior scleritis
- b) Necrotizing anterior scleritis
- c) Non-necrotizing posterior scleritis
- d) Necrotizing posterior scleritis

Question 10:

All the following statements are true for a 40-year-old diagnosed with scleritis except-

- a) Scleromalacia perforans is associated with systemic disease.
- b) Pain is not a prominent feature.
- c) Glaucoma may occur.
- d) Retinal detachment is a known complication.

Question 11:

A 63-year-old woman presented with a two-week history of gradually worsening sharp and stabbing left eye pain with redness. The severe pain kept her awakened during the night. On investigation, B-scan ultrasound shows the 'T' sign due to fluid accumulation in sub-tenon space. What is the likely diagnosis?

- a) Optic neuritis
- b) Posterior scleritis
- c) Anterior scleritis
- d) Endophthalmitis

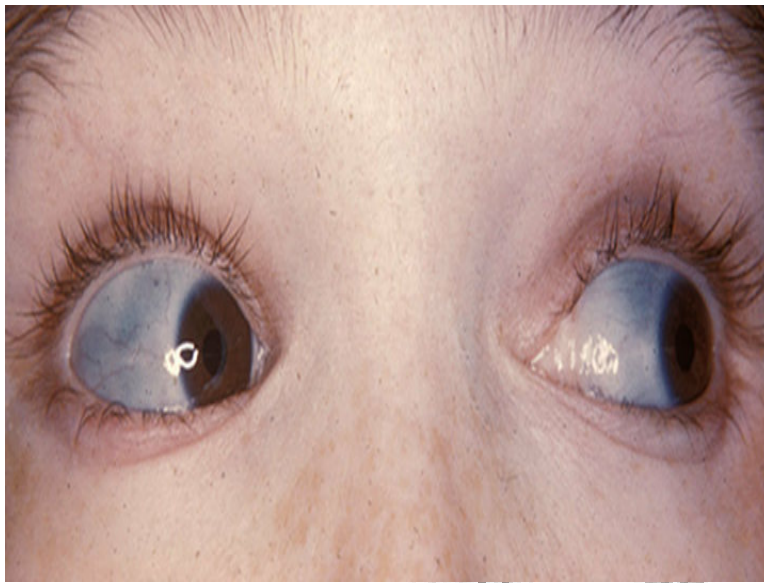
Question 12:

A patient develops surgically induced necrotising scleritis following an ocular surgery. The patient most likely underwent any of the following surgeries except:

- a) Pterygium excision
- b) Cataract surgery
- c) Trabeculectomy
- d) Tarso-frontalis sling surgery

Question 13:

As a first-year ophthalmology resident, you come across a child with the following presentation. Which of the following disorders is it associated with?



- a) Hemochromatosis
- b) Osteogenesis imperfecta
- c) Congenital ocular melanocytosis
- d) Rheumatoid arthritis

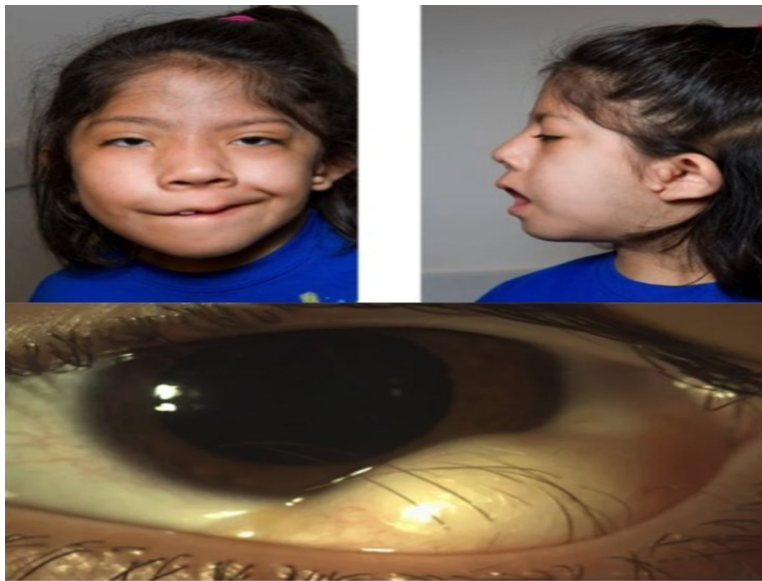
Question 14:

The sclera is thick in all the following conditions except-

- a) Nanophthalmos
- b) Phthisis bulbi
- c) Posterior scleritis
- d) Buphthalmos

Question 15:

An intern comes across a 7-year-old girl with the following features. All the following are components of her condition except-



- a) Disc coloboma
- b) Cardiac anomalies
- c) Hemivertebrae
- d) Lung hypoplasia

Answer Key

Question No.	Correct Option
1	b
2	d
3	d
4	a
5	c
6	d
7	d
8	c
9	a
10	b
11	b
12	d
13	b
14	d

Detailed Explanations

Solution to Question 1:

Lamina fusca is the innermost layer of the sclera.

Lamina fusca faces the choroid. It is the transition zone between the sclera and the choroid. It consists of a delicate meshwork of elastic fibres, fibrocytes and melanocytes, traversed by long posterior ciliary nerves and vessels.

Layers of the sclera from outside to inside include episclera, sclera proper, and lamina fusca, in that order.

Solution to Question 2:

The sclera is the thinnest posterior to the insertion of the rectus muscles.

The sclera is 1mm thick posteriorly, with maximum thickness around the optic nerve, 0.66 mm at the insertions of the rectus muscles and 0.33 mm beneath the rectus muscles and thinnest behind their insertions

Solution to Question 3:

The lateral limbus is the least likely site to be involved in a traumatic scleral rupture.

Traumatic scleral rupture, or globe rupture, occurs after blunt-force trauma. It usually occurs due to the compression force of the trauma. It therefore likely occurs at the insertion of rectus muscles, at the equator, or in an area parallel to the limbus opposite from the site of the impact. Rupture at the site of trauma, in this case lateral limbus, is rare.

The major signs of scleral rupture are conjunctival edema and subconjunctival hemorrhage, hyphema, ocular hypotony, impairment or dislocation of the lens, vitreous hemorrhage, and retinal detachment.

Solution to Question 4:

Collagen type I is the predominant type of collagen found in the sclera.

Collagen forms 50 to 75% of the dry weight of the sclera. Type II is absent in the sclera. It is found in the vitreous body of the eye.

Solution to Question 5:

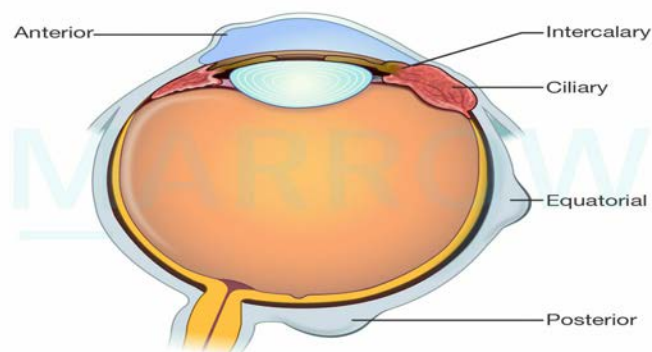
High intraocular pressure leads to the thinning of the outer tunics of the eyeball such as cornea and sclera, thereby resulting in their ectasia (bulging) which is known as staphyloma. Ciliary and equatorial staphylomas are most commonly associated with glaucoma.

Staphyloma is an abnormal protrusion of the uveal tissue through a weak and thin portion of the cornea or sclera. So, staphyloma is lined internally by uveal tissue and externally by weak cornea or sclera.

Types of Staphyloma:

Types	Pathology	Causes
Anterior staphyloma	Due to bulging of the pseudo cornea, internally lined by the iris & externally by the newly formed epithelium.	Sloughing corneal ulcer
Intercalary staphyloma	Due to bulging of weakened limbal tissue at the limbus, internally lined by the root of iris & the anterior most part of the ciliary body	Developmental glaucoma Secondary angle closure glaucoma Healing of perforating injury Peripheral corneal ulcer
Ciliary staphyloma	Due to bulging of the weakened sclera, internally lined by the ciliary body, can extend up to 8mm behind the limbus	Scleritis End stage glaucoma Perforating injuries
Equatorial staphyloma	Due to bulging of sclera with choroidal lining, seen around 14 mm behind the limbus, in the equatorial region	Scleritis Chronic uncontrolled glaucoma Scleral degeneration in pathological myopia
Posterior staphyloma	Due to bulging of sclera lined by the choroid, seen behind the equator	Posterior scleritis Pathological myopia Perforating injuries

Anatomical types of staphyloma



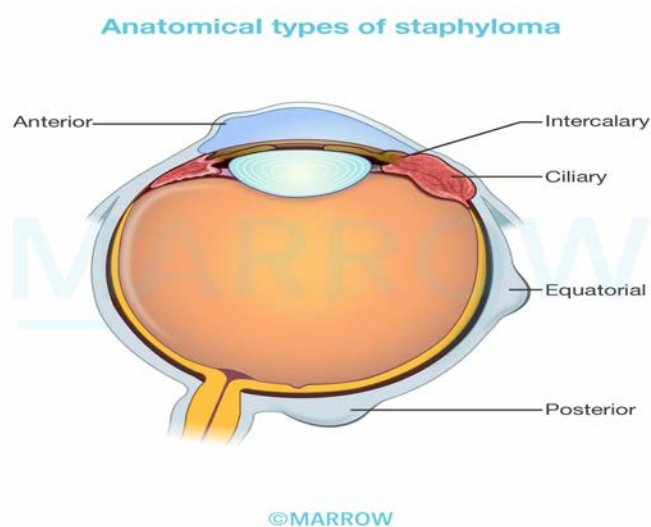
©MARRROW

Keratoconus (Option A) is a progressive bilateral corneal ectatic disorder due to defective synthesis of mucopolysaccharide and collagen tissue. It usually starts at puberty and progresses slowly, not associated with risk factors like glaucoma. Therefore, we can rule out keratoconus in this patient.

Solution to Question 6:

Staphyloma (a bunch of grapes) is a clinical condition characterized by ectasia (dilation or bulging) of the outer coats (cornea or sclera or both) of the eye with the incarceration (trapping) of the uveal tissue.

The image given below shows the anatomical types of staphyloma.



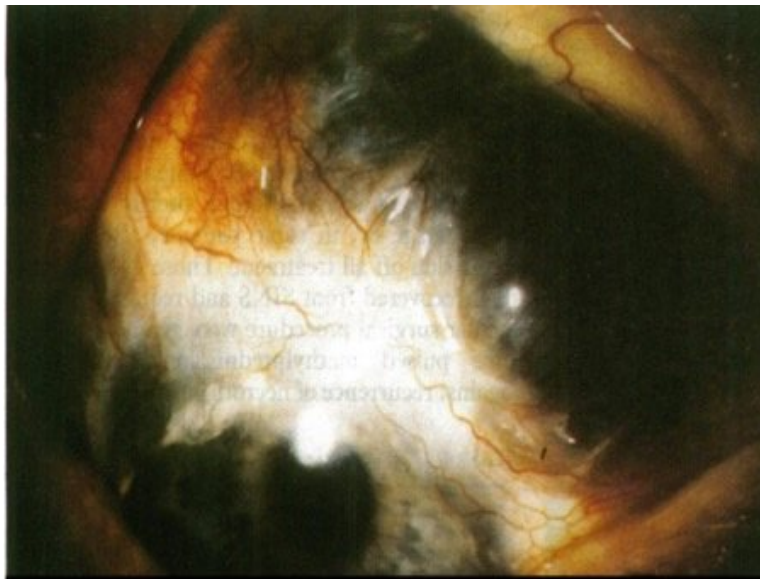
The above image shows an anterior staphyloma.

Solution to Question 7:

Ciliary staphyloma is due to scleritis.

Ciliary staphyloma is the ectasia (bulging) of weak sclera lined by the ciliary body. This affects the ciliary zone that includes the region up to 8 mm behind the limbus. It can be caused by scleritis, absolute glaucoma or trauma to the ciliary region of the eye.

The image below shows ciliary staphyloma with scleritis.



Solution to Question 8:

Irritation in the eye associated with areas of scleral thinning through which the uvea shines is suggestive of scleromalacia perforans. It is commonly associated with rheumatoid arthritis.

Scleromalacia perforans is a specific type of progressive scleral thinning without inflammation. Despite the term 'scleromalacia perforans', perforation is extremely rare as the integrity of the sclera is maintained by a thin layer of fibrous tissue.

Other collagen disorders associated with it are as follows:

- Wegener's granulomatosis (option B)
- Polyarteritis nodosa (option A)
- Systemic lupus erythematosus (SLE)
- Ankylosing spondylitis
- Relapsing polychondritis

Solution to Question 9:

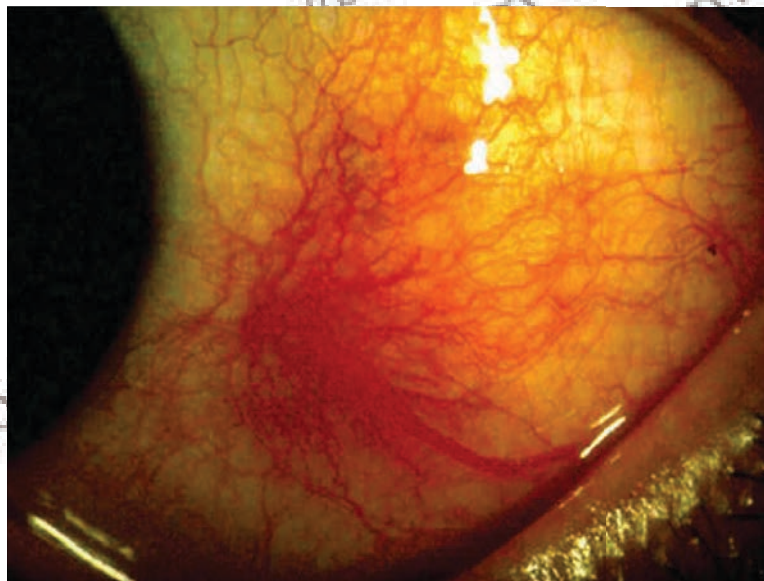
Non-necrotizing anterior scleritis is the most common type of scleritis.

Scleritis can be classified into anterior and posterior scleritis. Anterior scleritis affects the sclera in front of the equator, while posterior scleritis affects the sclera behind the equator of the eye.

Anterior scleritis is further classified into the following types:

- Non-necrotizing
- Diffuse: The most common variety
- Nodular: Scleral nodules are present. Annular scleritis is a variant in which the scleral nodules are arranged in a ring around the limbus.
- Necrotizing
- With inflammation
- Without inflammation: Also known as scleromalacia perforans. It is seen in elderly females with long-standing rheumatoid arthritis.

The image below shows anterior nodular scleritis.



Solution to Question 10:

Pain is a prominent feature of scleritis.

Scleritis is an inflammation of the sclera. Patients present with moderate to severe eye pain, redness, lacrimation and photophobia. Anterior scleritis affects the sclera in front of the equator and may lead to the following complications:

- Acute infiltrative stromal keratitis
- Sclerosing keratitis
- Peripheral ulcerative keratitis

- Uveitis
- Glaucoma
- Hypotony (rarely phthisis)
- Perforation of the sclera

Posterior scleritis involves the sclera behind the equator and may lead to uveal effusion with choroidal detachment, disc edema, proptosis or exudative retinal detachment.

Scleromalacia perforans is a type of anterior scleritis. It typically occurs in elderly women with rheumatoid arthritis (systemic disease). There is ischemia and necrosis of the sclera; however, there is no inflammation. Clinically, it is a painless condition with a white eye. Areas of thinned out sclera through which the uvea shines are seen in such patients.

Solution to Question 11:

The given clinical scenario of severe worsening eye pain with redness and a B-scan showing T sign due to fluid accumulation in sub-tenon space is suggestive of posterior scleritis.

Posterior scleritis is the inflammation involving the sclera behind the equator. Ultrasonography in posterior scleritis may show the following:

- Increased scleral thickness
- Scleral nodules
- Separation of tenon capsule from the sclera
- Disc oedema
- Choroidal folds
- Retinal detachment
- Fluid in the sub-tenon space may give a characteristic "T" sign. The stem of the "T" is formed by the optic nerve and the cross-bar by the fluid-containing gap.

The image given below shows a B-scan ocular ultrasound of the eye.



Solution to Question 12:

Surgically induced necrotizing scleritis (SINS) may occur following any ocular surgery involving the creation of a scleral wound. Tarso-frontalis sling surgery is a lid surgery and does not involve the sclera.

SINS is a condition of scleral inflammation and necrosis that may occur as a complication of the following surgeries:

- Pterygium excision
- Cataract surgery (especially with sclero-corneal tunnel)
- Trabeculectomy
- Vitreo-retinal surgeries
- Squint surgery

SINS is more likely to develop in those with rheumatoid arthritis, connective tissue disorders or following intra-operative use of mitomycin C. It is treated with NSAIDs and systemic steroids.

Solution to Question 13:

The child has blue sclera which is associated with osteogenesis imperfecta.

Blue scleral discoloration is due to scleral thinning. The underlying uvea thus becomes prominent and visible. Osteogenesis imperfecta (types I and IIA) and Ehlers-Danlos syndrome (type VI) are the most commonly associated diseases. Both diseases are inherited disorders of collagen formation.

Other options:

Option A: Hemochromatosis occurs due to iron deposition in various tissues. Rusty-brown perilimbal conjunctiva and scleral discoloration are seen.

Option C: Congenital ocular melanocytosis occurs due to an increase in the number, size and pigmentation of melanocytes. Sclera is multifocal slate-grey pigmented.

Option D: Rheumatoid arthritis may present with extraarticular manifestations such as secondary Sjögren's syndrome where keratoconjunctivitis sicca can be seen.

Solution to Question 14:

The sclera is not thick in buphthalmos.

The sclera is 1mm thick posteriorly, 0.66mm thick at the insertion of recti and 0.33mm thick beneath the recti. In buphthalmos, the sclera is stretched and thinned out, resulting in "blue sclera." Buphthalmos refers to an enlarged eyeball due to congenital glaucoma.

Solution to Question 15:

The given scenario of a young girl with facial asymmetry, deviation of angle of the mouth, flat nasal bridge, preauricular tags and limbal dermoid (lenticular yellow tumour at the limbus) of the eye is suggestive of Goldenhar syndrome. Lung hypoplasia is not a component of this syndrome.

Goldenhar syndrome or oculoauriculovertebral spectrum is a congenital developmental anomaly affecting structures derived from the first branchial arch. It has the following features:

- Systemic:
 - Hypoplasia of the malar, maxillary, and mandibular regions
 - Macrostomia and microtia
 - Preauricular and facial skin tags
 - Hemivertebrae (usually cervical)
 - Mental handicap
 - Cardiac, renal, and central nervous system (CNS) anomalies.
- Ocular:
 - Dermoid
 - Upper lid notching or coloboma
 - Microphthalmos
 - Disc coloboma

Basics of Cornea and Infectious Keratitis

Question 1:

What is the corneal epithelium normally composed of?

- a) Stratified keratinized epithelium
- b) Stratified non-keratinized epithelium
- c) Columnar epithelium
- d) Pseudostratified epithelium

Question 2:

Which of the following layers is responsible for maintaining the hydration and transparency of the cornea?

- a) Descemet's membrane
- b) Endothelial cells
- c) Stroma
- d) Corneal epithelium

Question 3:

What is the average density of corneal endothelial cells?

- a) 2000–2500 cells/mm²
- b) 1000–1500 cells/mm²
- c) 3500–4000 cells/mm²
- d) 2500–3000 cells/mm²

Question 4:

Below what horizontal diameter can a diagnosis of micro-cornea be made?

- a) 8 mm
- b) 9 mm

- c) 10 mm
- d) 11 mm

Question 5:

Which of the following organisms are capable of invading the intact cornea?

- a) Gonococcus
- b) Staphylococcus aureus
- c) Pneumococcus
- d) Pseudomonas

Question 6:

A patient with a history of uncomplicated cataract surgery 1 year back, presents with pain, redness, and purulent discharge from his right eye. A greyish-white serpiginous ulcer was noted over the lower third of the cornea. The anterior chamber showed a 4 mm hypopyon and a moderate flare reaction. A smear from the ulcerated area is likely to demonstrate which of the following organisms?

- a) Pseudomonas
- b) Pneumococcus
- c) Corynebacteria
- d) Gonococcus

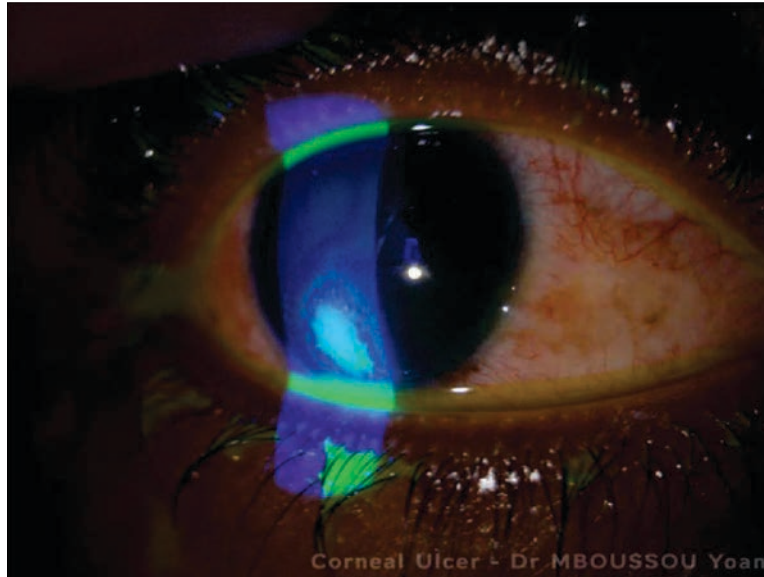
Question 7:

A young woman presents with sudden onset, rapidly progressing ocular pain, photophobia, and blurred vision in her right eye for 1 day. A deep, 4 mm diameter corneal ulcer, surrounded by stromal infiltrates was visualized. The anterior chamber has small amounts of mucopurulent discharge. She mentions using a cosmetic contact lens which she forgot to remove before sleeping 2 days ago. What is the most likely etiology?

- a) Staphylococcus
- b) Diphtheria
- c) Pseudomonas
- d) Aspergillus

Question 8:

A young woman comes with sudden onset ocular pain, tearing, and photophobia in the left eye. You suspect a bacterial corneal ulcer. What concentration of cephalazolin drops would you prescribe for this patient?



- a) 2%
- b) 3%
- c) 4%
- d) 5%

Question 9:

A non-sterile hypopyon is seen in_____.

- a) Pneumococcus infection
- b) Pseudomonas infection
- c) Fungal infection
- d) Gonococcal infection

Question 10:

A 30-year-old man presents with a history of injury to the eye caused by a leaf 5 days ago, resulting in pain, photophobia, and redness of the eye for 2 days. What would be the most likely pathology?

- a) Anterior uveitis
- b) Conjunctivitis
- c) Fungal corneal ulcer
- d) Corneal laceration

Question 11:

A farmer presents with blurring of vision in his left eye. He has a history of injury with a bush to the same eye while working. An ulcer is seen in his cornea with feathery margins. Slit-lamp examination shows infiltrates in the corneal stroma in a circular fashion. Which type of hypersensitivity reaction causes these infiltrates?

- a) Type 4
- b) Type 3
- c) Type 2
- d) Type 1

Question 12:

What is the characteristic feature of a fungal ulcer?

- a) Reverse hypopyon
- b) Dendritic ulcer on a fluorescein dye
- c) Ring abscess
- d) Satellite lesion

Question 13:

Steroids are contraindicated in_____.

- a) Vernal keratoconjunctivitis
- b) Fungal corneal ulcer
- c) Atopic dermatitis
- d) Stromal keratitis

Question 14:

A 41-year-old female presented with a one-month history of photophobia and foreign body sensation in her right eye. A slit-lamp examination showed a corneal ulcer. The culture of the corneal scrapings grew a filamentous organism identified as a fusarium species. What would be the drug of choice in this case?

- a) Natamycin
- b) Nystatin
- c) Fluconazole
- d) Amphotericin B

Question 15:

Which of the following is not a feature of the given condition?



- a) Superficial punctate keratitis
- b) Corneal guttata
- c) Geographical ulcer
- d) Immune stromal keratitis

Question 16:

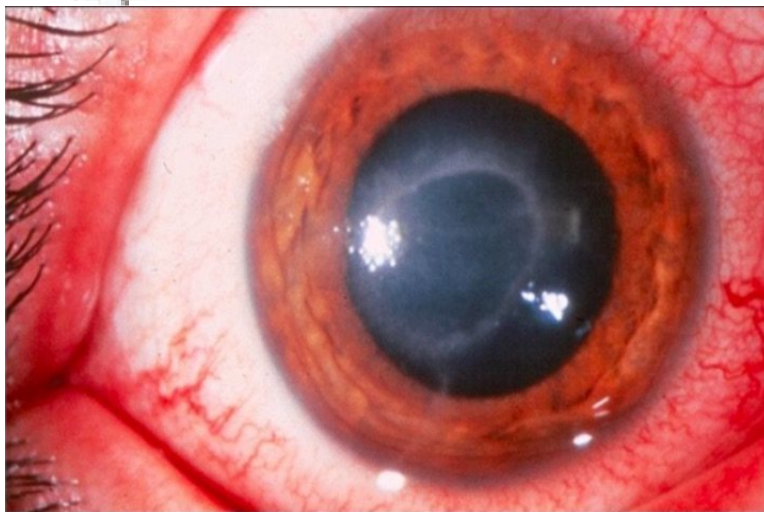
Which of the following statements is true about the given condition?



- a) It is related to Hutchinson's triad
- b) There is usually a history of primary HSV infection as a child
- c) Can lead to neurotrophic keratitis
- d) High dose topical acyclovir is the treatment of choice

Question 17:

A 17-year-old boy presented with blurred vision, photophobia, and increasing pain in his right eye. He has been using contact lenses for the past 2 years. He mentions swimming with his cousins 2 weeks back in a local pond. A close-up image of the eye is given below. What is the most likely explanation?



- a) Bacterial keratitis
- b) Fungal keratitis
- c) Simplex keratitis
- d) Acanthamoeba keratitis

Question 18:

A 39-year-old contact lens user with a corneal ulcer, showing no improvement to tobramycin drops, was referred to a tertiary care centre. Corneal biopsy revealed double-walled cysts within the stroma. Which of the following is the treatment of choice?

- a) Polyhexamethylene biguanide
- b) Neomycin
- c) Propamidine isothionate
- d) Nystatin

Question 19:

Which of the following does not cause interstitial keratitis?

- a) Tuberculosis
- b) Herpes simplex virus
- c) Syphilis
- d) Diphtheria

Question 20:

Which of the following infections is classically associated with salmon patch appearance of the cornea?

- a) TB
- b) HIV
- c) Syphilis
- d) HSV

Question 21:

A patient with recurrent corneal epithelial erosions is diagnosed with limbal stem cell deficiency. She has been enrolled on a research trial for corneal epithelial stem cell transplant. All are markers that can be used to identify stem cells from an ex-vivo culture except?

- a) ABCG2 transporter
- b) Nuclear protein p63
- c) Alpha-enolase
- d) Connexin 50

Question 22:

For her final year thesis, an ophthalmology resident is studying endothelial cell loss due to phacoemulsification surgery. The image is taken during the pre-op evaluation of one of the cataract patients. What does this show?



- a) Normal endothelium
- b) Polymegathism only
- c) Polymegathism and pleomorphism
- d) Corneal guttata

Question 23:

Which of the following is not a feature of herpetic keratitis?

- a) Stromal keratitis
- b) Dendritic ulcer
- c) Corneal guttata
- d) Disciform keratitis

Question 24:

A corneal foreign body was removed from a 53-year-old man. He was sent home with chloramphenicol and steroid eye drops. Two days later he came back with a corneal ulcer and a hypopyon. The ulcer scraping under microscopy revealed branched septate hyphae. What is the likely diagnosis?

- a) Candida
- b) Aspergillus
- c) Mucormycosis
- d) Histoplasma

Question 25:

Corneal sensation is lost in _____

- a) Conjunctivitis
- b) Fungal infection
- c) Trachoma
- d) Herpes simplex keratitis

Answer Key

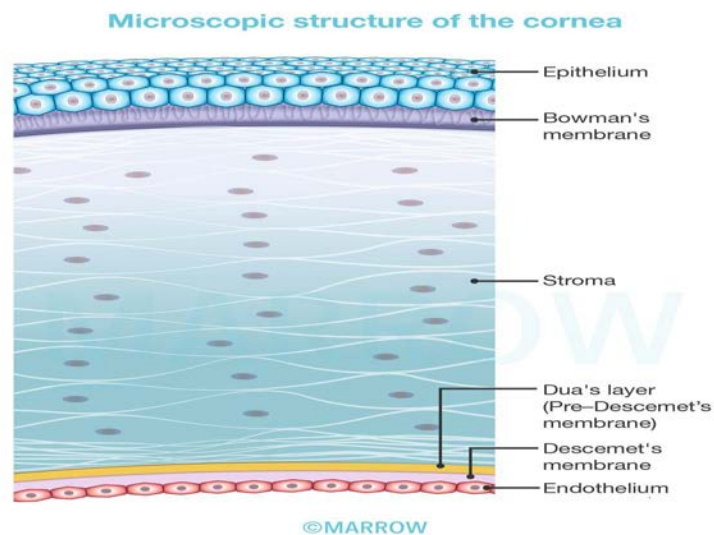
Question No.	Correct Option
1	b
2	b
3	d
4	c
5	a
6	b

7	c
8	d
9	c
10	c
11	b
12	d
13	b
14	a
15	b
16	c
17	d
18	a
19	d
20	c
21	d
22	c
23	c
24	b
25	d

Detailed Explanations

Solution to Question 1:

The corneal epithelium is made up of stratified squamous non-keratinized epithelium.



Layers of the cornea:

- Epithelium (Stratified squamous non-keratinized epithelium)
- Bowman's membrane - Acellular; does not regenerate
- Stroma (Thickest layer) - The stroma can scar, but cannot regenerate
- Dua's layer (Newly discovered, toughest layer)
- Descemet's membrane- Ends peripherally as Schwalbe's ring; has regenerative potential
- Endothelium (Most metabolically active layer)- does not regenerate

Note: Limbal stem cells are located at the corneoscleral limbus, in the palisades of Vogt. Deficiency may result in chronic epithelial defects and "conjunctivalization" of the cornea

Solution to Question 2:

The hydration level and transparency of the cornea is maintained mainly by the endothelial cells of the cornea.

The endothelium possesses Na⁺/K⁺ ATPase, which keeps the corneal stroma in a relatively dehydrated state. This helps maintain corneal transparency. Endothelial cells decrease in number with age. If the cornea becomes overhydrated, it begins to scatter light and gradually begins to lose its transparency

Other factors contributing to the transparency of the cornea are as follows:

- Regular arrangement of corneal lamellae (Maurice theory)
- The peculiar refractive index of corneal lamellae (Goldmann theory)
- Avascularity of the cornea
- A balance between the intraocular pressure (IOP) and swelling pressure

The different layers of the cornea are as follows:

- Epithelium
- Bowman's membrane
- Stroma (substantia propria)
- Dua's layer (pre-Descemet's layer)
- Descemet's membrane
- Endothelium

Solution to Question 3:

The density of corneal endothelial cells in adults is about 2500–3000 cells/mm².

Variation in corneal endothelial density with age:

- At birth- 3,500–4,000 cells/mm² at birth
- Young adults- 2500–3000 cells/mm²
- Old age (about 80 years)- 2000 cells/mm²

Corneas stored in eye banks are assessed for their quality and viability using a variety of parameters. Donor tissues are screened for stress lines, mechanical trauma, scarring, corneal edema, and infiltrates that could indicate poor tissue quality. They are further evaluated for thickness and endothelial cell density (a surrogate for endothelial function) which are important determinants of successful graft transplantation.

Endothelial cell density (ECD) and morphology are assessed using a specular microscope or by a bright-field/phase-contrast light microscope.

Note: The number of cells decreases at about 0.6% per year, and the neighboring cells enlarge to fill the space as the cells cannot regenerate. At a density of about 500 cells/mm², corneal edema develops and transparency is impaired. This is known as critical endothelial density.

Solution to Question 4:

To make a diagnosis of microcornea, in this child, the diameter of the cornea should be less than 10 mm.

Micro-cornea

- Horizontal corneal diameter is 10 mm or less (over 2 years of age) or less than 9 mm (in the newborn).
- There may be hypermetropia and a shallow anterior chamber but other dimensions are normal.

Megalocornea

- Horizontal corneal diameter is 12 mm or more (up to 2 years of age) or 13 mm or more (if age >2 years)
- There is typically high myopia and astigmatism, but normal corrected visual acuity.
- Usually X-linked recessive; 90% of affected individuals are male.

Cornea plana

- The cornea is flatter than normal, i.e., the radius of curvature is larger.
- There is a corresponding reduction in refractive power, resulting in high hypermetropia.

Keratectasia

- It is characterized by a protuberance between the eyelids caused by a severely opacified, and sometimes vascularized, cornea.
- It is often associated with raised intraocular pressure as well as intrauterine keratitis.
- Usually unilateral

Microphthalmos

- Microphthalmia is a condition in which the entire eye is small, with an axial length at least two standard deviations below the mean for age.
- It may or may not be associated with ocular dysgenesis.

Nanophthalmos

- In nanophthalmos (simple microphthalmos), the entire eye is small, but is structurally normal;
- It is usually associated with hypermetropia and strabismus

Anophthalmos: Anophthalmos (anophthalmia) refers to the complete absence of any visible globe structure.

Solution to Question 5:

Gonococcus can invade an intact cornea.

Organisms that can penetrate the intact epithelium of the cornea can be remembered with the mnemonic- CHaNNeLS

- Corynebacterium diphtheria
- Haemophilus influenza; Haemophilus aegypticus
- Neisseria gonorrhea; Neisseria meningitidis
- Listeria
- Shigella

Solution to Question 6:

Hypopyon corneal ulcer with a characteristic serpiginous description (ulcus serpens) is typically caused by Pneumococcus. The bacterial hypopyon is sterile.

Ulcus serpens is a greyish-white or yellowish disc-shaped ulcer occurring near the centre of the cornea. The ulcer has a tendency to creep over the cornea in a serpiginous fashion.

Risk factors include both surgical and non-surgical ocular trauma, dry eye, chronic corneal disease, immune alterations and contact lenses.

An examination may reveal the following signs:

- Swelling of lids and blepharospasm
- Conjunctival hyperemia
- Ciliary congestion, hypopyon and anterior chamber flare reaction
- Iridocyclitis

Image given below shows ulcus serpens.



Solution to Question 7:

The sudden onset and rapid progression of ocular symptoms and ulceration in this patient along with the history of improper contact lens usage are suggestive of a *Pseudomonas* infection.

Pseudomonas aeruginosa is the most common gram-negative organism isolated from corneal ulcers and is a frequent cause of contact lens-associated keratitis. Other risk factors include chlorinated swimming pools, hot tubs, ventilators, nebulizer, and vaporizer solutions, and ophthalmic solution bottles.

Pseudomonas keratitis has a rapid and fulminant course if left untreated and are associated with a worse visual prognosis than that caused by most other common bacterial pathogens. *Pseudomonas* spp. can cause corneal perforation in just 48 hours. Treatment is mainly by topical NSAIDs and topical antipseudomonal antibiotic eye drops such as tobramycin 1.3% and fluoroquinolone eye drops.

Solution to Question 8:

The concentration of fortified cephalazolin used to treat uncomplicated bacterial keratitis is 5% (50 mg/mL)

Treatment of uncomplicated bacterial keratitis:

- Antibiotic eyedrops (broad-spectrum)
- If not responding, fortified antibiotic eye drops are used
- Fortified cephalazolin (5%)
- Fortified tobramycin (1.3%)
- Fortified vancomycin (5%)

- Cycloplegic drugs- Preferably 1% atropine eye ointment or drops to prevent pain due to ciliary spasm. Also, it prevents the development of posterior synechiae.
- Oral vitamin A and vitamin C help in the healing of the epithelium
- Systemic analgesics and anti-inflammatory drugs

Management of non-healing ulcers:

- Debridement
- Chemical cauterization- 10–20% trichloroacetic acid, carbolic acid
- Conjunctival flapping

Management of impending perforation:

- Decrease intraocular pressure (IOP) by antiglaucoma drugs
- Tissue adhesive glue such as cyanoacrylate
- Bandage contact lens
- No strain

Solution to Question 9:

Fungal keratitis can cause non-sterile hypopyon.

Unlike bacterial ulcer, the hypopyon may not be sterile as the fungi can penetrate into the anterior chamber without perforation.

Aspergillus spp. are the most common cause.

Hypopyon in a fungal ulcer is fixed, non-sterile and contains fungal filaments.

Solution to Question 10:

The given clinical vignette of the patient presenting with pain, photophobia, and redness of the eye following trauma by a vegetative matter is suggestive of a fungal corneal ulcer.

Fungal corneal ulcer also known as mycotic keratitis is usually caused by trauma from vegetative matter such as branches of trees, leaves and animal tails. Injudicious use of topical steroids and antibiotics increases the predisposition to develop secondary fungal keratitis.

Mycotic keratitis can be caused by

- Filamentous fungi such as *Aspergillus* and *Fusarium*
- Yeasts such as *Candida* and *Cryptococcus*
- Dimorphic fungi such as *Histoplasma* and *Blastomyces*

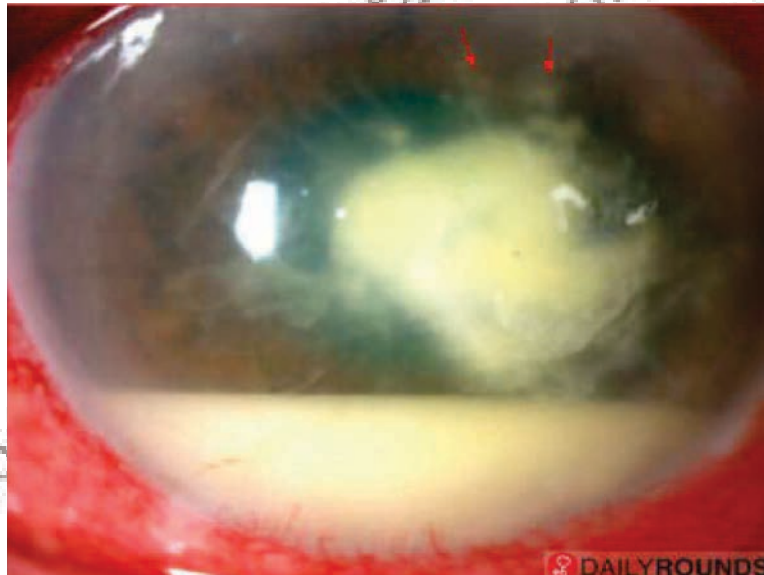
Typically in fungal corneal ulcers, the signs are more prominent than the symptoms. The symptoms of fungal keratitis are similar to that of bacterial ulcers which include pain, lacrimation, photophobia, blurred vision, and redness of the eyes.

A typical fungal ulcer has the following signs:

- Greyish white looking dry corneal ulcer with rolled out margin.
- Feathery finger-like extensions into the corneal stroma under intact epithelium
- Wessley ring which is a sterile immune ring that occurs due to the deposition of immune complexes
- Multiple small satellite lesions around the ulcer
- Hypopyon: The hypopyon in fungal ulcer is immobile and unsterile unlike bacterial ulcer (sterile hypopyon)
- Perforation is rare
- Corneal vascularization is usually absent.

A diagnosis of fungal ulcer is made by a typical history of trauma with vegetative matter followed by clinical examination and KOH mount or SDA agar cultures from corneal scrappings.

The image below depicts a fungal corneal ulcer with the arrowhead indicating a satellite lesion.



Other options:

Option A: Anterior uveitis is the inflammation of the uveal tissue from the iris up to the pars plicata of the ciliary body. Infectious uveitis can be caused by the exogenous entry of the organisms following penetrating injuries, perforation of corneal ulcer, and post-intraocular operations.

Acute anterior uveitis presents with symptoms of pain, photophobia, lacrimation, blurring of vision and redness of the eye. Signs of anterior uveitis include:

- Aqueous flare - It is the earliest sign of acute anterior uveitis, it occurs following the leakage of protein particles into the aqueous humor.
- Keratic precipitates(KP) - These are pathognomonic for anterior uveitis. They are proteinaceous cellular deposits at the back of the cornea arranged in a triangular fashion. These include mutton fat KP's as seen in granulomatous uveitis and granular KP's of non granulomatous uveitis.

- Iris nodules: typically occur in granulomatous uveitis. Koppe's nodules are at the pupillary border and Busacca's nodules are situated near the collarette.

Option C: Conjunctivitis is the inflammation of the conjunctiva of which the bacterial conjunctivitis is the most common. Acute bacterial conjunctivitis is characterized by marked conjunctival hyperemia, swelling of the conjunctiva(chemosis), edematous eyelids, sticking together of lid margins and mucopurulent discharge from the eye.

The image shows a patient with bacterial conjunctivitis.



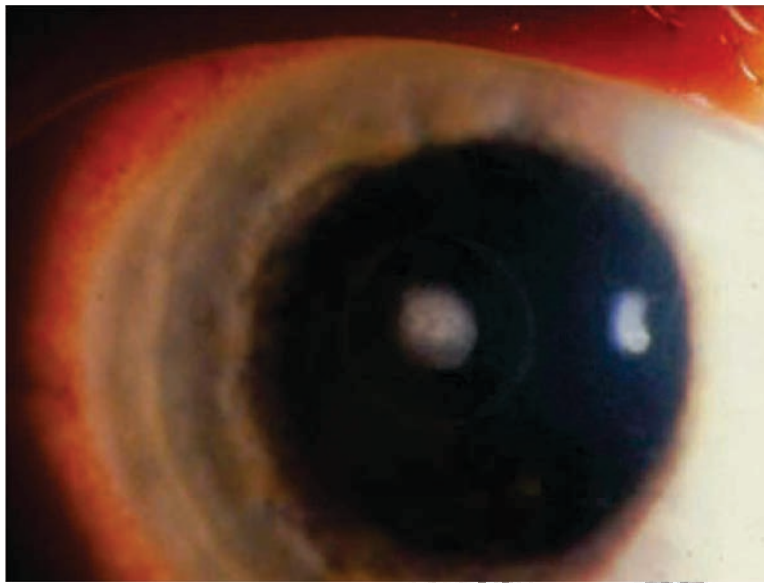
Option D: Corneal laceration: Partial corneal laceration or lamellar corneal laceration occurs from a blunt trauma and is treated by patching and antibiotics.

Solution to Question 11:

The given clinical situation points towards a fungal corneal ulcer and the infiltrates of stroma in a circular fashion are suggestive of the immune ring of Wessely. This immune ring is due to type 3 hypersensitivity reaction (antigen-antibody complex).

The immune ring of Wessely can be made out as a deep stromal haze in a slit lamp. It occurs as a result of the deposition of antigen-antibody complexes in the stroma. Once deposited it draws the immune cells, inciting further inflammation.

The given shows an ulcer in the cornea and the adjacent immune ring of Wessely.



Solution to Question 12:

Satellite lesions are characteristic of fungal ulcer.

The signs of a fungal ulcer include:

- Irregular margins of the ulcer.
- Presence of satellite lesions surrounding the ulcer.
- An immune ring (Wessely) may be seen around the ulcer due to deposition of the immune complexes.
- Slough is usually dry, with feathery borders surrounded by a yellow line of demarcation.
- Hypopyon may be present or absent. If present, it is thick, immobile due to the direct invasion of the fungal hyphae into the anterior chamber.

Solution to Question 13:

Steroids are contraindicated in fungal corneal ulcer

Steroids make the fungi facultative pathogens when they are otherwise symbiotic saprophytes. Therefore, the use of these drugs predisposes the patients to fungal infections and perforation.

Solution to Question 14:

This patient with a one-month history of eye foreign body sensations and photophobia and slit lamp exam showing an ulcer that had fusarium growth most likely has a fungal corneal ulcer. Natamycin 5% suspension is the drug of choice for filamentous fungal keratitis caused by

fusarium species.

Surface debridement is also done as it helps to remove slough, reduce the load of infection and enhances drug penetration. Amphotericin B is particularly effective against yeasts but less effective against filamentous fungi. Also, its penetration through an intact epithelium is less than natamycin.

Amphotericin B is the first agent of choice against yeasts. Amphotericin B (0.15%) drops can be considered alone or in combination with natamycin in refractory cases.

Treatment of specific fungal corneal ulcers:

Topical antifungal drops

- Aspergillus and Fusarium- Natamycin 5%, amphotericin B, fluconazole (0.2%)
- Candida- Amphotericin B (0.15%) or econazole (1%)

Systemic antifungals such as itraconazole and voriconazole are used for severe non-responding cases or if the location of the ulcer is near the limbus or if there is a risk of perforation leading to endophthalmitis.

Solution to Question 15:

The above image shows a dendritic ulcer caused by herpes simplex virus. Corneal guttata are not seen in HSV infection. They are droplet-like accumulations of non-banded collagen on the posterior surface of Descemet's membrane. It is a feature of corneal dystrophies.

Solution to Question 16:

The image shows a unilateral painful, vesicular rash suggestive of herpes zoster ophthalmicus (HZO) caused by reactivation of the varicella-zoster virus (VZV) within the ophthalmic nerve. Neurotrophic keratitis is a complication of HZV infection.

Keratitis occurs due to corneal nerve damage by the virus leading to decreased sensation, frequent infections, persistent epithelial defect, thinning and even perforation.

Corneal lesions in HZO (zoster keratitis):

- Epithelium- superficial punctate keratitis and pseudo-dendritic ulcer
- Stroma- nummular keratitis and disciform keratitis

Other ocular lesions in herpes zoster:

- Uveitis
- Episcleritis and scleritis
- Conjunctivitis
- Acute retinal necrosis
- Secondary glaucoma

- Anterior segment necrosis and pthisis bulbi

Other options:

Option A: Hutchinson's rule is related to herpes zoster ophthalmicus. It states that if the tip of the nose is involved, there will be a significant risk of ocular involvement. This is due to the involvement of the nasociliary nerve. Hutchinson's triad is seen in congenital syphilis.

Option B: There is primary HZV infection (chickenpox/varicella) not HSV infection.

Option D: Systemic acyclovir is the treatment of choice; not topical antiviral drops. Oral acyclovir 800 mg five times a day for 7-10 days, ideally initiated within 72 h of the onset of symptoms is the usual regimen.

The image shows Hutchinson's sign in a patient with herpes zoster ophthalmicus.



Solution to Question 17:

The history of a patient with contact lens use with swimming in freshwater ponds with the image depicting a ring lesion (stromal infiltrate) makes *acanthamoeba keratitis* the likely diagnosis.

Acanthamoeba keratitis is a rare protozoal infection of the cornea. It is more common in soft contact lens users. The organism exists either as active, motile trophozoites or dormant, double-walled cysts. Suspicious sources of infection can be swimming or use of the hot tub while wearing contact lenses, homemade saline solution to wash contact lenses, exposure to standing groundwater etc.

Pain is disproportionate to the degree of keratitis due to perineural invasion. It presents as ring lesion (most common), pseudo-dendrites and as a reticular pattern.

Diagnosis:

- Confocal microscopy - may reveal bright trophozoites with pseudopodia in the basal epithelium and double-walled cyst forms

- Histopathology - double-walled cysts are seen on H&E. Other stains for Acanthamoeba include calcofluor white, acridic orange, Giemsa-Wright, Wheatly trichrome or periodic acid-Schiff
- Culture - Acanthamoeba can be grown on a bed of E. coli plated on a non-nutrient agar

Solution to Question 18:

This patient with a corneal ulcer whose biopsy showed double walled cysts in the stroma most likely has acanthamoeba keratitis. Polyhexamethylene biguanide (PHMB) is the treatment of choice for acanthamoeba keratitis.

Combination therapy with chlorhexidine is most effective against trophozoites and cysts. This is recommended as the first line of therapy.

Other topical anti-amoebic agents:

- Diamidines- Propamidine isethionate (0.1 %), Hexamidine
- Aminoglycosides- Neomycin, Paramomycin
- Imidazoles- Clotrimazole, Miconazole

Solution to Question 19:

Diphtheria is not a cause of interstitial keratitis.

Interstitial keratitis is a non-ulcerating, inflammatory reaction of the corneal stroma that is characterized by non-suppurative infiltration and often stromal vascularization. The epithelium and endothelium are not affected. Scarring due to stromal inflammation and vascularization, in interstitial keratitis, can result in decreased vision.

Interstitial keratitis (IK) is a common endpoint of any keratitis that predominantly involves the corneal stroma. So there is a wide range of potential etiologies.

Treatment for IK typically involves the use of topical corticosteroids, as well as appropriate systemic medications for the associated systemic disease.

Immune-mediated etiologies	Infectious etiologies
Cogan syndrome: Interstitial keratitis + deafness + vertigo Sarcoidosis Mycosis fungoides	Bacterial causes: Syphilis; Lyme disease Tuberculosis; Leprosy Brucellosis; Leptospirosis. Viral causes: Herpes simplex (HSV), Herpes zoster (VZV) Epstein-Barr (EBV), mumps, rubella (measles) influenza, and vaccinia Parasitic causes: leishmaniasis onchocerciasis trypanosomiasis plasmodium acanthamoeba

Solution to Question 20:

Salmon patch appearance of the cornea is classically associated with syphilitic infection.

This is because, syphilis causes deep vascularization of the cornea, leading to a radial bundle of brush-like vessels.

Stages of interstitial keratitis in congenital syphilis:

- Progressive stage - ground-glass appearance
- Florid stage - salmon patch appearance
- Stage of regression - ghost vessels



Solution to Question 21:

Connexin 50 is not a limbal stem-cell marker.

The major markers proposed for epithelial stem cells can be categorized into at least three groups:

- Nuclear proteins such as transcription factor p63
- Cell membrane or transmembrane proteins
- Integrins (integrin $\beta 1$, $\alpha 6$, and $\alpha 9$),
- Receptors (epidermal growth factor [EGF] receptor and transferrin receptor CD71)
- Drug-resistance transporters (ABCG2—a universal stem cell marker)
- Cytoplasmic proteins such as
- Cytokeratins (cytokeratin 14 - exclusively at LSC)

- Alpha-enolase

Solution to Question 22:

The given image is a specular microscopy view of endothelial cells of different shapes (pleomorphism) and large surface area (polymegathism). These are indicators of endothelial dysfunction.

A normal endothelium has a regular hexagonal shape. Because these cells are terminally differentiated cells, they do not divide, and when they die, the remaining cells have to stretch to cover more area, resulting in polymegathism and pleomorphism.

Taking an endothelial cell count can be helpful before taking patients up for cataract or refractive surgery so that any potential endothelial dysfunction or thinning is recognized before they have the surgical intervention.

Solution to Question 23:

Corneal guttata are seen in corneal dystrophy.

Peripheral guttata are seen in old age—Hassall–Henle bodies. Central guttata, on the other hand, are found in Fuch's endothelial dystrophy.

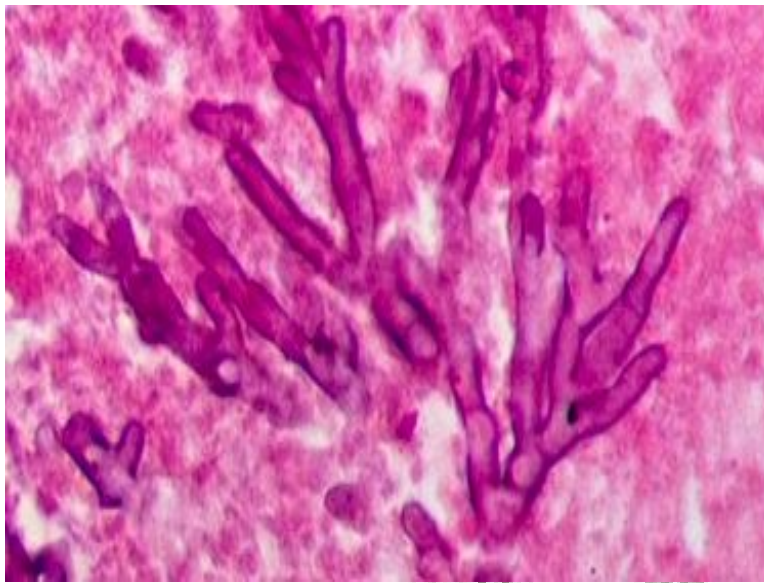
Solution to Question 24:

The given history of trauma and worsening with antibiotic-steroid drops suggests a fungal corneal ulcer. Out of the given options, *Aspergillus* is the fungus with branched septate hyphae.

Clinically, fungi are classified on a morphological basis into filamentous, yeast, and dimorphic forms:

- Filamentous organisms are multicellular with branched hyphae. It can be septate or aseptate.
- Septate filamentous organisms, such as *Fusarium* and *Aspergillus*, have hyphae that are divided by cell walls.
- Non-septate filamentous fungi, include *Mucor* and *Rhizopus*.
- Yeasts, such as *Candida* and *Cryptococcus*, are unicellular fungi that reproduce by budding, but in tissue, they might develop elongated buds (pseudohyphae).
- Dimorphic fungi such as *Histoplasma*, *Coccidioides*, and *Blastomyces* demonstrate both a yeast phase that occurs in tissues and a mycelial phase that appears on culture media and saprophytic surfaces.

Image: Branched septate hyphae of *Aspergillus* spp.



Solution to Question 25:

Corneal sensation is lost in herpes simplex virus (HSV) infection.

The pathognomic feature of viral keratitis is decreased corneal sensation. Decreased corneal sensation is also seen in diabetes mellitus and leprosy.

Non-infectious Disorders of Cornea

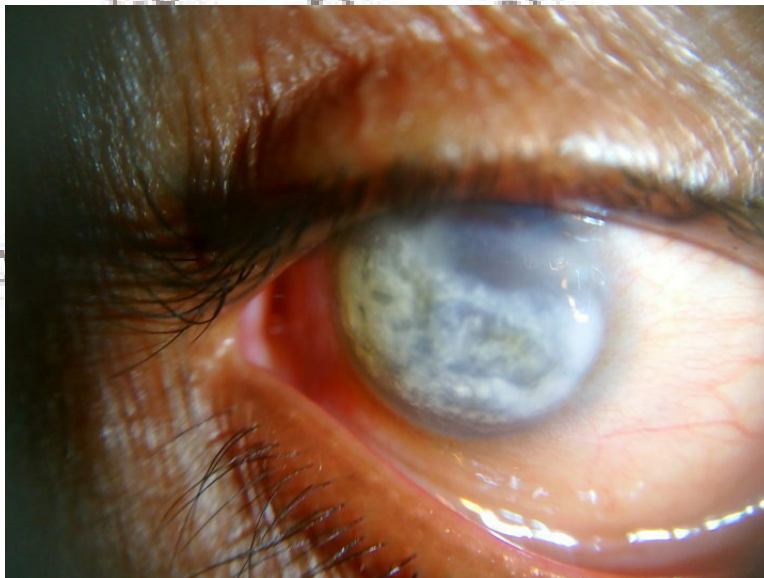
Question 1:

Deposition of which of the following is responsible for band-shaped keratopathy?

- a) Amyloid
- b) Calcium
- c) Monopolysaccharide
- d) Cholesterol

Question 2:

A patient was with the given finding has been having painless and slowly progressive vision loss for the past 10 years. Which of the following diseases is not associated with the given condition?



- a) Systemic lupus erythematosus
- b) Sarcoidosis
- c) Juvenile rheumatoid arthritis
- d) Chronic renal failure

Question 3:

A 64-year-old woman was brought by her son, during a cataract outreach camp. She gives a history of trauma to her left eye, one year back and a complete loss of vision in that eye since then. Her image is given below. Which of the following statement is true about her condition?



- a) Corneal tattooing is a treatment option in this lady.
- b) The given lesion is an adherent leucoma.
- c) It is due to scars involving Bowman's layer and superficial stroma.
- d) Excimer laser phototherapeutic keratectomy is the current treatment of choice.

Question 4:

Which of the following conditions is least likely to be associated with bilateral corneal opacities in a neonate?

- a) Sclerocornea
- b) Birth trauma
- c) Peter's anomaly
- d) Keratoglobus

Question 5:

What causes neurotrophic keratopathy?

- a) Facial nerve palsy
- b) Bell's palsy

- c) Trigeminal nerve palsy
- d) Third nerve palsy

Question 6:

Paralysis of which of the following nerves is responsible for exposure keratopathy?

- a) Trigeminal nerve
- b) Optic nerve
- c) Facial nerve
- d) Oculomotor nerve

Question 7:

Which of the following drugs is least likely to cause the given condition?



- a) Amiodarone
- b) Erythromycin
- c) Tamoxifen
- d) Chloroquine

Question 8:

A 30-year-old albino woman presents with a characteristic finding as shown below. Which of the following statements are true regarding the suspected condition?



- a) 1, 3, 5
- b) Only 2
- c) 2 and 4
- d) 1 and 3

Question 9:

A young male labourer presents to the casualty with severe bilateral eye pain that woke him from sleep. He reports welding the previous afternoon without a face shield and symptoms of photophobia, blurry vision, tearing, and worsening pain in bright light. Examination reveals mild conjunctival injection and diffuse punctate epithelial erosions on the cornea. What is the most likely diagnosis?

- a) Severe retinal bleaching
- b) Photophthalmia
- c) Infra-red keratitis
- d) Macular burn

Question 10:

Which of the following is a method for long-term preservation of cornea?

- a) Cryopreservation
- b) MK medium
- c) Moist chamber
- d) Organ culture method

Question 11:

Which of the following is the most common cause of infection following corneal transplantation in India?

- a) Staphylococcus epidermidis
- b) Streptococcus pyogenes
- c) Streptococcus pneumoniae
- d) Staphylococcus aureus

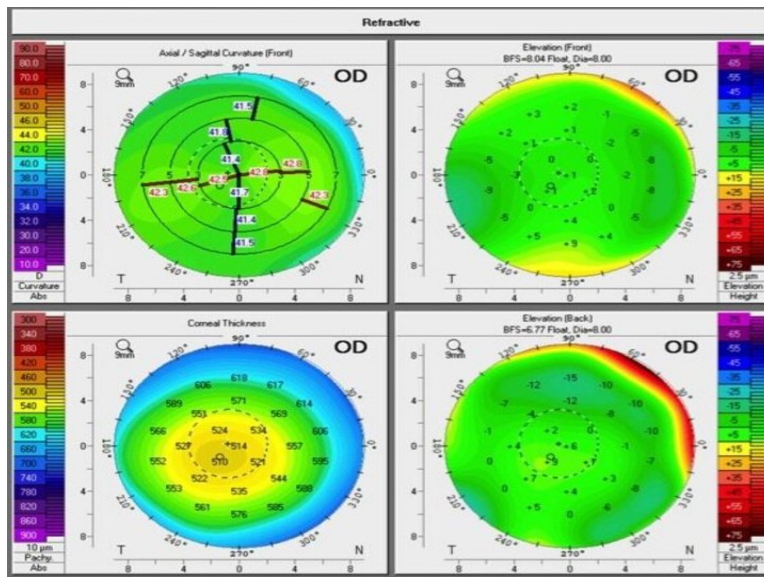
Question 12:

Which of the following techniques is the gold standard for assessing corneal thickness?

- a) Confocal microscopy
- b) Binocular keratometry
- c) Ultrasound pachymetry
- d) Corneal topography

Question 13:

Your aunt recently shared that it is becoming cumbersome for her to frequently change her glasses. You suggested a referral and the rapid change in power was diagnosed to be due to keratoconus. The following image shows the results of one of the investigations done on her. What is the name of the investigation?



- a) Pachymetry
- b) Keratometry
- c) Specular microscopy
- d) Corneal topography

Question 14:

A 20-year-old girl with high myopia was referred by the optometrist for further evaluation. She has a history of recurrent allergic conjunctivitis and persistent eye rubbing. An image is given below. Which of the following statements are true about the given condition?



- a) 2, 4, 5

- b) 1, 3, 4
- c) 1 and 2
- d) 3 and 5

Question 15:

Which of the following is not a treatment option in advanced keratoconus?

- a) Toric soft contact lens
- b) Vaulting scleral contact lens
- c) Rigid gas permeable contact lenses
- d) Collagen cross-linking

Question 16:

A 40-year old male with progressive loss of vision in his left eye was diagnosed with against-the-rule astigmatism. Examination revealed peripheral corneal thinning, but there was no inflammation. Which of the following is least likely to be the cause?

- a) Terrien's marginal degeneration
- b) Pellucid marginal degeneration
- c) Dellen
- d) Peripheral ulcerative keratitis

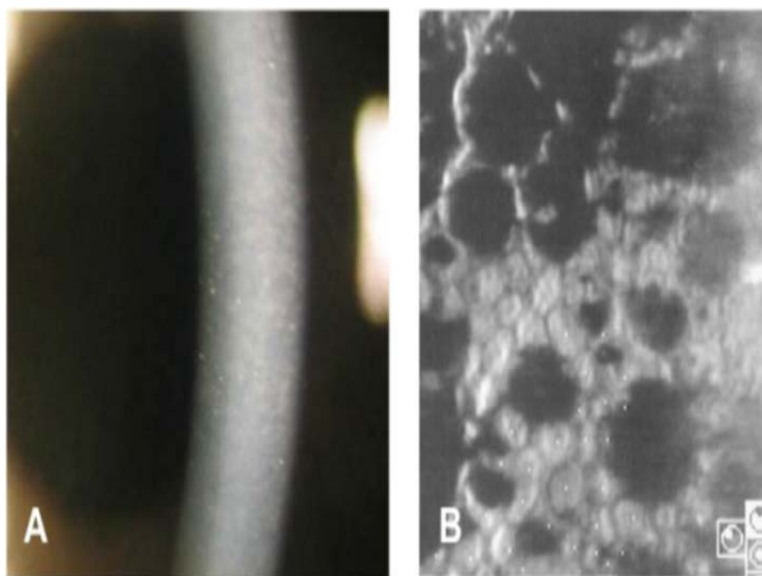
Question 17:

A 47-year-old woman presented with tearing, irritation, and photophobia of the left eye. On examination, a paracentral, crescent-shaped corneal melt was noted. Peripheral ulcerative keratitis (PUK) was suspected. What is the most likely cause of this condition?

- a) *Pseudomonas aeruginosa*
- b) *Staphylococcus epidermidis*
- c) Rheumatoid arthritis
- d) Ankylosing spondylitis

Question 18:

A 54-year-old school teacher presented with a chief complaint of pain in the eyes and intermittent blurry vision which was worse in the mornings. Generalized corneal haze was noted in both eyes but was more significant in the left. Examination findings are given below. What is the likely diagnosis?



- a) Lattice dystrophy
- b) Meesmann dystrophy
- c) Fuch's dystrophy
- d) Reis-Buckler's dystrophy

Question 19:

Which of the following is not true about macular corneal dystrophy?

- a) Spares the peripheral limbus
- b) Cause profound visual impairment
- c) Associated with mucopolysaccharidoses
- d) Autosomal recessive

Question 20:

Which of the following stains is used in identifying granular dystrophy of the cornea?

- a) Congo-red
- b) Periodic acid Schiff

- c) Alcian blue
- d) Masson's trichrome

Question 21:

Which of the following is implicated in the pathogenesis of oil-droplet keratopathy?

- a) Deposition of lipid in the anterior stroma
- b) Exposure to UV rays
- c) Lagophthalmos
- d) Trigeminal nerve palsy

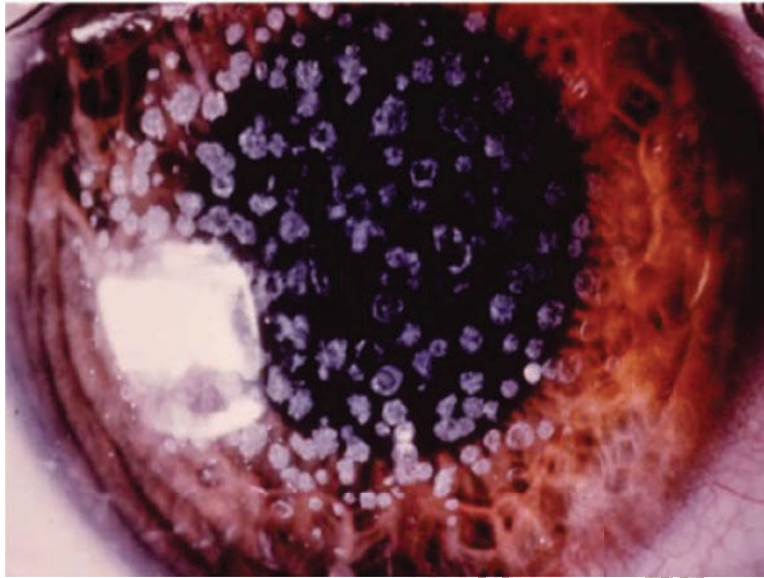
Question 22:

Which of the following statements regarding intracorneal ring segment (ICRS) implantation is false?

- a) The procedure is reversible.
- b) It is used in the management of keratoconus.
- c) Corneal perforation is a very common complication.
- d) It is made of polymethyl methacrylate.

Question 23:

A 29-year-old woman with normal visual acuity undergoes a routine eye exam. Slit-lamp examination reveals numerous crumb-like centrally dense deposits in the anterior corneal stroma, bilaterally symmetrical, with clear intervening stroma. Which of the following statements is likely to be true?



- a) Histopathology specimen will stain positive with alcian blue
- b) Histopathologic review will reveal amyloid deposits on congo red staining
- c) Histopathologic review will reveal hyaline deposits on the Masson trichrome stain
- d) Endothelium will show "beaten metal" appearance

Question 24:

A 6-month-old infant is brought with a history of diffuse, ground glass-like corneal haze in both eyes. The corneas are thickened with normal diameters and associated nystagmus. Which one of the following statements is true regarding this condition?

- a) Intraocular pressure (IOP) is likely to be elevated.
- b) The corneal stroma shows abnormal deposits that stain with amyloid.
- c) It could be a bilateral choristoma.
- d) Descemet's membrane may be thickened due to an abnormal endothelium.

Question 25:

Which of the following statements is true about Mooren's ulcer?

- a) Painless superficial ulcer that can perforate
- b) Associated with autoimmune and rheumatological conditions
- c) Characterized by a steep, overhanging edge
- d) Ulcerative process begins centrally and then moves circumferentially

Question 26:

Which layer of the cornea is affected in Reis–Buckler dystrophy?

- a) Epithelium
- b) Stroma
- c) Bowman's membrane
- d) Endothelium

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	d
5	c
6	c
7	b
8	b
9	b
10	a
11	a
12	c
13	d
14	a
15	a
16	d
17	c
18	c
19	a
20	d
21	b
22	c

23	c
24	d
25	c
26	c

Detailed Explanations

Solution to Question 1:

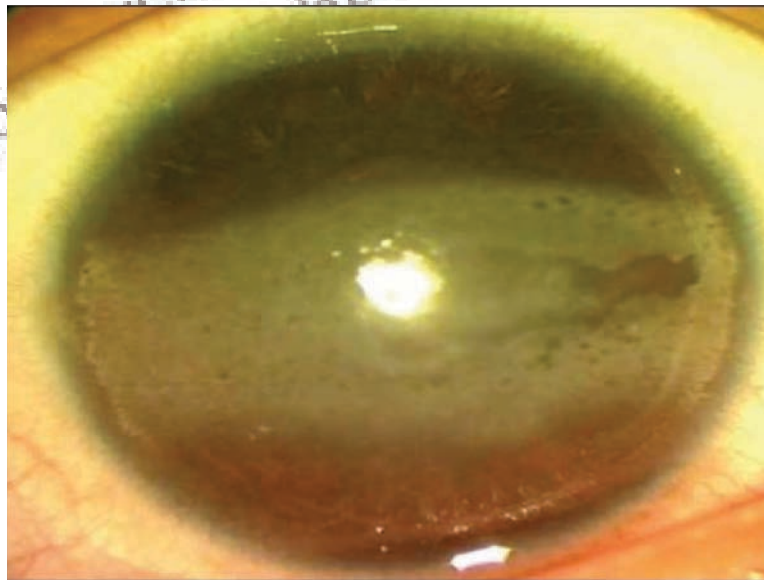
Band-shaped keratopathy is caused by pathological calcium deposition on the cornea in the form of a band.

On exam, there is a white band-like formation of calcium across the corneal surface with irregular borders and a peripheral clearing of cornea between the calcium deposits and the limbus. The calcium deposition is typically found beneath the epithelial layer within Bowman's layer but may extend into the anterior stroma.

Once the visual axis starts to get involved, there will be progressive loss of vision. Recurrence is common unless the predisposing cause is identified and treated.

Local causes such as chronic ocular irritation or inflammatory conditions and systemic diseases that elevate serum calcium levels are the main causes for band-shaped keratopathy. It can rarely have hereditary causes as well.

Image given below shows band-shaped keratopathy.



Solution to Question 2:

Based on the given clinical scenario and ocular finding depicted here, it is suggestive of band-shaped keratopathy, and systemic lupus erythematosus is not associated with this condition.

Band-shaped keratopathy (BSK) is due to calcium deposition, primarily in the form of hydroxyapatite, in the epithelial basement membrane, Bowman's layer, and the anterior stroma. Initially, these deposits are grey but may become chalky-white as the degeneration progresses.

It typically presents as a band-shaped opacity in the interpalpebral zone with a clear interval between the ends of the band and the limbus. The condition begins at the periphery and gradually progresses towards the centre. The surface of this opaque band is stippled due to holes in the calcium plaques in the area of nerve canals of Bowman's membrane.

Causes of band-shaped keratopathy include local causes such as chronic ocular irritation or inflammatory conditions and systemic diseases that elevate serum calcium levels:

Treatment:

- Medical management- Chelation using topical ethylenediaminetetraacetic acid (EDTA) 1.5–3.0%
- Surgical management- Photorefractive keratectomy (PRK) using excimer laser or manual superficial keratectomy

Local causes	Systemic diseases
Chronic anterior uveitis (especially in Juvenile Idiopathic Arthritis) Interstitial keratitis Superficial keratitis Phthisis bulbi End-stage glaucoma Silicone oil left in the aphakic eye	Hyperparathyroidism Sarcoidosis Multiple myeloma Hypophosphatemia Paget disease Chronic renal failure

Solution to Question 3:

The above image shows a dense white opacity, which results from the scarring of more than half of the stroma, characteristic of leucoma.

Corneal tattooing is a treatment option in patients with corneal opacities, who have lost their vision or who do not expect to recover it. It improves cosmetic appearance by altering the white colour of the opacity and blending it to normal eye colour.

Depending on the density, corneal opacities are graded as nebula, macula and leucoma.

- Nebular corneal opacity (fog or mist):
- Hard-to-see corneal scar, due to superficial scars involving Bowman's layer and superficial stroma.
- Slit-lamp detection is required.
- A thin, diffuse nebula covering the pupillary area interferes more with vision than the localised leucoma away from the pupillary area. Nebula produces more discomfort due to blurred image (irregular astigmatism).

- Macular corneal opacity:
- Semi-dense opacity that can be seen with proper illumination.
- Due to scarring involving about half the corneal stroma.
- Leucomatous corneal opacity (leucoma simplex):
- White scar that is easily seen just by looking at the eye.
- Due to scarring of more than half of the stroma
- Adherent leucoma:
- It results when healing occurs after perforation of the cornea with incarceration of iris

Secondary changes in corneal opacity which may be seen in long-standing cases include hyaline degeneration, calcareous degeneration, pigmentation and atheromatous ulceration.

Treatment:

- Keratoplasty is the treatment of choice in leucomatous corneal opacities.
- Optical iridectomy may be performed in cases with central adherent leucomas
- Phototherapeutic keratectomy (PTK) performed with excimer laser is useful only in superficial (nebular) corneal opacities.

Solution to Question 4:

Keratoglobus is least likely to be associated with bilateral corneal opacities in a neonate. Here, the cornea is clear.

Keratoglobus is a rare noninflammatory, familial disorder marked by generalised thinning and globular protrusion of the cornea. It can be congenital or acquired. The thinning is usually greatest at corneal periphery or midperiphery. It can cause refractive errors, otherwise, cornea is clear, normal angle of anterior chamber, normal intraocular pressure with no optic disc cupping.

Intrastromal ring segments and cross-linking can be used for management.

Other options:

Option B: Birth injuries resulting in Descemet's membrane tears can also cause corneal opacities.

Option C: Peter's anomaly, also known as iridocorneal adhesions or keratolenticular adhesions, is a posterior corneal defect with an overlying stromal opacity.

Option D: Sclerocornea is a congenital disorder in which the cornea is opaque and resembles the sclera, making the limbus indistinct. The central cornea is clearer than the periphery.

Solution to Question 5:

Neurotrophic keratopathy is caused by trigeminal nerve palsy.

Lesions of the fifth cranial nerve from the trigeminal nucleus to the cornea may lead to abnormalities of the normal corneal sensation and trophic stimulation.

Causes:

- Varicella-zoster (most common cause)
- Herpes simplex keratitis
- Trauma
- Irradiation to the eye
- Aneurysm/stroke
- Multiple sclerosis

Mackie characterized three stages of neurotrophic keratitis.

- Stage I is characterized by an irregular corneal surface, which later develops into an easily recognized punctate keratitis.
- Stage II is characterized by a frank epithelial defect, which is typically associated with mild anterior stromal oedema and inflammation.
- Stage III involves stromal melting and occasionally perforation.

Solution to Question 6:

Exposure keratopathy is caused by paralysis of the facial nerve.

It is the result of incomplete lid closure (lagophthalmos) resulting in drying of the cornea (despite normal tear production).

Causes:

- Neuroparalytic, e.g., facial nerve palsy
- Reduced muscle tone as in parkinsonism
- Mechanical such as lid scarring
- Proptosis

Note: Neurotrophic keratopathy is due to loss of trigeminal nerve innervation resulting in partial or complete loss of sensation.

Solution to Question 7:

Erythromycin is least likely associated with cornea verticillata/vortex keratopathy, a whorl-like opacity in corneal epithelium, as shown in the image above.

Vortex keratopathy occurs due to corneal deposits in the basal epithelium. They are almost always bilateral and are readily recognisable by their distinct appearance as fine, golden-brown or grey opacities that branch out from a central whorl, usually across the inferior cornea. Opacity due to

drugs is reversible upon stopping the drug.

Causes:

- Amiodarone (most common) (Option A)
- Tamoxifen (Option C)
- Indomethacin
- Chloroquine (Option D)
- Chlorpromazine (rarely)
- Fabry's disease

Solution to Question 8:

A 30-year-old individual with corneal changes, as shown above, known as gerontoxon or arcus lipoides, may require screening for underlying lipid disorder.

Arcus senilis (AS), also known as gerontoxon, arcus lipoides or arcus cornealis, is usually an age-related lipid degeneration of the peripheral cornea that forms a light grey or bluish ring. It is the most common peripheral corneal opacity, typically benign, occurring bilaterally and prevalent among the elderly.

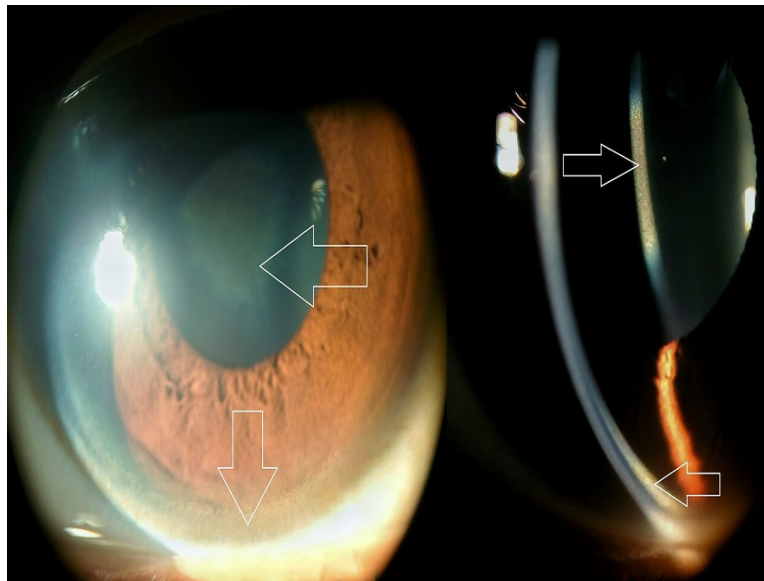
However, when observed in individuals under 50, termed arcus juvenilis, it may indicate underlying lipid disorder and an increased risk of cardiovascular disease, warranting lipid profile screening. Unilateral cases of arcus senilis may indicate contralateral carotid artery stenosis due to decreased ophthalmic blood flow on the ipsilateral side, resulting in protection from forming arcus in the ipsilateral eye.

The condition results from the leakage of lipoproteins (phospholipids and cholesterol) from limbal capillaries into the corneal stroma, initiating deposition at the superior and inferior cornea. Over time, the ring progresses circumferentially, sparing a clear corneal zone adjacent to the limbus called the interval of Vogt.

The following opacities and rings can resemble arcus senilis:

- Kayser- Fleischer (KF) ring: A dark brown ring caused by the deposition of excess copper in Descemet's membrane, classically seen in Wilson's disease. It may also be seen in certain chronic, cholestatic liver conditions such as primary biliary cirrhosis.
- Limbus sign: Caused by dystrophic calcification, seen as a "milky precipitate" at the corneal limbus. It can sometimes be confused with AS in the elderly population.
- Pseudo-gerontoxon: A greyish peri-limbal band of superficial scarring adjacent to a previously inflamed segment of the limbus resembling AS, seen in vernal keratoconjunctivitis.

The image below depicts the Kayser–Fleischer ring and anterior capsule of the crystalline lens (Sunflower cataract) seen in Wilson's disease.



Solution to Question 9:

The described corneal findings, with his classic history of welding less than 12 hours prior to presentation, are suggestive of photophthalmia.

Photophthalmia is produced due to exposure to short-wavelength ultraviolet (UV-B) rays. The cornea absorbs the majority of UV light (especially 311 to 290 nm). It results in desquamation of the epithelium leading to multiple painful corneal erosions. Causes of UV keratitis:

The symptoms appear characteristically after a latent period of 4- 5 hours of exposure to ultraviolet light. There is intense burning pain, foreign body sensation, and watering of the eyes. There may be associated marked photophobia and blepharospasm. On slit-lamp examination, corneal epithelium shows a breach in continuity and takes up fluorescein stain.

Management comprises cold compresses, lubricant eye drops and bandaging eyes for 24 hours with antibiotic eye ointment and cycloplegic. Avoid local surface topical anaesthesia. Crooke's glass is a prophylactic aid consisting of a spectacle lens combined with metallic oxides to absorb ultraviolet or infrared rays and should be used by those who are prone to exposure.

Note: Photoretinitis and macular burns are often a result of infrared radiation.

Artificial causes	Natural causes
Arc weldingFlash from short-circuits,Damaged metal halide lampsTanning bedsUV germicidal lamps.	Exposure to strong solar radiation at high altitudes especially on snow-covered terrain (snow blindness)Prolonged viewing of a solar eclipse without proper eye protection.

Solution to Question 10:

Cryopreservation is used for long-term preservation of cornea, up to 1 year.

Solution to Question 11:

Staphylococcus epidermidis is the most common cause of infectious keratitis after corneal transplantation in developing countries.

Pneumococcus spp. and *Staphylococcus aureus* have been found to be the most common causes of post-corneal transplant infections in developed countries.

Solution to Question 12:

The gold standard technique for assessing corneal thickness is ultrasound pachymetry, which uses a corneal pachymeter.

Measurement of the corneal thickness (pachymetry) indirectly reflects endothelial function. The average central corneal thickness is around 0.5 mm, which gradually increases toward the periphery to around 0.7 mm. Normally, as a diurnal variation, corneas tend to be slightly thicker just after a person awakes in the morning.

Other options:

Option A: Confocal microscopy is an imaging technique that allows in vivo examination of corneal structures at high magnification and resolution. It is used to assess corneal nerve morphology and corneal endothelium.

Option B: Keratometry measures the anterior corneal curvature and is traditionally performed with a manual keratometer, also known as an ophthalmometer.

Option D: Corneal topography is used to assess the anterior corneal surface, its shape and thickness.

The image below shows a corneal pachymeter.



Solution to Question 13:

The investigation shown in the image is corneal topography, and it is used for conditions like keratoconus, as seen in this clinical scenario.

Corneal topography captures the shape of the cornea by projecting concentric rings of light onto its anterior surface, forming a Placido disc pattern. The reflected light is analyzed with specialised software to create a detailed map of the corneal curvature.

Uses:

- Detection of corneal ectasia, mainly keratoconus
- To measure corneal thickness
- Prior to refractive surgery
- Sometimes prior to contact lens fitting

The image shows a Placido or Keratoscopic disc which is used for corneal surface assessment.



Solution to Question 14:

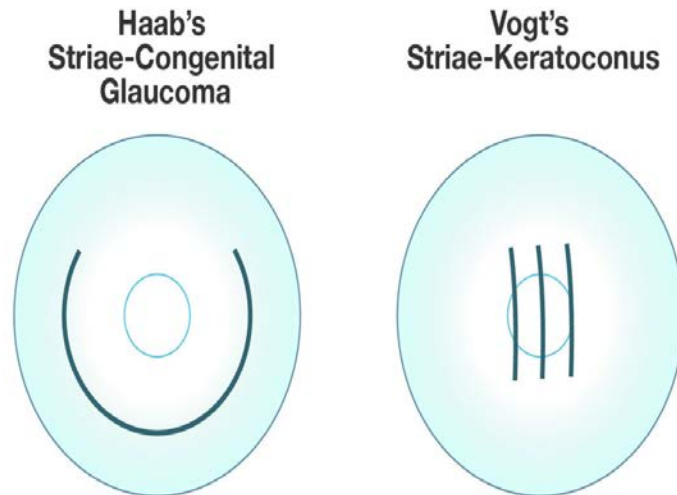
Keratoconus is a progressive bilateral corneal ectatic disorder. It manifests as characteristic cone-like steepening of the cornea associated with irregular stromal thinning, resulting in a cone-like bulge (protrusion) and significant loss of vision.

The pathogenesis is not fully known. Several associations have been identified, which include rigid contact lens wear, chronic eye rubbing, Down's syndrome, atopic disease, Leber's congenital amaurosis, and connective tissue disease.

It is a non-inflammatory disease. With the exception of the significant loss of vision, other classic signs of inflammation are not usually present. The asymmetrical thinning and the highly irregular corneal topography often causes significant myopia and irregular astigmatism.

Corneal topography is the primary diagnostic tool for KCN detection.

Whereas keratoplasty, historically has been the mainstay of treatment for keratoconus, it is now generally reserved for eyes that are unable to achieve functional vision with other surgical or non-surgical modalities and is only considered only as a last resort, in very advanced disease. It is the definitive treatment for keratoconus, not the treatment of choice.



Solution to Question 15:

Vision correction with spectacles with spherical/toric soft contact lenses is suboptimal and only applicable to the early stages of keratoconus, not in the advanced stages.

Keratoconus is a progressive, non-inflammatory, bilateral corneal ectasia characterized by cone-like steepening due to irregular stromal thinning. This leads to a conical corneal protrusion, significant vision loss, myopia and irregular astigmatism.

The available options for the management of keratoconus are highly dependent on the stage of the disease and its progression. If the disease is stabilized (no progression), the emphasis is given to correcting the vision. If the disease is progressing, the emphasis is on slowing (arresting) the progression.

Treatment options in moderate-to-advanced keratoconus:

- Rigid gas-permeable contact lenses and vaulting scleral lenses are the mainstays of vision treatment for moderate to advanced KCN. Their main advantage is the creation of a tear pool between the lens and the cornea, which naturally neutralizes the ocular aberrations associated with keratoconus. (Option B and C)
- Intra-corneal ring segments (INTACS)- small curved PMMA ring or set of rings that are implanted in the cornea to help flatten corneal curvature to improve vision.
- Corneal transplantation- Deep anterior lamellar keratoplasty (DALK) and penetrating keratoplasty (PKP)

Corneal collagen cross-linking with riboflavin (CXL or C3R) and ultraviolet-A rays arrest the progression of the disease. (Option D)

Solution to Question 16:

The examination in a 40-year-old male revealed peripheral corneal thinning with no inflammation, making peripheral ulcerative keratitis, an inflammatory condition, the least likely cause.

Peripheral ulcerative keratitis is a destructive inflammatory disease involving the peripheral cornea. It is associated with signs and symptoms of inflammation. The other options are causes of peripheral thinning in a quiet eye.

Other options:

Options A and B: Terrien's and Pellucid marginal degeneration are noninflammatory corneal degenerations. Pellucid degeneration causes bilateral, inferior corneal thinning that leads to marked, irregular, against-the-rule astigmatism. Terrien's degeneration usually begins in the superior cornea.

Option C: Dellen is a localized thinning of the peripheral cornea associated with the drying of a focal area. Usually associated with an adjacent elevated lesion (e.g., pinguecula or a large subconjunctival haemorrhage) that impairs physiological lubrication.

Solution to Question 17:

Rheumatoid arthritis is commonly associated with peripheral ulcerative keratitis (PUK). It is also the most common cause.

PUK is a group of destructive inflammatory diseases involving the peripheral cornea, characterized by the sloughing of corneal epithelium and keratolysis (corneal melting). It can lead to corneal perforation. The causes can be divided into:

- Infectious- bacterial, viral, protozoal, fungal.
- Non-infectious- rheumatoid arthritis, Wegener's granulomatosis, SLE, rosacea, psoriasis etc.

Ophthalmologic manifestations of rheumatoid arthritis (due to type III and IV hypersensitivity reactions):

- dry eye
- scleritis
- stromal keratitis
- peripheral ulcerative keratitis
- paracentral keratolysis

Treatment of PUK includes protection of the surface from drying and exposure; systemic corticosteroids and immunomodulatory; use of adjuncts such as amniotic membrane grafts to promote healing; and in some situations, tectonic keratoplasty.

Solution to Question 18:

The given clinical scenario along with beaten metal appearance on slit lamp and reduced endothelial cell density with coalescing guttata on specular microscopy is suggestive of Fuch's

endothelial dystrophy.

Fuchs endothelial dystrophy is a bilateral, slowly progressive, often asymmetric corneal disease. It is characterized by deterioration of endothelial cells and development of guttata, which are excrescences of Descemet's membrane (seen on specular or confocal microscopy). Females are affected more than males. It is more common in the 4th decade or later.

Endothelial layer dysfunction causes loss of corneal deturgescence, and bilateral oedema of the stroma, leading to ocular pain, glare, halos, and decreased visual acuity. Patients have a worse vision in the morning because of the eyes being closed during the night causing a buildup of corneal oedema.

Stages of Fuch's endothelial dystrophy:

- Stage 1: Central corneal guttata that spreads peripherally
- Stage 2: Corneal oedema - beaten metal-like appearance
- Stage 3: Bullous keratopathy
- Stage 4: Subepithelial scarring and superficial vascularization

Treatment:

- Conservative option- topical hypertonic (saline sodium chloride 5% drops or 6% ointment) and bandage contact lenses.
- The definitive treatment is endothelial keratoplasty: Descemet's stripping endothelial keratoplasty (DSEK/DSAEK) or Descemet membrane Endothelial keratoplasty (DMEK).

Solution to Question 19:

Macular dystrophy does not spare the peripheral cornea.

It is the least common corneal dystrophy. It has an autosomal recessive inheritance. Most other corneal dystrophies are autosomal dominant. It is associated with mucopolysaccharidoses and is characterized by the deposition of glycosaminoglycans in the stroma and staining by Alcian blue. It develops in childhood and is progressive, resulting in profound visual impairment in early adult life.

Clinical examination reveals superficial whitish fleck-like opacities. These opacities extend peripherally to the limbus and the deep stroma down to Descemet's membrane. The intervening corneal stroma develops a progressive and diffuse haze.

Note:

- Most common dystrophy overall- Microcystic/fingerprint/map-dot dystrophy
- Most common stromal dystrophy- Lattice dystrophy
- Dystrophy with corectopia- Posterior polymorphous dystrophy
- Autosomal recessive dystrophy- Macular dystrophy

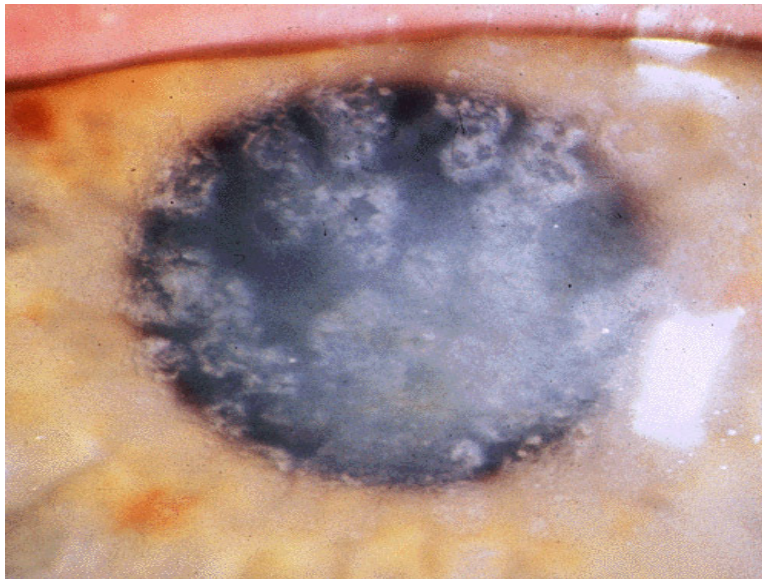
Solution to Question 20:

Masson trichrome stains stromal hyaline deposits in granular dystrophy of the cornea.

Granular corneal dystrophy belongs to the stromal dystrophy of the cornea. It is of autosomal dominant inheritance and is caused by a defect in the TGFB1 gene.

Granular corneal dystrophy is characterized by milky, granular, hyaline deposits in the anterior stroma, which may coalesce into irregular shapes. The peripheral cornea and the area between the opacities/deposits remain clear. Early symptoms may involve light sensitivity, glare, ocular discomfort, pain, and watering. As opacification progresses with age, visual acuity decreases.

The following image shows numerous irregularly shaped discrete crumb-like corneal opacities of Granular dystrophy, which stain bright red with Masson trichrome.

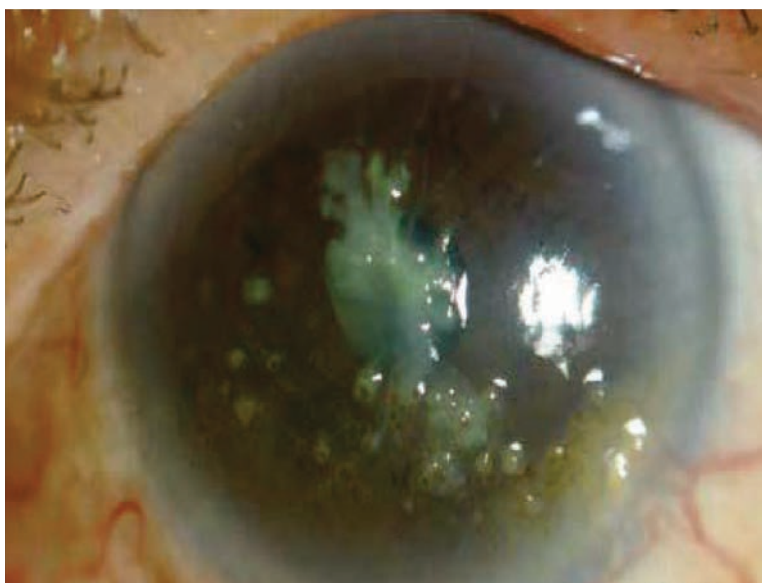


Solution to Question 21:

Oil-droplet keratopathy occurs due to exposure to ultraviolet (UV) rays.

Climatic droplet keratopathy (CDK), alternatively known as oil droplet keratopathy/spheroidal degeneration/Labrador keratopathy, is an acquired corneal degenerative disorder. Clumps of variably sized golden-yellow globules accumulate in the superficial corneal stroma and are thought to be derived from photochemical degradation of plasma proteins, by UV radiation.

The image shows Labrador keratopathy.



Other options:

Option A: Lipid keratopathy involves the accumulation of whitish or yellow fat deposits, mainly cholesterol and fatty acids, which begin intracellularly and later move extracellularly due to necrosis of corneal stromal cells. It can be primary (rare, in the cornea free of vascularization) or secondary (common, in the vascularized cornea).

Option C: Exposure keratopathy results from lagophthalmos (incomplete eyelid closure), causing corneal dryness despite normal tear production. Causes include neuromuscular conditions (like facial nerve palsy), reduced muscle tone in Parkinsonism, lid scarring, post-blepharoplasty and proptosis.

Option D: Neurotrophic keratopathy results from loss of trigeminal nerve innervation to the cornea, affecting both sensation and healing. This disruption can cause corneal dryness, ulceration, exfoliation, and edema. It is commonly linked to conditions such as trigeminal ganglion surgical ablation for neuralgia, stroke, diabetes, herpes simplex and herpes zoster keratitis.

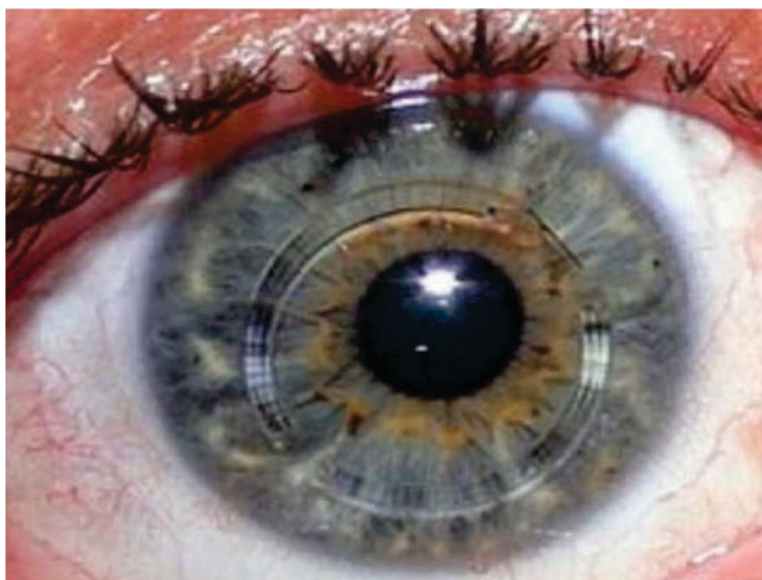
Solution to Question 22:

Corneal perforation may occur but is quite rare in intracorneal ring segment (ICRS) implantation.

Salient points on ICRS:

- These segments are made of polymethyl methacrylate (PMMA) and are implanted into the mid-peripheral corneal stroma.
- These are used in the management of keratoconus and low myopia.
- Improvement in visual acuity and refraction after ICRS implantation is accomplished by a shortening of collagen lamellae along the arc length of the ring. There is a redistribution of corneal stress due to a change in the curvature of the cornea after implantation of the ICRS.
- Decreased corneal sensation can be a postoperative complication.

- More than 90% of patients have ≥ 1 D change in refraction at 1-year postoperative follow-up.
- The image shows the intrastromal corneal ring segment.



Solution to Question 23:

The given clinical scenario, with the given image showing multiple breadcrumb-like deposits, is suggestive of granular corneal dystrophy, where histopathology reveals hyaline deposits on the Masson trichrome stain.

Granular corneal dystrophy belongs to the stromal dystrophy of the cornea. It is of autosomal dominant inheritance and is caused by a defect in the TGFB1 gene.

In granular corneal dystrophy, opacities form due to the abnormal accumulation of hyaline protein, assuming a discrete granular form and subsequently coalescing into various irregular shapes. The cornea between the opacities and the peripheral cornea remains clear. The condition is characterised by milky granular hyaline deposits in the anterior stroma.

In many patients, no symptoms occur, whereas some patients develop recurrent erosions. In the fifth decade or later, some patients develop sufficient visual difficulties to require keratoplasty. Treatment is usually keratoplasty for severe visual handicaps.

Note: The beaten metal appearance on the slit lamp and reduced endothelial cell density with coalescing guttata on specular microscopy is suggestive of Fuch's endothelial dystrophy.

Solution to Question 24:

The given clinical scenario most likely suggest congenital hereditary endothelial dystrophy (CHED), where Descemet's membrane is abnormally thickened.

CHED is a rare disorder characterized by focal or diffuse thickening of the Descemet membrane and endothelial degeneration. The key features of this condition include corneal thickness, which

is two to three times the normal, normal intraocular pressure (IOP) and normal corneal diameter. Associated features are corneal pannus, nystagmus, and esotropia.

Treatment usually involves corneal transplantation (lamellar or penetrating keratoplasty) for visual rehabilitation.

Other options:

Option A: As the corneal diameters are normal, and there are no typical symptoms, congenital glaucoma is ruled out.

Option B: Amyloid deposits in the corneal stroma are seen in Lattice stromal dystrophy.

Option C: Choristomas are congenital lesions representing normal tissue(s) in an abnormal location. They are the most common epibulbar and orbital tumors in children. It usually arises at the limbus of the eye and may include dermoid tissue, lipodermoid or other tissues.

Note: The combination of congenital endothelial dystrophy and progressive neural deafness, is known as Harboyan syndrome. This disorder must be distinguished from another autosomal recessive disorder, congenital endothelial dystrophy, in which deafness does not occur.

Solution to Question 25:

Moorens's ulcer/ulcus roden is characterized by a steep, overhanging edge.

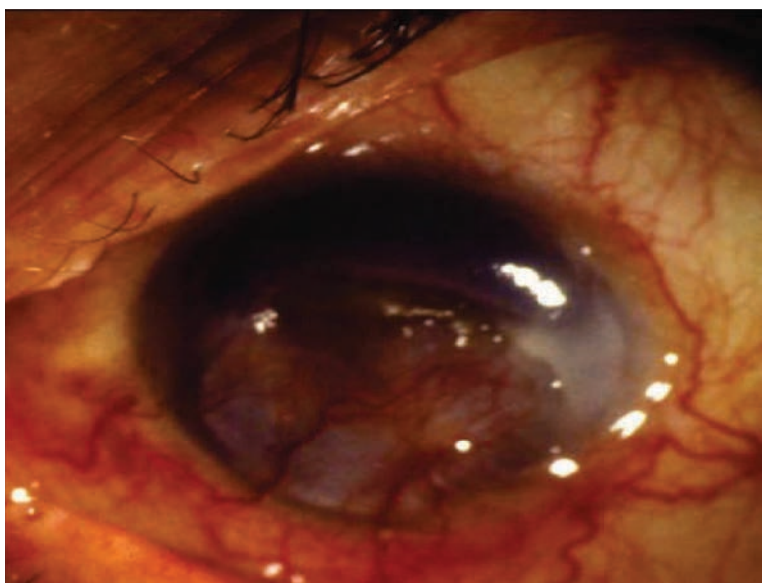
It is described as a chronic serpiginous ulcer of the cornea. It is a subtype of peripheral ulcerative keratitis (PUK). It is defined as an idiopathic condition without associated systemic or rheumatological conditions and hence is a diagnosis of exclusion. The pathophysiology is thought to be due to autoimmune.

Clinical features include redness, tearing, and photophobia, but pain is typically the outstanding feature. The pain often is incapacitating and may well be out of proportion to the inflammation. There may also be a complaint of decreased visual acuity, which may be secondary to associated iritis or central corneal involvement.

Initially, the ulcerative process spreads circumferentially. Later, it moves centrally to involve the entire cornea. In contrast to other forms of PUK, the limbus is involved. The end-stage result is typically a scarred, vascularized cornea that may be thinned to less than half of its original thickness.

Treatment involves corticosteroids and excision of a strip of adjacent conjunctiva, which eliminates the conjunctival sources of collagenase and proteoglycanase.

The image shows a perforated Mooren's ulcer.



Solution to Question 26:

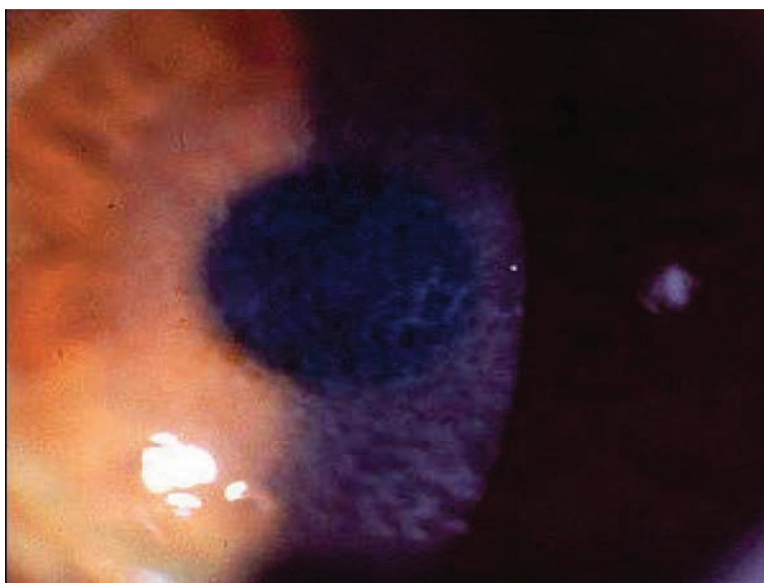
The corneal layer involved in Reis–Buckler dystrophy is Bowman's membrane.

Corneal dystrophies are primarily a bilateral, inherited, non-inflammatory condition specific to the cornea.

Reis–Buckler dystrophy, an autosomal dominant disorder, is an anterior variant of granular stromal dystrophy, characterised by the replacement of the Bowman layer with connective tissue bands, primarily affecting the Bowman layer and superficial stroma. Several recurrent corneal erosions may occur in childhood along with visual impairment.

In this condition, there are grey-white, geographic subepithelial opacities (that are most dense centrally and increase over time, eventually forming a reticular pattern. Additionally, corneal sensation is reduced as the disease progresses.

The image below shows Reis–buckler corneal dystrophy on slit-lamp examination:

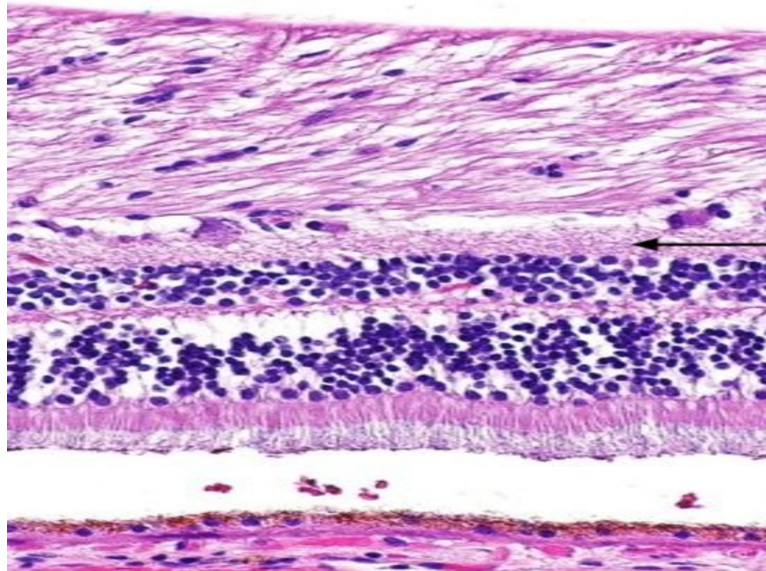


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Retinal vascular disorders and Retinal detachment

Question 1:

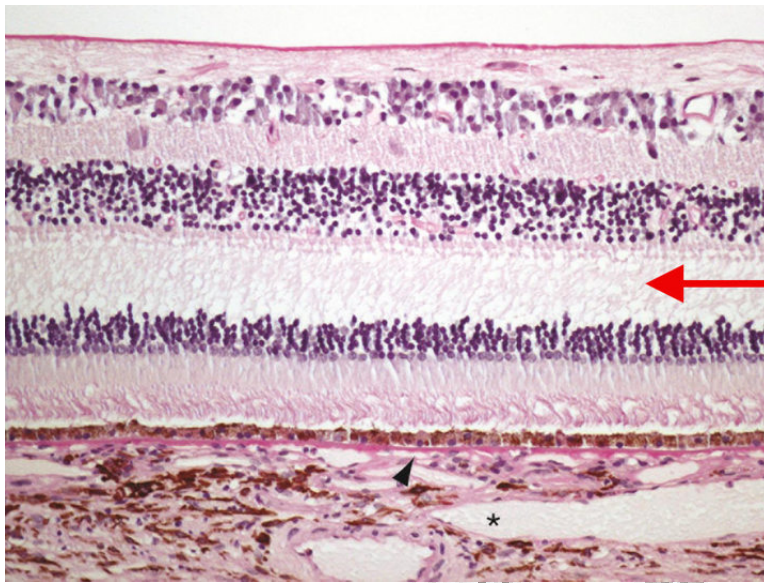
Identify the marked layer of the retina.



- a) Inner plexiform layer
- b) External limiting layer
- c) Internal limiting layer
- d) Outer plexiform layer

Question 2:

Identify the structure marked with red arrow



- a) Outer limiting membrane
- b) Inner plexiform layer
- c) Outer plexiform layer
- d) Outer nuclear layer

Question 3:

The distance of the fovea from the temporal margin of the optic disc is _____.

- a) 1 disc diameter
- b) 2 disc diameters
- c) 3 disc diameters
- d) 4 disc diameters

Question 4:

Causes for expanding blind spot are all except _____

- a) POAG
- b) Hypoplasia of optic disc
- c) Papilledema
- d) Medullated nerve fibres

Question 5:

Which statement regarding Müller cells is false?

- a) They are modified glial cells
- b) Their nuclei lie in the outer nuclear layer
- c) Their foot processes form the inner & outer limiting membranes
- d) They contribute to inner blood retinal barrier

Question 6:

Which of the following statements is false regarding the development of retinal vasculature?

- a) Grow to temporal ora by 34 weeks of gestation
- b) Grow to nasal ora by 36 weeks of gestation
- c) Originate from mesenchymal tissue during week 16 of gestation
- d) Radiate centrifugally from optic nerve

Question 7:

A preterm baby treated for respiratory distress syndrome is found to have white pupillary reflex on ophthalmologic examination. What is the likely diagnosis?

- a) Posterior persistent primary hyperplastic vitreous
- b) Coat's retinopathy
- c) Retrolental fibroplasia
- d) Congenital cataract

Question 8:

Which of the following is included under threshold disease when screening neonates for retinopathy of prematurity?

- a) ROP stage 1 extending from 6'o clock to 11'o clock in zone 1 with plus disease
- b) ROP stage 3 extending from 2'o clock to 7'o clock in zone 2 with plus disease
- c) ROP stage 3 extending from 1'o clock to 4'o clock in zone 1 without plus disease
- d) ROP stage 2 extending from 3'o clock to 6'o clock in zone 2 with plus disease

Question 9:

All of the following are true statements regarding retinopathy of prematurity except:

- a) First screening should be done 32 weeks PMA or 4 weeks PNA, whichever is later
- b) For infants less than 28 week, screen at 3 week of postnatal age to detect APROP
- c) Neonates are best examined at ophthalmology OPD and hence should be transported
- d) Screening is done using indirect ophthalmoscopy by an expert ophthalmologist

Question 10:

A preterm neonate admitted to NICU who received supplemental oxygen is being screened for retinopathy of prematurity. Fundus examination reveals a subtotal retinal detachment. How will you stage the disease?

- a) Stage 2
- b) Stage 3
- c) Stage 4
- d) Stage 5

Question 11:

A 14-day-old neonate born at 28 weeks of gestation had developed respiratory distress syndrome at birth. You want to screen for retinopathy of prematurity. How many weeks from now would you schedule the screening?

- a) 4 weeks
- b) 2 weeks
- c) 8 weeks
- d) 12 weeks

Question 12:

Which of the following is the most important factor for the development of diabetic retinopathy?

- a) Duration of diabetes
- b) Glycemic control

- c) Association with hypertension
- d) Age of the patient

Question 13:

Which of the following findings on a fundus examination would suggest severe non-proliferative diabetic retinopathy?

- a) Significant venous beading in any of the 4 quadrants
- b) Micro-aneurysms in any 2 quadrants
- c) Severe hemorrhages in all 4 quadrants
- d) Macular star

Question 14:

A person is diagnosed to have diabetes on his 45th birthday. When will you recommend a dilated fundus examination for him?

- a) Immediately
- b) Before his 50th birthday
- c) After his 50th birthday
- d) When he complains of decrease in vision

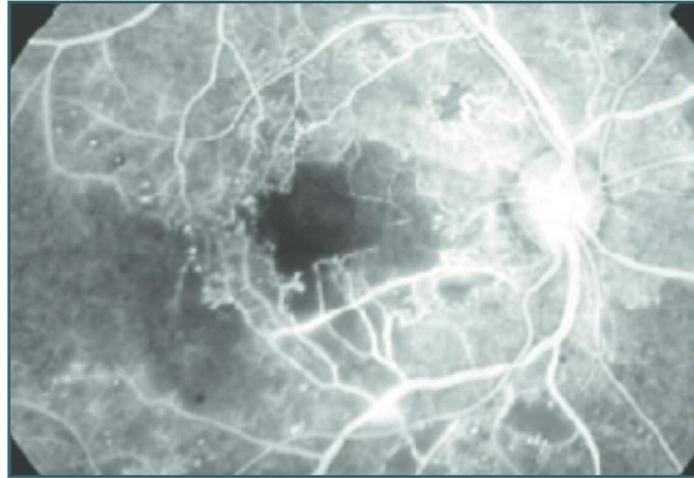
Question 15:

ETDRS study was related to?

- a) Central retinal artery occlusion
- b) Diabetic macular edema
- c) Primary open angle glaucoma
- d) Non-infectious uveitis

Question 16:

A patient with a history of diabetes and hypertension underwent fundus fluorescent angiography (FFA) to evaluate his decreasing vision. The FFA yields an image as shown below, which is characteristic of:



- a) Macular ischemia
- b) Macular edema
- c) Submacular haemorrhage
- d) Premacular haemorrhage

Question 17:

A 69-year-old patient with hypertension has silver wiring, papilledema, and arteriovenous nipping visible on fundoscopy, but no other changes. How would you grade his disease?

- a) Grade I retinopathy
- b) Grade IV retinopathy
- c) Grade III retinopathy
- d) Grade II retinopathy

Question 18:

A 40-year-old male, newly diagnosed with systolic hypertension, presents with a three-week history of bilateral blurry vision. His visual acuity is 20/30 and 20/120 in the right and left eye, respectively. The rest of the examination is unremarkable. Which of the following findings is least likely to be seen on his fundoscopy?

- a) Arteriolar narrowing
- b) Intraretinal microvascular abnormalities
- c) Cotton wool spots

d) Papilledema

Question 19:

The following image shows the picture of the fundus in a patient taken 48 hours after sustaining severe head trauma. What is the diagnosis?



- a) Hypertensive retinopathy
- b) Diabetic retinopathy
- c) Purtschers retinopathy
- d) Central retinal vein occlusion

Question 20:

Which of the following statements is not true regarding sickle cell retinopathy?

- a) Sea-fan appearance is seen in supero-temporal quadrant.
- b) HbSS has more severe ocular manifestations than HbSC.
- c) Angioid streaks are seen.
- d) Central retinal vein occlusion may be seen.

Question 21:

Roth spots are seen in _____.

- a) Acute leukemia
- b) Uveal melanoma
- c) Uveal lymphoma
- d) All of the above

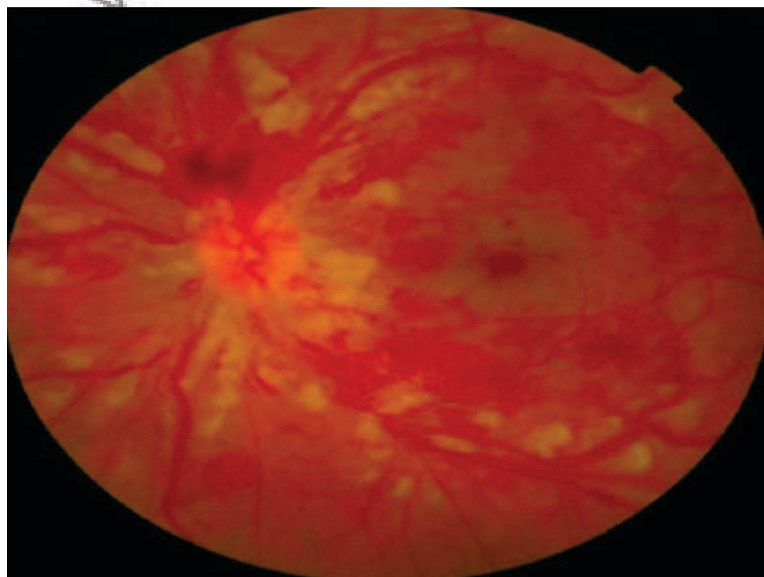
Question 22:

A 7-year-old boy presents with 6/6 vision in the right eye and vision for only hand movements in the left eye. On fundoscopy, the right eye appears normal. The left eye shows retinal detachment, subretinal yellowish exudates, and telangiectatic vessels. What is the most probable diagnosis?

- a) Sympathetic ophthalmitis
- b) Familial exudative vitreoretinopathy
- c) Retinopathy of prematurity
- d) Coat's disease

Question 23:

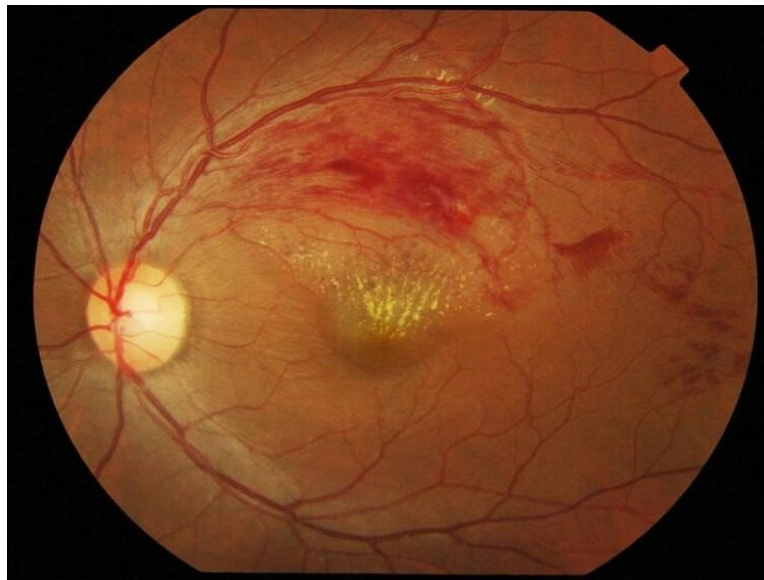
A 64-year-old female presented with complaints of blurring of vision and pain in her left eye. She also gave a history of redness in her left eye about 3 months back, which she passed off as normal conjunctivitis. On tonometry, the IOP is 55 mm Hg. The slit-lamp revealed rubeosis iridis in all four quadrants. The fundus picture of the patient is given below. What is the most probable diagnosis?



- a) Central retinal vein occlusion (CRVO)
- b) Branched retinal vein occlusion (BRVO)
- c) Central retinal arterial occlusion (CRAO)
- d) Branched retinal artery occlusion (BRAO)

Question 24:

A man with history of hypertension and loss of vision in the left eye has the following fundoscopy findings. What is the likely diagnosis?



- a) Central retinal arterial occlusion (CRAO)
- b) Central retinal venous occlusion (CRVO)
- c) Branched retinal arterial occlusion (BRAO)
- d) Branched retinal venous occlusion (BRVO)

Question 25:

A 67-year-old male presents to the emergency department with sudden painless vision loss in his left eye. He has had a similar episode before but his vision was quickly restored. On physical exam, a carotid bruit is heard and a relative afferent pupillary defect is noted on ocular examination. His fundus appears as shown below. Which of the following is the best immediate step in the management of this condition?



- a) Ocular massage
- b) Steroids
- c) Laser photodisruption
- d) Fibrinolytic therapy

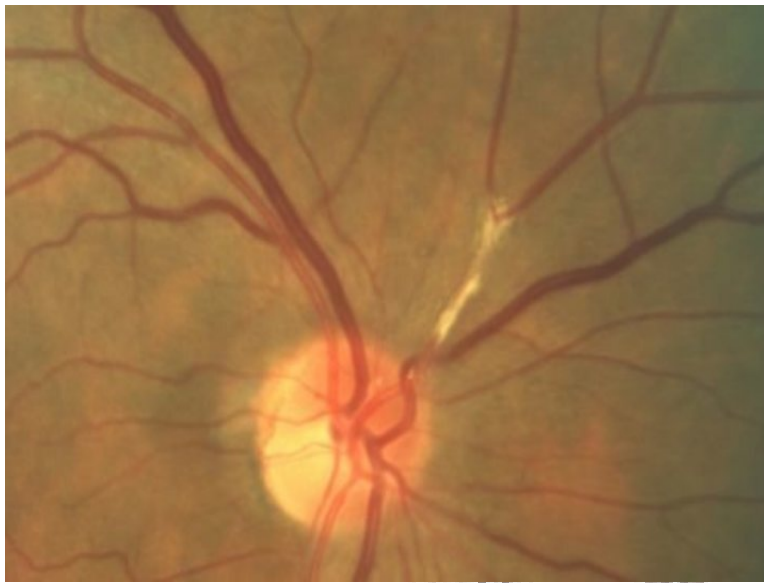
Question 26:

A 52-year-old woman presents with complaints of sudden painless loss of vision in the left eye since yesterday. She also gives a history of temporal headaches, myalgias, and pain in the jaw while chewing. Examination of the left eye reveals a visual acuity of HM+ and an RAPD. Fundus examination of the left eye shows cattle trucking of blood within the retinal veins on applying gentle pressure. Which of the following is the likely cause of fundal findings?

- a) Ischemic central retinal venous occlusion
- b) Central retinal artery occlusion
- c) Non- ischemic central retinal venous occlusion
- d) Branched retinal artery occlusion

Question 27:

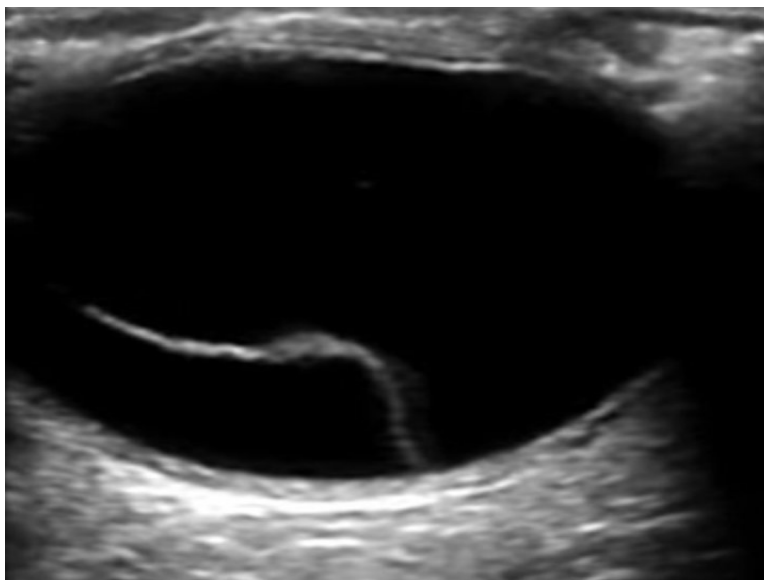
A 60-year-old man presents with sudden painless loss of vision. Fundoscopy findings are shown below. Which of the following findings do you expect in this patient?



- a) Relative afferent pupillary defect
- b) Peripheral visual field loss
- c) Redness of conjunctiva
- d) Increased intraocular pressure

Question 28:

A 61-year-old university professor presents with a loss of vision in his left eye. He describes the sensation that a curtain was coming up over his affected eye. In addition, 7 days ago the patient noticed what he described as flashing lights and floaters. He gives a history of having cataract surgery in the same eye 3 weeks ago. A B-scan image of his left eye is shown below. Which of the following statements best describes the pathology?



- a) Fluid accumulation below retina
- b) Fibrovascular membranes in the vitreous causing traction
- c) Fluid accumulation between NSR and RPE due to retinal tear
- d) Fluid accumulation between NSR and RPE due to choroidal exudation

Question 29:

Pneumatic retinopexy is an outpatient procedure where retinal detachment is sealed with air insufflation. Which of the following gases would you use while performing the said procedure?

- a) Carbon dioxide
- b) Sulphur hexafluoride
- c) Nitrogen oxide
- d) Selenium hexafluoride

Question 30:

A 24-year-old woman who is 8 months pregnant, myopic, and diabetic presents to you with sudden painless loss of vision in her right eye. She is a known case of Eales' disease. Her blood pressure is 170/110 mmHg. Which of her conditions is responsible for a fundus finding of retinal detachment with a positive shifting fluid sign?

- a) Pathological myopia
- b) Type 1 diabetes
- c) Eales disease
- d) Toxemia of pregnancy

Question 31:

Shifting fluid sign is seen in?

- a) Exudative RD
- b) Traction RD
- c) Rhegmatogenous RD
- d) Retinal dialysis

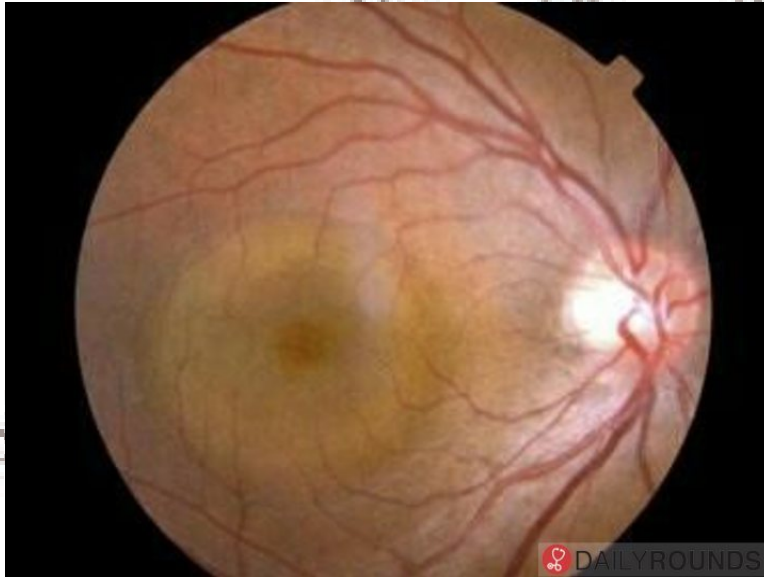
Question 32:

Hollenhorst plaques are seen in _____.

- a) Central retinal arterial occlusion (CRAO)
- b) Central retinal venous occlusion (CRVO)
- c) Cystoid macular edema
- d) Central serous retinopathy

Question 33:

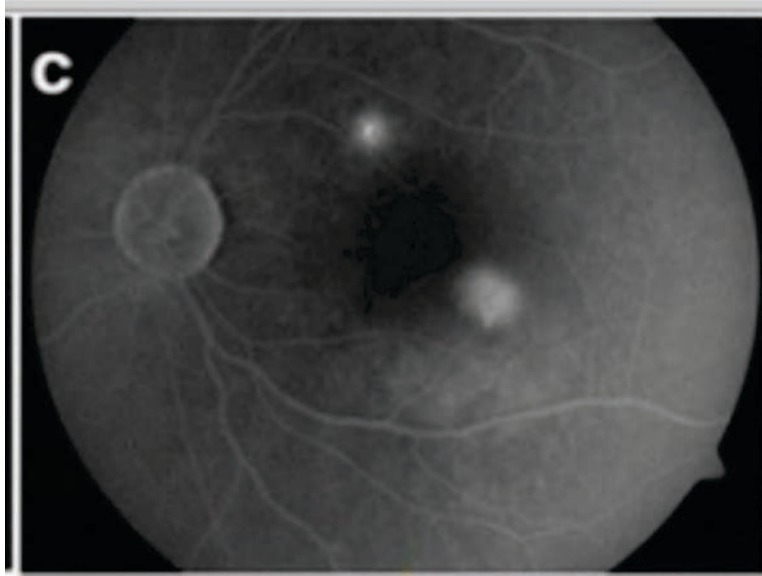
Identify the condition.



- a) Foveal detachment
- b) Macular hole
- c) Macular edema
- d) Macular Drusens

Question 34:

In which of the following conditions is the given sign seen?



- a) Cystoid macular edema
- b) Central serous retinopathy
- c) Significant macular edema
- d) Coat's disease

Question 35:

The ETDRS chart is used for visual evaluation in diabetic patients. What does ETDRS stand for?

- a) Extended treatment for diabetic retinopathy study
- b) Early treatment for diabetic retinopathy study
- c) Emergency treatment for diabetic retinopathy study
- d) Emerging treatment for diabetic retinopathy study

Answer Key

Question No.	Correct Option
1	a
2	c
3	b
4	b

5	b
6	a
7	c
8	b
9	c
10	c
11	b
12	a
13	c
14	a
15	b
16	a
17	b
18	b
19	c
20	b
21	a
22	d
23	a
24	d
25	a
26	b
27	a
28	c
29	b
30	d
31	a
32	a
33	a
34	b
35	b

Detailed Explanations

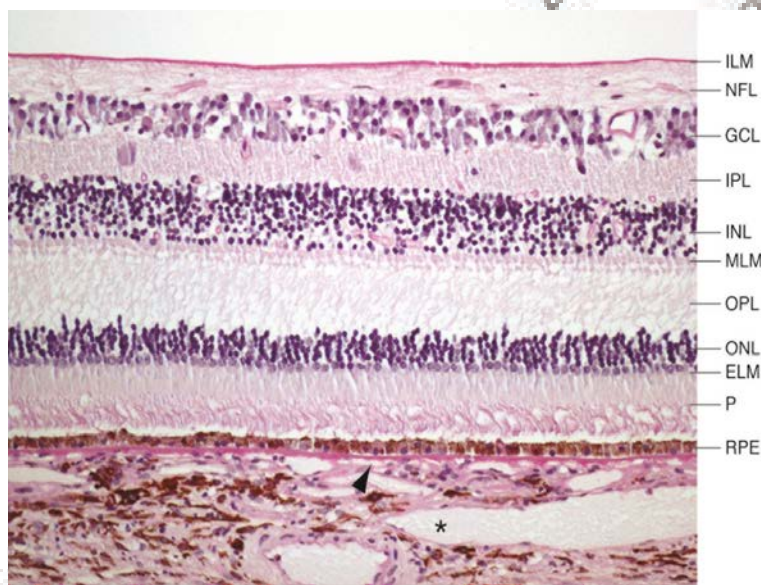
Solution to Question 1:

The inner plexiform layer is the marked layer in the given histological image of retina, identified as the fourth layer from the uppermost (innermost) layer.

The retina consists of 10 layers. The retinal pigment epithelium (RPE) is the outermost layer - below the Bruch's membrane of the choroid - and the internal limiting membrane is the innermost layer - separating the retina from the vitreous.

The retinal pigment epithelium consists of a single layer of cells containing pigment. It is separated from the neurosensory retinal layers by the potential space between them which is known as interphotoreceptor matrix.

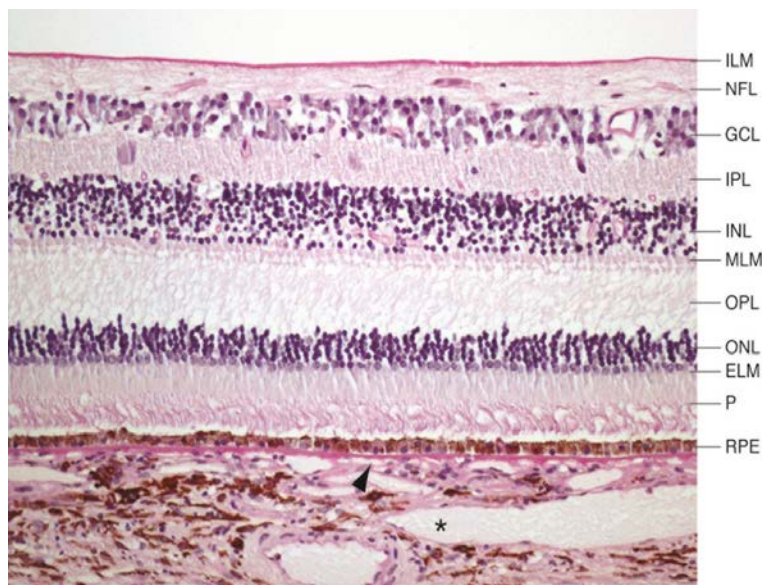
The inner plexiform layer consists of synapsing nerve fibers. The axons of bipolar cells synapse with dendrites of ganglion cells; and both these processes synapse with processes of amacrine cells. It also contains some horizontally placed internal plexiform cells; and also a few ganglion cells.



Solution to Question 2:

The marked structure in the slide is the outer plexiform layer of the retina.

The outer plexiform layer (outer synaptic layer) is present between the outer nuclear layer and the inner nuclear layer. This layer stains lightly. This layer consists of the axons of rods and cones which synapse with dendrites of bipolar neurons and horizontal cells.

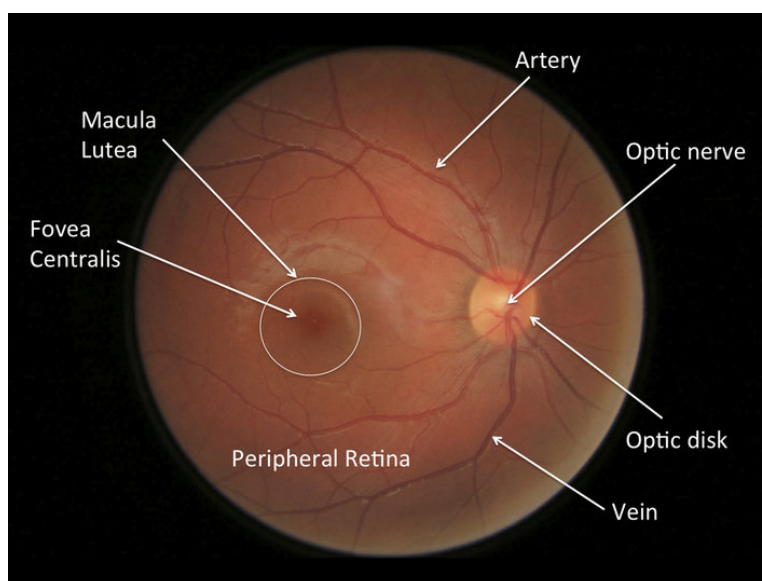


Solution to Question 3:

The distance of the fovea from the temporal margin of the optic disc is 2 disc diameters.

The fovea is an excavation in the retinal surface at the center of the macula, with a diameter of 1.5 mm, which is about the same as that of the optic disc. The foveola forms the bottom of the fovea and has a diameter of 0.35 mm. It is devoid of ganglion cells. The umbo is the central floor the foveola, which corresponds to the foveolar light reflex. The foveal avascular zone (FAZ) is a central area (including foveola and surrounding areas) which contains no blood vessels but is surrounded by a continuous network of capillaries.

The macula is a round area at the posterior pole, lying inside the temporal vascular arcades. It measures 5.5 mm in diameter. The inner layers of the macula contain the yellow xanthophyll carotenoid pigments lutein and zeaxanthin in far higher concentrations than the peripheral retina (hence, the full name "macula lutea", which means yellow plaque).



Solution to Question 4:

Optic disc hypoplasia does not cause enlargement of the blind spot.

The physiological blind spot is an area of absolute scotoma in the visual field, seen in the region of optic disc. It is due to the absence of photoreceptors (rods and cones) in that region. Enlarged blind spot is seen in primary open-angle glaucoma (POAG) (Option A), papilledema (Option C), medullated nerve fibers (Option D), optic disc drusen, coloboma of optic disc and myopic disc with a crescent.

Optic nerve hypoplasia is the most common congenital optic disc anomaly. There is underdevelopment of the optic disc in the disease and it typically presents with a small disc that is surrounded by a yellowish pigmented ring, called the 'double ring sign'.

Option A: In primary open-angle glaucoma (POAG), the optic disc enlargement is seen secondary to increased intra-ocular pressure (IOP) in the eye.

Option C: Papilledema is a passive disc swelling seen with raised intracranial pressure. The blind spot is enlarged due to the edema of the optic disc.

Option D: Medullated or opaque nerve fibers are the abnormal myelination of nerve fibers of the retina. Normally, optic nerve myelination stops at the level of lamina cribrosa. If the myelination continues beyond the optic disc it leads to the expansion of the blind spot. On ophthalmoscopic examination, these are seen as a whitish patch with feathery margins.

Solution to Question 5:

The nuclei of Müller cells lie in the inner nuclear layer.

Müller cells are a type of macroglial cells of the retina. The human retina contains 8–10 million Müller cells. Their nuclei lie in the inner nuclear layer and their processes extend across the retinal layers to form the inner and outer limiting membranes. Their functions include:

1. Providing structural stability to the retina
2. Interacting with other neurons and their functional and metabolic support
3. Providing neurons with trophic substances and removing metabolic waste
4. Playing a critical role in regulating the extracellular space volume as well as ion and water homeostasis and in maintaining the inner blood-retinal barrier

Astrocytes are another type of macroglial cells of the retina. They play a crucial role in retinal vascularization. In the mature retina, they are restricted to the nerve fiber and ganglion cell layers.

Solution to Question 6:

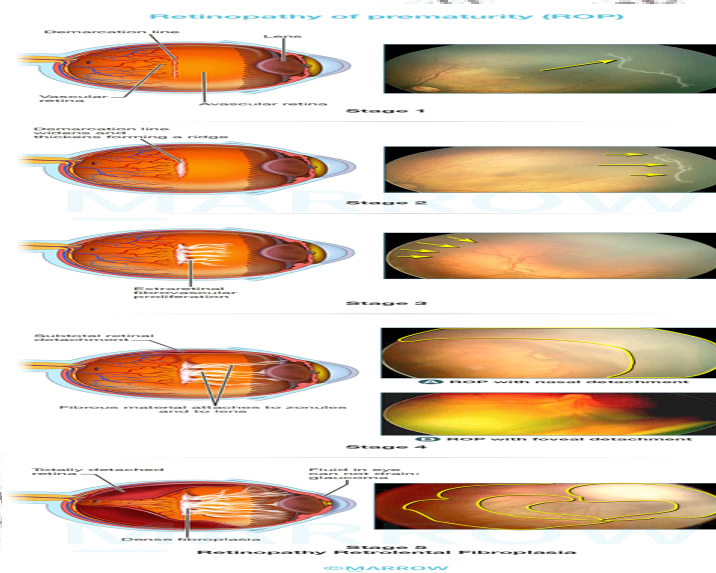
Retinal vessels reach temporal ora at approximately 40 weeks of gestation.

Retinal vessels develop de novo from mesenchymal tissue, beginning during the 4th month of gestation. They radiate outward (centrifugally) from the optic nerve, reaching the nasal ora by 36 weeks.

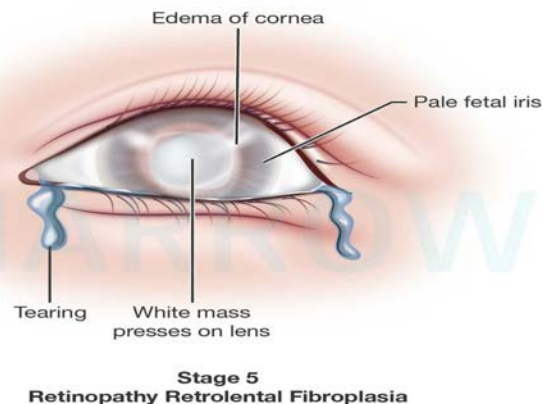
Solution to Question 7:

A preterm baby, treated for respiratory distress syndrome, with leucocoria (white pupillary reflex) most likely has retrolental fibroplasia.

Retrolental fibroplasia, also known as retinopathy of prematurity, is a proliferative retinopathy. Risk factors include prematurity, very low birth weight and supplemental oxygen therapy (given for respiratory distress syndrome). ROP can be mild and may resolve spontaneously, but may lead to blindness in serious cases.



Retinopathy of prematurity (ROP)



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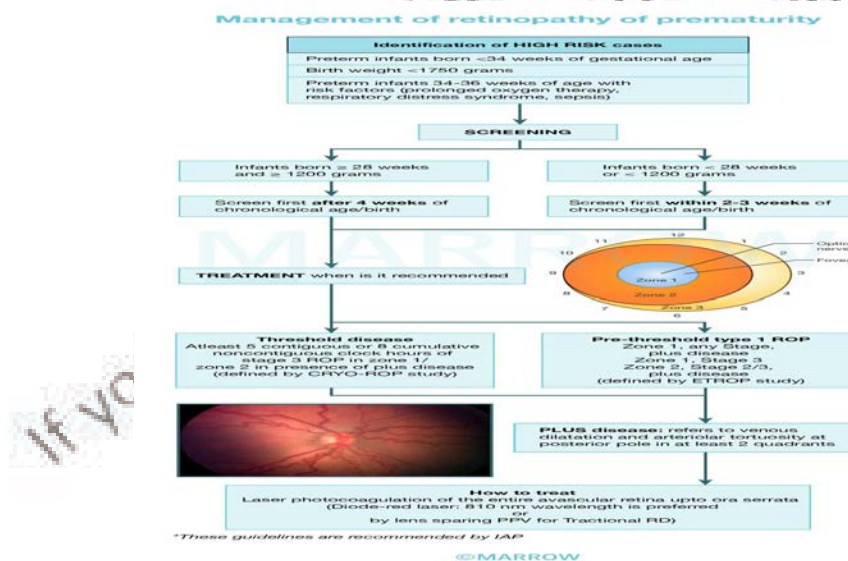
Solution to Question 8:

ROP stage 3 extending from 2'o clock to 7'o clock (6 clock hours) in zone 2 with plus disease fits into the definition of threshold disease.

Threshold disease of retinopathy of prematurity is defined by the CRYO-ROP study as:

- Stage 3 disease: Ridge with extraretinal fibrovascular proliferation into the vitreous
- Involvement of zones 1 or 2
- Involvement of either at least 5 contiguous (continuous) clock hours or 8 non-contiguous clock hours
- Associated with "plus disease": Dilatation and tortuosity of posterior pole vessels in at least two quadrants at the posterior pole.

Treatment of threshold disease is laser photocoagulation of the involved part within 72 hours of diagnosis.



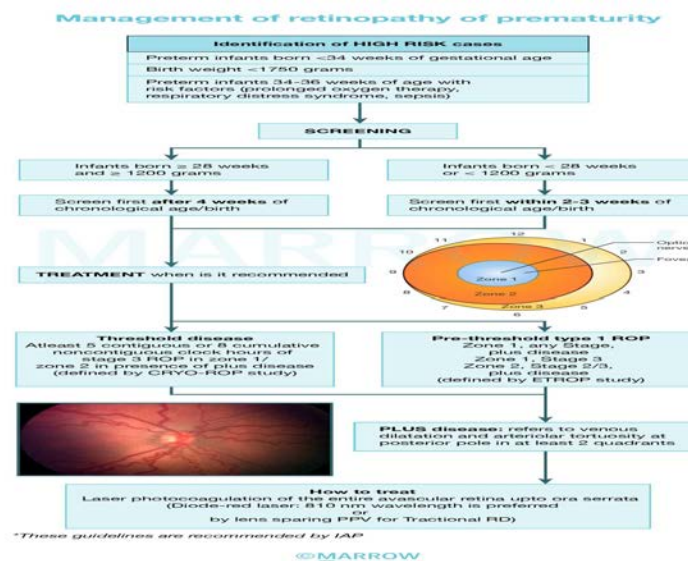
Treatment of the various stages of ROP includes:

- Pre-threshold type 1 disease - Treatment is now recommended within 48 hours and is done by laser photocoagulation of avascular retina.
- Pre-threshold type 2 disease (Any ROP not following under type 1)- Observation for progression
- Threshold disease - Laser photocoagulation of avascular retina
- Aggressive posterior-ROP- Intravitreal anti-VEGF agents, eg., bevacizumab
- Tractional retinal detachment (stage 4A) - Pars plana vitrectomy

Solution to Question 9:

Neonates are best examined in the neonatal unit itself under the supervision of the attending neonatologist. It is not recommended to transport infants to the ophthalmic outpatient department for examination.

The first screening examination should be carried out by indirect ophthalmoscopy at 32 weeks PMA or 4 weeks PNA, whichever is later. For infants born at less than 28 weeks, screening is done at 3 weeks PNA to detect APROP. Aggressive posterior ROP ("rush" disease, AP-ROP), if untreated, usually progresses to total retinal detachment (stage 5) within a few days. It is characterized by its posterior location, prominence of plus disease and ill-defined nature of the retinopathy.



Solution to Question 10:

Subtotal retinal detachment is considered to be stage 4 of retinopathy of prematurity (ROP).

The clinical staging of ROP is as follows:

- Stage 1 - Formation of a demarcation line at the edge of vessels, dividing the vascular from the avascular retina.
- Stage 2- Formation of an intraretinal ridge
- Stage 3- Ridge with extraretinal fibrovascular proliferation into the vitreous
- Stage 4- Subtotal retinal detachment
- Stage 5- Total retinal detachment



gestational age of the premature infant is 2 weeks, and 4 weeks i.e., 4 weeks of postnatal age, as per the guidelines for ROP (ROP).

First examination should be performed between 31–33 weeks of post-conceptual age (calculated as gestational age + postnatal age), whichever is later. Ideally, 2 examinations must be performed to ensure complete retinal vascularisation.

IF NO

st examination should be performed between 31–33 weeks of post-conception, whichever is later. Ideally, 2 exams complete retinal vascularisation.



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The most important factor for the development of diabetic retinopathy is the duration of diabetes. Diabetic retinopathy constitutes the retinal changes seen in patients with type 2 diabetes mellitus. The most important factor for its development is duration of diabetes after the onset of puberty, followed by glycemic control. It is more common in type 1 diabetes mellitus. The presence of hypertension increases the risk of diabetic retinopathy.

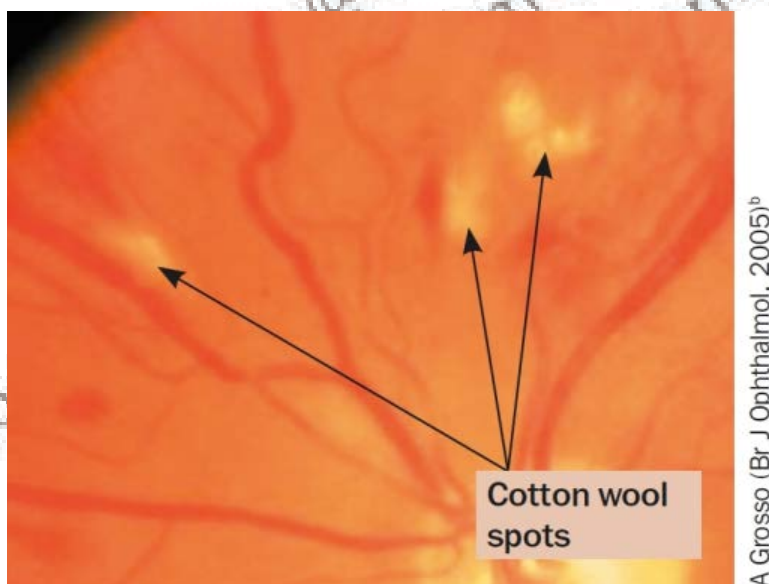
First fundus examination for type 1 diabetes is done after 5 years after diagnosis and for type 2 is done at the time of diagnosis. The frequency of follow-up for types 1 and 2 diabetes mellitus is every year. Diabetes in pregnancy is followed up every 3 months.

The earliest feature of diabetic retinopathy is microaneurysms, which are present in the inner nuclear layer of the retina.

Solution to Question 13:

Severe hemorrhages in all 4 quadrants is a feature of severe NPDR.

Note: Macular star is a feature of malignant hypertensive retinopathy.



Solution to Question 14:

A 45 year old diagnosed with diabetes mellitus is likely to have type 2 diabetes and thus, he should undergo screening for diabetic retinopathy immediately.

Duration of diabetes is the most important risk factor for developing diabetic retinopathy, especially proliferative diabetic retinopathy. The recommendations for first fundus examination in diabetic retinopathy is:

- Within 3-5 years of diagnosis in type 1 diabetes
- Immediately in type 2 diabetes.

Later, the screening is continued once every year in both Type 1 and Type 2 diabetes. In the event proliferative diabetic retinopathy develops, screening is done more frequently - once every 2 months.

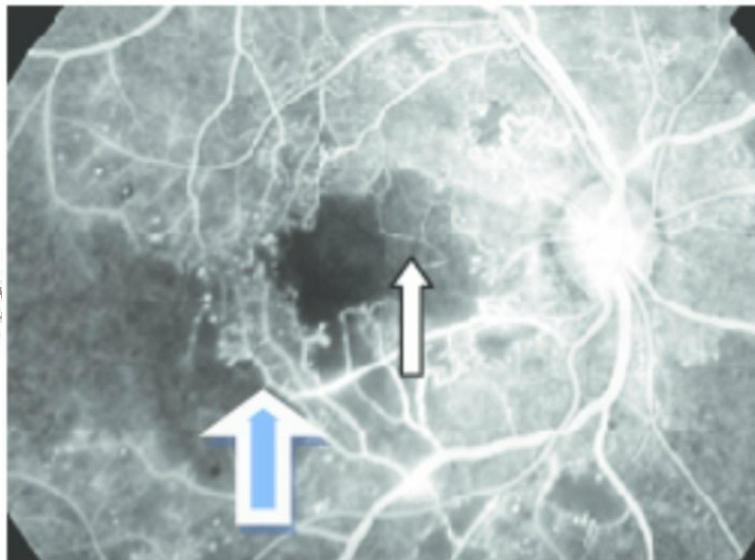
Solution to Question 15:

Early Treatment Diabetic Retinopathy Study (ETDRS) was a randomized clinical trial that studied the effect of photocoagulation on diabetic macular edema.

The results of the study demonstrated the beneficial effects of focal photocoagulation on diabetic macular edema. The study resulted in the recommendation that all eyes with clinically significant diabetic macular edema should be considered for focal photocoagulation.

Solution to Question 16:

The fundus fluorescent angiography (FFA) image shows enlarged foveal avascular zone, seen as a hypofluorescent patch in the macula (white arrow), with surrounding capillary non perfused areas (blue arrow). Both of these are suggestive of macular ischemia.



The above mentioned findings when combined with the history of diabetes in this patient points towards the diagnosis of diabetic macular ischemia.

FFA images in a case of diabetic retinopathy are given below:

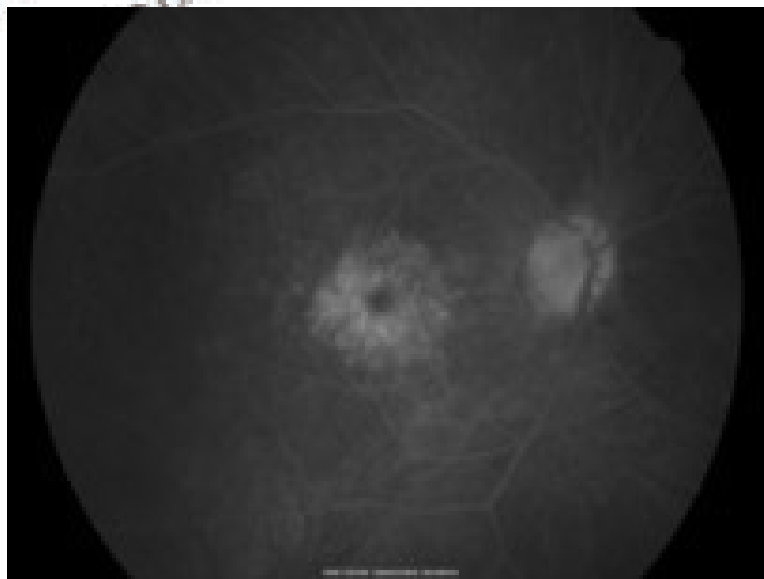
- Non proliferative diabetic retinopathy (NPDR)
- Punctate or pinpoint areas of hyperfluorescence - microaneurysms
- Abnormal branching of arteriovenous shunt - intraretinal microvascular abnormalities (IRMA)
- Patchy filling defects of ischemia due to capillary atrophy or non perfusion

- Increase in the size of hypoperfusion in foveal avascular zone in macular ischemia
- 2. Proliferative diabetic retinopathy(PDR)
- Fluorescein leakage - in retinal neovascularization.
- Leakage in the disc region suggests neovascularization of the disc (NVD) and leakage in other regions suggests neovascularization elsewhere (NVE).

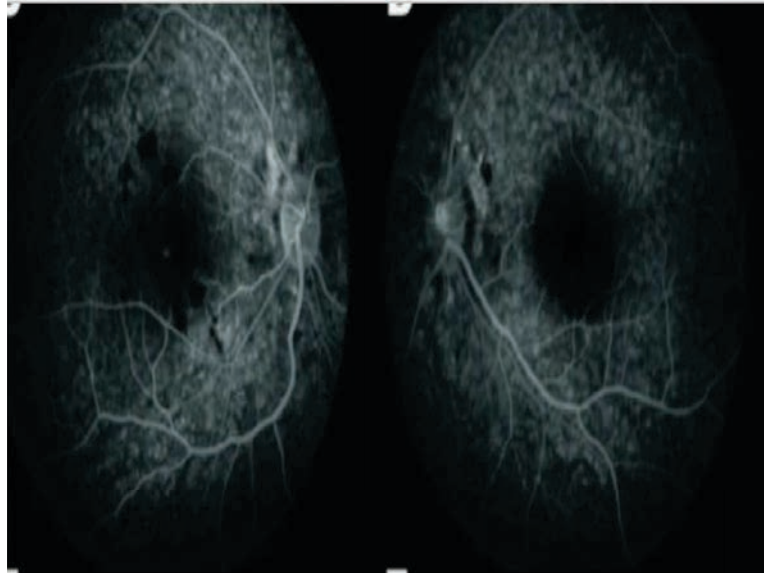
The image below shows pinpoint areas of hyperfluorescence and areas of fluorescein leakage in the fundus.



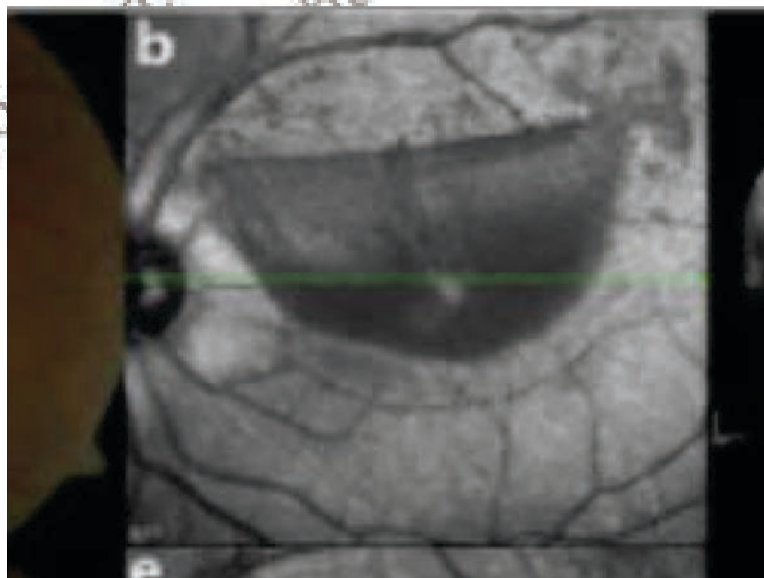
Option B: Macular edema also known as cystoid macular edema (CME) is a collection of fluid between the outer plexiform layer and the inner nuclear layer of the retina following a break in the blood-retina barrier. On funduscopy, a honeycomb appearance is seen, and on FFA, flower petal pattern can be seen. The below image shows the flower petal pattern on FFA as seen in CME.



Option C: Submacular hemorrhage also known as subretinal hemorrhage will obscure the choroidal dye visualization and appears as ill defined areas of hypoperfusion in the corresponding area. The retinal vessels overlying the area would, however, be visualized. There are no associated features of pinpoint hyperfluorescence and vascular leakage that are usually seen with diabetic macular ischemia. The below given images show fluorescein angiography which reveal blocked fluorescence in the maculae of both eyes due to subretinal hemorrhages.

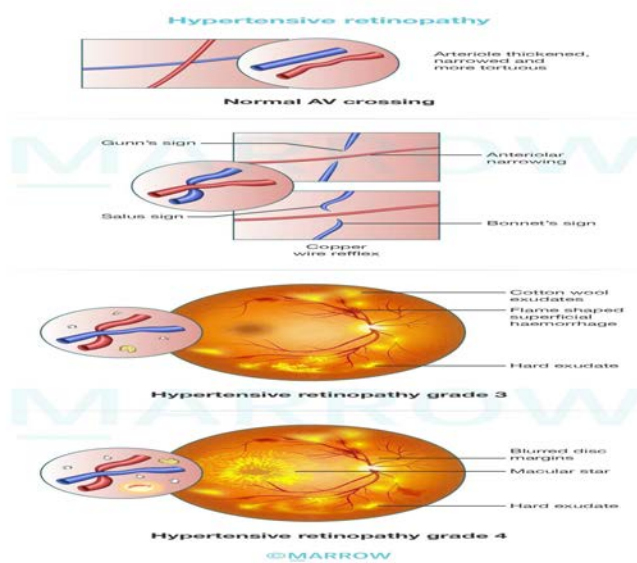


Option D: Premacular hemorrhage is a term used to define vitreous hemorrhage that occurs in the subhyaloid space. It is seen as a region of hypofluorescence with no overlying retinal vasculature visualization on FFA.



Solution to Question 17:

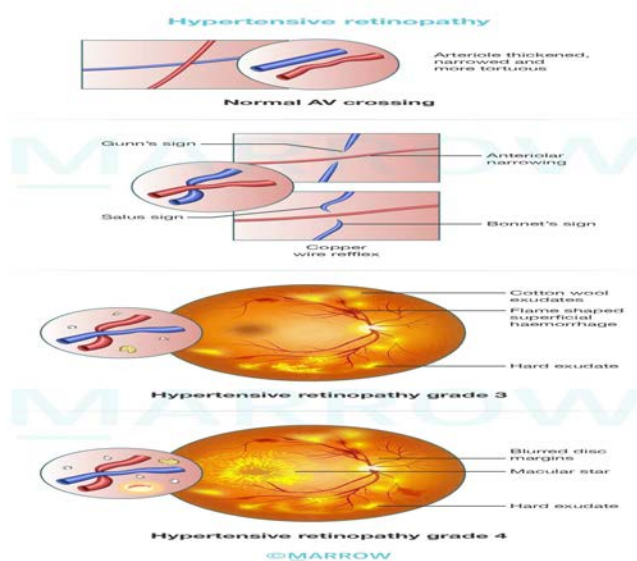
A patient with hypertension having "silver wiring," papilledema, and arteriovenous nipping visible on fundoscopy has Grade IV retinopathy.



Solution to Question 18:

Systolic hypertension (systolic blood pressure over 200 mm Hg) with bilateral blurry vision and unremarkable examination findings point to a diagnosis of hypertensive retinopathy. Intra-retinal microvascular abnormalities (IRMA) are not seen in hypertensive retinopathy. IRMA is a feature of diabetic retinopathy.

Arteriolar narrowing occurs due to vasoconstriction and sclerosis of retinal vessels due to hypertension. Cotton wool spots are areas of nerve fibre layer infarcts. They occur due to capillary obliteration in hypertension.



Solution to Question 19:

The image shows multiple cotton wool spots around the disc. This is Purtschers' retinopathy, which is classically associated with severe head injury.

It is also seen in pancreatitis and chest trauma due to fat/air embolism. Ophthalmoscopy reveals multiple, variably sized cotton-wool spots and intraretinal hemorrhages around the optic nerve. There is no effective treatment.

Solution to Question 20:

Hemoglobin SC and S-Thal disease have the most severe ocular manifestations, although sickle SS disease has the most severe systemic manifestations.

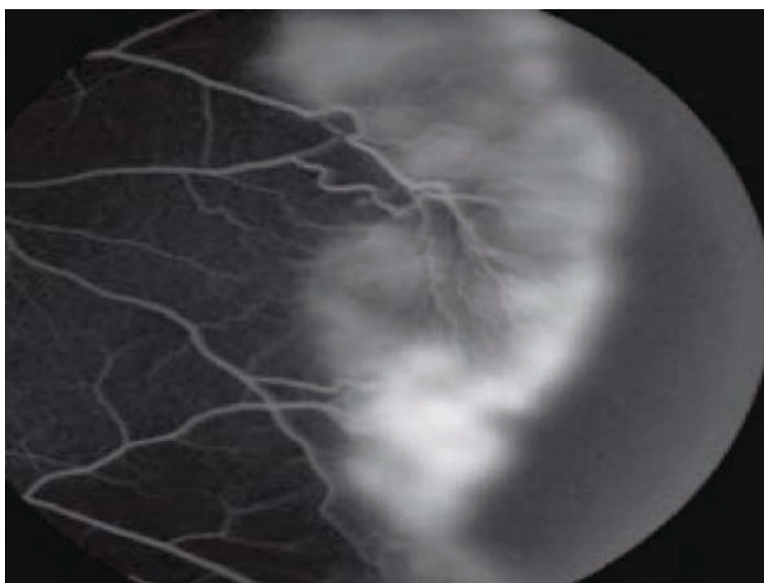
The incidence of proliferative retinopathy in SC disease (Hb SC) is 33%, sickle cell thalassemia (Hb S-Thal) is 14%, SS disease (Hb SS) is 3%. In sickle cell retinopathy, elevated blood viscosity may occlude venules and arterioles, as well as capillaries. The following fundus findings can be noted as a result:

- Retinal vascular peripheral non-perfusion
- Retinal hemorrhages: "salmon patches," "iridescent spots," and "black sunbursts" are hallmarks of nonproliferative sickle cell retinopathy.
- Peripheral retinal neovascularization in a "sea fan" configuration is the hallmark of proliferative sickle cell retinopathy. They are also described as "parachute" lesions. They are most commonly found in the supero-temporal vascular arcades.
- Angioid streaks (Rupture of Bruch's membrane)

Conjunctival "comma-shaped" capillaries and iris atrophy are other ocular findings seen.

Complications of sickle cell retinopathy include central retinal vein occlusion (CRVO) (occurs due to retinal hypoxia), vitreous hemorrhage and tractional retinal detachment.

The image below shows 'sea fans' as seen on fluorescein angiography.



Solution to Question 21:

Roth's spots are seen in acute leukemia.

Roth's spots are retinal hemorrhages with white or pale center. The white centre due to coagulated fibrin-platelet plug is surrounded by red rim that is caused by hemorrhage. Roth's spots may be seen in:

- Subacute bacterial endocarditis
- Leukemia
- Diabetes
- Sickle cell retinopathy
- Hypertensive retinopathy
- HIV retinopathy
- Systemic lupus erythematosus

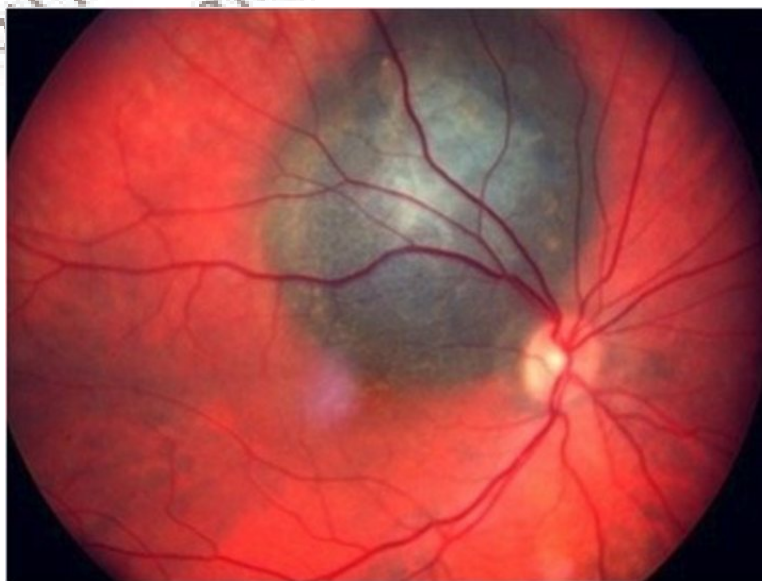
The fundus image below shows Roth's spots.



Option B: Uveal melanoma develops due to malignant transformation of the normal melanocytes present in the uveal tissue. Primary posterior uveal melanoma (choroidal melanoma) which forms the majority of uveal melanoma manifests as:

- A solitary elevated subretinal grey-brown dome shaped mass
- "Collar stud" appearance on B-scan - when the tumor breaks through the Bruchs membrane and protrudes into subretinal space
- "Double circulation" sign

Primary anterior uveal melanoma frequently involves the iris and presents as nodular melanocytic mass on the iris. The below given fundus image shows the "double circulation sign" of choroidal melanoma.



Option C: Uveal lymphoma is characterized by localized or diffuse uveal infiltrates of malignant lymphoma cells. Fundus examination in these patients show diffuse creamy thickening of posterior uvea with nodular accentuation.

Solution to Question 22:

Unilateral decreased vision with retinal detachment, subretinal yellowish exudates and telangiectatic vessels in a young boy points to the diagnosis of Coats' disease.

Coats' disease, described by Coats in 1908, affects boys three times more often than girls, has no reported racial or ethnic predilection, and is usually unilateral. The average age at diagnosis is 8–16 years. The clinical course is generally progressive.

It is characterized by discrete zones of retinal vascular abnormalities like aneurysmal dilatation and leakage. It is associated with localized lipid deposition (hard exudates) and varying degrees of subretinal exudates. Vessels may appear telangiectatic, and they may have aneurysms that are grape-like, clustered, or light bulb-shaped.

The image below depicts the typical vascular abnormalities with aneurysmal dilatation, telangiectasia, exudation, and severe lipid deposition in the macula which are seen in Coats' disease.



Solution to Question 23:

Monocular blurring of vision and pain with "splashed sauce" appearance on fundoscopy points towards a diagnosis of central retinal vein occlusion (CRVO).

Ischemic central retinal vein occlusion (CRVO) is characterised by sudden visual loss. There is splashed sauce appearance on fundoscopy due to retinal hemorrhages. Risk factors include high intraocular pressure (IOP), hypertension and blood viscosity syndromes.

Its complications include neovascularisation of the iris (rubeosis iridis) and neovascular glaucoma (NVG). NVG typically occurs 3 months after the occlusive event. Up to 50% of eyes develop NVG following ischaemic CRVO. Pan-retinal photocoagulation is the treatment of choice.

CRVO is classified into two types:

- Ischemic CRVO: Characterized by relative afferent pupillary defect (RAPD), markedly reduced visual acuity, and many cotton-wool spots
- Non-ischemic CRVO: Characterized by normal pupillary reaction, reduced visual acuity, and few cotton-wool spots

Treatment usually involves treating the underlying cause and complications.

Solution to Question 24:

The image above demonstrates characteristic flame-shaped hemorrhages, localized to the superotemporal sector of the retina. The image, along with loss of vision and history of hypertension, points to a diagnosis of branched retinal vein occlusion (BRVO), specifically affecting the superotemporal branch of the retinal vein.

Occlusion of a single branch of the retinal vein leads to branched retinal vein occlusion. This occlusion leads to impaired venous drainage, resulting in the accumulation of blood within the retina. The visual defect in BRVO tends to be partial.

Note: The different types of retinal vascular obstructions are branched retinal vein obstruction (BRVO), central retinal vein obstruction (CRVO), branched retinal artery obstruction (BRAO) and central retinal artery obstruction (CRAO). Retinal vein occlusion (RVO) is the second most common retinal vascular disorder after diabetic retinopathy.

In CRVO, tortuosity and engorgement of all branches of the central retinal vein are seen, affecting all quadrants of the retina.

Solution to Question 25:

Sudden painless unilateral vision loss with carotid bruit (carotid artery disease) and the fundus picture showing a cherry-red spot is suggestive of central retinal artery occlusion (CRAO). The best immediate step in management is ocular massage among the given options.

Central retinal artery occlusion (CRAO) occurs due to obstruction of the retinal artery at the level of lamina cribrosa. Risk factors include heart disease, carotid artery disease, mucormycosis, varicella infection, radiation exposure and trauma. It may be caused by the following:

- Emboli of carotid artery or cardiac origin are the most common cause of CRAO. Other emboli are cholesterol emboli, calcium emboli, and platelet fibrin emboli
- Atherosclerosis-related thrombosis
- Retinal arteritis with obliteration
- Angiospasm
- Raised intraocular pressure
- Thrombophilic disorders

Patients often present with sudden painless monocular loss of vision with history of a similar episode that resolved spontaneously (amaurosis fugax). On examination, direct pupillary reflex is absent and relative afferent pupillary defect (RAPD) is positive. Fundus examination shows marked attenuation of arteries (thread like arteries), interrupted blood flow in vein (cattle track appearance), cherry red spot, severe edema and retinal pallor.

A cherry-red spot of the macula is typical and arises in this area because of the absence of ganglion cells at the foveola resulting in the transmission of the normal choroidal color. This in contrast to the surrounding area where intense retinal whitening blocks the transmission of the normal choroidal color.

Treatment is to dislodge the emboli by suddenly decreasing the IOP by IV acetazolamide or ocular massage. Ocular massage is considered the most effective method. Other treatment options include IV mannitol, vasodilators (nitrates), fibrinolytic therapy, or laser photodisruption in some cases. In cases of giant cell arteritis, steroids are beneficial.

Solution to Question 26:

A 52-year-old woman temporal headaches, myalgias and pain in the jaw while chewing likely has giant cell arteritis (GCA). Given that she has sudden painless monocular loss of vision with RAPD and points to a diagnosis of central retinal artery occlusion (CRAO).

GCA is the most common primary vasculitis in adults. It is a risk factor for the development of CRAO. CRAO occurs due to obstruction of the retinal artery at the level of lamina cribrosa. When obstruction to the blood flow is not complete, the flow may be partially restored. Thus, gentle pressure upon the globe may break the column of venous blood into red beads separated by clear interspaces. This gives the characteristic 'cattle truck' appearance.

Option A: Ischemic CRVO shows a characteristic "splashed tomato sauce" appearance on fundoscopy.

Option C: Non-Ischemic CRVO does not present with such a severe loss of visual acuity.

Option D: Branched retinal artery occlusion presents with hemifield visual defects.

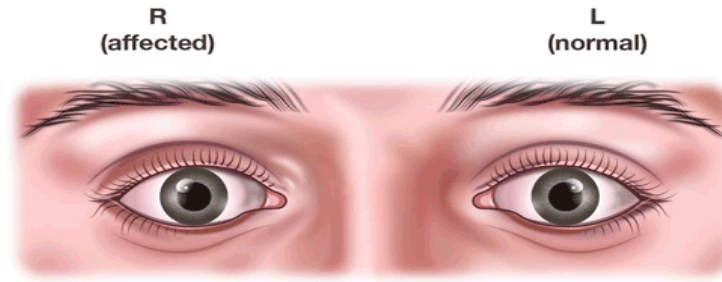
Solution to Question 27:

Sudden painless monocular loss of vision with fundoscopy showing Hollenhorst plaque is suggestive of central retinal artery occlusion (CRAO). Relative afferent pupillary defect is seen in these cases.

Hollenhorst plaques are cholesterol emboli that are seen in retinal arterioles. On fundus examination, it gives an appearance of a refractile yellow-white color substance in the retinal vessels.

Note: Although elevated intraocular pressure has been implicated as a cause of central retinal artery obstruction, it is unlikely that intraocular pressure can be raised high enough to block arterial inflow to the eye

Marcus-Gunn Pupil



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Solution to Question 28:

The prodromal symptoms of flashing lights, floaters followed by loss of monocular vision (curtain over the eye) and the B-scan suggest a diagnosis of rhegmatogenous retinal detachment. Rhegmatogenous retinal detachment is the result of the accumulation of fluid between NSR and RPE due to a retinal tear.

Retinal detachment is the separation of the neurosensory retina from the retinal pigment epithelium. Rhegmatogenous retinal detachment is the most common type of retinal detachment. It occurs due to fluid accumulation after a retinal break (tear). High myopia, trauma, pseudophakia, aphakia are predisposing factors. Patients present with photopsia (flashes of light due to vitreoretinal traction at the site of the break), floaters and visual field defect (black curtain/veil in front of the eye).

Signs include hypotony, as the liquefied vitreous in the subretinal space is absorbed through the RPE and Shafers sign or tobacco dusting (pigments in the anterior vitreous). The detached retina is convex in configuration. The surface is corrugated with undulations. The most dangerous type of retinal break is horseshoe tear and the most dangerous location of retinal break is the supero-temporal retina.

Treatment is by prophylactic laser barrage, scleral buckling, pneumatic retinopexy and pars plana vitrectomy.

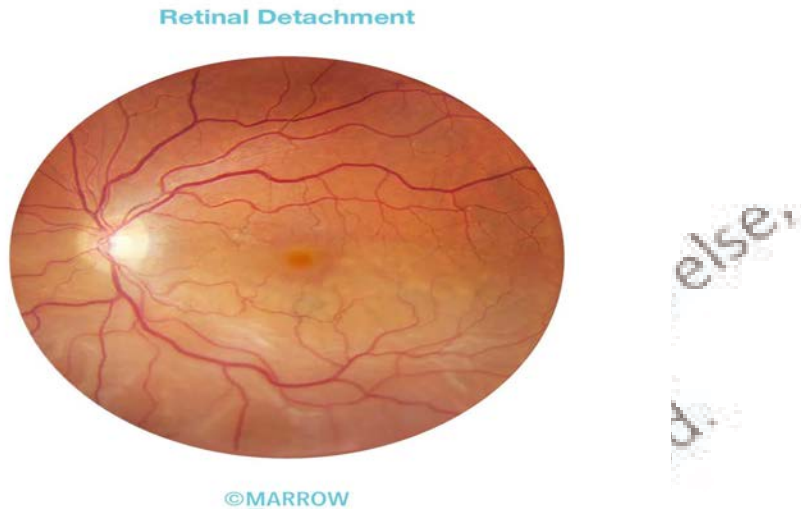
Other types of retinal detachment are tractional and exudative type:

Tractional retinal detachment occurs when fibro-vascular membranes in the vitreous due to long-standing retinal neovascularization exert traction on the retina. It is seen in proliferative diabetic retinopathy, central retinal venous occlusion, Eales disease and ROP. The detached retina is concave in configuration. No breaks are seen. Treatment is with pars plana vitrectomy with endo photocoagulation.

Exudative retinal detachment is seen when exudative fluid from the choroid collects in the subretinal space. Patients present with visual field defects, floaters due to vitritis but no

photopsia. The detached retina is convex in configuration. The surface is smooth. No break is seen. Shifting fluid is seen.

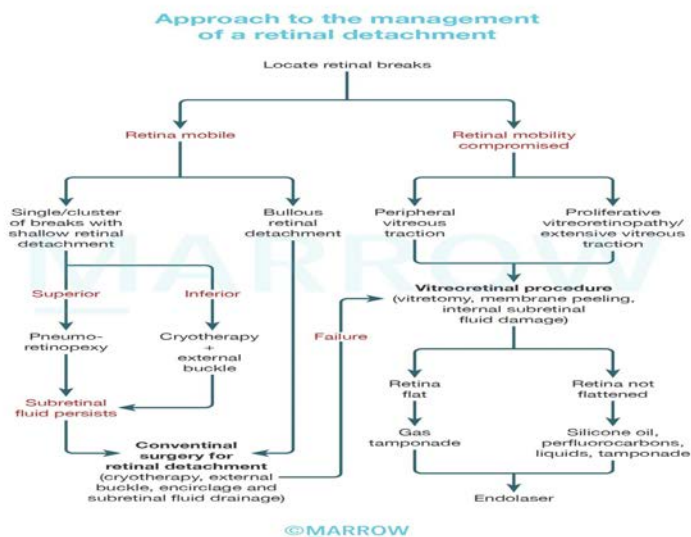
The image below shows a fundus which reveals retinal detachment inferiorly from 3-9 o'clock with tear at 3:30.



Solution to Question 29:

Sulphur hexafluoride is the gas used in pneumatic retinopexy for the treatment of retinal detachments

Pneumatic retinopexy is used for the treatment of fresh retinal detachments with a single retinal break in the superior two-thirds of the fundus. In this procedure, a bubble is injected intravitreally through the conjunctiva and postoperatively the patient is positioned so that the bubble tamponades the retinal break against the retinal pigment epithelium.



Solution to Question 30:

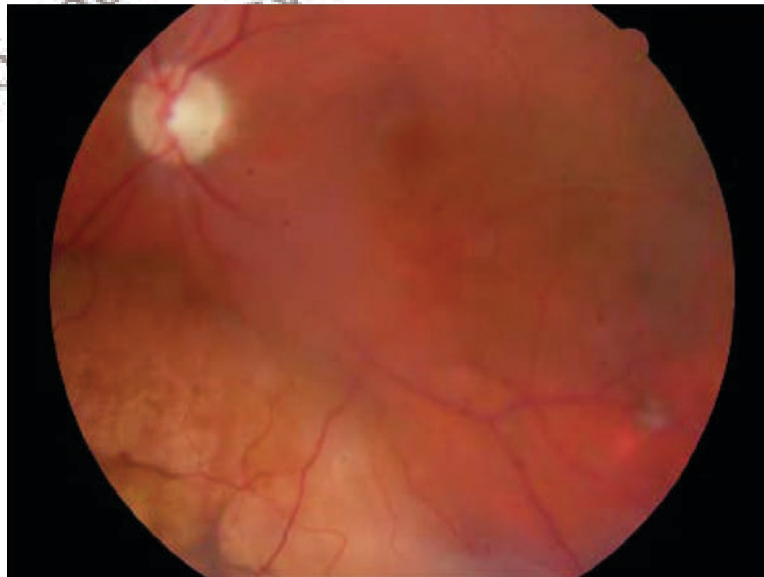
Exudative retinal detachment is characterized by positive shifting fluid sign. Among her conditions, toxemia of pregnancy is one of the causes of exudative retinal detachment.

Exudative retinal detachment occurs due to the presence of subretinal fluid in the absence of tear or traction. It is caused by:

- Malignant melanoma of the choroid
- Retinoblastoma
- Vogt–Koyanagi–Harada (VKH) syndrome
- Uveal effusion syndrome
- Posterior scleritis
- Coat's disease
- Malignant hypertension
- Pregnancy-induced hypertensive retinopathy
- Sympathetic ophthalmitis

Patients present with sudden and painless diminution of vision, visual field defects and grey reflex (detached retina looks grey). Indirect ophthalmoscopy is the investigation of choice.

The image depicts retinal detachment showing a smooth, detached convex retina with subretinal fluid.



Solution to Question 31:

The presence of shifting fluid is a hallmark feature of serous or exudative retinal detachment.

In serous retinal detachment, there is a subretinal accumulation of the serous fluid mostly due to the breakdown in the blood-retinal barrier. When the position of the head is altered, the fluid shifts to the dependent portion. Serous retinal detachment has a smooth convex configuration. A retinal tear or preretinal traction is absent.

Other options:

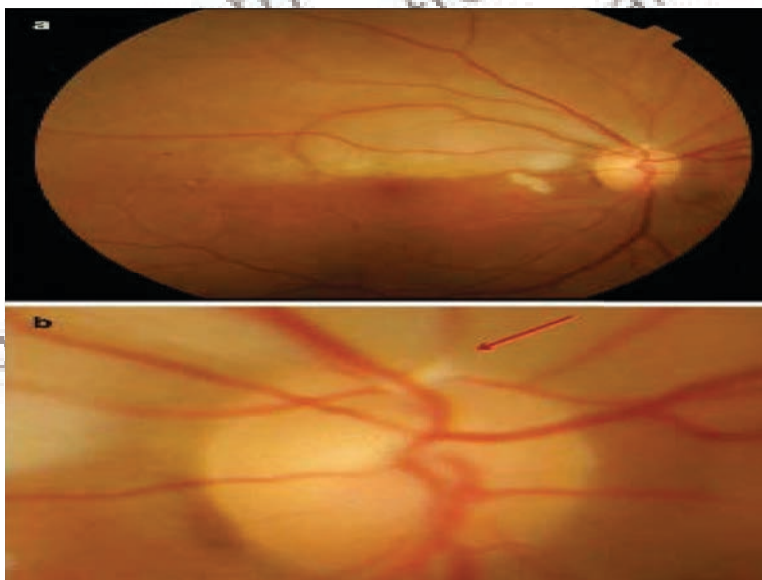
Option B: Traction RD has a concave configuration and breaks are absent. Retinal mobility is severely reduced and shifting fluid is absent.

Option C: Rhegmatogenous RD presents with tobacco dust in anterior vitreous with substantial vitreous blood or inflammatory cells. Retinal breaks are also seen.

Solution to Question 32:

Hollenhorst plaques are seen in central retinal arterial occlusion (CRAO).

They are cholesterol emboli seen in retinal arterioles. They originate from atherosclerotic plaques in larger arteries. The image below shows Hollenhorst plaques (arrow).



Solution to Question 33:

The above image shows a loss of foveal reflex and central serous retinopathy, therefore showing focal foveal detachment.

There is oedema at the fovea signifying foveal detachment from underlying retinal pigment epithelium. Macular oedema is also present (grey dull retina with a ring reflex around the site of oedema). Thus, the most appropriate choice would be foveal detachment.

Central serous retinopathy is the spontaneous serous detachment of neurosensory retina in the macular region. It occurs due to increased permeability of choriocapillaris, leading to fluid accumulation in the subretinal space. Therefore, it is now known as central serous choriopathy (CSC).

If there is serous retinal detachment at macula, there is sudden painless loss of vision. Patients also have relative positive scotoma, micropsia and metamorphopsia. On ophthalmoscopy, a circular ring reflex present. The foveal reflex is absent. On fundus fluorescein angiography (FFA), an enlarging ink-blot pattern or smoke-stack pattern is seen.

Treatment is generally not required. It is usually self-limiting, but often recurrent. Steroids are contraindicated.

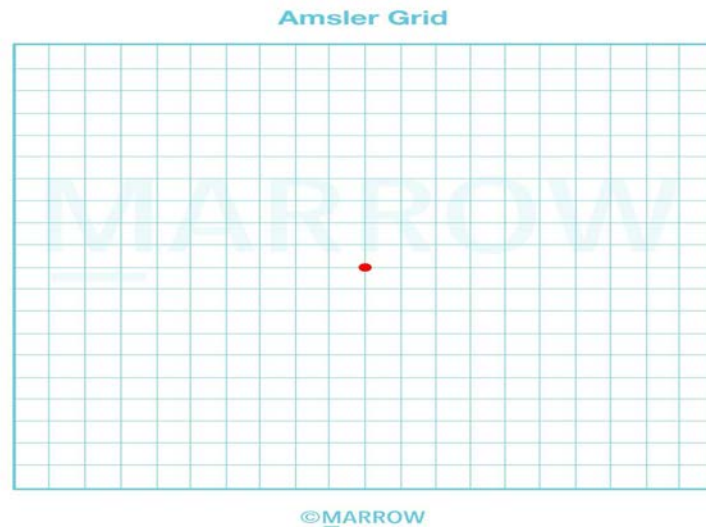


Solution to Question 34:

The image depicts fundus fluorescein angiography (FFA) showing an enlarging ink-blot pattern/enlarging dot sign, which is seen in central serous retinopathy (CSR). Smokestack appearance is another pattern of fundus seen in CSR and is a characteristic sign of this condition.

Amsler grid is used for:

- Assess macular function and detecting maculopathy.
- Evaluate the visual field of 10°
- Good for follow-up in CSR



Solution to Question 35:

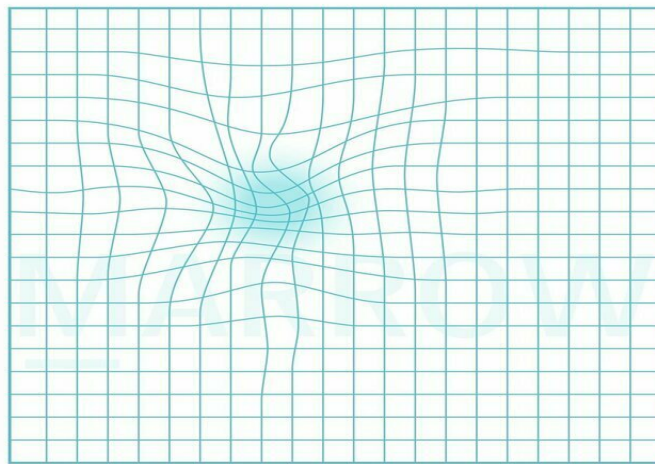
ETDRS stands for Early Treatment for Diabetic Retinopathy Study.

ETDRS was a clinical trial designed to evaluate if argon laser photocoagulation or aspirin treatment can reduce the risk of visual loss or slow the progression of diabetic retinopathy in patients with mild-to-severe nonproliferative or early proliferative diabetic retinopathy. During the study, three new visual acuity charts were created in place of Snellen's visual chart.

Macular disorders, Retinal dystrophies, and Vitreal disorders

Question 1:

A 25-year-old artist visits an art gallery. A painting of checkered squares appears distorted only to him, as shown below. He decides to go see an ophthalmologist who after initial evaluation, suggests an FFA. The FFA shows a characteristic smoke stack appearance. Breakdown of which of the following barriers is responsible for his condition?



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- a) Inner blood retinal barrier
- b) Outer blood retinal barrier
- c) Both the barriers are disrupted
- d) No disruption of either barriers

Question 2:

Where does fluid accumulate in central serous retinopathy?

- a) Between the Bruch membrane and retinal pigment epithelium
- b) In the subretinal space
- c) Between the outer plexiform layer and inner nuclear layer
- d) Along the nerve fiber layer

Question 3:

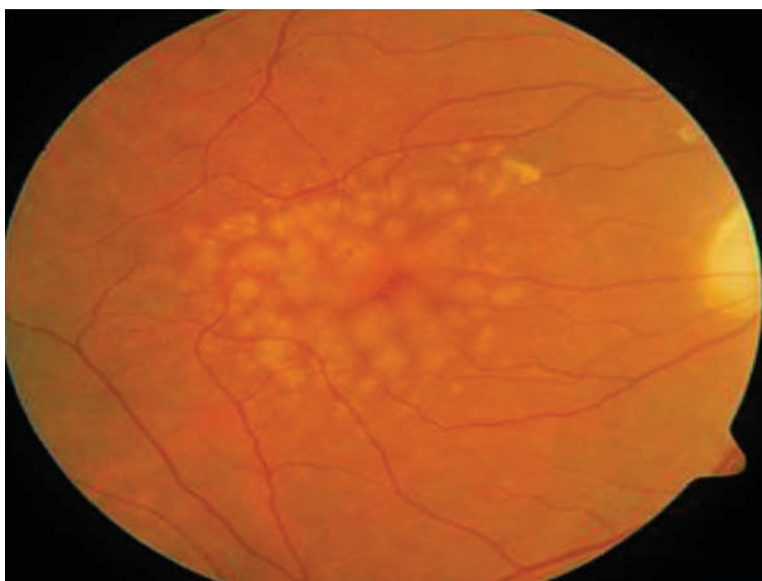
A 60-year-old female presented with acute loss of vision and metamorphopsia. There is a history of the positive scotoma. Fundal changes are given below. What will be the fundus fluorescein angiography finding?



- a) Ink blot appearance
- b) Petaloid appearance
- c) Stars in the sky appearance
- d) Dual circulation

Question 4:

A 69-year-old man who is a chronic smoker, hypertensive, and has a BMI of 35 kg/m² complains of diminished vision in the right eye. His visual acuity is 6/36 BE and the anterior segment appears normal. His fundus is as shown below. What is the most likely diagnosis?



- a) Diabetic retinopathy
- b) Age related dry macular degeneration
- c) Hypertensive retinopathy
- d) Age related wet macular degeneration

Question 5:

Indocyanine green angiography (ICGA) is primarily indicated in which of the following conditions?

- a) Minimal classic CNVM
- b) Occult CNVM
- c) Clinically significant macular edema
- d) Coats disease

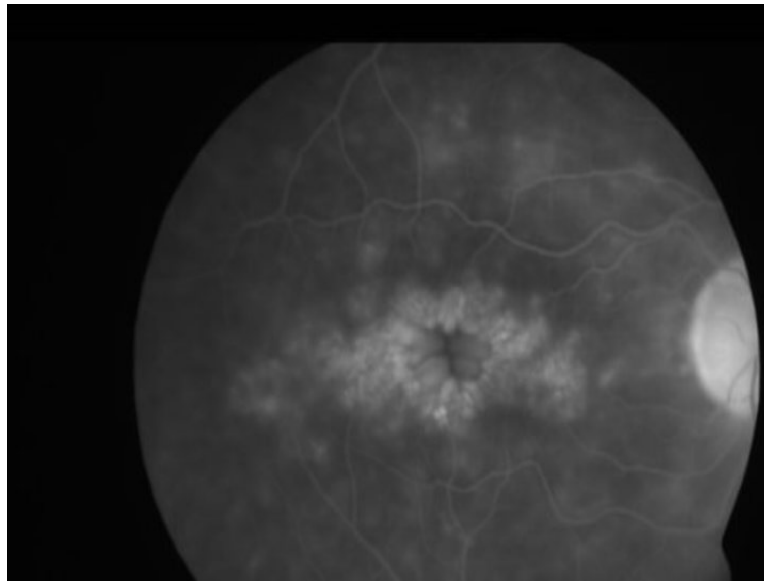
Question 6:

Intravitreal injection of anti-VEGF is used in the treatment of which of the following conditions?

- a) Vitreous haemorrhage
- b) Wet ARMD
- c) Glaucoma
- d) Dry ARMD

Question 7:

An old cab driver whose livelihood depends on his driving is distressed. He complains that everything appears distorted and he perceives objects to smaller than their usual size. His anterior segment appears normal. Examination of the fundus reveals a honeycomb appearance of the macula. FFA is shown below. What is the most likely diagnosis?



- a) Age related macular degeneration
- b) Macular hemorrhage
- c) Cystoid macular edema
- d) Central Serous retinopathy

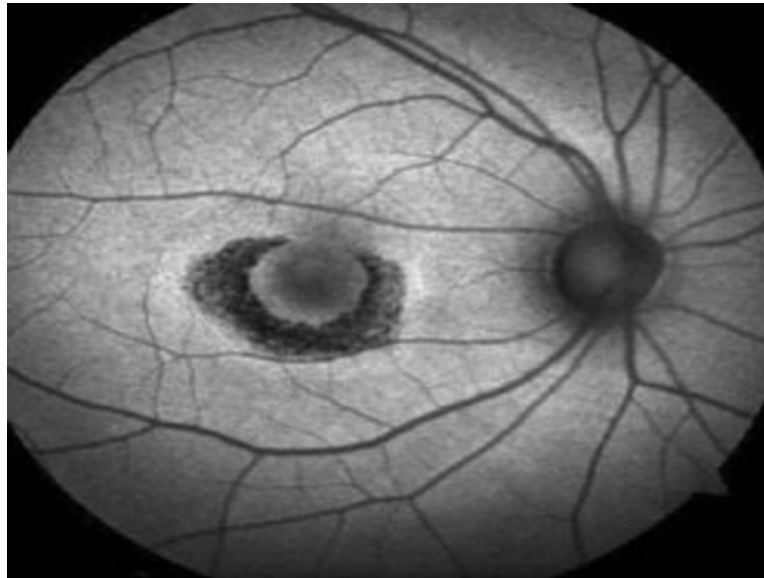
Question 8:

A patient presents to the emergency department and is found to have a corneal abrasion, corneal ulcer, corneal perforation, and foreign body. Which of the following will not cause hypotonic maculopathy in this patient?

- a) Suprachoroidal haemorrhage
- b) Cyclodialysis of uvea
- c) Corneal perforation
- d) Filtration site leak

Question 9:

A 45-year old patient on systemic treatment for rheumatoid arthritis develops bilateral painless decrease in vision. What does the image show?

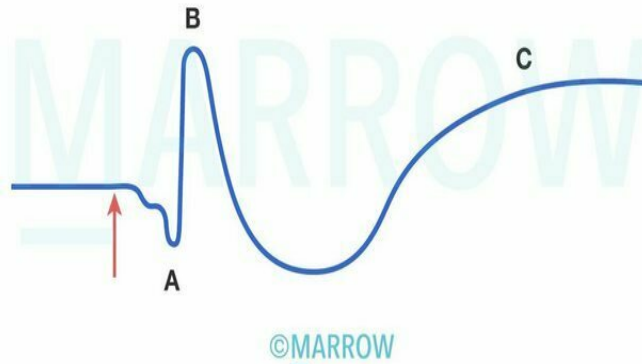


- a) Fluorescein angiography showing bull's eye maculopathy
- b) Autofluorescence showing changes of bull's eye maculopathy
- c) OCT showing bull's eye maculopathy
- d) Fluorescein angiography showing crystalline retinopathy

Question 10:

A study used electroretinography(ERG) in military conscripts whose visual abnormalities could not be explained by the findings of ophthalmologic, neurological, and psychiatric examinations. The ERG of pure malingerers was found to be, as shown below. The activity of which cells of the retina is depicted by the a-wave?

Electroretinogram (ERG)



- a) Rods and cones
- b) Muller cells
- c) Pigment epithelium
- d) Optic nerve

Question 11:

Which area of the retina has the highest concentration of rods?

- a) Mid-periphery of retina
- b) Optic disc
- c) Fovea
- d) Ora serrata

Question 12:

A researcher is studying mutations in the rhodopsin gene which are responsible for some forms of autosomal dominant retinitis pigmentosa. On which chromosome is this gene located?

- a) 3q
- b) 3p
- c) 13p
- d) 13q

Question 13:

Which of the following statements is false regarding the retinal pigment epithelium (RPE)?

- a) RPE forms the inner blood retinal barrier
- b) Electroculogram is an assessment of RPE health
- c) Autofluorescence helps evaluate RPE function
- d) Bruch's membrane is the basement membrane of RPE

Question 14:

Which of the following disorders is not associated with retinitis pigmentosa?

- a) Refsum's disease
- b) Kearn-Sayre syndrome
- c) Neuropathy, ataxia, and retinitis pigmentosa (NARP)
- d) Abetaglobulinemia

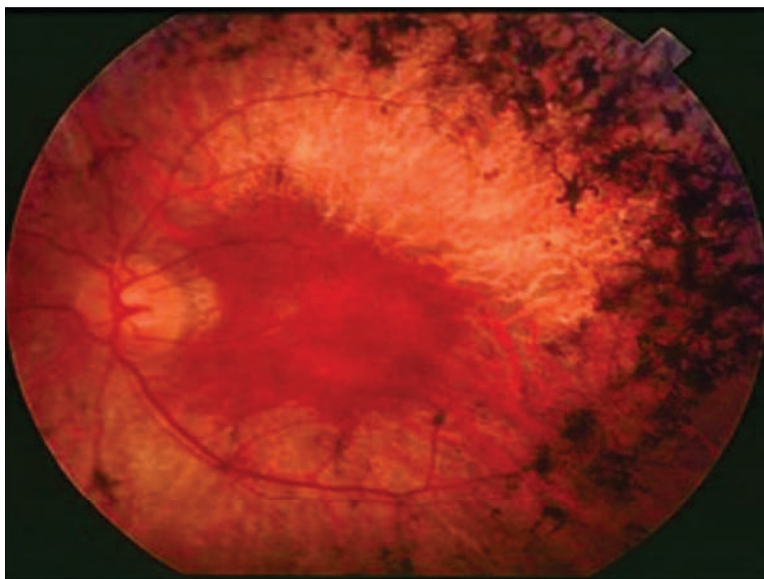
Question 15:

Which of the following is not true about retinitis pigmentosa?

- a) Retinal pigmentation
- b) Pale waxy disc
- c) Narrowing of vessels
- d) ERG – normal

Question 16:

What is the diagnosis from the image given below?



- a) Retinitis pigmentosa
- b) Cystoid macular oedema
- c) Central serous retinopathy
- d) Retinal detachment

Question 17:

In which of the following conditions is the Mizuo phenomenon seen?

- a) Oguchi's disease
- b) Fundus albipunctatus
- c) Fundus flavimaculatus
- d) Retinitis pigmentosa

Question 18:

Flat electrooculogram (EOG) is an Arden ratio of _____.

- a) Less than 1.85
- b) Less than 1.50
- c) Less than 1.25
- d) Less than 1.20

Question 19:

A young patient presents with loss of central vision. ERG is normal, but EOG is abnormal. What is the likely diagnosis?

- a) Retinitis pigmentosa
- b) Rod cone dystrophy
- c) Stargardts disease
- d) Best's disease

Question 20:

A young patient presents with loss of central vision. There is a family history of a similar condition. Both EOG and ERG are normal. What is the most likely diagnosis?

- a) Retinitis pigmentosa
- b) Rod-cone dystrophy
- c) Stargardts disease
- d) Best's disease

Question 21:

Silent choroid on FFA is a feature of which of the following disorders?

- a) Stargardt's disease
- b) Best's disease
- c) Age related macular degeneration
- d) Cystoid macular edema

Question 22:

A neonate with leukocoria and strabismus is diagnosed to have persistent hyperplastic primary vitreous (PHPV). Which of the following congenital disorders is it associated with?

- a) Down syndrome
- b) Tuberous sclerosis
- c) Patau syndrome
- d) Sturge Weber syndrome

Question 23:

When compared to blood, vitreous humor has a high concentration of which of the following?

- a) Ascorbate
- b) Glutathione
- c) Glucose
- d) Sodium

Question 24:

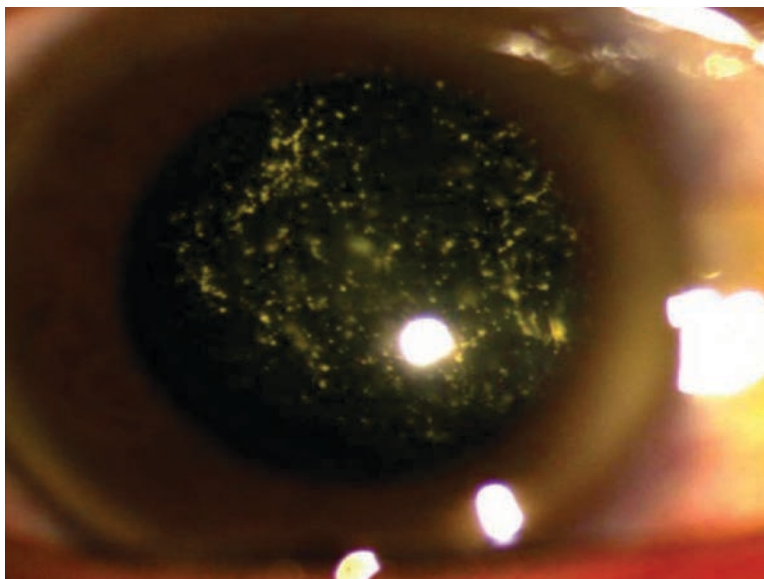
A 40-year-old woman complains of discomfort in her left eye. There is a past history of blunt trauma in her childhood which made her left eye blind. Slit-lamp examination of her left eye is shown below. What is your diagnosis?



- a) Mittendorf dots
- b) Bergmeister papilla
- c) Synchysis scintillans
- d) Asteroid hyalosis

Question 25:

A slit-lamp examination reveals the following finding. It is also noted that these yellow-white vitreous opacities do not settle down when the eye is immobile. All of the following statements about this condition are true except:



- a) Mostly unilateral
- b) Rarely causes symptoms
- c) Associated with diabetes
- d) Common in myopes

Question 26:

A young male presents with recurrent vitreous hemorrhage. What is the probable diagnosis?

- a) Eale's disease
- b) Central retinal vein occlusion
- c) Coats disease
- d) Proliferative vitreoretinopathy

Question 27:

Which of the following is characteristic of Eale's disease?

- a) Retinal hemorrhage
- b) Vitreous hemorrhage
- c) Conjunctival hemorrhage

d) Choroidal hemorrhage

Question 28:

What is the investigation of choice for the diagnosis of a vitreous hemorrhage?

- a) ERG
- b) USG A scan
- c) USG B-scan
- d) Optical coherence tomography

Question 29:

Sub-hyaloid hemorrhage is _____.

- a) Boat-shaped
- b) Crescent-shaped
- c) Round
- d) Flame-shaped

Question 30:

What is a reverse hypopyon?

- a) Collection of pus in the vitreous
- b) Collection of emulsified silicone oil in the anterior chamber
- c) Abscess in the orbit
- d) Perforated corneal ulcer

Question 31:

Which anti-fungal can be administered intravitreally?

- a) Fluconazole
- b) Voriconazole
- c) Ketoconazole
- d) Clotrimazole

Answer Key

Question No.	Correct Option
1	b
2	b
3	a
4	b
5	b
6	b
7	c
8	a
9	b
10	a
11	a
12	a
13	a
14	d
15	d
16	a
17	a
18	c
19	d
20	c
21	a
22	c
23	a
24	c
25	d
26	a
27	b
28	c
29	a
30	b
31	b

Detailed Explanations

Solution to Question 1:

The given clinical scenario with a characteristic smoke stack pattern on FFA is suggestive of a diagnosis of central serous retinopathy. In CSR, the outer blood-retinal barrier is disrupted.

CSR is the serous detachment of the neurosensory retina with focal and multifocal areas of leakage at the level of the retinal pigment epithelium (RPE), mainly affecting the macular area. That is why the patient who is usually a young male presents with macular symptoms such as blurring of central vision, metamorphopsia.

The blood-retina barrier (BRB) is composed of inner and outer barriers. The outer BRB refers to the barrier formed at the retinal pigment epithelial (RPE) cell layer and regulates the movement of solutes and nutrients from the choroid to the sub-retinal space. By contrast, the inner BRB, similar to the blood-brain barrier (BBB) is located in the inner retinal microvasculature and comprises the microvascular endothelium, which lines these vessels.

The image given below shows the smoke-stack pattern seen in CSR.

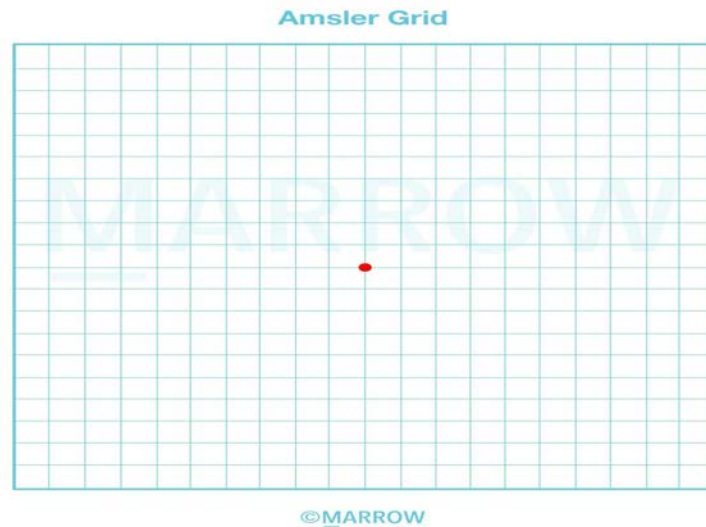


Amsler grid is used in detecting maculopathy.

Uses:

- Assess macular function
- Evaluate the visual field of 10°
- Good for follow-up in CSR

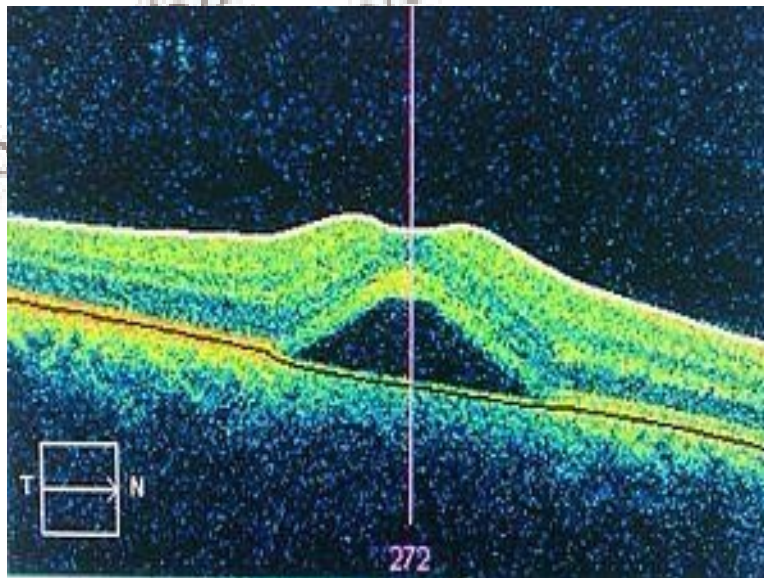
The image below shows Amsler's grid.



Solution to Question 2:

Fluid accumulation in central serous retinopathy (CSR) is in the subretinal space due to abnormal permeability of the choriocapillaris.

The image depicts optical coherence tomography showing fluid accumulation in the subretinal space, with shallow retinal detachment.



Solution to Question 3:

The above clinical scenario and the fundal image depicting focal fundal detachment are suggestive of central serous retinopathy. An enlarging inkblot pattern or smoke stack pattern is seen in FFA.

Central serous retinopathy refers to the spontaneous serous detachment of the neurosensory retina in the macular region. Recent studies implicate an abnormal choroidal circulation as the cause of CSR.

Clinical features:

- Retinal detachment - Sudden painless loss of vision
- Relative positive scotoma
- Micropsia
- Metamorphopsia

Investigations:

- Ophthalmoscopy - circular ring reflex present and foveal reflex absent
- FFA - enlarging inkblot pattern or smoke stack pattern

Spontaneous resolution is seen in most cases and hence observation is enough. Steroids are contraindicated. The disease is usually self-limiting but may recur.

The image depicts fundus fluorescein angiography (FFA), showing an enlarging ink-blot pattern/enlarging dot sign, which is seen in central serous retinopathy (CSR).



Other options:

Option B- Petaloid appearance is seen in cystoid macular edema

Option C- Stars in the sky appearance is suggestive of drusen

Option D- Dual circulation is a pathognomonic feature of choroidal melanoma

Solution to Question 4:

The clinical scenario along with the image which shows drusen (druplets) are a feature of dry or non-exudative age-related macular degeneration (ARMD).

Drusen are the earliest visible clinical signs of ARMD. They are composed of lipid, amyloid, complement factors, and associated cellular components.

There are two types of ARMD:

- Dry (non-exudative, non-neovascular) ARMD is the most common form, comprising around 90% of the diagnosed cases. Initially characterized by macular Drusens, which later progress to geographical atrophy. Geographic atrophy (GA) is the advanced stage of dry ARMD. Antioxidants like the AREDS2 supplements have proven effective.
- Wet (exudative, neovascular) ARMD is much less common than the dry type, but it is associated with more rapid progression to advanced sight loss.

Clinical classification:

Category	Definition(based on the presence of lesions within two disc diameters of the fovea in either eye)
No apparent age-related changes	No drusen No AMD pigmentary abnormalities
Normal age-related changes	Only drusen No AMD pigmentary abnormalities
Early ARMD	Medium Drusen (>63 μ m but <125 μ m) No AMD pigmentary abnormalities
Intermediate ARMD	Large Drusen (>125 μ m) Any ARMD pigmentary abnormalities
Late ARMD	Neovascular ARMD and/or geographic atrophy

Solution to Question 5:

Indocyanine green angiography (ICGA) is primarily indicated in an occult choroidal neovascular membrane (CNVM).

ICGA is used to visualize the choroidal vasculature.

Its main use is to differentiate between classic and occult CNVM when the clinical picture is not clear.

Solution to Question 6:

Intravitreal injection of anti-vascular endothelial growth factor (VEGF) is used as the treatment of wet age-related macular degeneration (ARMD).

Wet ARMD is characterized by retinal pigment epithelial detachment and choroidal/subretinal neovascularization.

Treatment of ARMD depends on the location of the choroidal neovascularization membrane:

- For extrafoveal membrane - Laser photocoagulation
- For subfoveal and juxta foveal membranes, treatment options are as follows:
- Intravitreal injection of anti -VEGF agents
- Photodynamic therapy - the dye used is verteporfin followed by exposure to light from a diode
- Transpupillary thermotherapy

Solution to Question 7:

The image is that of fluorescein angiography showing the "flower petal pattern," which is seen in cystoid macular edema (CME). The characteristic flower-petal appearance is due to the accumulation of dye in the outer plexiform layer and inner nuclear layer at the macula

The accumulation of fluid in the outer plexiform layer and the inner nuclear layer leads to retinal thickening at macula and loss of foveal depression, formation of cystic spaces (honey-comb macula). Risk factors include retinitis pigmentosa, uveitis, niacin, PGE2 analogues, epinephrine in aphakia, diabetic retinopathy, retinal venous occlusion and Irvin-Gass syndrome (CME post-cataract surgery).

Symptoms may include blurring of vision, distortion, and micropsia.

Therapeutic approaches for treating CME depend on the underlying etiology. This includes steroids, carbonic anhydrase inhibitors, anti-VEGF agents, PPV for associated vitreomacular traction.

Solution to Question 8:

Suprachoroidal hemorrhage will not cause hypotonic maculopathy in this patient.

Maculopathy is common in eyes developing hypotony, which is defined as IOP ≤ 6 mmHg. The most common cause is excessive drainage following glaucoma filtration surgery (Option D). Other causes include trauma such as cyclodialysis of the uvea (Option B), penetrating injury (Option C), chronic uveitis, retinal detachment, and other systemic causes such as dehydration, uncontrolled diabetes, uremia, and treatment with hyperosmotic agents. They usually cause bilateral hypotonic maculopathy.

Suprachoroidal hemorrhage is a bleed into the suprachoroidal space that results from the rupture of the posterior ciliary artery. When the bleed is severe it causes extrusion of intraocular contents that is termed as expulsive hemorrhage. It is a rare but dreaded complication of the phacoemulsification procedure. Signs seen in suprachoroidal hemorrhage are:

- Progressive increase in intraocular pressure
- Prolapse of iris

- Loss or partial obscuration of the red reflex
- In severe cases there can be extrusion of the posterior segment contents into the anterior chamber and through the incision such as vitreous extrusion

Immediate treatment involves rapid closure of the incision followed by administration of IOP-lowering medications such as oral acetazolamide and topical or systemic steroids to reduce inflammation. Standard post-operative antibiotics can be added in these patients.

Solution to Question 9:

The investigation shown in the image is an autofluorescence showing bull's eye maculopathy.

Hydroxychloroquine, which is generally used for the treatment of rheumatoid arthritis, toxicity causes bull's eye maculopathy. Chloroquine binds to melanin in the retinal pigment epithelium, causing cytotoxic effects. RPE atrophy occurs in the macula, resulting in dark areas on the autofluorescence image.

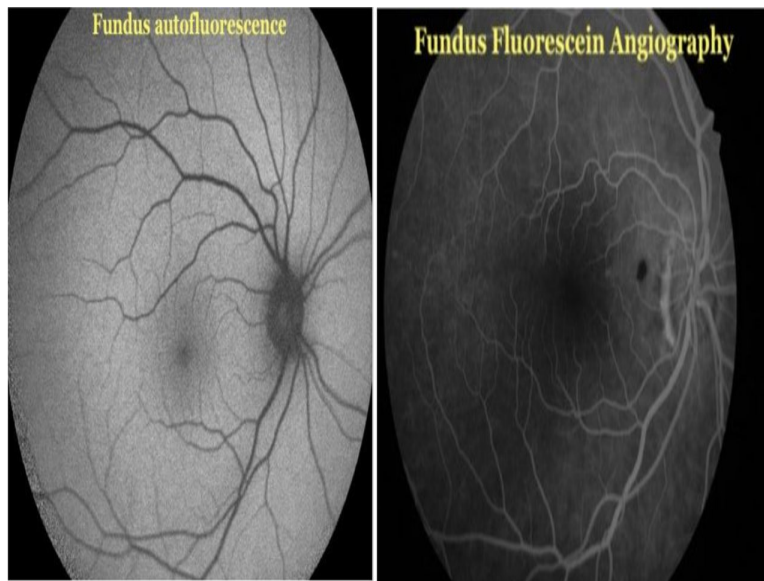
Other causes of Bull's eye maculopathy include chloroquine, hydroxychloroquine, congenital cone dystrophy, juvenile X-linked retinoschisis, and juvenile Batten's disease.

Fundus autofluorescence is a non-invasive imaging modality. It reflects the morphological changes associated with the metabolism of lipofuscin and does not require the injection of any fluorescent dye.

A fundus image of bull's eye maculopathy is shown below:



The following image shows the difference between fundus autofluorescence and fluorescence angiography



Other ocular effects of chloroquine include:

- Chloroquine keratopathy—vortex keratopathy
- Ocular palsy, ptosis
- Electroretinographic depression
- Optic neuritis
- Posterior subcapsular cataract

Solution to Question 10:

"a" wave in an electroretinogram (ERG) represents rods and cones.

Electroretinography is used to evaluate retinal functions by examining the electrical response to a light stimulus.

Electroretinograms (ERGs) show the amplitudes of currents that flow within the eye following exposure to a point of light. The amplitudes are analyzed into waves, which are composed of the responses of neuronal, glial, and retinal pigment epithelial cells.

ERG:

- The a-wave originates from photoreceptors (rods and cones)
- The b-wave originates from Muller and bipolar cells, but is directly dependant on photoreceptors
- The c-wave originates from retinal pigment epithelial cells and retinal glial cells
- Oscillatory potentials (OPs) originate in the inner retina

Scotopic ERGs are used to evaluate responses starting from rod photoreceptors exposed to flash-light in darkness, and photopic ERGs are used to evaluate responses starting from cone photoreceptors exposed to flash-light under constant light exposure.

Solution to Question 11:

Mid-periphery of the retina has the highest concentration of rods.

The following table lists the differences between rods and cones

Property	Rods	Cones
Number	Most numerous (120 million)	Far fewer in number (6 million)
Location	Mid-peripheral retina	Fovea
Function	Night vision Motion sense Peripheral vision	Day vision Colour vision Central and fine vision
Dysfunction symptoms	Nyctalopia Peripheral field loss	Poor central vision Dyschromatopsia Hemeralopia

Solution to Question 12:

The gene for rhodopsin is located at 3q22.1

Retinitis pigmentosa (RP) is a dystrophy of rods and cones, with rods being primarily affected. It is a genetic disease. The non-hereditary/sporadic type is most common. Among the hereditary types, it can occur in an autosomal dominant (AD), autosomal recessive (AR), or X-linked recessive (XLR) pattern. AD disease generally has the best prognosis.

The presenting feature is night blindness.

Investigations:

- Perimetry - the initial stage will have ring scotoma, which further develops into tunnel vision.
- Electroretinography - reveals a decreased amplitude of both "a" and "b" waves.

Retinitis pigmentosa is a condition with no cure. There are drugs used to slow down the progression such as acetazolamide for concomitant cystoid macular edema. Recently, vitamin A has been recommended, although the efficacy has not been established.

Solution to Question 13:

Tight junctions between retinal pigment epithelium (RPE) form the outer blood-retinal barrier. The inner blood-retinal barrier is formed by the tight junctions between the endothelial cells of retinal capillaries.

RPE serves many functions, like

- Development of photoreceptors during embryogenesis

- Tight junctions between RPE form the outer blood-retinal barrier
- Absorption of fluid from the subretinal space and adhesion of the underlying neurosensory retina
- Selective transport of metabolites to and from the retina
- Uptake, transport, storage, metabolism, and isomerization of vitamin A
- Regeneration of the visual pigment rhodopsin
- Phagocytosis of photoreceptor outer segment tips
- Absorption of stray light by melanin

If the RPE is damaged, this is seen as a loss of autofluorescence (RPE-deficient areas appear darker on autofluorescence images).

Bruch's membrane is the basement membrane of RPE.

Solution to Question 14:

Retinitis pigmentosa (RP) is associated with abetalipoproteinemia (Bassen–Kornzweig syndrome) and not with abetalglobulinemia.

Syndromes associated with RP (Atypical RP)	
Usher syndrome	Sensory neural deafness + RP
Bardet–Biedl syndrome (Previously called Lawrence–Moon–Biedl–Bartum Syndrome)	Polydactyly, obesity, mental handicap, and kidney disease + RP
Kearn–Sayre syndrome	Ocular myopathy and heart disease + RP
Refsum Syndrome	Varied skin as well as neurological and visceral features + RP
Bassen–Kornzweig syndrome	Abetalipoproteinemia and acanthocytosis + RP
Neuropathy, ataxia, and retinitis pigmentosa (NARP)	Neuropathy and Ataxia + RP

Solution to Question 15:

Electroretinogram (ERG) is abnormal in retinitis pigmentosa (RP).

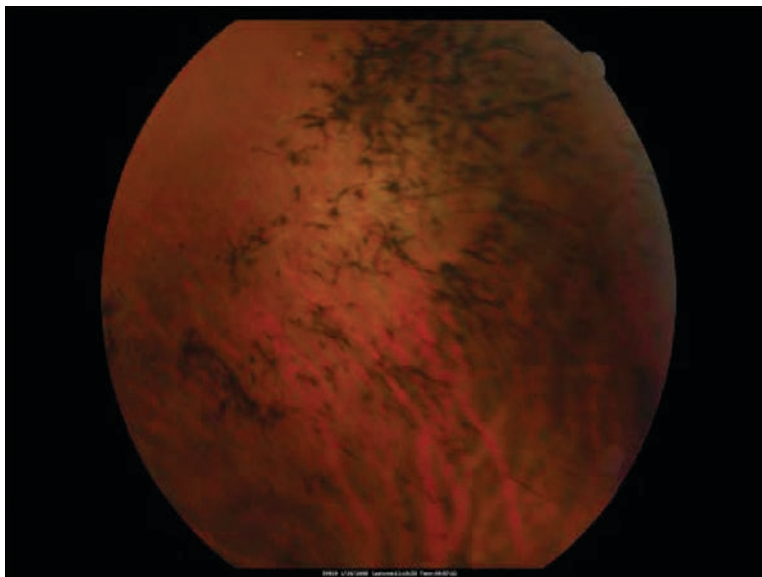
ERG is a sensitive diagnostic test in retinitis pigmentosa that shows reduced scotopic, subnormal responses in early disease and becomes completely extinguished in advanced disease. Electrooculogram (EOG) is also abnormal in retinitis pigmentosa.

Retinitis pigmentosa (RP) is an inherited or acquired pigmentary retinal dystrophy affecting the rods more than the cones. The earliest manifestation is worsening of night blindness and delayed adaptation which gradually progresses to tunnel vision and ultimately loss of central vision.

The classic triad of findings in RP comprises:

- Bone-spicule retinal pigmentation
- Arteriolar attenuation
- 'Waxy' disc pallor

The image below shows the classical triad findings of retinitis pigmentosa.



Solution to Question 16:

The fundus shows a pale waxy disc, attenuated arteries, and bony spicules, which implies retinitis pigmentosa (RP).

Retinitis pigmentosa (RP) is a dystrophy of rods and cones, with rods being primarily affected. It is a genetic disease. The non-hereditary/sporadic type is most common. Among the hereditary type, it can occur in an autosomal dominant (AD), autosomal recessive (AR), or X-linked recessive (XLR) pattern. AD disease generally has the best prognosis.

The presenting feature is night blindness.

Investigations:

- Perimetry - the initial stage will have ring scotoma, which further develops into tunnel vision.
- Electroretinography - reveals a decreased amplitude of both "a" and "b" waves.

Atypical types of RP are as follows:

- Sectoral RP
- Pericentric/Inverse RP

- RP sine pigmento
- RP punctata albescens

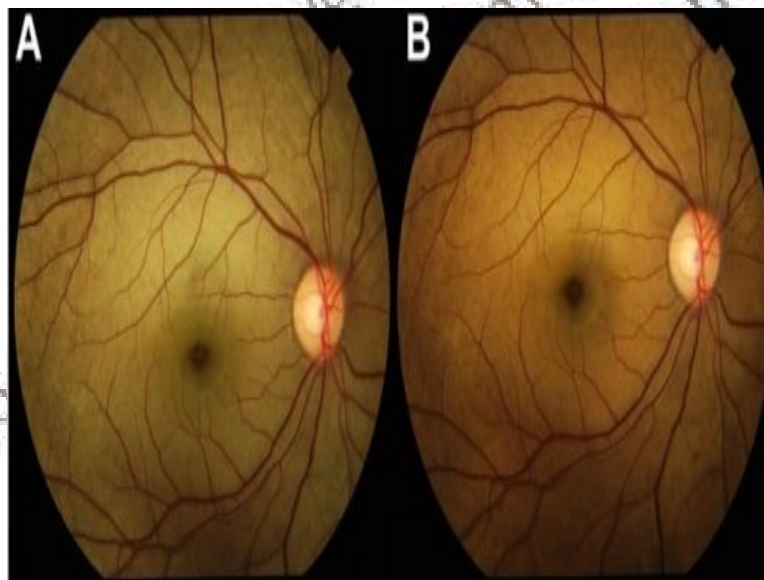
There is no effective treatment for RP.

Solution to Question 17:

Mizuo phenomenon is seen in Oguchi's disease.

Oguchi's disease is a type of congenital stationary night blindness with abnormal fundus. Congenital stationary night blindness refers to a group of disorders characterized by infantile-onset nyctalopia but non-progressive retinal dysfunction. The fundus has an unusual golden yellow color in the light-adapted state, which becomes normal after prolonged dark adaptation (Mizuo or Mizuo–Nakamura phenomenon).

The following image shows a comparison in the fundus appearances ("A" shows the golden yellow color of the fundus in a light-adapted state; "B" shows normal-appearing fundus after prolonged dark adaptation of 3 to 4 hours)



Rod function is absent even after 30 minutes of dark adaptation (normal dark adaptation time is 20–30 mins) but recovers to a near-normal level after a long period of dark adaptation (4 hours).

Solution to Question 18:

A flat electrooculogram (EOG) is an Arden ratio of less than 1.25.

An EOG measures the standing potential of the eye. It reflects the activity of the RPE and the photoreceptors. This means that an eye blinded by disease proximal to the photoreceptors will have a normal EOG. Arden ratio is the ratio between the light peak and the dark trough in an EOG.

The normal Arden ratio is 1.85 and above. A flat EOG occurs when the Arden ratio is less than 1.25.

Best disease is an autosomal-dominant childhood disease characterized by the dystrophy of the retinal pigment epithelium. In Best disease, the Arden ratio is less than 1.50.

Solution to Question 19:

The diagnosis based on the above scenario is Best's disease.

Best's disease/Vitelliform macular dystrophy is an inherited macular dystrophy in which lipofuscin accumulates in the central macula causing progressive central vision loss. It is autosomal-dominant and linked to the mutation in the bestrophin (VMD2) gene.

It is characterized by bilateral yellow, yolk-like (vitelliform) macular lesions.

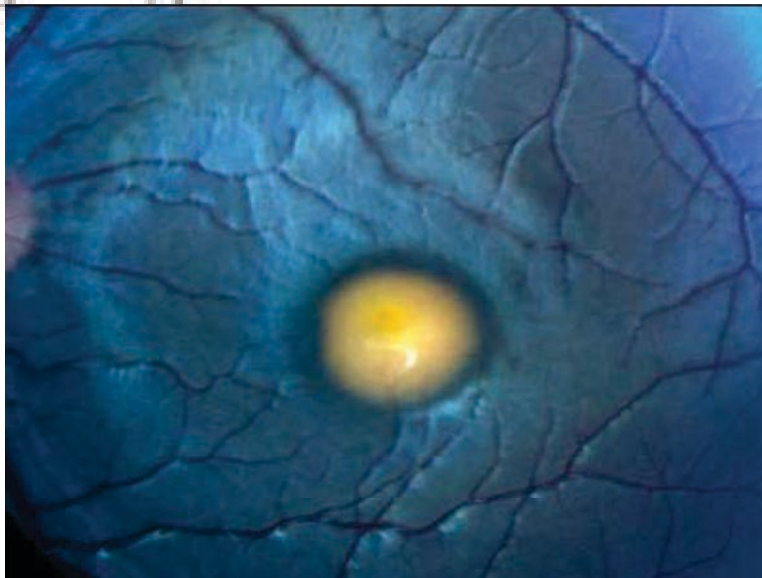
The stages of evolution of the macular lesions are described as progressing from:

- Pre-vitelliform stage
- Vitelliform stage - scrambled egg stage with or without hypopyon
- Vitelliruptive - the lesion breaks and visual acuity drops
- The atrophic stage

Patients with Best's disease usually have good visual acuity when the "yolk" remains intact.

Diagnosis: This is the only disease with relatively normal ERG results associated with an abnormal EOG. Optical coherence tomography (OCT) reveals that the vitelliform material appears as a dome-shaped, hyperreflective, and homogenous lesion, located below the hyperreflective photoreceptor layer.

The image below shows Best's disease - the typical vitelliform lesion in an 11-year-old girl.



Solution to Question 20:

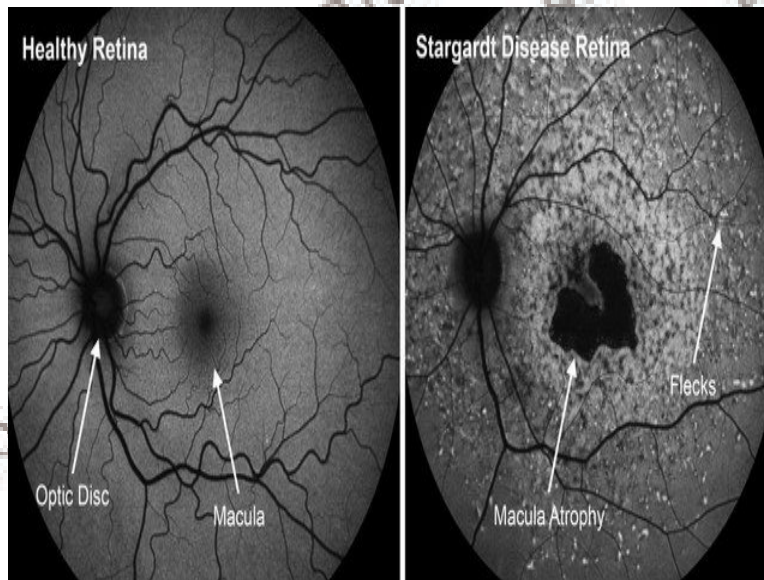
The most likely diagnosis based on the given scenario is Stargardt's disease.

Stargardt's disease also known as Stargardt macular dystrophy or fundus flavimaculatus is the most common macular dystrophy. It most commonly is inherited in an autosomal-recessive manner caused by a mutation on an adenosine triphosphate-binding cassette (ABCA4) gene.

Characterized by macular "pisciform" flecks that accumulate at the level of the retinal pigment epithelium (RPE), sometimes resulting in central macular atrophy early in life. Patients develop a macula with a "beaten bronze" appearance caused by atrophic changes in the RPE. In addition, a "dark" or "silent" choroid is seen on the fluorescein angiography that appears as a prominent retinal circulation against hypofluorescent choroid.

The electroretinogram (ERG) and electro-oculography (EOG) are normal early in the disease but may be reduced in more advanced cases. Of note, the ERG findings do not directly correlate with clinical findings.

There is no known proven treatment for this disease. Because ABCA4 plays a role in vitamin A processing in the visual cycle, additional vitamin A is suspected to make the disease worse.



The image below shows macular pisciform flecks.



Other options:

Option A: Retinitis pigmentosa: abnormal electroretinogram (ERG).

Option B: Rod–cone dystrophy: abnormal ERG.

Option D: Best's disease: Normal ERG with abnormal electrooculogram (EOG).

Solution to Question 21:

Silent/Dark choroid on FFA is a feature of Stargardt's disease. The classic feature of 'silent choroid' is due to the masking of background choroidal fluorescence by lipofuscin deposits in the subretinal space.

Stargardt disease constitutes the most common type of macular dystrophy. It has mutations in a photoreceptor cell-specific ATP-binding cassette gene (ABC4) that is usually inherited in an autosomal recessive manner. The disease usually starts in the juvenile group with progressive central vision loss and color vision loss that occurs due to the atrophy of the macular retinal pigment epithelium with an accumulation of discrete pisciform flecks of lipofuscin.

Signs of Stargardt's disease:

- Beaten bronze appearance of the macula that is caused by the atrophic changes in the RPE.
- Silent/Dark choroid on FFA.
- Visual field testing shows central vision loss.
- The electroretinogram (ERG) is normal early in the disease but may be reduced in more advanced cases.

Fundus flavimaculatus is genetically considered to be a part of the same spectrum as Stargardt's disease with late onset i.e. usually after 40 years of age.

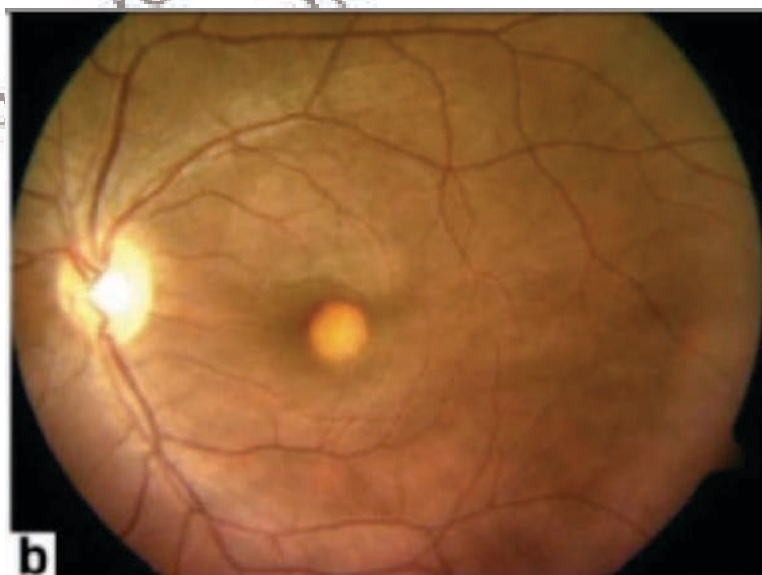


Other options:

Option B: Best's disease also known as vitelliform macular dystrophy is an autosomal dominant macular dystrophy in which lipofuscin accumulates in the central macula, causing progressive central vision loss. This disease is linked to mutations in the bestrophin (VMD2) gene and they do not exhibit silent choroid on FFA. Findings seen in Best's disease are:

- Vitelliform(yolk like) appearance of the fundus
- Relatively normal ERG results associated with abnormal EOG

The given below is the fundus picture of Best's disease showing an egg yolk lesion in the macula.

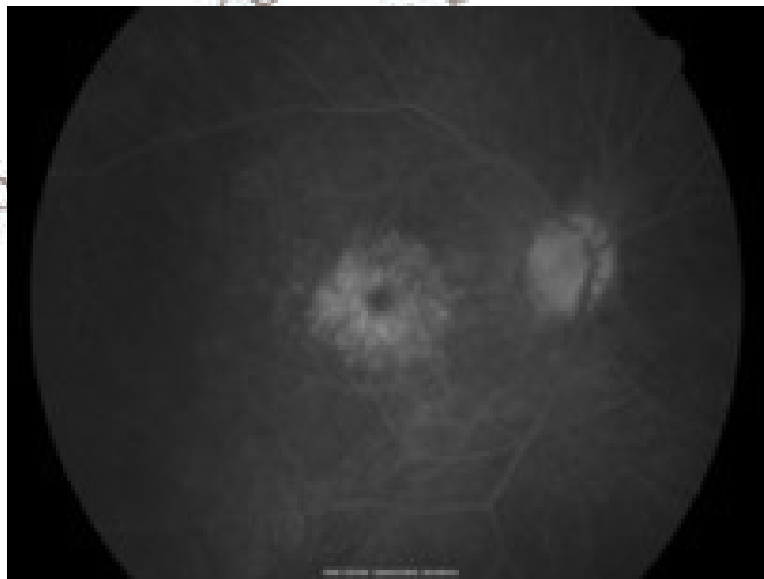


Option C: Age related Macular degeneration(ARMD) also known as senile macular degeneration is a chronic and progressive bilateral degenerative disorder of the macula that affects older individuals >50 yrs. ARMD is classified as dry and wet ARMD where the hallmarks are the presence of drusen and choroidal neovascularization respectively. Drusen are extracellular eosinophilic deposits between RPE and Bruch's membrane.

Clinical features include loss of central vision, distorted vision (lines seen as wavy), and sudden painless loss of vision as seen in wet ARMD. The image below shows Drusens in the macular area in dry macular degeneration.



Option D: Cystoid Macular edema (CME) is a collection of fluid between the outer plexiform layer and the inner nuclear layer of the retina following a break in the blood-retinal barrier. In CME on fundoscopy, a honeycomb appearance and on FFA flower petal pattern can be seen. The below image shows the flower petal pattern on FFA as seen in CME.



Solution to Question 22:

Persistent hyperplastic primary vitreous (PHPV) is associated with Patau syndrome (Trisomy 13).

PHPV is a developmental disorder of the vitreous humor. In this condition, there is persistence of the embryonic hyaloid vessel.

- Anterior PHPV - Here, the remnant vessel is attached to the back of the lens but does not reach the optic nerve.
- Posterior PHPV- In this form of the disease, the vessel arises from the optic nerve.

Clinically, PHPV presents with leucocoria and microphthalmia, and there can be complications such as tractional retinal detachments.

This condition has a poor prognosis regardless of the time of intervention.

Solution to Question 23:

When compared to blood, vitreous humor has a high concentration of ascorbate. The vitreous: plasma ratio of ascorbic acid concentration is 9:1.

This high concentration of ascorbic acid is maintained by the active transport by the ciliary body epithelium, and it is attributed to the ability of the ascorbate to absorb ultraviolet rays and act as a free radical scavenger.

Solution to Question 24:

These are synchysis scintillans, which are floaters composed of cholesterol bodies, found in the vitreous (extending into the anterior chamber in this picture).

Synchysis scintillans are vitreous opacities that occur as a consequence of chronic vitreous haemorrhage, often in a blind eye. This condition is usually discovered, when frank haemorrhage is no longer present. They are composed of cholesterol derived from plasma cells or degraded products of erythrocytes.

Vitreous opacities can be caused by a variety of pathological processes and the following are few examples of vitreous opacities:

- Vitreous haemorrhage
- Asteroid hyalosis
- Synchysis scintillans
- Amyloidosis
- Vitreous cyst
- Persistent fetal vasculature.

Solution to Question 25:

The examination findings are suggestive of asteroid bodies/hyalosis. It is not common in myopes.

Asteroid hyalosis is a common degenerative process in which calcium pyrophosphate particles collect within the vitreous gel. It is seen clinically as numerous tiny round yellow-white opacities of varying size and density. These move with the vitreous during eye movements but do not sediment inferiorly when the eye is immobile.

Only one eye is affected in 75% of patients; it rarely causes visual problems and the majority of patients are asymptomatic. The cause of asteroid hyalosis is unknown, but it has been associated with diabetes mellitus, hypertension, and hypercholesterolemia. The prevalence of asteroid hyalosis increases with age and affects 3% of those aged 75–86 years. It is more common in men than in women. While treatment of asteroid hyalosis is usually unnecessary, vitrectomy may occasionally be indicated.

Solution to Question 26:

For a young male presenting with recurrent vitreous hemorrhage, the diagnosis is Eales's disease.

Eales disease is an idiopathic occlusive peripheral periphlebitis. It is characterized by three overlapping stages: inflammatory, occlusive, and retinal neovascular. Visual prognosis is good in the majority of cases. It is an important cause of visual morbidity in young males from the Indian subcontinent.

Diagnosis:

- Symptoms - Floaters or sudden visual reduction due to vitreous hemorrhage. There can be neurological deficits as well.
- Mild anterior uveitis
- Fundus - The condition is typically bilateral, although often asymmetrical.
- Peripheral periphlebitis
- Branch retinal vein occlusion
- Peripheral capillary non-perfusion; microaneurysms, tortuosity, vascular shunts, and neovascularization with recurrent vitreous hemorrhage

Treatment:

- Steroids: Periocular, systemic, topical, and intravitreal steroids seem to be helpful in the inflammatory stage.
- Antitubercular treatment

Solution to Question 27:

Eales's disease is an idiopathic inflammatory venous occlusion (phlebitis) that primarily affects the peripheral retina that is periphlebitis and is characterized by recurrent bilateral vitreous hemorrhage.

Superficial retinal hemorrhages and cotton-wool spots can be sometimes seen, but they are not characteristic of Eales disease.

Solution to Question 28:

The investigation of choice for vitreous hemorrhage is ultrasonography (USG) B-scan. Mild hemorrhage causes sudden onset floaters and diffuse blurring of vision, but may not affect visual acuity, whilst a dense bleed may result in sudden painless loss of vision.

B-mode ultrasonography (B-scan) in unclotted vitreous hemorrhage generally shows a uniform appearance, and once cellular aggregates develop, small particulate echoes become visible.

Causes of vitreous hemorrhage:

- Acute posterior vitreous detachment associated either with a retinal tear or avulsion of a peripheral vessel
- Proliferative retinopathy
- Diabetic
- Retinal vein occlusion
- Sickle cell disease
- Eales disease
- Vasculitis
- Miscellaneous retinal disorders
- Macroaneurysm
- Telangiectasia
- Capillary hemangioma
- Trauma
- Systemic
- Bleeding disorders
- Terson syndrome- refers to the combination of intraocular and subarachnoid hemorrhage secondary to aneurysmal rupture, most commonly arising from the anterior communicating artery.

Treatment is dictated by severity and cause, but an increasingly low threshold is being adopted for early vitrectomy.

Solution to Question 29:

Sub-hyaloid hemorrhage is dome-shaped or boat-shaped.

Subhyaloid hemorrhage is the collection of blood in the potential space between the posterior vitreous and internal limiting membrane of the retina.

The image depicts a sub-hyaloid hemorrhage which is boat-shaped.



Solution to Question 30:

Reverse hypopyon is the collection of emulsified silicone oil in the anterior chamber.

Silicone oil is the vitreous substitute that is injected into the vitreous cavity after vitrectomy. When it gets emulsified, it enters the anterior chamber and collects in the upper part.

Complications of silicone oil in the anterior chamber:

- Glaucoma
- Band-shaped keratopathy

The image below shows reverse hypopyon.



Solution to Question 31:

Voriconazole and amphotericin-B can be given intravitreally.

Vancomycin and ceftazidime is the preferred combination for endophthalmitis (intravitreal).

Amikacin (0.4 mg in 0.1 ml) or gentamicin (0.4 mg in 0.1 ml) are used less frequently in endophthalmitis owing to the risk of macular infarction.

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Lens - Introduction, Types of Cataract and Clinical Features

Question 1:

What is the embryonic origin of the lens?

- a) Surface ectoderm
- b) Neural ectoderm
- c) Neural crest
- d) Mesoderm

Question 2:

Where are the youngest fibers of the lens present?

- a) Nucleus
- b) Cortex
- c) Equator
- d) Pre-equatorial zone

Question 3:

At a routine ophthalmological check-up, a patient was found to have bilateral, Y-shaped lens opacities as shown below. He was diagnosed with suture cataracts. Which zone of the lens nucleus is not involved here?



- a) Fetal nucleus
- b) Embryonic nucleus
- c) Infantile nucleus
- d) Adult nucleus

Question 4:

What is the ligament of Weigert?

- a) Attachment of the vitreous to the posterior capsule of the lens
- b) Medial palpebral ligament
- c) Attachment of superior oblique tendon
- d) Attachment of the lens to the ciliary muscles

Question 5:

Where does the crystalline lens derive its nutrition from?

- a) Blood vessels
- b) Connective tissue
- c) Aqueous humor
- d) Zonules

Question 6:

While studying the histological anatomy of the lens, you are measuring the thickness of the lens capsule in various areas. At which of the following zones would you get the lowest measurement?

- a) Anterior pole
- b) Posterior pole
- c) Pre-equator region
- d) At equator

Question 7:

You are observing the given slide in a histology examination. Where is the marked structure absent?



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- a) Anterior surface
- b) Anterior pole
- c) Posterior surface
- d) Zonular attachment

Question 8:

Ascorbate and alpha-tocopherol are maintained in the lens in the reduced state by _____.

- a) Glucose
- b) Glycoprotein
- c) Glutathione
- d) Fatty acids

Question 9:

As part of a public health initiative, screening for cataract is being carried out. What is the most common cause of this healthcare issue in the population?

- a) Congenital
- b) Diabetes mellitus
- c) Corticosteroids
- d) Ageing

Question 10:

You are compiling a literature review on congenital cataracts diagnosed in the last few years. Which of the following is the most common type?

- a) Blue-dot cataract
- b) Lamellar cataract
- c) Posterior pole cataract
- d) Total congenital cataract

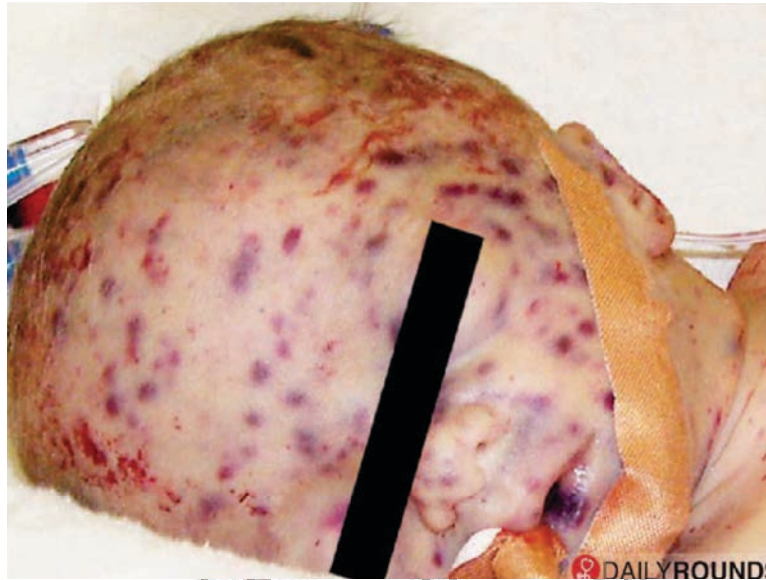
Question 11:

The parents of a child have been informed at a PTA meeting that he has a lot of incomplete classwork. On being questioned, the child complains that he is unable to see things clearly and is having difficulty in reading and writing. On ophthalmic examination, he is found to have a cataract. Which type of congenital cataract most commonly presents with these findings?

- a) Blue-dot cataract
- b) Lamellar cataract
- c) Posterior polar cataract
- d) Total congenital cataract

Question 12:

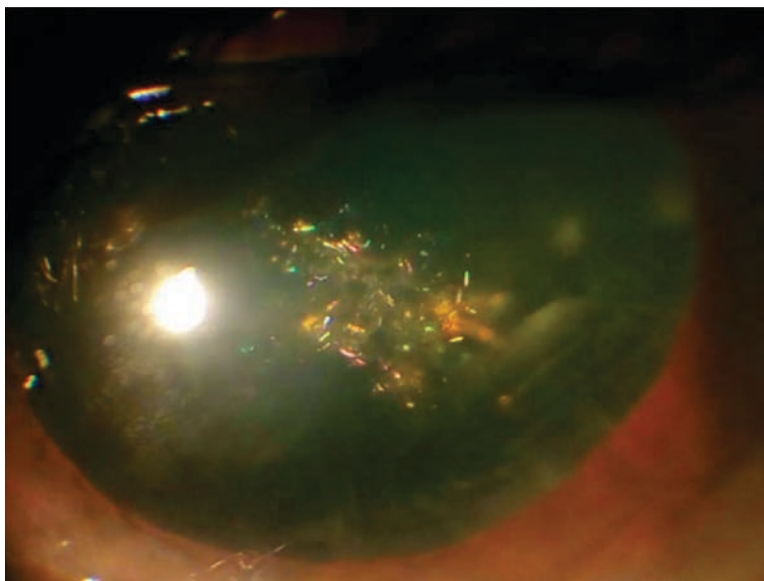
A 33-year-old woman delivers a baby girl with a congenital rash, as shown below, and patent ductus arteriosus. What type of cataract is commonly associated with this condition?



- a) Nuclear pearly cataract
- b) Lamellar cataract
- c) Blue-dot cataract
- d) Posterior polar cataract

Question 13:

The following finding is found on the ophthalmologic examination of an 18-year-old girl. What is the most likely diagnosis?



- a) Type 1 Diabetes mellitus
- b) Alport's syndrome
- c) Lowe's syndrome
- d) Myotonic dystrophy

Question 14:

A 65-year-old woman presents to the ophthalmology OPD with the findings shown below. She has been diagnosed with a right-sided senile cataract. Identify the stage of the cataract.



- a) Intumescent cataract

- b) Mature senile cataract
- c) Morgagnian cataract
- d) Sclerotic type of hypermature cataract

Question 15:

Where is polychromatic luster noted?

- a) Complicated cataract
- b) Diabetic cataract
- c) Post-radiation cataract
- d) Congenital cataract

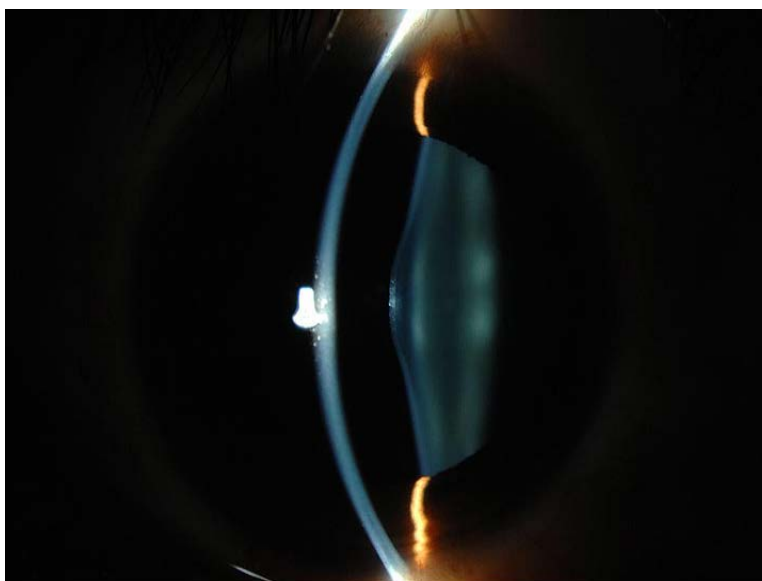
Question 16:

Who among the following patients can potentially develop drug-induced cataracts?

- a) 55 year old female on Zoledronate
- b) 40 year old male on Hydroxychloroquine
- c) 36 year old male on Dexamethasone
- d) 42 year old on Ethambutol

Question 17:

The ophthalmologic examination on a 23-year-old college student revealed the following image. What disease is commonly associated with this ocular finding?



- a) Crohn's disease
- b) Galactosemia
- c) Myotonic dystrophy
- d) Alport's syndrome

Question 18:

A 60-year-old man, diagnosed with bilateral cataracts two months ago, reports being able to read better without prescription glasses. However, he complains that his distant vision has become blurrier. Which type of cataract typically causes this phenomenon?

- a) Nuclear cataract
- b) Cortical cataract
- c) Zonular cataract
- d) Metabolic cataract

Question 19:

A 38-year-old lady is a known case of diabetes and is on insulin for the last 8 years. Her sugars are poorly controlled and now she complains of hazy vision. What type of cataract will be characteristically noted in her eyes?

- a) Sunflower cataract
- b) Breadcrumb cataract
- c) Polychromatic lustre

d) Snowflake cataract

Question 20:

A 55-year-old woman presents with an inability to see objects in bright light. On examination, she is diagnosed with a cataract. In which of the following cataract is this characteristic finding not seen?

- a) Posterior subcapsular cataract**
- b) Cuneiform cataract**
- c) Nuclear cataract**
- d) Steroid-induced cataract**

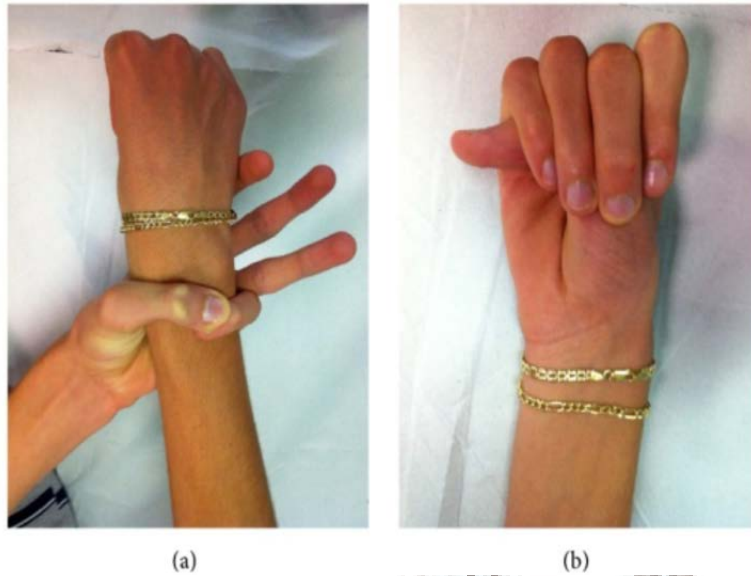
Question 21:

You are counseling the parents of Nisha, a 1-month-old child diagnosed with a congenital lamellar cataract in both eyes. What is the course of treatment you would advise?

- a) Pharmacotherapy**
- b) Combined cataract surgery and goniotomy**
- c) Cataract surgery with intraocular lens (IOL) implant**
- d) Pars plana lensectomy with no IOL implant**

Question 22:

A 28-year-old girl is being evaluated for an aortic root aneurysm. She says her mother died of an aortic dissection at the age of 43. She is tall and slim. A musculoskeletal system finding is shown below. What is the most common lens subluxation noted in the genetic disease she is suffering from?



- a) Superonasal
- b) Superotemporal
- c) Inferonasal
- d) Inferotemporal

Question 23:

Which of the following diseases does not show a specific pattern of cataract?

- a) Juvenile diabetes
- b) Leprosy
- c) Myotonic dystrophy
- d) Wilson disease

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	a
5	c

6	b
7	c
8	c
9	d
10	a
11	b
12	a
13	d
14	c
15	a
16	c
17	d
18	a
19	d
20	b
21	c
22	b
23	b

Detailed Explanations

Solution to Question 1:

The lens of the eye originates embryonically from the surface ectoderm.

Surface ectoderm gives rise to lens, conjunctival and corneal epithelium and epithelial lining of lacrimal glands and tarsal glands.

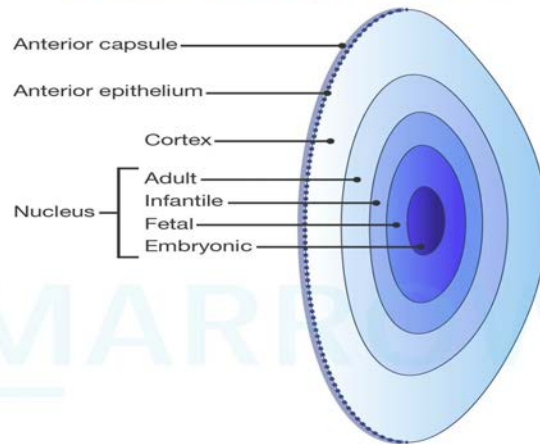
Solution to Question 2:

The youngest fibers of the lens are the peripheral part, the cortex.

The nucleus of the lens is the central part containing the oldest fibers.

The lens fibers are produced by mitosis of the epithelial cells in the pre-equatorial zone. As the lens fibers elongate and the new ones are formed, the older ones are pushed towards the depth of the lens so that the youngest fibers are the most superficially located.

Structure of the crystalline lens



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Solution to Question 3:

The primary fibers in the embryonic nucleus of the lens do not form Y-shaped sutures.

The image shows bilateral cataracts in the Y-shaped sutures of the lens. These are very slowly progressive congenital cataracts and do not require any surgical intervention.

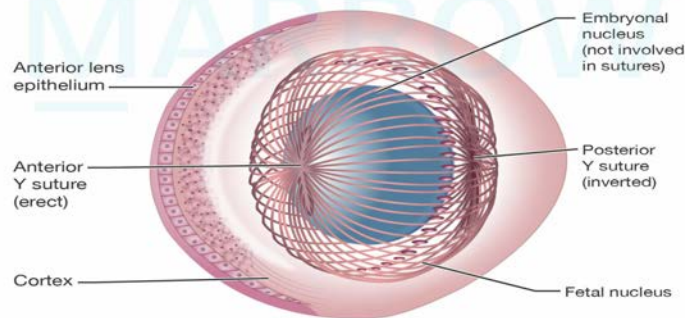
Types of lens nucleus and their age of development:

- Embryonic nucleus: Innermost nucleus formed at around 1–3 months of gestation
- Fetal nucleus: From 3 months of gestation till birth
- Infantile nucleus: From birth to puberty
- Adult nucleus

As lens fibers grow anteriorly and posteriorly, a pattern emerges where the ends of the fibers meet. These patterns of cell association are known as sutures.

Y-shaped sutures are recognizable at about 8 weeks of gestation.

Fibres and Sutures of the Lens



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Solution to Question 4:

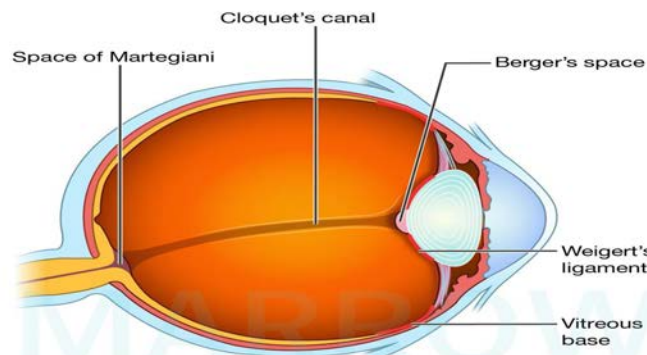
The hyaloideo-capsular ligament of Weigert is the attachment of the vitreous to the posterior capsule of the lens.

It is the annular region (1–2 mm in width and 8–9 mm in diameter) where the vitreous is attached to the posterior aspect of the lens.

Erggelet's or Berger's space is at the centre of the hyaloideo-capsular ligament, a potential space between the vitreous and the lens. The canal of Cloquet arises from this space and courses posteriorly through the central vitreous, which is the former site of the hyaloid artery in the embryonic vitreous.

Posteriorly, Cloquet's canal opens into a funnel-shaped region anterior to the optic disc, which is known as the area of Martegiani.

Posterior Chamber of Eye



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Solution to Question 5:

The crystalline lens derives its nutrition from the aqueous humor.

The lens is an avascular structure. It receives nutrition mostly from the aqueous humor and, to some extent, from the vitreous humor as well. Glucose is the main energy source. Major intrinsic protein 26 (MIP 26)/Aquaporin 0 in the lens is responsible for maintaining transparency by transporting water and glycerol across the lens.

Anti-oxidant mechanisms in the lens:

- Reduced glutathione (most important)
- Glutathione peroxidase
- Superoxide dismutase
- Catalase
- Vitamin C, Vitamin E, carotenoids.

Solution to Question 6:

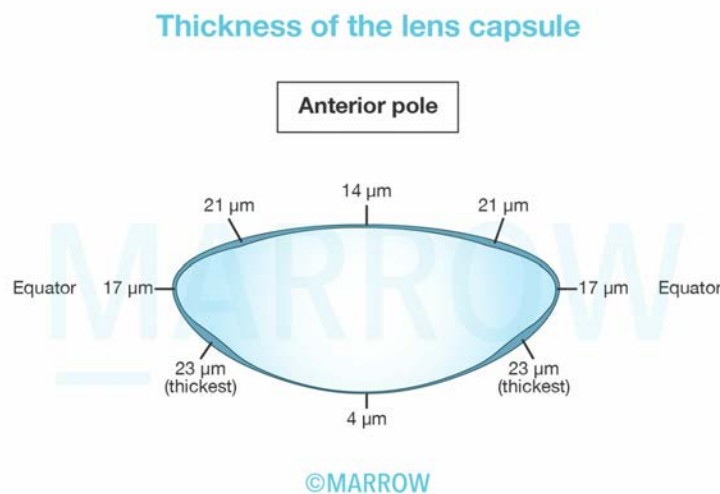
The lens capsule is the thinnest at the posterior pole, measuring 4 μm .

Other options:

Option A: At the anterior pole, the thickness of lens capsule is 14 μm .

Option C: In the pre-equator region, lens capsule is the thickest, measuring 23 μm .

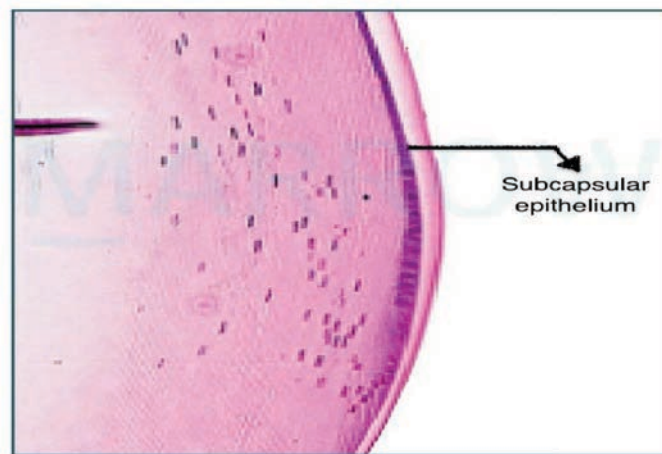
Option D: At the equator, as shown in the illustration below, the thickness is 17 μm .



Solution to Question 7:

The slide shows the histology of the lens and the structure marked in the subcapsular epithelium. The epithelium of the lens is absent on its posterior surface.

The lens epithelium is a single layer of cells on the anterior surface of the lens. There is no epithelium on the posterior surface, as these cells are used up in filling the central cavity of the lens vesicle during the development of the lens. The anterior epithelial lining creates new lens fibers throughout a person's life.



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Solution to Question 8:

Ascorbate and alpha-tocopherol are maintained in the lens in the reduced, functional state by glutathione.

Glutathione is present in high concentrations in the lens, particularly in the epithelial layer. One of its key functions is to maintain protein thiols in their reduced state, which preserves the lens transparency by preventing the formation of high molecular weight crystalline aggregates and protects the lens from oxidative stress.

The lens is highly prone to photochemical reactions from continuous exposure to ultraviolet rays (295–400 nm), leading to reactive oxygen species (ROS) generation. Its defence relies heavily on superoxide dismutase, ascorbate, catalase and glutathione peroxidase.

With ageing or cataracts, levels of glutathione and ascorbic acid (in reduced form) significantly decrease.

Solution to Question 9:

The most common cause of cataracts is ageing, specifically senile (age-related) cataract.

Important causes of cataract:

- Aging (Most common)
- Congenital cataract (crystallin gene - CRYAA, CRYAB, CRYGC, CRYGS) (Option A)
- Diabetes mellitus (Option B)
- Syndermatotic- atopic dermatitis is the most common skin condition associated with cataract
- Galactosemia- reversible cataract
- Radiation- UV rays, X-rays, gamma rays, infrared
- Toxic agents- Corticosteroids, miotics (long-acting anticholinesterases, e.g., echothiophate) (Option C)
- Obesity

Senile cataract is the most common type of acquired cataract and is also known as age-related cataract. The two types of senile cataracts are cortical (soft cataract)-75% and nuclear (hard cataract)-25%. In cortical cataract, the cuneiform type (70%) is the most common, and the cupuliform type is the most visually handicapping.

Solution to Question 10:

The most common type of congenital cataract is the blue-dot cataract.

Cataracta-punctata-caerulea, commonly known as blue dot cataract, is a prevalent type of congenital cataract. It typically develops during the first two decades of life. This condition is characterized by distinctive rounded bluish opacities located in the peripheral region of the adolescent nucleus and the deeper cortical layers of the lens. While these opacities are generally stationary and do not impair vision, larger ones may cause slight visual disturbance.

Solution to Question 11:

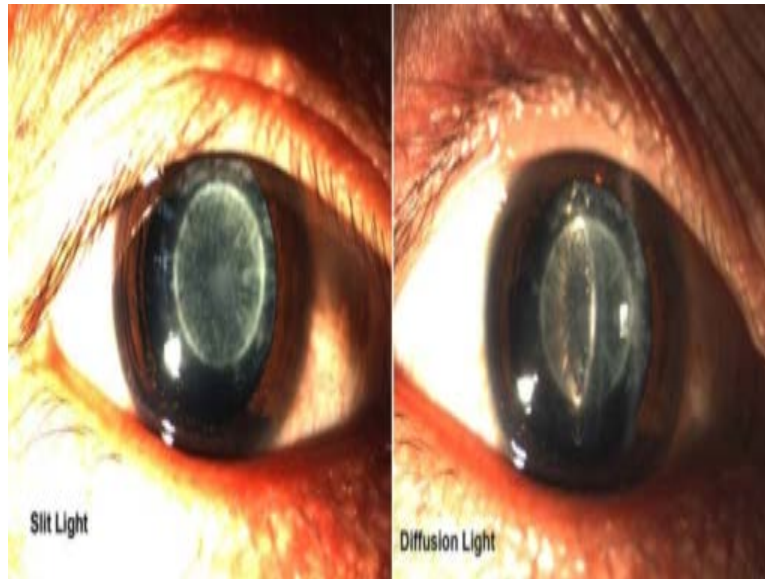
Lamellar or zonular cataract is the most common type of congenital cataract presenting with visual impairment.

The etiology may be either genetic or environmental in origin.

- Genetic- familial autosomal-dominant
- Environmental form is associated with:
 - Vitamin D deficiency
 - Hypocalcemia
 - Sometimes, maternal rubella infection contracted between the 7th and 8th week of gestation may also cause lamellar cataract.

The main mass of the lens internal and external to the zone of cataract is clear.

Small linear opacities like spokes of a wheel (riders) may be seen towards the equator.



Solution to Question 12:

The clinical scenario showing blueberry muffin rash, combined with patent ductus arteriosus in a baby girl, suggests a diagnosis of congenital rubella syndrome, which is typically associated with a progressive, pearly white nuclear cataract.

Ocular effects of rubella:

- Microphthalmia
- Rubella keratitis
- Angle-closure glaucoma
- Nuclear pearly cataract
- Salt and pepper fundus (a most common ocular feature of rubella) due to diffuse chorioretinitis.

Solution to Question 13:

The image above characteristically depicts a Christmas tree cataract, featuring multiple refractile colors commonly observed in myotonic dystrophy.

Myotonic dystrophy is an autosomal dominant disorder characterised by difficulty relaxing skeletal muscles after contraction. The initial symptoms typically manifest during adolescence or early adulthood.

Cataracts are an early and prominent feature in 90% of patients with myotonic dystrophy. They typically present as fine, dust-like opacities interspersed with iridescent, polychromatic spots in the cortical layer beneath the anterior and posterior lens capsule, resembling a Christmas tree. These cataracts may remain stable or progress.

Surgical outcomes are generally favourable.

Solution to Question 14:

The above image represents an advanced stage of cataract progression characteristic of a Morgagnian hyper-mature cataract.

Morgagnian hyper-mature cataracts are characterized by liquefaction of the entire cortex. The lens is converted into a bag of milky fluid. The small, brownish nucleus settles at the bottom, its position altering with a change in the position of the head.

Senile cataract is the most common type of acquired cataract and is also known as age-related cataract. The two types of senile cataracts are cortical (soft cataract)-75% and nuclear (hard cataract)-25%.

Cortical cataract is further divided into the cuneiform and cupuliform types.

Nuclear cataracts: These may be pigmented due to the deposition of urochrome pigments. These are cataracta brunescens (brown), cataracta nigra (black), cataracta rubra (red).

Cuneiform(70%)	Posterior subcapsular/Cupuliform(5%)
Radial spoke-like opacity starting in the periphery and extending centrally	Central saucer-shaped opacity in the posterior cortex just below the capsule
Late visual loss	Early loss of visual acuity; most visually handicapping senile cataract

Solution to Question 15:

Polychromatic luster is seen in complicated cataracts.

Due to disturbances in nutrition of the lens due to inflammatory/degenerative disease of other parts of the eye like iridocyclitis and ciliary body tumors. In inflammations/degenerations affecting the posterior segment, a characteristic opacification is seen in the posterior part of the cortex in the axial region (posterior subcapsular cataract).

These cataracts present with a breadcrumb appearance and polychromatic luster. Even in the early stages, vision is impaired.

Solution to Question 16:

Patients taking dexamethasone (corticosteroid) are prone to drug-induced cataracts, predominantly presenting as discoid posterior subcapsular cataracts.

Drugs causing cataracts:

Other options:

Option A: Zoledronate, a bisphosphonate drug can cause orbital inflammation, uveitis, and scleritis.

Option B: Hydroxychloroquine, a drug used in malaria and rheumatoid arthritis, does not cause cataracts, chloroquine does. It can rarely cause retinal toxicity.

Option D: Ethambutol is used in anti-tubercular therapy and its use has been shown to cause optic neuritis.

Drugs	Type of cataract
Corticosteroids	Discoid, posterior subcapsular
Anticholinesterases	Anterior subcapsular
Chlorpromazine	Yellow-brown anterior capsular granules/stellate/anterior polar
Busulfan	Posterior subcapsular
Chloroquine	White, flaky, posterior subcapsular
Amiodarone	Anterior subcapsular

Solution to Question 17:

The given picture of a slit-lamp biomicroscope examination depicts the anterior lenticonus, which is associated with Alport syndrome.

In Alport's syndrome, both anterior and posterior lenticonus are seen, but anterior lenticonus is more common. The diagnosis of lenticonus is typically confirmed through slit-lamp examination.

Note: Lowe's syndrome is an oculo-cerebro-renal syndrome characterized by congenital cataract, glaucoma, and posterior lenticonus

Solution to Question 18:

The phenomenon described is the second sight of old age, seen in nuclear cataracts.

Due to nuclear sclerosis, there is an increase in the refractive index, causing progressive myopia. A patient with presbyopia may be able to read again, without the aid of spectacles. This improvement in vision is termed second sight.

Solution to Question 19:

The typical appearance of a diabetic cataract is a snowflake cataract.

Hyperglycemia is reflected in a high level of glucose in the aqueous humor, which diffuses into the lens. Here, glucose is metabolized into sorbitol, which accumulates within the lens, resulting in secondary osmotic overhydration.



Solution to Question 20:

The inability to see objects in bright light is not a feature of cuneiform cataract, as the lens opacity is peripheral, allowing visibility in bright light during miosis.

The loss of the ability to see objects in bright light is a feature of posterior subcapsular cataract (visually handicapping), cupuliform cataract and nuclear cataract as these present with central opacities.

When a person is exposed to bright light, the pupil undergoes miosis (constriction). These central opacities come to lie directly in the path of axial rays, allowing no transmission of rays. Therefore, there is an inability to see objects in bright light. However, in dim light, when there is mydriasis (dilatation of the pupil), some rays may pass around the opacity, leading to an improvement in dim-light vision.

Furthermore, the complicated cataract, steroid-induced cataract, radiation-induced mainly present with a posterior subcapsular cataract and therefore have a similar clinical picture.

Solution to Question 21:

Cataract surgery with an intraocular lens (IOL) implantation is the treatment of choice in congenital cataracts.

It involves an extracapsular cataract extraction (ECCE) with primary posterior capsulorhexis and anterior vitrectomy. After removal of the cataractous lens, a posterior chamber intraocular lens (PCIOL) is implanted.

The critical period for the development of the fixation reflex in both unilateral and bilateral visual deprivation disorders is 2–4 months of age. Thus, any cataract dense enough to impair vision must be dealt with before this age, and the earliest possible time is preferred.

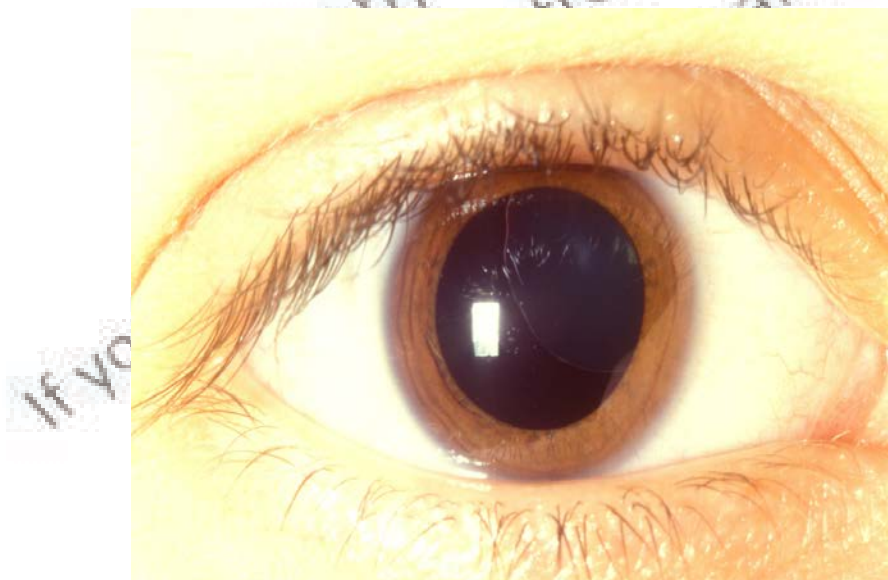
- Bilateral dense cataract- early surgery by 4-6 weeks of age
- Unilateral dense cataract or involving visual axis- urgent surgery within days

Intracapsular cataract extraction (ICCE) is contraindicated in children.

Solution to Question 22:

The clinical vignette of a girl with tall, slim stature, an aortic arch aneurysm, Steinberg sign (positive thumb sign), and similar family history suggest a diagnosis of Marfan's syndrome. It is most commonly associated with a superotemporal subluxation of the lens.

The image below shows a subluxated lens in a patient with Marfan's syndrome.



Congenital subluxation of the lens is called ectopia lentis.

Types of ectopia lentis are as follows:

- Simple type
- Ectopia lentis et pupillae (pupil also displaced in opposite direction)
- Ectopia lentis associated with syndromes
- Marfan's syndrome- superotemporal subluxation of lens
- Homocystinuria- inferonasal subluxation of the lens
- Weill–Marchesani- downward, forward subluxation of the lens.

- Ehler–Danlos syndrome

Solution to Question 23:

A specific pattern of cataract is not seen in leprosy.

A specific pattern of cataract is seen in the following:

- Juvenile diabetes: Snowflake cataract
- Myotonic dystrophy: Christmas tree cataract
- Wilson's disease: Sunflower cataract

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Lens - Cataract Surgery, Complications and IOLs

Question 1:

A child has congenital cataract involving the visual axis, which was detected at birth. When should the child be operated up on?

- a) At 4-6 weeks of age
- b) At 5 months of age
- c) At 1 year of age, when the globe attains normal size
- d) At 4 years, when ocular and orbital growth is complete

Question 2:

In which of the following cataract surgeries is a self-sealing sclerocorneal tunnel incision used?

- a) Intracapsular cataract extraction (ICCE)
- b) Conventional extracapsular cataract extraction (ECCE)
- c) Small incision cataract surgery (SICS)
- d) Anterior capsulotomy

Question 3:

In which of the following patients is an intracapsular cataract extraction indicated?

- a) 2-month-old with congenital cataract
- b) 78-year-old with Morgagnian cataract
- c) 25-year-old with a subluxated lens
- d) 60-year-old with secondary cataract

Question 4:

What is the length of the incision used in phacoemulsification?

- a) 10 - 12 mm
- b) 2 - 2.5 mm

- c) 12 - 14 mm
- d) 4 - 6 mm

Question 5:

What is the most common late complication following cataract surgery?

- a) Cystoid macular edema
- b) Glaucoma
- c) Posterior capsule opacification
- d) Uveitis

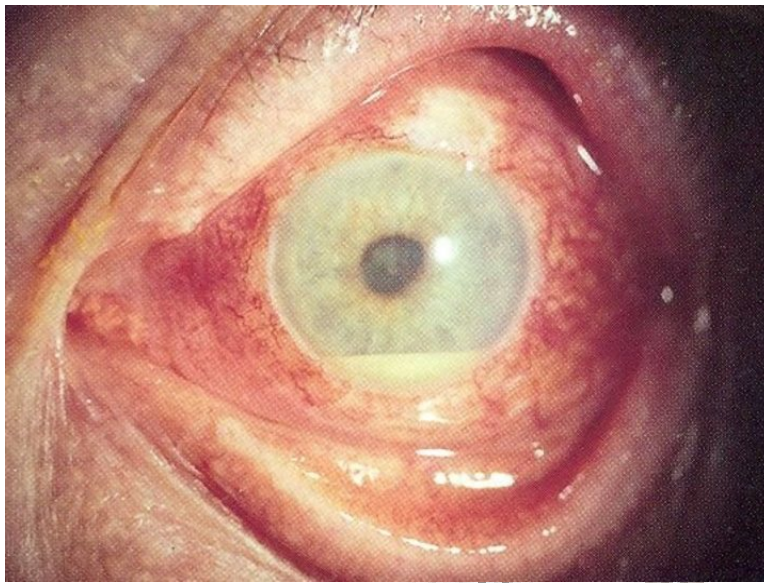
Question 6:

In a patient undergoing cataract surgery, which of the following is the most important strategy in the prevention of postoperative endophthalmitis?

- a) Preoperative preparation with povidone iodine
- b) One week antibiotic therapy prior to surgery
- c) Trimming of eyelashes
- d) Use of intravitreal antibiotics

Question 7:

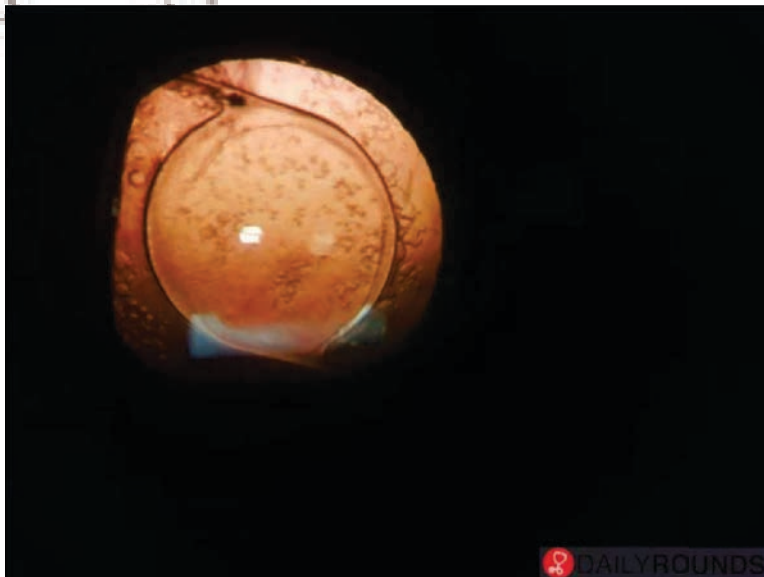
A patient, 36 hours after cataract surgery, presents with complaints of pain and visual disturbance. His eye is as shown below. What is the management of this condition?



- a) Intravenous steroids
- b) Intravenous antibiotics
- c) Bandage and padding of the eye
- d) Intravitreal antibiotics

Question 8:

A 62-year-old presented with diminished vision 1 year after cataract surgery. The findings are shown below. What is the most likely diagnosis?



- a) After cataract

- b) Irvine-Gass syndrome
- c) End ophthalmitis
- d) Uveitis-glaucoma-hyphema syndrome

Question 9:

What is the treatment of choice for after cataract?

- a) Intra-capsular cataract extraction
- b) Intravitreal steroids
- c) Anterior vitrectomy
- d) Nd:YAG laser posterior capsulotomy

Question 10:

What is the cause of late-onset endophthalmitis after cataract surgery?

- a) Staphylococcus epidermidis
- b) Pseudomonas
- c) Streptococcus pyogenes
- d) Propionibacterium acne

Question 11:

An elderly male residing in an old-age home is brought to the hospital with senile cataracts in both eyes. He says he had difficulty in seeing objects for a few years, but never went to the clinic. He is posted for surgery in 3 weeks. Which of the following is a complication is he unlikely to develop in the 3 weeks until his surgery?

- a) Anaphylactic uveitis
- b) Open angle glaucoma
- c) Lens subluxation
- d) Hypopyon

Question 12:

What is the best irrigating fluid for cataract surgery?

- a) Normal saline
- b) Ringer lactate
- c) Balanced salt solution
- d) Balanced salt solution plus glutathione

Question 13:

How many weeks following surgery are spectacles prescribed?

- a) 6 weeks
- b) 10 weeks
- c) 12 weeks
- d) 14 weeks

Question 14:

A 72-year-old diabetic who underwent uncomplicated cataract extraction with PCIOL implant 15 days back presents with painless, decreased vision in the operated eye. His visual acuity is 6/12 and does not improve with a pinhole. On examination, the anterior chamber is deep, and PCIOL is well-centered in the capsular bag. Dilated fundus exam reveals clear media, and optic nerve with a cup-to-disc ratio of 0.4, a few peripheral microaneurysms, and a dull foveal reflex. The rest of the fundus exam is unremarkable. Which test would be of least benefit?

- a) Optical coherence tomography
- b) Fluorescein angiography
- c) Gonioscopy
- d) Amsler's grid

Question 15:

Two months after undergoing cataract surgery, a 45-year-old man is found to have after cataract. He is planned for an Nd:YAG laser posterior capsulotomy. What is the principle behind the use of this laser?

- a) Photoablation
- b) Photodisruption
- c) Photocoagulation
- d) Photoemulsification

Question 16:

Which of the following complaints in a patient presenting to your OPD would prompt you to perform a Fincham's test?

- a) Colored halos
- b) Polyopia
- c) Scotoma
- d) Flashes of light

Question 17:

As part of the pre-operative evaluation of a patient posted for cataract surgery, you are asked to calculate the power of the IOL to be implanted. What investigation do you perform?

- a) Biometry
- b) Pachymetry
- c) Tonography
- d) Tonometry

Question 18:

In the SRK formula for intraocular lens power calculation ' $E=A-2.5L-0.9K$ ', the value of L is estimated using _____

- a) Slit lamp biomicroscopy
- b) Humphrey field analyser
- c) Keratometry
- d) A-scan ultrasonographic biometry

Question 19:

What is a rigid intraocular lens made of?

- a) Polymethyl methacrylate (PMMA)
- b) Silicone
- c) Hydrogel

d) Collamer

Question 20:

A patient, on follow-up after cataract surgery, has the lens shown below. What is the type of lens implanted in the patient?



- a) Kelman multiflex anterior chamber IOL
- b) Modified C loop posterior chamber IOL
- c) Iris claw lens
- d) Four-loop posterior chamber IOL

Question 21:

An 80-year-old man is posted for cataract surgery. The given lens is chosen to be used for implantation. Identify the lens.



- a) Kelman multiflex anterior chamber IOL
- b) Modified C loop posterior chamber IOL
- c) Singh and Worst's iris claw lens
- d) Four-loop posterior chamber IOL

Question 22:

Identify the intraocular lens shown below.



- a) Kelman's AC-IOL
- b) Diffractive IOL

- c) Accommodating IOL
- d) Toric IOL

Answer Key

Question No.	Correct Option
1	a
2	c
3	c
4	b
5	c
6	a
7	d
8	a
9	d
10	d
11	d
12	d
13	a
14	c
15	b
16	a
17	a
18	d
19	a
20	c
21	b
22	d

Detailed Explanations

Solution to Question 1:

The child should be operated at 4-6 weeks of age.

When a child has unilateral or bilateral cataract impairing vision (involving the visual axis), the cataract surgery must be performed before 2-4 months of age. This is because 2-4 months is the critical period for visual maturation and development of the fixation reflex in children. A unilateral cataract is more dangerous than a bilateral cataract because the risk of amblyopia is greater. Under-correction of intraocular lens (IOL) power is needed in children.

In children < 2 years of age:

- In unilateral cases, primary implantation is indicated as soon as the patient is fit for anesthesia, ideally between 2 and 3 months of age. The power is undercorrected by 20%.
- The earlier the surgery is done, the better the chance that deep amblyopia can be overcome.
- After the cataract has been aspirated, an elective posterior capsulectomy–anterior vitrectomy is performed. The posterior chamber IOL is inserted through a scleral tunnel, which is prepared in advance.
- Sulcus implantation is indicated in neonates and infants less than 1 year of age.

In children between 2 and 8 years of age:

- An IOL is implanted but power is undercorrected by 10%.
- An in-the-bag IOL is chosen for children above 1 year of age because they are less likely to need an IOL exchange.

Beyond 8 years, IOL power is the same as the calculated value.

Solution to Question 2:

The self-sealing sclerocorneal tunnel incision is made in small-incision cataract surgery (SICS).

SICS is a type of extracapsular cataract extraction techniques (ECCE). In these techniques, the major portion of the anterior capsule with the epithelium, nucleus, and cortex are removed, leaving behind the intact posterior capsule.

Note: Newer cataract surgeries include femtosecond laser-assisted cataract surgery (FLACS), micro-incision cataract surgery (MICS) and Phacovit. FLACS is a microincision cataract surgery or a micro-phacoemulsification where a femtosecond laser is used to make the corneal incisions, perform capsulorhexis and fragment the lens. MICS refers to the phacoemulsification techniques that can be performed through a microincision (<2 mm). Phacovit is the term for the bimanual micro-phacoemulsification done with the needle incision technique through an incision of <1mm.

Solution to Question 3:

At present, the only indication of intracapsular cataract extraction (ICCE) is markedly subluxated and dislocated lens.

In ICCE, the entire cataractous lens along with the intact capsule is removed by rupturing the zonules. Therefore, weak and degenerated zonules are a prerequisite for this method. Methods of

lens delivery include cryoextraction (best method), forceps extraction (Arruga forceps), wire vectis method and Indian Smith method. ICCE is no longer used for cataract treatment due to the large incision required, inability to implant an IOL and a high rate of complications like vitreous loss, retinal detachment and cystoid macular edema.

Solution to Question 4:

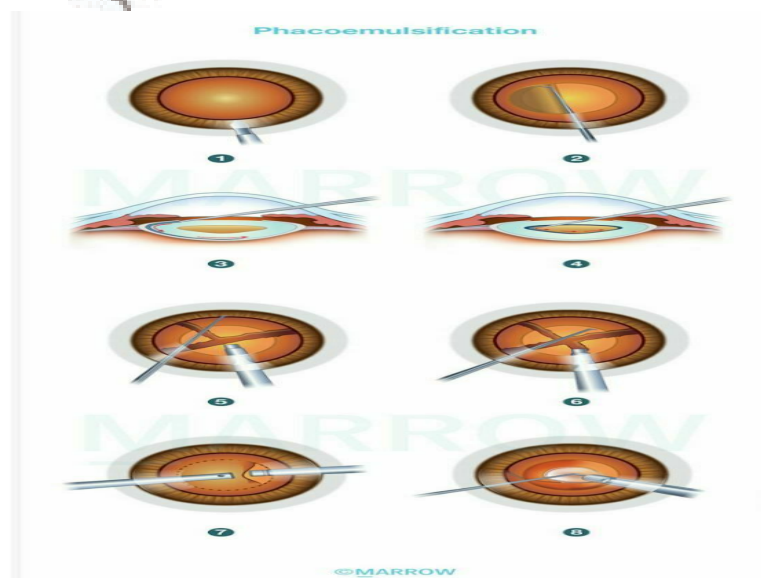
The length of the incision in phacoemulsification varies from 1.5 to 2.4 mm. The option closest to this answer is B, i.e., 2 to 2.5 mm.

In phacoemulsification, the nucleus of the lens is emulsified and aspired. This is done by the phacoemulsifier. It acts through a hollow 1-mm titanium needle, which vibrates by at an ultrasonic speed of 40000 times a second. The steps of phacoemulsification are given below:

- Step 1 - Small clear corneal incision
- Step 2 - Continuous curvilinear capsulorhexis
- Step 3 - Hydro dissection
- Step 4 - Hydro delineation is done in some cases.
- Steps 5 and 6 - Emulsification and aspiration of the nucleus
- Step 7 - Removal of the cortical lens matter using an irrigation-aspiration technique
- Step 8 - Implantation of a foldable lens

Length of the incision in various cataract surgeries:

- Phacoemulsification- 1.5–2.4 mm.
- Micro-phacoemulsification- 1–1.5 mm.
- Intracapsular cataract extraction (ICCE)- 12–14 mm.
- Extracapsular cataract extraction (ECCE)- 8–12 mm.



Solution to Question 5:

The most common late complication following cataract surgery is posterior capsule opacification. Posterior capsule opacification, known as after or secondary cataract, is the most common late postoperative complication. It may be caused by residual opaque lens matter or proliferation of left out anterior epithelial cells.

Types of after-cataract



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Solution to Question 6:

The most important factor in the prevention of postoperative endophthalmitis is preoperative preparation with povidone-iodine.

Postoperative endophthalmitis is most commonly caused by *Staphylococcus epidermidis*. Flora of the eyelids and conjunctiva are the most frequent source. Other potential sources include contaminated solutions and instruments, environmental air, and the surgeon and other operating room personnel.

Prophylaxis for postoperative endophthalmitis includes:

- Instillation of 5% povidone-iodine into the conjunctival fornices and leaving this undisturbed for at least 3 minutes prior to surgery. This is the most important factor.
- Scrupulous preparation of the surgical site
- Treatment of pre-existing infections such as blepharitis
- Antibiotic prophylaxis
- Intracameral cefuroxime (1 mg in 0.1 mL) injected into the anterior chamber (AC) at the end of surgery.

- Post-operative subconjunctival injection can achieve bactericidal levels in the AC for at least 1–2 hours.
- Preoperative topical fluoroquinolone antibiotics are frequently given in regimens from 1 hour to 3 days before surgery

Patients present with ocular pain, diminished vision, circumciliary congestion and hypopyon. The differential diagnosis for postoperative endophthalmitis includes retained lens material in the AC, vitreous hemorrhage, postoperative uveitis and toxic reaction to the use of inappropriate or contaminated irrigating fluid or viscoelastic.

Solution to Question 7:

A post cataract surgery patient presenting within 36 hours of surgery with inflammation, congestion and hypopyon as seen in the image, is suggestive of acute bacterial endophthalmitis. The treatment of acute bacterial endophthalmitis is intravitreal antibiotics.

Endophthalmitis is an intraocular inflammation affecting the inner structures of the eyeball. Acute postoperative bacterial endophthalmitis occurs within 1-7 days after cataract surgery and the most important factor to prevent it is preoperative administration of 5% povidone iodine. The cardinal features of endophthalmitis are pain, swelling of the lid, and a decrease in vision. The patient classically shows inflammation of the eye, corneal edema and a hypopyon due to the pouring of exudates in the vitreous cavity, anterior chamber, and posterior chamber.

Once the endophthalmitis sets in it is important to treat all the cases empirically with a combination of antibiotics and steroids from the beginning. The various drugs, routes and surgical management used to treat endophthalmitis are as follows:

- The mainstay of treatment is intravitreal injections of antibiotics.
- First choice: Vancomycin 1 mg in 0.1 mL plus ceftazidime 2.25mg in 0.1ml.
- Second choice: Vancomycin 1 mg in 0.1 mL plus amikacin 0.4mg in 0.1ml.
- If there is no improvement, a repeat intravitreal injection should be given after 48 hours
- Topical antibiotics are started immediately and used frequently.
- Vancomycin (50mg/ml) or cefazolin (50mg/ml)
- Amikacin (20mg/ml) or tobramycin
- Systemic antibiotics have a limited role in the treatment but are commonly used in practice.
- Steroid therapy includes:
 - Intravitreal injection of dexamethasone (0.4mg in 0.1ml) which is started along with antibiotics.
 - Topical dexamethasone (0.1%)
 - Oral steroids such as prednisolone are started after 24 hours of intensive antibiotic therapy.
- Supportive therapy includes:
 - Cycloplegics such as atropine 1%

- Anti-glaucoma drugs such as oral acetazolamide or timolol in patients with raised intraocular pressure
- Pars plana vitrectomy is done if the patient does not improve with the above therapy in 48-72 hours or when the visual acuity is reduced to hand movements near the eye

Other options:

Option A: Steroids are given in endophthalmitis to reduce the inflammation of the tissues. The preferred route of administration is either by intravitreal injections, topical concentrations or oral route. Intravenous route (option A) of administration of steroids is not used in endophthalmitis.

Option B: Systemic administration (intravenous) of antibiotics have a limited role in the treatment but are commonly used in practice as an add on.

Solution to Question 8:

Diminished vision 1 year after cataract surgery, supported with the image of vacuolated clusters seen behind IOL is suggestive of Elschnig's pearls. It is a type of after cataract.

After cataract, or secondary cataract or posterior capsular opacification, is a late postoperative complication that follows extracapsular extraction of the lens. It is the most common late postoperative complication. It may be caused by residual opaque lens matter or proliferation of left out anterior epithelial cells. Different types of after cataract include:

- Elschnig's pearls: The subcapsular epithelial cells proliferate and develop into vacuolated soap bubble like clusters along the posterior capsule.
- Soemmering's ring: A peripheral dense ring is formed behind the iris and is enclosed between layers of the capsule.
- Dense membranous after-cataract: It is formed in the pupillary area impairing the vision.
- Thin membranous after-cataract: It is thickened posterior capsule.

Types of after-cataract



Dense, membranous after-cataract



Soemmering's ring

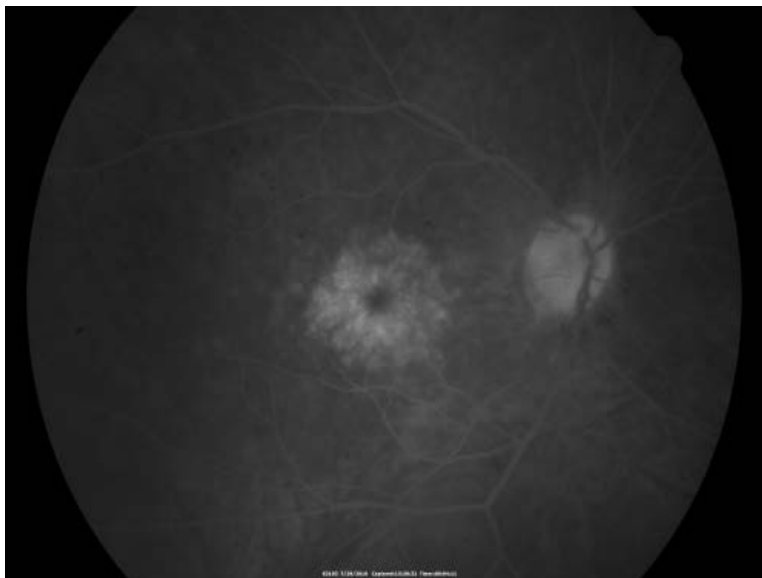


Elschnig's pearls

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Treatment of after cataract is by Nd-YAG laser capsulotomy of the posterior capsule, which works on the principle of photodisruption, or surgical posterior capsulotomy.

Option B: Irvine–Gass syndrome is cystoid macular edema (CME) that develops in post-cataract surgery patients. Though it is mostly clinically insignificant, some cases can present with blurring of vision, one to three months after surgery. In CME, on fundoscopy a honeycomb appearance and on fundus fluorescein angiography(FFA) flower petal pattern can be seen. The below image depicts the flower petal pattern seen on FFA.



Option C: Post operative endophthalmitis presents as ocular pain, diminished vision and hypopyon within months to years after the surgery. Fungal endophthalmitis can rarely occur with puff ball exudates in the vitreous cavity.

Option D: UGH syndrome is where uveitis, glaucoma and hyphema occurs simultaneously. It is seen commonly with rigid anterior chamber intraocular lens (IOL) implantation.

Solution to Question 9:

Nd:YAG laser is the treatment of choice for after cataract.

Thin membranous after-cataract, thickened posterior capsule and Elschnig's pearls are treated with Nd:YAG laser capsulotomy. Dense membranous after-cataract requires capsulotomy using a vitreous cutter. Sommering's ring, if the central posterior capsule is clear, requires no treatment.

Other options:

Option A: Intra-capsular cataract extraction (ICCE) is currently used for the treatment of markedly subluxated and dislocated lens.

Option B: Intravitreal steroids may be used for the treatment of acute endophthalmitis, an early postoperative complication of cataract surgery.

Option C: Anterior vitrectomy is performed for the treatment of cystoid macular edema with vitreous incarceration. This is a late post-cataract surgery complication.

Solution to Question 10:

Late-onset endophthalmitis after cataract surgery is caused by *Propionibacterium acnes*.

Late or delayed-onset endophthalmitis following cataract surgery develops when an organism of low virulence such as *P. acnes*, becomes trapped within the capsular bag (saccular endophthalmitis). Organisms can become sequestered within macrophages, protected from eradication but with continued expression of bacterial antigens. Patients present with symptoms between 4 weeks to years (mean 9 months) postoperatively following an uneventful cataract surgery.

Solution to Question 11:

Hypopyon is not a complication of senile cataracts.

Hypermaturation, or senile cataracts, can lead to the following complications.

- Phacoanaphylactic uveitis: The lens proteins may leak into the anterior chamber. They may act as an antigen and induce antigen-antibody reaction, leading to phacoanaphylactic uveitis.
- Lens-induced glaucoma:
- Phacomorphic glaucoma: Most common, seen in intumescent cataract. The lens swells and causes secondary angle closure glaucoma.
- Phacolytic glaucoma: Seen in hypermature cataract. The leaked lens proteins are engulfed by macrophages, which clog the trabecular meshwork and cause secondary open-angle glaucoma.
- Phacotopic glaucoma: Seen in hypermature cataract. The lens may subluxate or dislocate and block the angle of the anterior chamber.
- Subluxation or dislocation of the lens: It may occur due to degeneration of zonules in the hypermature stage.

Solution to Question 12:

The best irrigating fluid for cataract surgery is balanced salt solution plus glutathione (BSS plus).

In the early days of phacoemulsification, the only irrigating solutions available were normal saline, plasmalyte, and lactated Ringer's solution. Their main adverse effects were endothelial cell toxicity. The buffer in BSS Plus is bicarbonate, which is an improvement over the sodium acetate and citrate buffers in BSS.

Solution to Question 13:

Spectacles are prescribed 6 weeks after cataract surgery after the course of steroids is over. It is the time required for the incision to heal. Then visual acuity is checked and glasses are prescribed.

Solution to Question 14:

In the above case scenario, gonioscopy would be the least helpful.

A painless decrease in vision after cataract surgery could be due to:

- Posterior capsular opacification (most commonly, after 1-3 months)
- Cystoid macular edema (early)
- Rhegmatogenous retinal detachment (delayed)

In the given question, the media is clear (rules out posterior capsular opacification). Also, there are no symptoms nor signs of retinal detachment (RD), hence it is ruled out. Decreased vision not improving with a pinhole in the presence of clear media is attributable to macular pathology. Also, there is a dull foveal reflex. The diagnosis in this patient is cystoid macular edema (CME).

The incidence of clinical CME is about 1% to 2% following uncomplicated cataract surgeries. The incidence of angiographic evidence of CME is 10 times greater at around 10% to 20%.

Optical coherence tomography (option A) is useful in identifying cystoid macular edema and evaluating the vitreomacular interface. It is also useful in the measurement of retinal thickening, thus allowing the assessment of the therapeutic response. On fluorescein angiography (option B), the classic petaloid leakage in the fovea is seen in the case of CME. Amsler's grid (option D) is a macular function test.

Hence, the least useful in this scenario is gonioscopy.

Solution to Question 15:

The principle of the Nd:YAG laser (1064 nm) is photodisruption.

Nd:YAG laser capsulotomy is used for treatment of thin membranous after-cataract, posterior capsular opacification and Elschnig's pearls. The plasma produced at the laser target expands and disrupts the tissues, creating an opening. Double frequency Nd:YAG laser is used in posterior segment photocoagulation.

Solution to Question 16:

Fincham's test is used to distinguish whether the colored halos are due to cataract or glaucoma.

Colored halos are rainbow-colored rings seen around lights at nighttime. It may be caused by cataracts, acute congestive glaucoma and mucopurulent conjunctivitis. These conditions lead to prismatic disruption of light.

Solution to Question 17:

The power of the intraocular lens (IOL) to be implanted after cataract surgery is calculated by a biometer and the procedure is termed as biometry.

Biometry consists of keratometry and axial length measurement. The most common method of determining IOL power uses a regression formula called modified Sanders–Retzlaff–Kraff (SRK) formula, which is given below:

$$E = A - 2.5L - 0.9K$$

Where,

- E is the emmetropic power designated for the eye
- A is a predetermined constant, which is specific for each intraocular lens type
- L is the axial length of the eyeball in millimeter, determined by using the "A" scan ultrasonography
- K is the average corneal curvature, which is determined by keratometry.

The image given below shows a biometer.



Solution to Question 18:

In the SRK formula, the value of L is calculated using the A-scan ultrasonographic biometry.

The most widely used formula for intraocular lens power calculation is the modified Sanders-Retzlaff-Kraff (SRK) formula, given by:

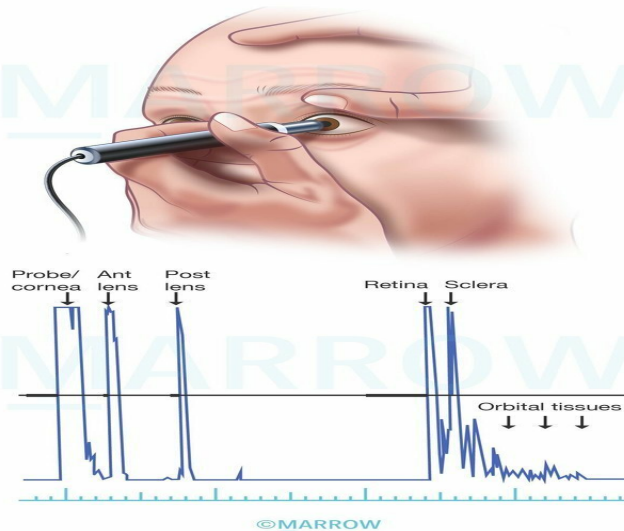
$$E = A - 2.5L - 0.9K$$

Where,

- E: Emmetropic power designated for the eye
- A: Predetermine constant which is specific for each intraocular lens type.

- L: Axial length of eyeball in millimeters, determined using the A scan ultrasonography.
- K: Average corneal curvature determined by keratometry.

Given below is an image of A scan ultrasonography.



Solution to Question 19:

Rigid IOLs are made of polymethyl methacrylate (PMMA).

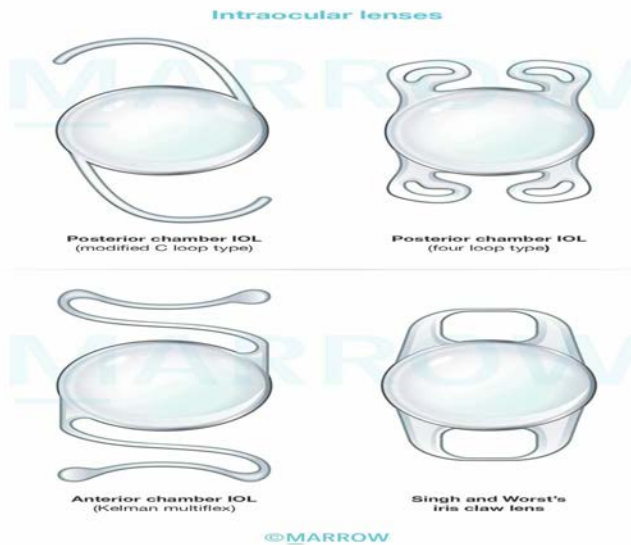
Depending on the type of material of manufacturing, 3 types of posterior chamber intraocular lens (PCIOLs) are available:

- Rigid IOLs are made entirely from polymethyl methacrylate (PMMA).
- Foldable IOLs are made of silicone, hydrogel, acrylic, and Collamer. These are implanted through a small incision.
- Rollable IOLs are ultra-thin IOLs made of acrylic or hydrogel. These are implanted through a 1-mm incision after micro-incision cataract surgery technique.

Solution to Question 20:

The image depicts Singh and Worst's iris claw lens.

This lens is an iris-supported lens, which is fixed with the help of sutures, loops or claws. They are associated with a high incidence of postoperative complications and are thus only used if posterior chamber IOLs cannot be implanted.



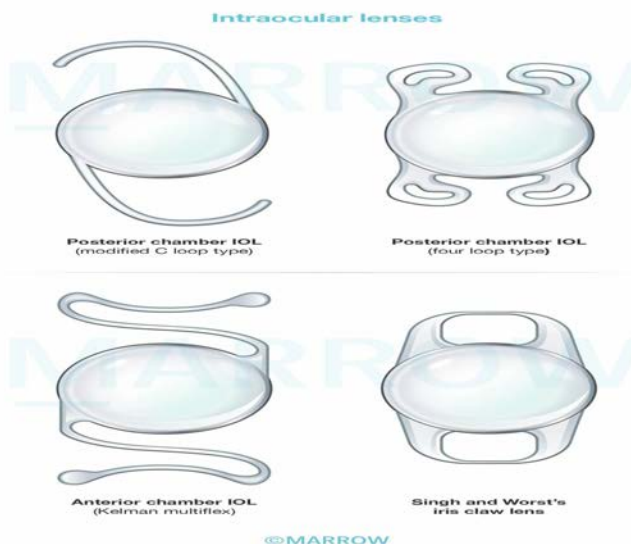
Note: Kelman multiflex type of anterior chamber intraocular lens (ACIOL) is the most commonly used ACIOL. Anterior chamber IOLs lie entirely in front of the iris and are supported in the angle of the anterior chamber. These are associated with a higher incidence of bullous keratopathy. Thus, they are currently used only for correction of aphakia in cases where posterior chamber IOLs cannot be used (eg: rupture of posterior capsule).

Solution to Question 21:

The above image depicts a modified C-loop type of posterior chamber intraocular lens (PCIOL).

A PCIOL rests entirely behind the iris. They may be supported and fixed by capsular bag fixation of PCIOL (most ideal method), ciliary sulcus fixation or scleral fixation.

The PCIOL consists of a central part overlying the optic axis is called the optic and the peripheral arms used for placement and stabilization called the haptics.



Solution to Question 22:

The given image is of a toric IOL used to correct pre-existing astigmatism. The dotted markings on the optic of the IOL indicate the axis of IOL, and it is aligned to coincide with the steep axis of the patient's eye.

For every, one degree of misalignment, about 3 percent of the lens cylinder power is lost.

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you may have been scammed.

Glaucoma

Question 1:

What is the normal aqueous production rate?

- a) 2 mL/min
- b) 5 mL/ min
- c) 2 μ L/ min
- d) 5 μ L/ min

Question 2:

What is the critical angle of the air-cornea interface?

- a) 46 degrees
- b) 64 degrees
- c) 24 degrees
- d) 36 degrees

Question 3:

Which statement regarding the depth of the anterior chamber is false?

- a) Depth is less in women than men
- b) Depth increases with age
- c) Depth is more in myopes
- d) Depth is less in hypermetropes

Question 4:

The child in the picture has large corneas, photophobia, and lacrimation. What is the likely diagnosis?



- a) Congenital glaucoma
- b) Retinoblastoma
- c) Congenital hereditary endothelial dystrophy
- d) Megalocornea

Question 5:

A neonate is brought with complaints of photophobia and watering. Clinical examination shows normal tear passages and clear but large cornea. Investigation reveals Haab's striae seen in the eye. What is the most likely diagnosis?

- a) Buphthalmos
- b) Keratoconus
- c) Trachoma
- d) Keratoglobus

Question 6:

A patient presents to the clinic for a change of her presbyopic glasses. This is the third time that she is coming in 2 months. What is the likely underlying diagnosis in this patient?

- a) Senile cataract
- b) Retinal detachment
- c) Intumescent cataract

d) Open-angle glaucoma

Question 7:

Normal cup-to-disc ratio is up to_____.

- a) 0.6
- b) 0.4
- c) 0.5
- d) 0.3

Question 8:

A 61-year-old man, who is a known case of diabetes mellitus, presents with headache and eye pain. He also reports difficulty with driving at night. Examination showed a lamellar dot sign. What is the likely diagnosis?

- a) Uveitis
- b) Glaucoma
- c) Retinal detachment
- d) ARMD

Question 9:

What is the last visual field to be affected in glaucoma?

- a) Nasal
- b) Inferior
- c) Superior
- d) Temporal

Question 10:

Which of the following is a late visual field defect in open-angle glaucoma?

- a) Siedel scotoma
- b) Bjerrum scotoma

- c) Paracentral scotoma
- d) Baring of blind spot

Question 11:

A 70-year-old patient presents with progressive deterioration of vision. The pupillary reaction is sluggish and the IOP is elevated. Fundoscopy shows a large and deep cup and the visual field reveals paracentral scotoma. What is the first line of treatment for this diagnosis?

- a) Timolol
- b) Pilocarpine
- c) Epinephrine
- d) Carbonic anhydrase inhibitor

Question 12:

A patient with a known case of MI on aspirin comes with shortness of breath. She has an active history of asthma and uses inhalational bronchodilators, her last attack was 7 months ago. She was recently prescribed medications for increased intraocular pressure. What is the likely cause of her current symptoms?

- a) Aspirin
- b) Betaxolol
- c) Latanoprost
- d) Timolol

Question 13:

As a first-year ophthalmology resident, you have been referred a case of glaucoma. He complains that his current medication is causing excessive drowsiness and is interfering with his work. Which of the following anti-glaucoma medications can cause drowsiness?

- a) Latanoprost
- b) Brimonidine
- c) Timolol
- d) Dorzolamide

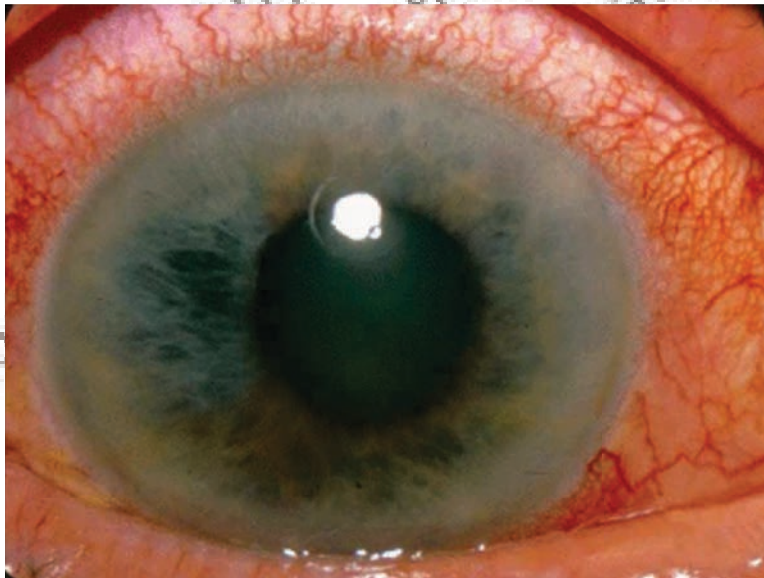
Question 14:

A 55-year-old female, who is a known case of sickle cell anemia, comes to the casualty with a history of severe eye pain, redness, and diminution of vision. On examination, the visual acuity is 6/60, and there is circumcorneal congestion, corneal edema, and a shallow anterior chamber. What is the drug of choice from the given options?

- a) Atropine ointment
- b) Intravenous mannitol
- c) Ciprofloxacin eye drops
- d) Oral acetazolamide

Question 15:

A 50-year-old female presents with an acute painful red-eye. An image of her eye is shown below. What is the most likely diagnosis?



- a) Acute retrobulbar neuritis
- b) Acute angle-closure glaucoma
- c) Acute anterior uveitis
- d) Severe keratoconjunctivitis

Question 16:

A 36-year-old woman develops pain in the eye after dark-room/prone-provocative test. Which of the following drugs should be avoided?

- a) Atropine
- b) Pilocarpine
- c) Acetazolamide
- d) Timolol

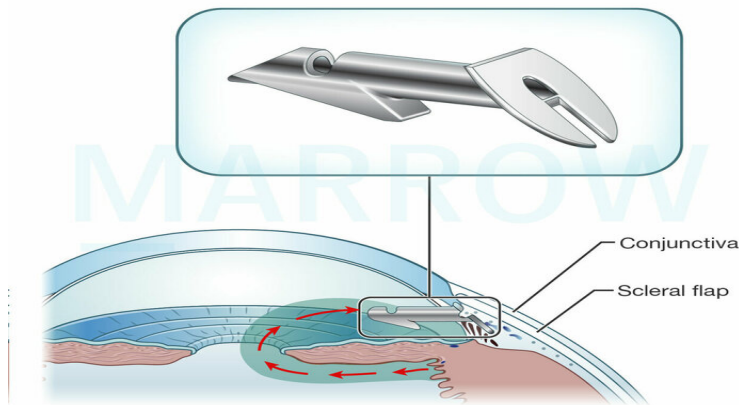
Question 17:

A young female presents with sudden onset severe pain in the left eye after watching a movie. She rushes to the hospital as there is a progressive loss of vision. She has ciliary injection and a shallow anterior chamber. The pupil is semi dilated and the cornea appears oedematous. What is the procedure of choice for this condition?

- a) Laser iridotomy
- b) Trabeculectomy
- c) Trabeculotomy
- d) Trabeculoplasty

Question 18:

A 50-year-old male presented with progressive blurred vision, photophobia, and eye pain. On examination, there was conjunctival redness, a mid-dilated pupil, and a shallow anterior chamber in the right eye. He has been treated both medically and surgically for the condition but showed no improvement. The following procedure incorporates a device to aid drainage. Identify the device.



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- a) Ex-Press™ implant
- b) Ahmed glaucoma valve
- c) Molteno implant
- d) Baerveldt implant

Question 19:

What is Glaukomflecken?

- a) Acute uveitis due to glaucoma
- b) Lens opacity due to glaucoma
- c) Retinal detachment due to glaucoma
- d) Corneal opacity due to glaucoma

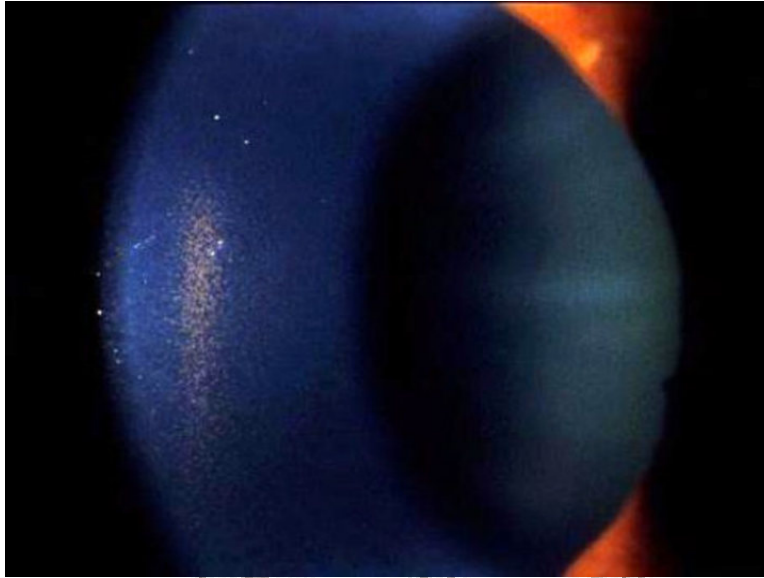
Question 20:

A 25-year-old patient presents with painless white eye with an intraocular pressure of 60 mmHg. What is the most likely diagnosis?

- a) Chronic papilledema
- b) Acute angle-closure glaucoma
- c) Glaucomatocyclitic crisis
- d) Acute anterior uveitis

Question 21:

An ophthalmologist evaluating a patient with glaucoma identifies the following finding on slit-lamp examination. What is the finding?



- a) Ferry line
- b) Fleischer ring
- c) Hudson-Stahli line
- d) Krukenberg spindles

Question 22:

Malignant glaucoma is seen in _____.

- a) After intra-ocular surgery
- b) Intraocular malignancy
- c) Trauma
- d) Central retinal arterial occlusion

Question 23:

What is iridocorneal endothelial syndrome associated with?

- a) Progressive atrophy of the iris stroma
- b) Bilateral stromal edema of the iris

- c) Primary angle-closure glaucoma
- d) Deposition of glycosaminoglycans in Descemet's membrane

Question 24:

What is seen in hypersecretory glaucoma?

- a) Marfans syndrome
- b) Hypertension
- c) Diabetes
- d) Epidemic dropsy

Question 25:

100-day glaucoma is seen in which of the following conditions?

- a) Central retinal vein occlusion
- b) Eale's disease
- c) Central retinal artery occlusion
- d) Steroid-induced glaucoma

Question 26:

Which of the following statements is false regarding the progression of ocular hypertension to primary open-angle glaucoma?

- a) Reducing IOP is effective in delaying the onset of POAG
- b) Thin cornea is a risk factor for POAG development
- c) Family history is not a risk factor for POAG development
- d) Increased cup disc ratio is a risk factor for POAG development

Question 27:

Identify the instrument given in the image below.



- a) Express implant
- b) Gonioscope
- c) Baerveldt implant
- d) Ozurdex implant

Question 28:

What is the drug of choice for inverse glaucoma?

- a) Pilocarpine
- b) Prostaglandin analogues
- c) Timolol
- d) Atropine

Question 29:

An elderly male who is a known case of diabetes presents with blurry vision near the center of his field of vision. He also informs that colours appear washed out or faded. His IOP was found to be elevated. Which of the following drugs should not be used to control IOP in this diabetic patient?

- a) Prostaglandin analogue
- b) Dorzolamide
- c) Pilocarpine

- d) Alpha agonists

Question 30:

In a 55-year-old patient presenting with secondary open-angle glaucoma, what is the most likely etiology?

- a) Phacolytic glaucoma
- b) Pseudo exfoliation syndrome
- c) Pigment dispersion syndrome
- d) Inflammatory glaucoma

Question 31:

Which of the following drugs increases the uveoscleral outflow?

- a) Timolol
- b) Latanoprost
- c) Pilocarpine
- d) Carbonic anhydrase inhibitor

Question 32:

True regarding POAG:

- a) A, B, D and E
- b) A, B and C
- c) B and E only
- d) B, C and D

Question 33:

A 48-year-old lady was started on an ocular medication that she was instructed to take daily. She develops redness of the eye. Examination reveals a colour change of the iris and orbital fat atrophy. Which of the following medications was she likely prescribed?

- a) Timolol
- b) Pilocarpine

- c) Latanoprost
- d) Scopolamine

Question 34:

First drug to be given in acute angle-closure glaucoma is _____.

- a) Atropine
- b) Timolol
- c) Pilocarpine
- d) Acetazolamide

Question 35:

The normal extent of the visual field superiorly is _____.

- a) 60°
- b) 50°
- c) 70°
- d) 90°

Answer Key

Question No.	Correct Option
1	c
2	a
3	b
4	a
5	a
6	d
7	a
8	b
9	d
10	b
11	a

12	d
13	b
14	b
15	b
16	a
17	a
18	a
19	b
20	c
21	d
22	a
23	a
24	d
25	a
26	c
27	c
28	d
29	a
30	b
31	b
32	c
33	c
34	d
35	b

Detailed Explanations

Solution to Question 1:

The rate of formation of aqueous humor is 2-3 $\mu\text{L}/\text{min}$.

Formation of aqueous humor involves the following:

- Ultrafiltration
- Secretion- Na/K ATPase of non-pigmented epithelium of the ciliary process
- Diffusion

Solution to Question 2:

The critical angle of the air–cornea interface is 46 degrees. Gonioscopy is a method of evaluating the anterior chamber (AC) angle.

Direct gonioscopes include the Koeppe, Medical Workshop, Barkan, and Swan–Jacob. Indirect gonioscopes include Goldmann, Zeiss, and Posner.

Structures seen on gonioscopy from anterior to posterior are:

- Schwalbe line: represents the posterior limit of Descemet's membrane
- Trabecular meshwork
- Scleral spur
- Ciliary body and root of the iris

Shaffer's grading of angle width:

Grade 4 ($35\text{--}45^\circ$) is the widest angle, characteristic of myopia and pseudophakia; the ciliary body can be visualized without tilting the lens.

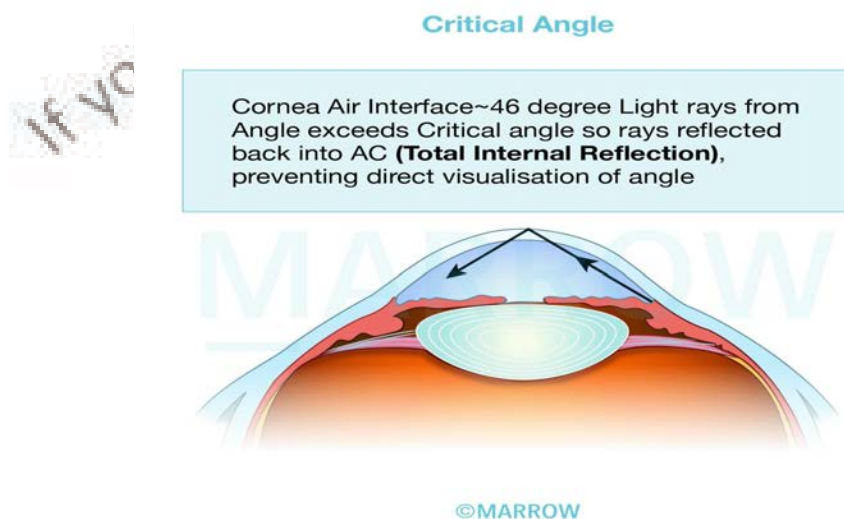
Grade 3 ($25\text{--}35^\circ$) is an open-angle in which the scleral spur is visible.

Grade 2 (20°) is an angle in which the trabeculum but not the scleral spur can be seen.

Grade 1 (10°) is a very narrow-angle in which only the Schwalbe line, and perhaps the top of the trabeculum, can be identified.

A slit angle is one in which there is no obvious iridocorneal contact but no angle structures can be identified.

Grade 0 (0°) is closed due to iridocorneal contact.



Solution to Question 3:

The depth of the anterior chamber decreases with age.

Anterior chamber depth

- The anterior chamber is shallow in very young and in old age.
- At other periods of life, it is normally about 2.5 mm deep.
- Depth is less in women than in men.
- Depth is more in myopes.
- Depth is less in hypermetropia and in angle-closure glaucoma

Solution to Question 4:

The given clinical scenario of an infant with large corneas, lacrimation and photophobia supported with the image is suggestive of congenital glaucoma.

Primary congenital glaucoma (PCG) also known as developmental glaucoma refers to abnormally high intraocular pressure (IOP) which results from the developmental anomaly of the angle of the anterior chamber.

It can be classified as:

- True congenital or newborn glaucoma in which IOP is elevated during intrauterine life and the baby is born with ocular enlargement.
- Infantile glaucoma: which manifests prior to age 3 years.
- Juvenile glaucoma: IOP rises between 3 and 16 years of age.

Clinical findings include:

- Blepharospasm
- Buphthalmos: Large eyeball due to stretching of the globe from elevated IOP prior to the age of 3 years.
- Corneal enlargement:
- Corneal haze is due to corneal edema.
- Blue sclera - due to the shining of the underlying uveal tissue through the thinned-out sclera
- Haab striae are healed breaks in Descemet's membrane

Treatment:

- Goniotomy - when the angle of the anterior chamber is visible through the clear cornea.
- Trabeculectomy when the corneal clouding prevents visualization of the angle.
- Filtration procedures such as Trabeculectomy or combined trabeculectomy with trabeculectomy if the above procedures fail.
- Glaucoma drainage devices in unresponsive cases.

Other options:

Option A: Retinoblastoma is the most common primary intraocular malignancy of childhood that is seen between 1-2 years.

Findings of Retinoblastoma include:

- Leukocoria (white pupillary reflex)- the most common manifestation
- Strabismus
- Painful red eye with secondary glaucoma
- An endophytic tumor grows into the vitreous as a white mass that may 'seed' into the gel. If calcification is present it gives a "cottage cheese appearance". An exophytic tumor grows outwards and causes overlying exudative retinal detachment.

The image below shows the leucocoria of the left eye.



Option C: Congenital hereditary endothelial dystrophy (CHED) is an inherited corneal dystrophy that manifests since birth with endothelial degeneration. It appears as diffuse bilateral corneal edema in the absence of elevated IOP. The clinical features of the disease include corneal clouding, visual impairment, and nystagmus.

Option D: Megalocornea is a bilateral non-progressive X-linked condition that most commonly affects males. In this condition, the horizontal corneal diameter is $\geq 12\text{mm}$ (< 2 years) and is $\geq 13\text{mm}$ (> 2 years). It is characterized by a clear cornea, normal corneal thickness, and no breaks in the Descemet's membrane. Systemic associations of Megalocornea include Marfan's syndrome, Apert syndrome, Ehlers-Danlos, and Down syndrome.

Solution to Question 5:

This neonate with complaints of photophobia and watering with large cornea and Haab's striae is suggestive of buphthalmos. Haab's striae are horizontal/circumferential breaks in Descemet's membrane seen in buphthalmos.

Buphthalmos (Bull-like eyes) is an enlargement of the eyeball seen in true congenital and infantile glaucoma, usually when the disease onset is before 3 years.

The image given below shows Haab's striae and a child having buphthalmos in congenital glaucoma.



Note: Vogt's striae are vertical, fine, whitish lines (folds/tears) in Descemet's membrane commonly found in patients with keratoconus.

The image given below shows Vogt's striae.



Solution to Question 6:

Frequent changes in presbyopic glasses in this patient is a symptom of open-angle glaucoma (OAG).

This is due to increasing accommodative failure as a result of constant pressure on the ciliary muscle and nerve supply.

In intumescent cataracts, patients report difficulty with distance vision and frequent change of spectacle prescription. However, it is not associated with frequent changes of presbyopic glasses.

Solution to Question 7:

Normal cup-to-disc ratio is up to 0.6. Although the average cup-to-disc ratio is around 0.3 to 0.4, a ratio of up to 0.6 may be found in healthy individuals. This larger ratio is frequently seen in cases of large discs, myopic discs or physiological cupping. A cup-to-disc ratio of > 0.6 is suggestive of glaucoma.

Solution to Question 8:

The given clinical scenario of a patient with headache and eye pain along with the finding of a lamellar dot sign is suggestive of glaucoma.

The lamellar dot sign is an advanced glaucomatous change in the optic disc. This occurs because the pores in the lamina cribrosa are slit-shaped and are visible up to the margin of the disc.

Glaucomatous changes in the optic disc

Early glaucomatous changes:

- Slit defects of the retinal nerve fiber layer
- Splinter hemorrhages on disc margin
- Large cup - 0.7 or more
- Asymmetry of the cups: A difference equal to or more than 0.2 between two eyes is significant
- Vertically oval cup

Advanced glaucomatous changes:

- Marked cupping (Beanpot cupping)
- Thinning of neuroretinal rim
- Bayonetting sign - Sharp angulation of vessel at the margin of the excavated cup
- Laminar dot sign

Solution to Question 9:

The temporal visual field is the last field to be affected in glaucoma.

The pattern of visual field progression in glaucoma from early to severe is predictable in light of the pattern of thinning of the retinal nerve fiber layer, which usually begins with the superior and inferior arcuate fibers, then the papillomacular bundle and finally, the nasal fibers.

Since the nasal fibers are the last to be involved, the temporal field is the last to be affected in glaucoma.

Solution to Question 10:

Bjerrum scotoma (arcuate) is a late and clinically significant visual field defect.

Progression of visual field changes seen in glaucoma:

Early non-specific changes	Isopter contraction Baring of a blind spot Exclusion of blind spot from the central field
Early significant changes	Peripheral nasal step (may be early or late) Paracentral scotoma in Bjerrum's area Siedel's scotoma Paracentral scotoma joins the blind spot
Late significant changes	Arcuate or Bjerrum scotoma Ring or Double arcuate scotoma When 2 arcuate scotomas join together Roenne's central nasal step

Early non-specific changes	Isopter contraction Baring of a blind spot: Exclusion of blind spot from the central field
Advanced glaucomatous field defects	Tubular vision No light perception: Complete visual field defect.

Solution to Question 11:

The given clinical scenario of deterioration of vision, sluggish pupils, elevated IOP and paracentral scotoma is suggestive of primary open-angle glaucoma (POAG). Timolol is one of the first-line agents used in the treatment of primary open-angle glaucoma.

First-line agents for the treatment of glaucoma are:

- Prostaglandin analogs such as latanoprost and bimatoprost
- Topical beta-blockers such as timolol, betaxolol, and levobunolol
- Selective alpha 2 adrenergic agonist brimonidine

Adjunctive therapy: Carbonic anhydrase inhibitors

Laser trabeculoplasty: In cases where intraocular pressure (IOP) is uncontrolled despite maximal tolerated medical therapy.

Note: Since prostaglandin (PG) analogs are not included in the options, the next best option to choose is timolol.

Historically, beta-blockers were the most common first-line agents used in the medical treatment of POAG. However, with an increase in our experience with newer agents, specifically the prostaglandin analogues that facilitate outflow, they were recognized as being more efficacious in lowering IOP and systemically safer than beta-blockers. As a result, now they are the first-line agents in glaucoma treatment.

Solution to Question 12:

This patient with a history of asthma now presenting with shortness of breath after starting medications for increased intraocular pressure was most likely diagnosed with glaucoma. The drug she was likely prescribed was timolol, a non-selective beta blocker.

Timolol is a non-selective beta-blocker that is used in glaucoma. Non-selective beta-blockers can cause precipitation of bronchial asthma in susceptible individuals (beta-2 block on bronchi). They cause bronchospasms and are therefore contraindicated in patients with asthma (e.g. timolol, levobunolol). Ocular beta-blockers reduce intraocular pressure by decreasing aqueous formation by blockade of β_2 receptors. Timolol is specifically associated with allergic blepharoconjunctivitis and punctal stenosis

Side effects: Systemic: Bronchospasm, bradycardia, decreased cardiac output, hypotension, depression, impotence, and altered lipid profile.

Contraindications: Asthma, chronic obstructive pulmonary disease, congestive heart failure, sinus bradycardia, and atrioventricular blocks

Other options:

Option A: The patient has been stable on aspirin for the past three years which is unlikely to be the cause of her current symptoms.

Option B: Betaxolol is a beta 1 selective blocker that is used in glaucoma and has a lower propensity to precipitate bronchial asthma.

Option C: Latanoprost is a prostaglandin (PGF₂ alpha) analog and does not precipitate bronchial asthma.

Solution to Question 13:

Brimonidine is an anti-glaucoma medication that can cause drowsiness.

It is a selective alpha-2 agonist. It can cross the blood-brain barrier and, hence, causes drowsiness and CNS depression; it is contraindicated in infants and children.

A new class of glaucoma drugs, Rho kinase (ROCK) inhibitors, promises to act specifically on the trabecular meshwork, a main outflow and blockage site in glaucoma. ROCK inhibitors are not yet approved and available for glaucoma patients. Two US companies, Aerie and Altheos, are currently in early clinical research development with topical ROCK inhibitors to lower IOP.

Solution to Question 14:

This patient with complaints of severe headache, corneal congestion, and edema with a shallow anterior chamber most likely has acute angle closure glaucoma. Since this patient has a past history of sickle cell disease, intravenous mannitol is the drug of choice for this patient

Acetazolamide is contraindicated in cases of sickle-cell anemia since these drugs tend to increase sickling of erythrocytes.

Solution to Question 15:

The given clinical scenario, supported with the image of the mid-dilated and vertically oval pupil is suggestive of acute angle-closure glaucoma. The pupil is non-reactive to both light and accommodation.

Acute primary angle-closure glaucoma refers to the occlusion of the trabecular meshwork by the peripheral iris (iridotrabecular contact [ITC]), obstructing aqueous outflow.

Angle closure can be primary, when it occurs in an anatomically predisposed eye or secondary to another ocular or systemic factor.

Classification:

Symptoms: Pain, redness, photophobia, impaired vision, watering

Signs:

- Corneal edema- colored halos
- Conjunctiva- congestion, chemosis
- IOP- elevated 40–70 mm Hg
- AC very shallow
- Pupil- vertically oval, mid-dilated
- Angle- completely closed, Grade 0

Treatment:

- Immediate- Intravenous mannitol, intravenous acetazolamide followed by oral dose, topical beta-blockers, brimonidine, latanoprost
- Definitive- Laser peripheral iridotomy, filtration surgery

Primary angle-closure suspect (PACS)	Gonioscopy shows posterior meshwork ITC in three or more quadrants but no peripheral anterior synechiae (PAS). Intraocular pressure (IOP), optic disc, and visual field are normal. No PAS are seen. The risk of PACG at 5 years may be around 30%.
Primary angle closure (PAC)	Gonioscopy shows three or more quadrants of ITC with raised IOP and/or PAS or excessive pigment smudging on the TM. Normal optic disc and field.
Primary angle-closure glaucoma (PACG)	ITC in three or more quadrants, with glaucomatous optic neuropathy. Optic nerve damage from an episode of severe IOP elevation, such as acute angle closure, may not appear as typical glaucomatous cupping.

Solution to Question 16:

Pain after dark-room prone-provocative test indicates occludable angles. Hence, atropine is contraindicated as it is a long-acting mydriatic and can precipitate angle closure.

Dark-room/prone-provocative test (DRPPT)- The patient sits in a dark room, face down for one hour without sleeping (sleep induces miosis). The intraocular pressure (IOP) is checked

(immediately after the test, as IOP can normalize very rapidly), and an IOP rise of 8 mmHg or more is frequently considered significant.

Solution to Question 17:

The given clinical scenario of sudden onset severe pain in the eye, progressive loss of vision, shallow anterior chamber, semi-dilated pupil, and edematous cornea is suggestive of acute angle-closure glaucoma. Laser iridotomy is the procedure of choice especially in emergency situations.

The attack may have been precipitated by pupillary dilation, which may result from activities such as going to a movie theater. Laser iridotomy re-establishes the communication between the posterior and anterior chamber, bypassing the pupillary block; hence, it immediately relieves crowding of the angle.

Various surgical and laser modalities used in glaucoma:

Angle-closure glaucoma:

- Laser iridotomy by Nd:Yag along with prophylactic iridotomy of the other eye → Treatment of choice.
- Trabeculectomy → Filtration surgery

Congenital glaucoma:

- Goniotomy
- Trabeculotomy

Primary open-angle glaucoma:

- Laser trabeculoplasty: Argon or diode (if not controlled medications)
- Trabeculectomy: Filtration surgery (if not controlled medications)

Absolute glaucoma:

- Cyclocryotherapy
- Nd:Yag lasercyclo destruction

Solution to Question 18:

The clinical vignette points to a diagnosis of glaucoma not responsive to medical treatment and trabeculectomy and the device used is an ex-Press™ glaucoma implant (mini shunt).

It is a form of minimally invasive glaucoma surgery (MIGS) to increase aqueous drainage.

Ex-PRESS glaucoma implant is a small stainless steel device. It is inserted under a scleral flap during a modified trabeculectomy. Complications such as hypotony are lower than with standard trabeculectomy.

Glaucoma drainage devices are devices that allow aqueous outflow by establishing communication between the anterior chamber and sub-Tenon's space/sub-conjunctival space. While these devices were traditionally reserved for high-risk patients, or after trabeculectomy had failed, they are becoming increasingly popular.

The commonly used devices are as follows:

1. Shunts using episcleral explants
 - Ahmed glaucoma valve: Silicone
 - Molteno implant - Polypropylene plate and silicone plate
 - Baerveldt implant - Silicone
2. Mini-shunts
 - Ex-Press™ Mini-Shunt: Stainless steel
 - iStent® - Titanium

Solution to Question 19:

Glaukomflecken is anterior subcapsular lenticular opacity and a component of Vogt's triad seen in patients after an acute angle closure glaucoma attack.

The Vogt's triad comprises of:

- Glaukomflecken
- Patches of iris atrophy
- Slightly dilated nonreactive pupil

Solution to Question 20:

The given scenario fits the description of glaucomatocyclitic crisis.

Glaucomatocyclitic crisis/Posner–Schlossman Syndrome [PSS] describes recurrent attacks of unilateral acute raised intraocular pressure (IOP) associated with mild anterior uveitis. The mechanism is thought to be acute trabeculitis, and there is evidence that infection, possibly cytomegalovirus (CMV) or Helicobacter pylori, may play a role. PSS typically affects young to middle-aged adults. Males are affected more frequently than females.

Signs:

- Slit-lamp biomicroscopy typically shows a few anterior chamber cells and one to several fine white central keratic precipitates.
- Mydriasis is common; posterior synechiae are not a feature.
- IOP is typically raised to over 40 mmHg, out of proportion to iritis severity.
- Gonioscopy shows an open-angle.

Treatment:

- Topical steroids are used to control inflammation, with aqueous suppressants to lower the IOP.
- Topical or oral non-steroidal anti-inflammatory agents may also be beneficial.

Note: In PSS, the rise in IOP is out of proportion to the pain felt. There is often a mild discomfort and pain may even be absent.

Solution to Question 21:

The given image shows a pattern of the deposition of pigments on the back of the cornea in pigmentary glaucoma. They are called Krukenberg spindles.

Pigmentary glaucoma is an open-angle glaucoma having the following features:

- Krukenberg spindles
- Gonioscopy shows pigment accumulation along the Schwalbe line inferiorly, and this accumulation is called the Sampaolesi line.
- Radial transillumination defect in the iris is a pathognomonic feature.
- Pigment granules may be deposited on the anterior surface. There may be a Scheie line or Zentmayer ring of pigment on the peripheral/equatorial surface around the zonular insertions.

Other options:

Option A: Ferry line: Corneal epithelial iron line at the edge of filtering blebs of glaucoma surgery

Option B: Fleischer line/ring: Epithelial iron deposits at the base of the cone in keratoconus

Option C: Hudson–Stahli line: Iron deposition line in the corneal epithelium, which is commonly seen in the junction between middle and lower third cornea. This finding is mainly age-related.

Solution to Question 22:

Malignant glaucoma is seen after intra-ocular surgery (usually after trabeculectomy).

Ciliary block glaucoma is also called malignant glaucoma/inverse glaucoma. It can be seen after any intraocular surgery. Forward flow of the aqueous is blocked as tips of ciliary processes rotate forward and press against the lens/intraocular lens (IOL)/anterior hyaloid of the vitreous. Hence, aqueous humor starts collecting in the vitreous cavity.

Signs:

- High intraocular pressure (IOP) and absent bleb.
- Negative Seidel test.
- Shallow anterior chamber (AC)

Treatment:

- Initial treatment is with cycloplegics (atropine 1%) to dilate the ciliary ring and push the vitreous-lens-iris posteriorly.
- Subsequent treatment if medical therapy fails is with Nd:YAG laser hyaloidotomy.
- Pars plana vitrectomy can be considered in conditions that don't respond to the above treatment modalities.

Solution to Question 23:

Progressive atrophy of iris stroma is a clinical entity of iridocorneal endothelial syndrome (ICE). ICE syndrome is a proliferative endotheliopathy, which leads to secondary angle-closure glaucoma.

It includes three clinical entities:

- Progressive iris atrophy
- Chandler's syndrome - corneal edema
- Cogan–Reese syndrome - nodules and nevus on the iris; hence, it is known as iris nevus syndrome.

Solution to Question 24:

Hypersecretory glaucoma is associated with epidemic dropsy.

Epidemic dropsy occurs due to Argemone mexicana poisoning. Here, glaucoma is due to excessive production of aqueous humor.

Solution to Question 25:

100-day glaucoma is a neovascular glaucoma that occurs in ischemic type of central retinal vein occlusion (CRVO).

It is known as 100-day glaucoma because it occurs 3 months after the episode of CRVO. Although neovascular glaucoma occurs as a complication in Eale's disease (Option B) and central retinal artery occlusion (Option C) and long-term use of topical steroid therapy causes steroid-induced glaucoma (Option D) none of them are termed as 100-day glaucoma.

CRVO is further classified into two types:

- Non-ischemic CRVO: occurs secondary to venous stasis. It is the most common type of CRVO of the two.
- Ischemic CRVO: occurs due to acute occlusion of the central retinal vein. It is also termed as hemorrhagic retinopathy.

Ischemic CRVO presents with sudden loss of vision and Relative afferent pupil defect (RAPD).

Fundus examination shows massive enlargement of retinal veins and massive retinal hemorrhages giving a "splashed tomato appearance". Other findings include cotton wool spots, optic disc edema and neovascularization at the disc and in the periphery.

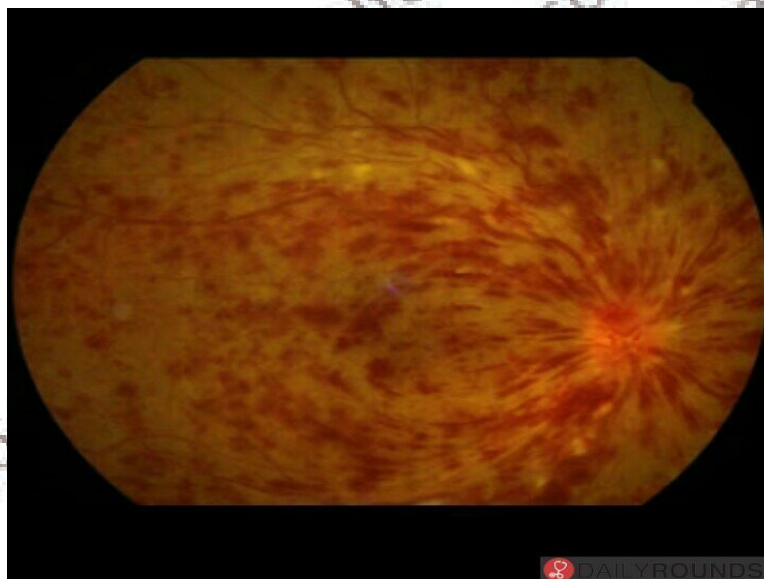
Pathognomic features of ischemic CRVO that differentiates it from non-ischemic CRVO are:

- RAPD
- Reduced amplitude of B-wave on electroretinogram
- Visual field defects

Complications of the disease:

- Neovascular Glaucoma known as 100 day glaucoma
- Neovascularisation of iris known as Rubeosis iridis
- Vitreous hemorrhage
- Proliferative retinopathy

The below-given fundus image is of the "splashed tomato" appearance of CRVO.



Option B: Eales disease is an idiopathic inflammatory disease of the peripheral retinal veins. It is considered to be a hypersensitivity reaction to the tubercular proteins and it is commonly seen in young males. It is characterized by recurrent vitreous hemorrhage.

Complications include:

- Neovascular glaucoma
- Proliferative vitreoretinopathy
- Tractional retinal detachment,
- Rubeosis iridis

Option C: Central retinal artery occlusion (CRAO) is most often caused due to an embolus from an atherosclerotic plaque. It presents as sudden painless loss of vision and amaurosis fugax (transient monocular blindness that is due to ischemia). Fundus examination shows an opaque,

milky-white retina with a cherry-red spot.

Neovascular glaucoma may occur as a delayed complication of CRAO but it is not termed as 100 day glaucoma.

The below image depicts the fundus image seen in CRAO with the cherry red spot.



Option D: Steroid induced glaucoma is a type of secondary open angle glaucoma that usually follows weeks-months of topical steroid therapy.

Solution to Question 26:

Family history is a risk factor for the development of primary open-angle glaucoma.

Ocular hypertension (OHT) is characterized by only raised intraocular pressure (IOP). Optic disc and normal visual field are normal, with no evidence of angle closure.

Risk factors for the progression of ocular hypertension to primary open-angle glaucoma include the following:

- African American race
- Thin cornea
- Increasing age
- Increased cup-disc ratio
- Increased IOP
- Family history of POAG

Solution to Question 27:

The given image shows Baerveldt implant which is a glaucoma drainage device (GDD)/seton/aqueous shunt.

It consists of a scleral plate that is sutured to the sclera and a tube connecting it into the anterior chamber. It is valveless, and hence, it carries the risk of postoperative hypotony due to excessive aqueous drainage.

Glaucoma drainage devices are devices that allow aqueous outflow by establishing a communication between the anterior chamber and sub-Tenon's space/sub-conjunctival space. The glaucoma valve implant is also known as a seton.

The commonly used devices are as follows:

1. Shunts using episcleral explants

- Ahmed glaucoma valve: Silicone
- Molteno implant- Polypropylene plate and silicone plate
- Baerveldt implant- Silicone

2. Mini-shunts

- Ex-Press™ Mini-Shunt: Stainless steel
- iStent®- Titanium

Solution to Question 28:

The drug of choice for inverse glaucoma is atropine.

Inverse glaucoma is a type of secondary angle-closure glaucoma caused by the flow of aqueous humour posteriorly to the posterior segment instead of the posterior chamber. The increased pressure in the posterior segment pushes the lens forward leading to the shallowing of the anterior chamber.

Miotics are contraindicated as they can cause the ciliary zonules to slack, the lens to move forward and further shallowing of the anterior chamber. So mydriatic and cycloplegic agents such as atropine are administered.

Inverse glaucoma is also called:

- Malignant glaucoma
- Ciliary block glaucoma
- Aqueous misdirection.

Note:

Pilocarpine is the drug of choice in angle-closure glaucoma but is contraindicated in inverse glaucoma.

Whereas, atropine is the drug of choice in inverse glaucoma but is contraindicated in angle-closure glaucoma.

Solution to Question 29:

This diabetic patient with new onset central loss of vision and reports of colors appearing faded or washed out most likely has macular edema. Prostaglandin analogs have a high propensity to cause and further exacerbate cystoid macular edema and hence are not preferred in this patient.

Drugs associated with cystoid macular edema (CME):

- Prostaglandin analogues
- Epinephrine
- Dipivefrine (only the non-selective alpha agonists, in aphakic eyes)
- Tamoxifen
- Timolol
- Docetaxel
- Interferons
- Fingolimod
- Thiazolinediones
- Niacin (pseudo-CME)

Solution to Question 30:

Pseudoexfoliation (PEX) syndrome is the most likely cause of secondary open-angle glaucoma.

Pseudoexfoliation syndrome is an ocular manifestation of an age-related systemic disorder of the extracellular matrix, characterized by the production and progressive accumulation of fibrillar extracellular material in many ocular and systemic tissues. The clinical hallmark of PEX syndrome is the deposition of whitish abnormal pseudoexfoliative debris on the anterior lens capsule.

A deposition is also noted on:

- Zonules (causing zonulolysis and phacodonesis, lens subluxation)
- Pupillary margin
- Trabecular meshwork (causing the secondary open-angle glaucoma)

Solution to Question 31:

Latanoprost is a prostaglandin drug, which acts by increasing the uveoscleral outflow and thereby lowers the intraocular pressure.

Mechanism of action: Latanoprost is a prodrug and gets activated during its passage through the cornea. It acts by stimulating the prostanoid receptors on the ciliary muscles. This leads to the release of matrix metalloproteinases, which increases the uveoscleral outflow.

Solution to Question 32:

In primary open-angle glaucoma (POAG), family members are likely to develop steroid responsiveness (statement B) and visual field defect can exist with a normal C:D ratio (statement E).

Statement A: Gonioscopy cannot visualize the specific abnormalities that occur in the trabecular meshwork in POAG. Gonioscopy diagnoses POAG because of the visibility of all the structures of the angle of the anterior chamber in contrast to the non-visibility of different structures in angle closure glaucoma.

Statement C: Primary angle-closure glaucoma is associated with exacerbation of IOP with pupillary dilatation and not POAG.

Statement D: There is a 10% increased risk of POAG in siblings and a 4% increased risk in children of patients with POAG.

POAG also known as chronic simple glaucoma of adult-onset is characterized by the presence of any 2 of the following 3 categories of changes:

- Increased intraocular pressure:
- IOP ≥ 21 mmHg on more than one occasion, and/or
- Circadian variation in IOP of ≥ 8 mmHg, or
- Asymmetry in IOP ≥ 5 mmHg between both eyes
- Optic nerve head changes suggestive of glaucomatous damage. These may include one or more of the following changes:
 - Vertically oval cup
 - Cup-disc (C:D) ratio ≥ 0.6 (average cup: disc ratio 0.3-0.4)
 - Asymmetry in C:D ratio ≥ 0.2 between both eyes.
 - Narrowing, notching, or pallor of the neuroretinal rim.
 - Disc haemorrhages
 - Nasal shifting of retinal vessels-Bayonetting sign
 - Lamellar dot sign
- Visual field changes - isopter contraction, barring of blind spot, paracentral scotoma, Seidel's scotoma, Bjerrum's scotoma, double arcuate scotoma, and Roenne's central nasal step.

A patient presenting with just raised IOP and visual field defects with normal optic nerve head findings such as a normal C:D ratio can still be diagnosed as POAG (statement E).

Solution to Question 33:

The given clinical history of a patient that has iris pigmentation and redness of the eye, was most likely started on latanoprost.

Drug	Side Effects
Latanoprost	Allergic conjunctival hyperemia, permanent increase in iris pigmentation, orbital fat atrophy.
Timolol	Exacerbation of asthma and chronic obstructive pulmonary disease, bradycardia, depression, impotence
Scopolamine	Photosensitivity, blurred vision
Pilocarpine	Corneal edema, miosis, induced myopia, decreased vision, brow ache, retinal detachment

Solution to Question 34:

The first drug to be given in acute angle-closure glaucoma is oral acetazolamide or intravenous mannitol.

Acetazolamide reduces IOP by decreasing the production of aqueous.

The definitive drug is pilocarpine as it constricts the pupil and breaks the attack.

Finally, when the corneal edema resolves, the patient has to undergo laser iridotomy to prevent such attacks in the future. So, laser iridotomy is the definitive management for acute angle-closure glaucoma.

Solution to Question 35:

The extent of visual field superiorly is 50°

Normal extent of the visual field:

- Temporal: $90^\circ/100^\circ$
- Inferiorly: 70°
- Nasal: 60°
- Superiorly: 50°

Note: According to Parson's 22nd Ed., Page no. 101, the visual field extends 60° superiorly. However, most journals and other standard textbooks consider the answer to be 50° . Hence, we choose 50° as the most appropriate answer.

Uveitis - Anterior and Intermediate

Question 1:

Which of the following statements are false about the ciliary body?

- a) It consists of the pars plana and pars plicata.
- b) Contraction of the ciliary body helps in accommodation.
- c) It secretes aqueous humor.
- d) It derives its blood supply from short posterior ciliary arteries.

Question 2:

A 3-year-old child with the following finding has high myopia. What is the diagnosis?



- a) Corectopia
- b) Anisocoria
- c) Ectopia lentis
- d) Polycoria

Question 3:

A patient presents to the casualty after ocular trauma with a D-shaped pupil. What is the likely diagnosis?

- a) Uveitis
- b) Glaucoma
- c) Iridodialysis
- d) Endophthalmitis

Question 4:

Which of the following is the pathognomonic sign of acute anterior uveitis?

- a) Aqueous flare
- b) Keratic precipitates
- c) Festooned pupil
- d) Bussaca's nodules

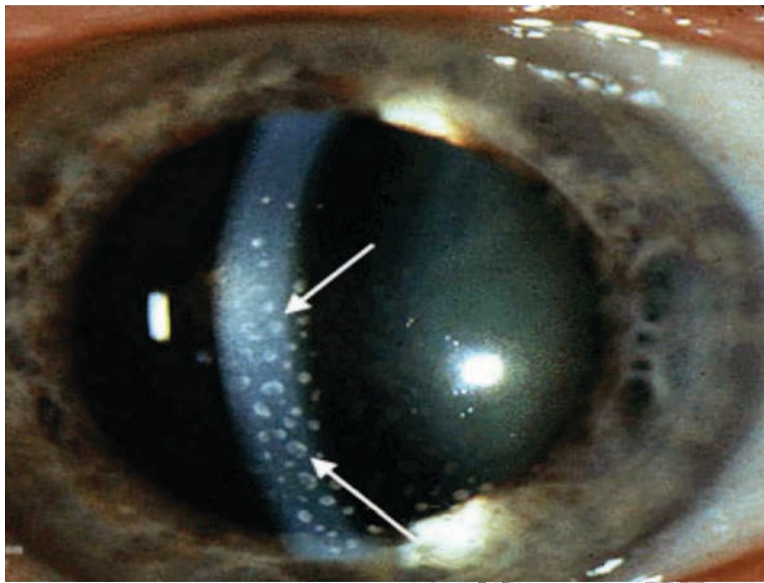
Question 5:

What is the first sign a patient with anterior uveitis may show?

- a) Keratic precipitate
- b) Aqueous flare
- c) Corneal edema
- d) Corneal opacity

Question 6:

A 34-year-old lady presents with eye pain, photophobia, and red-eye. The slit-lamp examination finding is shown below. On which layer of the cornea are these present?



- a) Epithelium
- b) Endothelium
- c) Stroma
- d) Bowman's membrane

Question 7:

You are evaluating a 45-year-old man admitted for acute anterior uveitis. What is the pupillary shape you expect to find in his eyes?

- a) Oval, regular
- b) Large, irregular
- c) Small, irregular
- d) Normal, regular

Question 8:

A patient with sarcoidosis shows Koeppe's nodules on ophthalmologic examination. Where are these seen?

- a) Cornea
- b) Conjunctiva
- c) Iris
- d) Retina

Question 9:

What is the type of synechiae formed in iris bombe?

- a) Ring
- b) Total
- c) Goniform
- d) Filiform

Question 10:

A 38-year-old woman has been having repeated episodes of red-eye, eye pain, and blurred vision. She is diagnosed with chronic granulomatous uveitis. Which of the following is not a cause?

- a) Sarcoidosis
- b) Tuberculosis
- c) Phacoanaphylactic
- d) Fuchs heterochromic iridocyclitis

Question 11:

Intermediate uveitis is also called as _____.

- a) Iridocyclitis
- b) Chorioretinitis
- c) Pars planitis
- d) Choroiditis

Question 12:

In a patient with intermediate uveitis, which of the following is a pathognomonic examination finding?

- a) Corneal oedema
- b) Snow banking
- c) Aqueous flare

- d) Keratic precipitates

Question 13:

While studying a cohort of patients with juvenile idiopathic arthritis, you are evaluating the risk of iridocyclitis among the different subtypes of JIA. Which subtype has the highest risk of iridocyclitis associated?

- a) Systemic JIA
- b) RF-positive oligoarticular JIA
- c) ANA-positive oligoarticular JIA
- d) ANA-positive polyarticular JIA

Question 14:

Where is Amsler's sign seen?

- a) Fuchs heterochromic iridocyclitis
- b) Posner-Schlossman syndrome
- c) Uveal effusion syndrome
- d) Cystoid macular edema

Question 15:

What is Posner-Schlossman syndrome?

- a) Ipsilateral optic atrophy with contralateral papilloedema
- b) Unilateral glaucomatous changes with mild anterior uveitis
- c) Granulomatous uveitis with iris heterochromia
- d) Non-granulomatous uveitis with bilateral cataracts

Question 16:

A 44-year-old patient with AIDS and opportunistic infections was started on treatment. He develops a sterile hypopyon anterior uveitis that is suspected to be drug-induced. Which of the following drugs may be implicated here?

- a) Rifabutin

- b) Zidovudine
- c) Indinavir
- d) Pentamidine

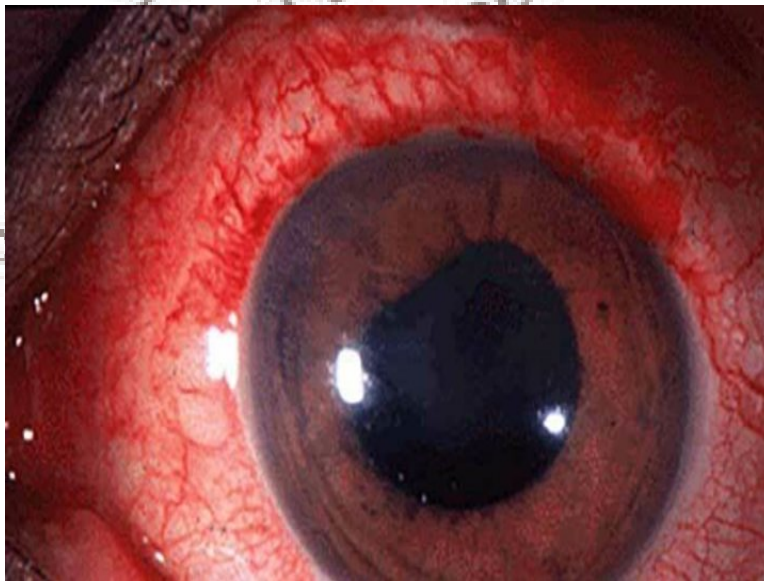
Question 17:

Which among the following is associated with transient hypopyon syndrome?

- a) Behcet's disease
- b) Reiter's syndrome
- c) Heerfordt's syndrome
- d) Still's disease

Question 18:

A patient presents with photophobia and blurring of vision. On examination, the eye appears as shown below. What is the mainstay of treatment?



- a) Antibiotics and cycloplegics
- b) Steroids and cycloplegics
- c) Antibiotics and steroids
- d) Immunosuppressants

Answer Key

Question No.	Correct Option
1	d
2	a
3	c
4	b
5	b
6	b
7	c
8	c
9	a
10	d
11	c
12	b
13	c
14	a
15	b
16	a
17	a
18	b

Detailed Explanations

Solution to Question 1:

The ciliary body derives its blood supply from the long posterior ciliary artery and anterior ciliary arteries, which form the major arterial circle at the root of the iris.

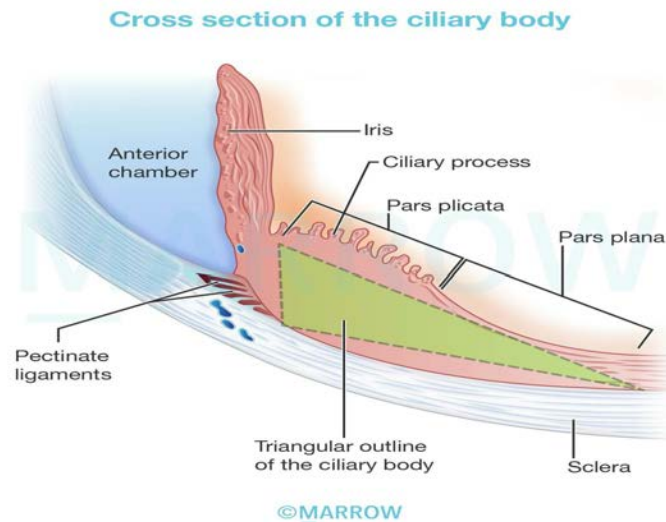
The ciliary body is a triangular structure that begins at the base of the iris and becomes continuous with the choroid at the ora serrata. It is approximately 6–6.5 mm in an anteroposterior dimension.

It consists of an anterior portion called the pars plicata and a posterior portion called the pars plana (option A). The pars plicata contains approximately 70 finger-like projections called ciliary processes, which are covered by the ciliary body epithelium. These processes secrete aqueous humor (option C). The pars plana is the flat part of the ciliary body and ends at the ora serrata.

The smooth muscle of the ciliary body is divided into three groups of fibers:

- Outer longitudinal portion that attaches anteriorly to the scleral spur and trabecular meshwork fibers

- Middle oblique portion
- Inner circular portion (responsible for accommodation) (option B)



Solution to Question 2:

The image shows corectopia, an abnormal eccentrically placed pupil.

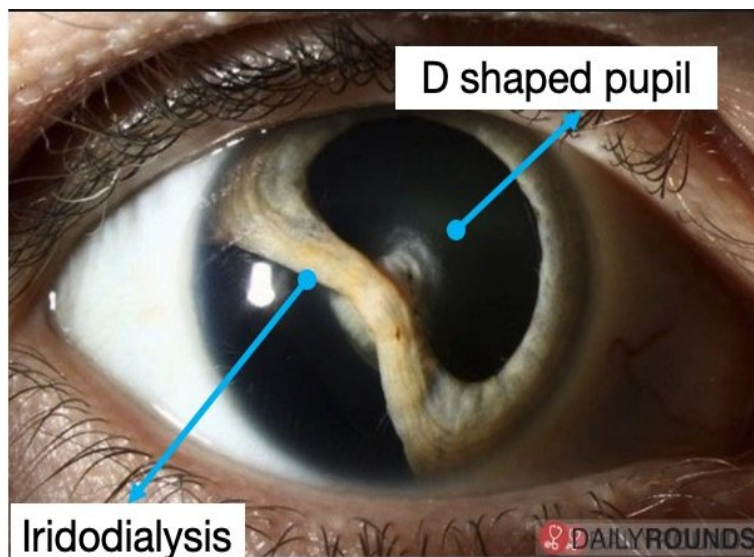
It can be a congenital anomaly or an acquired defect with essential iris atrophy. It may be associated with high myopia or ectopia lentis, among other conditions.

Solution to Question 3:

A D-shaped pupil is seen in iridodialysis.

Iridodialysis is the dis-insertion of the iris root from the ciliary body attachment. It may occur following ocular trauma. A red reflex can be obtained from the peripheral gap in the iris.

The given image depicts iridodialysis and a D-shaped pupil.



Solution to Question 4:

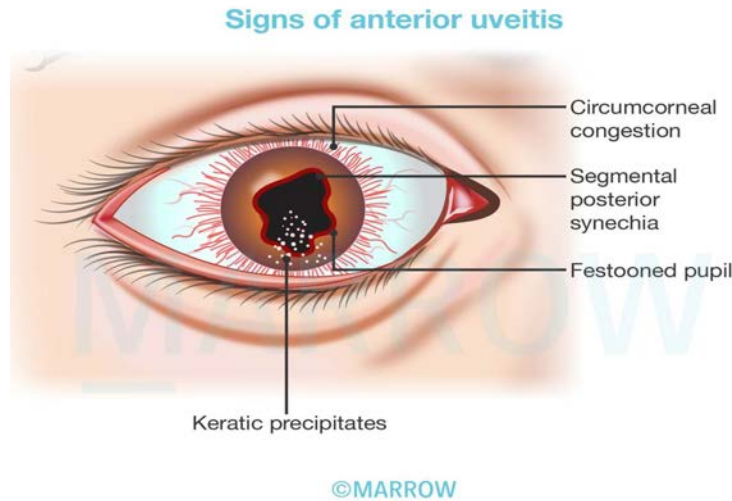
The pathognomonic sign of acute anterior uveitis is keratic precipitates (KP).

Keratic precipitates are proteinaceous cellular deposits at the back of cornea. The composition of KPs varies with the type of uveitis.

Anterior uveitis is the inflammation of the uveal tissue from the iris to the pars plicata of the ciliary body. Slit-lamp examination shows the following signs in anterior uveitis:

- Corneal signs
- Corneal edema
- Keratic precipitates (Pathognomonic)
- Posterior corneal opacity
- Anterior chamber signs
- Aqueous cells
- Aqueous flare (earliest sign)
- Hypopyon
- Deep and irregular anterior chamber due to posterior synechiae
- Iris signs
- Loss of normal pattern
- Change in color
- Iris nodules (Koeppe, Busacca)
- Posterior synechiae
- Neovascularization

- Pupillary signs
- Constricted pupil
- Irregular pupil shape
- Ectropion of pupillary margin
- Occlusion pupillae



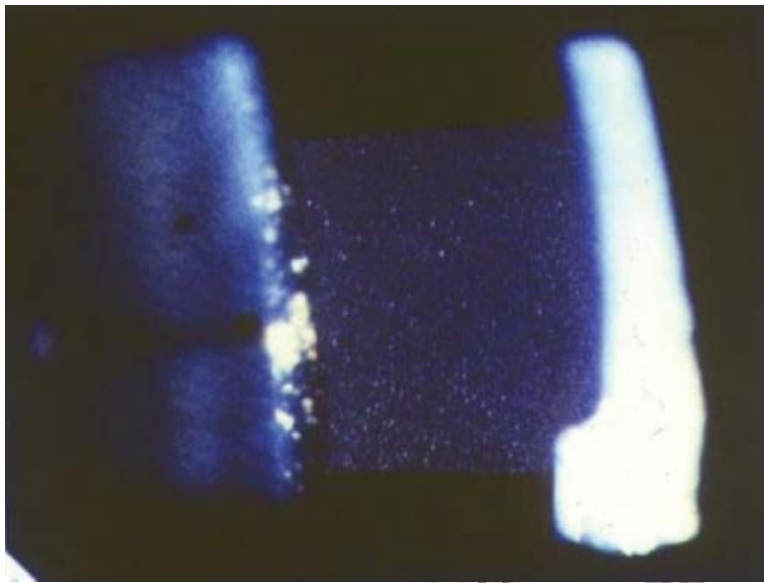
Solution to Question 5:

Aqueous flare is the earliest sign of acute anterior uveitis.

Aqueous flare is due to the leakage of protein particles into the aqueous humor from damaged blood vessels. It is demonstrated on the slit-lamp examination. It is based on the principle of Brownian movement or Tyndall phenomenon.

Aqueous flare is usually marked in non-granulomatous uveitis and minimal in granulomatous uveitis. The presence of cells in the anterior chamber denotes an active inflammatory process.

The image below is a high magnification slit-beam photograph 3×1 mm in size in a darkroom showing the presence of cells and flare.



Solution to Question 6:

Eye pain, photophobia and red-eye along with the slit lamp finding of keratic precipitates suggests a diagnosis of acute anterior uveitis. Keratic precipitates are proteinaceous cellular deposits occurring at the back of the corneal endothelium.

Keratic precipitates are arranged in a triangular fashion, occupying the center and inferior parts of the cornea (Arlt's triangle).

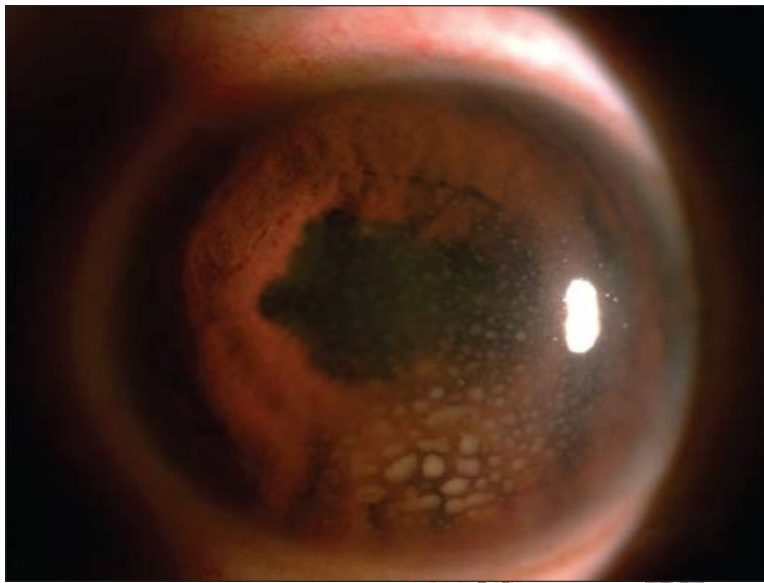
Solution to Question 7:

The pupil in acute anterior uveitis is constricted (pupillary sphincter spasm), irregular and sluggishly reactive to light

In an acute attack of uveitis, the pupil is irregular and narrow due to irritation of sphincter pupillae. Pupillary signs in chronic anterior uveitis is as follows:

- Festooned pupil results from the application of mydriatics in segmental posterior synechiae.
- Occlusio pupillae results when the pupil is completely occluded due to the organization of the exudates across the pupillary area.
- Seclusio pupillae due to ring synechiae, that prevent the circulation of aqueous humor from the posterior chamber to the anterior chamber.

The image below shows a festooned pupil.



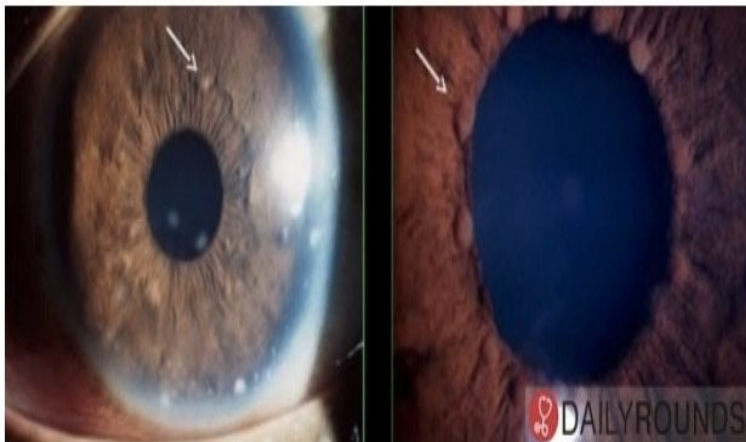
Solution to Question 8:

Koepple's nodules and Bussaca's nodules develop on the iris in granulomatous uveitis.

Granulomatous uveitis is a chronic proliferative inflammation seen in response to tuberculosis, leprosy, syphilis and sarcoidosis. Infiltration of lymphocytes and plasma cells, with proliferation of mononuclear cells leads to the formation of iris nodule. Koepple's nodules are seen at the pupillary margin. Bussaca's nodules are seen on the peripheral part of the anterior surface of iris.

Busacca's nodule

Koepple's nodules

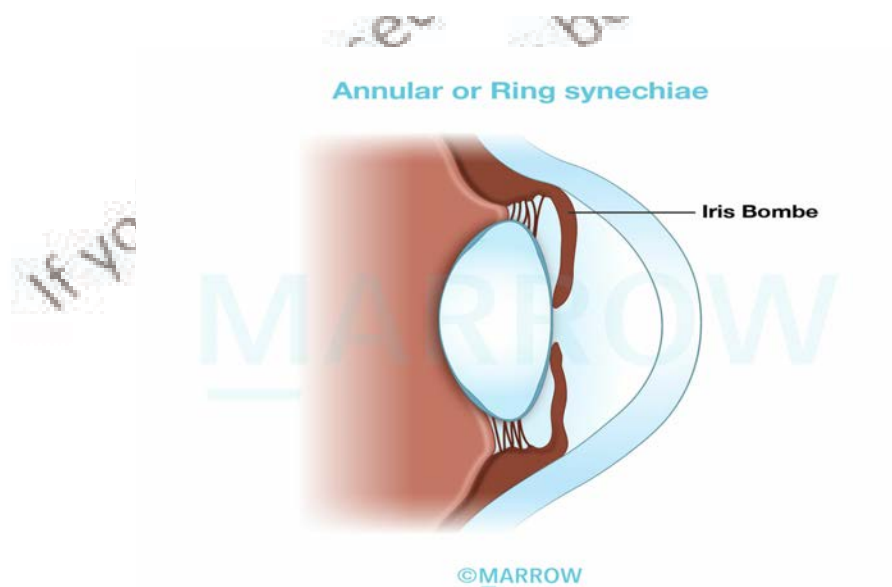
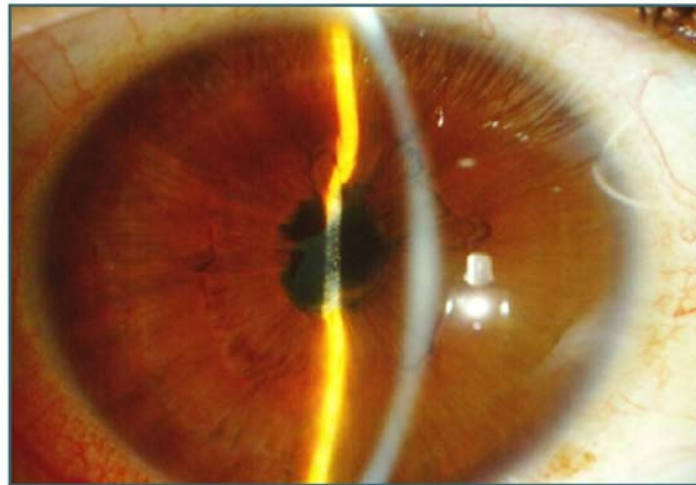


Solution to Question 9:

Iris bombe formation is due to the development of annular or ring synechiae.

Annular posterior/ring synechiae are 360° adhesions of the pupillary margin to the anterior capsule of the lens. They prevent the circulation of aqueous humor from the posterior chamber to the anterior chamber (seclusio pupillae). Thus, the aqueous collects behind the iris and pushes it anteriorly leading to iris bombe. This is usually followed by a rise in intraocular pressure. The anterior chamber is funnel-shaped when ring synechiae are present.

The image below shows iris bombe on slit lamp examination.



Solution to Question 10:

Fuchs heterochromic iridocyclitis is a type of non-granulomatous uveitis.

Fuchs heterochromic iridocyclitis is an insidious progressive anterior uveitis. It is mostly unilateral and associated with early cataract formation. Stellate keratic precipitates are seen and there is a notable absence of posterior synechiae.

Causes for chronic granulomatous uveitis include sarcoidosis, tuberculosis, herpes simplex, herpes zoster, syphilis, leprosy, brucellosis and phacoanaphylactic.

Solution to Question 11:

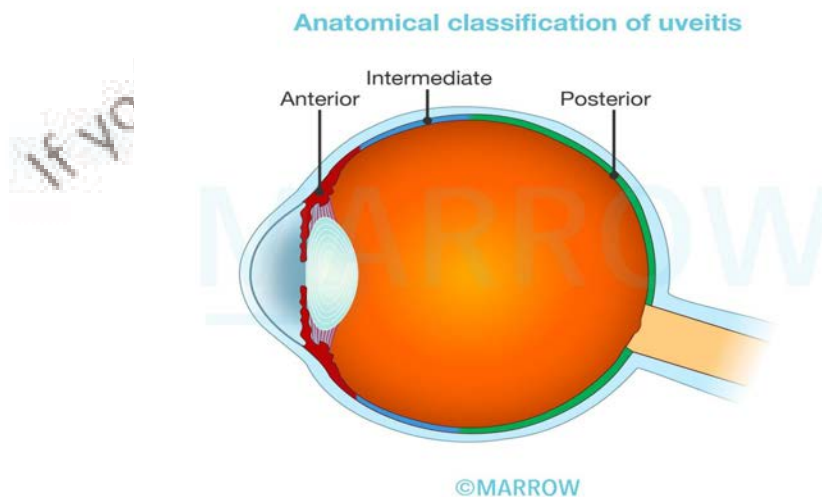
Intermediate uveitis refers to pars planitis. It involves inflammation of the pars plana.

The uvea consists of the iris, ciliary body and choroid. Inflammation of uveal tissue is called uveitis. It can be classified anatomically as follows:

- Anterior uveitis: Inflammation of iris and pars plicata, consisting of iritis, iridocyclitis and anterior cyclitis
- Intermediate uveitis: Inflammation of pars plana (pars planitis)
- Posterior uveitis: Inflammation of the retina with or without involvement of the choroid. It includes choroiditis, chorioretinitis, retinochoroiditis and neuroretinitis.
- Panuveitis

Pathological classification divides uveitis into suppurative and nonsuppurative uveitis. Nonsuppurative uveitis is further classified into nongranulomatous and granulomatous uveitis.

The image is a schematic representation of the anatomical classification of uveitis.



Solution to Question 12:

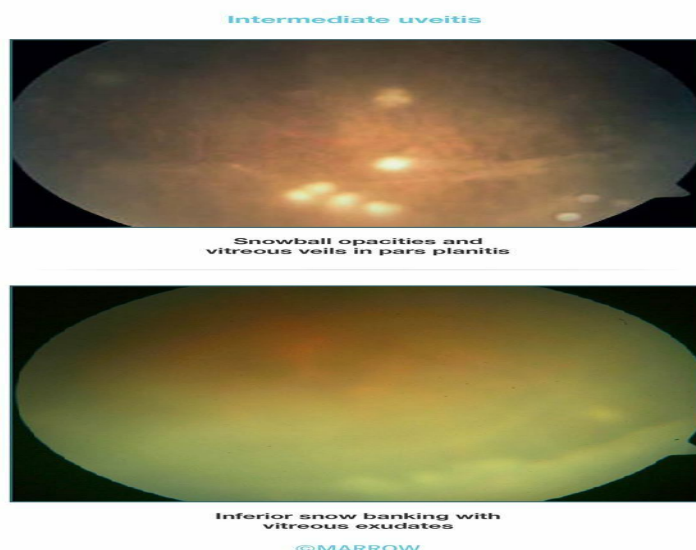
Snow banking is the hallmark of intermediate uveitis.

Snow banking refers to greyish white fibrovascular plaques over the pars plana, with or without peripheral retinal periphlebitis.

Intermediate uveitis is the inflammation of the pars plana and presents with retrolental flare, few keratic precipitates, vitreous cells and snow-ball opacities (Ant's egg opacities). These are typical exudates present near the ora serrata in the inferior quadrant.

Treatment of intermediate uveitis is by the Kaplan 4-step protocol:

- Periocular steroid: Sub tenon injection of triamcinolone acetate
- Systemic steroid
- Cryotherapy over snow banking
- Pars plana vitrectomy



Solution to Question 13:

ANA-positive oligoarticular juvenile idiopathic arthritis (JIA) is associated with the highest risk for iridocyclitis (uveitis).

A positive antinuclear antibody (ANA) test result confers increased risk for uveitis in children with JIA and requires periodic slit-lamp examination. Uveitis in JIA is characteristically chronic, non-granulomatous, and bilateral. It presents as a white eye without congestion- white uveitis. Complications include complicated cataracts, secondary glaucoma, and band-shaped keratopathy.

Note: Intraocular lens (IOL) implantation can be considered in most pediatric patients with well-controlled uveitis, but JIA-associated uveitis should be considered as a relative contraindication.

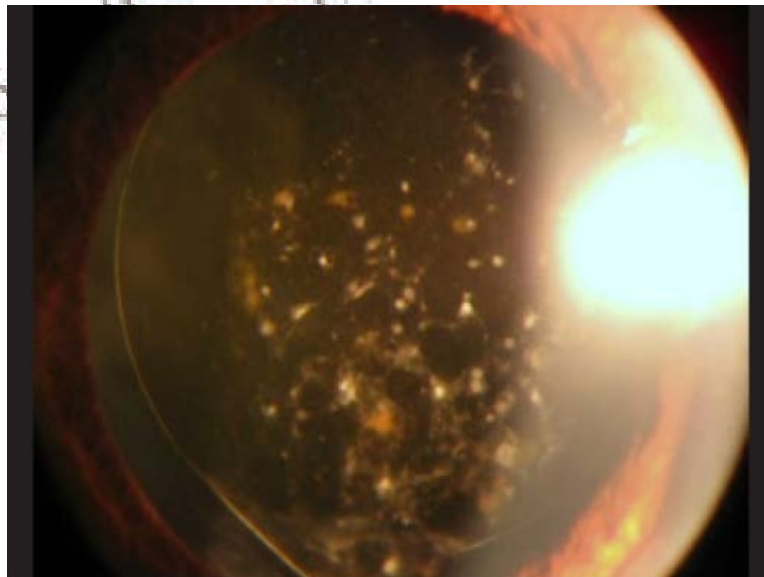
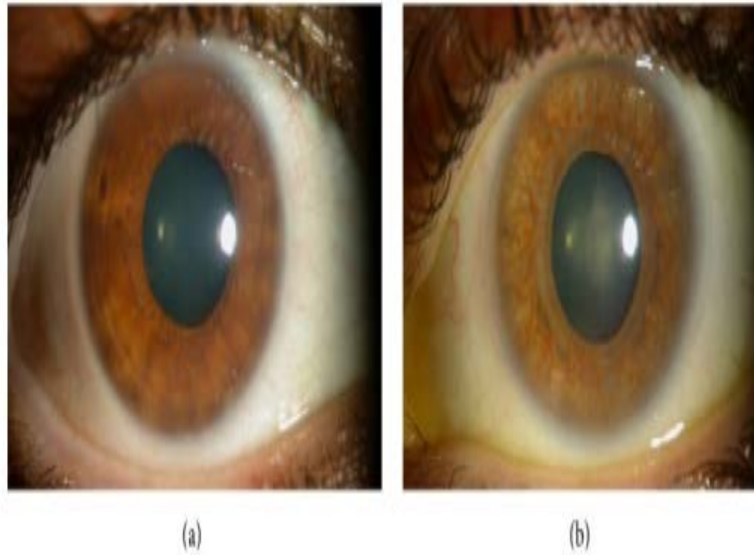
Solution to Question 14:

Amsler sign is seen in Fuch's heterochromic iridocyclitis (FHI).

Amsler sign is the hemorrhage caused by cataract surgery in FHI. During cataract surgery, fine filiform hemorrhages from the opposite angle are seen to occur as soon as the anterior chamber is opened. This is due to neovascularization.

Fuchs heterochromic iridocyclitis is an insidiously progressive, non- granulomatous anterior uveitis, that is usually unilateral. Stellate KPs are seen and tiny crystals (Russell bodies) may be present on the iris surface. There are no posterior synechiae. Rubeosis iridis may be present.

The 1st image shows Fuchs heterochromic uveitis. The right eye is normal. The left hypochromic eye is affected. The 2nd image shows stellate keratic precipitates.



Solution to Question 15:

Posner–Schlossman syndrome is characterized by unilateral glaucomatous changes with mild anterior uveitis.

This syndrome, also known as glaucomatocyclitic crisis or hypertensive uveitis, presents with recurrent unilateral attacks of acute raised intraocular pressure (IOP) (>40 mmHg) changes, fine white central keratic precipitates (KPs) and mydriasis. Treatment includes topical corticosteroids, cycloplegics, and aqueous suppressants.

Solution to Question 16:

Rifabutin causes uveitis in AIDS patients.

Causes of drug-induced uveitis include:

- Cidofovir
- Rifabutin
- Bisphosphonates
- Oral moxifloxacin

Solution to Question 17:

Behcet's disease presents with recurrent iridocyclitis associated with transient hypopyon.

Behcet's disease is a systemic vascular disease. It is associated with HLA B5 & B51. Patients have obliterative vasculitis due to circulating immune complexes, leading to a recurrent, bilateral non-granulomatous uveitis. There is acute recurrent iridocyclitis, transient hypopyon, periphlebitis, retinitis, vitritis and cystoid macular edema (CME).

Solution to Question 18:

The clinical scenario of photophobia and blurring of vision along with red-eye and irregularly shaped pupil suggests a diagnosis of anterior uveitis. Steroids and cycloplegics are the mainstay of the treatment of uveitis.

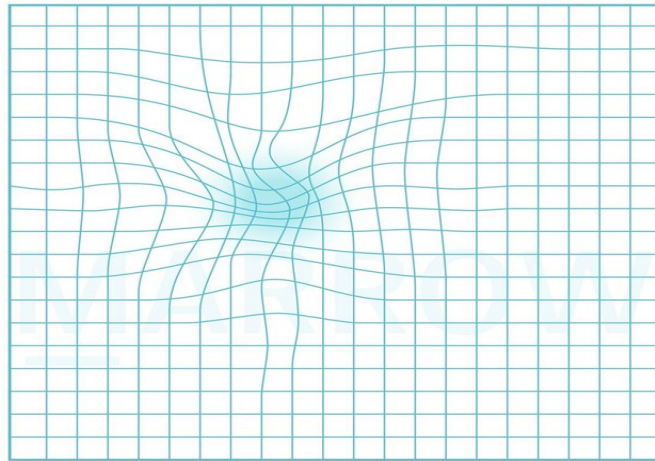
Steroids are the drug of choice in anterior uveitis. They reduce inflammation by their anti-inflammatory effect. Cycloplegics like 1% atropine relieve spasm of the iris sphincter and ciliary muscles. They prevent the formation of synechiae and reduce exudation. Systemic therapy includes steroids, NSAIDs, and immunosuppressants.

Immunosuppressants like voclosporin (calcineurin inhibitor) are used in severe cases. They are being studied under a program called LUMINATE.

Uveitis - Posterior and Panuveitis

Question 1:

A patient describes the image below on the Amsler grid test. In which of the given conditions is this phenomenon noted?



©MARROW

- a) Anterior uveitis
- b) Posterior uveitis
- c) Cataract
- d) Glaucoma

Question 2:

Which of the following features is unlikely to be noted in a patient with Vogt-Koyanagi-Harada syndrome?

- a) Vitiligo
- b) Vertigo
- c) Granulomatous uveitis
- d) Iris heterochromia

Question 3:

A middle aged male presents with recurrent oral ulcers and ulcers over his scrotum. He also has pain and redness of both his eyes. A positive pathergy test is noted. What is the HLA association of the disease this patient is most likely to be suffering from?

- a) HLA B27
- b) HLA DR1
- c) HLA B5
- d) HLA Bw54

Question 4:

A 28 year old man presents to the medicine OPD with cough with hemoptysis of 1 month duration. A sputum culture done is shown below. What is the most common ocular manifestation associated with this infection?



- a) Uveitis
- b) Phlyctenular keratoconjunctivitis
- c) Complicated cataract
- d) Vitreous haemorrhage

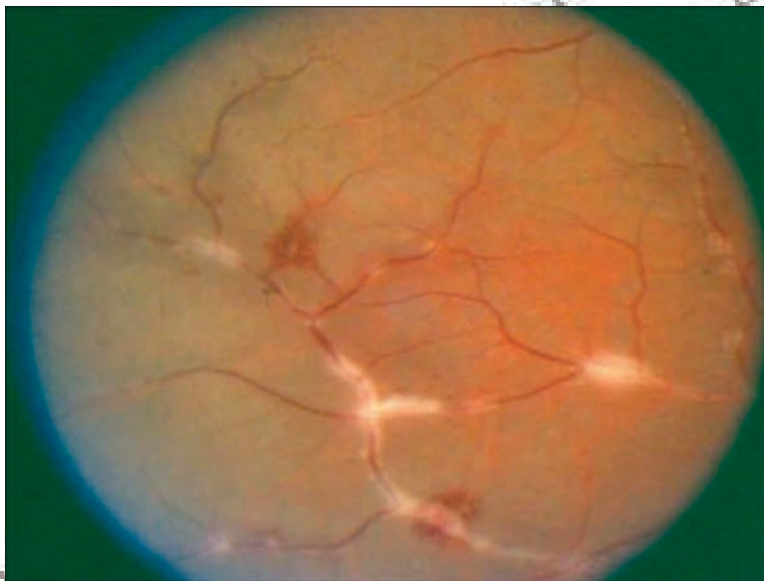
Question 5:

In a patient showing salt and pepper appearance of the fundus, which of the following is not a differential diagnosis?

- a) Rubella
- b) Syphilis
- c) Refsum's disease
- d) Tuberculosis

Question 6:

A 32-year-old lady presented with dry, itchy eyes, blurred vision, and sensitivity to light. She says there are black dots floating around in her visual field. She denies wearing any spectacles or contact lenses. On fundoscopy, the following image is obtained. What is the diagnosis?



- a) CMV retinitis
- b) Sarcoidosis
- c) Toxoplasmosis
- d) Behcet's disease

Question 7:

What is the most common ocular manifestation in patients infected by the human immunodeficiency virus?

- a) Microangiopathy
- b) Complicated cataract
- c) Vitreous hemorrhage

d) Choroiditis

Question 8:

An HIV positive patient presented to OPD with painless blurring of vision. On fundus examination, the following findings were seen. What is the diagnosis?



- a) Eales disease**
- b) CMV retinitis**
- c) Toxoplasmosis**
- d) Sympathetic ophthalmia**

Question 9:

A newly diagnosed HIV-positive lady with CD4+ count of 35/ μ L presented with decreased visual acuity and floaters. On fundus examination, there were dense, white areas of retinal necrosis with interspersed areas of retinal hemorrhage. What is the most rapid and accurate method to confirm the diagnosis?

- a) Virus isolation in intraocular fluid**
- b) Virus antigen detection in vitreous**
- c) Viral nucleic acid detection in intraocular fluid**
- d) Viral antibody detection in blood by ELISA**

Question 10:

A 32-year-old man presented with blurring of vision in the right eye. Examination reveals mild iritis, severe vitritis, and a focal necrotic lesion on the macula. What is the most likely diagnosis?

- a) Multiple evanescent white dot syndrome
- b) Ocular toxoplasmosis
- c) Multifocal choroiditis
- d) Ocular sarcoidosis

Question 11:

In which disease is the "headlight in fog" appearance seen?

- a) Toxoplasmosis
- b) Syphilis
- c) HIV
- d) CMV

Question 12:

A 45-year-old man presented with red-eye, ocular pain, and acute loss of vision to the casualty. On fundoscopy, areas of retinal ischemia and necrosis were seen, leading to a diagnosis of acute retinal necrosis. Which of the following statements is false regarding this condition?

- a) Severe arteritis is common.
- b) It is seen in immunocompetent individuals.
- c) Posterior pole is involved initially with centrifugal spread.
- d) It may lead to retinal detachment.

Question 13:

A 65-year-old male presents with a painful decrease in vision of his right eye for the past 3 days. He underwent right phacoemulsification with PC-IOL insertion 6 months ago. Examination revealed a mild granulomatous anterior uveitis with minimal hypopyon and mild vitritis. There is a white plaque seen on the posterior lens capsule. Which one of the

following is the most appropriate therapy for this condition?

- a) Vitrectomy and intravitreal injection of amphotericin B
- b) Observation
- c) Vitrectomy, removal of the IOL, and intravitreal injection of gentamicin
- d) Vitrectomy, posterior capsulotomy, and intravitreal injection of vancomycin

Question 14:

Sympathetic ophthalmia is seen after _____.

- a) Penetrating trauma
- b) Blunt trauma
- c) Chemical injury
- d) Urinary tract infection

Question 15:

What is the most common cause of endophthalmitis in a patient following eye trauma?

- a) Staphylococcus aureus
- b) Staphylococcus epidermidis
- c) Aspergillus
- d) Bacillus cereus

Question 16:

Which of the following drugs should not be used in a patient with uveitis and raised IOP?

- a) Atropine
- b) Timolol
- c) Pilocarpine
- d) Acetazolamide

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	a
5	d
6	b
7	a
8	b
9	c
10	b
11	a
12	c
13	d
14	a
15	b
16	c

Detailed Explanations

Solution to Question 1:

The image shows metamorphopsia seen in posterior uveitis.

Metamorphopsia is a condition in which patients perceive distorted images of the object. It is measured using the Amsler grid, which tests the central 10 degrees of vision. In posterior uveitis, metamorphopsia occurs due to macular edema.

Posterior uveitis is a painless inflammation of the choroid. Patients present with various visual symptoms like defective vision, black spots floating in front of the eye, micropsia/macropsia and positive scotomas.

Solution to Question 2:

Iris heterochromia is not a feature of Vogt–Koyanagi–Harada syndrome.

Vogt–Koyanagi–Harada syndrome is an idiopathic, multi-system disorder characterised by autoimmunity against melanocytes. It is associated with HLA-DR1 and HLA-DR4 across different racial groups.

Patients may present with cutaneous, neurological and ocular features. Cutaneous features include alopecia, poliosis and vitiligo (option A). Neurological features include meningism, encephalopathy, tinnitus, deafness and vertigo (option B). Ocular features include bilateral chronic granulomatous panuveitis (option C), posterior uveitis, exudative retinal detachment, perilimbal vitiligo (Suguiura's sign) and sunset glow appearance of the fundus.

The modified diagnostic criteria for Vogt–Koyanagi–Harada syndrome are as follows:

- Absence of a history of penetrating ocular trauma
- Absence of other ocular disease entities
- Bilateral uveitis
- Neurological and auditory manifestations
- Integumentary findings, not preceding the onset of the central nervous system or ocular diseases, such as alopecia, poliosis, and vitiligo.

The treatment is high-dose (1–2 mg/kg/day) oral prednisolone, tapered over 3–6 months. This may be preceded by intravenous methylprednisolone pulse therapy (500–1000 mg/day).

Solution to Question 3:

Recurrent oral ulcers, ulcers over his scrotum with binocular pain and redness and a positive pathergy test point to a diagnosis of Behcet's disease. HLA B5 (HLA B51) is associated with Behcet's disease.

Behcet's disease is a systemic vasculitis that can cause recurrent oral ulcers, genital ulcers, skin lesions and ocular inflammation. Pathergy test is usually positive. Ocular inflammation occurs in about 70% of patients and tends to be more severe in men. A classical presentation is relapsing/remitting acute onset panuveitis with retinal vasculitis which resolves spontaneously even without treatment. It is usually bilateral and non-granulomatous. Patients present with a transient mobile hypopyon in a relatively white eye. They may develop visual impairment if retinal vascular disease (vasculitis and occlusion) occurs.

Other options:

Option A: HLA-B27 is associated with Reiter's syndrome.

Option B: HLA DR1 and HLA-DR4 are associated with Vogt–Koyanagi–Harada's disease.

Option D: HLA Bw54 is associated with Possner–Schlossman syndrome.

Note: The association between HLA-A29 and birdshot retinochoroidopathy is the most consistent of any medical disorder. Approximately 95% of patients with birdshot retinochoroidopathy are HLA-A29-positive.

Solution to Question 4:

The image shows the growth of Mycobacterium tuberculosis (MTB) colonies on the Lowenstein-Jensen medium. In tuberculosis (TB), granulomatous panuveitis is the most common

ocular feature.

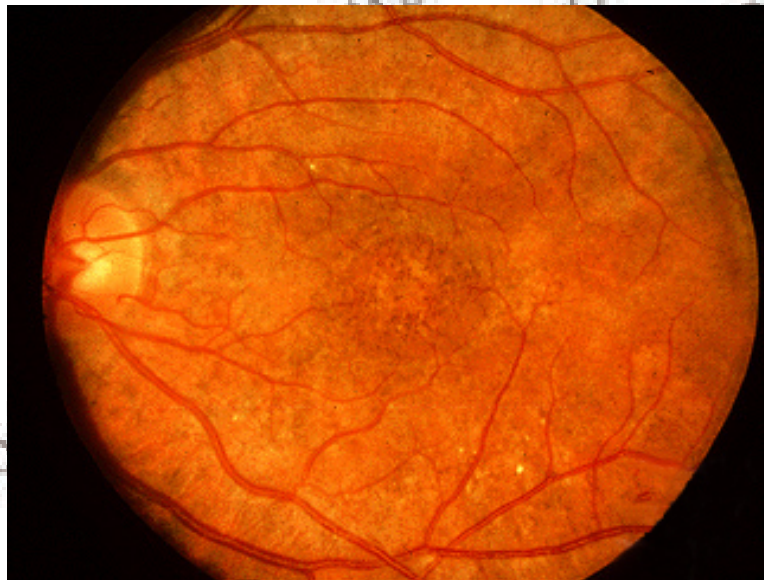
Ocular TB commonly occurs due to hematogenous spread from pulmonary TB. It can have multiple presentations like lupus vulgaris of eyelid, tubercular conjunctivitis, interstitial keratitis, granulomatous uveitis and endophthalmitis. Phlyctenular keratoconjunctivitis is caused by a hypersensitivity reaction to the MTB antigen.

Treatment of granulomatous panuveitis is anti-tubercular therapy (ATT) plus steroids.

Solution to Question 5:

Salt and pepper fundus is not seen in tuberculosis.

Salt and pepper fundus is seen when there are multiple small black and white spots present on the retina. It is also known as pseudo-retinitis pigmentosa. Its causes include congenital syphilis, rubella, Refsum's disease, carriers of albinism, Leber's congenital amaurosis and retinitis pigmentosa sine pigmentosa.



Solution to Question 6:

Dry, itchy eyes, blurred vision, sensitivity to light with candle wax dripping on fundus examination points to a diagnosis of sarcoidosis.

Sarcoidosis is a chronic granulomatous multi-system disease. Ocular inflammation occurs in 25–70% of sarcoid patients, which includes uveitis, conjunctival and lacrimal gland involvement. Chronic granulomatous panuveitis may be seen. Anterior uveitis is the most common manifestation.

In anterior uveitis due to sarcoidosis, "mutton fat" keratic precipitates (KPs) and/or small granulomatous KPs with iris nodules (Koeppe and/or Busacca) are seen. Trabecular meshwork (TM) nodules and/or tent-shaped peripheral anterior synechiae (PAS) is also seen.

In intermediate uveitis, vitreous cells with snowbanking and snowball opacities/string of pearl appearance are seen. Posterior uveitis presents with candle-wax dripping (thick venous sheathing and periphlebitis), pre-retinal nodules (Lander's sign), choroidal granulomata and sarcoid infiltration. Other ocular features seen are interstitial keratitis and band-shaped keratopathy. Steroids are the mainstay of treatment in sarcoidosis.



Solution to Question 7:

Microangiopathy is the most common ocular feature of HIV infection.

Microangiopathy of the retina in HIV consists of the hemorrhages, micro-aneurysms and cotton-wool spots or soft exudates (axonal debris). Cotton wool spots are the most common finding of HIV retinopathy.

HIV leads to multiple opportunistic infections of the eye like cytomegalovirus (CMV) - most common cause of posterior uveitis in HIV - toxoplasmosis, Pneumocystis carinii and herpes zoster. These infections present as keratitis, choroiditis, retinitis and endophthalmitis.

The most common ocular adverse effect in patients receiving highly active antiretroviral therapy (HAART) is immune recovery uveitis (IRU). It is a non-granulomatous uveitis (usually anterior uveitis and vitritis) associated with a paradoxical or exaggerated reactivation of the immune system and an increase in intraocular inflammation. It occurs after receiving HAART for CMV chorioretinitis. Visual complications of IRU include cataract, cystoid macular edema (CME), glaucoma, and epiretinal membrane formation.

Solution to Question 8:

Painless blurring of vision in an HIV positive patient with "sauce and cheese" or "pizza pie" retinopathy (shown in the image) is characteristic of cytomegalovirus (CMV) retinitis.

CMV retinitis is the most common ocular opportunistic infection in patients with HIV. It is often asymptomatic and can be classified as central or peripheral CMV retinitis. Central CMV retinitis is characterised by dense, white areas of retinal necrosis and retinal hemorrhage in the areas of necrosis - pizza pie or sauce and cheese appearance.

Peripheral CMV retinitis is less demarcated. The infection spreads along the vascular arcades in a brush fire-like fashion.

Solution to Question 9:

An HIV-positive patient with CD4+ count of $35/\mu\text{L}$ presenting with decreased visual acuity, floaters and pizza pie retinitis (dense, white areas of retinal necrosis with interspersed areas of retinal hemorrhage) likely has CMV retinitis. The most rapid and accurate method to diagnose cytomegalovirus (CMV) retinitis is viral nucleic acid detection in the intraocular fluid.

CMV retinitis is an opportunistic infection seen in HIV patients with a CD4+ count less than $50/\mu\text{L}$. Patients present with reduced vision due to macular involvement and floaters due to vitritis. One eye is usually affected initially. Retinitis is seen. Frosted branch angiitis (FBA), which is marked vascular sheathing that occurs in about 6% patients, may be seen. Cataract is common late-stage manifestation of CMV retinitis.

Highly active antiretroviral therapy (HAART) is the mainstay of management, restoring the patient's innate ability to suppress CMV activity. Valganciclovir is a prodrug of ganciclovir that is taken orally. Ganciclovir slow-release intravitreal implant can also be used.

Screening of patients with low CD4+ counts:

- 3-monthly, if $<50/\mu\text{L}$
- 6-monthly, if $50-100/\mu\text{L}$
- Yearly, if $>100/\mu\text{L}$

Solution to Question 10:

Monocular iritis, severe vitritis with a focal necrotic lesion on the macula is a classical description of ocular toxoplasmosis.

Ocular toxoplasmosis is caused by *Toxoplasma gondii*, an obligate intracellular protozoan. It causes severe recurrent attacks of posterior uveitis in adults, due to reactivation at previously inactive cyst-containing scars. A single inflammatory focus of fluffy white retinitis or retinochoroiditis associated with a pigmented scar ("satellite lesion") is typical. "Headlight in the fog" is the classic description of a white retinal inflammatory nidus viewed through vitritis. "Spill-over" anterior uveitis is common.

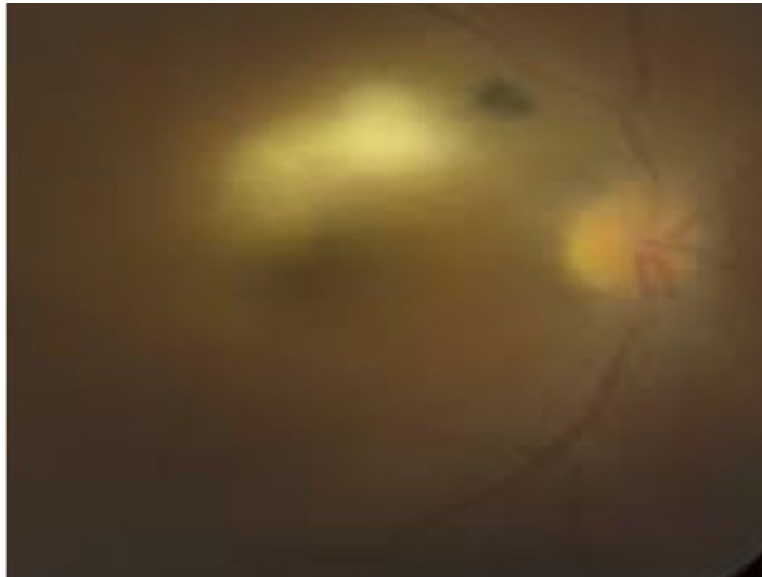
Treatment:

- Prednisolone (1 mg/kg) is given initially and tapered according to clinical response.
- Pyrimethamine is administered as a loading dose of 75–100 mg for 1–2 days followed by 25–50 mg daily for 4 weeks in combination with oral folinic (not folic) acid 5 mg three times a week to

retard thrombocytopenia, leukopenia, and folate deficiency.

- Sulfadiazine 1 g four times daily for 3–4 weeks is usually given in combination with pyrimethamine.
- Intravitreal therapy with clindamycin (1 mg) and dexamethasone (400 µg) may be as effective as triple therapy in reactivated infection.

The image shows the characteristic "headlight in the fog" appearance:

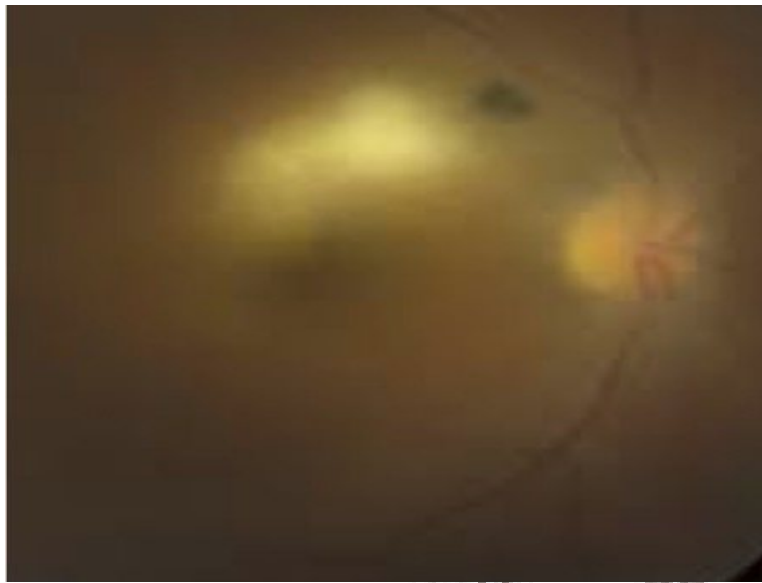


Solution to Question 11:

The "headlight in fog" appearance is seen in toxoplasmosis.

Ocular toxoplasmosis presents with iridocyclitis, which can be granulomatous or non-granulomatous. There is focal superficial necrotizing retinochoroiditis along with vitritis, which gives the "headlight in fog" appearance (image). Fundoscopy shows deep retinitis and papillitis with granulomas. Old choroid lesions of toxoplasmosis are punched-out lesions with pigmented borders.

The treatment of choice for ocular toxoplasmosis is triple therapy with systemic steroids, pyrimethamine, and sulfadiazine. Clindamycin is also used.



Solution to Question 12:

In acute retinal necrosis, the posterior pole is typically spared until late.

Acute retinal necrosis (ARN) occurs in immunocompetent individuals. It is a severe posterior uveitis causing retinal necrosis. Herpes zoster ophthalmicus is the most common cause of acute retinal necrosis in adults. Cytomegalovirus (CMV) and herpes simplex virus (HSV) are the commonest causes of acute retinal necrosis in children.

Patients present with confluent peripheral necrotizing retinitis, retinal arteritis with vaso-occlusion, and vitritis. Rhegmatogenous retinal detachments are common due to retinal thinning from large areas of necrosis. The posterior pole is involved only in the later stages.

Aciclovir is administered intravenously (10 mg/kg, every 8 hours) for 10–14 days, and then orally (800 mg, five times daily) for 6–12 weeks. Oral valaciclovir or famciclovir may be substituted for oral aciclovir, with similar outcomes but better tolerability. Intravitreal ganciclovir or foscarnet may enhance the prognosis. Systemic steroids may be started 24 hours after the initiation of antiviral therapy, especially in severe cases. Laser retinopexy is performed around necrotic areas.

Progressive outer retinal necrosis is a manifestation of HSV or varicella zoster virus (VZV) that occurs in immunocompromised individuals. It may begin at the posterior pole or the periphery. It has a rapid course and is not generally associated with vasculitis.

Solution to Question 13:

Painful decrease in vision following cataract surgery 6 months ago with granulomatous anterior uveitis, hypopyon, vitritis and a white plaque on the posterior lens capsule suggests a diagnosis of delayed-onset postoperative endophthalmitis caused by *Propionibacterium acnes*. The best management for the above clinical scenario would be vitrectomy and posterior capsulotomy, followed by intravitreal injection of vancomycin.

Delayed-onset endophthalmitis, a complication of cataract surgery, has a slow course and panuveitis like picture. There is low grade anterior chamber inflammation and vitritis. When caused by *P. acnes*, it typically presents from 3 months to 2 years after surgery with a characteristic white retro-lenticular plaque on the posterior capsule of lens. Endophthalmitis caused by *Candida* occurs within 3 months of surgery.

Management includes anterior vitrectomy, IOL explantation, posterior capsulectomy, intravitreal vancomycin, and a scleral-fixated intraocular lens (IOL) implantation.

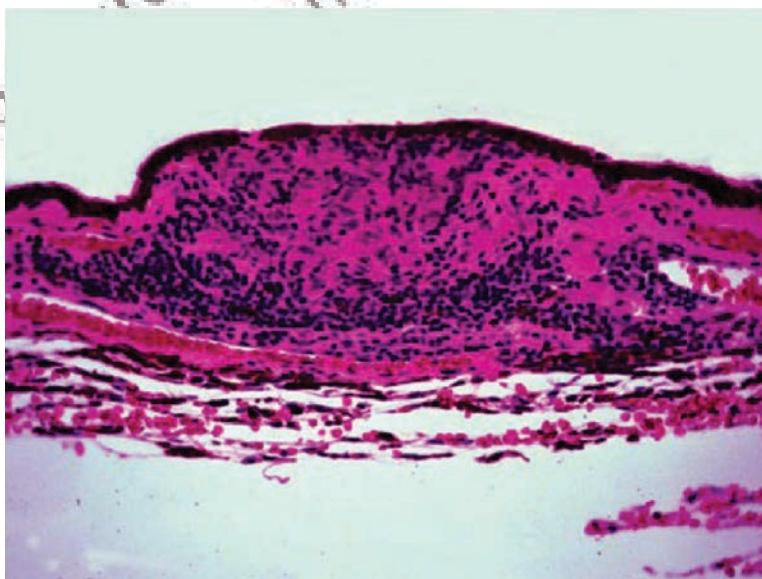
Solution to Question 14:

Sympathetic ophthalmia is seen after a penetrating trauma.

Sympathetic ophthalmia is a bilateral granulomatous panuveitis seen after a penetrating trauma or less frequently, following intraocular surgery, usually multiple vitreoretinal procedures. Patients present between 2 weeks and 3 months after initial injury.

The earliest symptom is blurred near vision in the sympathizing (uninjured) eye. The earliest sign is a retrolental flare in the sympathizing eye. Anterior uveitis develops in both eyes. The fundus shows multifocal choroidal infiltrates in the mid-periphery. Sub-retinal pigment epithelium (RPE) infiltrates corresponding to Dalen–Fuchs nodules (image)(a granuloma situated between Bruch membrane and the RPE) are also present.

Steroids are the basis of treatment. High-dose oral prednisolone is given for several months. Immunosuppressants such as azathioprine, cyclosporine, and methotrexate can be used in resistant cases or as steroid-sparing agents. Enucleation of a severely injured eye with no visual potential within 2 weeks of injury prevents the occurrence of sympathetic ophthalmia.



Solution to Question 15:

Staphylococcus epidermidis is the most common cause of post-traumatic endophthalmitis, followed by Bacillus cereus.

However, if the trauma involves soil contamination, Bacillus cereus is the most common cause of post-traumatic endophthalmitis.

Solution to Question 16:

Pilocarpine should not be used in cases of uveitis with raised intraocular pressure.

Hypertensive uveitis or inflammatory glaucoma occurs when uveitis is associated with raised IOP. It is treated with topical cycloplegics (atropine), topical steroids and IOP-lowering drugs like topical timolol or systemic acetazolamide. Pilocarpine is contraindicated since it causes miosis and ciliary spasm. It also increases intraocular inflammation.

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Disorders of the Eyelid

Question 1:

Which of the following is defined as loss of eyelashes?

- a) Tylosis
- b) Madarosis
- c) Trichiasis
- d) Distichiasis

Question 2:

What is the term used to describe an extra row of eyelashes behind the grey line?

- a) Trichiasis
- b) Distichiasis
- c) Tylosis
- d) Madarosis

Question 3:

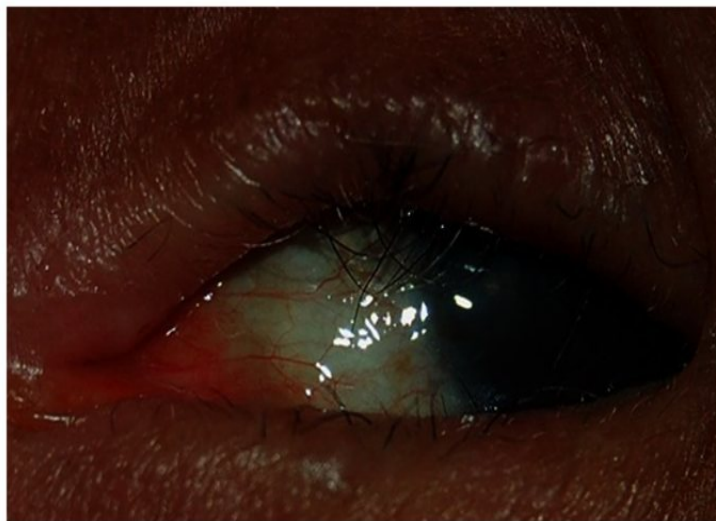
A term neonate is referred to the district hospital for an ophthalmic exam. Partial fusion of her right upper and lower eyelids by a central narrow band of tissue is noted as shown below. The left eye examination is normal. What is the most likely diagnosis?



- a) Symblepharon
- b) Ankyloblepharon
- c) Blepharophimosis
- d) Lagophthalmos

Question 4:

A male patient complains of a foreign body sensation, irritation, and lacrimation in the right eye. During the evaluation, you notice the following findings with normal lid margins. What is this condition known as?



- a) Distichiasis

- b) Trichiasis
- c) Pseudotrichiasis
- d) Pseudodistichiasis

Question 5:

What defines telecanthus?

- a) Widened inter-pupillary distance
- b) Widely separated medial orbital wall
- c) Widened inter-orbital distance
- d) Widened root of the nose with normal inter-pupillary distance

Question 6:

Weiss operation is done in which of the following condition?

- a) Entropion
- b) Ectropion
- c) Trichiasis
- d) Ptosis

Question 7:

For which of the following is medial conjunctivoplasty the surgical modality of treatment?

- a) Entropion
- b) Ectropion
- c) Blepharophimosis
- d) Ptosis

Question 8:

Which among the following is not a feature of blepharophimosis syndrome?

- a) Narrow palpebral fissure
- b) Ptosis

- c) Telecanthus
- d) Entropion

Question 9:

The physician noticed the following finding in a patient that came to the OPD for an eye exam. Which of the following pair of muscles are likely to be affected in this patient?



- a) Muller's muscle and superior rectus
- b) Levator palpebrae superioris and superior oblique
- c) Superior oblique and superior rectus
- d) Levator palpebrae superioris and Muller's muscle

Question 10:

Which of the following conditions is a cause of pseudoptosis?

- a) Lambert–Eaton syndrome
- b) Big chalazion
- c) Blepharophimosis syndrome
- d) Brow ptosis

Question 11:

A 45-year-old patient comes with complaints of progressive weakness as the day progresses. He is found to have antibodies against nicotinic acetylcholine receptors. Which is the most common ocular finding in this disease?

- a) Diplopia
- b) Lagophthalmos
- c) Exophthalmos
- d) Ptosis

Question 12:

Which of the following conditions will not have a finding of bilateral ptosis?

- a) Myasthenia gravis
- b) Marfan's syndrome
- c) Myotonic dystrophy
- d) Kearns-Sayre syndrome

Question 13:

During OPD hours, a patient came with the finding shown below. On examination, the upper 4 mm of the cornea is covered by the upper eyelid. What is the grade of ptosis in this patient?



- a) Mild
- b) Moderate

- c) Severe
- d) Profound

Question 14:

A child with a high risk of amblyopia is planned for surgical correction of unilateral simple congenital ptosis. He has poor levator function. What is the treatment of choice in this case?

- a) Levator resection
- b) Frontalis sling
- c) Fasanella-Servat operation
- d) Mullerectomy

Question 15:

A patient presents to you with anhidrosis and enophthalmos. Examination reveals miosis along with mild ptosis. You decide to correct the ptosis by performing a Fasanella-Servat operation, by resecting which of the following muscles?

- a) Levator palpebrae superioris muscle
- b) Frontalis muscle
- c) Superior rectus muscle
- d) Muller's muscle

Question 16:

A couple brought their 11-month-old child for an eye exam when they noticed that his left upper eyelid was in a lower position than his right one. The mother also noticed that his left upper eyelid goes "up and down" when she breastfeeds him. His delivery was unremarkable. When the baby opens his mouth or sucks his thumb, the left eyelid ptosis disappears. What is the pathology behind this condition?

- a) Superior division of the third cranial nerve is misdirected
- b) Sympathetic nerves to Muller's muscles are misdirected
- c) Ophthalmic division of the fifth cranial nerve is misdirected
- d) Mandibular division of the fifth cranial nerve is misdirected

Question 17:

Which of the following is the causative agent for blepharitis acarica?

- a) Moraxella
- b) Staphylococcus
- c) Propionibacterium
- d) Demodex folliculorum

Question 18:

What is the most common malignant tumor of the eyelid?

- a) Sebaceous gland carcinoma
- b) Basal cell carcinoma
- c) Squamous cell carcinoma
- d) Malignant melanoma

Question 19:

Which of the following systemic drugs is used specifically in the management of periocular basal cell carcinoma?

- a) Imiquimod
- b) Vismodegib
- c) Siponimod
- d) Interferon

Question 20:

A 6-month-old girl was brought by her worried parents due to a swelling on the eyelid as shown below. They were told by a local village doctor that the swelling will disappear on its own but recently it has been increasing rapidly. What would be the most likely management of this baby's condition?



- a) Reassurance and observation
- b) Treatment with intralesional steroids at 2 years of age
- c) Immediate treatment with radiotherapy
- d) Immediate treatment with oral propranolol

Question 21:

Identify the clinical condition.



- a) Marfan's syndrome
- b) Neurofibromatosis

- c) Tuberous sclerosis
- d) Mikulicz's syndrome

Question 22:

Which of the following sign is used to describe the fullness of eyelids seen in Grave's ophthalmopathy?

- a) Mobius' sign
- b) von Graefe's sign
- c) Stellwag's sign
- d) Enroth's sign

Answer Key

Question No.	Correct Option
1	b
2	b
3	b
4	b
5	d
6	a
7	b
8	d
9	d
10	d
11	d
12	b
13	a
14	b
15	d
16	d
17	d
18	b
19	b

20	d
21	b
22	d

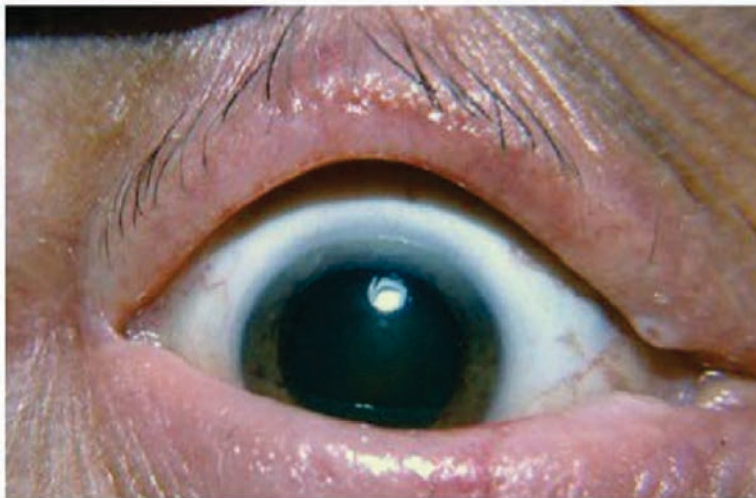
Detailed Explanations

Solution to Question 1:

Madarosis is the loss of eyelashes or eyebrows.

- Loss of eyebrows is seen in leprosy and myxedema.
- Loss of eyelashes is seen in chronic blepharitis.

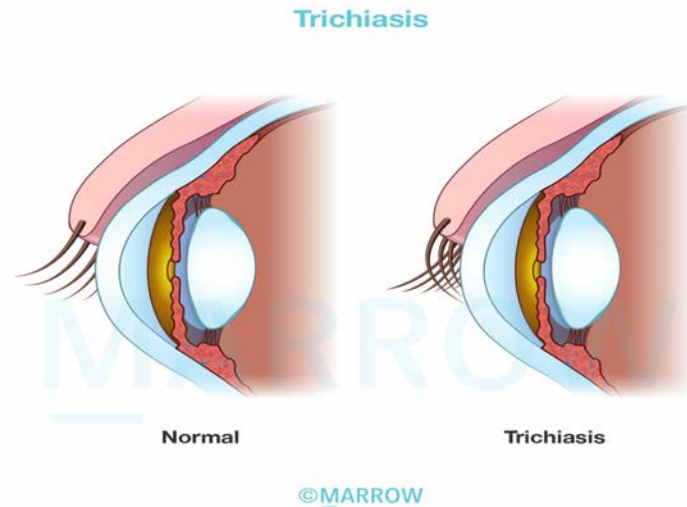
Madarosis



Other options:

Option A: Tylosis is a thickening of the tarsal border of the eyelid.

Option C: Trichiasis is an inward misdirection of cilia, with the normal position of the lid margin.



Option D: Distichiasis is the presence of a double row of eyelashes.

Distichiasis



Solution to Question 2:

The extra row of eyelashes behind the grey line is called distichiasis.

Distichiasis can occur due to congenital or acquired causes. These lashes may irritate the cornea. Management is by lid splitting with cryotherapy to the base of the hair follicles.

Other options:

Option A: Trichiasis is inward misdirection of cilia, with the normal position of the lid margin.

Option C: Tylosis is the thickening of the tarsal plate.

Option D: Madarosis is the loss of eyelashes or eyebrows.

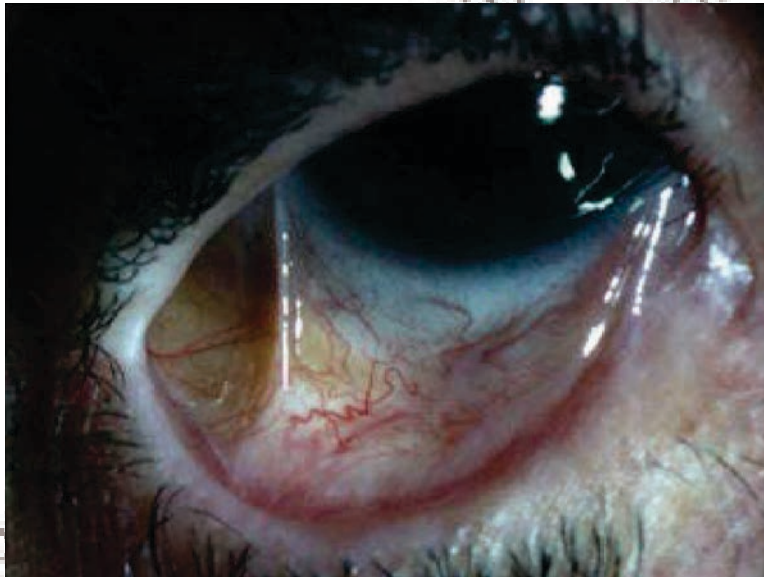
Solution to Question 3:

The given image shows an adhesion between the margins of the upper and lower eyelids, which suggests ankyloblepharon.

It may occur as a congenital anomaly or as acquired adhesions after healing of chemical burns, ulcers, or traumatic wounds of the lid margins. Treatment is the excision of the adhesions between lid margins. In severe cases where the adhesions involve the angle of the eyes, epithelial grafts can be used.

Other options:

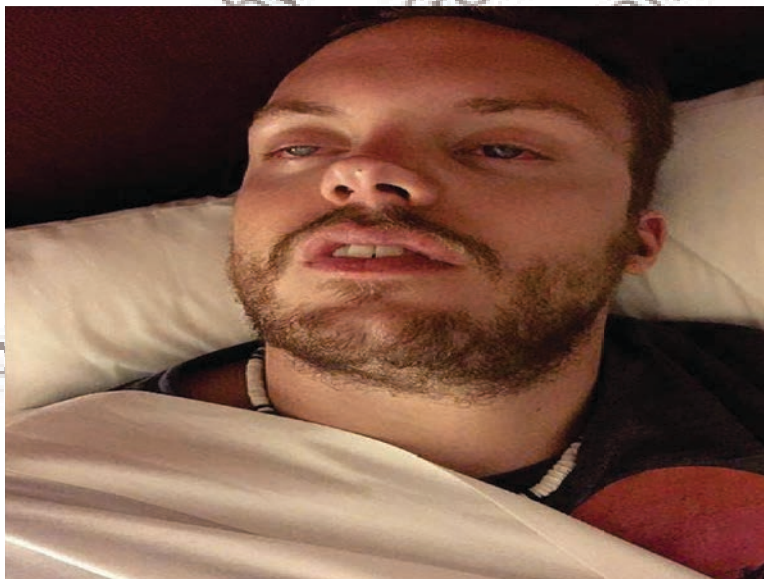
Option A: Symblepharon is the adhesion between the palpebral and bulbar conjunctiva, whereby the eyelids become adherent to the eyeball.



Option C: Blepharophimosis is a congenital anomaly in which the extent of the palpebral fissure is decreased. It appears contracted at the outer canthus.



Option D: Lagophthalmos is characterized by the inability to close the eyelids voluntarily. It may be due to a malfunction of the facial nerve. The image below shows nocturnal lagophthalmos (inability to close eyelids during sleep).



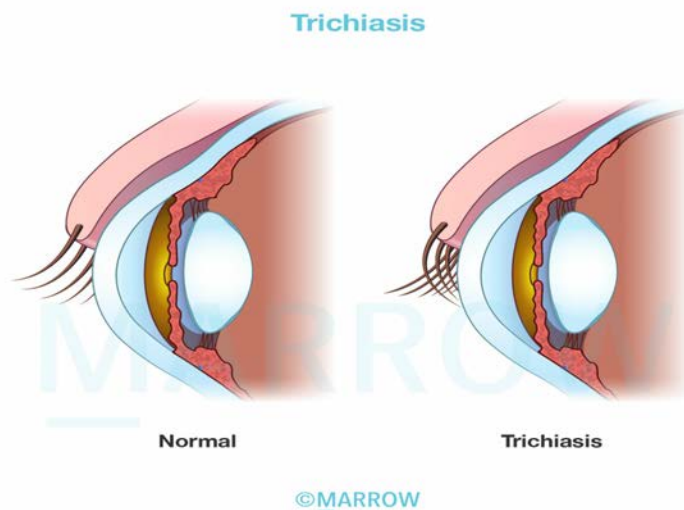
Solution to Question 4:

The given clinical scenario and the above image is suggestive of trichiasis.

Trichiasis refers to the inward misdirection of cilia, which usually rub against the eyeball with the normal position of the lid margin. These patients present with foreign body sensations and photophobia. They experience troublesome irritation, pain, and lacrimation. On examination, one or more misdirected cilia touching the cornea are seen. The conjunctiva may be congested.

Treatment includes:

- Epilation: Mechanical removal with forceps
- Electrolysis: Method of destroying the lash follicle by electric current.
- Cryoepilation
- Surgical correction.

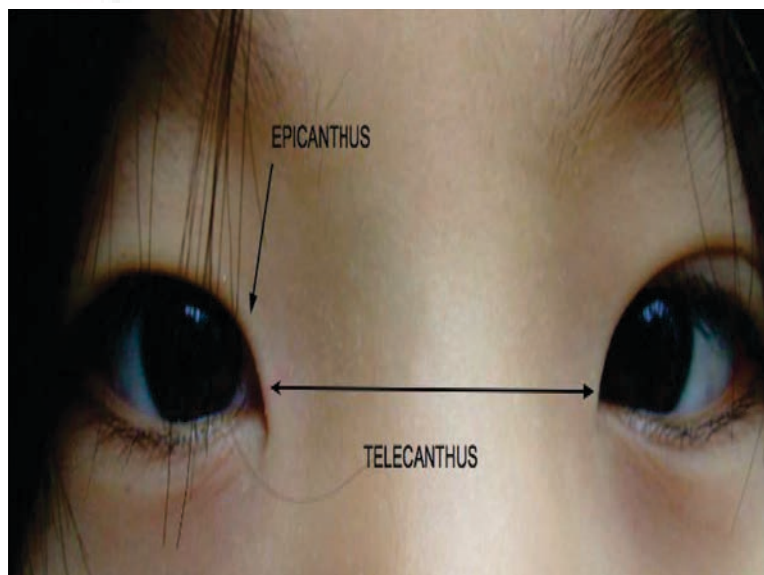


Solution to Question 5:

Telecanthus means an increased distance between medial canthi (widened root of the nose) with normal inter-pupillary distance.

In hypertelorism, there is a wide bony separation of the orbits.

In epicanthus, there are bilateral vertical folds of skin that extend from the upper or lower lids towards the medial canthi. They may give rise to pseudoesotropia.



Solution to Question 6:

Weiss operation is done in senile entropion. In this operation, the transverse lid split, and everything sutures are placed.

Entropion refers to the inward rolling and rotation of the lid margin towards the globe. It is more common in the lower lid.

Causes include:

- Horizontal lid laxity
- Vertical lid instability
- Over-riding of the pretarsal by the preseptal orbicularis during lid closure
- Orbital septum laxity



Solution to Question 7:

Medial conjunctivoplasty is done in involutional ectropion.

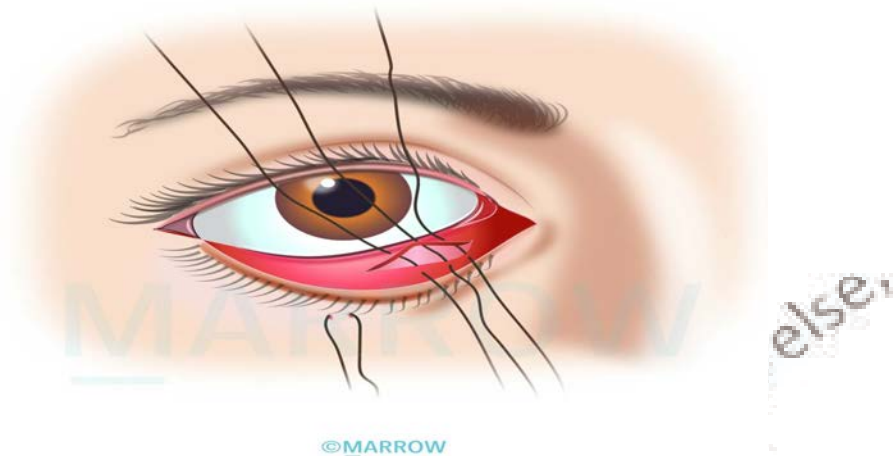
Ectropion is the eversion of the lid margin and eyelashes away from the globe.

Types:

- Involutional/senile - affects the lower lid of elderly patients.
- Cicatricial - caused by the scarring or contracture of the skin and underlying tissues
- Paralytic - caused by ipsilateral facial nerve palsy
- Mechanical - caused by tumors on or near the lid margin that mechanically evert the lid

- Congenital

Medial conjunctivoplasty for ectropion



Solution to Question 8:

Entropion is not a feature of blepharophimosis syndrome.

Blepharophimosis, ptosis, and epicanthus inversus syndrome (BPES) consists of the following:

- Narrow palpebral fissure
- Ptosis
- Congenital ectropion of the lower lid
- Telecanthus
- Epicanthus inversus: an extra fold of skin arising from the lower lid

The image below shows blepharophimosis.



Solution to Question 9:

The patient has mild right eye ptosis. Injury or weakness of levator palpebrae superioris and Muller's muscle leads to ptosis.

Levator palpebrae superioris is the important upper eyelid retractor. Frontalis and Muller's muscle assist in the elevation of the upper eyelid.

Solution to Question 10:

Brow ptosis is a type of pseudoptosis, not ptosis.

Ptosis is abnormal drooping of the upper eyelid to a level that covers more than 2mm of the superior cornea.

Types of ptosis:

- Mechanical ptosis - Can be due to lid tumors, multiple/big chalazions, and conjunctival scarring
- Aponeurotic ptosis - Due to dehiscence of the aponeurosis and in postoperative patients
- Involutional ptosis seen in old age
- Neurogenic ptosis due to a 3rd nerve palsy, Horner syndrome and synkinetic ptosis/Marcus–Gunn jaw-winking ptosis
- Myogenic ptosis
- Congenital: most common myogenic ptosis
- Simple
- Blepharophimosis syndrome
- Congenital synkinetic ptosis

- Acquired
- Myasthenia gravis
- Lambert–Eaton syndrome
- Ocular myopathy

Causes of pseudoptosis (apparent ptosis):

- Lack of support of the lids by the globe e.g. artificial eye, microphthalmos, phthisis bulbi, or enophthalmos.
- Contralateral lid retraction
- Ipsilateral hypotropia
- Brow ptosis occurs due to excessive skin on the brow, or seventh nerve palsy, which is diagnosed by manually elevating the eyebrow
- Dermatochalasis

Solution to Question 11:

The given clinical scenario is suggestive of myasthenia gravis and ptosis is the most common ocular feature of the disease.

Myasthenia gravis (MG) is an autoimmune disorder of the neuromuscular junction most commonly caused by anti-acetylcholine receptor antibodies. The disorder is characterized by weakness and fatigability of skeletal muscles where the weakness increases during repeated use and may improve following rest.

Ptosis and diplopia are common initial complaints due to early involvement of the lids and extraocular muscles (EOMs). Of them, Ptosis is a more common manifestation than diplopia. Ptosis in myasthenia gravis is insidious, bilateral, and frequently asymmetrical.

Investigations of myasthenia gravis include:

- Ice pack test -This tests for a transient improvement of the ptotic eyelid on placing an ice pack for 2 minutes, as the cold inhibits the breakdown of acetylcholine by acetylcholinesterase.
- Tensilon (Edrophonium) test - This tests for a transient improvement in the weakness on administration of a short-acting anticholinesterase such as edrophonium.
- Antibody testing - for Acetylcholine receptor (AChR) antibodies, MuSK protein antibodies helps support a diagnosis of myasthenia gravis.

Solution to Question 12:

Bilateral ptosis is not seen in Marfan's syndrome.

Causes of unilateral ptosis:

- Horner's syndrome
- Oculomotor nerve palsy

Causes of bilateral ptosis:

- Congenital
- Senile/involutional/aponeurotic
- Blepharophimosis syndrome
- Myasthenia gravis
- Chronic progressive external ophthalmoplegia (CPEO) - seen in Kearns-Sayre syndrome
- Myotonic dystrophy
- Ocular myopathies

The image below shows a case of botulinum toxicity with bilateral ptosis.



Solution to Question 13:

In this question, 4 mm of the cornea is covered by the upper eyelid. Subtracting 2 mm from this implies that there is 2 mm of ptosis, which is classified as mild grade.

In bilateral cases, the grade of ptosis can be determined by measuring the amount of cornea covered by the upper lid and then subtracting 2 mm. Depending upon this value, the ptosis is graded as follows:

- Mild: 2 mm
- Moderate: 3 mm
- Severe: 4 mm

Solution to Question 14:

Frontalis sling procedure is the treatment of choice for congenital ptosis with poor levator function.

Solution to Question 15:

The given clinical scenario is suggestive of Horner's syndrome and the associated mild ptosis in this condition can be corrected by resecting Muller's muscle.

Ptosis in Horner's syndrome is due to the paralysis of Muller's muscle.

Fasanella-Servat operation is indicated in cases having mild ptosis(< 2 mm) and good levator function. In this surgery, the upper tarsal border along with its attached Muller's muscle and conjunctiva are resected.

Solution to Question 16:

The given clinical scenario is suggestive of Marcus–Gunn jaw-winking phenomenon. In this condition, the mandibular division of the fifth cranial nerve is misdirected to the levator muscle.

Some cases of congenital ptosis are associated with the Marcus–Gunn jaw-winking phenomenon. The vast majority are unilateral. Signs include retraction of the ptotic lid in conjunction with stimulation of the ipsilateral pterygoid muscles by chewing, sucking, opening the mouth, or contralateral jaw movement. Less common stimuli to winking include jaw protrusion, smiling, swallowing, and clenching of teeth. Jaw-winking does not improve with age although patients may learn to mask it.

The following image shows a woman with Marcus–Gunn jaw-winking phenomenon. See how her right eyelid ptosis retracts with the opening of the mouth.





Solution to Question 17:

Blepharitis acarica is caused by *Demodex folliculorum*.

Blepharitis is a chronic inflammation of the margins of the lids.

The two types are

- Anterior
- Seborrheic and ulcerative types
- Caused by staphylococci, mainly
- Posterior – Meibomianitis

Occasionally caused by

- *Demodex folliculorum* – Blepharitis acarica
- Crab louse – Phthiriasis palpebrarum

The following image shows swollen and inflamed eyelid suggestive of blepharitis.



Solution to Question 18:

Basal cell carcinoma constitutes 85–90% of all malignant epithelial eyelid tumors. It is locally malignant and most commonly involves the lower lid, followed by the medial canthus, upper lid, and outer canthus.

Type	Common location	Clinical features	Spread
Basal cell carcinoma	Lower lid/Medial canthus	NoduleCentral ulceration with pearly surface, telangiectasia	local
Squamous cell carcinoma	Lower lid, from previous actinic keratosis	Ulcer with thickened margins, keratosis papillomatous	Local, lymph nodes
Sebaceous cell carcinoma	Upper lid	Nodule-like chalazionMulti focalFemales > malesRecurrences common	Local intraepithelial lymph nodes
Malignant melanoma	Previous nevi	>6 mm pigmented, Vascularization inflammation	Local, vascular lymph nodes

Solution to Question 19:

Vismodegib is the systemic drug used for the management of periocular basal cell carcinoma (BCC).

Vismodegib is an eye-sparing adjuvant therapy of periocular and orbital BCC for patients:

- Who cannot tolerate surgery or radiotherapy, or
- With relapsed locally advanced disease after surgery or

- With metastatic disease.

Vismodegib is a Hedgehog pathway inhibitor, given 150 mg orally twice a day.

Treatment options for BCC consist of surgery, radiotherapy, and chemotherapy. Surgical excision using Mohs micrographic surgery or wide surgical excision with frozen section margin control is the first-line treatment of periocular BCC.

Other options:

Option A: Imiquimod 5% topical cream can also be used in patients with BCC. But, it is not a systemic drug.

Option C: Siponimod is the drug approved for secondary progressive multiple sclerosis.

Option D: Interferons are generally not used in periocular BCC.

Solution to Question 20:

The history and image are suggestive of capillary hemangioma involving the left upper lid and periocular region causing severe mechanical ptosis. Immediate management with oral propranolol is the treatment of choice in this case.

The ptosis is so severe that it obstructs the pupillary area (visual axis) with the risk of sensory deprivation amblyopia. This warrants immediate treatment to reduce the bulk of the tumor.

Capillary hemangioma regresses spontaneously very often. It may be managed by oral propranolol, oral prednisolone, or intralesional steroid.

Indications for treatment of patients with capillary hemangioma with pharmacologic therapy are:

- Visual symptoms (amblyopia, astigmatism, occlusion, strabismus)
- Cosmesis
- Optic nerve compression
- Exposure keratopathy

Solution to Question 21:

The image shows plexiform neurofibroma of the upper lid causing an S-shaped deformity of the upper lid and mechanical ptosis. This is seen particularly in neurofibromatosis type 1.

NF1 is caused by the autosomal dominant loss-of-function mutations in the NF1 gene on chromosome 17. It is a tumor suppressor gene that encodes neurofibromin, a negative RAS regulator. The disease is clinically diagnosed when any two of the following seven features are present:

- Six or more café-au-lait macules (CALMs) - They are the hallmark of neurofibromatosis and are present in almost 100% of patients. They are present at birth but increase in size, number, and pigmentation, especially during the first few years of life, and are scattered over the entire body.
- Axillary or inguinal freckling

- Two or more iris Lisch nodules - They are iris hamartomas that are best identified by a slit-lamp examination. They are present in the majority of patients with NF1.
- Two or more neurofibromas or 1 plexiform neurofibroma
- A distinctive osseous lesion such as sphenoid dysplasia or cortical thinning of long bones such as the tibia
- Optic gliomas (most frequently observed CNS tumor in NF1)
- A first-degree relative with NF-1

Solution to Question 22:

The fullness of the eyelids is described as the Enroth's sign in Graves ophthalmopathy.

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Disorders of Lacrimal Apparatus and Glands of the Eye

Question 1:

Moll glands are examples of _____

- a) Apocrine sweat gland
- b) Eccrine sweat gland
- c) Sebaceous gland
- d) Apoeccrine gland

Question 2:

A patient presented with a painless swelling and watering in his left eye. Examination revealed the following finding. Which of the following statements is false regarding his diagnosis?



- a) It needs to be surgically treated.
- b) It affects the meibomian glands.
- c) It does not have malignant potential.
- d) It is a chronic lipogranulomatous inflammation.

Question 3:

A recurrent chalazion should be subjected to histopathological evaluation to exclude the possibility of _____.

- a) Squamous cell carcinoma
- b) Basal cell carcinoma
- c) Sebaceous gland carcinoma
- d) Adenosquamous carcinoma

Question 4:

A patient presented to the clinic with severe left eye pain, watering, and swelling of the lid for 2 days. He says it's difficult to see the laptop screen while working and he is unable to complete his daily targets. On examination, tender swelling is noted as shown below. What is the cause of this finding?



- a) Acute infection of Zeis gland
- b) Acute infection of Moll gland
- c) Acute infection of meibomian gland
- d) Chronic infection of meibomian gland

Question 5:

What is the length of the nasolacrimal duct?

- a) 10-14 mm
- b) 12-18 mm
- c) 18-24 mm
- d) 8-12 mm

Question 6:

Where does the nasolacrimal duct open?

- a) Supreme meatus
- b) Superior meatus
- c) Middle meatus
- d) Inferior meatus

Question 7:

A 60-year-old woman came for an eye exam due to a gritty sensation that has been causing her a lot of discomfort. On examination, Rose Bengal staining is positive. She is diagnosed with keratoconjunctivitis sicca. Which of the following layer of the tear film is deficient in this condition?

- a) Aqueous layer
- b) Lipid layer
- c) Mucin layer
- d) All of the above

Question 8:

Which of the following statements is true regarding the phenol red test for dry eye?

- a) It requires topical anesthesia.
- b) It measures tear production based on the length of colour change in the thread
- c) If color changes to blue, it depicts mucin deficiency.
- d) It requires a pH meter.

Question 9:

Identify the test shown below.



- a) Schirmer test
- b) Ishihara test
- c) Jones test
- d) Goldmann's tonometry

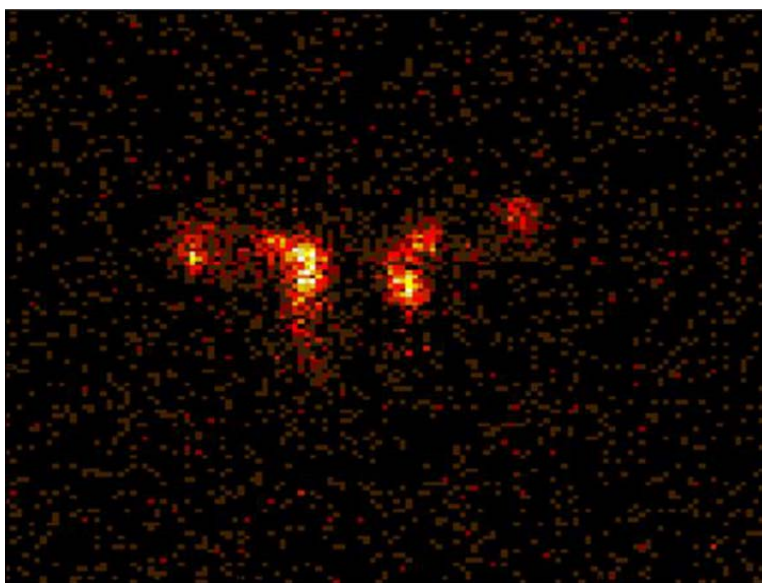
Question 10:

You are conducting retrospective research on factors responsible for dry eye disorder. As a part of your study, you are looking through the files in the clinic of dry eye patients. Which among the following patients would not be included in your study?

- a) A 26 year old patient showing decreased levels of lactoferrin in tear samples
- b) A 20 year old patient showing tear film break up at 18-20 seconds
- c) A 24 year old patient showing A pattern stain in Rose bengal staining
- d) A 22 year old patient showing Schirmer test tear strip wet upto 9 mm

Question 11:

The findings of a test done in a patient under evaluation for dry eye are shown below. Identify the test.



- a) Regurgitation test
- b) Dacryocystography
- c) Jones dye test
- d) Lacrimal scintigraphy

Question 12:

What is the site of blockage in congenital dacryocystitis?

- a) Lacrimal canaliculi
- b) Nasolacrimal duct
- c) Punctum
- d) Lacrimal sac

Question 13:

A mother noticed the following finding in her 3-month-old child. There is a history of watering and discharge from his right eye since birth. Which of the following is the best next step in the treatment?



- a) Syringing
- b) Probing with syringing
- c) Sac massage with topical antibiotics
- d) Dacryocystorhinostomy

Question 14:

A 60-year-old man presents with watering from the right eye for 1 year. Syringing revealed a patent drainage system. Rest of the ocular examination was normal. A provisional diagnosis of lacrimal pump failure was made. Which of the following test will help in the confirmation of the diagnosis?

- a) Dacryocystography
- b) Dacryoscintigraphy
- c) Regurgitation test
- d) Canaliculus irrigation test

Question 15:

A 55-year-old woman is being counseled for surgery for her long-standing dacryocystitis. Which of the following is the surgery of choice?

- a) Dacryocystorhinostomy
- b) Dacryocystectomy
- c) Conjunctivo-cystorhinostomy

- d) Balloon catheter dilation

Question 16:

A patient with chronic dacryocystitis, refractory to the medical treatment, has been scheduled for dacryocystorhinostomy. Following the surgery, the sac would be draining into the _____.

- a) Superior meatus
- b) Inferior meatus
- c) Middle meatus
- d) Medial wall of nasal cavity

Question 17:

What is the most common ocular manifestation in mumps?

- a) Dacryocystadenitis
- b) Chorioretinitis
- c) Anterior uveitis
- d) Membranous conjunctivitis

Question 18:

Which is the most common lacrimal gland tumor?

- a) Adenoid cystic carcinoma
- b) Pleomorphic adenoma
- c) Mucoepidermoid carcinoma
- d) Malignant mixed tumor

Answer Key

Question No.	Correct Option
1	a

2	c
3	c
4	c
5	b
6	d
7	a
8	b
9	a
10	b
11	d
12	b
13	c
14	b
15	a
16	c
17	a
18	b

Detailed Explanations

Solution to Question 1:

Moll glands are examples of apocrine sweat gland.

Moll glands are present in the eyelid. They are a type of modified apocrine sweat glands opening either into a lash follicle or directly onto the anterior lid margin between the lashes.

Zeis glands are a type of modified sebaceous glands associated with lash follicles. Meibomian glands are modified sebaceous glands found in the tarsal plates.

Solution to Question 2:

A painless swelling on the lower lid is suggestive of a chalazion (tarsal cyst/meibomian cyst). It has the malignant potential to become an adenocarcinoma of the meibomian gland.

Chalazion is the chronic lipogranulomatous inflammation (option D) of the meibomian gland (option B). It presents as painless swelling of the eyelid. It is more common in adults than children. Treatment includes incision and curettage (option B) and intralesional injection of triamcinolone acetate.

Solution to Question 3:

A recurrent chalazion should be subjected to histopathological evaluation to exclude the possibility of sebaceous gland carcinoma.

Note: A chalazion can undergo malignant change into sebaceous gland carcinoma (adenocarcinoma of meibomian gland).

Recurrences of chalazia are not uncommon. However, the physician should entertain the possibility of malignancy in such cases and should biopsy a lesion that recurs or appears atypical. The pathologist should be alerted to the suspicion of sebaceous gland carcinoma and frozen sections and lipid stains should be requested.

Solution to Question 4:

The patient has acute presentation and painful swelling in the palpebral conjunctiva as shown in the image, suggestive of internal hordeolum. It is an acute suppurative inflammation of the meibomian gland associated with blockage of the duct.

Other options:

Options A and B: Hordeolum externum or sty is the acute staphylococcal infection of the hair follicle/Zeis gland/Moll gland.



Option D: Chalazion is a chronic non-suppurative lipogranulomatous inflammation of the Meibomian gland.



Solution to Question 5:

The nasolacrimal duct is 12–18 mm long.

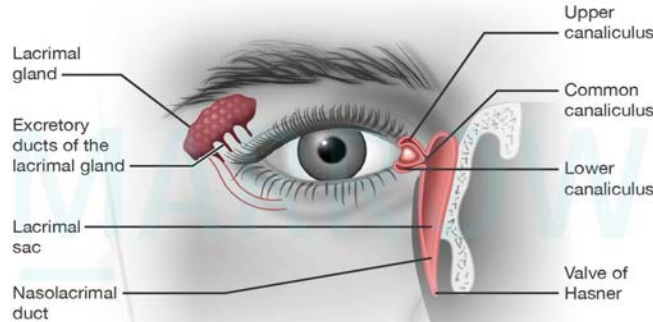
The lacrimal drainage system consists of the lacrimal sac, which is 10–12 mm long. It lies in the lacrimal fossa, between the anterior and posterior lacrimal crests. The lacrimal bone and the frontal process of the maxilla separate the lacrimal sac from the middle meatus of the nasal cavity.

The nasolacrimal duct is the inferior continuation of the lacrimal sac. It is 12–18 mm long and is directed downwards, backward, and laterally. The opening of the duct into the inferior meatus is partially covered by a mucosal fold (valve of Hasner).

The puncta are openings on the lid margin. There are two canaliculi from the puncta. Each canaliculus (superior and inferior) has two parts: Vertical (2 mm) and horizontal (8 mm). The superior and inferior canaliculi usually unite to form the common canaliculus, which opens into the lateral wall of the lacrimal sac. A small flap of mucosa (Rosenmüller valve) overhangs the junction of the common canaliculus and the lacrimal sac and prevents the reflux of tears into the canaliculi.

The image below shows structures of the lacrimal apparatus.

Lacrimal Apparatus



©MARROW

Solution to Question 6:

The nasolacrimal duct opens into the inferior meatus.

The nasolacrimal duct is the inferior continuation of the lacrimal sac. It is 12–18 mm long and is directed downwards, backward, and laterally. The opening of the duct into the inferior meatus is partially covered by a mucosal fold, called the valve of Hasner. The valve of Hasner prevents reflux from the nose.

Solution to Question 7:

Keratoconjunctivitis sicca is due to the deficiency of the aqueous layer of the tear film.

In keratoconjunctivitis sicca, patients present with irritation and foreign body sensation. Marginal tear strip is reduced. The conjunctiva become dry and lustreless with positive Rose Bengal staining. Causes of keratoconjunctivitis sicca:

- Sjogren's syndrome (primary keratoconjunctivitis sicca)
- Non-Sjogren's keratoconjunctivitis sicca
- Primary age-related hyposecretion (most common)
- Lacrimal gland deficiencies - congenital, infiltration by tumors or sarcoidosis, surgical removal
- Lacrimal gland duct obstruction - old trachoma, burns, SJS
- Reflex hyposecretion (neurogenic causes) - Parkinson disease, 7th nerve palsy, reflex sensory block

Solution to Question 8:

Phenol red test measures the tear production according to the length of colour change in the thread.

In phenol red test, the alkalinity of tears changes its colour to bright orange on contact. It doesn't require anaesthesia (unlike Schirmer's test). Three millimetres of a fine dye-impregnated 75-mm cotton thread is placed under the lateral one-fifth of the inferior palpebral lid margin for 15 seconds.

Based on length of colour change,

- >20 mm - normal tear production
- 10 to 19 mm - borderline dry eye
- <10 mm - severe dry eye

Solution to Question 9:

The image shows a tear strip placed in the lower lid of the eye. This is the Schirmer test which is used to check for dry eye.

Dry eye is a multifactorial disease of the eye characterised by loss of homeostasis of the tear film. Patients present with irritation, foreign body sensation and dryness. Treatment of dry eye includes medical treatment (lubricating eyedrops, most commonly methylcellulose derivatives) or surgical treatment (lacrimal punctum occlusion using silicone plugs).

Other tests for dry eye include:

- Tear film break-up test is used to determine the interval between a complete blink and appearance of first randomly distributed dry spot on the cornea. A drop of fluorescein is instilled in the eye, following by examination in the cobalt-blue light of a slit-lamp.
- Rose Bengal staining is used for the detection of dry, dead epithelium in dry eye.
- Tear osmolarity test is performed with an osmometer.

Other options:

Option B: Ishihara plates are used to test colour vision. They help detect red-green colour blindness.

Option C: Jones test is used in watering eye. It uses fluorescein dye.

Option D: Goldmann's tonometry is used to measure intra-ocular pressure. It consists of a double prism mounted on a slit-lamp.

Solution to Question 10:

A patient with tear film break up at 18-20 seconds is considered to be normal. All the other patients have dry eye.

Tear film break-up (BUT) time is the interval between a complete blink and the appearance of a first randomly distributed dry spot on the cornea. It is an indicator of the adequacy of the mucin component of tears.

Inference:

- 15-35 seconds - normal
- < 10 seconds- unstable tear film

Other options:

Option A: Tear samples can be assayed for the presence of markers known to be decreased (e.g. lactoferrin) or elevated (e.g. matrix metalloproteinase-9) in dry eye. This is known as tear constituent measurement.

Option C: A patient with A pattern on Rose Bengal staining has severe dry eye.

Option D: A patient with 9mm wet Schirmer strip has moderate dry eye.

Solution to Question 11:

The image above depicts lacrimal scintigraphy.

Lacrimal scintigraphy (dacryoscintigraphy) is used to assess the lacrimal drainage function. A radioactive tracer (technetium-99m) is instilled into the conjunctival sac. Its path is then visualised with an Anger gamma camera. It is useful for the evaluation of watering eye.

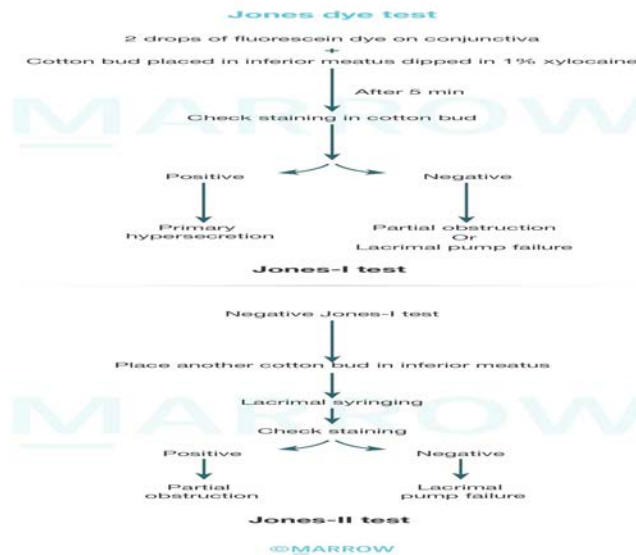
Other tests for evaluation of watering eye include regurgitation test, syringing, fluorescein dye disappearance test (FDDT), Jones dye test and dacryocystography.

Other options:

Option A: In regurgitation test, steady pressure with the index finger is applied above the medial palpebral ligament.

Option B: In dacryocystography, a radioopaque material is inserted into the lacrimal sac, following which X-rays are taken.

Option C: The image shows the Jones dye test.



Solution to Question 12:

The site of blockage in congenital dacryocystitis is the nasolacrimal duct.

Congenital dacryocystitis occurs due to the stasis of secretions in the lacrimal sac due to congenital blockage in the nasolacrimal duct. Membranous occlusion at its lower end near the valve of Hasner is the commonest cause.

The image below shows a neonate with congenital dacryocystitis.



Solution to Question 13:

The image showing right-sided swelling in the lacrimal sac area along with watering of the eye is suggestive of congenital dacryocystitis. Treatment for a 3-month-old child includes sac massage with topical antibiotics.

Congenital dacryocystitis is chronic inflammation of the lacrimal sac in newborn infants. It occurs due to congenital blockage of the nasolacrimal duct. More than 90% of patients with undergo spontaneous resolution by 1 year of age.

Treatment of congenital dacryocystitis depends on the age of the child:

- Less than 9 months: Massaging (Crigler's massage which increases the hydrostatic pressure within the sac) and antibiotic eye drops
- 9 months to 4 years: Probing
- Older children (> 4 years): Dacryocystorhinostomy (DCR) gives the best result. In DCR, an opening is made into the nasal cavity from the lacrimal sac.

Solution to Question 14:

Lacrimal pump failure can be confirmed with dacryoscintigraphy.

Lacrimal pump failure means an inability of the lid muscles to push tears towards the punctum for drainage. It is also called functional obstruction and may be secondary to lower lid laxity or orbicularis muscle weakness (eg. facial nerve palsy). It is confirmed by dacryoscintigraphy, in which a radioactive tracer (technetium-99m) is instilled into the conjunctival sac and its path visualised with an Anger gamma camera. An atonic lacrimal sac is seen.

Note: In cases of partial obstruction of the lacrimal apparatus, forced injection of fluid into the canalicular system (dacryocystography) can overcome the obstruction sites and point to a diagnosis of lacrimal pump failure. Thus, dacryoscintigraphy is better for the confirmation of lacrimal pump failure.

Solution to Question 15:

The treatment of choice for chronic dacryocystitis is dacryocystorhinostomy (DCR)

DCR involves removal of the bone adjacent to the nasolacrimal sac and suturing the lacrimal sac with the lateral nasal mucosa. It re-establishes the lacrimal drainage and is hence the preferred surgery for chronic dacryocystectomy.

Chronic dacryocystitis is a long-standing inflammation of the lacrimal sac. Streptococcus hemolyticus and Staphylococcus aureus are the most common etiological agents. Initial treatment is with antibiotics and anti-inflammatory agents. Sequelae include lacrimal fistula formation and mucocele formation, which when infected, leads to pyocele formation and finally lacrimal fibrosis. Treatment of lacrimal sac fibrosis is dacryocystectomy. Treatment of lacrimal fistula is dacryocystorhinostomy (DCR).

Solution to Question 16:

After dacryocystorhinostomy, the lacrimal sac drains into the middle meatus.

Dacryocystorhinostomy (DCR) surgery involves removal of the bone adjacent to the nasolacrimal sac and suturing the lacrimal sac with the lateral nasal mucosa. This allows tears to drain directly from the sac via a new low-resistance pathway into the middle meatus.

Solution to Question 17:

The most common ocular manifestation in mumps is dacryocystadenitis.

Dacryocystadenitis, or inflammation of the lacrimal gland, is seen in mumps. It presents as a painful swelling in the lateral part of the upper lid.

Solution to Question 18:

Pleomorphic adenoma is the most common lacrimal gland tumor. It is also the most common epithelial tumor of the lacrimal gland.

Pleomorphic adenoma, or benign mixed tumor, presents commonly in middle age as a slowly progressive painless swelling in the upper outer quadrant of the eyelid. The bone is usually not invaded in these tumors. They are treated by surgical removal along with their intact pseudo-capsule.

Malignant tumors of the lacrimal gland such as adenoid cystic carcinoma (option A), mucoepidermoid carcinoma (option C) and malignant mixed tumor (option D) generally occur in the elderly with a short history of appearance, rapid progression and pain. These tumors also present in the upper eyelid and are known to invade the adjacent bones. Radical surgical procedures and high dose radiotherapy are necessary for the removal of the tumor. Adenoid cystic carcinoma is the most common malignant epithelial tumor of the lacrimal gland.

Both malignant and benign tumors are known to push the eyeball downwards and inwards while restricting the upwards and outwards movement of the eyeball.

The most common epithelial tumors, in order of frequency, are as follows:

- Benign mixed tumor (BMT), also known as a pleomorphic adenoma (PA)
- Adenoid cystic carcinoma (ACC)
- Pleomorphic adenocarcinoma (ACA)
- Mucoepidermoid carcinoma

Orbit Anatomy and Ocular Injuries

Question 1:

Which of the following structures does not pass through the superior orbital fissure?

- a) Optic nerve
- b) Lacrimal nerve
- c) Frontal nerve
- d) Nasociliary nerve

Question 2:

You are volunteering in a medical camp held in an orphanage. You see a kid with hypoplasia of the zygoma and mandible along with antimongoloid slant of palpebral fissures as shown below. These findings correspond to which of the following syndrome?



- a) Crouzon syndrome
- b) Treacher Collins syndrome
- c) Apert syndrome
- d) Median facial cleft syndrome

Question 3:

A patient with a history of trauma, presented with diplopia, epistaxis, and restricted eye movements. Looking at the CT image given below, what will be your diagnosis?



- a) Le- fort fracture
- b) Fracture maxilla
- c) Fracture zygomatic
- d) Blow out fracture

Question 4:

What is the most common site of an orbital blowout fracture?

- a) Medial wall
- b) Lateral wall
- c) Roof
- d) Floor

Question 5:

Which of the following statements regarding the clinical sequelae expected in a patient with orbital blowout fracture is false?

- a) Anesthesia of the cheek is very common.
- b) Emphysema of eyelids is more common with medial wall fracture.
- c) Enophthalmos develops about 10 days after the injury.

- d) Severe ocular damage is seen in most cases.

Question 6:

Two people who got in a brawl are brought to the emergency department because of severe facial injuries. In one of those patients, there is periorbital ecchymosis and anesthesia over the cheek. Orbital blowout fracture is suspected. What X-ray finding will confirm this diagnosis?

- a) Steeple sign
- b) Lightbulb sign
- c) Hanging drop sign
- d) Signet ring sign

Question 7:

All of the following are characteristic clinical findings that are associated with a history of blunt trauma to the eye, except:

- a) Berlin's edema
- b) Vossius ring
- c) Soemmering's ring
- d) Rosette cataract

Question 8:

What is the most common site of globe rupture following trauma?

- a) Superonasal limbus
- b) Inferonasal limbus
- c) Superotemporal limbus
- d) Inferotemporal limbus

Question 9:

In which of the following conditions is the following clinical finding seen?



- a) Unilateral anterior cranial fossa fracture
- b) Bilateral anterior cranial fossa fracture
- c) Unilateral posterior cranial fossa fracture
- d) Bilateral posterior cranial fossa fracture

Question 10:

Clinical examination patients with which of the following conditions would show Dalen Fuchs Nodules?

- a) Corneal ulcer
- b) Retinitis pigmentosa
- c) Sympathetic ophthalmitis
- d) After cataract

Question 11:

Which of the following views would you advise, so as to visualize the optic foramina?

- a) Caldwell view
- b) Water's view
- c) Lateral view
- d) Rhese view

Question 12:

Heterochromia iridis may be caused by which of the following intraocular foreign bodies?

- a) Copper
- b) Iron
- c) Aluminium
- d) Glass

Question 13:

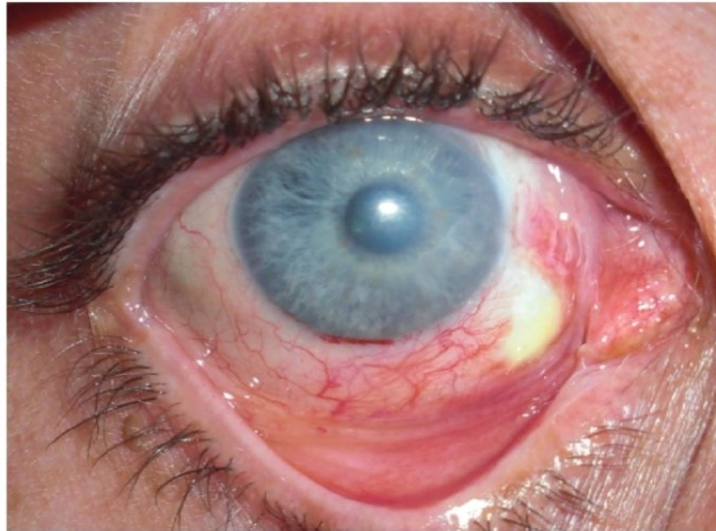
Which of the following conditions is most likely to produce this finding on CT?



- a) Retinoblastoma
- b) Optic nerve glioma
- c) Sphenoid wing meningioma
- d) Neuroblastoma

Question 14:

A 30-year-old construction worker presented to the emergency department with complaints of pain and decreased vision over his right eye. He gives a history of accidental exposure with a cleaning agent that contains ammonia. Ocular examination finding is given below. According to the Roper-Hall system, what is the grade of chemical injury?



- a) Grade I
- b) Grade II
- c) Grade III
- d) Grade IV

Answer Key

Question No.	Correct Option
1	a
2	b
3	d
4	d
5	d
6	c
7	c
8	a
9	b
10	c
11	d
12	b
13	c
14	b

Detailed Explanations

Solution to Question 1:

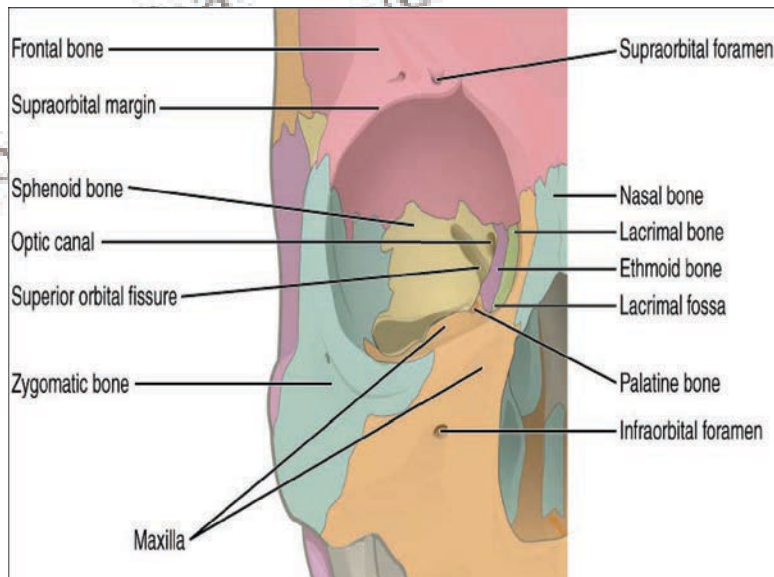
The optic nerve (2nd cranial nerve) passes through the optic canal, not the superior orbital fissure.

Anatomy of the orbit

Orbit is a pear-shaped cavity, the stalk of which is the optic canal. The volume of the orbit is 30 cc.

The image below shows the bones forming the orbit.

Walls of the orbit	Bones forming them
Roof	Lesser wing of the sphenoid The orbital plate of the frontal bone
Lateral wall (strongest wall)	Greater wing of the sphenoid Zygomatic bone
Floor (most frequently fractured wall)	Zygomatic bone Maxillary bone Palatine bone
Medial wall (thinnest wall)	Maxillary bone Lacrimal bone Ethmoid bone Sphenoid bone



The inferior orbital fissure lies between the greater wing of the sphenoid and the maxilla, connecting the orbit to the pterygopalatine and infratemporal fossae. Through it run the maxillary nerve, the zygomatic nerve, and branches of the pterygopalatine ganglion, as well as the inferior ophthalmic vein.

Solution to Question 2:

The given findings correspond to Treacher–Collins syndrome.

Treacher–Collins syndrome/Mandibulofacial dysostosis refers to a condition resulting from the hypoplasia of the zygoma and mandible.

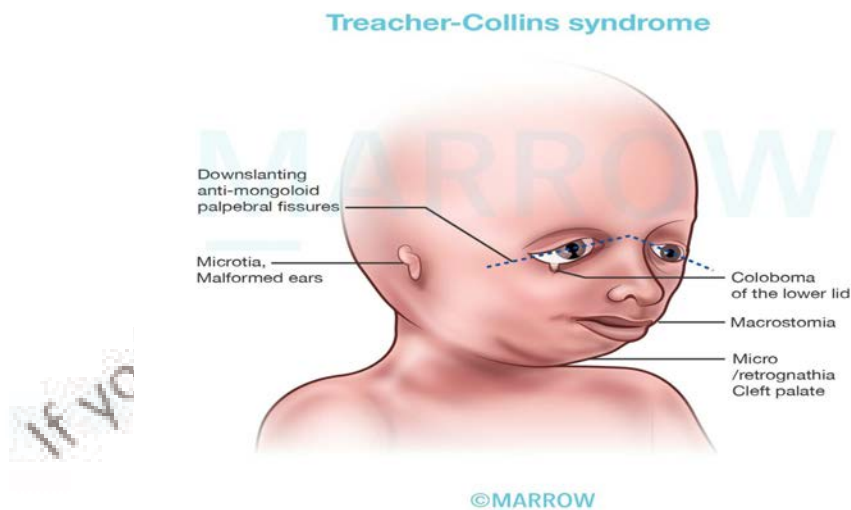
Ocular features:

- Indistinct inferior orbital margin
- Coloboma of the lower eyelid
- Anti-mongoloid slant of palpebral fissures

Systemic features

- Macrostomia with a high-arched palate
- External ear deformity
- Bird-like face

The image below shows features of Treacher–Collins syndrome.



Craniofacial dysostosis/Crouzon syndrome refers to the premature closure of the coronal and sagittal sutures (brachycephaly—clover-leaf skull) associated with maxillary hypoplasia. Ocular associations include the following:

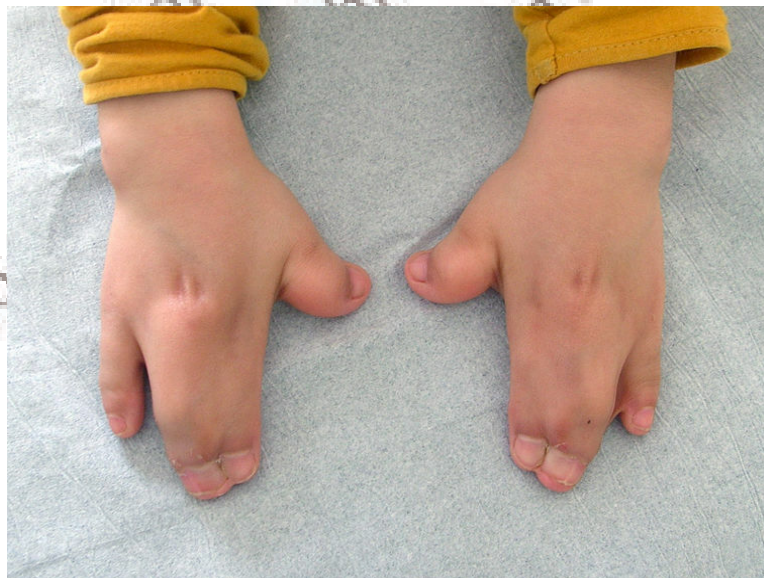
- Blue sclera
- Megalocornea
- Aniridia
- Glaucoma
- Cataract
- Ectopia lentis
- Optic nerve hypoplasia

The image below shows a baby with Crouzon syndrome.



Apert syndrome is the most severe form of craniosynostosis, involving premature closure of multiple cranial sutures (acrocephaly), along with syndactyly of the fingers and toes.

The image below shows the hands of a patient with Apert syndrome.



Solution to Question 3:

The given clinical scenario plus the supportive CT image showing the "hanging drop sign" is suggestive of a blowout fracture of the orbital wall.

Blow-out fractures are isolated comminuted fractures caused from direct trauma to the orbit that results in a backward displacement of the eyeball and an increase in the intraorbital pressure. This causes a fracture at the weakest point of the orbital wall which is usually the posteromedial

portion of the maxillary bone in the orbital floor. The second most common site for orbital blowout fracture is the medial wall of the orbit.

The patient with blow out fracture usually presents with immediate periorbital edema and blood extravasation in and around the orbit. Proptosis of the eyeball secondary to orbital edema or hemorrhage with paresthesia along the infraorbital nerve are also very common.

Imaging by a plain x-ray in water's view or by CT demonstrates depression of bony fragments and orbital contents herniating through the orbital floor into the maxillary antrum. This is called as the "hanging drop sign" or "tear drop sign".

Definitive treatment is by surgical repair 10-14 days after injury with initial supportive management with analgesics and antibiotics.

Le fort fracture- Is the term used to describe different types of maxillary fractures.

Solution to Question 4:

The most common site of an orbital blowout fracture is the floor of the orbit.

Blow-out fractures are isolated comminuted fractures caused from direct trauma to the orbit that results in a backward displacement of the eyeball and an increase in the intraorbital pressure. This causes a fracture at the weakest point of the orbital wall which is usually the posteromedial portion of the maxillary bone in the orbital floor.

The second most common site for orbital blowout fracture is the medial wall of the orbit (Option A).

The fractures of the roof (Option C) are relatively rare and most often do not require any treatment. The lateral wall (Option B) of the orbit is the most solid orbital wall and a fracture here is usually seen in injuries with extensive facial damage.

The patient with blow out fracture usually presents with immediate periorbital edema and blood extravasation in and around the orbit. Proptosis of the eyeball secondary to orbital edema or hemorrhage with paresthesia along the infraorbital nerve are also very common.

Imaging by a plain x-ray in water's view or by CT demonstrates depression of bony fragments and orbital contents herniating through the orbital floor into the maxillary antrum. This is called as the "hanging drop sign" or "tear drop sign".

Definitive treatment is by surgical repair 10-14 days after injury with initial supportive management with analgesics and antibiotics.

The below given CT image shows hanging drop sign.



The below given X-ray image in water's view shows "hanging drop sign".



Solution to Question 5:

Severe ocular damage associated with a "blowout" fracture is rare.

This is because a "blowout fracture" is nature's way of protecting the globe from injury.

Clinical features of blow out fracture are as follows:

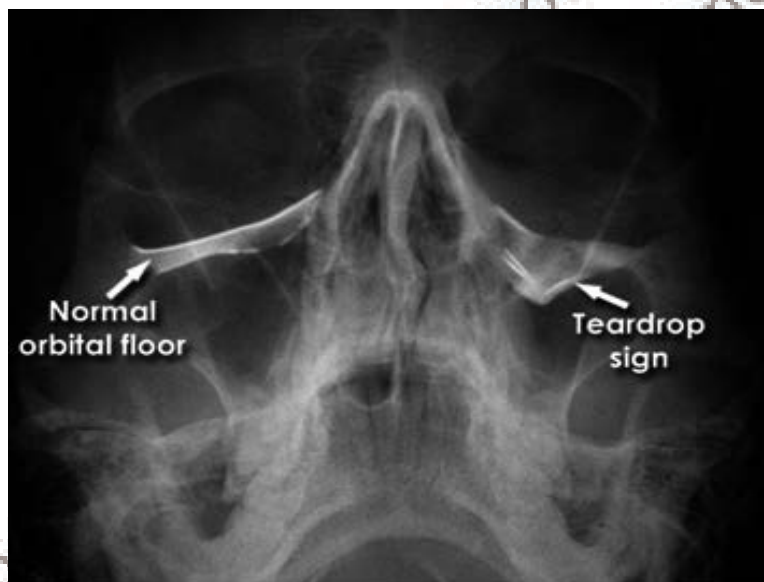
- Periorbital edema
- Paresthesia and anesthesia in the distribution of the infraorbital nerve (lower lid, cheek, side of the nose, upper lip, and upper teeth)
- Emphysema of the eyelids-more frequent with medial wall fractures

- Proptosis-seen initially due to orbital edema and hemorrhage.
- After about 10 days, as the edema decreases, the eyeball sinks backward and somewhat inferiorly, resulting in enophthalmos and mechanical ptosis.
- Double diplopia- diplopia in upward as well as downward gaze

Solution to Question 6:

The X-ray finding seen in an orbital blowout fracture is hanging drop/tear-drop sign.

Hanging drop/tear-drop sign is characteristic of an orbital "blow-out" fracture, which represents the orbital contents herniating through the floor of the orbit into the maxillary antrum. The most useful projection for detecting an orbital floor fracture is Water's view.



The optimal time for surgery following a blowout fracture is 10–14 days.

Indications for surgery in a blowout fracture are as follows:

- Diplopia not resolving significantly in the early days after trauma
- A fracture with a large herniation of tissues into the antrum
- Incarceration of tissues in the fracture, resulting in globe retraction and increased applanation tension on attempted upward gaze
- Enophthalmos greater than 2 mm

Solution to Question 7:

Vossius ring, not Soemmering's, is seen after blunt trauma to the eye

Soemmering's ring refers to a thick ring of after cataract, also known as "secondary cataract." It is the opacity that persists or develops after extracapsular lens extraction.

Blunt trauma to the eye produces:

- Vossius ring - It is a circular ring of brown pigment seen on the anterior capsule. It occurs due to the striking of the contracted pupillary margin against the crystalline lens.
- Rosette cataract - It is the most typical form of concussion cataract.
- Commotio retinae (Berlin's edema) - It manifests as milky white cloudiness involving a considerable area of the posterior pole with a "cherry-red spot" at foveola.

Solution to Question 8:

The superonasal limbus is the most common site of globe rupture.

This occurs due to the contrecoup effect—the lower temporal quadrant is the most exposed to trauma; hence, the effect is seen in the superonasal limbus, which lies on the opposite side.

Solution to Question 9:

The image shows bilateral periorbital ecchymoses, which is called the panda sign, and it is seen in bilateral anterior cranial fossa fracture. It is also known as panda eyes/raccoon eyes.

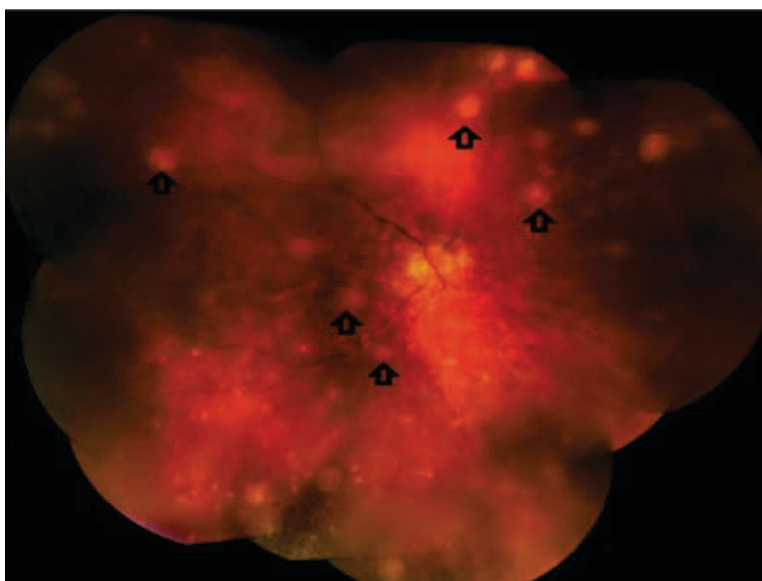
The pooling of blood surrounding the eyes is most commonly associated with fractures of the anterior cranial fossa. This finding is typically not present during the initial evaluation and is delayed by 1 to 3 days. If bilateral, this finding is highly predictive of a basilar skull fracture.

Solution to Question 10:

Dalen Fuchs nodules are seen in sympathetic ophthalmitis and Vogt-Koyanagi-Harada disease (VKH) syndrome.

These are local granulomas due to the proliferation of lymphocytes and epithelioid cells and are located between RPE and Bruch's membrane.

The image below shows Dalen Fuchs nodules.



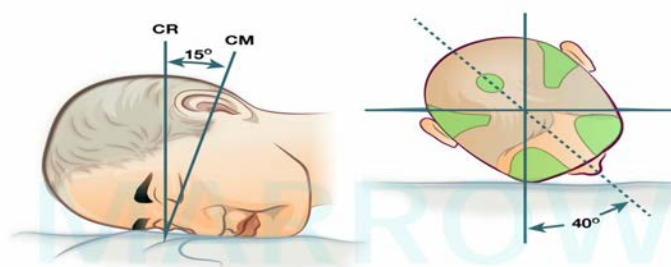
Solution to Question 11:

The Rhese view in an X-ray scan helps in the visualization of the optic foramina.

The optic foramen/optic canal transmits the optic nerve and ophthalmic artery.

Rhese view: for optic foramen

Rhese view



The Zygoma, nose and chin should touch the cassette. The x-ray beam is directed posterior anteriorly at 40 degrees to the midsagittal plane

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Other X-ray views of the orbit:

1. Caldwell's view: Occipitofrontal

Structures seen:

- Superior orbital fissure
- Greater and lesser wing of the sphenoid

- Ethmoid and frontal sinus

- Floor of the sella

2. Water's view:

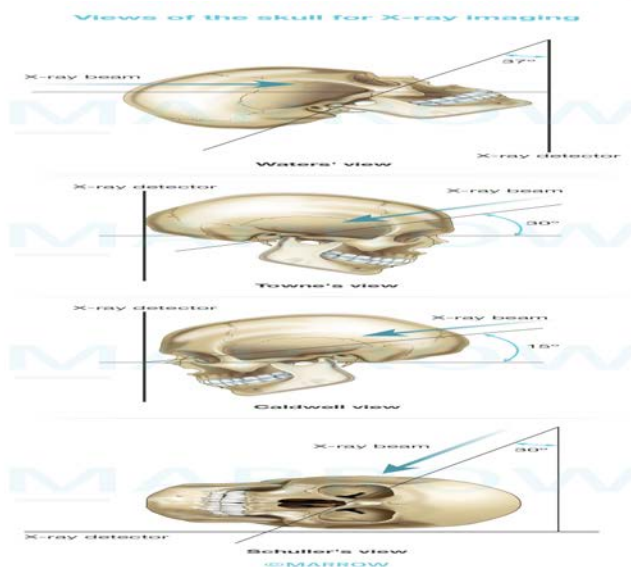
- Occipitomeatal: floor of the orbit is seen (important in "blow-out" fractures)

- Occipitomenal: maxillary sinus and floor of the orbit

3. Towne's view:

Structures seen

- Dorsum sellae of the sphenoid bone
- Inferior orbital fissure



Solution to Question 12:

Heterochromia iridis may be caused by iron deposition.

Siderosis bulbi refers to the ocular degenerative changes produced by an iron foreign body. In siderosis bulbi, the iris is first stained greenish and later turns reddish brown, causing heterochromia iridis.

Chalcosis refers to the specific changes produced by the alloy of copper in the eye. Clinical features include the following:

- Kayser–Fleischer ring- a golden brown ring that occurs due to the deposition of copper in the Descemet's membrane of the cornea.
- Sunflower cataract- produced by the deposition of copper under the anterior capsule of the lens. It is brilliant golden green in color and arranged like the petals of a sunflower.

The image below shows siderosis bulbi.



Solution to Question 13:

The CT image shows hyperostosis of the lateral wall of the left orbit.

Hyperostosis causing lesions are:

- Sphenoid wing meningioma
- Fibrous dysplasia
- Metastasis from cancer prostate

Solution to Question 14:

The image shows a grade II chemical injury. There is limbal ischaemia in the lower left quadrant involving $\frac{1}{3}$ of the limbus and a hazy cornea with visible iris details.

The Roper–Hall system is used for the classification of chemical injuries to the eye.

Chemical injuries to the eye:

- Alkali burns are twice as common as acid burns.
- Alkalis tend to penetrate more deeply than acids, as the latter coagulate surface proteins, forming a protective barrier.
- Most commonly involved alkalis are ammonia, sodium hydroxide, and lime.

Grade	Limbal ischemia	Cornea	Iris	Prognosis
I	Absent	Clear	Visible	Excellent

Grade	Limbal ischemia	Cornea	Iris	Prognosis
II	<1/3 of limbus	Hazy	Visible	Good
III	1/3 - 1/2 of limbus	Hazy	Obscured	Guarded
IV	>1/2 limbus	Opaque	Obscured	Poor

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Diseases of the Orbit

Question 1:

What is the most common cause of bilateral proptosis in children?

- a) Craniofacial dysostosis
- b) Thyroid ophthalmopathy
- c) Neuroblastoma
- d) Orbital cellulitis

Question 2:

Which of the following conditions does not presents as pulsatile proptosis?

- a) Caroticocavernous fistula
- b) Meningocele
- c) Neurofibromatosis
- d) Hemangioma

Question 3:

A patient presents with unilateral non-pulsatile proptosis, which is compressible and increases on bending forward. No thrill or bruit is present. MRI shows a retrobulbar mass with enhancement. What is the most likely diagnosis?

- a) Carotid cavernous fistula
- b) Orbital varix
- c) Orbital encephalocele
- d) Neurofibromatosis

Question 4:

A 26-year-old man with a history of blunt trauma to the head 2 months earlier presented with rapidly progressive swelling, pain, and diminishing vision in the left eye. The physical examination of the left eye revealed chemosis and pulsating proptosis. A bruit could be heard

with a stethoscope placed on the left temple. What is the most probable diagnosis?

- a) Carotid-cavernous fistula
- b) Retro-orbital hematoma
- c) Superior orbital fissure syndrome
- d) Orbital apex syndrome

Question 5:

Which of the following conditions will present with pseudoproptosis?

- a) Buphthalmos
- b) Pseudotumors
- c) Orbital hemorrhage
- d) Cavernous sinus thrombosis

Question 6:

A 48-year-old lady presents with unilateral mild axial proptosis. On examination, the following nail and skin changes were seen as shown. Which of the following is the most appropriate next step of investigation?



- a) CT scan to rule out meningioma
- b) USG to rule out orbital pseudotumor
- c) T₃ and T₄ measurement to check for thyrotoxicosis

- d) Doppler to rule out hemangioma

Question 7:

A middle-aged woman presented with bilateral proptosis, restriction of eye movements, and chemosis. Her thyroid profile showed she was euthyroid. What is the most probable cause?

- a) Orbital pseudotumor
- b) Thyroid ophthalmopathy
- c) Orbital lymphoma
- d) Orbital cellulitis

Question 8:

What is the most common clinical feature of Graves' ophthalmopathy?

- a) Dalrymple's sign
- b) Exophthalmos
- c) von Graefe's sign
- d) Mobius' sign

Question 9:

A 40-year-old male presents with heat intolerance and weight loss for the past few weeks. He has lost 10 kgs despite having an increased appetite. On examination, he has a fine tremor of the hands, and his thyroid is enlarged. Lab investigations reveal low TSH and elevated T₃ and T₄. What is the most common ocular motility defect seen in this condition?

- a) Elevator restriction
- b) Depressor palsy
- c) Adductor palsy
- d) Elevation palsy

Question 10:

What is known as the infrequent blinking seen in Graves' ophthalmopathy?

- a) Mobius' sign

- b) von Graefe's sign
- c) Enroth's sign
- d) Stellwag's sign

Question 11:

A lady is referred from the medicine department for an eye exam due to gritty sensation in her eyes. She is a known case of hyperthyroidism. Her ophthalmic examination shows the following finding along with the inability to look up. Which of the following is not a treatment modality for this condition?



- a) Radiation
- b) Steroids
- c) Beta-blockers
- d) Orbital decompression

Question 12:

A patient presented with unilateral proptosis and bilateral abducent palsy. What is the most probable cause ?

- a) Retinoblastoma
- b) Orbital pseudotumor
- c) Orbital cellulitis
- d) Cavernous sinus thrombosis

Question 13:

A 20-year-old man with a history of repeated pain over the medial canthus and chronic use of nasal decongestants presents to the OPD with high-grade fever and chills, headache, and bilateral proptosis. On further history, it is noted that the proptosis began in the right eye, and a few days later, the left eye was also affected. What is the most probable diagnosis?

- a) Orbital cellulitis
- b) Cavernous sinus thrombosis
- c) Panophthalmitis
- d) Carotico-cavernous fistula

Question 14:

Among the following, which muscle is affected first in cavernous sinus thrombosis?

- a) Superior oblique
- b) Inferior oblique
- c) Medial rectus
- d) Lateral rectus

Question 15:

MRI of a patient revealed features of right cavernous sinus thrombosis. Which of the following would not be seen in this patient?

- a) Constricted right pupil in response to light
- b) Engorgement of the retinal veins
- c) Ptosis of the right eyelid
- d) Right ophthalmoplegia

Question 16:

A 40-year-old male presented with a 1-day history of fever, headache, and bilateral periorbital and mid-facial swelling. He had attempted to burst a furuncle in the inner margin of his right nostril 2 days back. On examination, marked facial and periorbital swelling with chemosis and proptosis are noted. MRI is suggestive of bilateral cavernous sinus thrombosis. Which of

the following veins is responsible for this condition?

- a) Maxillary veins
- b) Retromandibular veins
- c) Superficial temporal vein
- d) Ophthalmic veins

Question 17:

A child is brought to the OPD with fever, severe pain, and swelling of the left eye. He was bitten by a dog over the eye a few days back. On examination, there is mild proptosis with restriction of ocular movements. What is the most likely diagnosis?



- a) Orbital cellulitis
- b) Endophthalmitis
- c) Panophthalmitis
- d) Preseptal cellulitis

Question 18:

An 8-year-old child presents with fever and painful swelling of the left eye. He is being treated for acute ethmoiditis for 5 days. On examination, the visual acuity in the left eye is 6/12, there is chemosis, axial proptosis, limitation of ocular motility in all directions, and RAPD. What is the most likely diagnosis?

- a) Optic neuritis
- b) Cavernous sinus thrombosis
- c) Orbital cellulitis
- d) Rhabdomyosarcoma

Question 19:

What is the surgery of choice for panophthalmitis?

- a) Evisceration
- b) Enucleation
- c) Exenteration
- d) Debulking

Question 20:

A 72-year-old female presents with normal visual acuity, diminished corneal sensation, and global impairment of ocular motility of the right eye. The patient also has a severe boring headache. The MRI shows an enhancing mass within the right cavernous sinus. The pain and ophthalmoplegia respond dramatically to corticosteroid therapy. What is the most likely diagnosis?

- a) Carotid cavernous sinus fistula
- b) Meningioma of the cavernous sinus
- c) Internal carotid aneurysm
- d) Tolosa- Hunt syndrome

Question 21:

The clinical distinction between a cavernous sinus syndrome and an orbital apex syndrome is best made by the dysfunction of which cranial nerve?

- a) CN II
- b) CN III
- c) CN V
- d) CN VI

Question 22:

A patient was being evaluated for retro-orbital pain in the ward. After multiple investigations, he was diagnosed with Tolosa-Hunt syndrome. Which of the following findings will not be seen in this patient?

- a) Sensory loss over forehead
- b) Afferent pupillary defect
- c) Ptosis
- d) Contralateral lateral rectus palsy

Answer Key

Question No.	Correct Option
1	c
2	d
3	b
4	a
5	a
6	c
7	b
8	a
9	a
10	d
11	c
12	d
13	b
14	d
15	a
16	d
17	a
18	c
19	a
20	d
21	a
22	d

Detailed Explanations

Solution to Question 1:

The most common cause of bilateral proptosis in children is neuroblastoma.

Most common cause:

The image below shows the bilateral proptosis.

Unilateral proptosis	Bilateral proptosis
In children: Orbital cellulitis	In children: Neuroblastoma and leukemia (chloroma)
In adults: Thyroid ophthalmopathy	In adults: Thyroid ophthalmopathy



Solution to Question 2:

Among the given options, hemangioma will not present with pulsatile proptosis.

Causes of pulsatile proptosis:

- Pulsating vascular lesions:
- Carotidocavernous fistula- most common
- Saccular aneurysm of ophthalmic artery
- Transmitted cerebral pulsations in conditions with a deficient orbital roof:

- Congenital meningocele/meningoencephalocele
- Neurofibromatosis
- Traumatic or operative hiatus

Causes of intermittent proptosis:

- Orbital varix- most common
- Periodic orbital edema

Causes of reducible proptosis:

- Orbital varix
- Hemangioma
- Caroticocavernous fistula

Solution to Question 3:

The likely diagnosis from the above clinical scenario is orbital varix.

The type of proptosis seen in orbital varix is non-pulsatile, intermittent, and reducible proptosis. On activities such as forward bending or lowering of the head, turning the head forcefully, hyperextension of the neck, and coughing, the varices fill up, causing proptosis, and then, they regress on shifting to a non-dependent position.

All the other conditions listed, namely, carotid cavernous fistula, orbital encephalocele, and neurofibromatosis, present with pulsatile proptosis.

Solution to Question 4:

The given clinical scenario is suggestive of carotid–cavernous fistula.

All the given options are causes of proptosis following trauma, but among these, only a carotid-cavernous fistula presents with the classic triad:

- Pulsatile proptosis
- Chemosis
- Bruit

Selective carotid angiography is the investigation of choice.

A carotid-cavernous fistula requires a definitive treatment, which currently involves the endovascular technique. The fistula can usually be occluded with a balloon or coil, without the sacrifice of the carotid artery.

Solution to Question 5:

Buphthalmos causes pseudoproptosis.

The other listed conditions, namely, pseudotumors, orbital hemorrhage, and cavernous sinus thrombosis, produce actual proptosis.

Pseudoproptosis is a condition in which the eyeball appears to be proptosed, but there is no actual forward displacement.

Causes of pseudoproptosis:

- Buphthalmos is the term used when the eyeball enlarges in children who develop congenital glaucoma before the age of 3 years.
- Axial high myopia- increase in the size of the eyeball in myopia causes pseudoproptosis
- Retraction of the upper lid
- Enophthalmos of the opposite eye

Solution to Question 6:

The given clinical scenario along with the image showing thyroid acropachy and dermopathy is suggestive of Graves' disease. The most appropriate next step of the investigation is T₃ and T₄ measurement to check for thyrotoxicosis.

Thyroid ophthalmopathy is the most common cause of proptosis in adults, which may present as unilateral or bilateral, axial or non-axial, painful or painless proptosis.

The image given below shows unilateral proptosis in the right eye.



Solution to Question 7:

The given clinical scenario is suggestive of thyroid ophthalmopathy.

Thyroid eye disease (TED) is the most common cause of both bilateral and unilateral proptosis in an adult in the euthyroid state. The main features of TED include lid retraction, proptosis, periorbital edema, chemosis, and restrictive myopathy.

Other options-

Option A- Orbital pseudotumor presents with painful eye movements periorbital edema, erythema, proptosis, ptosis, diplopia.

Option C- Orbital lymphoma usually presents in the 7th decade, most commonly observed as a painless palpable mass in the superolateral quadrant.

Option D- Orbital cellulitis is usually a unilateral condition with pain, proptosis, and chemosis.

NOTE: Thyroid eye disease can also, though less commonly, occur in euthyroid patients.

Solution to Question 8:

Dalrymple's sign (retraction of the upper lids) is the most common clinical feature of Graves' ophthalmopathy. It is caused by the overaction of Muller's muscle. Levator palpebrae superioris (LPS) recession surgery may be done to correct upper lid retraction.

The image given below shows Dalrymple's sign (retraction of the upper lids) in the right eye:



NOSPECS classification by Werner reflecting the severity of thyroid ophthalmopathy.

Class	Definition
0	No physical signs or symptoms
1	Only signs, no symptoms (signs limited to upper lid retraction, stare, lid lag, and proptosis of up to 22 mm)

Class	Definition
2	Soft tissue involvement (symptom and signs)
3	Proptosis
4	Extraocular muscle involvement
5	Corneal involvement
6	Sight loss (optic nerve involvement)
NOSPECS: No signs or symptoms; Only signs; Soft tissue; Proptosis; Extraocular muscle; Cornea; Sight loss	

Solution to Question 9:

The given clinical scenario is suggestive of Grave's disease. The most common ocular motility defect seen in Grave's disease is elevator restriction.

It occurs due to the fibrotic contracture of the inferior rectus muscle. The inferior rectus muscle has to relax during the elevation of the eye. The fibrosis of the muscle restricts its relaxation and thus elevation of the eye is restricted.

There is an inability to look up (in both adducted and abducted positions) along with esotropia (fibrosis of the medial rectus).

Only the belly of the muscle is involved in thyroid myopathy; the tendon is characteristically spared.

Ocular motility restriction in order of frequency is as follows:

- Elevation defect
- Abduction defect, due to fibrosis of the medial rectus
- Depression defect, due to fibrosis of the superior rectus
- Adduction defect, due to fibrosis of the lateral rectus

Solution to Question 10:

Infrequent blinking is known as Stellwag's sign.

Solution to Question 11:

The clinical scenario of grittiness in the eyes (due to ocular dryness), hyperthyroidism and the image showing exophthalmos are suggestive of Graves ophthalmopathy.

Beta-blockers are not indicated in the treatment of graves ophthalmopathy, however, they are used in the symptomatic management of hyperthyroidism.

Treatment of thyroid ophthalmopathy:

(A) Non-surgical management

- Systemic steroids- These form the mainstay of the treatment. Oral, intravenous or orbital steroid injections can be used depending upon patient indications. (Option B)
- Radiotherapy- Orbital radiation given in a dose of 2000 rads over 10-14 days can be considered in cases where steroids are ineffective/ contraindicated. It is contraindicated in cases of extensive thyroid eye disease as it can worsen it. (Option A)
- Teprotumumab (IGF-1R inhibitor)- is used for persistent, corticosteroid-resistant disease or non-response to first-line treatment. It is effective in reducing proptosis and clinical activity in patients.

(B) Surgical management:

- Orbital decompression- It may be required for an acutely progressive optic neuropathy and/or exposure keratitis, in patients not responding to aggressive steroid therapy. (Option D)
- Extraocular muscle surgery- Inferior rectus recession is the most commonly performed surgery. These should be performed after the orbital decompression since decompression may alter extraocular motility. Extraocular muscle surgery helps in achieving a binocular single vision in the primary gaze and reading position.
- Eyelid surgery- This is done to correct eyelid retraction. Various surgeries that can be performed are mullerectomy, levator recession/ disinsertion, scleral grafts with LPS recession, recession of lower eyelid retractors, and blepharoplasty.

Solution to Question 12:

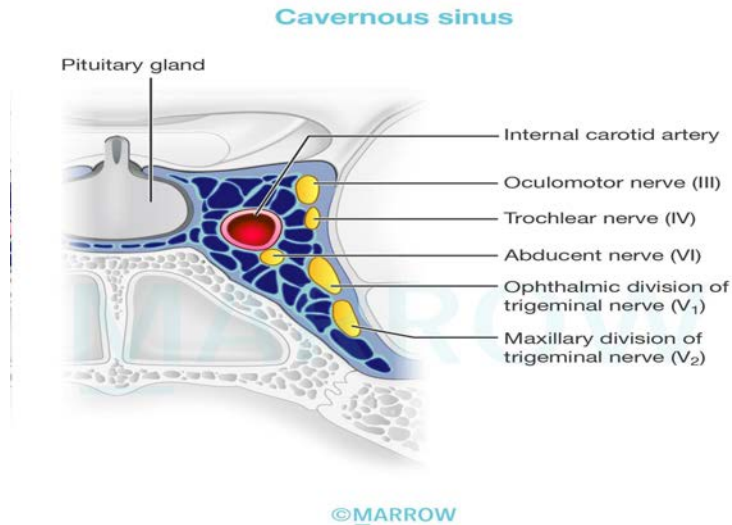
The given clinical scenario is suggestive of cavernous sinus thrombosis.

Features of cavernous sinus thrombosis include early involvement lateral rectus palsy (abducent nerve involvement) since it runs through the cavernous sinus. Oculomotor and trochlear nerve palsy occur later as they are related to the lateral wall of the cavernous sinus. Other signs are proptosis, severe supraorbital pain, edema in the mastoid region, absence of corneal sensation, and dilated pupil.

Options A- Retinoblastoma occurs in young children and presents with unilateral proptosis with leukocoria, buphthalmos, and squint.

Option B- Orbital pseudotumor presents with unilateral proptosis with diplopia, conjunctival chemosis. The inflammation is responsive to systemic corticosteroids.

Option C- Orbital cellulitis presents with unilateral proptosis with pain and decreased ocular motility.



Solution to Question 13:

The given clinical scenario is suggestive of cavernous sinus thrombosis.

Pain over the medial canthus and use of nasal decongestants indicate that the patient was suffering from sinusitis, which can spread to cause orbital cellulitis, cavernous sinus thrombosis, or panophthalmitis.

The differentiating feature here is the presentation pattern and bilaterality of the proptosis.

In cavernous sinus thrombosis, proptosis is initially unilateral but soon becomes bilateral due to the spread of infection to the contralateral sinus through intercavernous communication. Also, cavernous sinus thrombosis presents with high-grade fever with chills, rigor, and headache.

On the other hand, orbital cellulitis and panophthalmitis are unilateral conditions.

Differential diagnosis of acute inflammatory proptosis:

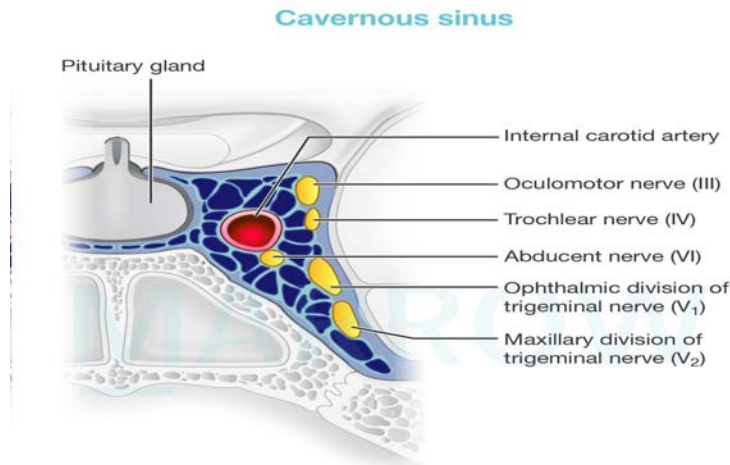
Clinical features	Cavernous sinus thrombosis	Orbital cellulitis	Panophthalmitis
Laterality	Initially unilateral but soon becomes bilateral	Unilateral	Unilateral
Vision	Not affected in early stage	affected	Complete loss of vision
Cornea and anterior chamber	Clear	Clear	Hazy
Edema in the mastoid region	Present (Pathognomonic of cavernous sinus thrombosis)	Absent	Absent
General symptoms with fever and prostration	Marked	Mild	Mild

Solution to Question 14:

The first muscle to be affected in cavernous sinus thrombosis is the lateral rectus.

Ophthalmoplegia in cavernous sinus thrombosis occurs in the following sequence:

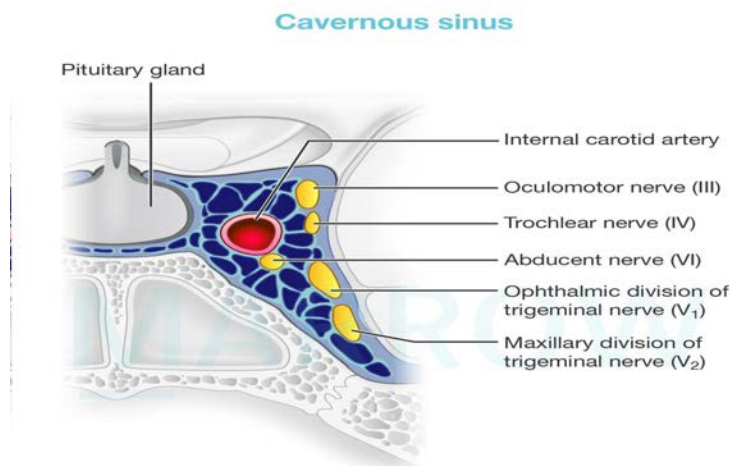
- 6th nerve palsy (lateral rectus palsy) occurring first of all as it passes through the cavernous sinus.
- 3rd and 4th nerves are involved later as they are related to the lateral wall of the cavernous sinus.



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Solution to Question 15:

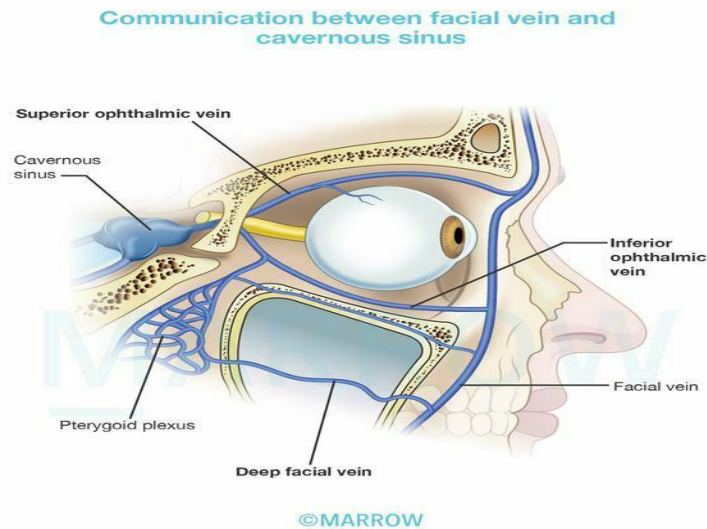
Dilation of the pupil will be seen in cavernous sinus thrombosis due to the involvement of the 3rd cranial nerve.



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Solution to Question 16:

Infection from the dangerous area of the face (the area from the corners of the mouth to the bridge of the nose, including the nose and maxilla) spreads to the cavernous sinus via the ophthalmic veins.



Solution to Question 17:

The given clinical scenario is suggestive of orbital cellulitis.

It is the acute inflammation of the soft tissues of the orbit behind the orbital septum. It can occur due to exogenous infection (penetrating injuries) or extension of an endogenous infection (dental infection).

It presents with fever, severe pain, and eye swelling which increases with movement, chemosis, axial proptosis, and restriction of ocular movements. Complications include subperiosteal abscess, orbital abscess, cavernous sinus thrombosis, and general septicemia.

The image shows orbital cellulitis in the left eye.



Option B: Endophthalmitis is defined as the inflammation of the inner structures of the eyeball, i.e., uveal tissue and retina, associated with the pouring of exudates into the vitreous cavity, anterior chamber, and posterior chamber.

Option C: Panophthalmitis is an intense purulent inflammation of the whole eyeball, including Tenon's capsule.

Option D: Preseptal cellulitis refers to infection of the subcutaneous tissues anterior to the orbital septum. In this condition, proptosis is absent and ocular movements are normal.

Solution to Question 18:

The most likely diagnosis of this patient is orbital cellulitis.

Orbital cellulitis is the most common cause of unilateral proptosis in children. Ethmoidal sinusitis is the most common cause of orbital cellulitis in children.

Orbital cellulitis presents with pain, lid edema, erythema, chemosis, and axial proptosis. Decreased ocular motility and elevated intraocular pressure can be seen. Systemic symptoms include malaise and fever. A rapid loss of vision is due to the involvement of the optic nerve. With posterior exenteration cavernous sinus thrombosis, subdural empyema, and intracranial abscess develop.

WARNING SIGNS: Dilated pupil, marked ophthalmoplegia, loss of vision, afferent pupillary defect, papilledema, and violaceous lids.

Solution to Question 19:

The surgery of choice for panophthalmitis is an evisceration. Evisceration gives a better cosmetic outlook after the surgery and is hence preferred. Enucleation is contraindicated in panophthalmitis due to the risk of meningitis through the exposed meningeal sheath of the optic

nerve stump.

Some definitions:

- Evisceration- removal of the contents of the eyeball, leaving behind the sclera. It is done in panophthalmitis.
- Enucleation- removal of the eyeball, with the maximum part of the optic nerve. Extraocular muscles are left behind. The length of the optic nerve removed is 10 to 15 mm. Absolute indications for enucleation are retinoblastoma and malignant melanoma.
- Exenteration- mutilating surgery in which all the contents of the orbits along with the periorbital are removed. It is indicated for malignant tumors arising from the orbital structures or spreading from the eyeball.
- Debulking- reduction of as much of the bulk (volume) of a tumor as possible. Nowadays, debulking of the orbit is preferred over exenteration.

Solution to Question 20:

The patient presents with acute ophthalmoplegia with the involvement of ophthalmic division of trigeminal nerve that responds promptly to steroid therapy. The most likely diagnosis is Tolosa-Hunt syndrome.

It is an idiopathic, sterile inflammation that primarily affects the cavernous sinus. It is characterized by painful acute ophthalmoplegia with or without the involvement of the optic nerve and the ophthalmic division of the trigeminal nerve.

The pain is described as gnawing or boring and may precede ophthalmoplegia.

The inflammation and pain in patients typically responds rapidly and dramatically to corticosteroid therapy. However, it is a diagnosis of exclusion.

Solution to Question 21:

The clinical distinction between a cavernous sinus syndrome and an orbital apex syndrome is made by the dysfunction of the optic nerve (CN II).

The optic nerve passes through the optic canal and does not enter the cavernous sinus. Patients with orbital apex syndrome manifest complete ophthalmoplegia, ptosis, decreased corneal sensation, and vision loss. They usually develop optic nerve signs such as relative afferent pupillary defect. Proptosis is often present, but patients do not always complain of pain.

Solution to Question 22:

Ipsilateral lateral rectus palsy due to 6th cranial nerve involvement is seen in Tolosa-Hunt syndrome, not contralateral.

Tolosa–Hunt syndrome refers to idiopathic, nonspecific, granulomatous inflammation, involving the superior orbital fissure and/or orbital apex and/or cavernous sinus. It is a unilateral eye condition.

Clinically, it manifests as painful ophthalmoplegia and depending on the extent of involvement, it may present as superior orbital fissure syndrome or orbital apex syndrome.

Superior orbital fissure syndrome is characterized by the following:

- Pain, typically labeled as retro-orbital ache
- Sensory loss/disturbances in the distribution of the ophthalmic division of the 5th cranial nerve
- Ipsilateral ophthalmoplegia due to the involvement of the 3rd, 4th, and 6th cranial nerve involvement
- Ptosis, due to 3rd nerve paralysis

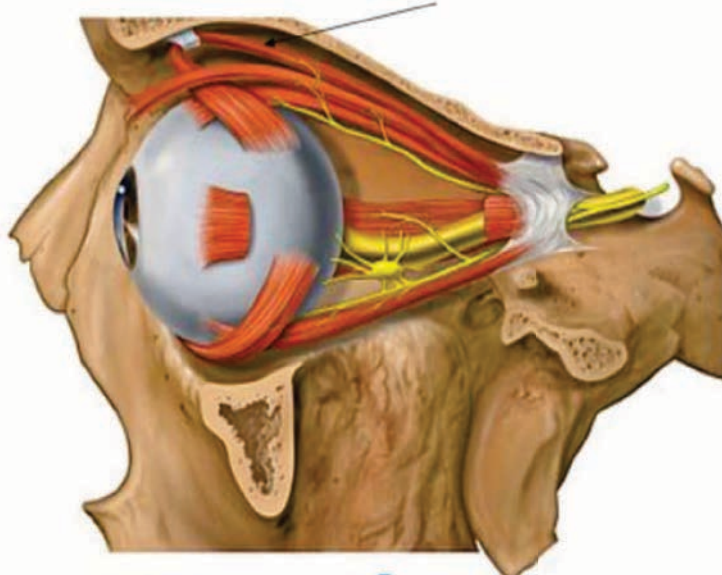
Orbital apex syndrome includes the following:

- Features of superior orbital fissure syndrome described above
- Features of optic nerve involvement, i.e., early visual loss and afferent pupillary defect.

Strabismus - Introduction, Symptomatology, and Evaluation

Question 1:

Which of the following best describes the actions of the muscle marked in the figure?



- a) Extorsion, Elevation, and Abduction
- b) Intorsion, Depression, and Abduction
- c) Extorsion, Depression, and Adduction
- d) Intorsion, Elevation, and Adduction

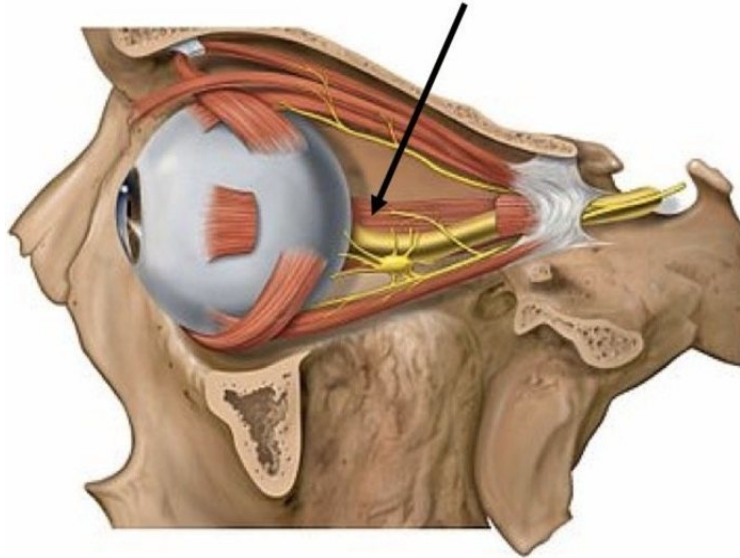
Question 2:

Which of the following pair of muscles are responsible for the intorsion of the eyeball?

- a) Superior rectus and superior oblique
- b) Inferior rectus and inferior oblique
- c) Inferior rectus and superior oblique
- d) Superior rectus and inferior rectus

Question 3:

Name the location of the nucleus supplying the muscle marked by the arrow in the image given below.



- a) Superior olivary
- b) Inferior colliculus
- c) Superior colliculus
- d) Facial colliculus

Question 4:

Which of the following pairs constitute yoke muscles?

- a) Right superior oblique (SO) and left inferior rectus (IR)
- b) Right superior oblique (SO) and left inferior oblique (IO)
- c) Right medial rectus (MR) and left medial rectus (MR)
- d) Right medial rectus (MR) and right lateral rectus (LR)

Question 5:

Yoke muscle of the right lateral rectus in dextroversion is the _____.

- a) Left superior oblique
- b) Left medial rectus
- c) Left inferior rectus
- d) Left superior rectus

Question 6:

During ocular motility, increased flow of innervation to the contracting muscle is accompanied by a decreased flow of innervation to the relaxing antagonist muscle. This statement describes which law?

- a) Herring's Law
- b) Sherrington's Law
- c) Laplace law
- d) Hick's Law

Question 7:

You are an intern posted in the ophthalmology department. You are asked to check the cardinal fields of gaze to check the range of extraocular movements. How many cardinal fields are there?

- a) 6
- b) 4
- c) 9
- d) 12

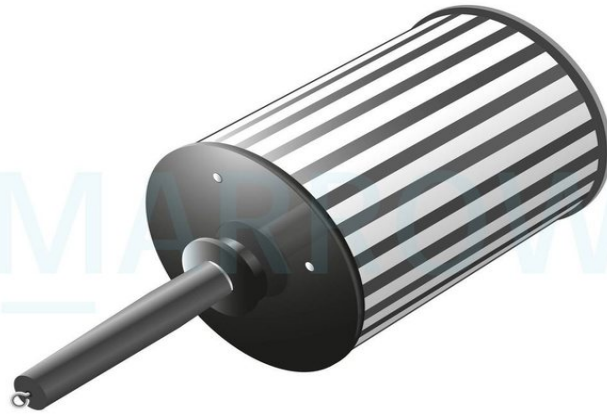
Question 8:

Which of the following best defines saccades?

- a) Sudden jerky conjugate eye movements
- b) Smooth slow eye movements
- c) Flicks
- d) Drifts

Question 9:

An ophthalmologist tests a particular phenomenon using the instrument shown below. Which of the following statements is false regarding the same?



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- a) The phenomenon is a physiological one
- b) Frontal lobe controls the slow pursuit movement
- c) Reversal of optokinetic nystagmus, in congenital motor nystagmus
- d) Used to check malingering

Question 10:

Which of the following is not a nystagmoid movement?

- a) Ocular flutter
- b) Ocular bobbing
- c) Opsoclonus
- d) Optokinetic

Question 11:

A 24-year-old woman presented with progressive diminution of vision, headache, and unsteadiness of gait. A comprehensive clinical evaluation revealed coarse, left-beating nystagmus with leftward gaze and a fine primary-position right-beating nystagmus which increased on rightward gaze, consistent with Brun's nystagmus. What is the most likely cause of her symptoms?

- a) Cerebellar lesion
- b) Cerebello pontine angle tumor
- c) Arnold Chiari malformation

d) Demyelination syndrome

Question 12:

Which of the following is a wrong statement regarding diplopia?

- a) Anisometropia is a cause for binocular diplopia.
- b) Crossed diplopia occurs in convergent squint.
- c) Incipient cataract is a cause for uniocular diplopia.
- d) Harmonious diplopia refers to uncrossed binocular diplopia.

Question 13:

Hirschberg test is used to detect which of the following?

- a) Diplopia
- b) Squint
- c) Refractive error
- d) Glaucoma

Question 14:

As an intern posted in the ophthalmology department, you are performing Hirschberg's test on a patient who is being evaluated for strabismus. The finding observed is shown below. What is the angle of deviation in this patient?



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- a) 90 degrees esotropia
- b) 90 degrees exotropia
- c) 90 Prism Dioptre esotropia
- d) 90 Prism Dioptre exotropia

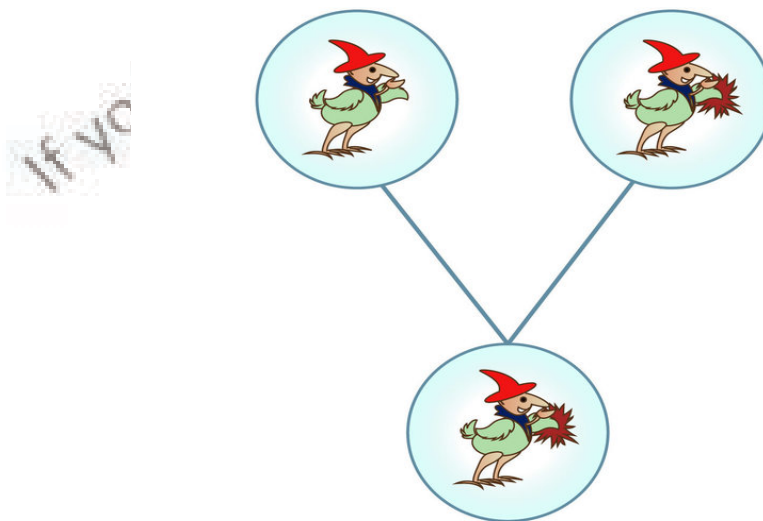
Question 15:

You are evaluating a case of strabismus in a 3-year-old child. Which of the following tests would you use to assess binocular single vision?

- a) Amsler grid
- b) Synoptophore
- c) Maddox rod
- d) Cardboard test

Question 16:

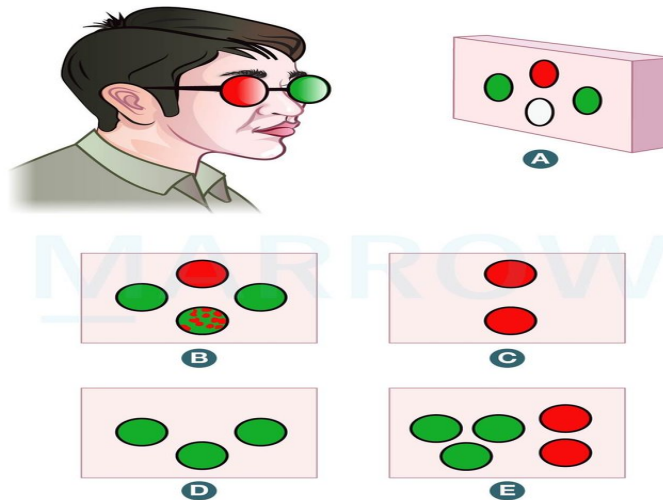
Which grade of binocular vision does the following image represent?



- a) Grade I
- b) Grade II
- c) Grade III
- d) Grade IV

Question 17:

You are an intern posted in the ophthalmology department where you are observing a patient subjected to the Worth's four-dot test. The patient is suffering from left suppression. Which of the following images would he most likely see?



- a) Image E
- b) Image B
- c) Image C
- d) Image D

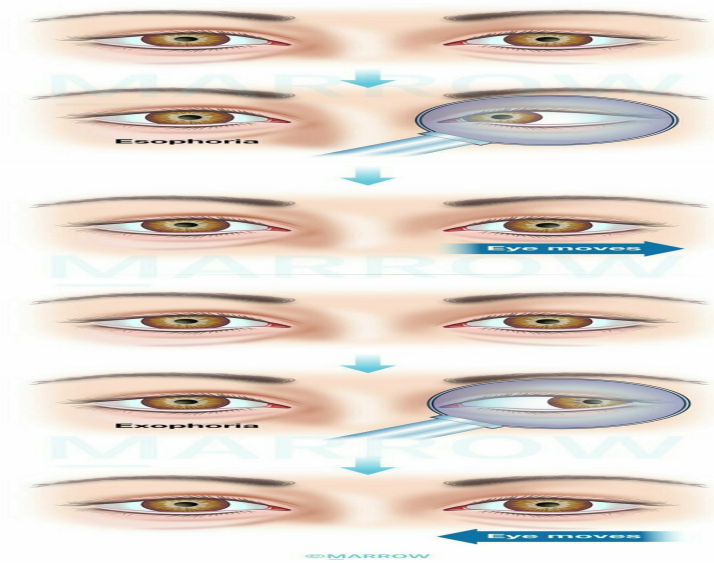
Question 18:

While evaluating a case of squint, an ophthalmologist does a forced duction test. What is the use of this test?

- a) To detect ocular muscle palsy
- b) To detect ocular muscle spasm
- c) To measure angle of deviation
- d) To quantify refractive error

Question 19:

Identify the test being shown below.



- a) Obscure test
- b) Prism test
- c) Cover uncover test
- d) Maddox rod test

Question 20:

Which of the following statements is true?

- a) Direct cover test is helpful in detecting presence of heterophoria.
- b) Alternate cover test helps to differentiate concomitant squint from paralytic squint.
- c) Cover-uncover test confirms the presence of manifest squint.
- d) In direct cover test, the squinted eye is covered and normal eye is uncovered.

Question 21:

A patient presented to an ophthalmologist with complaints of crowding of words, while reading for long hours. The results of the test that was performed are shown below What is the most likely diagnosis?



- a) Orthophoria
- b) Hypophoria
- c) Exophoria
- d) Esophoria

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	a
5	b
6	b
7	a
8	a
9	b
10	d
11	b
12	b
13	b
14	d

15	b
16	b
17	c
18	a
19	c
20	b
21	c

Detailed Explanations

Solution to Question 1:

The muscle marked in the above image is the superior oblique.

Actions of the superior oblique:

- Intorsion
- Depression
- Abduction

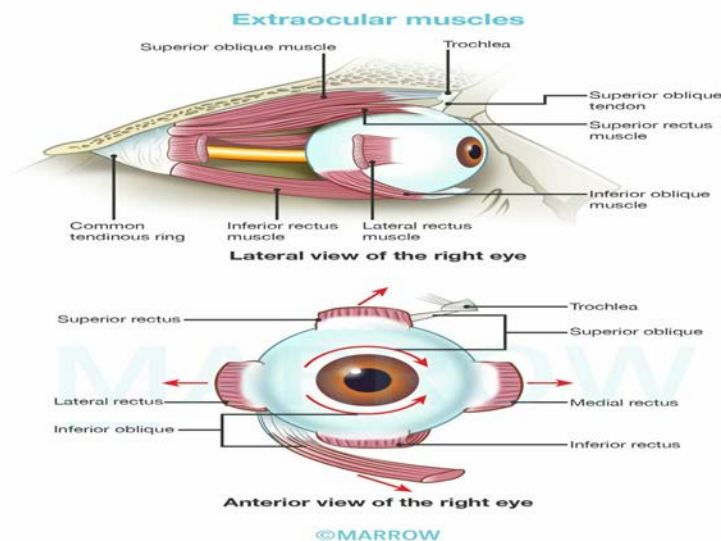
Longest extraocular muscle: Superior oblique

Shortest extraocular muscle: Inferior oblique

The primary action of a muscle is its major effect when the eye is in the primary position.

Subsidiary actions are the additional effects; these depend on the position of the eye.

The spiral of Tillaux is an imaginary line joining the insertions of the four recti and is an important anatomical landmark when performing surgery.



Solution to Question 2:

The muscles producing intorsion of the eyeball are the superior rectus and superior oblique.

Extorsion of the eyeball is due to the action of the inferior oblique (major) and the inferior rectus muscle.

Solution to Question 3:

The muscle marked is the medial rectus. The medial rectus is supplied by the oculomotor nerve, the nucleus of which is situated at the level of the superior colliculus in the posterior midbrain.

The oculomotor (3rd cranial) nerve nucleus consists of

- Single central nuclei for the levator palpebrae muscle of both the eyes
- Bilateral subnuclei with crossed projections supplying the contralateral superior rectus
- Bilateral subnuclei with uncrossed projections supplying the ipsilateral inferior rectus, inferior oblique, medial rectus in the dorso-ventral order.
- Bilateral subnuclei known as the Edinger-Westphal nucleus which provides parasympathetic input to the ciliary body and the iris sphincter.

Option A: Superior olivary nucleus is an important relay station in the auditory pathway, it is responsible for detection of direction and localization of sound.

Option B: Inferior colliculus- The trochlear(4th cranial) nerve nucleus lies in the midbrain at the level of the inferior colliculus. The trochlear nerves leave the midbrain from its posterior surface and decussate before supplying the contralateral superior oblique muscle of the eyeball.

Option D: Facial colliculus- the facial colliculus is located in the cranial part of the pons. It is produced by the facial nerve(7th cranial nerve) winding around the nucleus of the abducens nerve (6th cranial nerve).

Solution to Question 4:

Right superior oblique (SO) and left inferior rectus (IR) are the yoke muscle pair for levodepression.

The term "yoke muscles" refers to contralateral synergists. Includes a pair of muscles, one from each eye, which contract simultaneously during version movements.

Examples are:

- Right MR and Left LR for levoversion
- Right LR and Left MR for dextroversion

An easy method to determine other yoke muscle pairs:

Just take the opposite of each word-

Left → Right

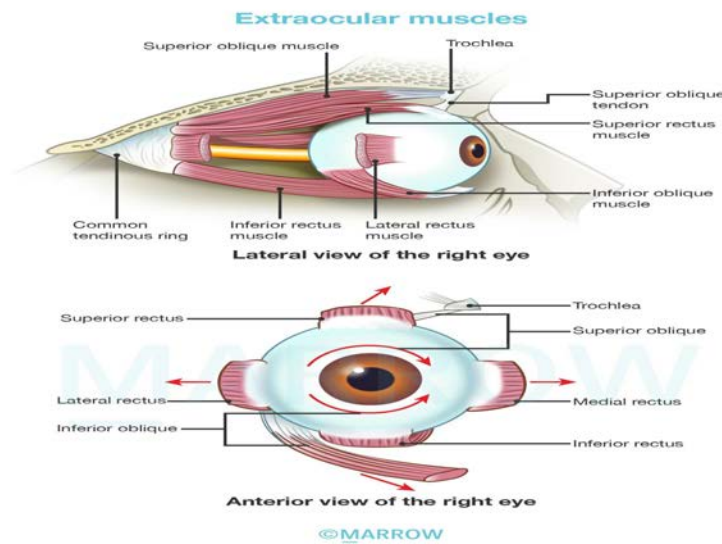
Superior → Inferior

Oblique → Rectus

“Left SO and Right IR” are a yoke muscle pair.

Similarly, the following are yoke muscle pairs:

- Right IO and Left SR - Levoelevation
- Right SR and Left IO - Dextroelevation
- Right IR and Left SO - Dextrodepression



Solution to Question 5:

The yoke muscle of the Right lateral rectus in right dextroversion is the left medial rectus.

Yoke muscles also known as contralateral synergists include a pair of muscles, one from each eye, which contract simultaneously during version movements.

Versions are simultaneous movements of both eyes in the same direction, such as dextroversion in which both the eyes move to the right.

Hence, in dextroversion, the right eye is in abduction due to the action of right lateral rectus and the left eye is in adduction by the action of the left medial rectus.

An easy method to determine yoke muscle pairs:

Just take the opposite of each word-

Left → Right

Superior → Inferior

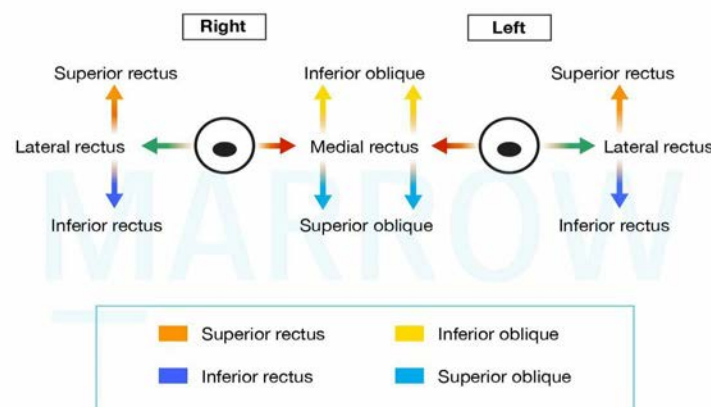
Oblique → Rectus

“Left SO and Right IR” are a yoke muscle pair.

Other examples of yoke muscles are:

- Right MR and Left LR for levoversion
- Right IO and Left SR - Levoelevation
- Right SR and Left IO - Dextroelevation
- Right IR and Left SO - Dextrodepression

[IO-inferior oblique;SO-superior oblique;SR-superior rectus;IR-inferior rectus;MR-medial rectus;LR-lateral rectus]



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Solution to Question 6:

The above statement elucidates Sherrington's Law of reciprocal innervation.

Solution to Question 7:

There are six cardinal positions (fields) of gaze. These are the positions that allow examination of each of the twelve extraocular muscles.

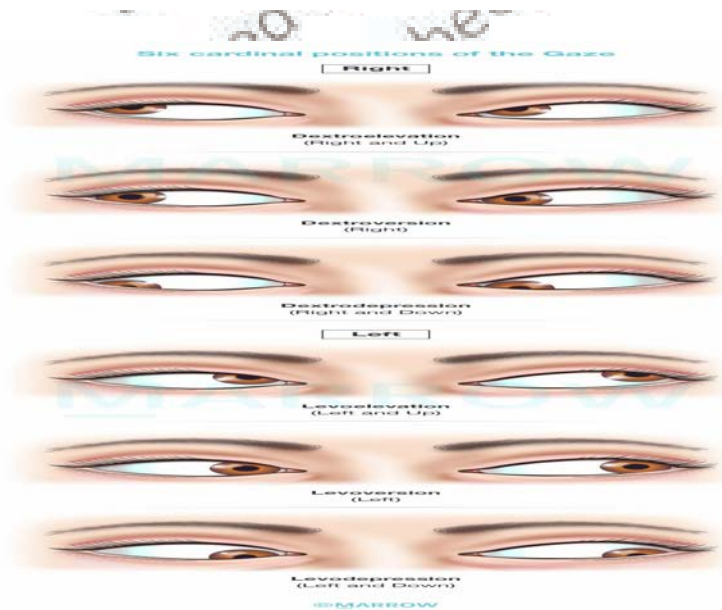
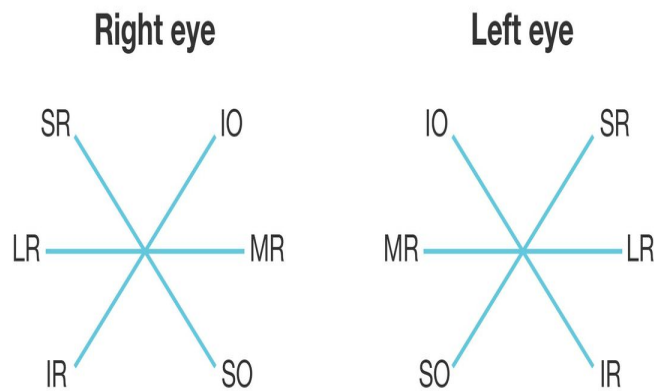
They are as follows:

- Dextroversion (right lateral rectus and left medial rectus)
- Laevoversion (left lateral rectus and right medial rectus)
- Dextroelevation (right superior rectus and left inferior oblique)
- Laevoelevation (left superior rectus and right inferior oblique)

- Dextrodepression (right inferior rectus and left superior oblique)
- Laevodepression (left inferior rectus and right superior oblique)

Note: There are 3 other positions, namely, normal primary position, supraversion, and infraversion. These are, however, not cardinal positions as the action of the muscles causing these positions can be elicited by the 6 cardinal positions themselves.

The image below illustrates the 6 cardinal positions of gaze.

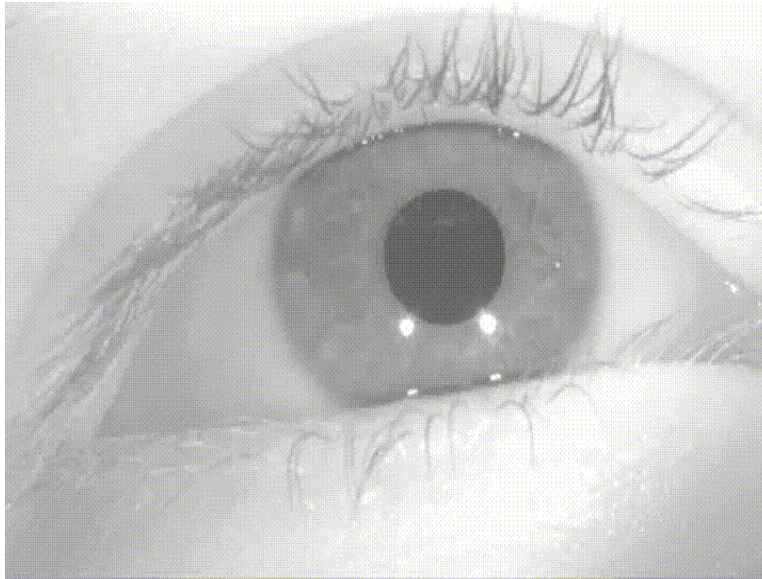


Solution to Question 8:

Saccades are sudden, jerky eye movements that occur as the gaze shifts from one object to another.

They are performed to bring the image of an object quickly on the fovea.

The image below shows saccadic movement of the eyeball



Eye movements that have a supranuclear control:

- Saccades
- Pursuits- smooth eye movements
- Vergence movement system- slow, disconjugate movements
- Vestibular eye movement- compensate for the effects of head movements
- Optokinetic system- during sustained head rotation
- Position maintenance system- consist of rapid micromovements such as flicks and drifts

Solution to Question 9:

The given image shows an optokinetic drum used to elicit optokinetic nystagmus (OKN). The parieto-occipital lobe controls the slow pursuit component, whereas the frontal lobes control the saccadic component.

OKN is normal physiological nystagmus in response to motion/continuously moving objects. It occurs in response to a repetitive visual stimulus moved across the visual field. The slow phase occurs in the direction of the moving stimulus; the fast phase occurs as a saccadic refixation movement in the opposite direction. In patients with congenital motor nystagmus, a reversal of the OKN response may occur. An OKN response elicited in a person who feigns visual loss indicates that visual input is present and is a malingerer.

Solution to Question 10:

All are nystagmoid movements except optokinetic.

Optokinetic nystagmus is a physiological nystagmus.

Nystagmoid movements:

- Ocular flutter
- Ocular bobbing
- Opsoclonus
- Superior oblique myokymia

Solution to Question 11:

Bruns nystagmus is an unusual type of bilateral nystagmus most commonly occurring in patients with cerebellopontine angle tumors.

It is caused by the combination of slow, large-amplitude nystagmus (gaze paretic nystagmus) when looking towards the side of the lesion, and rapid, small amplitude nystagmus (vestibular nystagmus) when looking away from the side of the lesion.

Option A- Cerebellar lesions and Arnold–Chiari malformation: downbeat nystagmus

Options C and D- Arnold–Chiari malformation and demyelination syndromes: periodic alternating nystagmus (left beating nystagmus->null phase->right beating nystagmus->null phase).

Solution to Question 12:

Crossed diplopia is seen in a divergent squint, as in medial rectus paralysis.

Binocular diplopia: Simultaneous perception of two images of a single object when seen by both eyes.

Causes:

- Anisometropia
- Paralysis of extraocular muscles; commonest
- Mechanical restriction of movements

Types:

Uncrossed—Harmonious

- False image on the same side of the deviation
- Seen in convergent squint

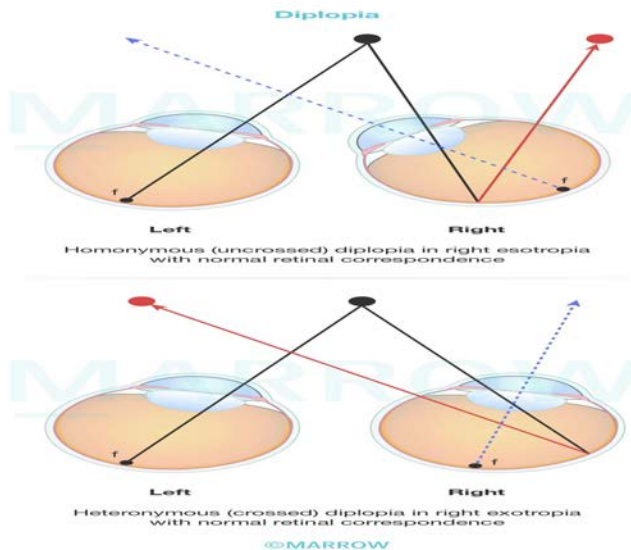
Crossed—Unharmonious

- False image on the opposite side of the deviation
- Seen in divergent squint

Unilateral diplopia: Simultaneous perception of two images of a single object when seen by a single eye.

Causes:

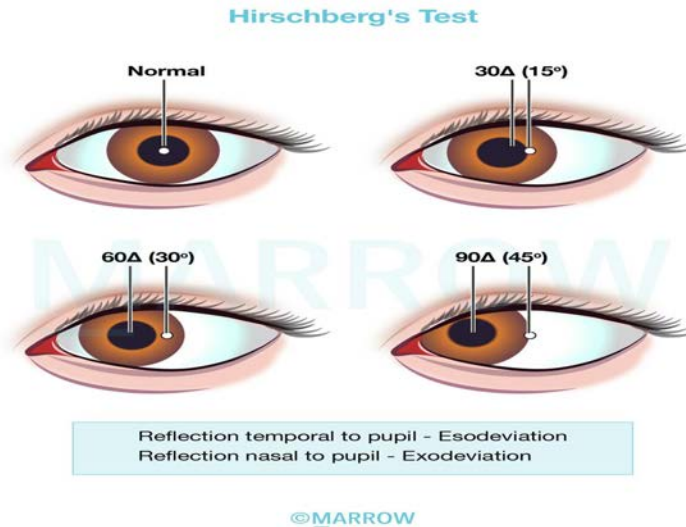
- Subluxated clear lens
- Subluxated intraocular lens (IOL)
- Double pupil
- Incipient cataract
- Keratoconus



Solution to Question 13:

The Hirschberg test is used to detect squint.

The Hirschberg test gives a rough objective estimate of the angle of manifest strabismus. A pen torch is shone into the eyes from arm's length, and the patient is asked to fixate on the light. The corneal reflection of the light will be (more or less) centered in the pupil of the fixating eye but will be decentred in a squinting eye, in the direction opposite to that of the deviation. The distance of the corneal light reflection from the center of the pupil is noted; each millimeter of the deviation is approximately equal to 7°.



Solution to Question 14:

From the given image of a Hirschberg test, it can be inferred that the patient has 90 prism dioptres (PD) of exotropia.

On the Hirschberg test, if the light reflection is on the nasal limbus, it means 45 degrees exotropia. One degree is equal to 2 PD (prism dioptre). Therefore, 45 degrees exotropia would mean 90 PD exotropia in the right eye.

Solution to Question 15:

Binocular single vision is tested by a synoptophore.

Binocular single vision is the state of simultaneous vision, which is achieved by the coordinated use of both eyes so that separate and slightly dissimilar images arising in each eye are appreciated as a single image by the process of fusion.

The synoptophore compensates for the angle of squint and allows stimuli to be presented to both eyes simultaneously. It can thus be used to investigate the potential for binocular function in the presence of a manifest squint and is of particular value in assessing young children (from age 3 years), who generally find the test process enjoyable. It can also detect suppression.



Solution to Question 16:

The given image depicts Grade II of binocular vision, fusion.

Solution to Question 17:

The patient will see 2 red lights as shown in image C as he is suffering from left suppression.

Worth's four-dot test is a sensory function test. A box is viewed which has four lights: one red, two green, and one white. The patient has to wear red-green goggles.

Inference:

- A: Normal—All 4 lights seen
- B: Abnormal retinal correspondence—All 4 lights seen in the presence of manifest squint
- C: Left Suppression—2 red lights seen
- D: Right suppression—3 green lights seen
- E: Diplopia—5 lights seen (2 red, 3 green)

Solution to Question 18:

Forced duction test (FDT) is done to find out ocular muscle palsy.

It is performed to differentiate between the incomitant squint due to paralysis of the extraocular muscle and that due to the mechanical restriction of the ocular movements.

Forced duction test:

- Positive in cases of incomitant squint due to mechanical restriction
- Negative in cases of extraocular muscle palsy

Solution to Question 19:

The test shown is the cover - uncover test.

Solution to Question 20:

Alternate cover test helps in differentiating concomitant squint from paralytic squint is a true statement. In a concomitant squint, the eye moves rapidly to fixate, whereas in a paralytic squint, the eye will not move to fixate.

Solution to Question 21:

The given test is a Maddox rod test, and the pattern of the appearance of white dot/light in relation to the red line denotes a case of left eye exophoria. In the given image, the above image refers to the examiner's view and the below image refers to the patient's view.

The Maddox rod can be used to test for horizontal and vertical deviations, and when used in conjunction with another Maddox rod, for cyclo-deviations.

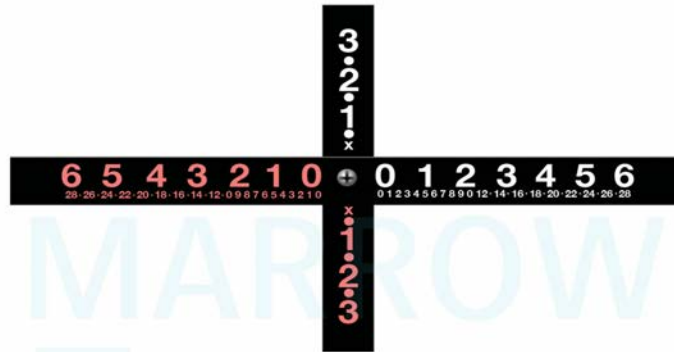
Interpretation:

To test for horizontal deviations, the Maddox rod is placed in front of one eye (for this example, the left eye) with the cylinders in the horizontal direction. The patient fixates a point source of light and then sees a vertical red line with the left eye and white dot/light with the right eye:

- Orthophoria: if the light/dot superimposes the line.
- Exo-deviation/exophoria: In this example, light/dot is on the left side of the line (rod placed on left eye)
- Eso-deviation/esophoria: if the light was on the right side of the line (rod placed on left eye)

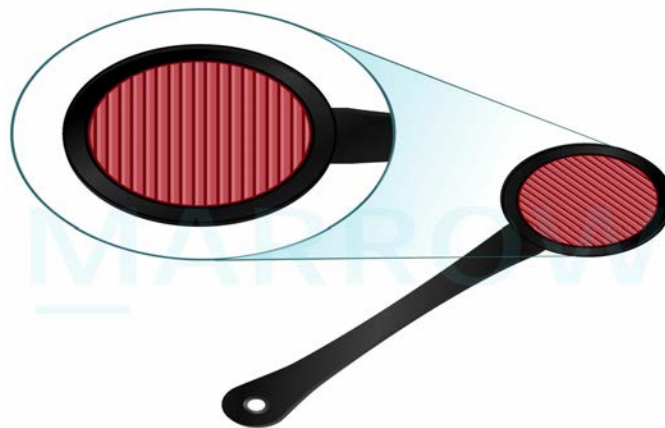
Maddox Rod Test involves the use of a Maddox rod and the Maddox tangent scale. It uses a specially constructed device, consisting of a series of parallel cylinders, that converts a point source of light into a line image. The optical properties of the cylinders cause the streak of light to be situated 90° to the orientation of the parallel cylinders.

Maddox tangent scale



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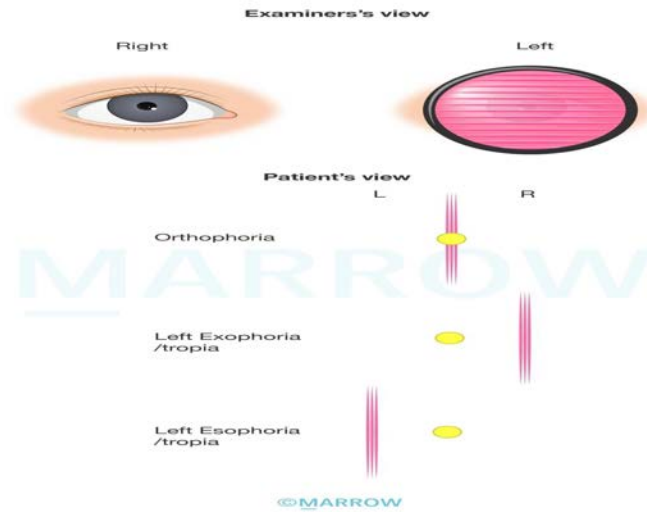
Maddox rod



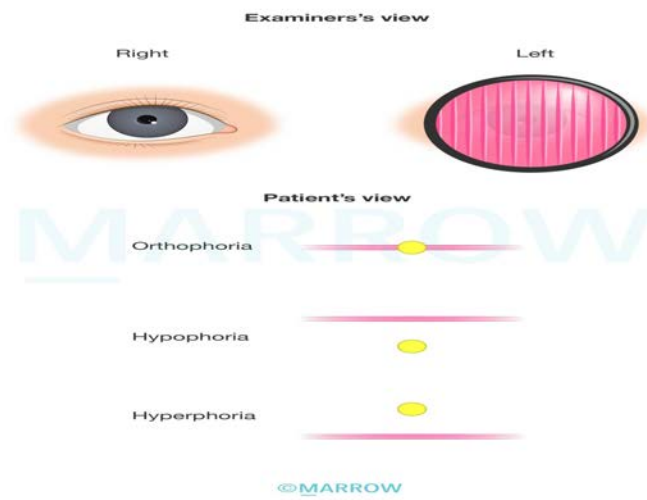
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In the image given below, the interpretation of the result is based on how the patient visualizes it. It would appear to be crossed to the patient. Pictorial representation of the eyes are as seen by the examiner. Imagine you are looking through a Maddox rod over your left eye while interpreting the images.

Maddox rod test



Maddox rod test



Strabismus - Types and Treatment

Question 1:

A child is brought by her parents to the OPD with complaints of crossed eyes as shown below. By doing a simple torchlight test you conclude that the child has pseudostrabismus and the inter-pupillary distance is normal. Which of the following can be associated with the child's condition?



- a) Negative angle kappa
- b) Hypertelorism
- c) Hypermetropia
- d) History of retinopathy of prematurity

Question 2:

A child presents to the OPD with a mild squint. The rest of the history is quite insignificant. On examination, Hirschberg's corneal reflex test is found to be normal. All other eye parameters are also normal except for hypertelorism. What is the most likely diagnosis?

- a) Pseudoesotropia
- b) Exophoria
- c) Esophoria
- d) Psuedoexotropia

Question 3:

Which of the following conditions is least likely to present with diplopia?

- a) Manifest squint
- b) Paralytic squint
- c) Latent squint
- d) Restrictive squint

Question 4:

Which of the following is the most common type of squint seen in children?

- a) Congenital esotropia
- b) Congenital exotropia
- c) Accommodative esotropia
- d) Primary exotropia

Question 5:

Which of the following is true regarding comitant squint?

- a) Primary deviation > Secondary deviation
- b) Primary deviation < Secondary deviation
- c) Primary deviation = Secondary deviation
- d) No co-relation between primary and secondary deviations

Question 6:

A mother noticed that her child has had malaligned eyes since birth but she couldn't afford treatment due to financial constraints. She now presents at 2-months of age and the child has the following finding on examination. An ophthalmoscopic examination under full cycloplegia reveals no abnormalities in refraction and media. Cranial nerves are intact. Which of the following statements is true regarding this condition?



- a) Angle of deviation is large and constant.
- b) Amblyopia develops in 90 % of the cases.
- c) It is associated with latent vertical nystagmus.
- d) Correction by spectacles is the treatment of choice.

Question 7:

A surgeon decides to resect a part of a muscle while operating on a case of strabismus. This action would lead to which of the following?

- a) Weakening of the muscle
- b) Strengthening of the muscle
- c) Paralysis of the muscle
- d) No change in strength of the muscle

Question 8:

Which of the following statements is false regarding accommodative esotropia?

- a) It is the most common type of squint.
- b) Miotics are used as the treatment strategy in the high AC/A ratio
- c) It is associated with high myopia.
- d) Refractive type can be fully corrected by use of spectacles.

Question 9:

Which of the following is the best line of treatment in a 3-year-old child presenting with a recent onset of 15-degree accommodative esotropia?

- a) Orthoptic exercise
- b) Surgery
- c) Relieving prism
- d) Refractive correction

Question 10:

Charlie, a 5-year-old boy with crossed eyes received his first eye exam from a primary care ophthalmologist. Examination, including cycloplegic refraction and dilated fundus exam, resulted in diagnoses of constant right esotropia, and amblyopia in the right eye. By what age should his amblyopia be corrected?

- a) 10 years
- b) 8 years
- c) 12 years
- d) 14 years

Question 11:

A 5-year-old female patient presented for her first complete eye examination after a school vision screening found unequal visual acuity. Her parents have not noted any eye problems. Visual acuity in the right eye was 20/20 and the left eye is 20/80. The rest of the examination was unremarkable and glasses were prescribed. She was diagnosed with anisometric hyperopia and amblyopia of the left eye when her visual acuity at a return appointment was found to be 20/20 RE and 20/60 LE at a distance while wearing her new glasses. What is the treatment of choice?

- a) Convergent exercises
- b) Prism correction
- c) Surgery
- d) Conventional occlusion

Question 12:

A 3-year-old child presents with a right convergent squint of 6 months duration. Choose the most appropriate treatment strategy.

- a) Immediate surgical correction followed by amblyopia therapy
- b) Proper refractive correction, amblyopia therapy followed by surgical correction
- c) Prescription of spectacles and defer surgery until 5 years
- d) Botulinum toxin injection followed by occlusion therapy

Question 13:

In which of the following conditions should an ophthalmologist prescribe miotics?

- a) Accommodative esotropia
- b) Intermittent exotropia
- c) Paralytic esotropia
- d) All of the above

Question 14:

Paralytic squint presents with all except

- a) Deviation
- b) Confusion
- c) Diplopia
- d) Stereopsis

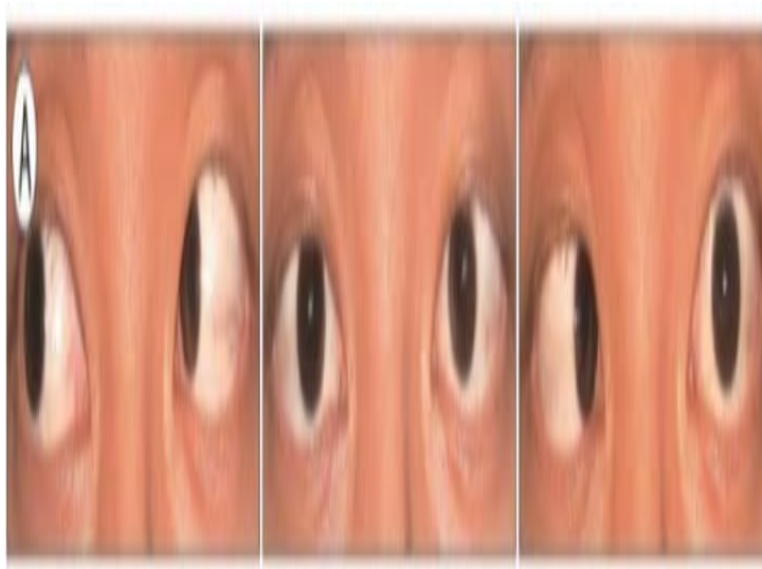
Question 15:

A 40-year-old diabetic patient presented with the ■squint. On examination, the secondary deviation was found to be more than the primary deviation. Forced duction test is negative. Which of the following types of squint is most likely to be present in this patient?

- a) Comitant squint
- b) Restrictive squint
- c) Paralytic squint
- d) Pseudosquint

Question 16:

Identify the condition.



- a) Left 3rd nerve palsy
- b) Right 6th nerve palsy
- c) Left 6th nerve palsy
- d) Right 3rd nerve palsy

Question 17:

A patient complains of difficulty in climbing down the stairs due to double vision. On examination, you notice his right eye has deviated upward and the deviation becomes more pronounced on the left gaze. The remaining extraocular movements are intact. Which nerve is most likely involved?

- a) Right oculomotor nerve
- b) Right trochlear nerve
- c) Left trochlear nerve
- d) Left oculomotor nerve

Question 18:

A patient suffers head trauma following which he complains of diplopia. His extraocular motility testing in the 9 gazes is as shown in the image. What is the most likely diagnosis?



- a) Internuclear ophthalmoplegia
- b) One and a half syndrome
- c) Trochlear nerve palsy
- d) Oculomotor nerve palsy

Question 19:

An 18-year-old destitute woman reports to a free eye camp with vision problems. Best-corrected visual acuity was 6/12 in the right eye and 6/6 in the left eye. Elevation in the right eye was absent. Her right eye showed 14–16 prism diopter (PD) hypotropia. Bell's phenomenon was absent in the right eye. Diagnosis of right eye double elevator palsy was made. Which of the following muscles are affected in her?

- a) Levator palpebrae superioris and Muller muscle
- b) Levator palpebrae superioris and superior rectus
- c) Muller muscle and Inferior oblique
- d) Superior rectus and Inferior oblique

Question 20:

A patient reports to the OPD with complaints that he feels dizzy on lateral gaze. His range of movements is shown below. Jerky movements are also observed in the left eye on the left gaze however convergence is normal. Which of the following structures is most likely affected?



- a) Occipital lobe
- b) Paramedian pontine reticular formation (PPRF)
- c) Medial longitudinal fasciculus
- d) Pretectal fibres

Question 21:

Identify the condition



A



B



C

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- a) Duane's retraction syndrome type 2
- b) Duane's retraction syndrome type 3

- c) Duane's retraction syndrome type 1
- d) Duane's restriction syndrome type 4

Question 22:

An eleven-year-old girl presented with a history of squint and poor vision. She had visual acuities of 6/24 and counting fingers in her right and left eyes respectively. There was a left hypotropia in the primary position of gaze with associated marked restriction of elevation in adduction and a positive forced duction test. A diagnosis of Brown syndrome was made. The pathology involves which of the following muscles?

- a) Superior oblique
- b) Superior rectus
- c) Levator palpebrae superioris
- d) Inferior oblique

Question 23:

Brown's syndrome simulates which muscle palsy?

- a) Superior oblique
- b) Superior rectus
- c) Inferior oblique
- d) Inferior rectus

Question 24:

Identify the condition.



- a) V pattern exotropia
- b) A pattern esotropia
- c) V pattern esotropia
- d) A pattern exotropia

Question 25:

A patient presents with left-sided head tilt which on straightening leads to right hypotropia. This increases on dextroversion and right head tilt. Paralysis of which of the following muscles is involved?

- a) Right superior oblique
- b) Right superior rectus
- c) Right inferior oblique
- d) Left inferior oblique

Answer Key

Question No.	Correct Option
1	a
2	d
3	c

4	c
5	c
6	a
7	b
8	c
9	d
10	b
11	d
12	b
13	a
14	d
15	c
16	c
17	b
18	d
19	d
20	c
21	c
22	a
23	c
24	b
25	b

Detailed Explanations

Solution to Question 1:

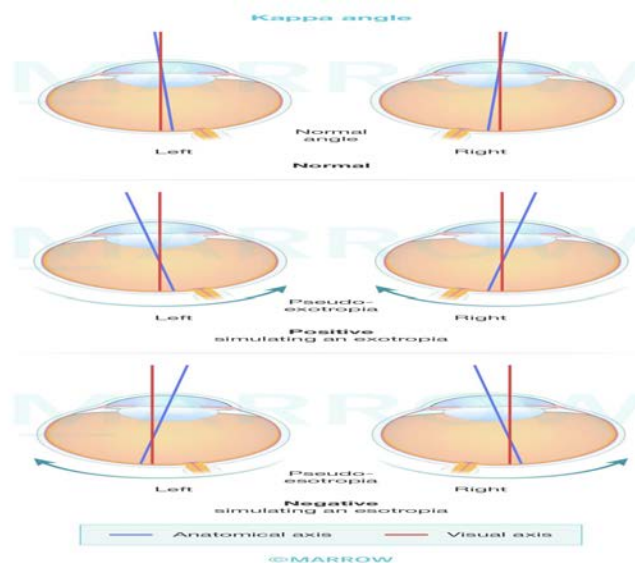
The child has pseudoesotropia (apparent convergent squint) which is associated with negative angle kappa and epicanthal folds.

Pseudostrabismus—apparent squint (visual axes are parallel, but eyes seem to have a squint)

Types:

- Pseudoesotropia
- Apparent convergent squint
- Prominent epicanthal fold
- Negative angle kappa
- Pseudoexotropia

- Apparent divergent squint
- Hypertelorism
- Positive angle kappa - due to retinopathy of prematurity, hypermetropia



Few definitions:

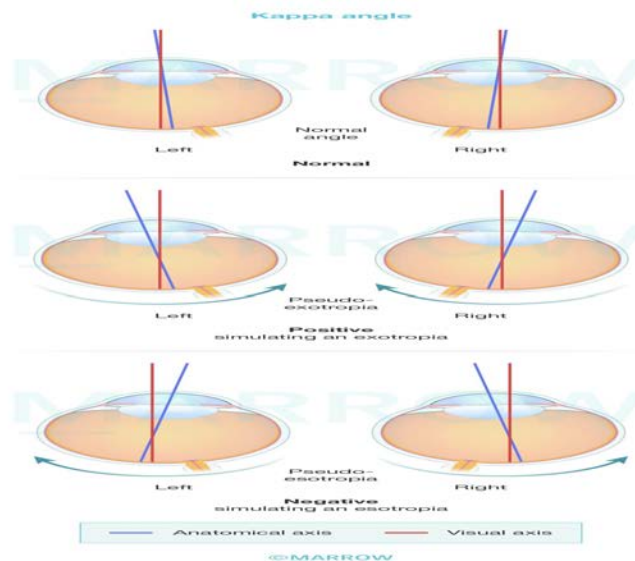
- Orthophoria- implies perfect ocular alignment in the absence of any stimulus for fusion
- Heterophoria ("phoria")- implies a tendency of the eyes to deviate when fusion is blocked (latent squint)
- Heterotropia ("tropia")- implies a manifest deviation in which the visual axes do not intersect at the point of fixation
- Hypertropia- upward displacement of one eye relative to the other is termed as hypertropia and a latent upward imbalance, a hyperphoria
- Hypotropia- downward displacement is termed a hypotropia and a latent imbalance, a hypophoria

Solution to Question 2:

The given clinical scenario is suggestive of pseudoexotropia.

Features of Pseudoexotropia:

- Normal Hirschberg's corneal reflex test
- Apparent divergent squint
- Associated with hypertelorism/a positive angle kappa.



Solution to Question 3:

Latent squint does not always present with diplopia.

Heterophoria (Latent squint) is a condition in which the tendency of the eyes to deviate is kept latent by fusion. When the influence of fusion is removed, the visual axis of one eye deviates away.

Types:

- Esophoria-convergent
- Exophoria-divergent
- Hyperphoria-deviation upwards
- Hypophoria-deviation downwards
- Cyclophoria-rotation around AP axis

Symptoms:

- Usually asymptomatic-compensated type.
- In cases where there are symptoms-decompensated type
- Headache
- Eye ache
- Photophobia
- Intermittent diplopia (only under fatigue – temporarily)

Solution to Question 4:

Accommodative esotropia is the most common type of squint seen in children.

Most common type of squint overall: accommodative esotropia

Second most common overall: congenital/Infantile esotropia

Most common type of exotropia: primary exotropia

Solution to Question 5:

In comitant squint (concomitant), the primary deviation is equal to the secondary deviation.

Primary deviation = Deviation of the squinting eye

Secondary deviation = Deviation of the normal eye under cover when the patient fixes with the squinting eye.

In incomitant (paralytic and restrictive squint), secondary deviation $>$ primary deviation due to Herring's Law of equal innervation.

Solution to Question 6:

The age at presentation, esotropia, no refractory error, no media opacities, and no cranial nerve palsies suggest a diagnosis of essential infantile esotropia. The angle of deviation in infantile esotropia is fairly large ($> 30^\circ$) and constant.

Infantile esotropia, also known as non-accommodative esotropia is the most common type of congenital esotropia.

Clinical features:

- Comitant convergent squint
- Age of onset- 1 to 2 months
- Angle of deviation- large and constant
- Amblyopia- develops in 25–40% of cases
- Associations include:
 - Inferior oblique overaction
 - Dissociated vertical deviation (70%)
 - Latent horizontal nystagmus

Treatment:

- Surgery- treatment of choice
- Amblyopia correction measures: Patching of normal eye
- Recession of both medial recti
- Timing: Preferably before 1 year

Solution to Question 7:

In squint surgery, muscle resection leads to the strengthening of the muscle.

The main types of procedure in squint surgery are as follows:

- Weakening to decrease the effective strength of action of a muscle.
- Recession slackens a muscle by moving it away from its insertion. It can be performed on any muscle except the superior oblique.
- Disinsertion (or myectomy) involves detaching a muscle from its insertion without reattachment. It is most commonly used to weaken an overacting inferior oblique muscle.
- Posterior fixation suture: The principle of this (Faden) procedure is to suture the muscle belly to the sclera posteriorly so as to decrease the pull of the muscle in its field of action without affecting the eye in the primary position.
- Strengthening, to enhance the pull of a muscle.
- Resection shortens a muscle to enhance its effective pull. It is suitable only for the rectus muscle.
- Tucking of a muscle or its tendon is usually confined to the enhancement of the action of the superior oblique muscle.
- Advancement of the muscle nearer to the limbus can be used to enhance the action of a previously recessed rectus muscle.
- Vector adjustment procedures have the primary aim of altering the direction of muscle action.
- Transposition refers to the relocation of one or more extraocular muscles to substitute for the action of an absent or severely deficient muscle. The most common indication is severe lateral rectus weakness due to acquired palsy of the 6th cranial nerve palsy.

Solution to Question 8:

Accommodative esotropia is associated with a high degree of hypermetropia (+4 to +7 D).

Accommodative esotropia is the most common type of squint. Typically presents around 2 to 3 years of age.

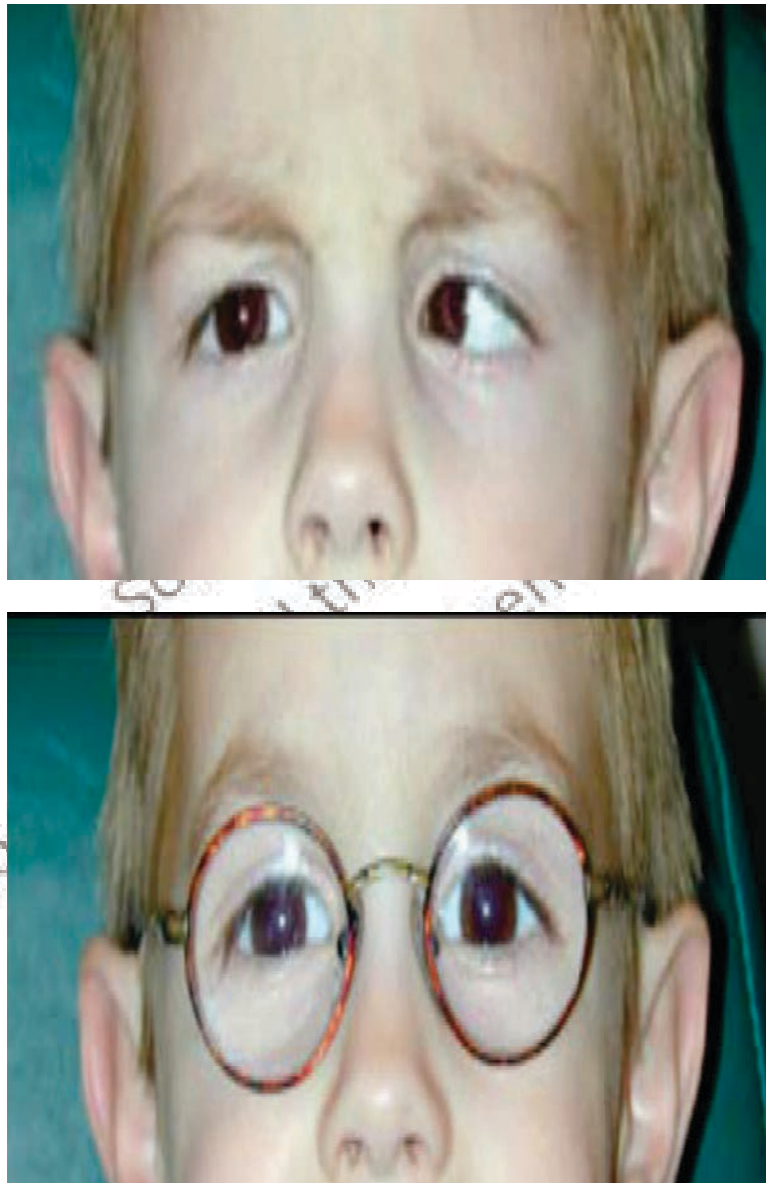
Types:

- Refractive
- Associated with high hypermetropia
- Fully correctable by spectacles
- Non-refractive type
- Abnormal accommodative convergence/accommodation (AC/A) ratio
- Fully corrected by bifocal glasses
- Miotics useful in cases where glasses are unsuitable
- Echthiophate causes iris cyst

- Prevention of iris cysts- Phenylephrine drops
- Mixed type

The accommodative convergence/accommodation (AC/A) ratio is defined as the amount of convergence measured in prism diopters per unit (diopter) change in accommodation.

The image below shows accommodative esotropia with and without correction.



Solution to Question 9:

The best line of treatment in accommodative esotropia is a refractive correction.

Esotropia for distance is corrected by the treatment of hypermetropia: convex glasses

Esotropia for near is corrected by +3DS lens.

Algorithm for the management of accommodative esotropia is as follows:

- R: Refractive correction: Glasses/Miotics
- O: Orthoptics/Occlusion: for amblyopia
- O: Operative correction: for non-accommodative component
- P: Prism correction: for remaining error

Solution to Question 10:

Strabismic amblyopia should be corrected before 8 years of age.

The sensitive period during which acuity of an amblyopic eye can be improved is usually up to 7–8 years in strabismic amblyopia and maybe longer (into the teens) for anisometropic amblyopia when a good binocular function is present.

In the absence of an organic lesion, a difference in best-corrected visual acuity (VA) of two Snellen lines or more (or >1 log unit) is indicative of amblyopia.

Treatment of amblyopia:

- Occlusion of the normal eye, to encourage the use of the amblyopic eye, is the most effective treatment.
- Penalization, in which vision in the normal eye is blurred with atropine, is an alternative method.

Solution to Question 11:

Occlusion of the normal eye is the treatment of choice for amblyopia.

Amblyopia is an uncorrectable decrease in vision in one or both eyes with no apparent structural abnormality seen to explain it. It is a diagnosis of exclusion. Generally, a difference of two lines or more on an eye chart test of visual acuity between the two eyes or a best-corrected vision of 20/30 or worse would be defined as amblyopia. For example, our patient has 20/20 vision with the right eye and only 20/80 with the left and the left eye even after best-corrected vision couldn't achieve $<20/30$ vision, the left eye is said to be amblyopic.

Clinical features:

- Decreased contrast sensitivity
- Diminution of vision
- Sunglass effect- brightness is less compared to the normal eye
- Crowding phenomenon- slipping of images

Treatment of choice:

- Occlusion of the normal eye (4 to 6 hours/day) using patches.
- Penalization: Blurring of vision using atropine.

Prism correction is not the right treatment as the child doesn't have strabismus.

Solution to Question 12:

The appropriate management in this case would be proper refractive correction, amblyopia therapy followed by surgical correction

The algorithm for management of accommodative squint is as follows:

- R: Refractive correction: Glasses/Miotics
- O: Orthoptics/Occlusion: For amblyopia
- O: Operative correction: For non-accommodative component
- P: Prism correction: For remaining error

Solution to Question 13:

Miotics are used in the management of accommodative esotropia.

The use of miotics in convergent strabismus is based on the peripheral spasm of accommodation caused by the drug. The accommodation triad involves convergence. An increased accommodative effort increases the amount of convergence. Pilocarpine would produce accommodation spasm and eliminate the accommodative convergence that would otherwise occur when the patient looks at a near object. It is useful in cases of an abnormal AC/A ratio as in, accommodative esotropia.

Solution to Question 14:

In squint, the deviation of the eye results in two vastly different images. The brain is unable to fuse them. Hence, there is no stereopsis.

Features of paralytic squint:

Symptoms:

- Diplopia
- Confusion
- Nausea, Vertigo
- Ocular deviation

Signs:

- Primary deviation & Secondary deviation
- Compensatory head posture—Ocular torticollis
- False projection or orientation

In adults- no amblyopia

In children <6 years- amblyopia can occur

Solution to Question 15:

The above clinical scenario is suggestive of paralytic squint. It is characterized by the presence of secondary deviation more than the primary deviation and a negative forced duction test.

The squint may be either comitant or incomitant.

- Comitant squint: The angle of squint measured is same in all gazes.
- Incomitant squint: The angle of squint measured is different in different gazes.

The differences between comitant and incomitant squint are shown in the table below:

The incomitant squint is further divided into two main types: Paralytic and restrictive.

Paralytic squint occurs due to the paralysis of extraocular muscle while restrictive squint is due to the mechanical restriction caused by tight or fibrosed muscle.

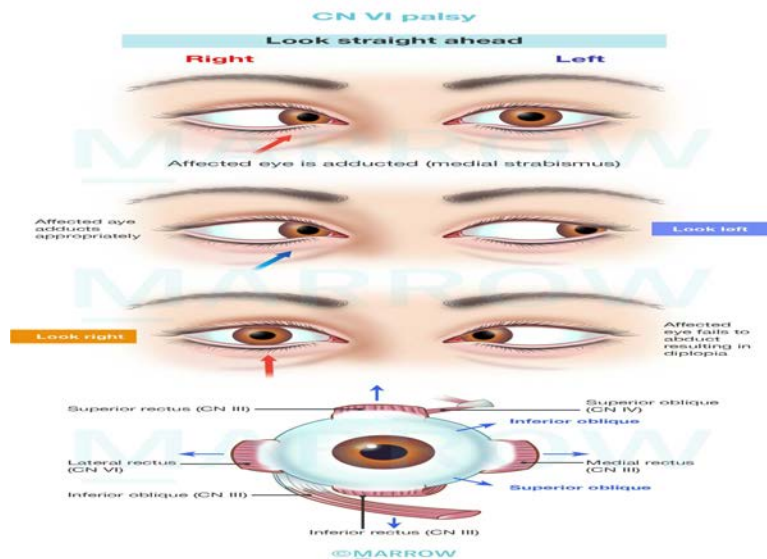
Forced duction test(FDT) differentiates between these two types. In this test, the eye is moved mechanically with the help of a forceps.

- The test is negative if there is no resistance to the movement. It is seen in a paralytic squint.
- The test is positive if there is resistance to the movement. It is seen in a restrictive squint.

Features	Comitantsquint	Incomitantsquint
Magnitude of squint	Same in all gazes	Varies with the gaze
Diplopia	Absent	Present
Ocular movements	Full	Restricted
False projection	Absent	Present
Abnormal head posture	Absent	Present
Secondary deviation	Equal to primary deviation	More than primary deviation

Solution to Question 16:

The patient has left eye esotropia in primary gaze which is suggestive of left abducent (6th) nerve palsy which innervates the left lateral rectus muscle.



Solution to Question 17:

The affected nerve in the given clinical scenario is the right trochlear nerve.

The right trochlear nucleus actually innervates the left superior oblique and vice versa. However, the nerve decussates at the midbrain at the level of the inferior colliculus. So, lesions of the trochlear nucleus lead to contralateral palsy. While that of trochlear nerve peripherally lead to ipsilateral palsy.

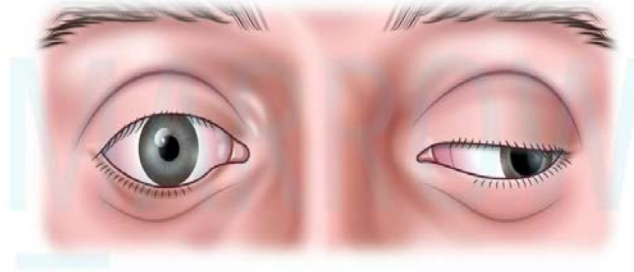
In this patient, the right-sided symptoms could be due to a lesion in the right trochlear nerve (ipsilateral) or left trochlear nucleus (contralateral). He complains of diplopia while climbing down the stairs because the normal eye can look down but the paralytic eye has deviated. There is also a limitation of depression due to superior oblique palsy, and hence the eye is deviated upward due to overaction of the inferior oblique. The obliques exert maximum force in an adducted position and hence the deviation becomes more prominent on the left gaze when the right eye is adducted.

Solution to Question 18:

The given scenario suggests a left oculomotor nerve palsy.

It is evident by the upper lid ptosis and down and out deviation of the left eye, as seen in the primary position of gaze (central image).

Left oculomotor nerve palsy



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Solution to Question 19:

In double-elevator palsy, the muscles involved are superior rectus and inferior oblique.

It is commonly associated with ptosis or pseudoptosis.

Solution to Question 20:

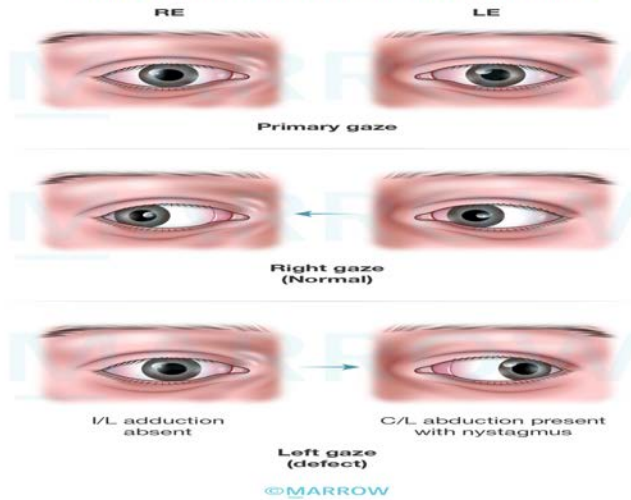
Lesion to the medial longitudinal fasciculus causes internuclear ophthalmoplegia (INO).

The right gaze is normal but on the left gaze, there is an inability of right eye adduction and in the left abduction is possible but with nystagmus. This along with normal convergence are all in favor of diagnosis of a lesion in right medial longitudinal fasciculus (MLF) leading to internuclear ophthalmoplegia.

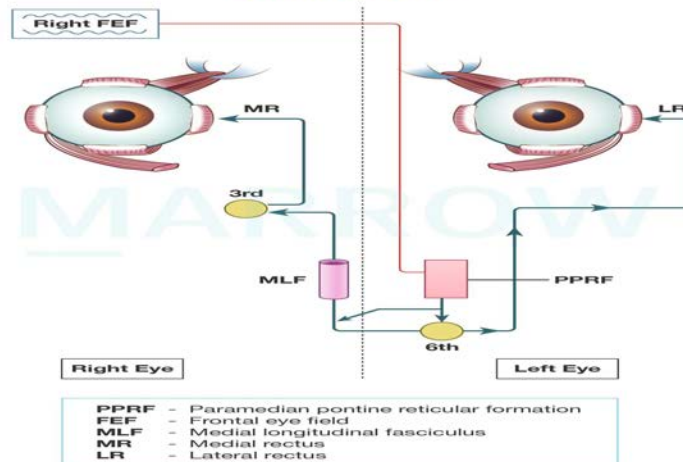
INO is seen in multiple sclerosis

The image below depicts internuclear ophthalmoplegia due to a lesion of the right MLF.

Internuclear ophthalmoplegia of Right Medial longitudinal fasciculus



Anatomical pathway for horizontal eye movements



Total ophthalmoplegia

- All extraocular and intraocular muscles are paralyzed.
- Paralysis of 3rd, 4th, and 6th nerve
- Seen in orbital apex syndrome, cavernous sinus thrombosis

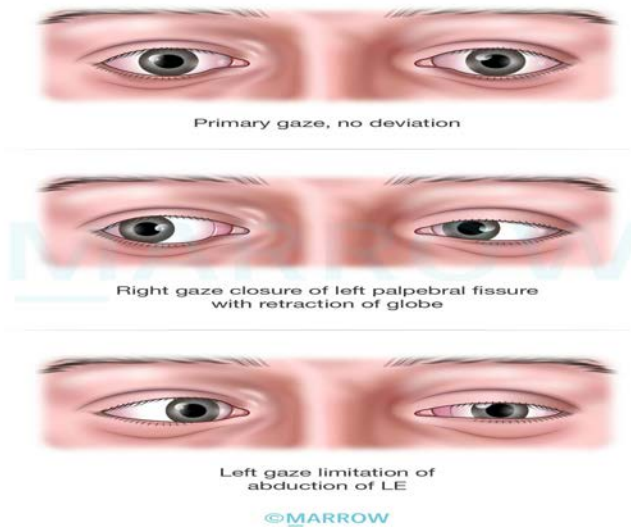
External ophthalmoplegia

- All extraocular muscles are paralyzed
- Lesions at the level of motor nuclei
- Sparing of Edinger Westphal nucleus

Solution to Question 21:

The image shows Duane's retraction syndrome type 1.

Paradoxical innervation of LR where it gets innervated by some fibres of III nerve instead of VI nerve. On impulse for abduction via VI nerve there is no abduction limitation. On impulse for adduction via III nerve both LR and MR contract thus causing retraction of globe (as the antagonist LR instead of relaxation, starts to contract)



Duane's retraction syndrome (DRS) is a cause of restrictive squint

There is a failure of innervation of the lateral rectus by a hypoplastic sixth nerve nucleus, with anomalous innervation of the lateral rectus by fibers from the third nerve. Retraction of the globe on attempted adduction. Widening of palpebral fissure on attempted abduction

Classification by Huber:

- Limitation of abduction- type 1
- Limitation of adduction- type 2
- Limitation of both- type 3

Note: DRS is an exception to Sherrington's law of reciprocal innervation.

Solution to Question 22:

Brown syndrome occurs due to mechanical restriction caused by a congenitally tight superior oblique sheath or by the impaired movement of the superior oblique tendon through the trochlea. It may be acquired in some cases.

Contracture of the superior oblique leads to restriction of its antagonistic muscle, the inferior oblique and that is why the patient presents with absence of elevation in adduction.

Solution to Question 23:

Brown's superior oblique tendon sheath syndrome simulates palsy of inferior oblique muscle. In both of these cases, there is a limited elevation of the eye in adduction.

Brown's superior oblique tendon sheath syndrome is a cause of restrictive squint. The forced duction test is positive.

Differential diagnoses of Brown's syndrome:

- Inferior oblique palsy- differentiated by a positive head-tilt test (negative in Brown's syndrome) and a negative forced duction test (positive in Brown's syndrome)
- Primary superior oblique overaction- differentiated by negative forced duction test (Positive in Brown's syndrome)

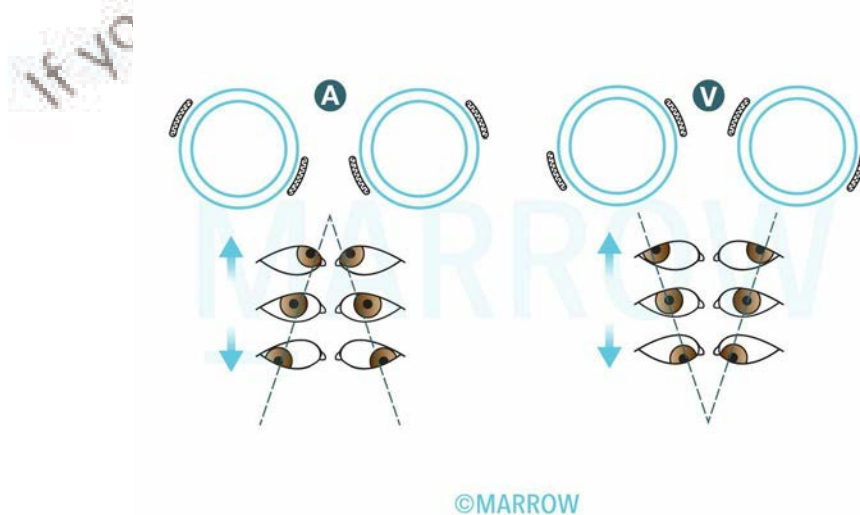
Solution to Question 24:

The given image classically depicts an A pattern esotropia. Relative convergence on upgaze (esotropia increases on upgaze) is a feature of A-pattern esotropia.

A or V pattern heterotropia:

- "V" or "A" patterns occur when the relative contributions of the superior rectus and inferior oblique to elevation, or of the inferior rectus and superior oblique to depression, are abnormal, resulting in the derangement of the balance of their horizontal vectors in up and downgaze.
- Oblique muscle dysfunction is the commonest
- "V" pattern is said to be significant when the difference between upgaze and downgaze is $\geq 15^\circ$.
- "A" pattern is considered significant if the difference between upgaze and downgaze is $\geq 10^\circ$.

The image below shows A and V pattern heterotropia



Solution to Question 25:

Right superior rectus palsy is the cause for the given clinical scenario.

The three-step test, also known as the Parks-Bielschowsky three-step test, is a diagnostic test used to identify which muscle is paralyzed in the case of an acquired hypertropia. This method uses the cover-uncover test, with or without the Maddox rod to measure the amount of deviation in different head positions.

The three-step test may also be used to diagnose the less common inferior oblique or vertical rectus muscle palsy. It is designed for the diagnosis of a single paretic vertical muscle and is unreliable when there are multiple paretic muscles or in restrictive strabismus. This test may also help in identifying if a superior rectus palsy is true or simulated in a patient with an inhibitional palsy of the contralateral antagonist.

Using the three-step test in the above clinical scenario:

Step 1. Which eye is hypotrophic?

The eye hypotrophic in the primary position is the right eye. The paretic muscles can be right eye elevator or left eye depressor.

- Right superior rectus
- Right inferior oblique
- Left inferior rectus
- Left superior oblique

Step 2. Does the hypotropia increase in the right gaze or left gaze?

The hypotropia increases in right gaze.

Therefore, the paretic muscle must be either (1) one of the muscles whose primary action is elevating the right eye on the right gaze, or (2) one of the muscles whose primary action is depressing the left eye in the right gaze. The left inferior oblique and the right superior rectus both have a primary action of elevation in right gaze. Therefore, one of these muscles could be the paretic muscle. Likewise, the left superior oblique and the right inferior rectus have a primary action of depression in the right gaze and could be paretic. But since the left inferior oblique and the right inferior rectus did not meet step 1, they can be eliminated from the possible paretic muscles. This leaves us:

- Left superior oblique
- Right superior rectus

Step 3. Does the hypotropia increase in the right head tilt or left head tilt?

The hypotropia increases in the right head tilt. If the deviation increases on right head tilt and decreases on left head tilt, as it does in this example patient, then the right superior rectus is implicated as the paretic culprit.

Disorders of Visual Pathway and Pupillary Reflexes

Question 1:

Which of the following forms the third-order neurons in the visual pathway?

- a) Bipolar cells
- b) From the lateral geniculate body
- c) Ganglion cells
- d) Optic tract

Question 2:

A 41-year-old woman with a known pituitary microadenoma presented with sudden-onset visual disturbances, headache, and hypotension during the second trimester of pregnancy. What is the visual field defect this patient is most likely to experience?

- a) Superio-temporal quadrantanopia
- b) Ipsilateral homonymous hemianopia
- c) Contralateral homonymous hemianopia
- d) Heteronymous hemianopia

Question 3:

Which of the following cannot be tested with monocular visual field testing using a pin with a red pinhead?

- a) Central scotoma
- b) Paracentral scotoma
- c) Temporal field defects
- d) Peripheral field defects

Question 4:

A lesion in which of the following structures is most likely to result in congruous homonymous hemianopia?

- a) Optic nerve
- b) Optic chiasma
- c) Optic tract
- d) Occipital cortex

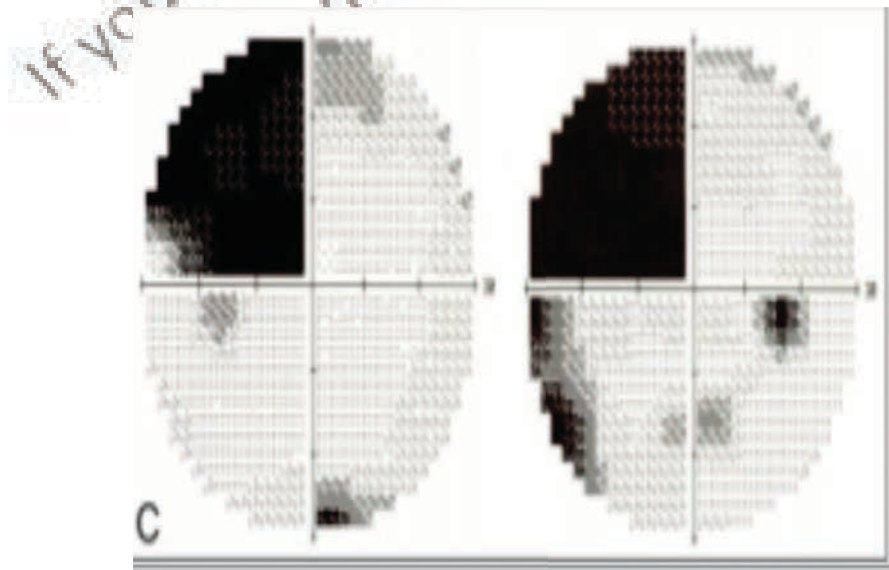
Question 5:

A 30-year-old woman presented with a persistent headache for the past few months. On evaluation, her vision on Snellen's chart was 6/6 in both eyes. The fundus also appears to be normal. Visual field charting shows a bitemporal visual field defect. Site of the lesion is _____

- a) Prelaminar optic nerve
- b) Optic nerve juxta lamina cribrosa
- c) Optic chiasma
- d) Optic tract

Question 6:

A young patient with visual disturbances was found to have a brain tumour. He was referred to ophthalmology for a complete visual field assessment before planning his resection. Humphrey analyzer revealed the following plot. What is the most likely site of the lesion?



- a) Optic tract
- b) Lateral geniculate body

- c) Parietal lobe
- d) Temporal lobe

Question 7:

The primary visual cortex is located in the _____.

- a) Superior temporal sulcus
- b) Parieto-occipital sulcus
- c) Posterior part of calcarine sulcus
- d) Central sulcus

Question 8:

A 72-year-old lady was brought by her son with bilateral loss of vision for 3 days, associated with slurred speech. It was preceded by an occipital headache. Despite her obvious blindness, illustrated by her walking into objects, the patient denied visual loss. MRI revealed bilateral occipital infarcts. Which of the following statement is true regarding the given condition?

- a) Direct and consensual reflexes are absent in both eyes.
- b) Direct and consensual reflexes are present in both eyes.
- c) Direct reflex is present but consensual reflex is absent.
- d) Consensual reflex is present but direct reflex is absent.

Question 9:

A neurology resident is demonstrating spinal reflexes in a comatose patient. When he pinches the skin over the side of the patient's neck, there is a rapid dilatation of the ipsilateral pupil. Which of the following forms the efferent limb of this reflex?

- a) Cholinergic fibres of oculomotor nerve
- b) Adrenergic fibres of oculomotor nerve
- c) Sympathetic fibres of cervical ganglion
- d) Motor fibres of trigeminal nerve

Question 10:

Which of the following is not part of the pupillary light reflex pathway?

- a) Edinger-Westphal nucleus
- b) Lateral geniculate body
- c) Pretectal nucleus
- d) Ganglion cells in retina

Question 11:

A patient presented with diplopia to the casualty. On examination, the left pupil was dilated. Both direct and consensual light reflexes were lost in the affected eye. What is the most probable diagnosis?

- a) II nerve palsy
- b) III nerve palsy
- c) V nerve palsy
- d) VII nerve palsy

Question 12:

A male patient came with a history of unprotected sexual intercourse around 20 years ago. Now, he complains of neurological symptoms and visual disturbances. Which of the following will be seen on exposure of his pupils to light?

- a) Pupils react but do not accommodate.
- b) Pupils accommodate but do not react.
- c) Pupils neither react nor accommodate.
- d) Pupils react and accommodate.

Question 13:

Which of the following would you suspect in a patient who presents with small pupils?

- a) Carbon monoxide poisoning
- b) Ethylene glycol poisoning
- c) Holmes–Adie pupil
- d) Pontine hemorrhage

Question 14:

A 67-year-old woman finally decides to get glasses, about thirty years after she actually needed them, and goes to an optometrist for an eyeglass prescription. Her eye examination reveals some unexpected findings. The pupils were small. When a pen-light is shone on to her eyes, neither pupil contracts. However, when the optometrist brings his fingers towards her face, both pupils constrict dramatically. What is the given condition?

- a) Anisocoria
- b) Adie's pupil
- c) Internal ophthalmoplegia
- d) Argyll-Robertson pupil

Question 15:

Wernicke's hemianopic pupil is due to a lesion in the _____.

- a) Optic radiation
- b) Optic tract
- c) Optic chiasma
- d) Lateral geniculate body

Question 16:

Which of the following pupillary abnormalities is most likely to be observed in a patient with primary open-angle glaucoma?

- a) Marcus Gunn pupil
- b) Adie's tonic pupil
- c) Argyll-Robertson pupil
- d) Hutchinson's pupil

Question 17:

The casualty intern noticed that a patient had unequal pupils. The anisocoria was greater under light. Which of the following drugs would be helpful in quickly differentiating between the potential causes?

- a) 1, 2, 3, 4 and 5

- b) 2 and 3 only
- c) 1, 4 and 5
- d) 1, 3 and 4

Question 18:

A 55-year-old lady with advanced spondylosis underwent cervical vertebral fusion surgery. In the post-operative room, the resident noted that her pupils were unequal. Which of the following features would rule out the possibility of Horner's syndrome in this patient?

- a) Enophthalmos
- b) Dilatation lag
- c) Mydriasis
- d) Reverse ptosis of lower lid

Question 19:

Where is the cell body of the second-order neuron in the sympathetic pathway for the pupil?

- a) Middle cervical ganglion
- b) Superior cervical ganglion
- c) Ciliospinal center of Budge
- d) Ciliary ganglion

Question 20:

A simultaneous lesion in both sympathetic and parasympathetic supply of the eye leads to which of the following?

- a) Mydriatic pupil that reacts to light
- b) Constricted pupil that reacts to light
- c) Mid-sized pupil that doesn't react to light
- d) Mid-sized pupil that reacts to light

Question 21:

During the neurological examination of a patient, on shining light in his left eye, the left pupil does not constrict but the right pupil does. On shining light in his right eye, the right pupil constricts and left doesn't. The defect is in:

- a) Right eye afferent
- b) Right eye efferent
- c) Left eye efferent
- d) Left eye afferent

Question 22:

Which of the following is the most common feature of third nerve palsy, in a case of long-standing, uncontrolled diabetes?

- a) Proptosis
- b) Pseudoptosis
- c) Ptosis
- d) Pupillary dysfunction

Question 23:

A 60-year-old man was brought to the OPD by his daughter with complaints of drooping of his left eyelid and double vision. The pupillary examination was normal. What is the likely cause?

- a) Aneurysmal rupture
- b) Trauma
- c) Diabetes
- d) Raised ICT

Question 24:

The patient presented with the following appearance when asked to look straight. What is the probable diagnosis?



- a) 3rd nerve palsy
- b) Horizontal gaze palsy
- c) 6th nerve palsy
- d) Internuclear ophthalmoplegia

Answer Key

Question No.	Correct Option
1	c
2	d
3	d
4	d
5	c
6	d
7	c
8	b
9	c
10	b
11	b
12	b
13	d
14	d

15	b
16	a
17	b
18	c
19	c
20	c
21	c
22	c
23	c
24	c

Detailed Explanations

Solution to Question 1:

Third-order neurons in the visual pathway are the ganglion cells.

The afferent neuronal impulses are carried in the following manner:

Photoreceptors (I) $\Rightarrow$ Bipolar cells (II) $\Rightarrow$ Ganglion cells (III) $\Rightarrow$ Lateral geniculate body (IV).

Solution to Question 2:

The given clinical scenario is suggestive of pituitary apoplexy in pregnancy. This acute onset event is almost always associated with compression at the level of optic chiasma causing heteronymous hemianopia (bitemporal hemianopia).

Pituitary apoplexy results from hemorrhagic infarction of a pre-existing pituitary adenoma or within a physiologically enlarged gland, which is well documented in pregnancy.

The nasal fibers of each retina cross through the chiasma. Slow-growing microadenomas compress lower nasal optic fibers causing superior temporal quadrantanopia initially, before progressing to bitemporal hemianopia. However, an abrupt infarction or hemorrhage of a pre-existing pituitary adenoma will lead to rapid progression of the visual defect.

Heteronymous hemianopia is the loss of half of the visual field on different sides in both eyes. It is separated into two categories:

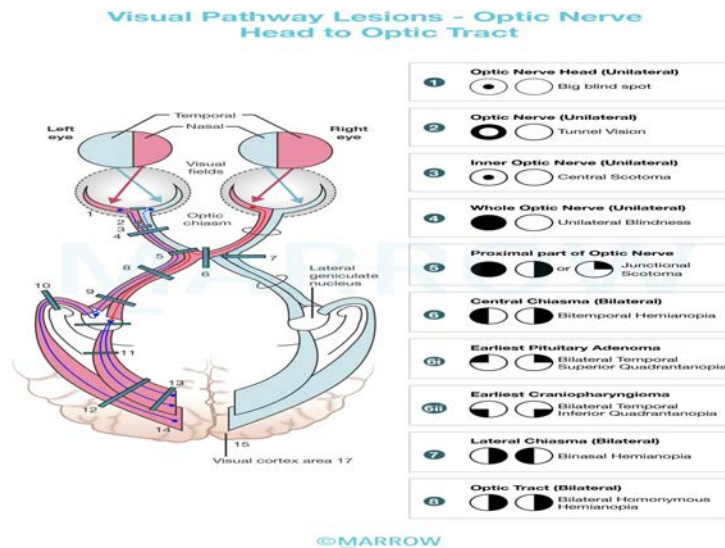
- Binasal hemianopia – the loss of the fields surrounding the nose
- Bitemporal hemianopia – the loss of the fields closest to the temples.

Common causes of bitemporal hemianopia due to central chiasmal lesions are as follows:

- Suprasellar aneurysms
- Tumors of the pituitary gland

- Craniopharyngioma
- Suprasellar meningioma
- Glioma of the third ventricle

Other visual defects according to the site of the lesion are as follows:



Solution to Question 3:

Central scotoma, paracentral scotoma and temporal field defects can be tested by monocular visual field testing with a red-headed neurological pin.

A patient who has central scotoma will either not see the pinhead in the centre of the field or perceive it black. This can be defined by moving the pin outwards in four different directions until the patient sees the pinhead as being red.

Placing the pin in each of the quadrants close to the centre and checking if the patient sees the red pinhead can detect a paracentral scotoma. This can also detect temporal field defects.

Solution to Question 4:

Congruous homonymous hemianopia is most suggestive of a lesion in the occipital cortex.

Homonymous hemianopia involves vision loss on the same side of the visual field in both eyes. This type of visual field loss is suggestive of a lesion posterior to the optic chiasma in the visual pathway.

Complete homonymous hemianopia involves the complete visual hemifields of both eyes and can occur with any large enough lesion posterior to the optic chiasma. Visual field examination alone cannot further localize the lesion in such cases.

Incomplete homonymous hemianopia is due to smaller lesions in the visual pathway posterior to the optic chiasm. They are further divided into congruous and incongruous depending on the extent to which the defects in the two half fields match each other exactly.

Because of a relative disorder among axon bundles in the immediate post-chiasmal region, diseases in this location produce rather incongruent lesions.

More posteriorly, the axon bundles of the corresponding areas of the two hemifields are more closely aligned and hence, lesions of the occipital cortex produce congruous homonymous hemianopia while incongruous homonymous hemianopia is more likely seen with optic tract lesions.

Solution to Question 5:

The site of the lesion is optic chiasma. Lesions at the chiasma characteristically produce bitemporal field defects due to the decussation of nasal fibres.

In retrochiasmal lesions, homonymous hemianopia is seen. Further, in cortical lesions, macular (central) vision is spared.

Solution to Question 6:

The visual field defect shown is the typical "pie in the sky" defect or superior quadrantic hemianopia. They are produced when the inferior fibers of the optic radiations in the temporal lobe (Meyer's loop) are involved.

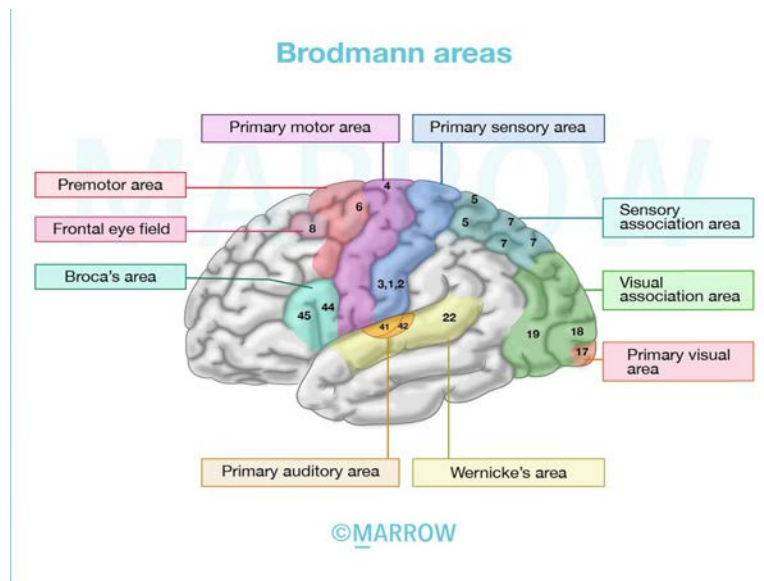
"Pie in the floor" defects refer to inferior quadrantic hemianopia. It is seen in the involvement of the superior fibers of optic radiations in the parietal lobe (Baum's loop).

Solution to Question 7:

The primary visual cortex(Brodmann area 17) is located in the posterior part of the calcarine sulcus in the occipital lobe.

Accessory visual areas are area 18 (secondary) and area 19 (tertiary).

Image: Brodman areas



Solution to Question 8:

The given clinical scenario is suggestive of cortical blindness with visual anosognosia (Anton syndrome). Since anterior visual pathways are not affected in cortical blindness, direct and consensual light reflexes are preserved.

Cortical blindness is usually associated with bilateral occipital infarcts where it is supplied by the posterior cerebral arteries and these infarcts usually involve both the primary visual cortex and visual association area. It is secondary to damage to the visual pathways posterior to the lateral geniculate nuclei; anterior visual pathways are intact. Visual field defects can vary depending on the extent and location of the infarct.

Direct and consensual reflexes are present in both eyes in cortical blindness as pupillary light reflex fibers exit the visual pathway at the level of the superior colliculus. They exit the optic tract before the lateral geniculate body (LGB) to reach the center for pupillary light reflex- pretectal nucleus at the midbrain.

Note: Anton syndrome is visual anosognosia (denial of loss of vision) associated with confabulation (defined as the emergence of memories of events and experiences which never took place) in the setting of obvious visual loss and cortical blindness.

Solution to Question 9:

The given scenario describes cilio-spinal reflex. Sympathetic fibers of cervical ganglion form the efferent limb of this reflex.

Cilio-spinal reflex:

- Afferent inputs of this reflex are carried by the trigeminal nerve or cervical pain fibers (lateral spinothalamic tract).

- It synapses at the ciliospinal center of Budge, located at C8 to T2 in the intermediolateral (IML) horn of the gray matter of the spinal cord.
 - The efferent branch of the reflex consists of:
 - Preganglionic sympathetic fibers from the upper thoracic and lower cervical spinal cord which goes to the cervical sympathetic chain and its three ganglia.
 - The postganglionic sympathetic fibers pass on the surface of the internal carotid artery and join the ophthalmic nerve at the cavernous sinus. Here, it passes through the nasociliary nerve to reach the ciliary body and innervates the dilator pupillae muscles, causing dilatation of the pupil.
- This reflex is absent in Horner's syndrome and in lesions involving the cervical sympathetic fibers and it is enhanced in patients with cluster headache and barbiturate-induced coma.

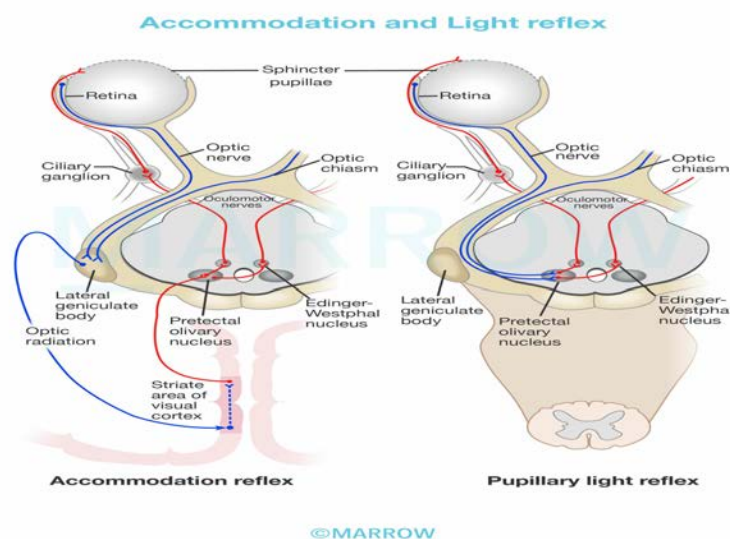
Solution to Question 10:

The lateral geniculate body is not a part of the pupillary reflex pathway. It is part of the accommodation reflex and the visual pathway.

Pupillary light reflex pathway:

- First neurons (sensory) connects each retina with both pretectal nuclei in the midbrain at the level of the superior colliculi.
- Second neurons (internuncial) connects each pretectal nucleus to both Edinger–Westphal nuclei.
- Third neurons (preganglionic motor) connects the Edinger–Westphal nucleus to the ciliary ganglion.
- Fourth neurons (postganglionic motor) leaves the ciliary ganglion and passes in the short ciliary nerves to innervate the sphincter pupillae.

Image: Pupillary reflex pathways



Solution to Question 11:

The given clinical situation suggests that the patient has III nerve (oculomotor nerve) palsy.

Absence of contraction of the pupil in both direct and consensual reflex suggests a lesion in the efferent pathway of the affected eye. This is III nerve/oculomotor palsy.

Light reflex consists of direct and consensual reflexes.

When light is shone in an eye,

- Constriction of the pupil of the same eye is called direct reflex.
- Constriction of the contralateral pupil is called consensual reflex/indirect reflex

For the light reflex,

- Afferent nerve fiber is the II cranial nerve (optic nerve).
- Efferent nerve fiber is the III cranial nerve (oculomotor nerve)-Bilaterally

In II nerve palsy, there will be a loss of direct reflex in the affected eye and loss of consensual light reflex in the opposite eye.

Solution to Question 12:

The patient most likely has acquired syphilis due to unprotected sex and has developed neurological and visual disturbances secondarily as a complication of neurosyphilis. Argyll-Robertson pupils(ARP) are seen associated with neurosyphilis where the pupils accommodate but do not react to light.

Argyll Robertson pupil is attributed to a dorsal midbrain lesion of neurosyphilis that affects the pupillary light pathway but spares the more ventrally placed pupillary near reflex pathway(Accommodation pathway), thereby causing a light-near dissociation.

Features of Argyll Robertson pupil (ARP) include:

- Bilateral small, irregular pupils in dim light
- Accommodation Reflex Present (ARP)- The eye accommodates by constricting the pupil for a near target.
- Light Reflex absent: pupils do not constrict when exposed to bright light.
- Cocaine eye drops(mydriatics) induced mydriasis.

Other clinical features in neurosyphilis:

- Fleeting and repetitive lancinating pains, primarily in the legs
- acute abdominal pain with vomiting(visceral crisis)
- Tabes dorsalis with Ataxia and gait disturbances due to loss of position sense
- Loss of reflexes in the legs
- Paresthesias, bladder disturbances

Other causes of light-near dissociation are:

In unilateral eye

- Adie pupil
- Herpes zoster ophthalmicus
- Aberrant regeneration of the third cranial nerve

In bilateral eyes

- Neurosyphilis
- Type 1 diabetes mellitus
- Myotonic dystrophy

Note: Adie's pupil more commonly manifests in young women with unilateral large regular pupil (Anisocoria) with absent light reflex and sluggish and tonic accommodation reflex.

It is caused by post-ganglionic parasympathetic pupillary supply damage and is usually associated with absent knee jerk.

Adie's pupil constricts with weak pilocarpine (0.1%) drops, while normal pupil and Argyll Robertson pupil does not.

Solution to Question 13:

Pontine hemorrhage is associated with small pupils. The remaining options, namely, carbon monoxide poisoning, ethylene glycol poisoning, and Holmes–Adie pupil, cause dilation of the pupils.

Solution to Question 14:

The given clinical situation of bilateral small irregular pupils with light-near dissociation suggests a diagnosis of Argyll Robertson pupil.

Features of Argyll Robertson pupil (ARP):

- Accommodation Reflex Present (ARP)
- Light Reflex absent
- Constricted pupil
- An example of light near dissociation.
- Seen in neurosyphilis

Option A: Anisocoria is when the pupils are not of the same size.

Option B: Adie's tonic pupil

- Light reflex absent or sluggish.
- Near reflex: Slow and tonic
- Pupil is larger

- Constricts with weak pilocarpine.

Option C: Ophthalmoplegia interna

- Dilated paralyzed pupil
- Does not respond to light or accommodation

Solution to Question 15:

Wernicke's hemianopic pupil is due to a lesion in the optic tract.

The optic tracts contain both visual and pupillomotor fibers.

The visual fibers terminate in the lateral geniculate body, but the pupillary fibers leave the optic tract anterior to the lateral geniculate body, projecting through the brachium of the superior colliculus to terminate in the pretectal nuclei.

An optic tract lesion may, therefore, give rise to an afferent pupillary conduction defect.

Characteristically, the pupillary light reflex will be normal when the unaffected hemiretina is stimulated and absent, when the involved hemiretina is stimulated (i.e., light is shone from the hemianopic side).

Solution to Question 16:

Primary open-angle glaucoma commonly causes bilateral and asymmetric optic nerve damage. Marcus Gunn pupil also known as relative afferent pupillary defect (RAPD) is the earliest sign of optic nerve disease as seen in optic nerve neuritis or primary open-angle glaucoma (POAG).

RAPD is caused due to a defect in the afferent pathway (optic nerve) of the pupillary light reflex in one eye.

It is diagnosed by the swinging flashlight test where the flashlight is swung to and fro, and the pupillary response is observed. Normally, both pupils constrict when any one eye is stimulated (due to direct and consensual light reflex). In the presence of RAPD, both pupils will abnormally dilate when the affected eye is stimulated. This paradoxical response of the pupil to light is called Marcus Gunn pupil.

Causes of RAPD:

- Optic nerve lesions (e.g. All types of optic neuritis, glaucoma)
- Optic tract lesions (e.g., infarction)
- Severe retinal disease (e.g., central retinal artery occlusion, extensive retinal detachment)
- Maculopathy (macular degeneration)

Other options:

Option B: Adie tonic pupil is a unilateral dilated pupil with light-near dissociation due to postganglionic parasympathetic pupillomotor nerve supply damage. In these patients the light reflex is absent and the accommodation (near) reflex is sluggish. It is commonly seen in young

females and is associated with absent knee jerk.

Option C: Argyll Robertson pupils are bilateral small pupils with light-near dissociation. It is commonly seen in patients with neurosyphilis. In these patients the pupillary reflex is absent but the accommodation reflex is present (ARP).

Option D: Hutchinson's pupil is a clinical sign where the pupil on the side of an intracranial mass lesion is dilated and unreactive to light, due to compression of the oculomotor nerve on that side.

Solution to Question 17:

Anisocoria greater in the light indicates the large pupil to be abnormal due to poor pupillary contraction from a parasympathetic abnormality. Among the given options, 1% pilocarpine and 0.1% pilocarpine can be used to differentiate fixed dilated pupil.

Mydriasis due to neurologic origin (3rd nerve palsy) responds readily to 1% pilocarpine, whereas there will be no response if the dilation is due to anticholinergic causes. Pupillary constriction to 0.1% pilocarpine diagnoses Adie's pupil (due to denervation hypersensitivity). Application of 0.1% pilocarpine shows no response in other conditions or in the normal eye.

Example: A person presenting with dilated pupil following accidental application or inadvertent rubbing of atropine or antimuscarinic alkaloids from wild plants into the eye will not constrict in response to topical application of 1% Pilocarpine. But the patient showing dilation due to 3rd nerve palsy shows pupillary constriction in response to pupillary instillation of 1% pilocarpine.

Note: Cocaine and phenylephrine are used in anisocoria that is greater in the dark to diagnose Horner's syndrome. In these patients, the small pupil fails to dilate even after the instillation of cocaine.

Solution to Question 18:

Horner's syndrome causes all the following except mydriasis. It causes miosis.

Horner's syndrome (oculosympathetic palsy)

The majority of the cases are unilateral.

Causes of bilateral involvement include cervical spine injuries and autonomic diabetic neuropathy.

Painful Horner syndrome, especially of acute onset, should raise the possibility of carotid dissection.

Clinical features

- Mild ptosis (usually 1–2 mm) as a result of weakness of Müller muscle
- Anisocoria is accentuated in dim light
- Pupillary constriction to light and near stimuli is normal.
- Hypochromic heterochromia (lighter color iris on affected side); in congenital cases.

- Mild ptosis (Mullers muscle is affected)
- Reduced ipsilateral sweating (only if the lesion is below the superior cervical ganglion)

Pharmacological tests

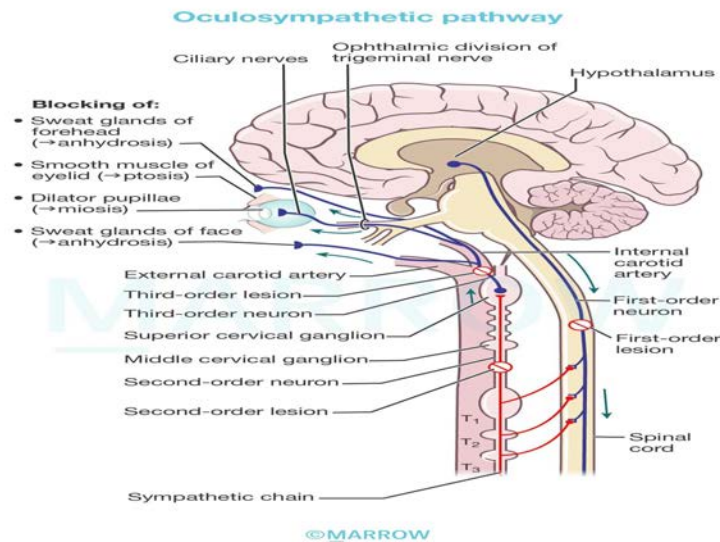
- Hydroxyamphetamine and adrenaline may be used to differentiate a preganglionic from a postganglionic lesion.
- Phenylephrine (1%)
 - Result: In an established (10 days) postganglionic lesion, the Horner pupil will dilate and ptosis may be temporarily relieved. A central or preganglionic Horner pupil and a normal pupil will not dilate or will dilate minimally.
 - Explanation: In postganglionic Horner's syndrome, the dilator pupillae muscle develops denervation hypersensitivity to adrenergic neurotransmitters due to its dysfunctional local motor nerve.
- Apraclonidine (0.5% or 1.0%)
 - Result: A Horner pupil will dilate, but a normal pupil is essentially unaffected.
 - Explanation: Alpha-1 receptors are upregulated in the denervated dilator pupillae.
- Cocaine (4%) is instilled into both eyes
 - Result: The normal pupil will dilate, but the Horner pupil will not.
 - Explanation: Cocaine blocks the re-uptake of noradrenaline secreted at the postganglionic nerve ending, which accumulates and causes dilatation of a normal pupil. In Horner's syndrome, there is no noradrenaline being secreted, so cocaine has no effect.

Solution to Question 19:

The cell body of the second-order neuron in the sympathetic pathway for the pupil lies at the ciliospinal center of Budge (C8–T1).

The sympathetic pathway to the iris muscles consists of a chain of three neurons:

- The first neuron arises from the hypothalamus along the brain stem to synapse in the intermediolateral column of the spinal cord—the ciliospinal center of Budge (C8–T1).
- From here (C8–T1), the second-order neuron arises. This joins the cervical sympathetic chain and synapses at the superior cervical ganglion.
- The third-order neuron arises from the superior cervical ganglion and travels through the cavernous sinus along with the internal carotid artery.
- Sympathetic fibers join the ophthalmic division of the trigeminal nerve and pass through the ciliary ganglion to the nasociliary and short ciliary nerves.



Solution to Question 20:

Simultaneous lesion in both sympathetic and parasympathetic supply of the eye leads to mid-sized pupil that doesn't react to light.

Pupillary dilation is achieved by sympathetic innervation of dilator pupillae. Therefore, any lesion of the sympathetic supply of the pupil leads to constricted pupil.

Pupillary constriction is a parasympathetic function. Lesions in this pathway leads to dilated pupil and failure to constrict.

A mid-sized unreactive pupil due to a lesion of both parasympathetic and sympathetic supplies is seen in aneurysms of the internal carotid artery within the cavernous sinus.

Solution to Question 21:

The above scenario is suggestive of left eye efferent (oculomotor nerve) defect.

The loss of pupillary light reflex (both direct and consensual) in the left eye associated with a preserved direct pupillary response in the right eye is characteristic of an efferent pupillary defect.

Solution to Question 22:

The most common feature seen in diabetic oculomotor palsy among the given options is ptosis.

In the anatomical arrangement of the nerve fibers of the 3rd cranial nerve, the fibers controlling the pupillary reflex are present superficially and are thus spared from diabetic ischemia-induced injury which starts from the central nerve fibres. Thus, pupil sparing is commonly seen in diabetic patients.

The etiology of diabetic neuropathy is hyperglycemia-induced damage to nerve cells and neuronal ischemic change due to microvascular angiopathy.

Involvement of the third cranial nerve is most common and it often presents with early features of ptosis and diplopia with pupil sparing.

The involvement of levator palpebrae superioris muscle of the eyelid manifests as ptosis and the involvement of extraocular muscles such as medial rectus, inferior rectus and inferior oblique presents as ophthalmoplegia with diplopia. All the muscles mentioned above are supplied by the Oculomotor nerve.

Note: Aneurysm of the posterior communicating artery is an important cause of isolated third nerve palsy with involvement of the pupil. This is because the aneurysm impinges on the superficial pupil nerve fibres of the 3rd cranial nerve.

Solution to Question 23:

Ptosis and double vision with no other features suggest an acquired, isolated 3rd nerve palsy with pupillary sparing. This is seen in ischemic conditions such as diabetes.

The vasoneurium supplies the nerve from outside to inside. As the somatic fibres are located in the inner part of the nerve, these fibres are affected more in the setting of ischemia. The parasympathetic pupillary fibres which run on the outside can remain unaffected (resulting in pupillary sparing). In compressive lesions, the parasympathetic fibres are affected before any disruption of the motor fibres. So pupillary sparing is not seen.

Isolated 3rd nerve palsy

Causes

- Microvascular disease associated with systemic risk factors such as hypertension and diabetes is the most common cause of third nerve palsy.
- Aneurysm of the posterior communicating artery at its junction with the internal carotid, both direct and secondary to subdural hematoma with uncal herniation.
- Episodic- migraine, compression, ischemia, and alterations in intracranial pressure
- Miscellaneous- tumour and inflammatory diseases such as syphilis, Lyme disease, and sarcoidosis; giant cell arteritis; and vasculitis associated with collagen vascular disorders.

Signs

- Profound ptosis due to the weakness of the levator muscle
- Abduction and depression in the primary position ("down and out")
- Normal abduction as the lateral rectus is intact
- Limited adduction due to medial rectus weakness
- Limited elevation due to the weakness of the superior rectus and inferior oblique

Solution to Question 24:

The above image shows adduction of the right eye due to paralysis of the right lateral rectus, indicating right abducens (6th) nerve palsy.

The nucleus of the sixth (abducens or abducent) nerve lies at the mid-level of the pons.

Nuclear lesion: A nuclear sixth nerve lesion also causes the failure of horizontal gaze towards the side of the damage due to the involvement of the adjacent horizontal gaze center (paramedian pontine reticular formation).

Foville (inferior medial pontine) syndrome is most frequently caused by vascular disease or tumors involving the dorsal pons. It is characterized by the following:

- Ipsilateral involvement of the fifth to eighth cranial nerves
- Central sympathetic fibers (Horner syndrome)
- Horizontal gaze palsy

Millard–Gubler (ventral pontine) syndrome involves the fasciculus as it passes through the pyramidal tract. As well as ipsilateral sixth nerve palsy, there is contralateral hemiplegia and often an ipsilateral lower motor neuron (LMN) facial nerve palsy.

Clinical features:

- Esotropia in the primary position due to relatively unopposed action of the medial rectus
- Limitation of abduction on the side of the lesion
- Normal adduction of the affected eye
- A compensatory face turn is towards the side of the paralyzed muscle in unilateral palsy

Disorders of Optic Nerve and Gaze Palsies

Question 1:

The longest part of the optic nerve measures _____.

- a) 30 mm
- b) 40 mm
- c) 50 mm
- d) 20 mm

Question 2:

Which of the following is not a feature of papilledema?

- a) Extracellular edema
- b) Axonal swelling
- c) Stasis of axoplasmic flow
- d) Disruption of neurofilament

Question 3:

A resident notes the following fundus picture in a patient referred to the ophthalmology OPD with visual complaints. What does it suggest?



- a) Retinal detachment
- b) Papilledema
- c) Primary optic atrophy
- d) Secondary optic atrophy

Question 4:

Which of the following conditions show ipsilateral optic atrophy and contralateral papilledema?

- a) Weber syndrome
- b) Foster-Kennedy syndrome
- c) Millard-Gubler syndrome
- d) Benedikt syndrome

Question 5:

A 45-year-old man presents to the casualty with a sudden loss of vision and painful ocular movements. On examination, the eye is white and there are no obvious signs on ophthalmoscopy. What is the most likely diagnosis?

- a) Optic nerve glioma
- b) Retrobulbar neuritis
- c) Eale's disease
- d) Neuroretinitis

Question 6:

A 33-year-old patient has come to the ophthalmology OPD with complaints that her vision becomes temporarily blurry after a hot bath in the evening. This resolves by itself after a few hours. She has a past history of an episode of painful unilateral optic neuritis, 3 months back which resolved with steroids. What is being described here?

- a) Pulfrich phenomenon
- b) Uhthoff phenomenon
- c) Lhermitte's sign
- d) Riddoch phenomenon

Question 7:

In a patient with multiple sclerosis, which of the following visual field defects is most likely?

- a) Bitemporal hemianopia
- b) Central scotoma
- c) Enlarged blind spot
- d) Tunnel vision

Question 8:

Optic neuritis is a common presentation in which of the following diseases?

- a) Myasthenia gravis
- b) Guillain-Barré syndrome
- c) Multiple sclerosis
- d) Amyotrophic lateral sclerosis

Question 9:

A 26-year-old female patient presents with 6/60 visual acuity in one eye and 6/18 in the other eye. She has recurrent episodes of diminution of vision which recovers on treatment with steroids. She also has a history of transverse myelitis. What is the most probable diagnosis?

- a) Hereditary spastic quadriplegia
- b) Subacute combined degeneration
- c) Neuromyelitis optica
- d) Multiple sclerosis

Question 10:

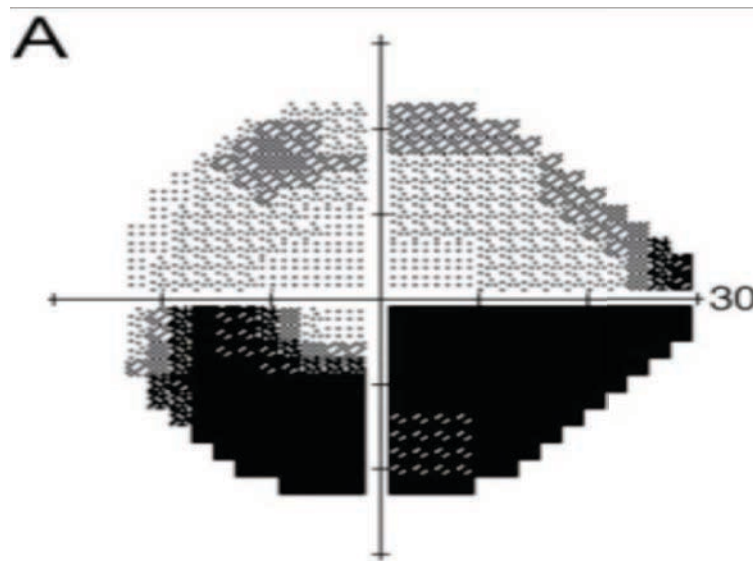
A middle-aged man with suspected optic neuropathy is being evaluated for visual field defects. Goldman perimetry reveals bilateral centrocecal scotomas. Which of the following is most likely to be the causative agent?

- a) Tobacco
- b) Ethanol

- c) Methanol
- d) Chloroquine

Question 11:

An anxious, young man came to the ER with complaints of having sudden trouble seeing with his left eye and a recent history of tripping over things. There was no associated pain or difficulty moving the eyes. An ophthalmology consultation was requested. On perimetry, the following visual field defect was picked up. What is the most likely diagnosis?



- a) Methanol intoxication
- b) Anterior ischemic optic neuropathy (AION)
- c) Suprasellar tumor
- d) Lesions to occipital cortex

Question 12:

What does the given fundoscopic picture suggest?



- a) Primary optic atrophy
- b) Secondary optic atrophy
- c) Glaucomatous optic atrophy
- d) Consecutive optic atrophy

Question 13:

Which of the following conditions is not associated with optic atrophy?

- a) Retinitis pigmentosa
- b) Leprosy
- c) Pituitary tumor
- d) Methanol intake

Question 14:

A 24-year-old man with a history of pernicious anemia was referred to ophthalmology for complaints of gradually progressing visual disturbances. On perimetry, which of the following defects is most likely to be seen in him?

- a) Enlarged blindspot
- b) Bitemporal hemianopia
- c) Heteronymous hemianopia
- d) Centrocecal scotoma

Question 15:

A 15-year-old boy has bilateral optic atrophy with diabetes mellitus and diabetes insipidus. What is the likely diagnosis?

- a) Kjer optic neuropathy
- b) Behr syndrome
- c) Wolfram syndrome
- d) Leber's congenital amaurosis

Question 16:

Which of the following is not a feature of amaurosis fugax ?

- a) Sudden onset
- b) Transient nature
- c) Monocular blindness
- d) Irreversible nature

Question 17:

Champagne cork disc is seen in which of the following conditions?

- a) Papilledema
- b) Papillitis
- c) Pseudopapillitis
- d) Anterior ischemic optic neuropathy

Question 18:

Which of the following defects is implicated in Leber's hereditary optic neuropathy?

- a) Post-translational defect
- b) Sporadic point mutation
- c) Mitochondrial DNA defect
- d) Chromosomal translocation

Question 19:

A 34-year-old man developed a painless diminution of vision in his right eye 6 months ago but did not seek medical attention. Two weeks ago, he started to notice mild blurring in his left eye and has now come for evaluation. According to him, his elder brother and maternal uncle also had similar vision problems. Which fundus finding would point towards Leber's hereditary optic neuropathy in this patient?

- a) Sectoral optic atrophy
- b) Disc leakage on fluorescein angiography
- c) Hyperemic optic disc with telangiectatic capillaries
- d) Papilledema and macular star

Question 20:

In order to diagnose a case as idiopathic intracranial hypertension, which of the following features is not necessary?

- a) Normal cerebrospinal fluid composition
- b) Elevated opening pressure on lumbar puncture
- c) Bilateral papilledema
- d) Normal neuroimaging studies

Question 21:

Which of the following is the most reliable sign to clinically differentiate between a case of papillitis and papilledema?

- a) Amount of disc elevation
- b) Degree of venous congestion
- c) Pupillary reaction
- d) Disc hyperemia

Question 22:

A patient evaluated for dizziness and visual disturbances was found to have down-beating nystagmus. A brain lesion was suspected and the patient was referred for an MRI. What is the

most likely site of the lesion?

- a) Parasellar region and optic chiasm
- b) Pineal region and the dorsal midbrain
- c) Cervico-medullary junction
- d) Posterior cerebral hemisphere

Question 23:

An RTA patient with severe head trauma is admitted to the surgical ICU for observation. On examination, nystagmus was noted. His CT scan reveals a transverse fracture of his left temporal bone. Which of the following statements is true regarding the nystagmus?

- a) Slow phase is towards the left and fast phase towards the right
- b) Slow phase is towards the right and fast phase towards the left
- c) It is an up-beating type of nystagmus
- d) It is a down-beating type of nystagmus

Question 24:

Which of the following is the site of lesion in internuclear ophthalmoplegia?

- a) Medial Longitudinal Fasciculus
- b) 6th Nerve nucleus
- c) Pontine paramedian reticular formation
- d) A & C

Question 25:

You are evaluating an elderly patient in the stroke ICU, who was admitted three days back with a history of sudden onset nausea, vomiting, and gait abnormalities. The visual examination showed conjugate horizontal left gaze palsy. He could maintain gaze near mid-position but could not cross the midline to the left side. His MRI is likely to show involvement of which of the following brain areas?

- a) Parapontine reticular formation
- b) Pretectal nucleus
- c) Medial longitudinal fasciculus

d) Occipital lobe

Question 26:

A patient has normal vertical saccades. On attempted left gaze as shown in the image below, the patient develops left jerk nystagmus. Where is the most likely location of the lesion?



- a) Left paramedian pontine reticular formation
- b) Right third nerve nucleus
- c) Left medial longitudinal fasciculus
- d) Right medial longitudinal fasciculus

Question 27:

Ophthalmoplegic migraine clinically presents as _____.

- a) Headache with irreversible loss of optic nerve function
- b) Headache associated with recurrent third nerve palsy
- c) Headache associated with third, fourth, and sixth nerve palsy
- d) Headache associated with optic neuritis

Answer Key

Question No.	Correct Option
1	a
2	d
3	b
4	b
5	b
6	b
7	b
8	c
9	c
10	a
11	b
12	c
13	b
14	d
15	c
16	d
17	a
18	c
19	c
20	c
21	c
22	c
23	a
24	a
25	a
26	d
27	c

Detailed Explanations

Solution to Question 1:

Intraorbital part is the longest part of the optic nerve.

It measures from 25 mm to 30 mm.

Length of the optic nerve ranges from 3.5 cm to 5.5 cm.

80% of the fibers originate from the macular region.

Parts of optic nerve:

Parts	Length
Intraocular part	1 mm
Intraorbital part	25–30 mm
Intracanalicular part	5–9 mm
Intracranial part	10–16 mm

Solution to Question 2:

Disruption of neurofilaments is not a feature of papilledema.

Papilledema is the swelling of the optic nerve head secondary to raised intracranial pressure (ICP).

"Disc swelling" and "disc edema" are non-specific terms that include papilledema but also indicate a disc swollen from other causes.

All patients with papilledema should be suspected of harboring an intracranial mass.

Features of papilledema:

- Raised ICP
- Edema of the optic nerve head
- Blockage of axoplasmic flow
- Axonal swelling
- Collection of extracellular fluid
- Painless progressive loss of vision
- Enlargement of the blind spot
- Color vision, visual acuity, and pupillary reaction are normal, until optic atrophy sets in.

Solution to Question 3:

The fundoscopic image shows dilated veins with a blurred disc margin which is suggestive of papilledema.

Ophthalmoscopic features of papilledema:

- Early-stage:

- Obscuration of the disc margins (nasal margins first)
- Absence of spontaneous venous pulsation at the disc
- Mild hyperemia of the disc
- The blurring of the peripapillary nerve fiber layer
- Established papilledema:
 - The disc appears to be enlarged and elevated (mushroom or dome-shaped)
- Macular fan appears
- Paton's lines- circumferential greyish white folds may develop due to the separation of nerve fibers
- Visual fields show enlargement of the blind spot
- Veins become tortuous and engorged
- Multiple cotton-wool spots and superficial hemorrhages
- Chronic or long-standing (vintage) papilledema:
 - The optic disc gives the appearance of the dome of a champagne cork
 - Small drusen-like crystalline deposits (corpora amylacea) may appear on the disc surface
- Atrophic stage:
 - Secondary optic atrophy

Solution to Question 4:

Ipsilateral optic atrophy and contralateral papilledema are seen in Foster–Kennedy syndrome.

Weber syndrome (superior alternating hemiplegia) is a form of stroke characterized by the presence of an ipsilateral oculomotor nerve palsy and contralateral hemiparesis or hemiplegia.

Millard–Gubler syndrome is a lesion of the pons. It is also called ventral pontine syndrome.

Benedikt syndrome, also known as paramedian midbrain syndrome, involves the fasciculus of oculomotor nerve as it passes through the red nucleus and is characterized by ipsilateral third nerve palsy and contralateral extrapyramidal signs such as hemitremor.

Solution to Question 5:

The given clinical scenario of a sudden painful loss of vision in the absence of fundoscopic changes is suggestive of retrobulbar neuritis.

There are 3 types of optic neuritis: papillitis, neuroretinitis, and retrobulbar neuritis. The common causes of retrobulbar neuritis include demyelinating disorders such as multiple sclerosis (MS) and neuromyelitis optica (NMO or Devic's disease).

Clinically, optic neuritis is characterized by sudden, progressive visual loss. Ocular movements are painful in retrobulbar neuritis and painless in papillitis and neuroretinitis. Glowing sensations known as phosphenes may be induced with sound or movement. Color vision may also be impaired.

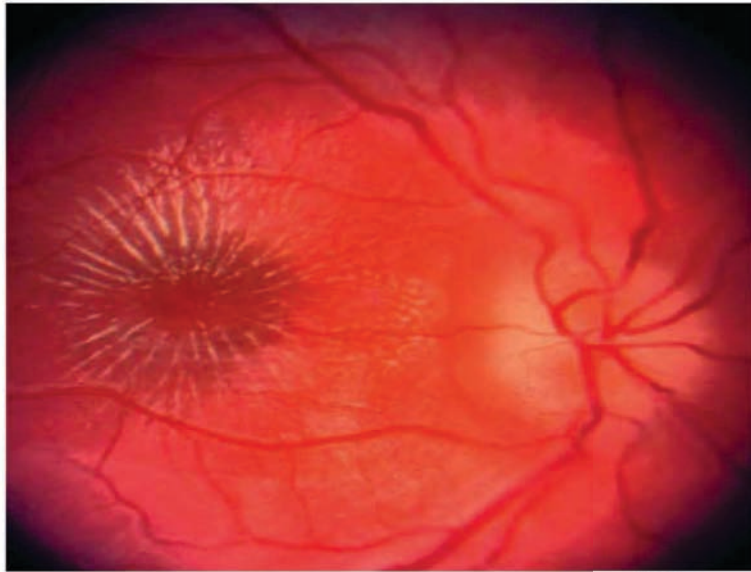
On examination, Marcus Gunn pupil or relative afferent pupillary defect (RAPD) is usually noted due to asymmetric involvement. The most common visual field defect is a central or centrocecal scotoma.

The image below shows papillitis.

Type	Retrobulbar neuritis	Papillitis	Neuroretinitis
Pathology	Inflammation of the optic nerve behind the disc	Inflammation of the optic nerve head	Inflammation of the optic nerve head and the macula
Fundus appearance	Optic disc appears normal	Hyperemia and oedema of the optic disc with or without peri-papillary flame-shaped hemorrhages	Macular star (due to exudates that form around macula)
Prevalence	Most common in adults	Most common in children	Least common type
Associated conditions	Strong association with demyelinating disorders like MS, NMO	Weak association with demyelinating disorders	Cat-scratch disease, Lyme disease, Syphilis etc.



The image below shows macular star appearance in neuroretinitis.



Solution to Question 6:

Episodic, transient obscuration of vision on exertion or on exposure to heat, which recovers on resting or moving away from the heat in an isolated optic neuritis patient is called Uhthoff phenomenon. It is commonly associated with multiple sclerosis (MS).

Option A: Pulfrich's phenomenon is an illusion of depth perception. It is seen in MS.

Option C: Lhermitte's sign refers to passive neck flexion causing an electric shock-like sensation radiating to the spine and shoulders. This is a result of the increased sensitivity of the myelin to stretch or traction; it is also seen in MS.

Option D: Riddoch phenomenon refers to ability to perceive kinetic but not static targets. It is seen in cortical blindness.

Solution to Question 7:

Central scotoma is more likely to occur with multiple sclerosis. It is also seen with retrobulbar neuritis and macular pathology.

Option A: Optic chiasma compression causes bitemporal hemianopia

Option C: Enlarged blind spot occurs with papilloedema

Option D: Tunnel vision occurs in glaucoma, retinitis pigmentosa and Retinal pan photocoagulation

Solution to Question 8:

One of the common presentations of MS is optic neuritis. Multiple sclerosis (MS) results from demyelinating plaques in the central nervous system.

Option A: Myasthenia gravis is the autoimmune destruction of the acetylcholine receptors at the neuromuscular junction.

Option B: Guillain–Barré syndrome is the postinfectious autoimmune-induced destruction of peripheral nerves, resulting in an ascending paralysis.

Option D: Amyotrophic lateral sclerosis (ALS) is a disease of the motor nerve fibers of the lateral columns and anterior horns of the spinal cord that results in progressive muscular atrophy.

Solution to Question 9:

The given clinical vignette of a patient presenting with decreased visual acuity and a history of transverse myelitis which is treated with steroids is indicative of Neuromyelitis optica (NMO).

Neuromyelitis optica also known as Devic's disease is an autoimmune demyelinating disease in which antibodies are formed against Aquaporin 4 present in the cell membranes of Astrocytes. It is characterized by the findings of unilateral or bilateral optic neuritis developing within a few weeks of transverse myelitis (demyelination of the spinal cord). The disease may be monophasic or recurrent.

Diagnostic Criteria for Neuromyelitis Optica Spectrum Disorder (NMOSD):

Diagnostic Criteria for NMOSD associated with AQP4-IgG

- At least 1 core clinical characteristic
- A positive test for AQP4-IgG using the best available detection method (cell-based assay)
- Exclusion of alternative diagnoses

Core Clinical Characteristics:

- Optic neuritis is usually bilateral and causes severe visual loss.
- Acute myelitis- may be severe, transverse and typically longitudinally extensive involving three or more contiguous vertebral segments.
- Area postrema syndrome: episodes of otherwise unexplained hiccups or nausea or vomiting
- Acute brainstem syndrome
- Symptomatic narcolepsy associated with NMOSD-typical diencephalic MRI lesions

For acute cases of NMO, high-dose glucocorticoids or plasma exchange is used. Mycophenolate mofetil, rituximab, and glucocorticoids with azathioprine are used for prophylaxis against relapses.

The FDA has approved satralizumab, a humanized IgG2 monoclonal antibody, for the treatment of NMO spectrum disorder with AQP4 antibody positivity.

Eculizumab, a complement inhibitor, and inebilizumab, a B-cell depletion agent, are two more new drugs that have recently been approved for use in NMO attack prevention.

A history of transverse myelitis with optic neuritis(decreased visual acuity) rules out Multiple sclerosis(option D). It is an autoimmune inflammatory and demyelinating disease of the CNS that affects both upper motor neurons (UMN) and lower motor neurons (LMN) with involvement of sensory system, ocular system and bowel and bladder function. Ocular manifestations of MS include Optic neuritis(usually retrobulbar)- decreased visual acuity, internuclear ophthalmoplegia and nystagmus.

Solution to Question 10:

Centrocecal scotoma is most commonly seen in tobacco-induced amblyopia. It is also seen in vitamin B12 deficiency.

Tobacco amblyopia is caused by excessive smoking and chewing of tobacco. The toxic agent involved is cyanide found in tobacco. It is characterized by bilateral and gradually progressive impairment of central vision. The classic visual field defect is a centrocecal scotoma, often involving both the blind spot and the macula.

Option B: Ethanol amblyopia is similar to tobacco amblyopia but often presents with peripheral neuritis and a "central scotoma".

Option C: Methanol amblyopia often results in complete blindness.

Option D: Quinine amblyopia results in near-complete blindness with associated deafness and tinnitus.

Solution to Question 11:

The perimetry result depicts altitudinal hemianopia, which is most commonly seen in anterior ischemic optic neuropathy (AION).

Altitudinal hemianopia can also be seen in suprasellar tumors and occipital lobe lesions, but the frequency is very low compared to that of AION. Hence, AION is the single best answer.

The two types of AION are as follows:

- Arteritic AION- caused by giant-cell arteritis
- Non-arteritic AION (NAAION)- more common

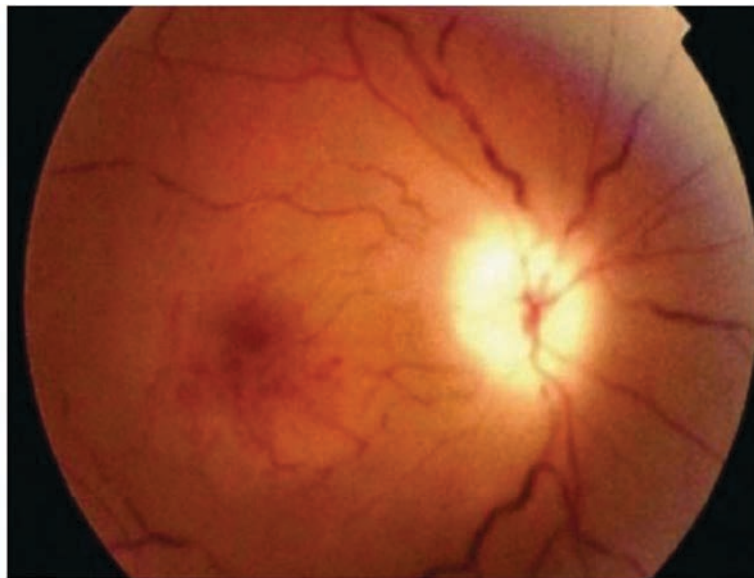
Clinical features:

Note: Corticosteroids are not proven to be effective in NAAION.

The image given below shows pale disc edema of Arteritic AION.

Features	AAION	NAAION
Age	>70 years	<70 years
Sex	F > M	F = M
HeadacheJaw claudication	Present	Absent

Features	AAION	NAAION
Visual loss	Marked	Less marked
Optic disc	Pale disc edema	Hyperemic disc edema
C reactive protein	Raised	Normal
ESR	Highly raised	Raised
FFA	Disc delay and choroidal delay	Only disc delay
Treatment	Corticosteroids	Levodopa–carbidopa combination for 3 weeks and then double dose for 1 month



Solution to Question 12:

The given fundoscopic image clearly depicts cupping, thinning of the neuroretinal rim, and nasal shifting of vessels. These are characteristic of glaucomatous optic atrophy.

Solution to Question 13:

Leprosy is not associated with optic atrophy.

All the other given conditions can cause optic atrophy.

Causes of optic atrophy:

- Primary optic atrophy
- Optic neuritis
- Compression by tumors and aneurysms

- Hereditary optic neuropathies
- Toxic and nutritional optic neuropathies
- Trauma
- Secondary optic atrophy
- Chronic papilledema
- Anterior ischemic optic neuropathy
- Papillitis
- Consecutive optic atrophy
- Extensive retinal photocoagulation
- Retinitis pigmentosa
- Prior central retinal artery occlusion
- Glaucomatous optic atrophy

Solution to Question 14:

The given scenario of pernicious anemia with gradually progressive visual disturbances is suggestive of vitamin B12 deficiency.

B12 deficiency is known to cause centrocecal scotoma and it is an important cause of nutritional optic atrophy.

Solution to Question 15:

The given clinical scenario suggests Wolfram syndrome. It is due to WFS1 gene mutation.

The presentation is usually between the ages of 5 and 21 years; diabetes mellitus is typically the first manifestation, followed by visual problems. It has a poor prognosis.

Clinical features are: (DIDMOAD)

- Diabetes insipidus
- Diabetes mellitus
- Optic atrophy
- Deafness (sensorineural type)

Option A: Kjer's optic neuropathy is an autosomal dominant condition. It presents with vision loss in early childhood and no significant systemic features.

Option B: Behr syndrome is an autosomal recessive condition. Clinical features include optic atrophy, pyramidal tract signs, ataxia, intellectual disability, urinary incontinence, and pes cavus

Option D: Leber's congenital amaurosis is an autosomal recessive condition. It usually presents with significant vision loss in infancy and nystagmus.

Solution to Question 16:

Amaurosis fugax is a sudden and transient monocular blindness, which is reversible.

It is a marker of advanced carotid artery atherosclerosis.

Solution to Question 17:

Champagne cork disc is seen in chronic papilledema.

The optic disc gives the appearance of the dome of a champagne cork. Small drusen-like crystalline deposits (corpora amylacea) may appear on the disc surface.

The most reliable sign to differentiate between papillitis and papilledema is pupillary reaction. Pupillary light reflex is normal in papilledema whereas it is sluggish in papillitis.

Solution to Question 18:

In Leber's hereditary optic neuropathy (LHON), the defect is a mutation of maternal mitochondrial DNA. Therefore the condition has a maternal inheritance.

LHON causes bilateral, subacute painless loss of vision. The characteristic visual field defect is a centrocaecal scotoma. Pupillary light reflex is relatively preserved until the late stage of the disease. Optic neuritis and optic atrophy start in one eye and goes on to involve the other eye subsequently. It typically affects males between the ages of 15 and 35 years. The diagnosis of LHON should be considered in any patient with bilateral optic neuropathy, irrespective of age.

Prognosis is poor, although some visual recovery may occur in a minority of cases even years later.

Note: LHON plus refers to rare variants with additional manifestations such as neuromuscular dysfunction.

Solution to Question 19:

Hyperemic optic disc with telangiectatic capillaries would be suggestive of LHON.

LHON is related to a mitochondrial DNA mutation. This results in impaired mitochondrial adenosine triphosphate production, which affects the highly energy-dependent optic nerve.

In the acute phase, the optic disc is hyperemic and swollen with telangiectatic capillaries.

However, the optic disc does not leak on fluorescein angiography. Later stages may show primary optic atrophy.

Solution to Question 20:

Papilledema is not a mandatory criterion for the diagnosis of idiopathic intracranial hypertension (IIH).

Idiopathic intracranial hypertension (pseudotumor cerebri) is a diagnosis of exclusion. Obstructive, compressive, and infiltrative CNS lesions must be excluded by neuroimaging studies. Papilledema, although present in 90% of the patients, is not necessary for diagnosis. Elevated CSF opening pressure on lumbar puncture is the hallmark of IIH.

Risk factors of IIH include:

- Female sex
- Obesity
- Nalidixic acid
- Tetracycline
- Vitamin A
- Oral contraceptives
- Use of or withdrawal from corticosteroids

Solution to Question 21:

The most reliable sign to differentiate between papillitis and papilledema is pupillary reaction.

Pupillary light reflex is normal in papilledema whereas it is sluggish in papillitis. Visual acuity and colour vision are also relatively more preserved in papilledema compared to papillitis.

The other three disc signs seen on ophthalmoscopy are present in both papillitis and papilledema.

Solution to Question 22:

Down-beat nystagmus is usually due to a lesion at the cervico-medullary junction (e.g. Arnold-Chiari malformation, syringobulbia) or as a manifestation of cerebellar degeneration.

Other causes of down-beat nystagmus:

- Drugs (phenytoin, lithium, barbiturates, carbamazepine)
- Wernicke's encephalopathy
- Demyelinating disorders

Acquired nystagmus can be attributed to disturbances of the three mechanisms that normally ensure steady gaze—visual fixation, the vestibulo-ocular reflex, and the mechanism that makes it possible to hold the eyes at an eccentric eye position (e.g. far-right gaze).

Other types of nystagmus:

Type of Nystagmus	Etiology
Slow-horizontal	Large posterior cerebral lesions (e.g. stroke)
Mixed horizontal (Torsional)	Usually associated with vertigo and seen in vestibular system lesions
Up-beat	Medulla, the ventral tegmentum and the anterior vermis of the cerebellum
Periodic alternating	Cerebellar disease (In PAN, the horizontal jerk nystagmus spontaneously reverses the direction of the quick phase every few seconds)
See-saw	Lesion involving parasellar region and optic chiasm (e.g. pituitary tumours, third ventricle)
Gaze-evoked	Side-effect of drugs, including sedatives, anticonvulsants and alcohol, as well as cerebellar disease

Solution to Question 23:

Fracture of the temporal bone causes vestibular injury resulting in a horizontal nystagmus with slow phase towards the affected side and fast phase towards the opposite side.

Peripheral vestibular nystagmus occurs in lesions of the labyrinth or vestibular nerve. Normally, there is tonic input to the brainstem vestibular nuclei from the periphery. Loss of the tonic input from the left vestibular nerve leads to deviation of the eyes towards the left side, which is the slow phase of the nystagmus. The cortical correction brings the eyes back to the centre in the primary gaze which is the fast-phase directed towards the right side.

Solution to Question 24:

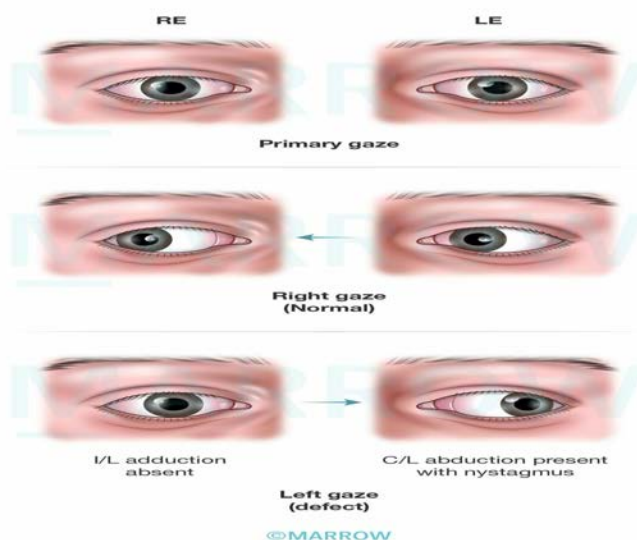
In internuclear ophthalmoplegia (INO) the lesion is at the medial longitudinal fasciculus (MLF).

In INO, there is impaired adduction on the ipsilateral side of the lesion, and nystagmus on abduction on the contralateral side. When the gaze is initiated in the motor cortex, the horizontal gaze center or the paramedian pontine reticular formation (PPRF) generates the eye movements.

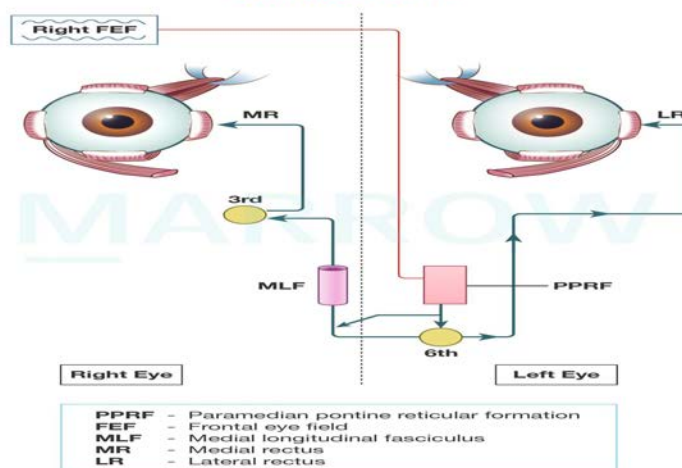
The PPRF has two outputs. The first is to the ipsilateral 6th cranial nerve nucleus which causes abduction of the ipsilateral eye. The second output is to the contralateral MLF which sends an impulse to the contralateral 3rd cranial nerve nucleus causing contralateral adduction. These two outputs from PPRF lead to conjugate movement of the eyes to the same side.

Therefore, a lesion in the right MLF leads to failure of the right eye adduction. The left eye, however, abducts normally. This is called internuclear ophthalmoplegia.

The following image shows the pathway for horizontal eye movements as described above.



Anatomical pathway for horizontal eye movements



Solution to Question 25:

Horizontal gaze palsy can occur due to a lesion in parapontine reticular formation (PPRF).

Horizontal gaze:

- It involves 2 muscles- the ipsilateral lateral rectus and the contralateral medial rectus.
- Center for this pathway is parapontine reticular formation (PPRF).
- From here, they move to the ipsilateral 6th nerve nucleus.

- Next, they pass through the medial longitudinal fasciculus (MLF) to the contralateral medial rectus subnucleus.

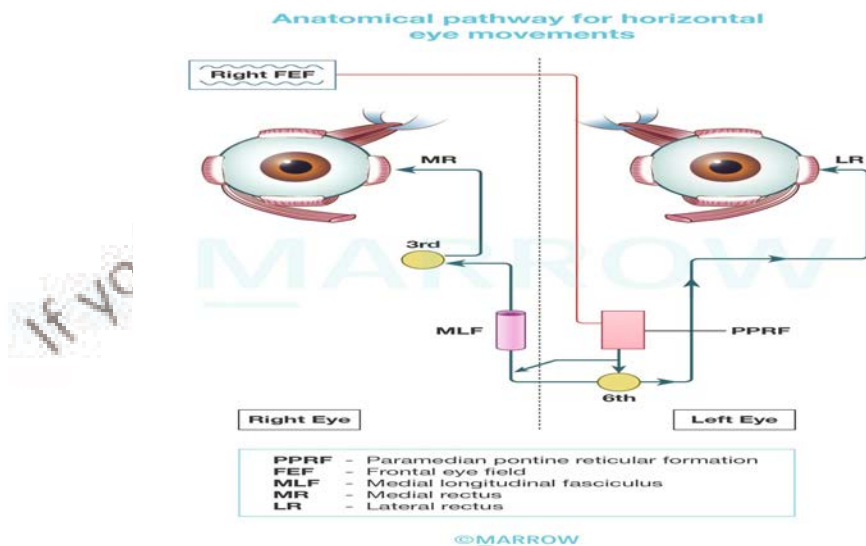
Lesions affecting the abducens nuclei will also cause ipsilateral horizontal gaze palsy, because of damage to motor neurons and internuclear neurons originating from the nuclei. Lesions involving the PPRF will lead to horizontal gaze palsy, as PPRF acts as a relay centre that connects the frontal eye fields in the cerebral cortex to the abducens nuclei. Conjugate horizontal gaze palsy can also be caused by lesion(s) involving both the abducens motor fibres and the MLFs preceding the abducens nucleus. Bilateral horizontal gaze palsy will be caused if lesion(s) involve both abducens nuclei or bilateral PPRF.

Isolated damage to the MLFs will cause the classic INO where eye abduction is preserved, however, the ipsilateral eye will not be able to be adducted on conjugate horizontal gaze.

Conjugate eye movements

- Binocular movements in which the eyes move synchronously and symmetrically in the same direction.
- Main types
- Saccades: They are fast movements controlled by the contralateral frontal lobe.
- Pursuits: They are slow following movements controlled by the ipsilateral occipital lobe.

Image: The Horizontal gaze pathway



Lesions in this pathway

- Horizontal gaze palsy: Due to lesion in the PPRF of the same side
- Internuclear ophthalmoplegia (INO): Caused by lesion in the medial longitudinal fasciculus (MLF)
- One and a half syndrome: Caused by lesions in both MLF and PPRF of the same side

Vertical gaze palsy

Center is located in the rostral interstitial nucleus of the dorsal midbrain (riMLF).

The important types of vertical gaze palsy are as follows:

- Parinaud's syndrome: Upgaze palsy
- Steel–Richardson's syndrome: Downgaze palsy

Solution to Question 26:

The most likely location of the lesion according to the given clinical scenario and the image is the right medial longitudinal fasciculus.

The image depicts an attempted left gaze and concludes that there is an adduction deficit in the right eye. This causes jerk nystagmus in the abducting left eye.

This is most likely due to a lesion of the medial longitudinal fasciculus (MLF) ipsilateral to the side of the adduction deficit. Hence, this is a case of a right MLF lesion or Right Internuclear Ophthalmoplegia (INO).

Solution to Question 27:

Ophthalmoplegic migraine refers to headache associated with third, fourth, and sixth nerve palsy.

Ophthalmoplegic migraine almost always involves the third nerve. Rare cases of sixth nerve involvement, and even rarer cases that involve the fourth nerve, have also been reported.

Clinical criteria for the diagnosis of ophthalmoplegic migraine include:

- A prior history of typical common or classic migraine headaches
- Ophthalmoplegia that occurs during (or, rarely, just before) an established migraine attack
- Exclusion of other reasonable causes by clinical and radiological studies

Tumors of Eye

Question 1:

Which among the following is the most common primary intraocular malignancy in adults?

- a) Retinoblastoma
- b) Malignant melanoma
- c) Hemangiosarcoma
- d) Neurofibroma

Question 2:

Which is the most common primary malignant tumor of the lid?

- a) Squamous cell carcinoma
- b) Basal cell carcinoma
- c) Malignant melanoma
- d) Merkel cell carcinoma

Question 3:

Which of the following is the most common lid malignancy in HIV patients?



- a) Basal cell carcinoma
- b) Kaposi sarcoma
- c) Squamous cell carcinoma
- d) Sebaceous gland carcinoma

Question 4:

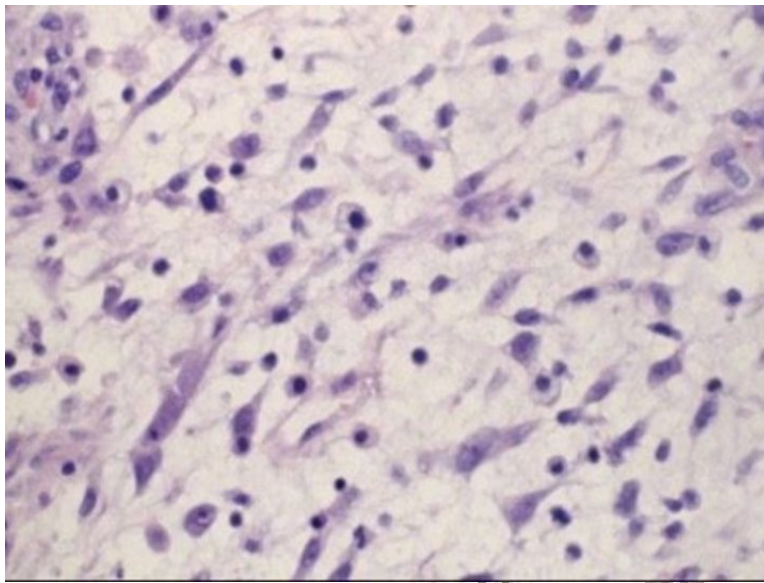
Which of the following cannot be used in the management of the clinical condition shown below?



- a) Oral propranolol
- b) Oral prednisolone
- c) Intralesional triamcinolone
- d) Interferon beta

Question 5:

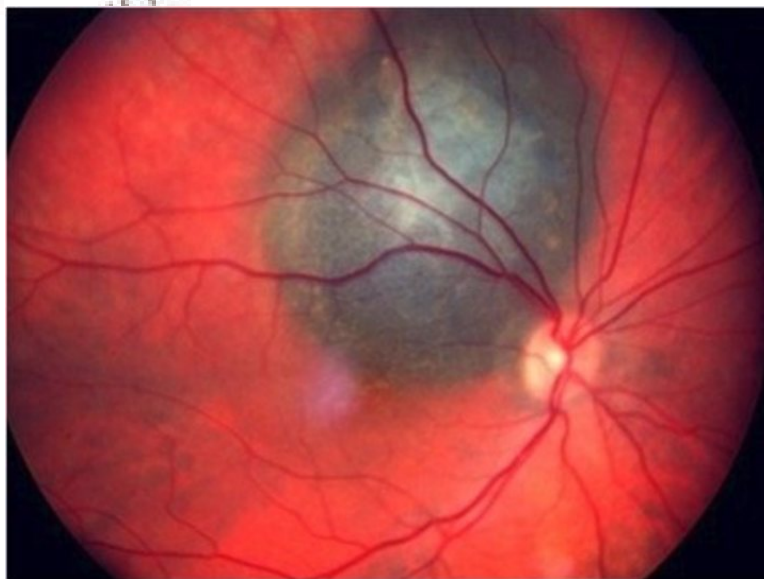
An 8-year old child presented with right-sided proptosis. Biopsy of the orbital mass was done and histopathological examination revealed this picture. What is the most likely diagnosis?



- a) Retinoblastoma
- b) Metastases from neuroblastoma
- c) Rhabdomyosarcoma
- d) Hemangiosarcoma

Question 6:

A patient presented with painless decrease in vision and metamorphopsia. Fundus examination revealed the lesion shown in the image. What is the expected finding on angiography?



- a) Smoke-stack appearance

- b) Inkblot appearance
- c) Collarstud appearance
- d) Double circulation

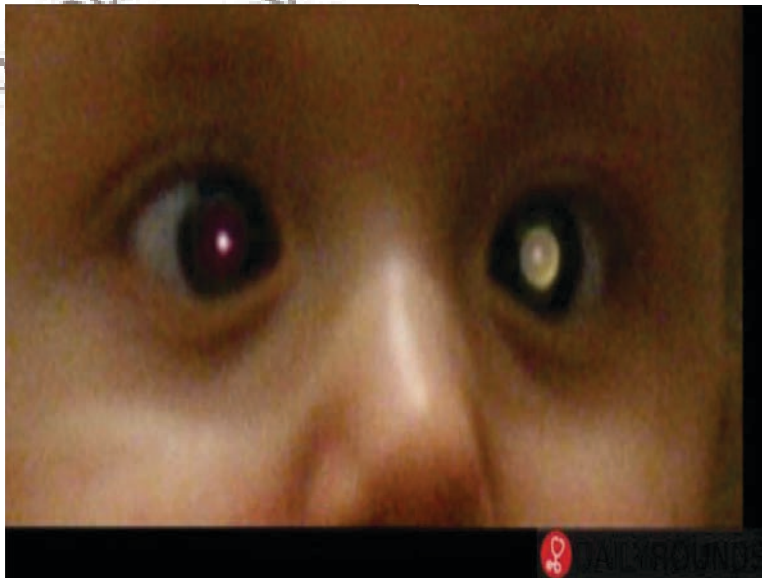
Question 7:

A 68-year-old man presents with jaundice. He was recently diagnosed with malignant melanoma of the uvea but did not receive proper treatment due to the prevalent pandemic situation. Investigations reveal liver secondaries. What would be the ideal line of management?

- a) Enucleation
- b) Exenteration + Chemotherapy + Immunotherapy
- c) Enucleation + Chemotherapy + Immunotherapy
- d) Exenteration + Radiotherapy

Question 8:

A father has noticed that his child's left pupil appears white as shown below. Worried about this, the father consults an ophthalmologist. All of the following should be included in the differential diagnosis except:



- a) Toxocara infestation
- b) Norrie disease
- c) Persistent hyperplastic primary vitreous

d) Congenital glaucoma

Question 9:

Retinoblastoma arises from which of the following structures?

- a) Pigmentary epithelium**
- b) Neurosensory retina**
- c) Optic nerve**
- d) Meninges**

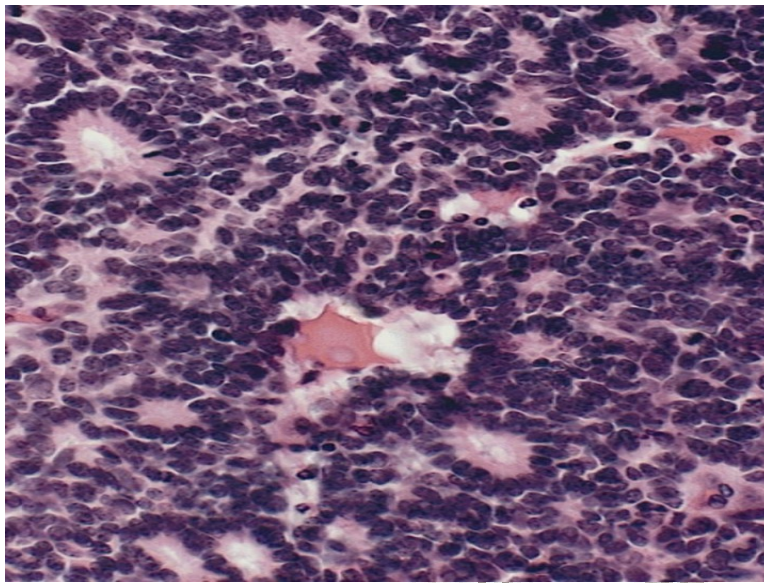
Question 10:

Which of the following is the most common clinical presentation of retinoblastoma?

- a) Leucocoria plus heterochromia iridis**
- b) Leucocoria plus pseudohypopyon**
- c) Leucocoria plus hyphema**
- d) Leucocoria plus strabismus**

Question 11:

A 3-year-old boy was brought by his parents with concern about being increasingly clumsy for the past 6 months. On examination, there is leukocoria with an absence of the red reflex in the left eye. As further workup reveals a tumor, the eye is enucleated and the microscopic appearance is as shown in the image. Molecular analysis of the tumor indicates loss of cell cycle control in the tumor cells. Where is the responsible gene located?



- a) 13p14
- b) 13q14
- c) 14p13
- d) 14q13

Question 12:

Knudson's two-hit hypothesis describes the occurrence of which of the following?

- a) Glaucoma
- b) Optic nerve glioma
- c) Retinoblastoma
- d) Meningioma

Question 13:

Why would an ophthalmologist use the Reese-Ellsworth classification for retinoblastoma?

- a) For the management of the tumor
- b) For the staging of the tumor
- c) For the assessment of the survival of the eye after local therapy
- d) For the assessment of the survival of the patient after surgery

Question 14:

A 2-year-old boy who presented with strabismus and leukocoria is suspected to have retinoblastoma. A B-scan detects multiple endophytic masses with calcifications, the largest measuring 6DD, and the smallest measuring 4 DD. CT scan reveals intraocular foci of calcifications and no other anomaly. The ophthalmologist decides to use chemo-reduction and local therapy to treat this child. According to Reese-Ellsworth classification, this belongs to which group?

- a) Group 1
- b) Group 2
- c) Group 3
- d) Group 4

Question 15:

What is the treatment of choice when retinoblastoma involves the optic nerve?

- a) Enucleation
- b) Enucleation + radiotherapy
- c) Enucleation + adjuvant chemotherapy
- d) Enucleation + adjuvant chemotherapy + radiotherapy

Question 16:

Which of the following regimen is commonly used in the treatment of retinoblastoma?

- a) Vinblastine, etoposide and bleomycin
- b) Vinblastine, vincristine and etoposide
- c) Vinblastine, vincristine and cisplatin
- d) Vincristine, carboplatin and etoposide

Question 17:

What is the treatment of choice for anterior lesions of retinoblastoma?

- a) Cryotherapy
- b) Photocoagulation

- c) Both a and b
- d) None of the above

Question 18:

The most common secondaries in choroid are from _____.

- a) Breast
- b) Brain
- c) Ovary/ Testis
- d) Lungs

Answer Key

Question No.	Correct Option
1	b
2	b
3	b
4	d
5	c
6	d
7	c
8	d
9	b
10	d
11	b
12	c
13	c
14	b
15	d
16	d
17	a
18	a

Detailed Explanations

Solution to Question 1:

Malignant melanoma is the most common primary intraocular malignancy in adults.

Retinoblastoma is the most common primary intraocular malignancy in children.

Solution to Question 2:

The most common primary malignant tumor of the lid is basal cell carcinoma.

The second most common tumor of the lid is squamous cell carcinoma.

Solution to Question 3:

Kaposi sarcoma is the commonest malignancy of the lid and conjunctiva in HIV patients.

It appears as a violaceous non-tender nodule on the eyelid or conjunctiva. It is a rapidly growing acquired neoplasm composed of malignant stromal cells densely vascularized by blood vessels of capillary size.

The typical lesion appears as an expanding, bright red subepithelial mass that projects through and replaces the overlying conjunctival epithelium. Spontaneous bleeding leading to bloody tears or a subconjunctival hematoma is relatively common.

Conjunctival Kaposi's sarcoma can be treated by excision, cryotherapy, radiation therapy (external beam or contact therapy), chemotherapy, or topical or intralesional interferon- α .

Solution to Question 4:

The image shows a capillary hemangioma involving the left upper lid and periocular area. Interferon beta is not used in the management of capillary hemangioma. However, note that interferon alfa-2a is used in resistant cases.

A capillary haemangioma is the most common tumor of the orbit and periorbital area in childhood.

It may present as small isolated lesions or large masses that can cause visual impairment (due to mechanical ptosis and the resulting sensory deprivation amblyopia), systemic effects (severe thrombocytopenia), or exist as part of PHACE syndrome (Posterior fossa anomalies, Hemangioma, Arterial lesions, Cardiac abnormalities, Eye or Endocrine abnormalities)

30% of lesions resolve by the age of 3 years and about 75% by the age of 7.

Indications for treatment are:

- Visual (amblyopia, astigmatism, occlusion, strabismus)

- Cosmesis
- Optic nerve compression
- Exposure keratopathy

Treatment options include:

- Beta blockers- oral propranolol, topical timolol.
- Steroids- intralesional triamcinolone or betamethasone, topical clobetasol, systemic steroids.
- Interferon alfa-2a and vincristine- for steroid-resistant sight-threatening lesions
- Local resection with cutting cautery or carbon dioxide laser.

Solution to Question 5:

The most likely diagnosis is rhabdomyosarcoma (RMS). Rhabdomyosarcoma is the most common soft tissue sarcoma of childhood.

40% of rhabdomyosarcomas develop in the head and neck, and they are the most common primary orbital malignancy in children but are still a rare condition. The older age of the child presenting with proptosis suggests rhabdomyosarcoma (average age 7 to 8 years) than retinoblastoma (average age at presentation is 18 months).

The histopathology shows elongated to round spindle cells with features of skeletal muscle at different stages of maturation with highly eosinophilic cytoplasm and hyperchromatic nuclei – the embryonal type of rhabdomyosarcoma

Immunohistochemistry is useful in identifying skeletal muscle proteins, such as alpha-actin, myosin, desmin, and myoglobin

Rhabdomyosarcoma shows a strong tendency for local invasion, local recurrence, and hematogenous and lymphatic metastases.

Decision-making for management is based on histopathologic confirmation as well as the staging of orbital RMS, which includes imaging (MRI of primary tumor, chest-CT, Tc bone scan), pathology, and work-up for metastases(bone marrow biopsy).

Treatment options include chemotherapy (vincristine, cyclophosphamide, and actinomycin D) and radiotherapy.

Solution to Question 6:

The clinical vignette is of a patient presenting with painless diminution of vision and metamorphopsia with the right fundus showing choroidal melanoma superotemporal to the optic disc and involving the macula.

In this patient a “Double circulation” pattern consisting of an internal circulation within the lesion and the normal vascularity of the overlying retina will be seen on angiography.

Although collar stud (Option C) appearance is pathognomonic for choroidal melanoma, it can be seen on B-scan and not on angiography.

Choroidal melanoma is a primary intraocular malignancy in adults that peaks around 60 years of age.

Symptoms include a range of visual disturbances such as painless decrease in vision and metamorphosia when macula is involved.

Other features of Choroidal melanoma includes:

- A solitary elevated subretinal grey-brown dome shaped mass
- Clumps of orange pigment are common over the mass
- Collar stud appearance on B-scan (option C): It is pathognomonic of choroidal melanoma. It is seen if the tumour breaks through the Bruch's membrane into the subretinal space.
- Associated haemorrhage

Smoke stack pattern (option A) and ink blot (option B) pattern is seen on FFA in central serous retinopathy (CSR).

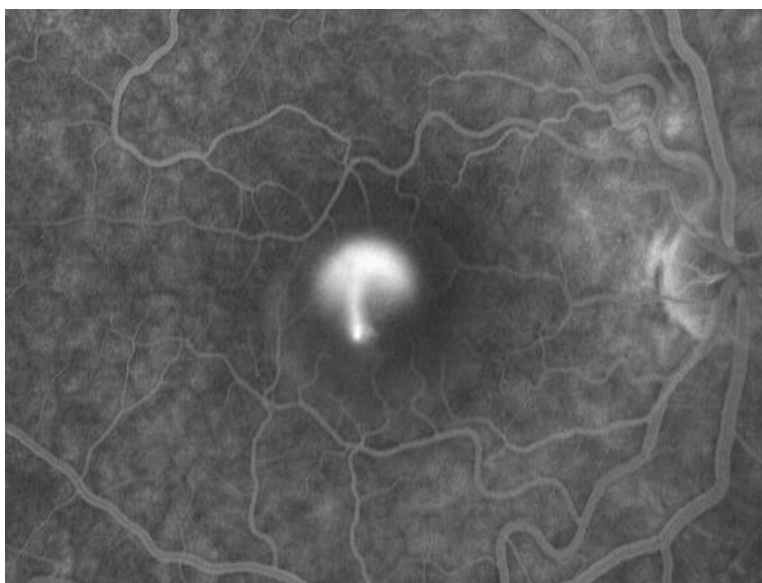
CSR is the spontaneous serous detachment of the sensory neuroretina in the foveal region with or without retinal pigment epithelium (RPE) detachment.

This occurs due to choroidal vascular hyperpermeability.

On FFA it is initially seen as an ink-blot pattern or smoke stack pattern of hyperfluorescence which then spreads to take the configuration of mushroom or umbrella pattern.

The below given images depict the smoke stack pattern and mushroom pattern respectively





Solution to Question 7:

The spread of malignant melanoma of the uvea outside the orbit is treated with enucleation + chemotherapy + immunotherapy.

Orbital spread of the malignant melanoma necessitates exenteration but metastasis elsewhere can be treated by enucleation, chemotherapy, and immunotherapy.

Solution to Question 8:

The child has leukocoria and congenital glaucoma is not to be included in the differential diagnosis.

Various conditions, other than retinoblastoma, which present as leukocoria are collectively called pseudoglioma.

Differential diagnosis of leukocoria:

- Retinoblastoma
- Congenital cataract
- Coloboma of the choroid
- Persistent hyperplastic primary vitreous
- Toxocara infestation
- Norrie disease
- Coat's exudative retinopathy

Intraocular pressure (IOP) is usually raised in retinoblastoma and lowered in pseudoglioma.

Solution to Question 9:

Retinoblastoma (also called glioma retinae) arises from the neurosensory retina. It is a proliferation of neural cells that have failed to evolve normally.

Tumours are composed of small basophilic cells (retinoblasts), with large hyperchromatic nuclei and scanty cytoplasm.

It is of 4 types:

- Endophytic (into the vitreous) with seeding of tumor cells throughout the eye
- Exophytic (into the subretinal space) leading to retinal detachment
- Mixed type
- Retinocytoma: retina may be diffusely infiltrated

Solution to Question 10:

The most common clinical presentation of retinoblastoma is leukocoria plus strabismus.

Clinical features of retinoblastoma:

- Present within the first two years of life
- Leukocoria
- Strabismus
- Painful red eye with secondary glaucoma
- Orbital inflammation mimicking orbital or pre-septal cellulitis.
- Orbital invasion or visible extraocular growth may occur in neglected cases.
- Metastatic disease involving regional lymph nodes and the brain before the detection of ocular involvement is rare.

Signs:

- An intraretinal tumor is a homogeneous, dome-shaped white lesion that becomes irregular, often with white flecks of calcification.
- An endophytic tumor projects into the vitreous as a white mass that may "seed" into the gel.
- An exophytic tumor forms multilobular subretinal white masses and causes an overlying retinal detachment.

Investigations:

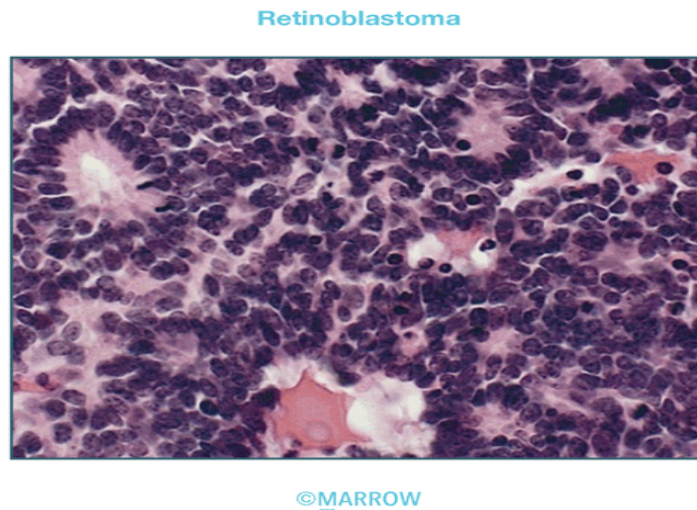
- Red reflex testing with a direct ophthalmoscope is a simple screening test for leukocoria.
- Dilated fundus examination using indirect ophthalmoscope
- MRI of the orbit + brain

Solution to Question 11:

The histology showing Flexner-Wintersteiner rosettes and clinical history of white reflex point towards a diagnosis of retinoblastoma. Retinoblastoma is due to mutation in Rb gene located on chromosome 13q14.

Retinoblastoma was first cancer known to be directly associated with a genetic abnormality.

The HPE image below shows Flexner-Wintersteiner rosettes, the clustering of small blue cuboidal or short columnar cells around a central lumen.



Solution to Question 12:

Knudson's two-hit hypothesis describes the occurrence of retinoblastoma.

Knudson's hypothesis can be stated as follows:

- Two mutations (hits), involving both alleles of RB are required to produce retinoblastoma.
- In familial retinoblastoma, one of the pair of alleles of RB1 is mutated in all the cells of the body. When a further mutagenic event ("second hit" according to the "two-hit" hypothesis proposed by Knudson) affects the second allele, the cell may then undergo malignant transformation. A large majority of these children develop bilateral and multifocal tumors.
- In sporadic cases, both RB alleles must undergo somatic mutation in the same retinoblast (two hits).

Familial retinoblastoma patients also have a predisposition to non-ocular cancers because of the presence of the mutation in all cells, such as pinealoblastoma (usually before the age of 5 years), osteosarcomas, soft tissue sarcomas, and melanomas.

Note:

Trilateral retinoblastoma = Pinealoblastoma + Bilateral retinoblastoma

Solution to Question 13:

Reese–Ellsworth classification is used in predicting which eye would survive the local therapy (cryotherapy, photocoagulation, plaque radiation therapy) and retain useful vision.

Cryotherapy is used for retinoblastoma lesions anterior to the equator and photocoagulation for lesions posterior to the equator.

Group	Prognosis for maintenance of sight	Features
1	Very favorable	Solitary tumor, <4DD, at or behind the equator Multiple tumors, all <4DD, all at or behind the equator
2	Favorable	Solitary tumor, 4–10 DD, at or behind the equator Multiple tumors, all 4–10DD all behind the equator
3	Possible	Any lesion anterior to the equator Solitary tumor, >10DD behind the equator
4	Unfavorable	Multiple tumors, some >10 DD Any lesion extending anteriorly to ora serrata
5	Highly unfavorable	Massive tumors involving more than one half of the retina Vitreous seeding

Solution to Question 14:

Multiple tumors of 4-10 DD in size behind the equator belong to group 2.

Group	Prognosis for maintenance of sight	Features
1	Very favorable	Solitary tumor, <4DD, at or behind the equator Multiple tumors, all <4DD, all at or behind the equator
2	Favorable	Solitary tumor, 4–10 DD, at or behind the equator Multiple tumors, all 4–10DD all behind the equator

Group	Prognosis for maintenance of sight	Features
3	Possible	Any lesion anterior to equator Solitary tumor, >10DD behind equator
4	Unfavorable	Multiple tumors, some >10 DD Any lesion extending anteriorly to ora serrata
5	Highly unfavorable	Massive tumors involving more than one half of the retina Vitreous seeding

Solution to Question 15:

Optic nerve involvement is treated by enucleation + adjuvant chemotherapy + radiotherapy

Indications for enucleation [Group E tumor - according to the International Classification of Retinoblastoma (ICRB)]:

- Tumor involves more than half of the retina
- Optic nerve involved
- Glaucoma is present and the anterior chamber is involved

Chemotherapy commonly used is a three-drug regimen containing vincristine, carboplatin, and etoposide.

Solution to Question 16:

Chemotherapy commonly used is a three-drug regimen containing vincristine, carboplatin, and etoposide.

Solution to Question 17:

Cryotherapy is used for anterior lesions of retinoblastoma.

Photocoagulation for posterior lesions of retinoblastoma.

Solution to Question 18:

The most common secondaries in choroid originate from the breast (overall and in females), and lungs (in males).

Examination shows the elevation of the retina, usually at the posterior pole. It is usually bilateral. Metastases are usually treated with systemic chemotherapy and immunotherapy. Radiotherapy may also be given.

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Practical Ophthalmology

Question 1:

For a patient presenting with gradual, painless diminution of vision, the differential diagnosis would not include _____.

- a) Optic atrophy
- b) Corneal dystrophy
- c) Primary open angle glaucoma
- d) Methyl alcohol amblyopia

Question 2:

What is the angle subtended by the largest alphabet on the Snellen's chart at 60m distance?

- a) 5
- b) 30
- c) 50
- d) 10

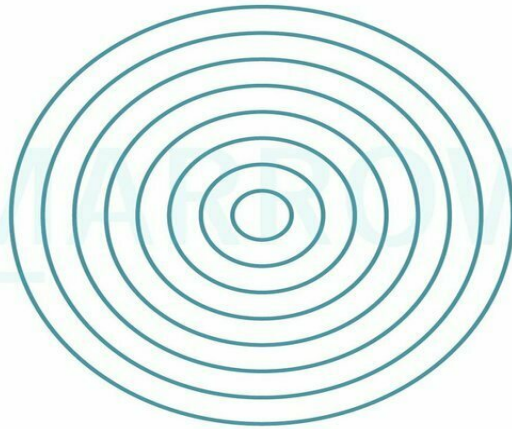
Question 3:

A 60-year-old man comes to the OPD for an eye exam. He is having difficulty reading newspapers but has no problem watching television. At what distance would you ask him to read a Jaeger's chart from?

- a) 20 cm
- b) 30 cm
- c) 35 cm
- d) 40 cm

Question 4:

A patient is being evaluated for a recent eye injury. The result of his Placido disc test is shown below. What is the correct interpretation?



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- a) Keratoconus
- b) Keratoglobus
- c) Normal
- d) Corneal abrasion

Question 5:

A patient presents with the following finding in his eye. Fluorescein staining is done and the eye is observed under cobalt blue light. What would be seen?

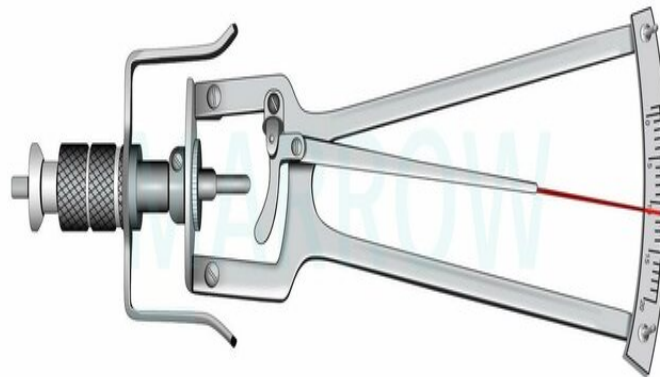


- a) Ulcer appears bright green.

- b) Ulcer appears opaque green.
- c) Normal area appears bright green.
- d) Normal area appears opaque green.

Question 6:

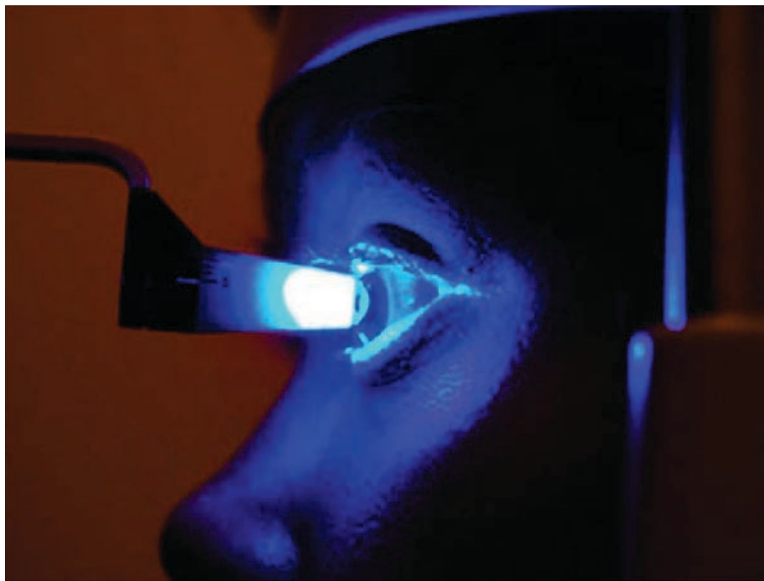
A 61-year-old man, known case of diabetes mellitus, presented with headache and eye pain. He also reports difficulty with driving at night. Visual field-testing indicates peripheral vision loss. The ophthalmologist uses the following instrument during his examination of this patient. What is he measuring by using the following instrument?



- a) Angle of anterior chamber
- b) Intraocular pressure
- c) Corneal density
- d) Corneal thickness

Question 7:

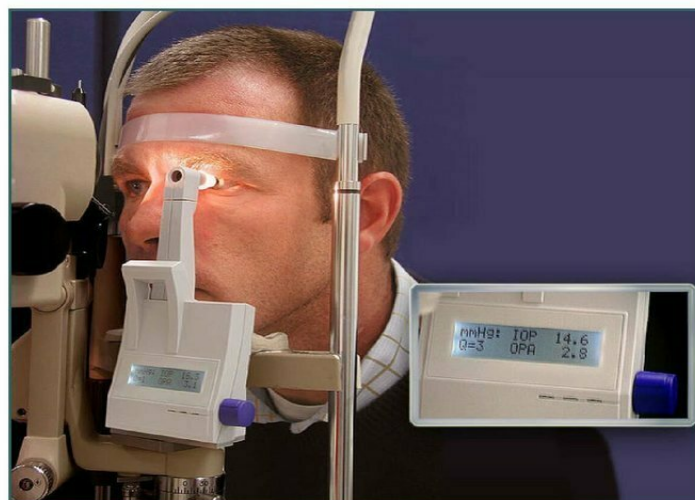
Which of the following statements is false regarding the procedure shown in the image?



- a) The concept of the procedure was introduced by Goldmann.
- b) It is used to treat after cataract
- c) It is based on the Imbert-Fick law
- d) It is used to investigate glaucoma

Question 8:

A patient recently underwent LASIK surgery for his myopia. He is now complaining of headaches and eye pain. Identify the test being performed on him.



- a) Pachymetry

- b) Tonometry
- c) Auto Refractometry
- d) A Scan

Question 9:

You can use any of the following to estimate the visual field of a patient except:

- a) Confrontation method
- b) Campimetry
- c) Octopus Fieldmaster
- d) Optical coherence tomography

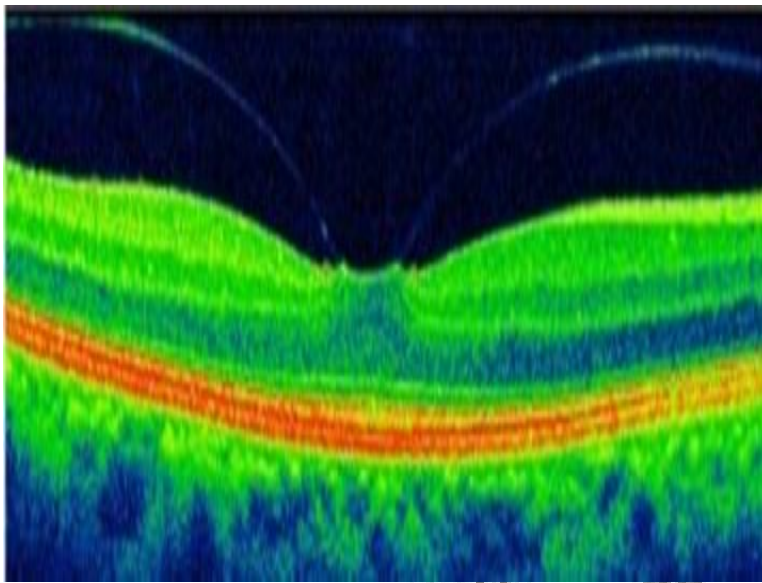
Question 10:

Which among the following phases indicate complete filling of the retinal arterioles and capillaries on fundus fluorescein angiography?

- a) Pre-arterial phase
- b) Arterial phase
- c) Arterio-venous phase
- d) Venous phase

Question 11:

A 55-year-old patient with diabetes presents with complaints of dark floating spots and flashes of light in her left eye. Optical coherence tomography of the left eye is given in the image below. Identify the pathology.



- a) Vitreomacular adhesion
- b) Partial thickness macular hole
- c) Full thickness macular hole
- d) Vitreomacular traction

Question 12:

A 50-year-old female, who is a known case of type 2 diabetes, is undergoing serial electroretinograms to monitor disease progression. The following finding was obtained from her recent test. What does the wave marked by the arrow indicate?



- a) Wave due to photoreceptors

- b) Wave from rods and cones
- c) Activity of bipolar cells
- d) Metabolic activity of pigment epithelium

Question 13:

You would classify the electrooculography (EOG) of a patient as subnormal if the Arden ratio is less than _____.

- a) < 185
- b) < 150
- c) < 125
- d) < 100

Question 14:

You can use 'A' scan for all of the following except:

- a) IOL power calculation
- b) Anterior chamber depth
- c) Thickness of intra ocular mass
- d) Assessment of vitreous hemorrhage

Question 15:

Which of the following is false regarding the instrument in the image?



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- a) It is used to test macular function
- b) It is used to measure vertical, horizontal tropias, and phorias
- c) It cannot be used to measure cyclodeviation
- d) It is based on diplopic projection

Question 16:

A woman came to the OPD due to blurring of vision while working on her laptop. Retinoscopy was performed at 1 m without cycloplegia. The red reflex moves in the same direction as the retinoscope. Which of the following refractive errors is not possible in this patient?

- a) 1D myopia
- b) Hypermetropia
- c) 0.5D myopia
- d) Emmetropia

Question 17:

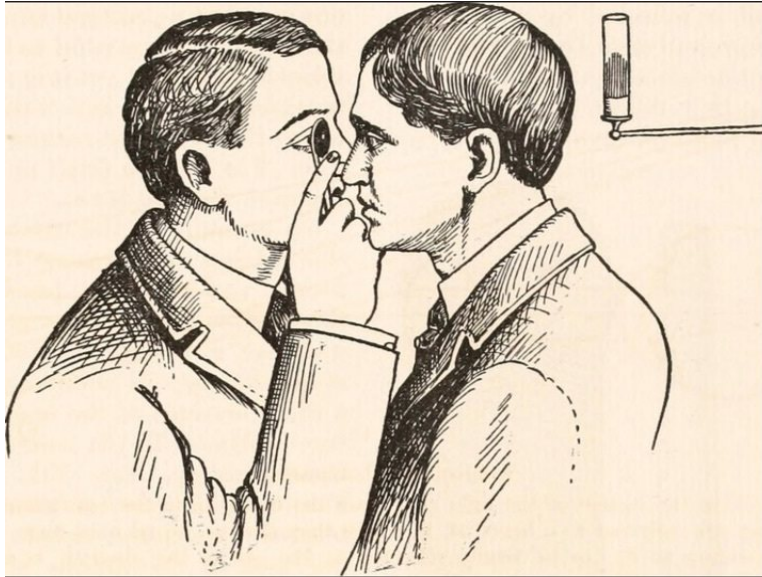
You can use keratometry for all of the following except:

- a) Estimating corneal astigmatism
- b) Measuring the curvature of cornea
- c) Calculating the power of the intraocular lens (IOL)

- d) Calculating the power of lens in spectacles

Question 18:

An old man comes to the OPD for an annual eye exam. The ophthalmologist examines his fundus as shown below. Identify the procedure.



- a) Direct ophthalmoscopy
- b) Distant direct ophthalmoscopy
- c) Indirect ophthalmoscopy
- d) Retinoscopy

Question 19:

Which of the following statements is false about direct ophthalmoscopy?

- a) Magnification is 15 times.
- b) There is no stereopsis.
- c) The type of image formed is real and inverted.
- d) A condensing lens is not needed.

Question 20:

Which among the following statements is false regarding the procedure shown in the image?



- a) The principle is to make eye emmetropic.
- b) Image is real, inverted, and magnified.
- c) +20 D lens used.
- d) It may be done even in patient's with a hazy media.

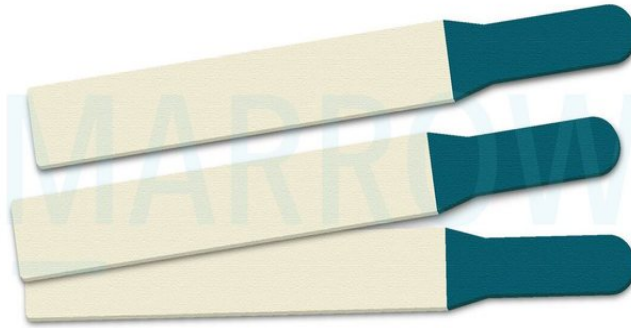
Question 21:

Which of the following can you assess using Holmgren's wool-matching test?

- a) Visual field
- b) Visual acuity
- c) Color vision
- d) Refraction

Question 22:

What is the name of the dye in the strip?



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- a) Fluorescein
- b) Rose Bengal
- c) Indocyanine Green
- d) Lissamine Green

Question 23:

The device shown below cannot be used in the management of which of the following?



- a) Retinoblastoma
- b) Retinal break

- c) Impending corneal perforation
- d) Pars planitis

Question 24:

The following image indicates:



- a) Trochlear nerve palsy
- b) Lateral rectus palsy
- c) Internuclear ophthalmoplegia
- d) Occulomotor palsy

Question 25:

Photostress test is used to differentiate between:

- a) Macula and optic nerve disease
- b) Cataract and glaucoma
- c) Lens and cornea
- d) Retina and vitreal pathologies

Answer Key

Question No.	Correct Option
1	d
2	a
3	c
4	c
5	b
6	b
7	b
8	b
9	d
10	c
11	d
12	c
13	b
14	d
15	c
16	a
17	d
18	a
19	c
20	a
21	c
22	d
23	c
24	d
25	a

Detailed Explanations

Solution to Question 1:

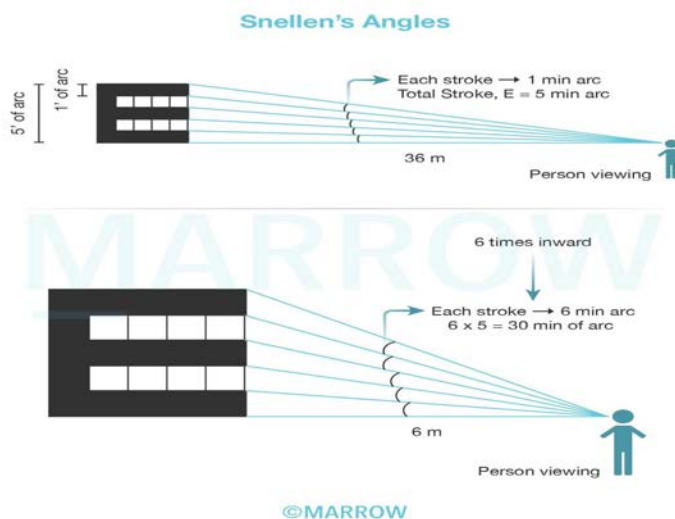
Methyl alcohol amblyopia causes sudden painless diminution of vision.
It is a toxic neuropathy of the optic nerve.

Solution to Question 2:

The angle subtended by the largest alphabet on the Snellens chart at 60m distance is 5 minutes.

Visual acuity is a measure of the spatial resolution of the eye or an estimation of its ability to discriminate between two points. Snellens chart is used for Visual acuity assesment.

In snellens chart the alphabets decrease in size as we move from above-downwards and these lines, from above-downwards should be read at 60, 36, 24, 18, 12, 9, 6, 5 and 4 m respectively. At these distances, the respective letters subtend a visual angle of 5 minutes. Hence, the largest alphabet that should be read at 60 m subtends a visual angle of 5 minutes in the eye.

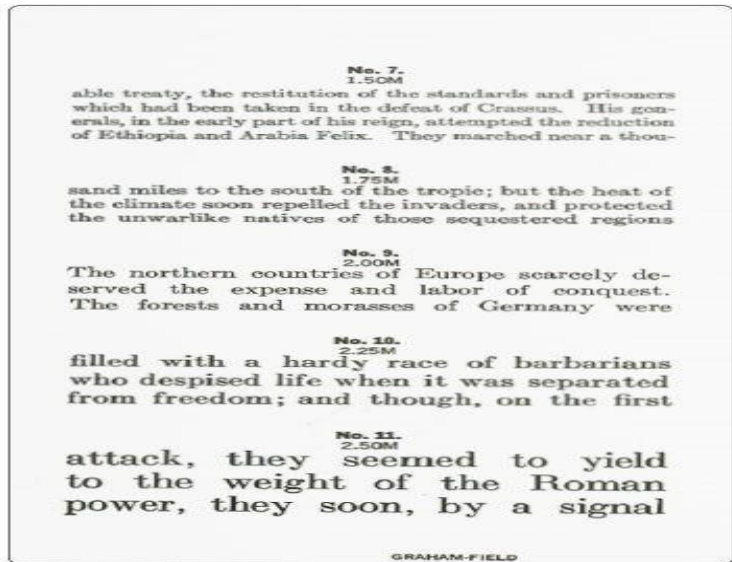


Solution to Question 3:

Jaeger's chart is held at a distance of 35 cm away from the patient.

It is used to measure near vision. The smallest letter is designated as N5 or J1 and the largest as N48. Snellen's near vision chart can also be used.

The image below shows a Jaeger's chart.

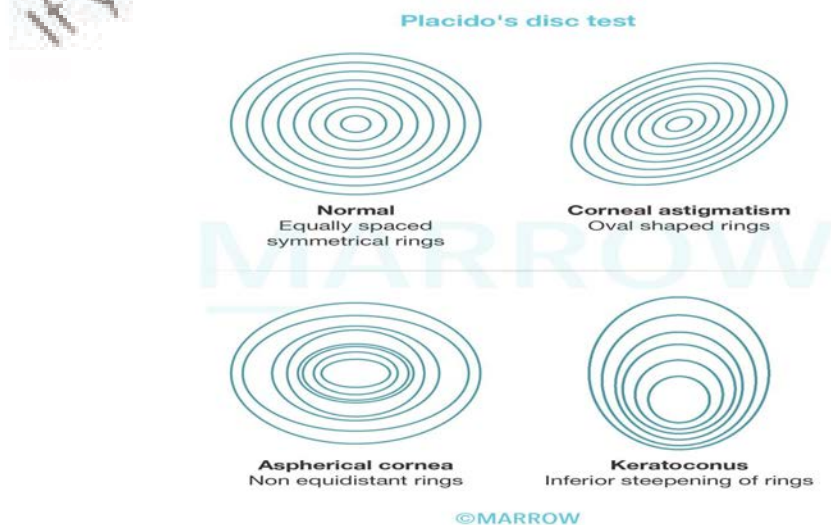


Solution to Question 4:

The given Placido's disc test result indicates a normal corneal surface.

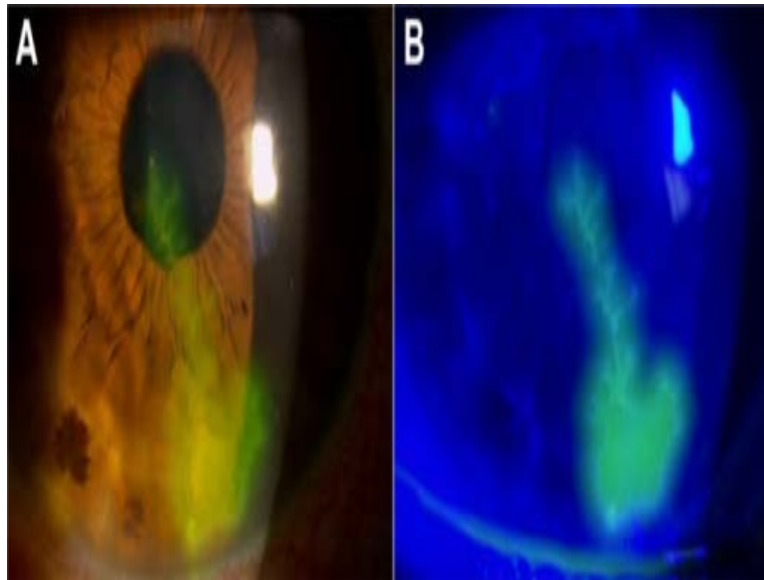
This test is used to detect an abnormal corneal surface. Irregularities of the corneal surface cause distortion of the circles.

Distortion of the circle is seen in keratoconus, keratoglobus, corneal ulcer, and corneal abrasion.



Solution to Question 5:

The image shows a corneal ulcer. The ulcer takes the stain and appears brilliant green, and when examined under cobalt blue light, it appears opaque green.



Solution to Question 6:

The instrument shown is the Schiotz tonometer, which is used to measure intraocular pressure (IOP).

This patient's complaints of headache, eye pain, difficulty in adapting to darkness, and examination findings of peripheral vision loss with increased intraocular pressure, is suggestive of glaucoma.

Schiotz tonometer is a type of indentation tonometer. The principle behind its use is that the plunger will indent a soft eye more than a hard eye.

Solution to Question 7:

The instrument shown is an applanation tonometer, which is used to measure intraocular pressure (IOP).

The concept behind the procedure was introduced by Goldmann. It is based on Imbert-Fick law, which states that the pressure inside a sphere (P) is equal to the force (W) required to flatten its surface divided by the area of flattening (A).

$$P = W/A$$

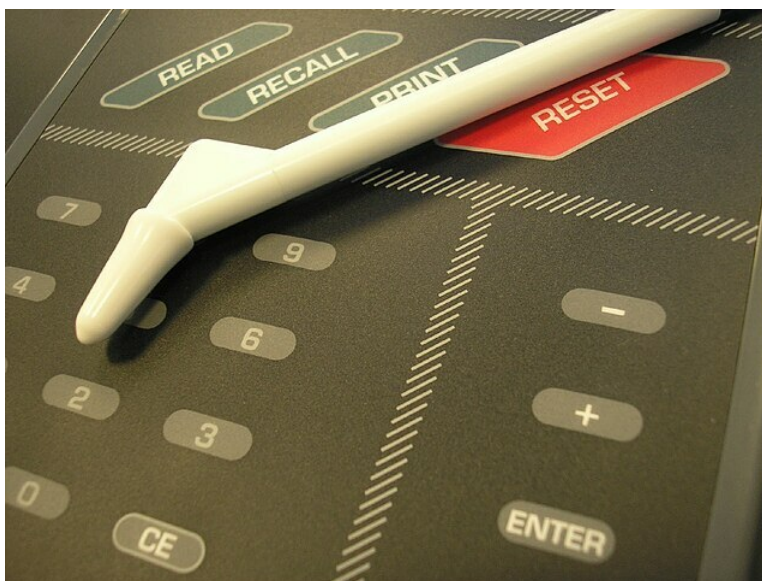
The different types are Goldmann tonometer, Perkin's tonometer, pneumatic tonometer, pulse air tonometer, tono-pen.

Solution to Question 8:

The above image is of the dynamic contour tonometer such as the Pascal tonometer. The dynamic contour tonometer(Pascal) offers a non-invasive and direct method of intraocular pressure (IOP) measurement with an additional measurement of the ocular pulse amplitude. This type of tonometer is unaffected by the corneal thickness and it is the tonometer of choice in post-Lasik patients.

Option A: Pachymetry is used to measure the corneal thickness.

The below image is of pachymetry.



Option C: Auto Refractometry is a computerized refractometer that is used to find out the error of refraction. This gives rapid information about the kind of refractive error in terms of sphere, cylinder with axis and inter pupillary distance.

Option D: A-Scan ultrasonography is a useful diagnostic tool for intraocular evaluation especially in the setting of opaque media. The report of the A scan is displayed as a unidimensional display of spikes along a baseline where the distance between the spikes gives the indirect measurement of the tissues. A-scan is used to measure

- The axial length of the eyeball for Intraocular lens (IOL) power calculation
- Anterior chamber depth
- Thickness of intraocular mass.

A-scan ultrasonography is shown in the image below.



Solution to Question 9:

Optical coherence tomography is not used to estimate the visual field. It is a non-invasive imaging technique to visualize cross-sections of the retina.

Perimetry is used to estimate the visual field.

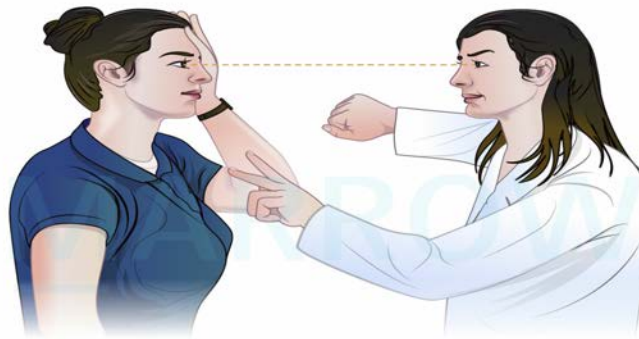
Types:

- Manual perimetry- Confrontation method, Lister's perimeter, Campimetry (Scotometry), Goldmann's perimeter
- Automated perimetry- Octopus Fieldmaster, Humphrey Field Analyzer

The images given below shows Lister's perimeter

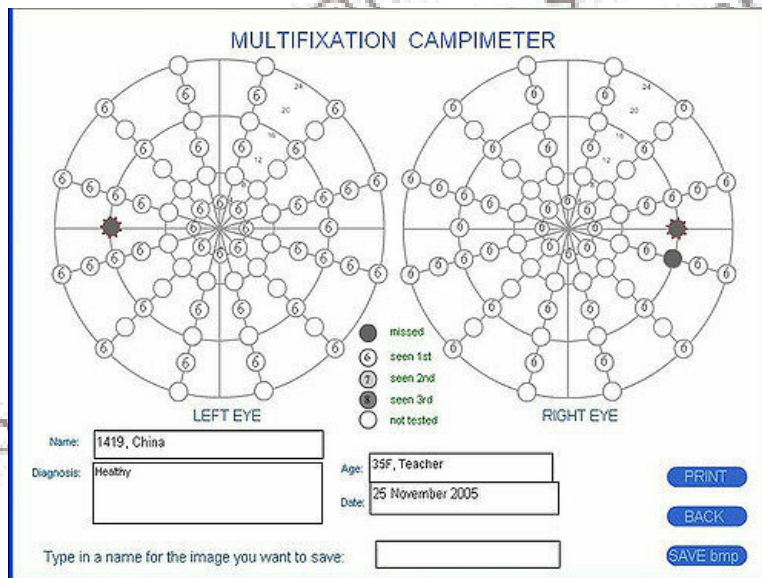


Confrontation method



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The image below shows a multifixation campimeter study result.



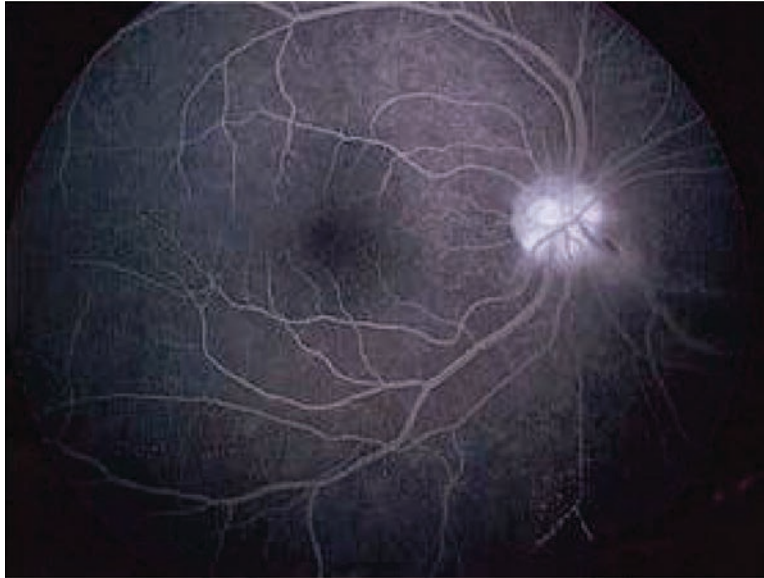
Solution to Question 10:

Complete filling of retinal arterioles and capillaries is seen in the arterio-venous phase.

In FFA, sodium fluorescein dye is injected into the antecubital vein. A fundus camera is then used to take photographs:

- 1st picture- after 5 seconds, then every second for 20 seconds, every 3–5 seconds for the next 1 minute
- Last picture- after 10 minutes
- Pre-arterial phase- choroidal circulation is filling

- Arterial phase- start filling of retinal arteriole till its completely filled
- Arterio-venous phase- complete filling of retinal arterioles and capillaries followed by laminar filling of veins
- Venous filling- arterioles are emptying and veins are filling



Indications for fundus fluorescein angiography:

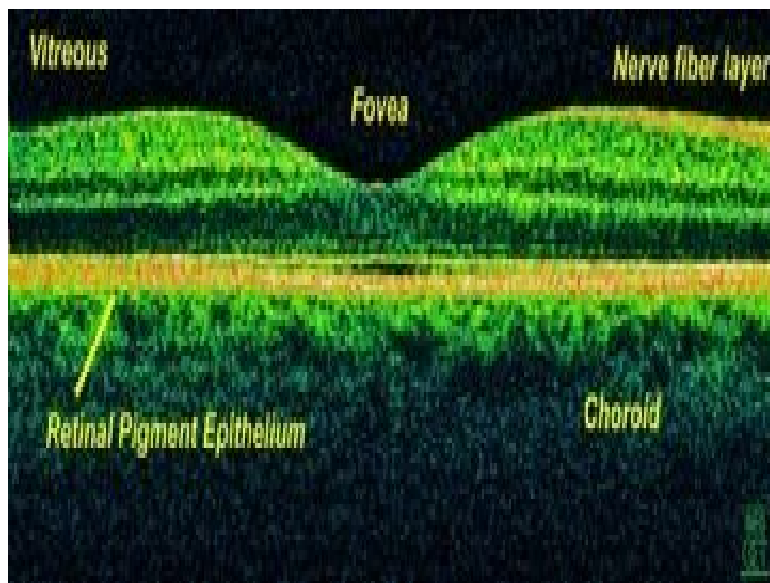
- Diabetic retinopathy
- Vascular occlusion
- Eale's disease
- Central serous retinopathy
- Cystoid macular edema

Solution to Question 11:

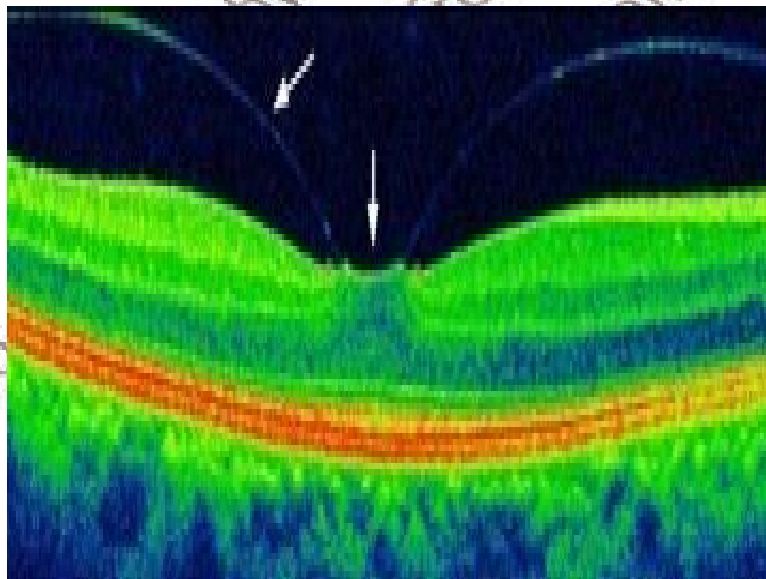
The image shows vitreomacular traction with detached posterior vitreous membrane and foveal traction.

OCT images:

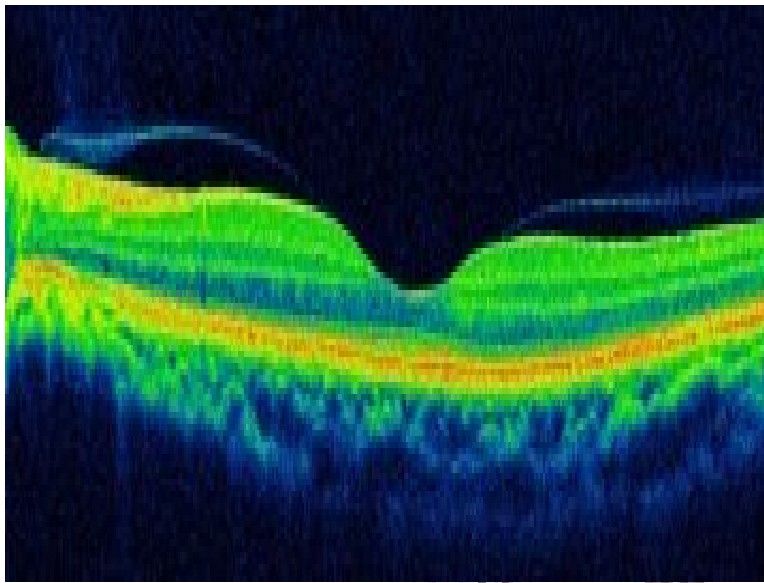
1. Normal OCT



2. Vitreomacular traction- The fovea is elevated from the retinal pigment epithelium (RPE), as indicated by the arrow. Pseudocysts and retinal morphology distortion may also be seen at the site of traction.

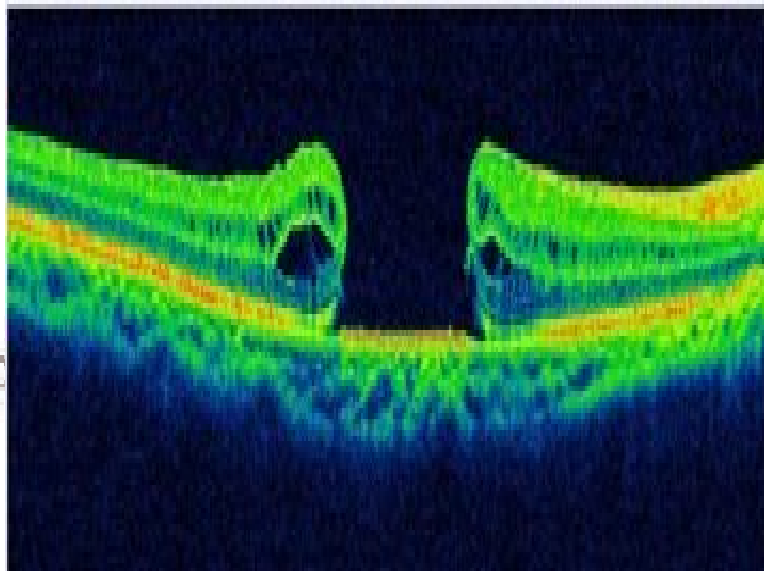


3. Vitreomacular adhesion- posterior vitreous detachment but no macular or foveal traction. The retinal morphology is normal.

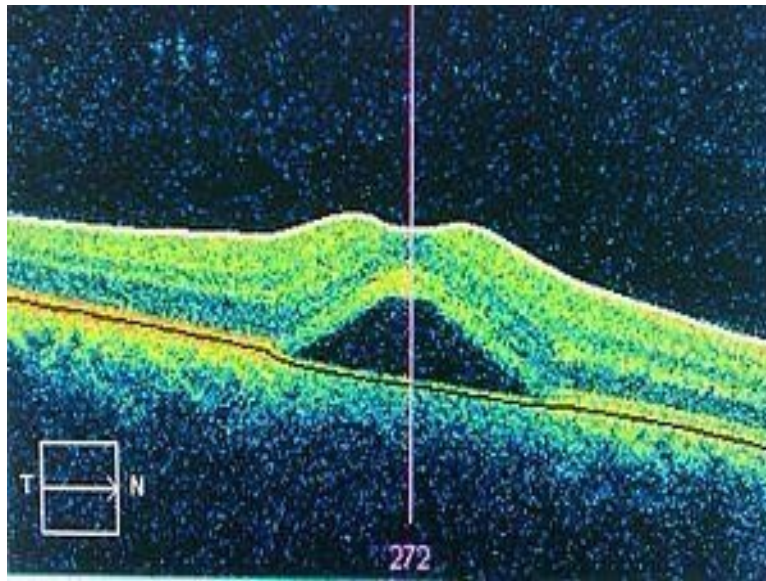


4. Macular hole - a partial or full-thickness retinal defect is seen.

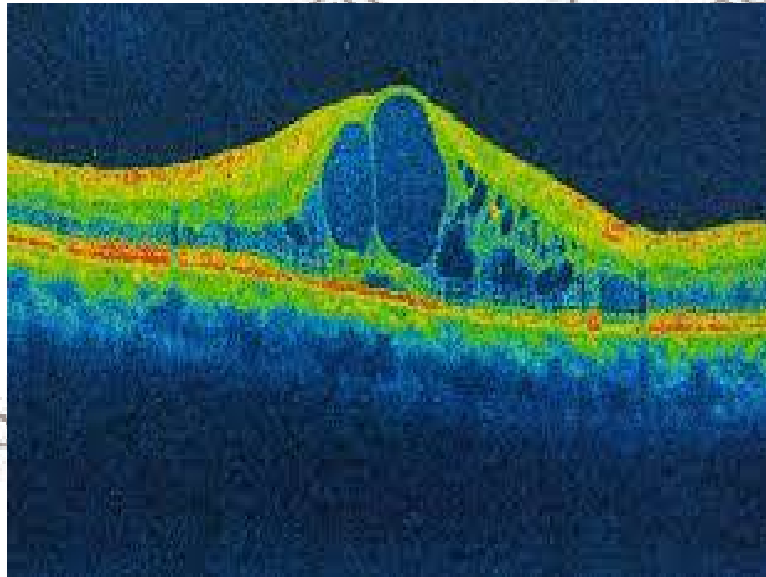
A full-thickness macular hole is seen in the image below.



5. Central serous retinopathy- fluid accumulation in the subretinal space, with shallow retinal detachment.



6. Cystoid macular edema - loss of foveal depression with cystoid spaces in the retina.



Solution to Question 12:

The wave marked is the "B" wave. It represents the activity of bipolar cells.

Electroretinogram (ERG) is used to detect the functional abnormality of the outer retina up to the bipolar cells.

Electroretinogram (ERG)



It has 3 waves:

A wave - arising from rods and cones (photoreceptors)

B wave - generated by Muller cells and represents the activity of the bipolar cells

C wave - represents the metabolic activity of pigment epithelium

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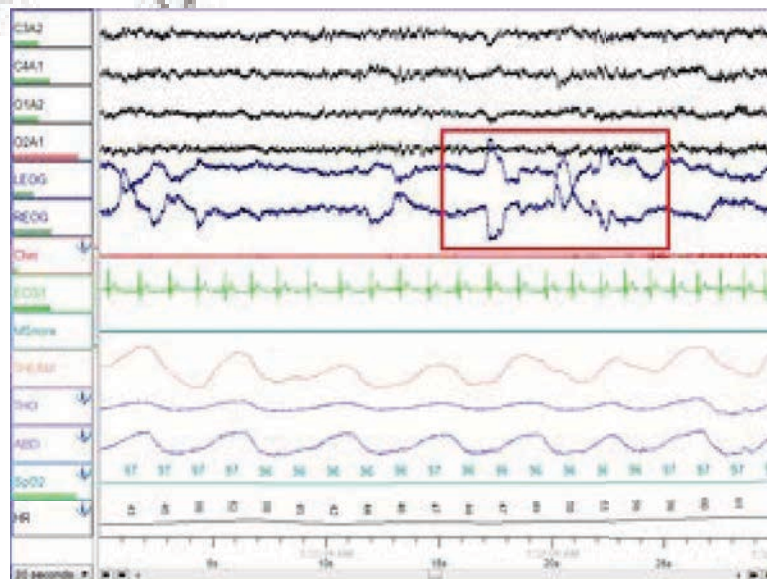
Solution to Question 13:

In electrooculography (EOG), the subnormal curve is <150 .

EOG is based on the measurement of the resting membrane potential of the eye between the cornea (+) and the back of the eye (-). Interpretation is based on the Arden ratio.

Arden Ratio = $\frac{\text{Max height of light peak}}{\text{Min height of dark trough}} \times 100$

- Normal curve: 185 or above
- Subnormal curve: <150
- Flat curve: <125



Solution to Question 14:

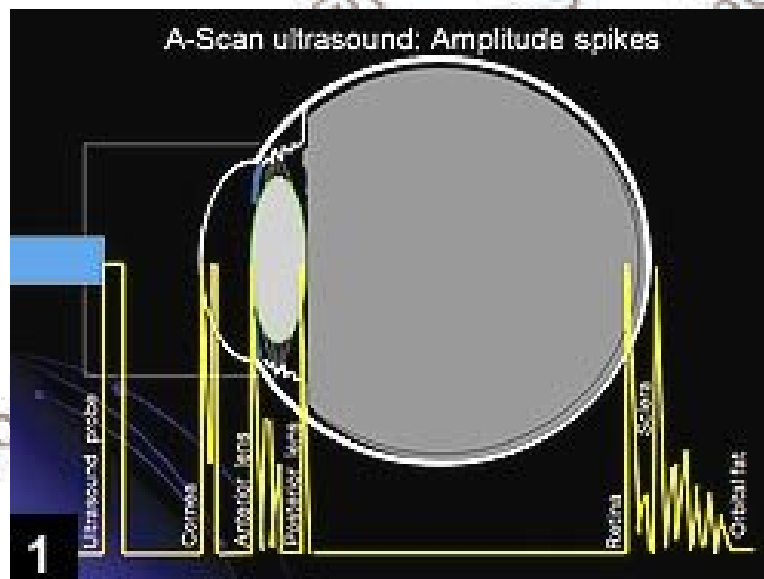
Assessment of vitreous hemorrhage is done by "B" scan.

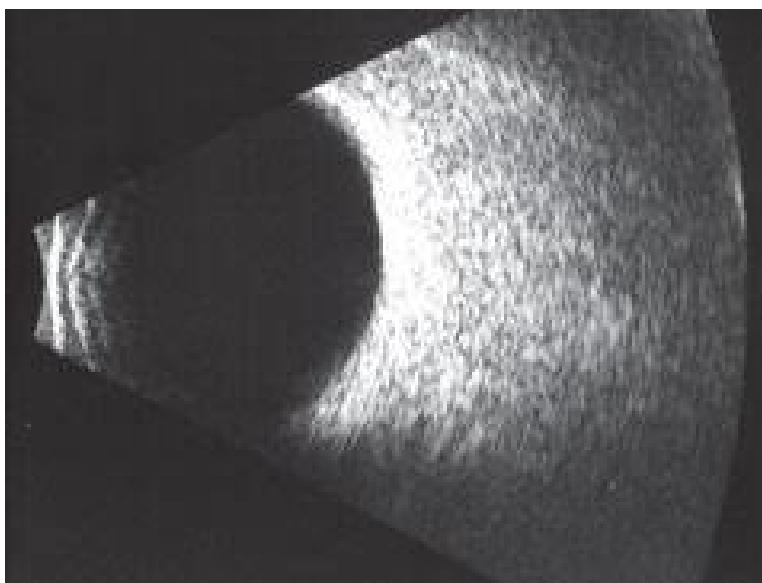
Both 'A' and 'B' scans are types of ultrasound (USG) scans.

The first image below shows "A" scan ultrasound amplitude spikes.

The second image shows a normal "B" scan of the eye.

A scan (time amplitude)	B scan (intensity modulation)
It produces a unidimensional image and echoes are plotted as spikes.	It produces a two-dimensional dotted section of the eyeball.
Uses: Axial length Intraocular lens (IOL) power calculation: Biometry Anterior chamber depth The thickness of intraocular mass	Uses: Assessment of the posterior segment in opaque media Study of intraocular tumors Localization of intraocular and intra-orbital foreign bodies





Solution to Question 15:

The image shows a Maddox rod.

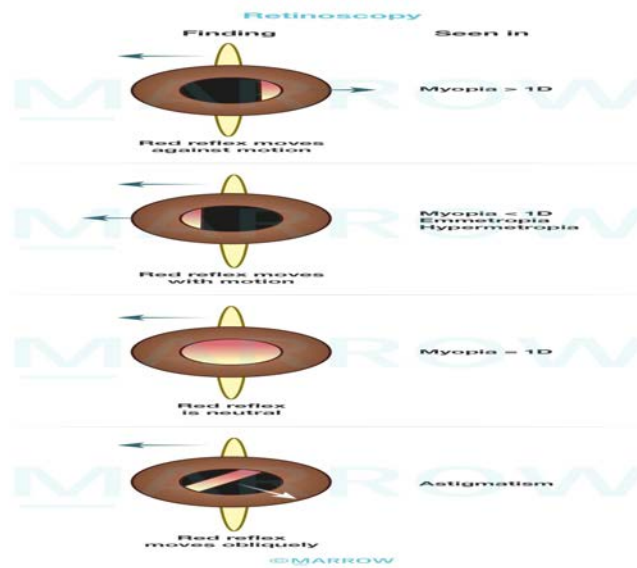
Cyclodeviation is measured with the double Maddox rod test.

The Maddox rod test uses a specially constructed device consisting of a series of parallel cylinders that converts a point source of light into a line image.

It can be used to test for horizontal and vertical deviations, and when used in conjunction with another Maddox rod, for cyclodeviations.

Solution to Question 16:

A patient with 1 D myopia would exhibit no movement of red reflex when retinoscopy is done from a distance of 1m without cycloplegia.



Solution to Question 17:

Keratometry or ophthalmometry is not used to calculate the power of the lens in spectacles.

Uses of keratometry:

- Estimate corneal astigmatism
- Measure the curvature of the cornea
- Calculate the power of the intraocular lens (IOL)

The image given below shows keratometry being performed on a patient.



Solution to Question 18:

The procedure shown is direct ophthalmoscopy. It is used to visualize the fundus.

Types:

- Direct ophthalmoscopy
- Distant direct ophthalmoscopy
- Indirect ophthalmoscopy

Solution to Question 19:

The type of image formed in direct ophthalmoscopy is virtual and erect.

Solution to Question 20:

The procedure is indirect ophthalmoscopy and its principle is to make the eye highly myopic.

A convex lens is used, commonly +20D. The image formed is real, inverted, and magnified. It is used for assessing and managing retinal detachment and other peripheral retinal lesions. It may be done with hazy media because of strong inbuilt illumination.

Solution to Question 21:

Holmgren's wool-matching test is used for the assessment of color vision.

Tests for color vision

- Ishihara pseudo-isochromatic chart
- There are 38 plates read at a distance of 75 cm.
- Drawback- Cannot detect a blue color defect
- Hardy–Rand–Rittler test
- City University test
- Holmgren's wool-matching test
- Naegels anamaloscope
- Farnsworth Munsell 100 hue test

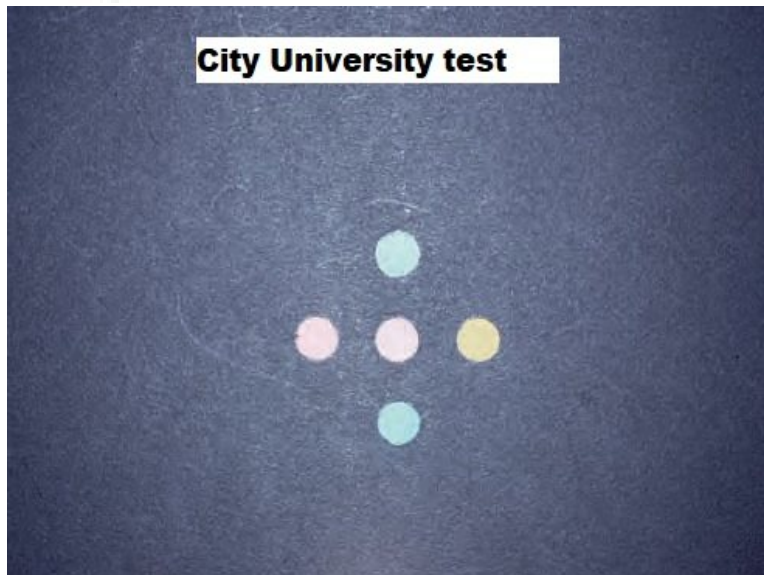
The image given below shows Ishihara pseudo-isochromatic chart

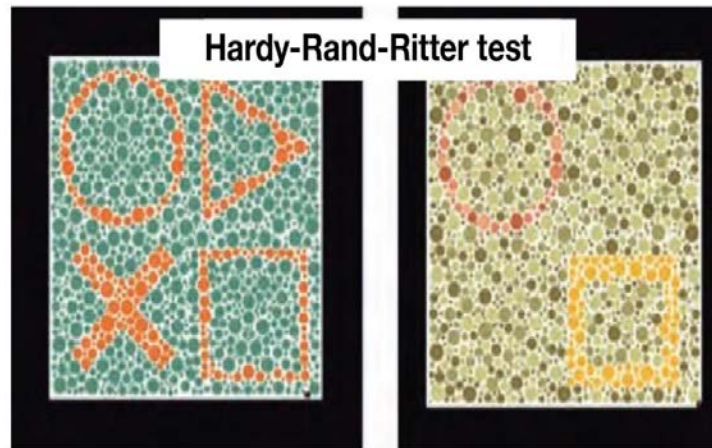


Farnsworth-Munsell 100-hue test



City University test





Solution to Question 22:

The given dye in the strip is lissamine green. It is used as 1% concentration.

Lissamine green stains dead and degenerate cells, but does not stain healthy epithelial cells. It is used to diagnose ocular surface disease.

It stains the conjunctiva. It does not cause stinging like Rose Bengal. The Green staining is best observed two minutes after instillation.

The image below shows the lissamine staining of the eye.



Solution to Question 23:

The image shows a cryoprobe. It is not used in the management of impending corneal perforation.

It is used in:

- Peripheral retinopexy for peripherally located retinal breaks and tears
- Treatment of anterior RB tumors
- Cyclocryotherapy in pars planitis and end-stage glaucoma

Solution to Question 24:

The given image is suggestive of oculomotor nerve palsy.

The central image shows the primary position of gaze where the patient's left eye is deviated down and out with ptosis of the left eyelid. From the eye movements in other positions, it can be seen that adduction and elevation are absent and depression is reduced in the left eye. All of these findings are indicative of oculomotor nerve palsy.

3rd Cranial nerve(oculomotor nerve) palsy presents with:

- Ptosis due to weakness of levator palpebrae superioris muscle (LPS)
- Abduction and depression in the primary position (down and out) due to unopposed action of the lateral rectus(6th nerve supply) and superior oblique muscles(4th nerve supply)
- Dilated pupil and defective accommodation due to parasympathetic palsy (involvement of Edinger Westphal parasympathetic supply)
- Limited adduction, elevation and depression due to ipsilateral paralysis of medial rectus, inferior oblique and inferior rectus .

Signs of Oculomotor Nerve Palsy

Description	Image
Profound ptosis due to weakness of levator	
Abduction & depression in 1° gaze due to unopposed action of lateral rectus & superior oblique	
Intorsion due to intact superior oblique	
Limited depression due to weak inferior rectus	
Limited elevation due to weak superior rectus & inferior oblique	
Dilated pupil due to parasympathetic palsy	
Normal abduction	

©Marrow

Option A: Trochlear(4th cranial) nerve palsy presents with superior oblique muscle paralysis.

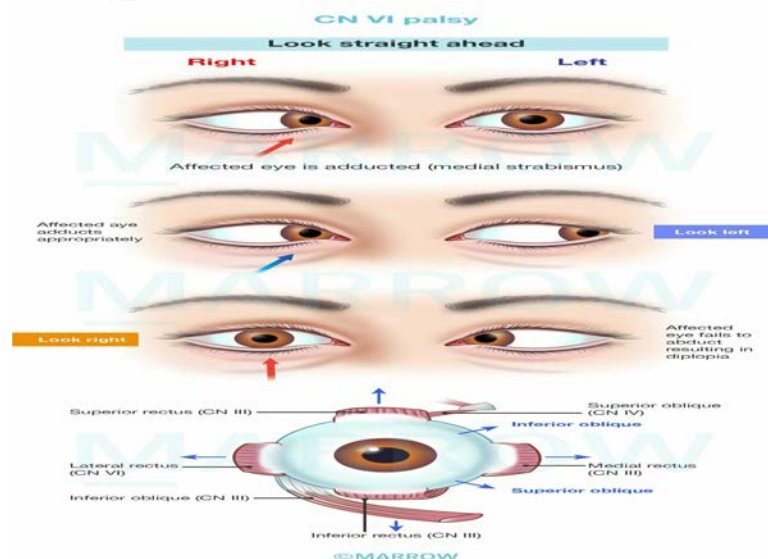
Clinical findings of trochlear nerve palsy include:

- Hypertropia (elevation) in the primary position
- Extorsion that is greatest in abduction
- Compensatory head posture with contralateral face turn and head tilt with chin slightly depressed. This position is assumed to avoid Diplopia.
- Limitation of depression
- Normal abduction and elevation.

The below given image depicts 4th cranial nerve palsy.

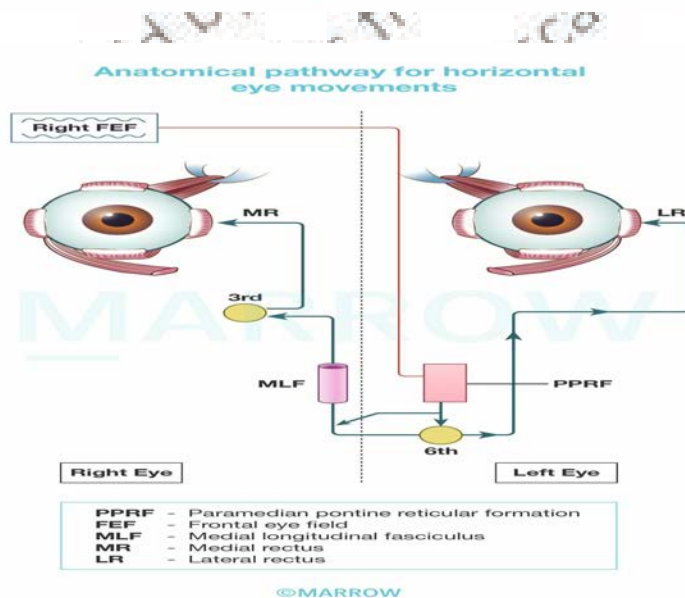
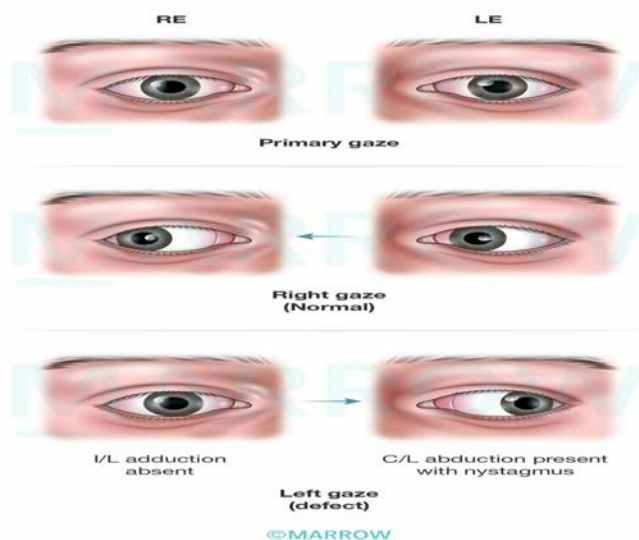


Option B: Lateral rectus palsy is seen in cases of isolated Abducens nerve palsy. These patients present with esotropia/convergent squint in primary position with limitation of abduction on the side of the lesion. This is due to unopposed action of the medial rectus muscle in the affected eye. There is also a compensatory head turn towards the side of the lesion to reduce the symptom of Diplopia. The adduction is normal in lateral rectus palsy.



Option C: Internuclear ophthalmoplegia occurs following an injury to the medial longitudinal fasciculus (MLF). A unilateral lesion to the MLF leads to the failure of adduction in the ipsilateral eye with a normal abduction of the contralateral eye.

The below given image depicts the pathogenesis of Inter nuclear ophthalmoplegia in detail.



Solution to Question 25:

Photostress test is used to differentiate between the disease involving the macula and optic nerve.

This test is of particular use in early macular diseases such as central serous retinopathy.

In this test, one eye is covered and the patient is asked to read the smallest line on the near chart. A bright light is then shone into the eye for 15 seconds, and the patient is asked to read the line again, and the time taken to recover is noted. The test is repeated with the other eye.

In normal people and in patients with optic nerve disease, the recovery time is not affected. Whereas, in patients with macular disease the recovery time is prolonged and is usually more than one-third of the normal recovery time. This is because, in macular disease, the photoreceptors are damaged, and there is a delay in the visual pigment regeneration process.

Sold by @Itachibot
If you purchased this from someone else,
you may have been scammed.

Eye Signs in Systemic Disease

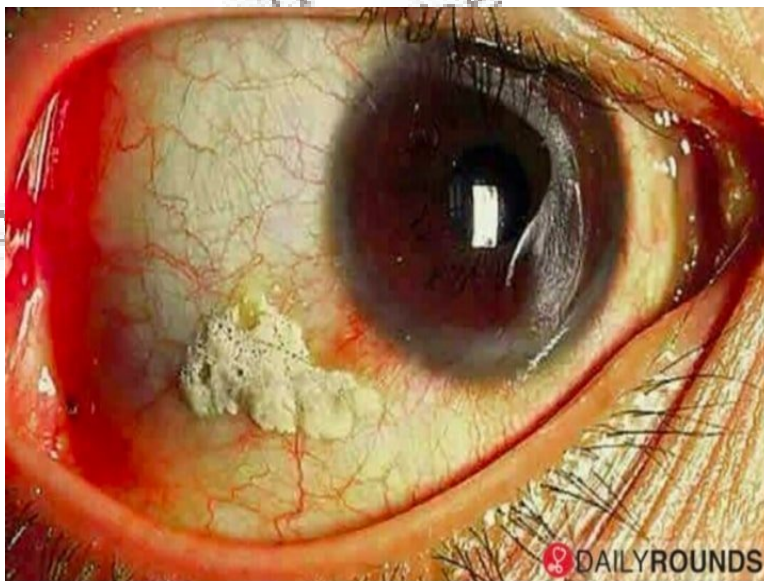
Question 1:

Which of the following deficiencies would you suspect in a patient with acute retrobulbar neuritis?

- a) Vitamin A deficiency
- b) Vitamin B1 deficiency
- c) Vitamin B2 deficiency
- d) Vitamin C deficiency

Question 2:

During a community outreach program in a rural area, you come across a girl with a conjunctival lesion as shown in the image. According to the WHO grading, which grade does this child falls under?



- a) XN
- b) X1A
- c) X1B
- d) X2

Question 3:

In which of the following conditions will Koplik's spots and conjunctivitis be seen together?

- a) Rubella
- b) Measles
- c) Mumps
- d) Whooping cough

Question 4:

A 45-year-old man presented to the emergency department with an extremely painful right foot. He stated that he had an alcohol binge the previous night. On examination, the right first metatarsophalangeal joint is red, swollen, and extremely tender to palpation. Which of the following ocular conditions is not associated with his underlying condition?

- a) Episcleritis
- b) Scleritis
- c) Uveitis
- d) Glaucoma

Question 5:

Where is snow flake cataract seen?

- a) Diabetes mellitus
- b) Galactosemia
- c) Wilson's disease
- d) Myotonic dystrophy

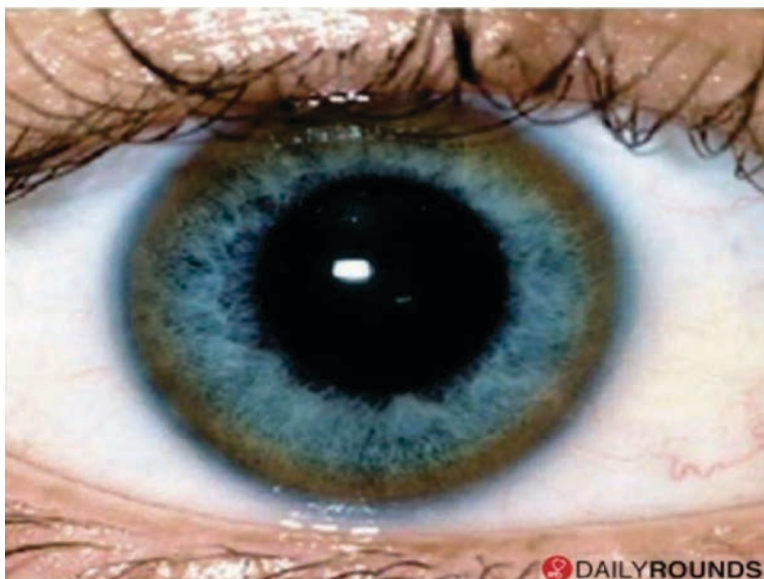
Question 6:

You would expect to see oil drop cataract in a patient with _____

- a) Diabetes
- b) Galactosemia
- c) Wilson's disease
- d) Homocystinuria

Question 7:

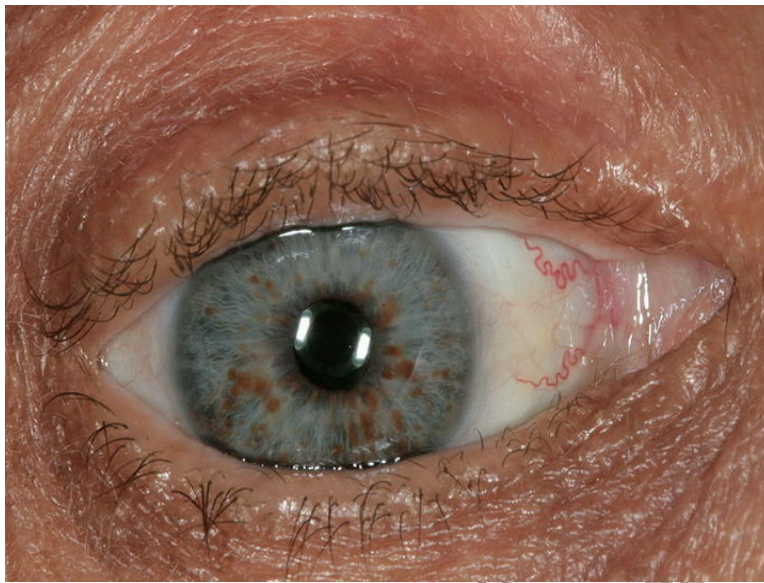
A 10-year-old girl with neurological manifestations was referred from the pediatric OPD. Her ocular examination findings is as given below. What is the likely diagnosis?



- a) Lysosomal storage disorder
- b) Galactosemia
- c) Wilson's disease
- d) Homocystinuria

Question 8:

A patient with a suspected neurocutaneous syndrome was referred to ophthalmology. His ocular findings are given. What is the most likely diagnosis?



- a) Sturge-Weber syndrome
- b) Neurofibromatosis
- c) Tuberous sclerosis
- d) von Hippel-Lindau syndrome

Question 9:

Which of the following conditions can occur as an ocular manifestation in a patient with dengue?

- a) 1, 2, 4, 5
- b) 1, 2, 3, 5
- c) 2, 3, 4, 5
- d) 1, 3, 4

Question 10:

A 6-year-old girl is admitted for evaluation for recurrent seizures. She has a large port-wine stain on the right side of her face as shown below. The MRI of the head shows a leptomeningeal angioma. What is the most common ocular manifestation of the underlying condition?



- a) Uveitis
- b) Retinitis pigmentosa
- c) Keratitis
- d) Glaucoma

Question 11:

A 70-year-old man presented with complaints of diminution of vision in both eyes for the last 6 - 8 months. On examination, bilateral cataracts were seen. Incidentally, you also note iris pearls. Which of the following conditions should you suspect in this patient?

- a) Candida endophthalmitis
- b) Leprotic uveitis
- c) Ophthalmia nodosum
- d) Sympathetic ophthalmia

Question 12:

A patient presents with areas of yellowish papular skin lesions and redundant and inelastic folds of skin on the neck and thighs. Which of the following ocular findings is not seen in this disease?

- a) Optic nerve drusen
- b) Arterial macroaneurysms
- c) Angioid streaks

d) Blue sclera

Question 13:

An elderly patient admitted with multiple comorbidities in the ICU, developed exposure keratitis and ulceration. An ophthalmology opinion was sought. Incidentally, on fundoscopy, the resident found the following picture. What is this most likely due to?



Clare Gilbert

- a) Chronic hypertension**
- b) Diabetes**
- c) Sickle cell disease**
- d) Central retinal vein occlusion**

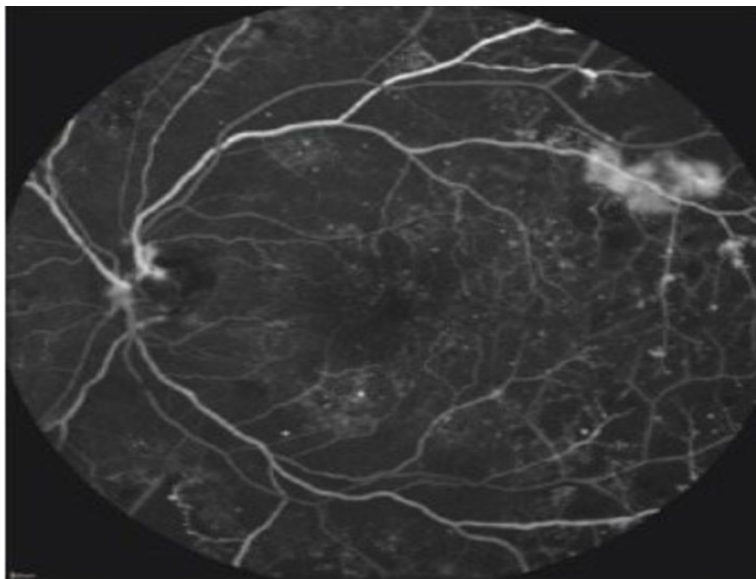
Question 14:

In patients with a long-standing history of Crohn's disease, which of the following ocular complications is the most likely to develop?

- a) Uveitis**
- b) Keratoconjunctivitis sicca**
- c) Episcleritis**
- d) Retinal detachment**

Question 15:

A patient with a history of decreased vision was evaluated with fundus fluorescein angiography. What is the most likely diagnosis?



- a) Sickle-cell retinopathy
- b) Hypertensive retinopathy
- c) Bird shot choroidopathy
- d) Diabetic retinopathy

Answer Key

Question No.	Correct Option
1	b
2	c
3	b
4	d
5	a
6	b
7	c
8	b
9	a
10	d
11	b

12	b
13	b
14	c
15	d

Detailed Explanations

Solution to Question 1:

Vitamin B1 deficiency can cause acute retrobulbar neuritis.

Nutritional optic neuropathy is characterized by bilateral, symmetric, cecocentral scotomas and disturbance of color vision. Fundus morphology is usually normal.

Deficiencies in the following vitamins are primarily responsible for the damage caused to the optic nerve: B1 (thiamine), B6 (pyridoxine), B9 (folic acid), and B12 (cobalamin). These vitamins are cofactors for many metabolic reactions and their deficiency leads to toxic byproduct accumulation within cells, which inhibits the electron transport chain and depresses mitochondrial function. Depletion of ATP leads to nerve dysfunction and eventual nerve death.

Chronic alcohol abuse is a commonly associated risk factor and a cause of malnourishment.

Solution to Question 2:

The given image shows Bitot's spot, which is seen in the X1B stage of vitamin A deficiency.

Solution to Question 3:

Koplik's spots are seen in measles.

Solution to Question 4:

The given clinical scenario is suggestive of acute gout. Glaucoma is not a known association of acute gout.

Ocular manifestations seen in acute gout are episcleritis, scleritis, and uveitis. Ocular structures have a relatively poor solvent ability, in addition to lower temperatures. Therefore, they are predisposed to tophi deposition. As a result of urate deposits, crystal-induced inflammation of the affected tissue is responsible for causing episcleritis, scleritis, and uveitis.

Solution to Question 5:

Snowflake cataract is seen in diabetes mellitus.

Ocular manifestations in diabetes mellitus

- Xanthelasma
- Telangiectasia
- Subconjunctival hemorrhage
- Punctuate keratopathy
- Descemet's folds
- Rubeosis iridis
- Snow-flake cataract
- Vitreous hemorrhage
- Diabetic retinopathy
- Neovascular glaucoma
- Recurrent hordeola
- Dry eye
- Tractional retinal detachment
- Diabetic maculopathy
- Papillopathy
- Central retinal vein occlusion

Solution to Question 6:

Oil droplet cataract is seen in galactosemia.

Characteristic associations are:

- Diabetes - Snowflake cataract
- Wilson's disease - Sunflower cataract
- Homocystinuria - Bilateral inferonasal subluxation of the lens

Solution to Question 7:

The image shows the Kayser–Fleisher (KF) ring, which is a characteristic finding of Wilson's disease.

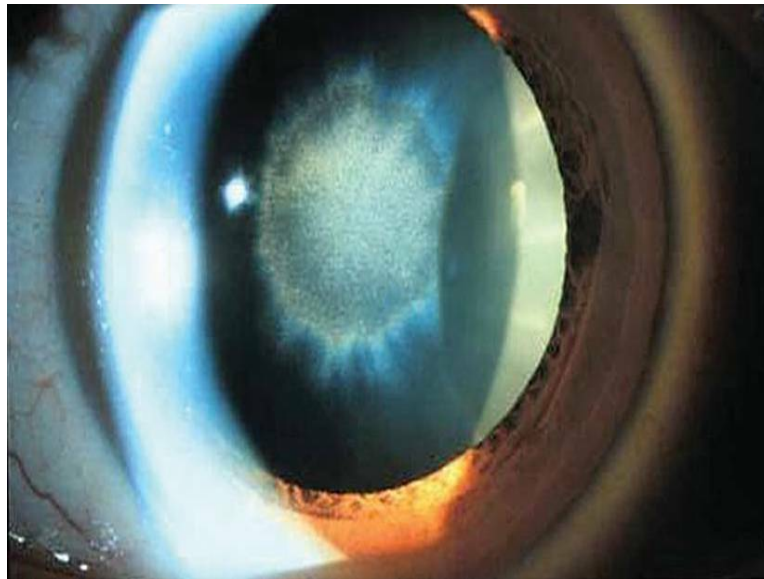
KF ring consists of the deposition of copper in the Descemet membrane of the cornea. The rings are usually golden-brown and are visible only under the slit-lamp. They do not impair vision. They

disappear on treatment with anti-copper drugs and reappear with disease progression. They are a characteristic sign of Wilson's disease but are not pathognomonic. They can also be seen in:

- Primary biliary cholangitis
- Primary sclerosing cholangitis
- Cholestasis.

Sunflower cataract is a rare finding of Wilson's disease. It occurs due to the deposition of copper in the centre of the lens. It consists of a greenish central opacity with spoke-like radial opacities and is visible only under the slit-lamp. It is also seen in chalcosis (deposition of intraocular copper-containing foreign body).

The image given below shows a sunflower cataract:



Solution to Question 8:

The image given shows Lisch nodules of the iris, which is suggestive of neurofibromatosis type 1.

Neurocutaneous syndrome	Ocular manifestations
Neurofibromatosis type 1	Eyelid plexiform neurofibroma gives a characteristic S-shaped deformity of the upper lid, and classically, it is texturally reminiscent of a "bag of worms." Optic nerve glioma - Apilo-cytic astrocytoma that typically occurs in young children. Bilateral Lisch nodules (at least 95%) are hamartomas that develop during the second-third decades. Congenital ectropion uveae Prominent corneal nerves Congenital glaucoma Sphenoido-orbital encephalocele - due to dysplasia of the greater wing of the sphenoid
Neurofibromatosis type 2	Posterior subcapsular cataract Hamartomas of the retinal pigment epithelium (RPE) and retina
Sturge-Weber syndrome	Choroidal hemangioma Iris heterochromia Ipsilateral glaucoma
Tuberous sclerosis	Atypical iris coloboma Iris hypopigmentation Retinal astrocytoma
von Hippel-Lindau syndrome	Retinal hemangiomas Optic nerve haemangioma

Solution to Question 9:

Cataract is not an ocular manifestation of dengue.

Dengue has varied manifestations, in different organ systems, mainly of inflammation and bleeding. Until recently, apart from retro-orbital pain, eye involvement was not recognized as a common feature of dengue infection. Potential pathogenic mechanisms include a direct viral effect on the eye, an immune-mediated vasculitis, and/or a variety of secondary effects related to dengue-induced vascular leakage and hemostatic derangements.

Ocular manifestations of dengue:

- Typically, patients complain of painless visual impairment within 1 week from illness onset, often around the time of the platelet nadir.
- Conjunctival injection and subconjunctival hemorrhage.
- Complications such as retinal hemorrhage, retinal edema, macular ischemia, macular edema, vitreous hemorrhage, vitritis, and optic neuritis or papillitis

- Depending on the severity of the ocular lesion, gradual improvement in vision occurs over several weeks, although in rare cases permanent visual impairment may result.

Solution to Question 10:

The given clinical picture of a young child presenting with unilateral port wine stain in the region of trigeminal nerve supply and leptomeningeal angioma as seen on MRI is indicative of Sturge-Webber syndrome (SWS). Glaucoma is the most common ocular manifestation of SWS.

Sturge weber syndrome is a neurocutaneous disorder(phakomatoses) that is characterized by unilateral capillary malformation at the region of trigeminal supply known as port wine stain or nevus flammeus, ipsilateral diffuse cavernous hemangioma of the choroid and ipsilateral leptomeningeal hemangiomatosis, which causes seizures and mental retardation.

Ocular manifestation of SWS include:

- Glaucoma - most common ocular manifestation. It develops anywhere between infancy to adulthood. When developed in infancy it presents as buphthalmos.
- Diffuse choroidal hemangioma
- Serous retinal detachment
- Visual field defects- homonymous hemianopia due to the leptomeningeal hemangiomas affecting the optic tract or occipital lobe.
- Heterochromia of the iris
- Episcleral and conjunctival lesions

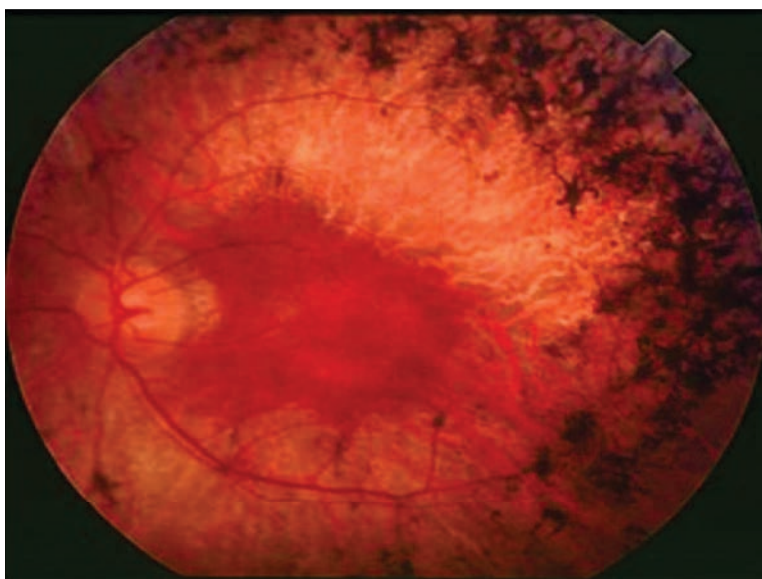
Option A: Uveitis is the inflammation of the uveal tissue of the eyes. It can be associated with a number of systemic conditions such as sarcoidosis, behcet's disease, ankylosing spondylitis, reiter's syndrome, Toxoplasmosis and Posner–Schlossman syndrome (PSS).

Option B: Retinitis pigmentosa is an inherited or acquired pigmentary retinal dystrophy affecting the rods more. The earliest manifestation is worsening of night blindness which gradually progresses to tunnel vision and ultimately loss of central vision. The classic triad of findings in RP comprises bone-spicule retinal pigmentation, arteriolar attenuation and waxy disc pallor.

Systemic Associations of RP include:

- Laurence-Moon-Biedl syndrome
- Cockayne's syndrome
- Refsum's syndrome
- Usher's syndrome
- Bassen-Kornzweig syndrome

The below given image shows the classical triad findings of Retinitis pigmentosa.



Option C: Keratitis is the inflammation of cornea which is characterised by corneal edema, cellular infiltration and ciliary congestion. It can be either infectious or non-infectious in nature which includes allergic keratitis, neurotrophic keratitis, exposure keratitis and peripheral ulcerative keratopathies such as Rosacea keratitis and Mooren's ulcer.

Solution to Question 11:

Iris pearls are pearly-white nodules seen on the iris in leprosy.

They are asymptomatic, 0.3 to 1.0 mm in diameter, dull yellowish-white in colour, and are seen at an early stage of the disease. It is not associated with any inflammatory changes in the anterior chamber. Histologically, it consists of lepromatous bacilli packed within mononuclear cells or acid-fast material with accompanying foreign body reaction. Inside the eye, leprosy bacilli preferentially lodge in the iris and ciliary body, due to the relatively cooler temperatures and high vascularity.

Solution to Question 12:

The given case suggests a diagnosis of pseudoxanthoma elasticum. Arterial macroaneurysm is not a feature of this disease.

Ocular findings in pseudoxanthoma elasticum:

- Blue sclera
- Angioid streaks
- Optic nerve head drusen
- "Peau-d'orange" appearance of the retina
- Choroidal neovascularization

Solution to Question 13:

The given image suggests a diagnosis of proliferative diabetic retinopathy.

The hallmark of Proliferative diabetic retinopathy (PDR) is neovascularization. Abnormal new vessel formation, which occurs on the inner (vitreous) surface of the retina and may extend into the vitreous cavity, can cause vitreous haemorrhage. Neovascularization is often accompanied by pre-retinal fibrous tissue, which, along with the vitreous, can contract, resulting in traction retinal detachment. In addition, tiny microaneurysms can also be seen.

Solution to Question 14:

The most common ocular manifestation of Crohn's disease is episcleritis.

Uveitis is associated with both Ulcerative colitis and Crohn's disease.

Solution to Question 15:

The given image depicts neovascularization, which is the hallmark of proliferative diabetic retinopathy.

Dye can be seen leaking through abnormal neovascular fronds. Both neovascularization of the disc (NVD), as well as neovascularization elsewhere (NVE i.e., 2 DD away from the optic disc margin), are seen.

The treatment available for proliferative diabetic retinopathy is photocoagulation of the ischaemic areas to reduce the metabolic demand and decrease or prevent the release of vasoproliferative factors.

Community Ophthalmology

Question 1:

What is the most common cause of blindness in India?

- a) Cataract
- b) Refractive error
- c) Glaucoma
- d) Corneal blindness

Question 2:

As part of a public outreach initiative under the National Programme for Control of Blindness and Visual Impairment, you are screening the population of a neighboring village. While preparing your report, what definition of blindness would you use?

- a) Presenting distance visual acuity less than 3/60 in the better eye
- b) Presenting distance visual acuity less than 1/60 in the better eye
- c) Presenting distance visual acuity less than 6/60 in the better eye
- d) Presenting distance visual acuity less than 2/60 in the better eye

Question 3:

Which is the most common cause of preventable childhood blindness?

- a) Refractive error
- b) Vitamin A deficiency
- c) Retinopathy of prematurity
- d) Injuries

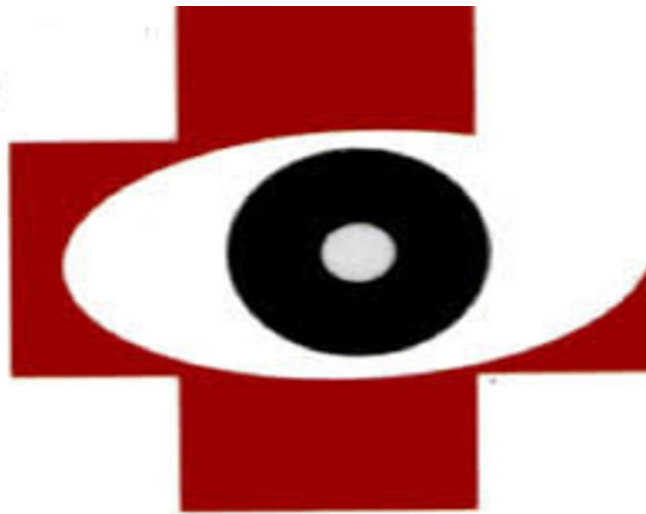
Question 4:

You have examined five patients(i-v) in the OPD this morning. Their visual acuity test results are as given below. Who amongst them would be described as blind as per the WHO Vision 2020 definition?

- a) Patients i and v
- b) Patients ii, iii and iv
- c) Patients iii, iv
- d) Patients i, ii and v

Question 5:

Which organization does the image given below stand for?



- a) National Programme for Control of Blindness
- b) National Programme for Trachoma
- c) Vitamin A Deficiency Control Programme
- d) National Programme for Control of Blindness and Visual Impairment

Question 6:

A 1-year-old child is brought to the well-baby clinic for immunization. What is the dose of prophylactic vitamin A to be given to her?

- a) 50,000IU
- b) 1 lakh IU
- c) 2 lakh IU
- d) 4 lakh IU

Question 7:

Match the findings of the given patients (i, ii, iii, iv) with the appropriate category of blindness they fall under.

- a) i-C; ii-B; iii-D; iv- A
- b) i-A; ii-D; iii-B; iv-C
- c) i-B; ii-C; iii-D; iv- A
- d) i-B; ii-D; iii-C; iv- A

Question 8:

Which of the following is not a disease that is included in the Global Vision 2020 Program?

- a) Cataract
- b) Refractive error
- c) Trachoma
- d) Glaucoma

Question 9:

What is the number of primary-level vision centers proposed to be developed across the country as per the Vision 2020 Program?

- a) 2,00,000
- b) 2000
- c) 20,000
- d) 2020

Question 10:

A 45-year-old Nigerian man hailing from an area endemic for river blindness presented with the given finding. What is the drug of choice for the treatment?



- a) Diethylcarbamazine (DEC)
- b) Ivermectin
- c) Suramin
- d) Fluconazole

Question 11:

You are appointed as the regional doctor in charge of overseeing the activities under Rashtriya Bal Swasthya Karyakram (RBSK). All of the following diseases are specifically covered under the purview of this program except?

- a) Congenital cataract
- b) Vision Impairment
- c) Retinopathy of prematurity
- d) Congenital Glaucoma

Answer Key

Question No.	Correct Option
1	a
2	a
3	b

4	c
5	d
6	c
7	c
8	d
9	c
10	b
11	d

Detailed Explanations

Solution to Question 1:

The most common cause of blindness in India is cataract (62.5%).

S. No	Common causes in India	Common causes World wide
1.	Cataract (62.6%)	Cataract (43%)
2.	Refractive error	Glaucoma
3.	Corneal blindness	Diabetic retinopathy
4.	Glaucoma	Trachoma
5.	Surgical complication	Vitamin A deficiency
6.	Posterior capsular opacification	Onchocerciasis
7.	Posterior segment disorder	

Solution to Question 2:

The definition of blindness under the National Programme for Control of Blindness and Visual Impairment (NPCBVI) is presenting distance visual acuity of less than 3/60 in the better eye and limitation of the field of vision to less than 10 degrees from the center of fixation.

Solution to Question 3:

The most common cause of preventable childhood blindness is vitamin A deficiency.

The most common cause of blindness (including adults), is however, cataract.

Solution to Question 4:

Patients iii and iv would be classified as blind.

In the Vision 2020 initiative, the WHO has revised its definition of blindness. As per the new definition,

"Blindness is defined as a presenting visual acuity of worse than 3/60 or a corresponding visual field loss to less than 10° in the better eye."

Note: Blindness is defined for the better eye, hence option (i) is insufficient information to classify as blind, as information about the better eye is not given.

Solution to Question 5:

The symbol stands for National Programme for Control of Blindness and Visual Impairment (NPCBVI), which was previously called the National Programme for Control of Blindness.

Solution to Question 6:

The dose of vitamin A prophylaxis for a 1-yr-old child would be 2 lakh IU orally.

Dose of oral vitamin A for prophylaxis:

- Less than 6 months of age: 50,000 IU
- 6–11 months of age: 1 lakh IU
- 12 months and more: 2 lakh IU

Solution to Question 7:

Type of Blindness	Description
Social Blindness	Presenting distance visual acuity of less than 3/60 (20/400) in the better eye and limitation of the field of vision to be less than 10 degrees from the center of fixation
Economic blindness	The inability of a person to count fingers from a distance of 6 meters
Manifest blindness	Vision 1/60 to just perception of light

Type of Blindness	Description
Absolute blindness	No perception of light
Curable blindness	That stage of blindness where the damage is reversible by prompt management, e.g., cataract
Preventable blindness	Loss of vision that could have been completely prevented by the institution of effective preventive or prophylactic measures e.g., xerophthalmia, trachoma, and glaucoma
Avoidable blindness	The sum total of preventable and curable blindness is often referred to as avoidable blindness.

Solution to Question 8:

Glaucoma is not included in the Global Vision 2020 Program.

Diseases targeted under Vision 2020

S. No	Globally	India
1.	Cataract	Cataract
2.	Refractive errors	Refractive errors
3.	Onchocerciasis	Childhood blindness
4.	Childhood blindness	Glaucoma
5.	Trachoma	Diabetic retinopathy
6.	Low vision	Corneal blindness

Solution to Question 9:

The number of primary-level vision centers proposed as per the Vision 2020 Program is 20,000.

Centers	No. of Service centers
Center of Excellence	20
Training centers	200

Centers	No. of Service centers
Service centers	2000
Primary level vision centers	20,000

Solution to Question 10:

The image shows sclerosing keratitis of onchocerciasis or river blindness. The drug of choice for the treatment of onchocerciasis is ivermectin.

Ivermectin: 6 mg, single-dose, repeated every 6-12 months, according to the weight of the patient.

Suramin and diethylcarbamazine (DEC) were given in the past but have been discontinued due to serious adverse effects.

Solution to Question 11:

Congenital glaucoma is not covered under Rashtriya Bal Swasthya Karyakram (RBSK)

Child health screening and early intervention services under RBSK envisages to cover 30 selected health conditions for screening, early detection, and free management in the age group birth to 18 years. The eye conditions that are covered are

- Vitamin A deficiency
- Congenital cataract
- Vision impairment
- Retinopathy of prematurity

Instruments in Ophthalmology

Question 1:

A 50-year-old woman comes to the out-patient with complaints of diminished vision along with floaters and flashes. The ophthalmologist uses the following instrument to assess the retinal lesions. Which of the following statements is true regarding the ophthalmoscopy done using the instrument shown in the image below?



- a) Area of field in focus is 2 disc diameters.
- b) The image produced is real and inverted.
- c) Condensing lens is not required.
- d) Stereopsis is absent.

Question 2:

Which of the following is a feature of indirect ophthalmoscopy?

- a) 15 times magnification
- b) Image is erect and virtual
- c) Binocular vision
- d) Smaller field of focus

Question 3:

On entering the Ophthalmology OPD for the first time, you notice multiple patients being examined using the following instrument. What is this instrument?



- a) Ophthalmoscope
- b) Refractometer
- c) Keratometer
- d) Slit lamp

Question 4:

Which of the following instruments can you use to measure corneal thickness?

- a) Keratometer
- b) Pachymeter
- c) Specular microscopy
- d) Ophthalmometer

Question 5:

In which surgery would you use the instrument shown below?



- a) Cataract
- b) Glaucoma
- c) Lid surgeries
- d) Corneal transplant

Question 6:

All of the following are true about the instrument shown in the image given below except-

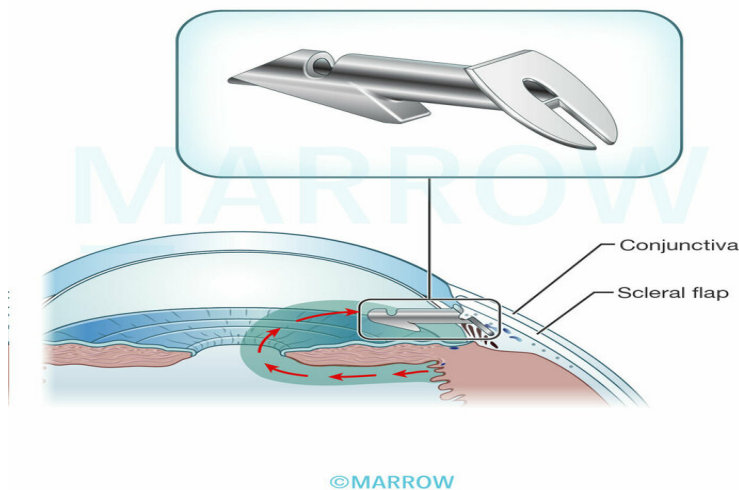


- a) It is used for the localization of a foreign body.
- b) It is used for the measurement of the angle of the posterior chamber.

- c) It is used for the demonstration of the extent of peripheral anterior synechiae.
- d) It provides a mirror image of the opposite angle.

Question 7:

A 50-year-old male presented with progressive blurred vision, photophobia, and eye pain. On examination, there was conjunctival redness, a mid-dilated pupil, and a shallow anterior chamber in the right eye. He has been treated both medically and surgically for the condition but showed no improvement. The following procedure incorporates a device to aid drainage. Identify the device.



- a) Ex-Press™ implant
- b) Ahmed glaucoma valve
- c) Molteno implant
- d) Baerveldt implant

Question 8:

During a visit to the ophthalmology department, you see the following test being performed for visual field charting of a patient. Identify the instrument used from the options given below.



- a) Octopus
- b) Field Master
- c) Humphrey Field Analyzer
- d) Lister's perimeter

Question 9:

Which among the following would you use for the measurement of intraocular pressure in a patient with a scarred cornea?

- a) Goldmann
- b) Perkins
- c) Tonopen
- d) Pulsair

Question 10:

A first-year ophthalmology resident is asked to counsel the parents of a 1-month-old infant diagnosed with a lamellar congenital cataract in both eyes. He discusses the treatment options and mentions that the following lens will be used in the surgery. This lens is a



- a) Anterior chamber intra-ocular lens
- b) Posterior chamber intra ocular lens (modified C-loop type)
- c) Singh and Worst's iris claw lens
- d) Posterior chamber intra ocular lens (four loop type)

Question 11:

As an intern posted in the department, you are assisting a senior surgical ophthalmologist for cataract surgery on a 72-year-old man. You notice the nursing head pass him the following lens for implantation. This is a _____



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- a) Anterior chamber intra-ocular lens (ACIOL)

- b) Posterior chamber intra ocular lens (modified C-loop type)
- c) Singh and Worst's Iris claw lens
- d) Posterior chamber intra ocular lens (four loop type)

Question 12:

Identify the given instrument.



- a) Maddox tangent scale
- b) Maddox wing
- c) Synoptophore
- d) Retinoscopy

Question 13:

You would identify the instrument shown in the image below as:



- a) Nagel's anomaloscope
- b) Jackson's cross cylinder
- c) Maddox rod
- d) Retinoscope

Question 14:

Identify the instrument.



- a) Maddox wing
- b) Maddox glass

- c) Maddox rod
- d) Phoropter

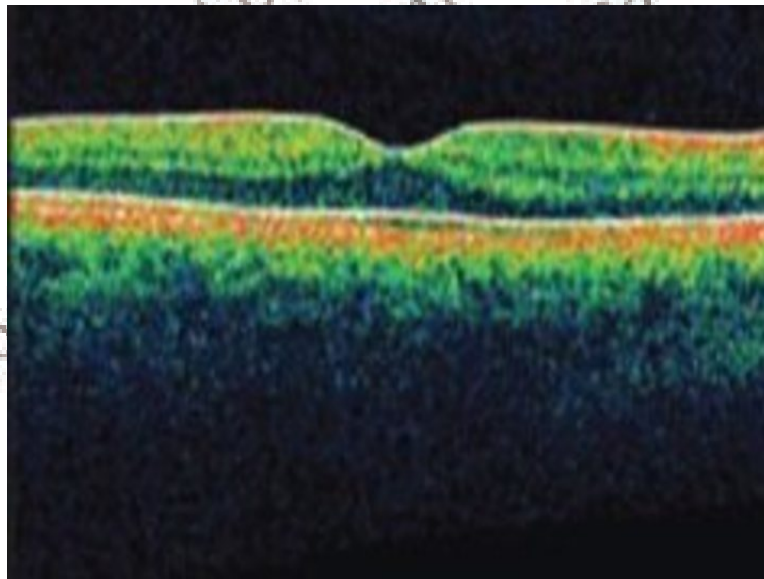
Question 15:

You can use a stenopaic slit for all of the following except _____.

- a) Fincham's test
- b) Corneal tattooing
- c) Optical Iridectomy
- d) To find out the axis in astigmatism

Question 16:

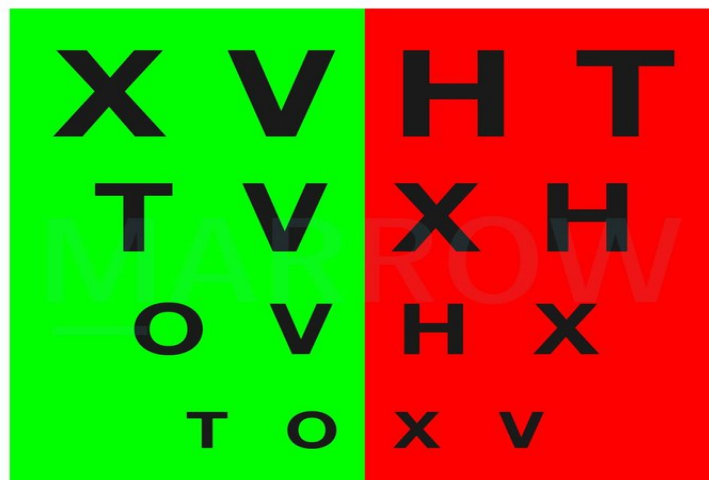
What is the investigation shown in the image below?



- a) Ultrasound Biomicroscopy
- b) Optical Coherence Tomography
- c) Corneal Topography
- d) Spectrometry

Question 17:

What is the principle of the test in which the patient is asked to read the chart shown in the image below?



- a) Spherical aberration
- b) Chromatic aberration
- c) Chromatic projection
- d) Diplopic projection

Question 18:

Which optical instrument provides an inverted and reversed image of the retina?

- a) Direct ophthalmoscope
- b) Indirect ophthalmoscope
- c) Hruby lens and slit lamp
- d) Goldmann contact lens and slit lamp

Question 19:

All are true about the instrument shown, except:



- a) It contains a titanium needle
- b) Frequency is 40000 Hz
- c) Contains a piezoelectric crystal
- d) Based on photodisruption

Question 20:

You see the following examination being performed in the ophthal OPD. What is the instrument being used?



- a) Castroviejo caliper

- b) Corneal trephine
- c) Hertel's exophthalmometer
- d) Schiottz tonometer

Question 21:

Identify the instrument shown in the image.



- a) Gonioscope
- b) Hruby lens
- c) Applanation tonometer
- d) Stenopic slit

Question 22:

While evaluating a child with suspected refractive error, you notice that the ophthalmology resident performs a retinoscopy. Which of the following is this test based on?

- a) Method of neutralization
- b) Imbert-Fick law
- c) Principle of bending of light
- d) Poiseuille law

Question 23:

Which of the following is true about the instrument shown in the image given below?



- a) Its is a convex lens.
- b) It provides a small field of vision.
- c) Power is +20 D.
- d) Fundus beyond the equator is visualized

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	b
5	a
6	b
7	a
8	d
9	c
10	b
11	a

12	b
13	b
14	c
15	b
16	b
17	b
18	b
19	d
20	c
21	a
22	a
23	b

Detailed Explanations

Solution to Question 1:

The instrument shown in the above image is an indirect ophthalmoscope, and the procedure is known as indirect ophthalmoscopy.

An indirect ophthalmoscope produces an inverted, or reversed, real image of 2 to 5 times magnification.

The area of field in focus is about 8 disc diameters.

It is a binocular, self-illuminated device where the condensing lens is usually required, and stereopsis is present.

Solution to Question 2:

Indirect ophthalmoscope needs a binocular vision.

Solution to Question 3:

The image shows a slit lamp. Slit-lamp biomicroscopy is the ideal method to visualize the fundus, especially for observing the optic disc changes seen in glaucoma.

The lenses used are as follows:

- Contact type: Goldmann three-mirror contact lens, modified Koeppel lens
- Non-contact type: Convex lens +60D, +78D, +90D (best lens)

- Hruby lens: planoconcave lens with dioptric power of -58.6 D.

Option B: The image given below shows a refractometer.



Option C: The image given below shows a keratometer



Option A: The image given below shows an ophthalmoscope



Solution to Question 4:

The pachymeter is used to measure corneal thickness.



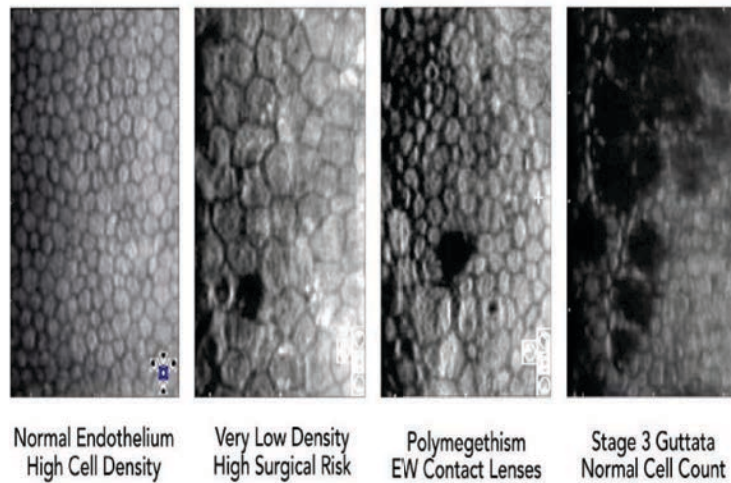
Options A and D: A keratometer (ophthalmometer) measures the curvature of the anterior surface of the cornea at two points about 1.25 mm on either side of its center.



Option C: Specular microscopy is the study of the corneal endothelium under a very high magnification (100 times greater than slit lamp biomicroscopy). Endothelium can be analyzed for cellular size, shape, density, and distribution. The healthy endothelial cell is a regular hexagon, and the normal cell density in a young adult is about 3000 cells/mm².



Cornea Endothelium



Solution to Question 5:

The image shows a Ziegler knife, which is used to open the lens capsule and aspirate the lens material in congenital cataract surgeries.

Solution to Question 6:

The given instrument is an indirect gonioscope, which is used to visualize the angle of the anterior chamber, i.e., for gonioscopy, and not the angle of the posterior chamber.

Types:

- Indirect gonioscope- provides mirror image of the opposite angle.
- Direct gonioscope- provides a direct view of the angle

Application:

- To view the angle of the anterior chamber
- Localization of the foreign body, abnormal blood vessels, or tumors in the angle
- Demonstration of the extent of peripheral anterior synechiae
- Direct gonioscope can be used during goniotomy

Solution to Question 7:

The clinical vignette points to a diagnosis of glaucoma not responsive to medical treatment and trabeculectomy and the device used is an ex-Press™ glaucoma implant (mini shunt).

It is a form of minimally invasive glaucoma surgery (MIGS) to increase aqueous drainage.

Ex-PRESS glaucoma implant is a small stainless steel device. It is inserted under a scleral flap during a modified trabeculectomy. Complications such as hypotony are lower than with standard trabeculectomy.

Glaucoma drainage devices are devices that allow aqueous outflow by establishing communication between the anterior chamber and sub-Tenon's space/sub-conjunctival space. While these devices were traditionally reserved for high-risk patients, or after trabeculectomy had failed, they are becoming increasingly popular.

The commonly used devices are as follows:

1. Shunts using episcleral explants

- Ahmed glaucoma valve: Silicone
- Molteno implant - Polypropylene plate and silicone plate
- Baerveldt implant - Silicone

2. Mini-shunts

- Ex-Press™ Mini-Shunt: Stainless steel
- iStent® - Titanium

Solution to Question 8:

The image given above shows a Lister's perimeter. It is used in manual perimetry for visual field charting

Classification:

- Manual: Confrontation, Listers, Campimetry, Goldmann perimeter
- Automated: Octopus, Field Master, Humphrey Field Analyzer
- Static: Friedman, Automated
- Kinetic: Confrontation, Listers, Scotometry, Goldmann

The best type is automated perimetry.

Solution to Question 9:

Tonopen is used for measuring intraocular pressure (IOP) in a scarred cornea.

It is a hand-held electronic contact tonometer (a modified version of the older Mackay–Marg tonometer). Besides portability, its main advantage is the ability to measure IOP reasonably accurately in eyes with distorted or edematous corneas and through a soft contact lens.

Types of tonometry:

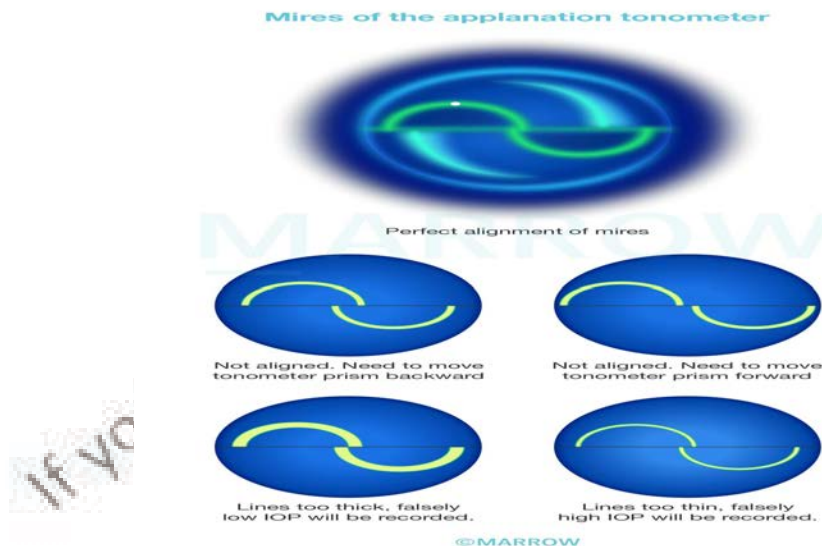
- Indentation tonometry- Reading depends on scleral rigidity, e.g., Schiotz tonometer

- Applanation tonometry- Based on the amount of force needed to flatten or applanate a known area of the cornea, Imbert–Fick principle. This is further divided into two types -
- Fixed force: IOP is proportional to the area, e.g., Maklakov, Barraquer
- Fixed area and variable force: IOP is proportional to force. The most reliable is Goldmann.
- Self-measurement: Rebound
- Non-contact type: Pulse Air, Air-Puff
- Irregular cornea: Tonopen, MacKay–Marg

Recent tonometers:

- Pascals tonometer (dynamic contour tonometry [DCT]): Contact; used for post-LASIK IOP measurement.
- Reichert Ocular Response Analyzer®: Non-contact

Note: Tonography is a non-invasive technique for determining the facility of aqueous outflow. The image given below shows fluorescein-stained semicircular mires.



Solution to Question 10:

Cataract surgery with an intraocular lens (IOL) implantation is the treatment of choice in congenital cataracts. The above image depicts a modified C-loop type of posterior chamber intraocular lens (PCIOL) used in such cases.

This surgery involves an extracapsular cataract extraction (ECCE) with primary posterior capsulorhexis and anterior vitrectomy. After removal of the cataractous lens, a posterior chamber intraocular lens (PCIOL) is implanted.

PCIOL rest entirely behind the iris. They may be supported and fixed as follows:

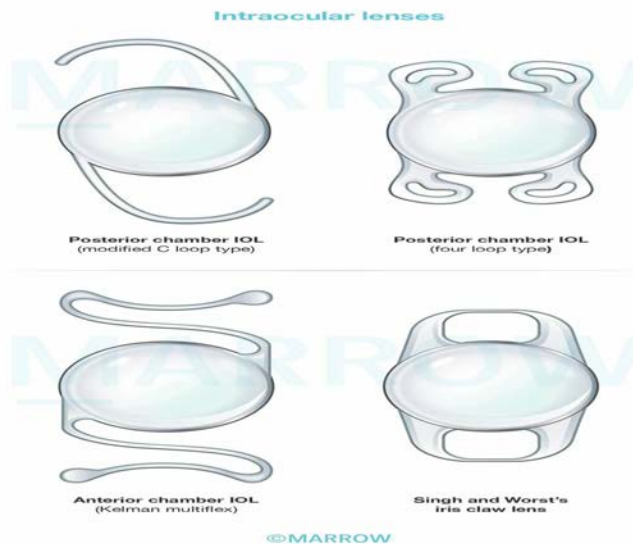
- Capsular bag fixation of PCIOL (most ideal method)

- Ciliary sulcus fixation
- Sclera fixation

Parts of the IOL:

- The central part overlying the optic axis is called the optic.
- The peripheral arms used for placement and stabilization are the haptics.

The image given below shows different types of intraocular lens.



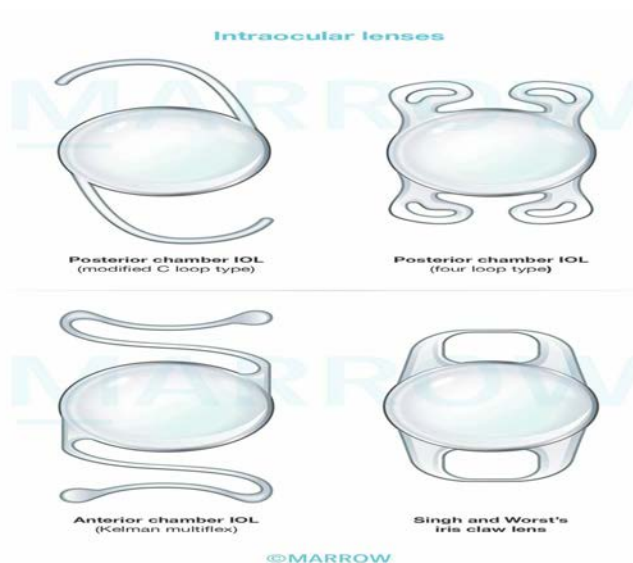
Solution to Question 11:

The image depicts Kelman multiflex type of anterior chamber intraocular lens (ACIOL). It is the most commonly used ACIOL.

Anterior chamber IOLs lie entirely in front of the iris and are supported in the angle of the anterior chamber. IOL implantation is the method of choice for correcting aphakia.

These are associated with a higher incidence of bullous keratopathy.

The image given below shows different types of intraocular lens.



Solution to Question 12:

The given image depicts a Maddox wing. It is an instrument by which the amount of heterophoria for near vision can be measured.

Methods to examine heterophoria:

- Testing for refractive error
- Cover–uncover test
- Prism bar cover test (PBCT)
- Maddox rod test
- Maddox wing test
- Maddox tangent scale: heterophoria for distance
- Measurement of the near point of convergence: Using RAF rule (Normally, it is 70 mm)
- Measurement of fusional reserve- done with a synoptophore or prism bar

Solution to Question 13:

The image shows Jackson's cross cylinder (flip-cross technique) which is used to refine the power and axis of the cylindrical lens.

Option A: Nagel's anomaloscope is used to test color vision.



Option C: Maddox rod consists of parallel red or colorless cylinders stacked together. Maddox rod test is a macular function test used to measure the degree of heterotropia or heterophoria in cases of torsional diplopia.



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Option D: Retinoscope

Streak retinoscope



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Solution to Question 14:

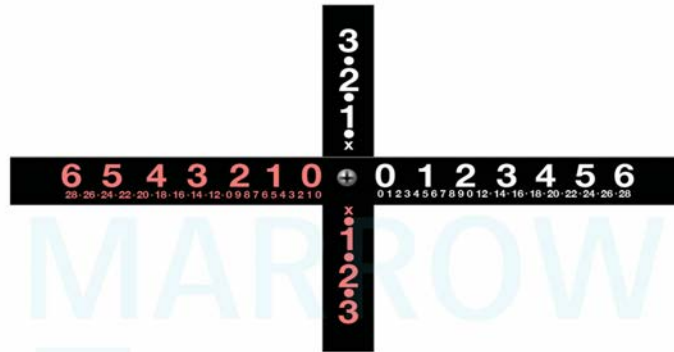
The above instrument is a Maddox rod.

Maddox Rod Test

A Maddox rod is a specially constructed device, consisting of a series of parallel cylinders, that converts a point source of light into a line image. It can be used to test for horizontal and vertical deviations, and when used in conjunction with another Maddox rod, for cyclo deviations.

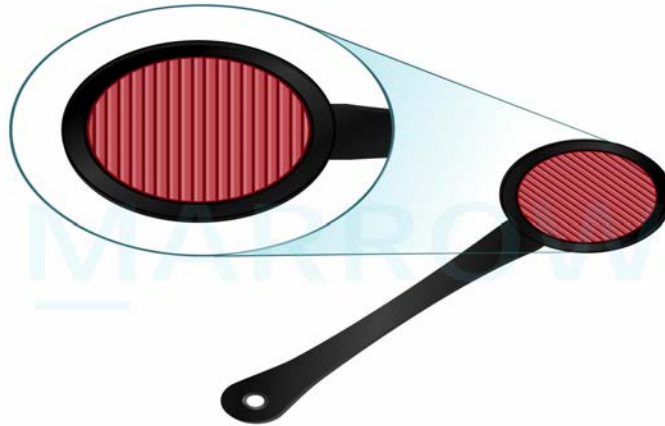
A patient is asked to fix on a point light in the center of Maddox tangent scale placed 6 meters away. A Maddox rod is placed in front of one eye with the axis of the rod parallel to the axis of deviation. The patient will see a point light with one eye and a red line with the other. Due to dissimilar images of the two eyes, fusion is broken and heterophoria becomes manifest. The number on the Maddox tangent scale where the red line falls will be the amount of heterophoria in degrees.

Maddox tangent scale



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Maddox rod



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Maddox rod test

Examiners's view

Right



Left



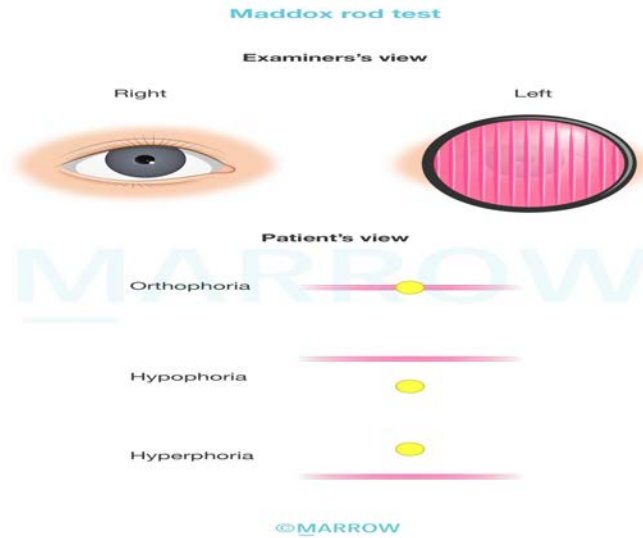
Patient's view

Orthophoria

Left Exophoria
/tropia

Left Esophoria
/tropia

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Solution to Question 15:

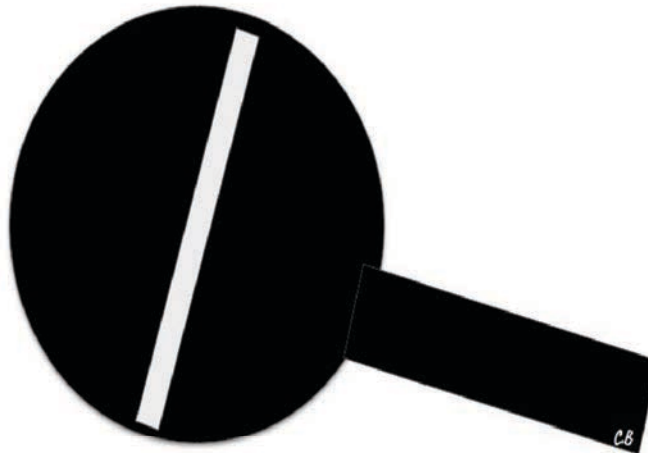
Stenopaeic slit is not used for corneal tattooing.

Stenopaeic slit is an ophthalmological instrument that is used in:

- Fincham's test (Option A) : This test is used to differentiate between the colored halos due to immature cataract and corneal abnormalities such as corneal edema due to glaucoma.
- To find out the axis in astigmatism (option D) for prescribing the correction glasses.
- To find out the optimal site for optical iridectomy (option C) for treating cases of central macular or leucomatous corneal opacities.

Corneal tattooing was done mainly in the past for cosmetic purposes. It is commonly done for firm scars of corneal opacities in a eye without any useful vision.

The below given image is of stenopaeic slit.



Solution to Question 16:

The given image depicts Optical Coherence Tomography (OCT) of a normal macula.

It is a non-invasive, noncontact imaging system providing high-resolution cross-sectional images of the posterior segment. It uses near-infrared light interferometry to do so.

Solution to Question 17:

The given image depicts a duo-chrome test based on chromatic aberration. It is used to refine the final sphere in subjective refraction.

The patient is asked to compare the clarity of the letters on the green and the red side. If a myopic sees the letters of the green side clearer, he is overcorrected. So, $+0.25$ D sphere is added or -0.25 D is removed. However, if a myopic sees the letters on the red side clearer, he is undercorrected and -0.25 D sphere is added. With optimal spherical correction, the letters on the red and green halves of the chart appear equally clear.

Solution to Question 18:

An indirect ophthalmoscope produces a real, inverted and laterally reversed image of the fundus.

All the other instruments mentioned including direct ophthalmoscope, Hruby lens and Goldman contact lens produce a virtual and erect image of the fundus.

Solution to Question 19:

The given image is of a phacoprobe showing the needle and the silicone irrigation sleeve. This is used in phacoemulsification to break the nucleus into fragments and aspirate them.

Solution to Question 20:

The image shows Hertel's exophthalmometer.

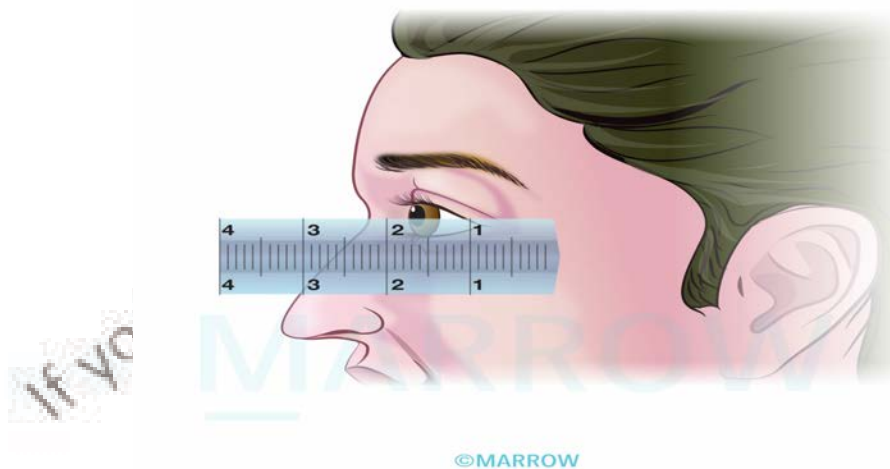
Exophthalmometry: measures protrusion of the apex of the cornea from the outer orbital margin (with the eyes looking straight ahead).

An absolute protrusion of >20 mm or a difference of >2 mm between the two eyes is labeled as proptosis.

Instruments used to measure proptosis:

- Luedde exophthalmometer
- Hertel's exophthalmometer- most common instrument used. Its advantage is that it measures both eyes simultaneously.

Luedde exophthalmometer



Solution to Question 21:

The given instrument is a gonioscope.

It is used to visualize the angle of the anterior chamber (gonioscopy).

Types:

- Indirect gonioscope- provides a mirror image of the opposite angle
- Direct gonioscope- provides a direct view of the angle

Application:

- To view the angle of the anterior chamber

- Localization of foreign body, abnormal blood vessels, or tumors in the angle
- Demonstration of the extent of peripheral anterior synechiae
- Direct gonioscopes used during goniotomy

Solution to Question 22:

Retinoscopy is an objective method used to determine the error of refraction by applying the method of neutralization. It is also called skiascopy or shadow test.

Principle: When light is reflected from a mirror into the eye, the direction in which light will travel across the pupil will depend upon the eye's refractive state.

Distance: Usually, the doctor sits at a distance of 1 meter or an arm's length.

Types of retinoscopes:

- Priestley–Smith retinoscope
- Streak retinoscope

The image given below shows a retinoscopy being performed on a patient.



Priestley-Smith retinoscope



©Marrow

Streak retinoscope



©Marrow

Solution to Question 23:

The given image shows a Hruby lens, and it provides a small field of vision.

It is a planoconcave lens. Dioptric power is $-58.6D$. It provides a small field of vision, with low magnification. Fundus beyond the equator cannot be visualized.